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Swayze et al.

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(54) **STAPLE CARTRIDGE ELECTRICAL CONTACTS**

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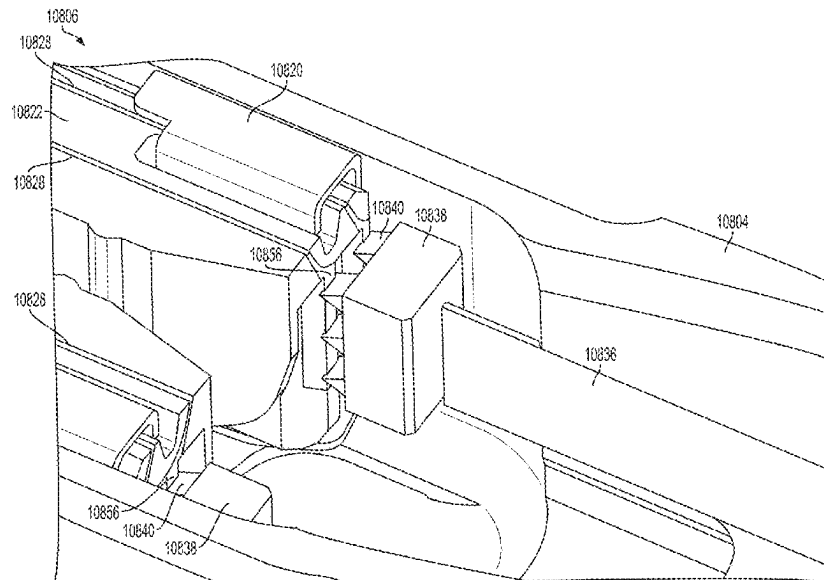
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(57) **ABSTRACT**

A staple cartridge assembly for use with a surgical stapler and surgical stapling systems are disclosed. The staple cartridge comprises a cartridge body comprising a proximal end. A plurality of staples. A plurality of electrical contacts positioned at the proximal end of the cartridge body and electrically coupleable to the electrical connector upon the installation of the replaceable staple cartridge assembly in the end effector along a distal-to-proximal installation motion and electrically decoupleable from the electrical connector upon removal of the replaceable staple cartridge assembly from the end effector along a proximal-to-distal removal motion.

14 Claims, 160 Drawing Sheets



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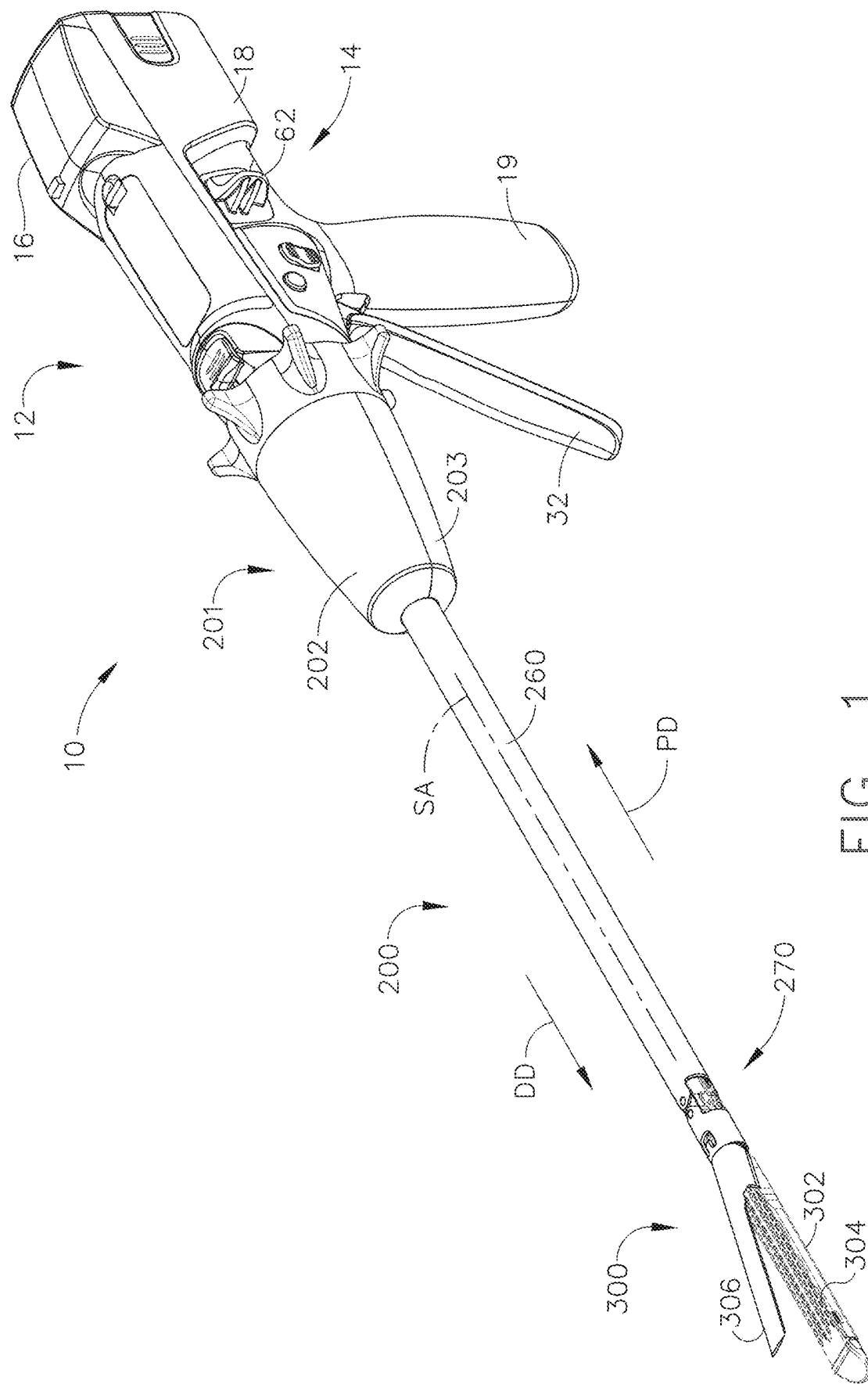


FIG. 1

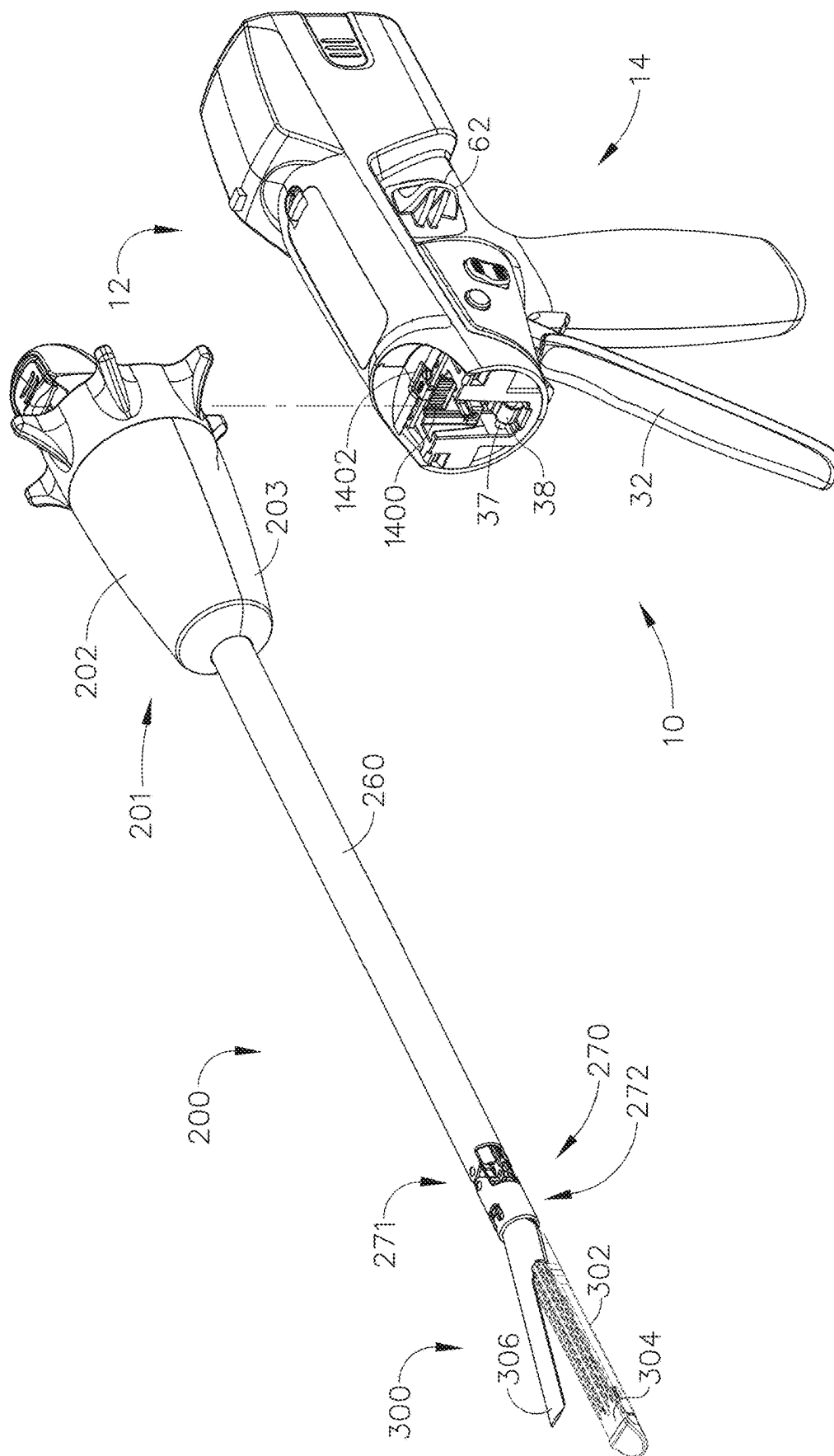
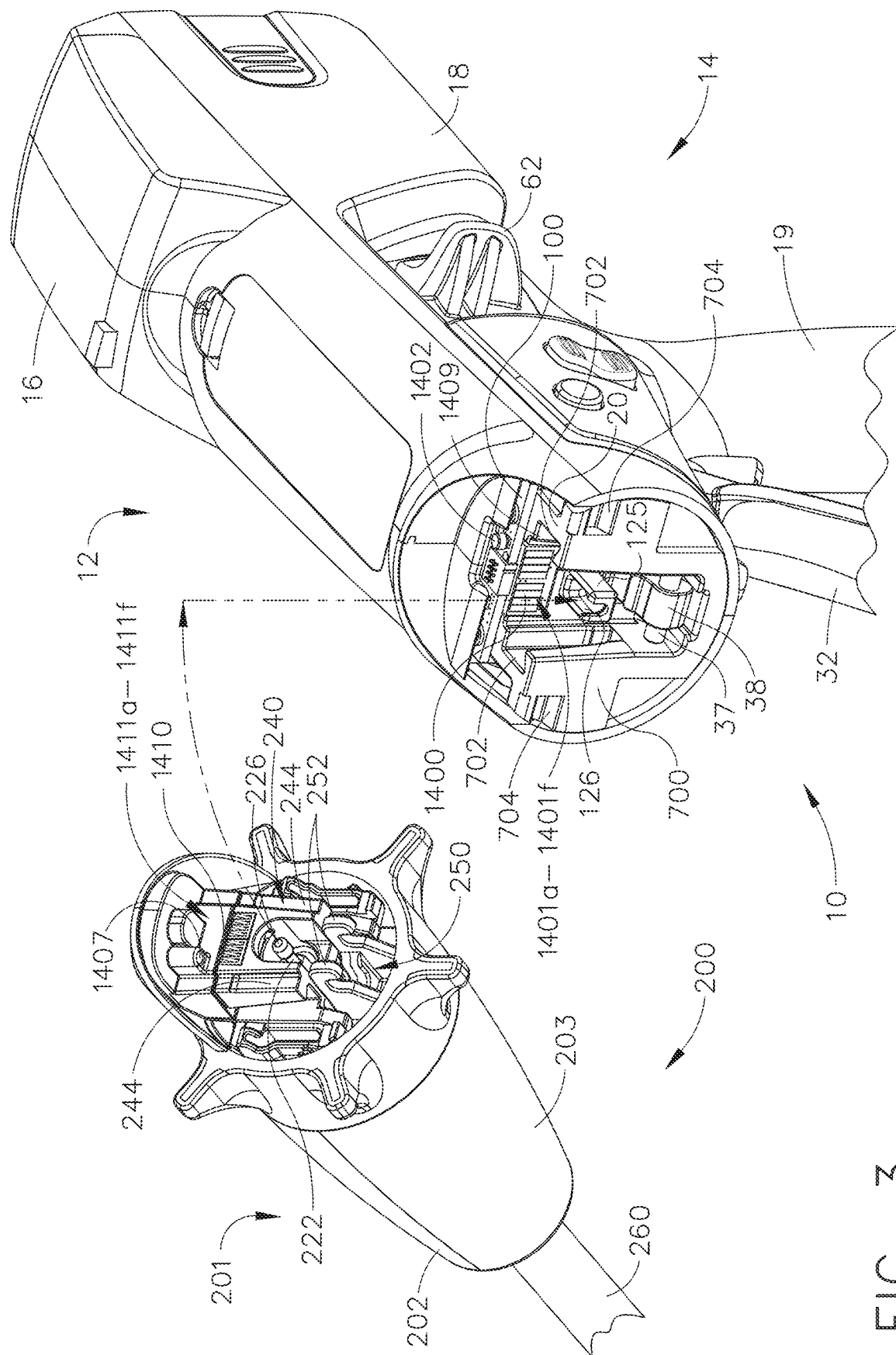


FIG. 2



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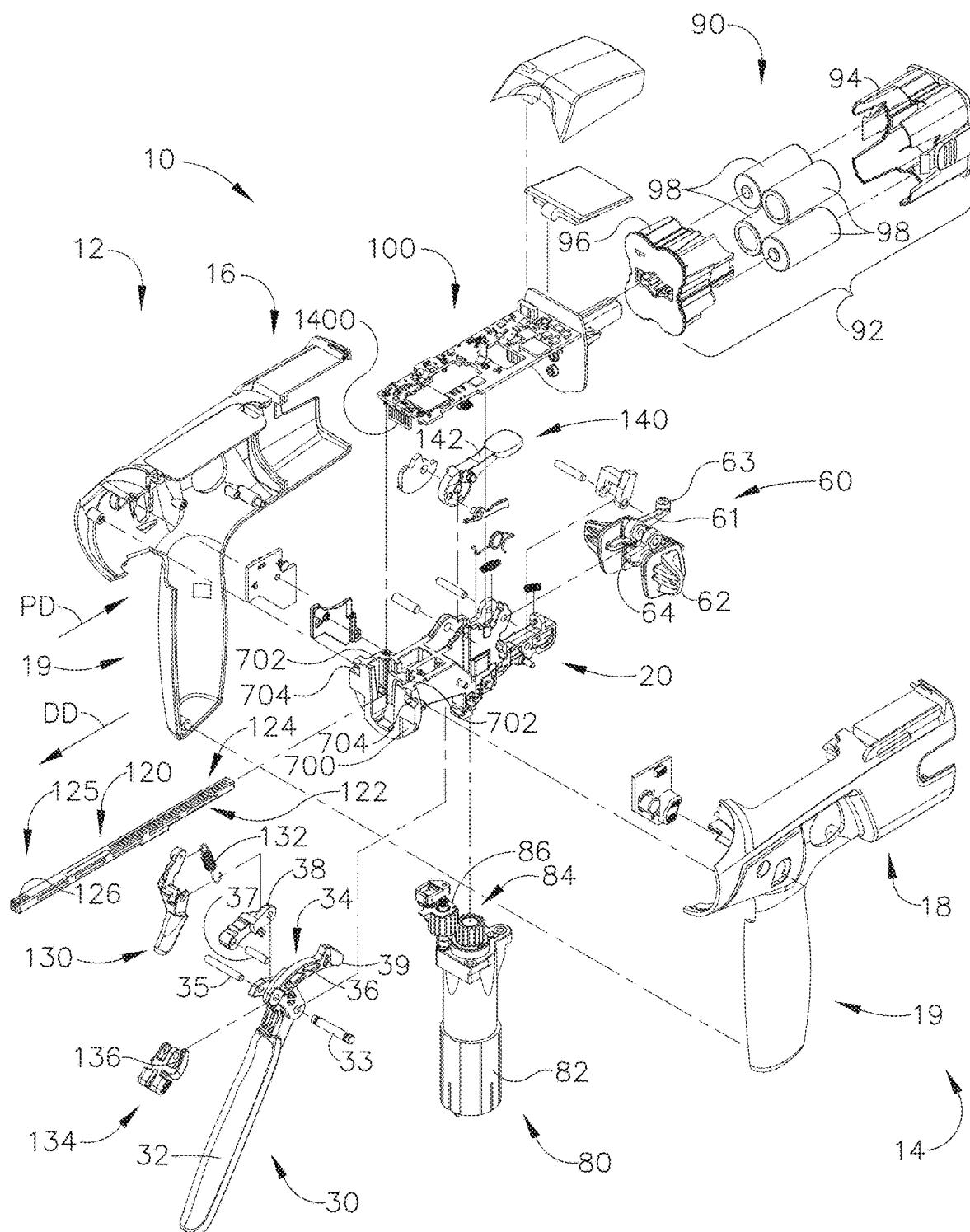


FIG. 4

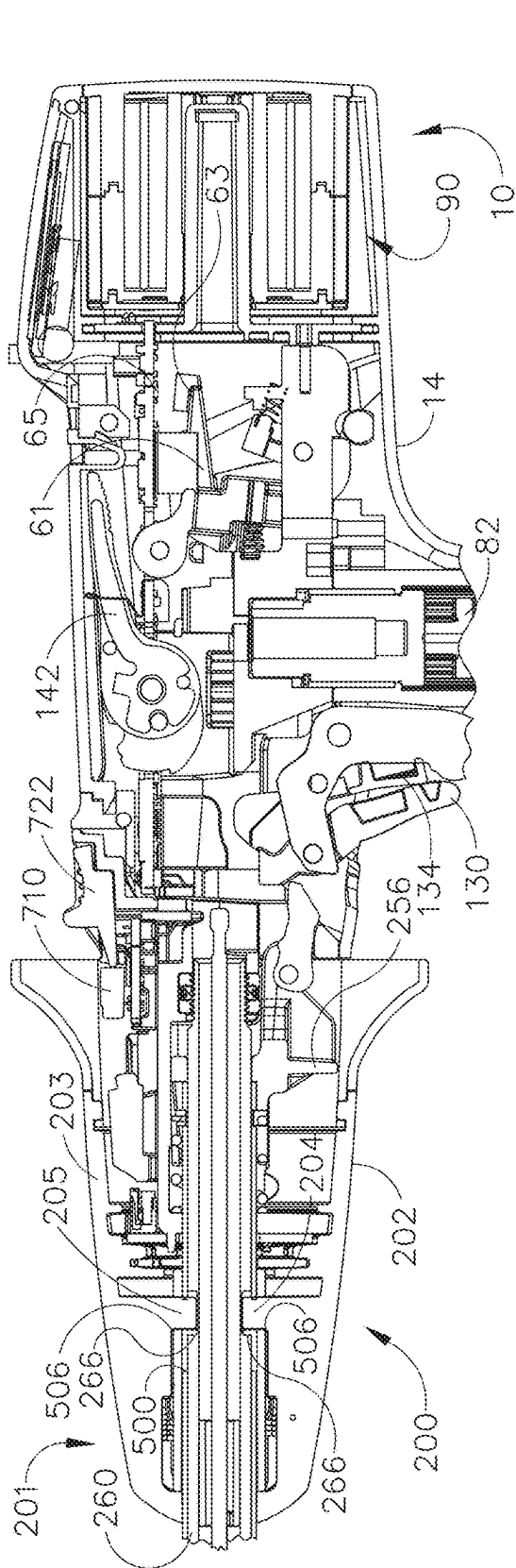


FIG. 5

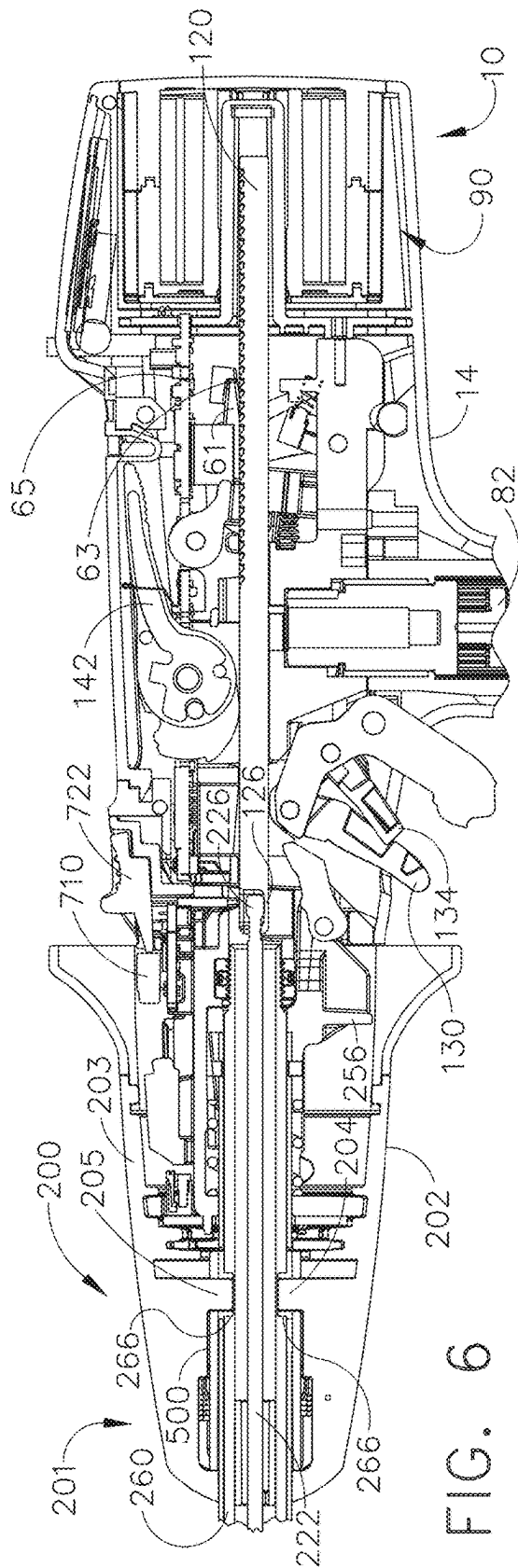


FIG. 6

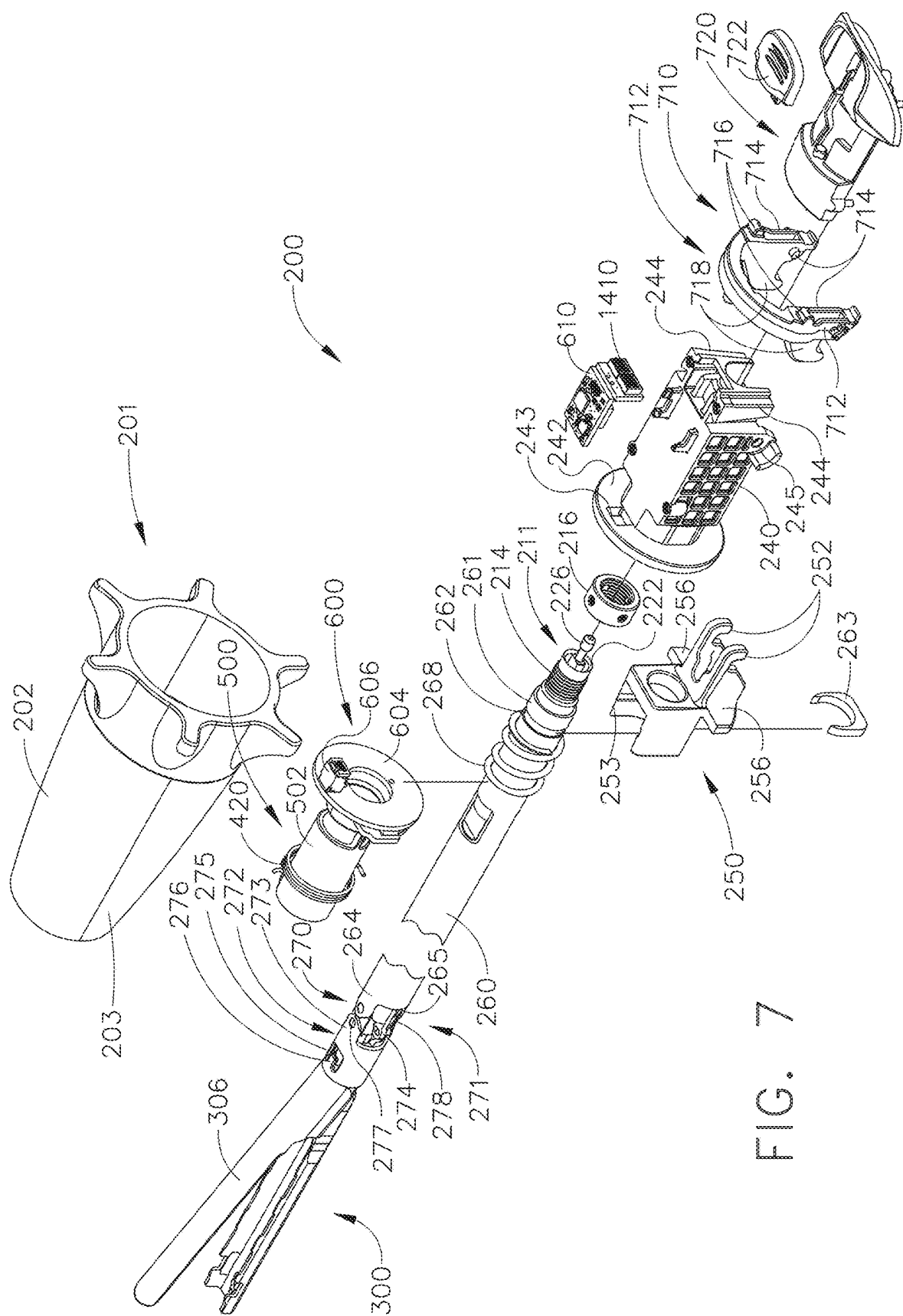


FIG. 7

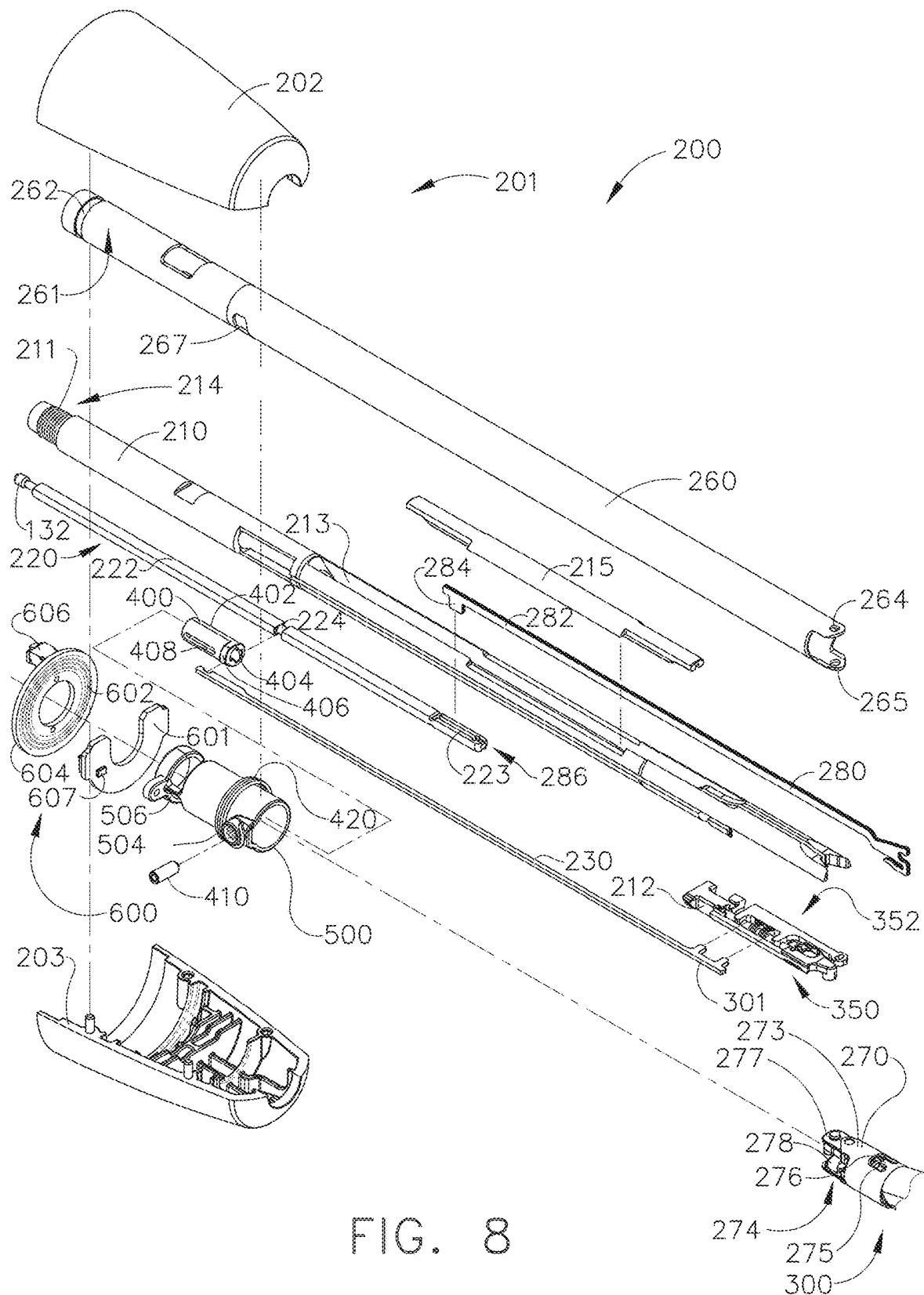


FIG. 8

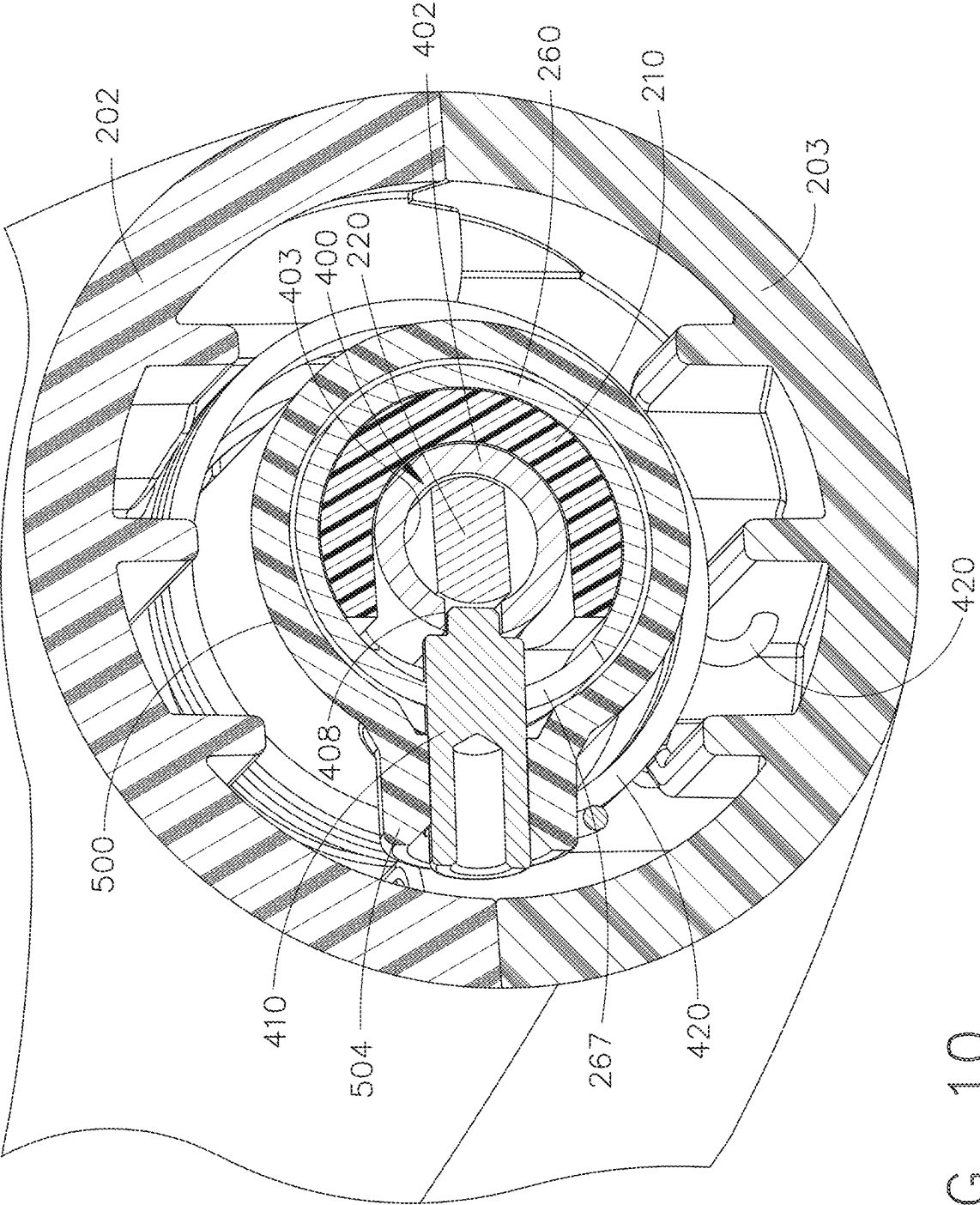


FIG. 10

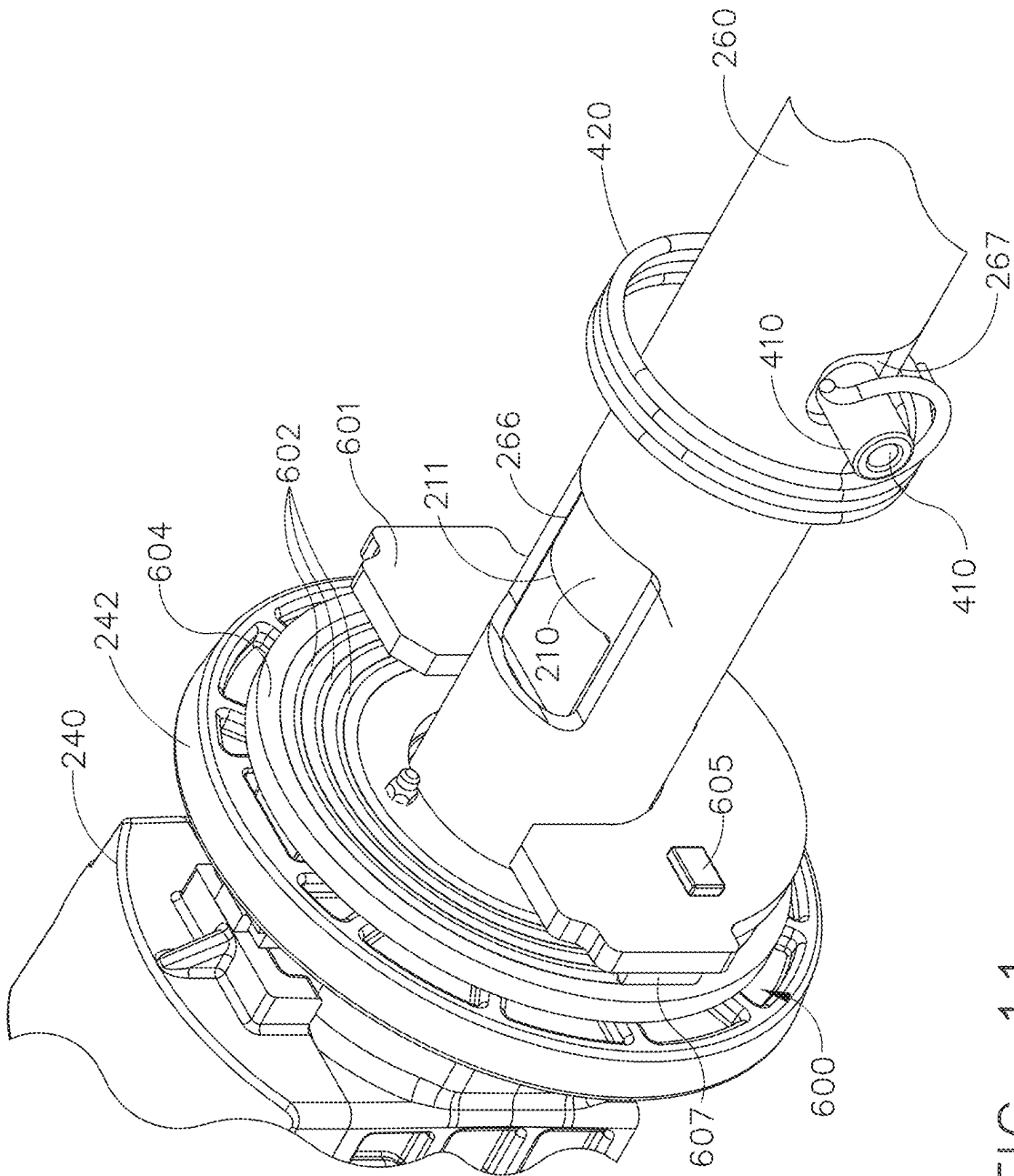


FIG. 11

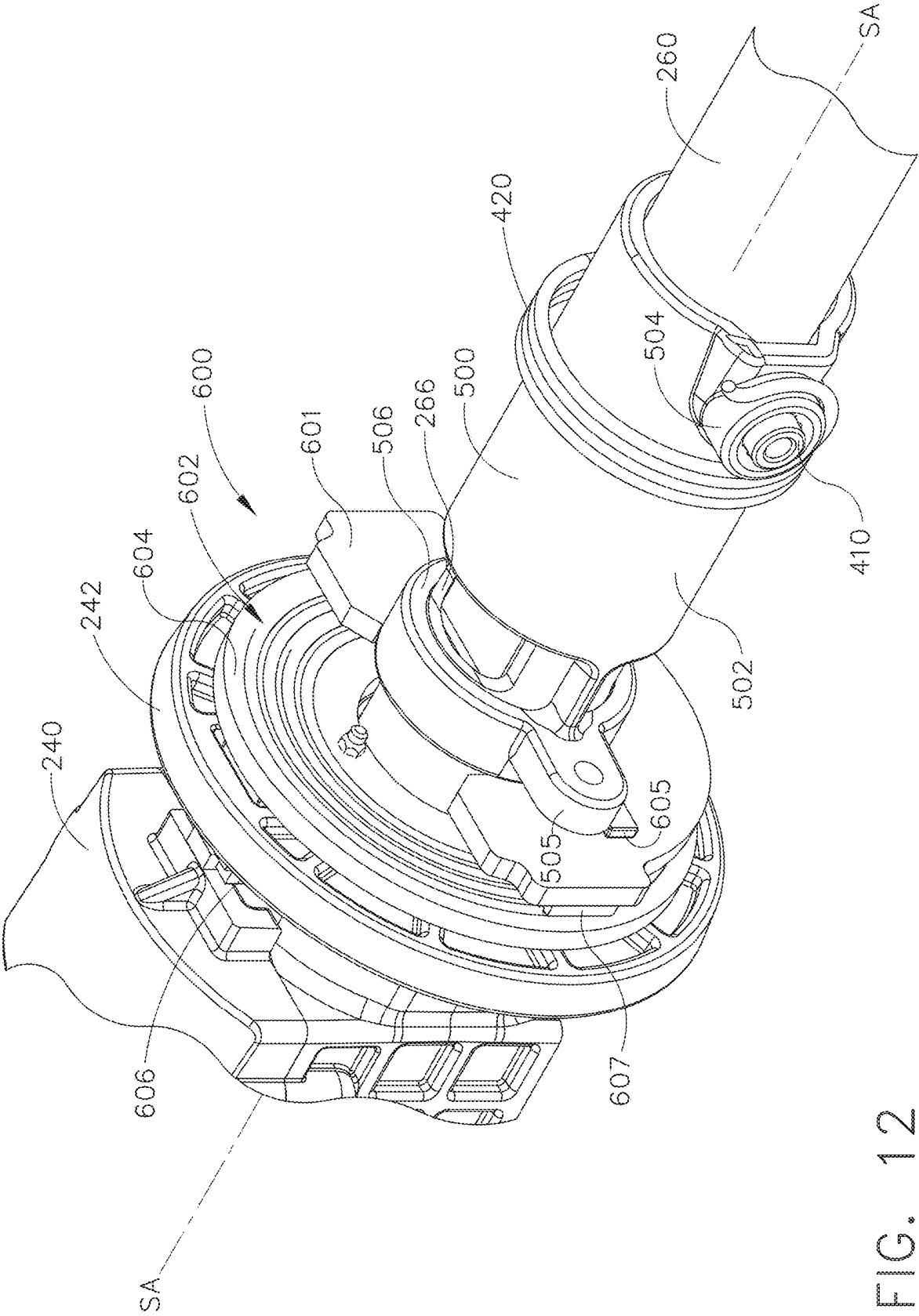


FIG. 12

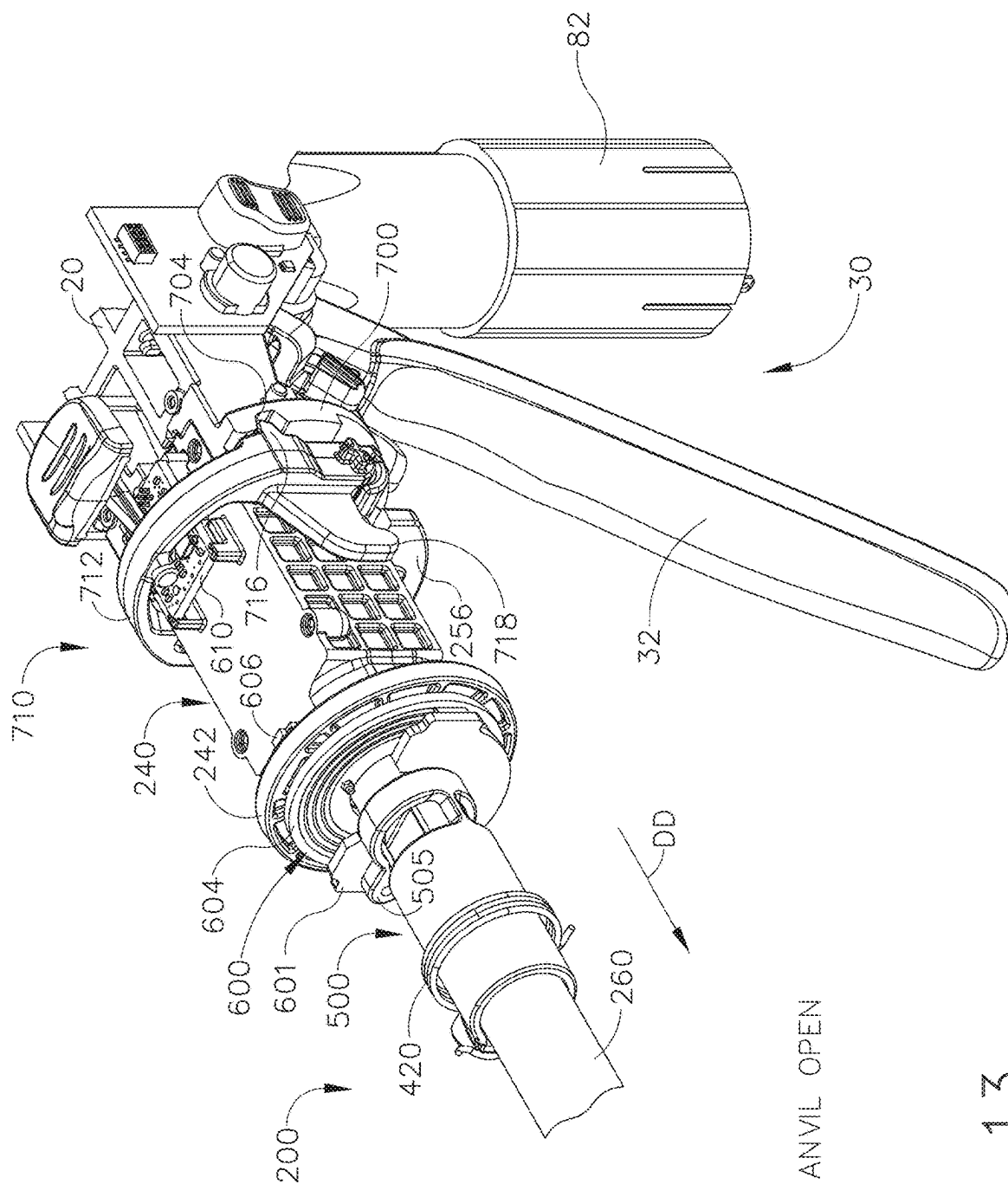
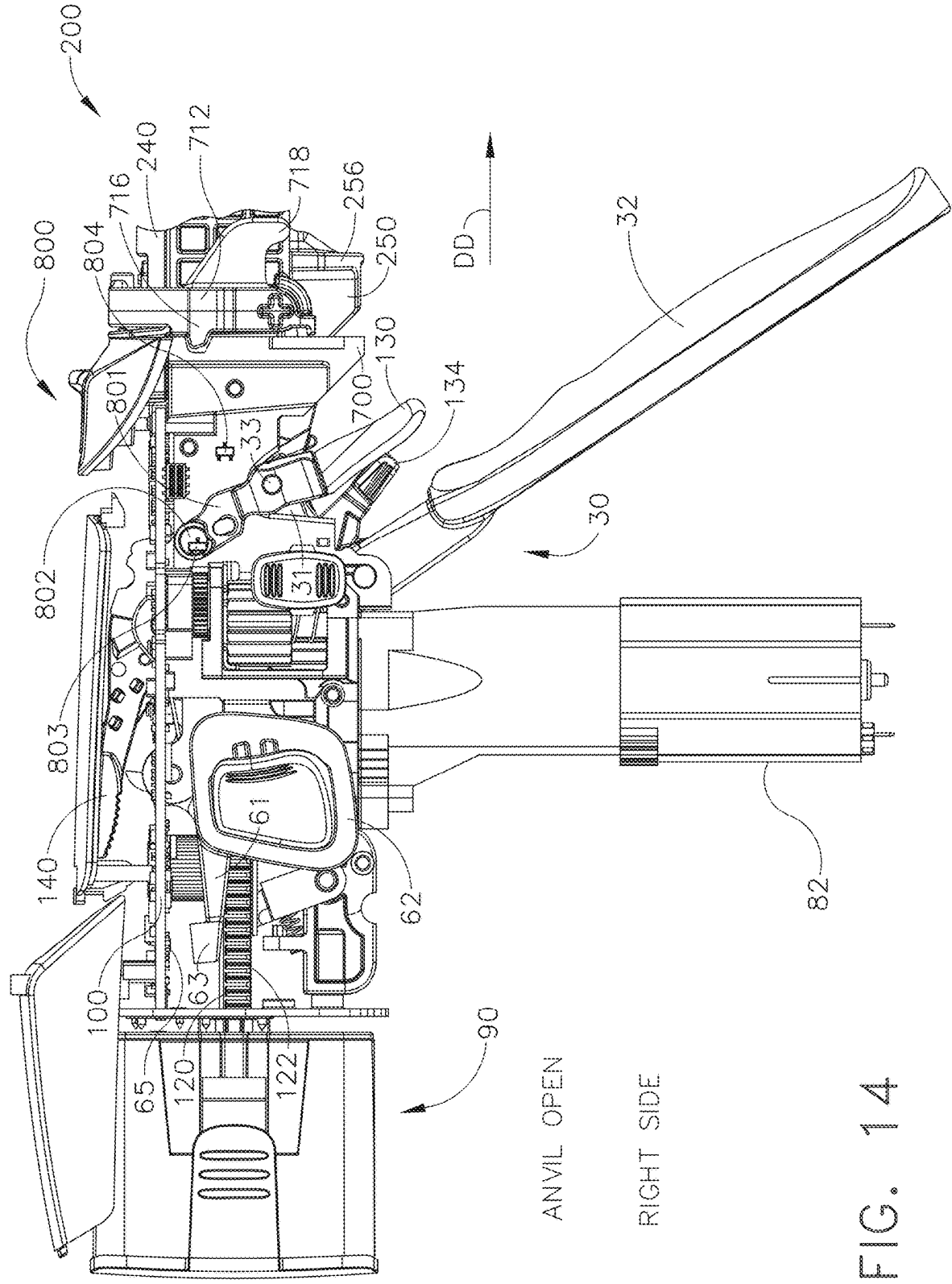
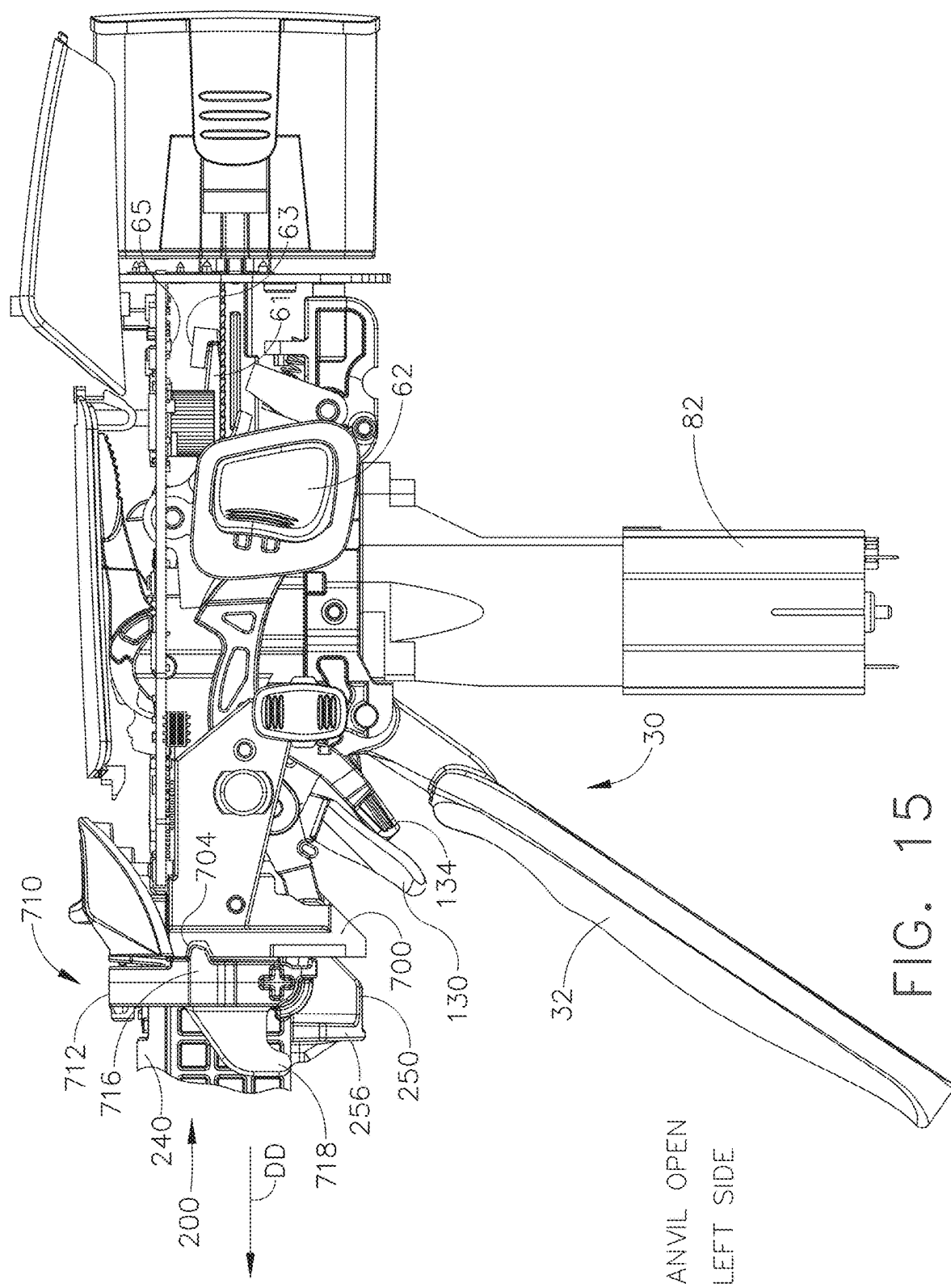
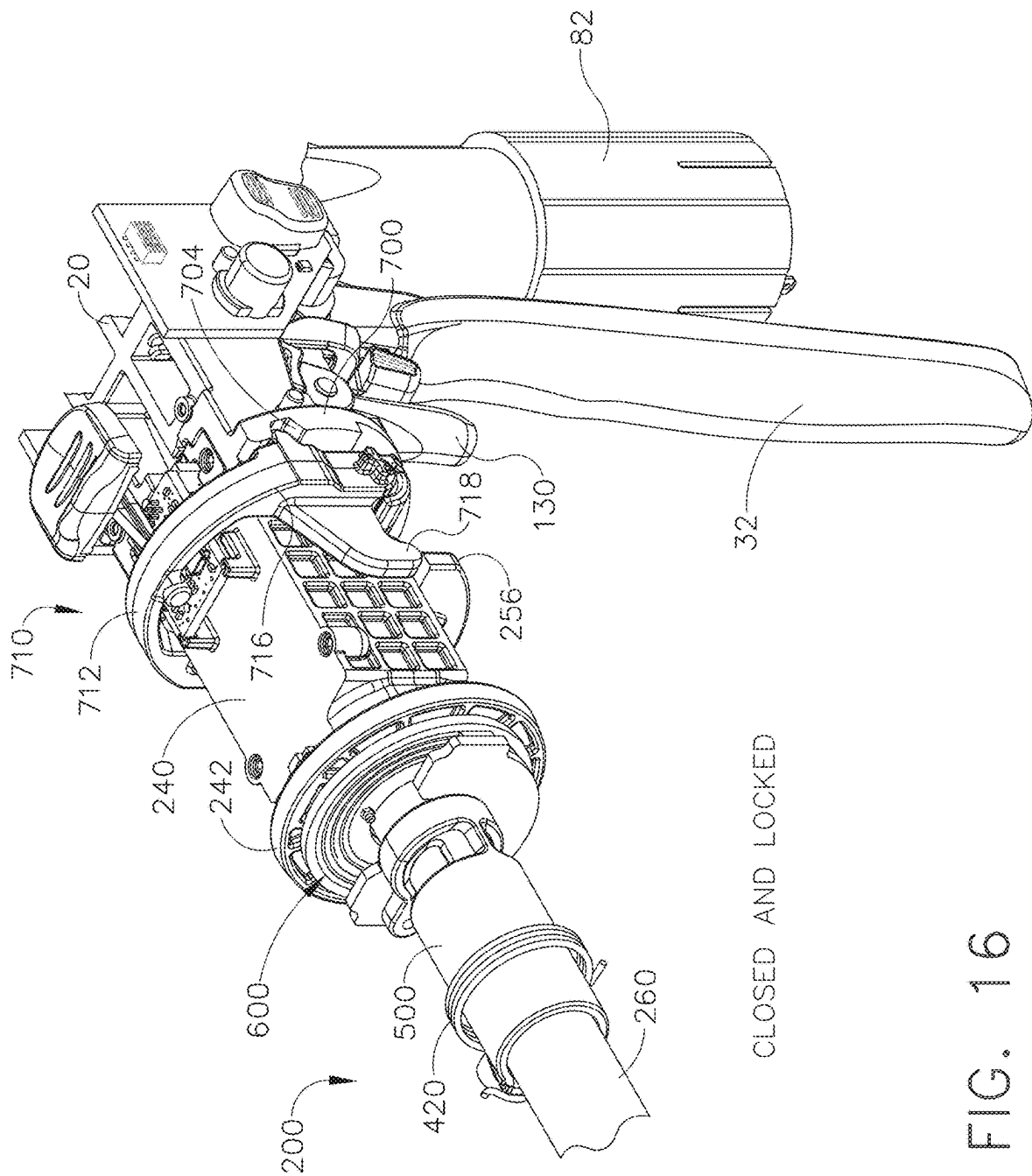


FIG. 13







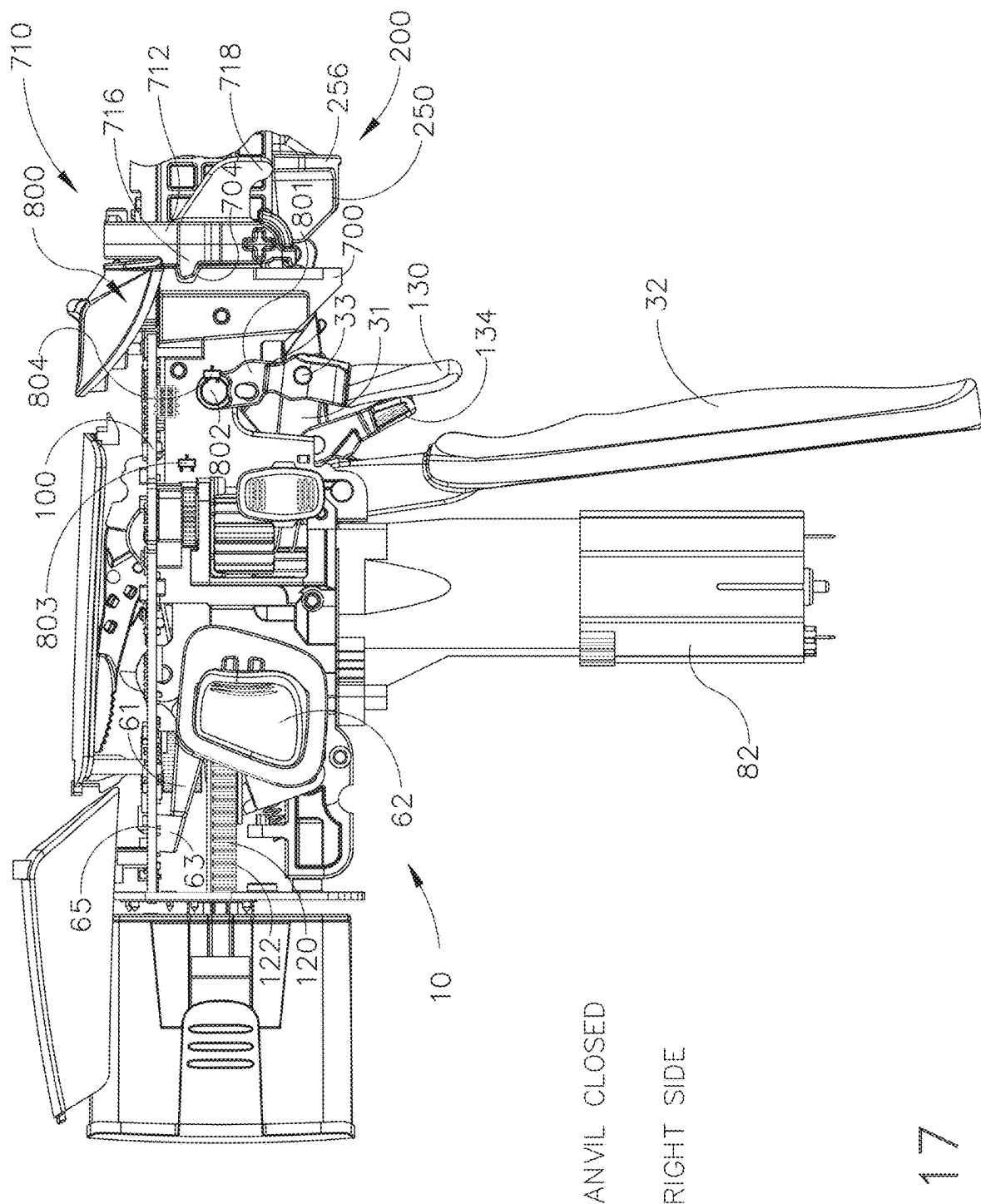
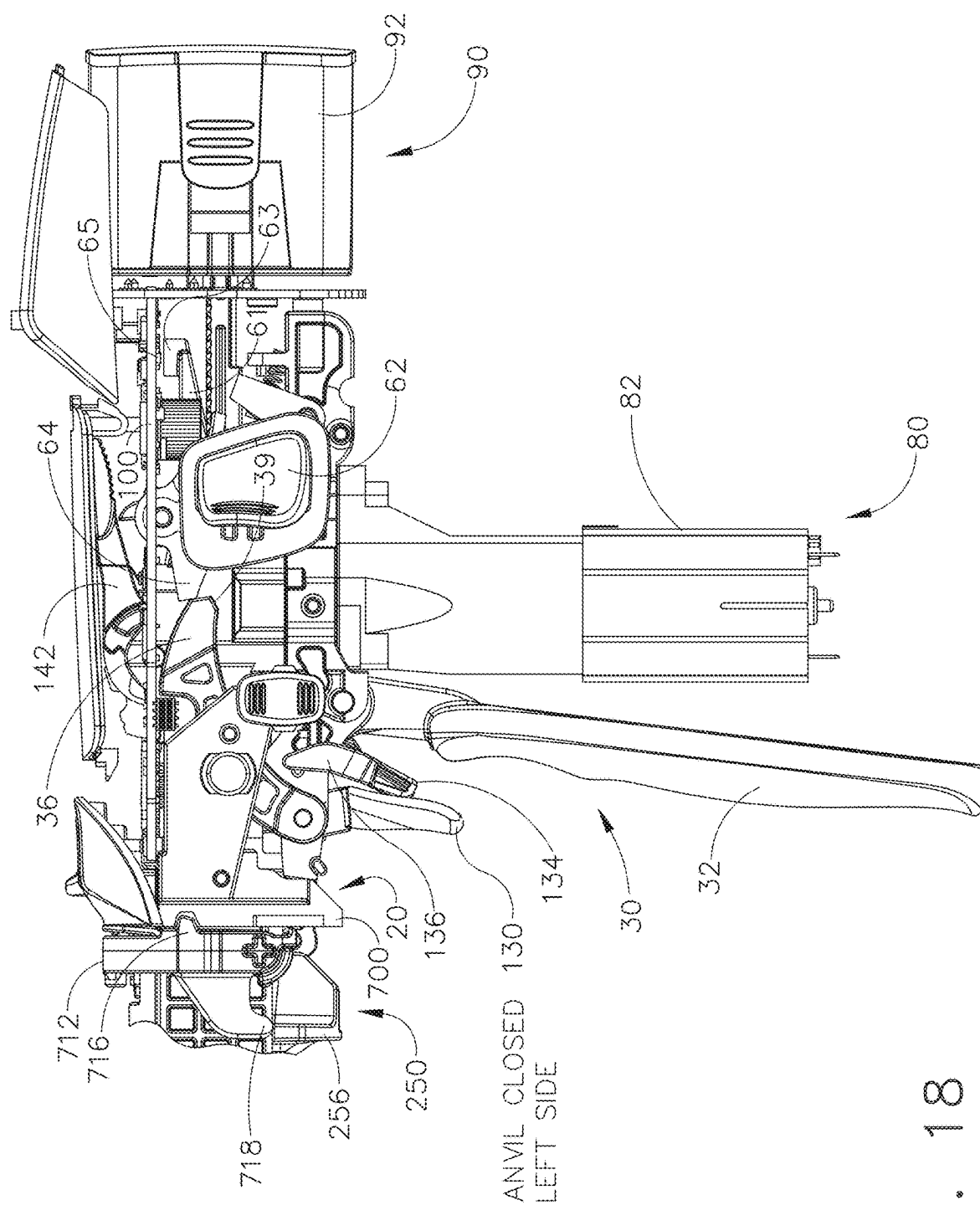


FIG. 17



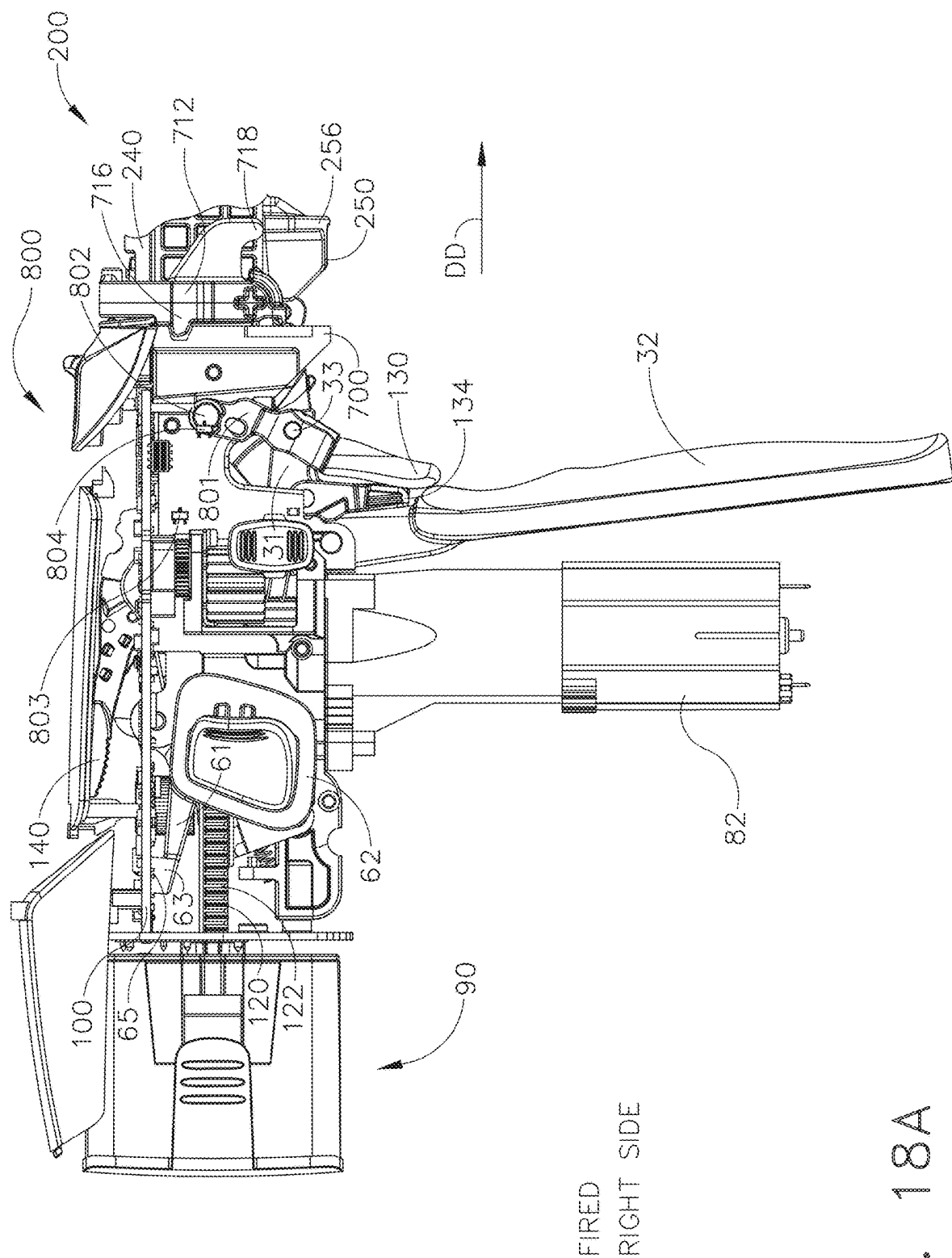
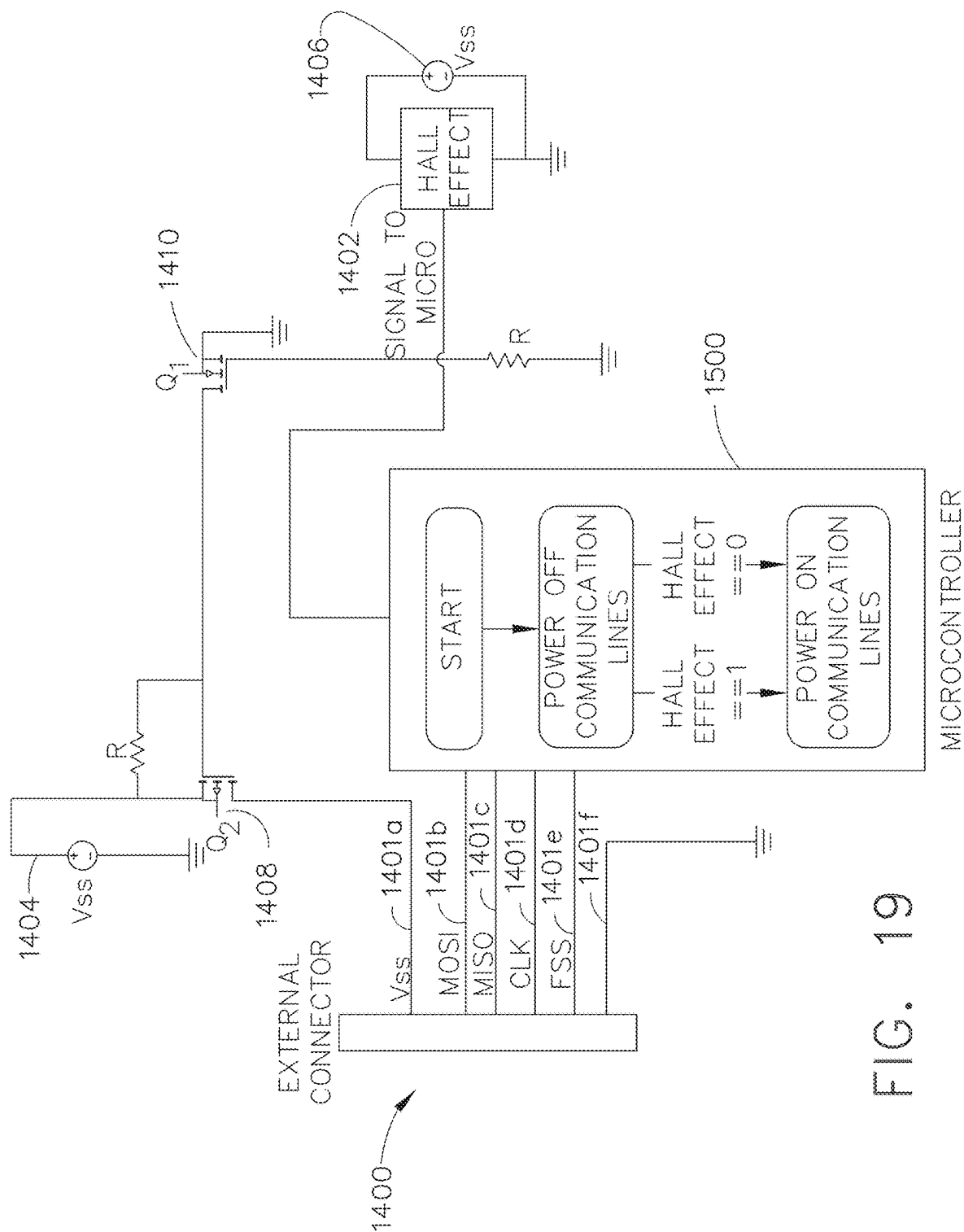


FIG. 18A



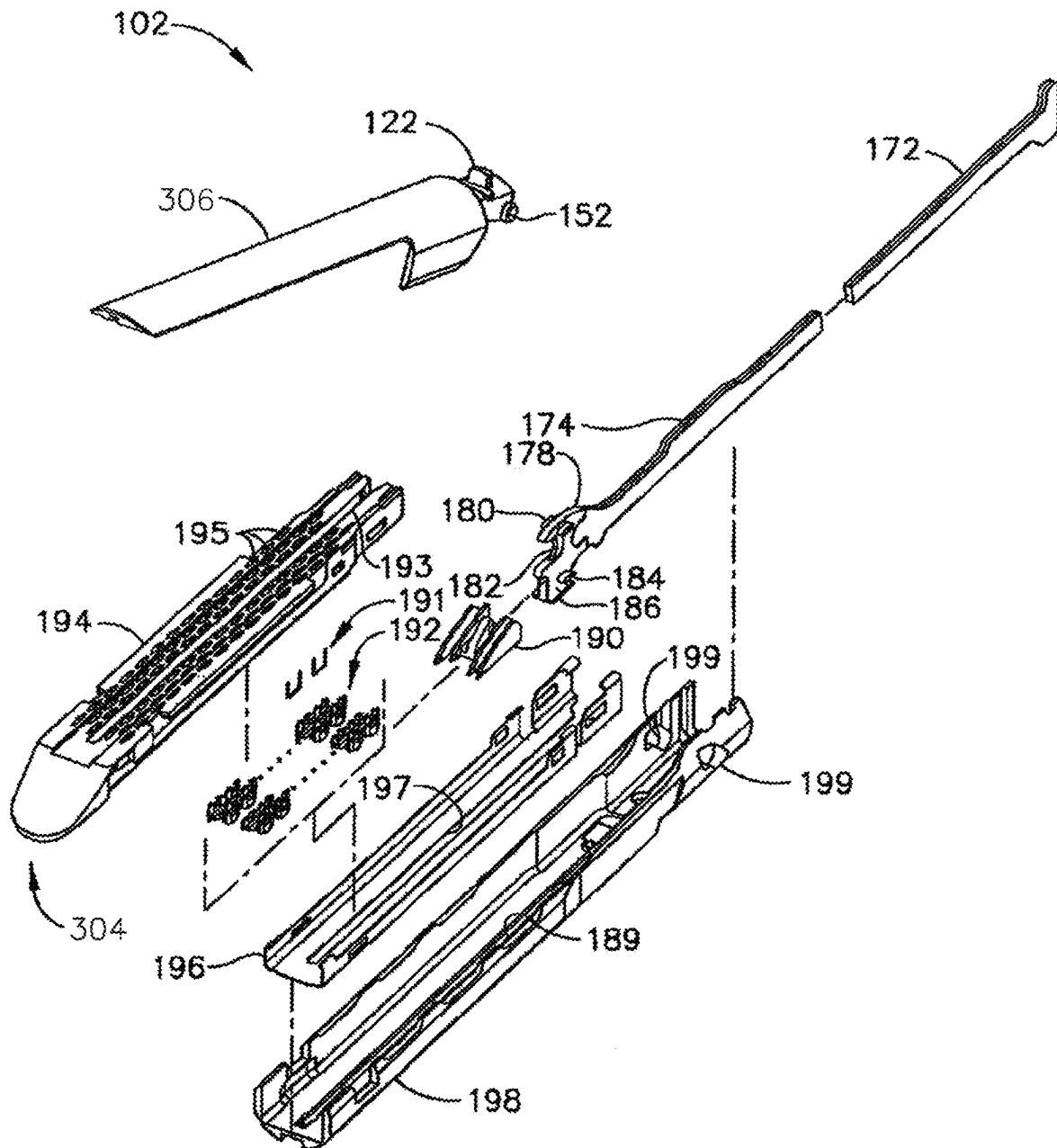
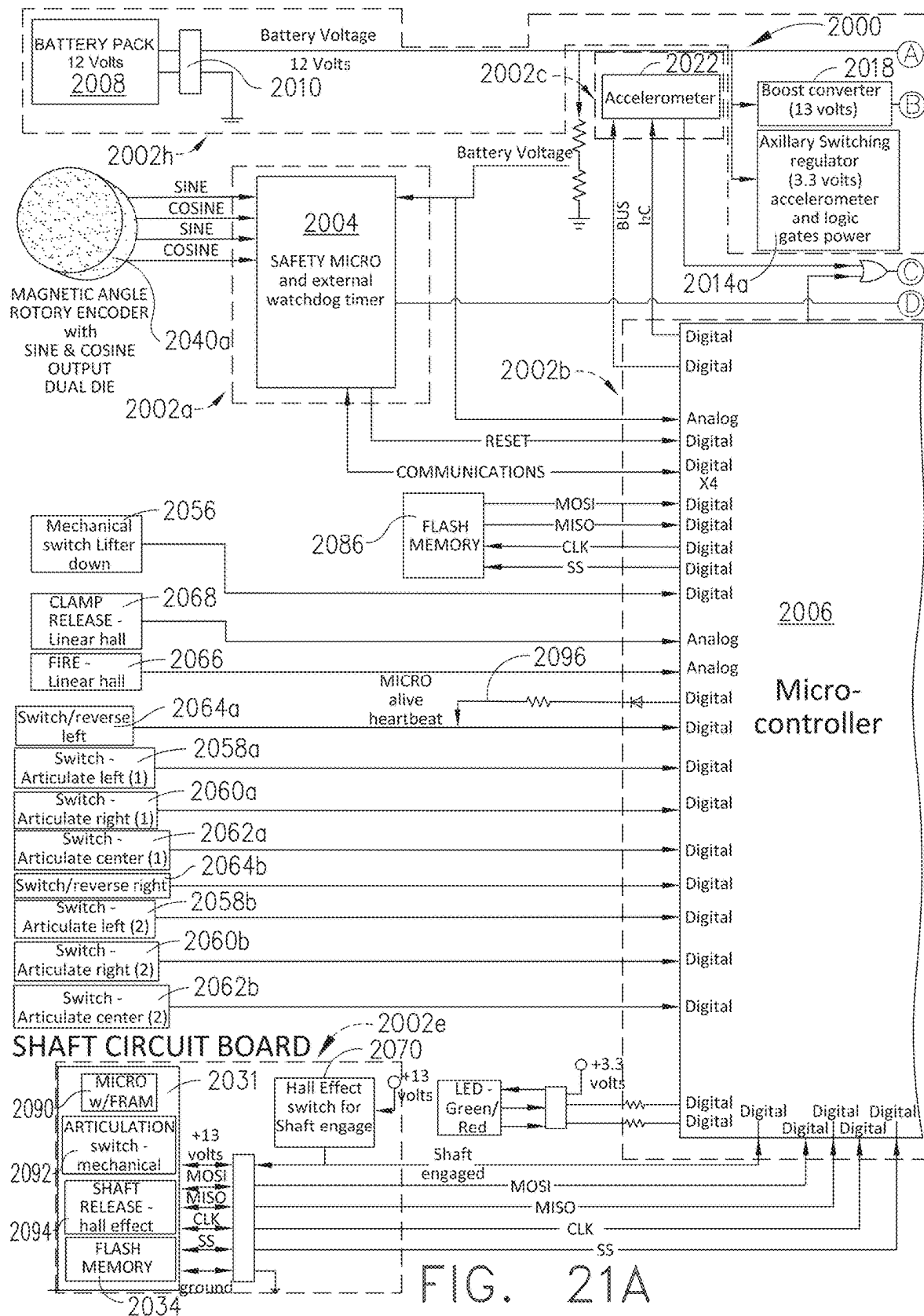
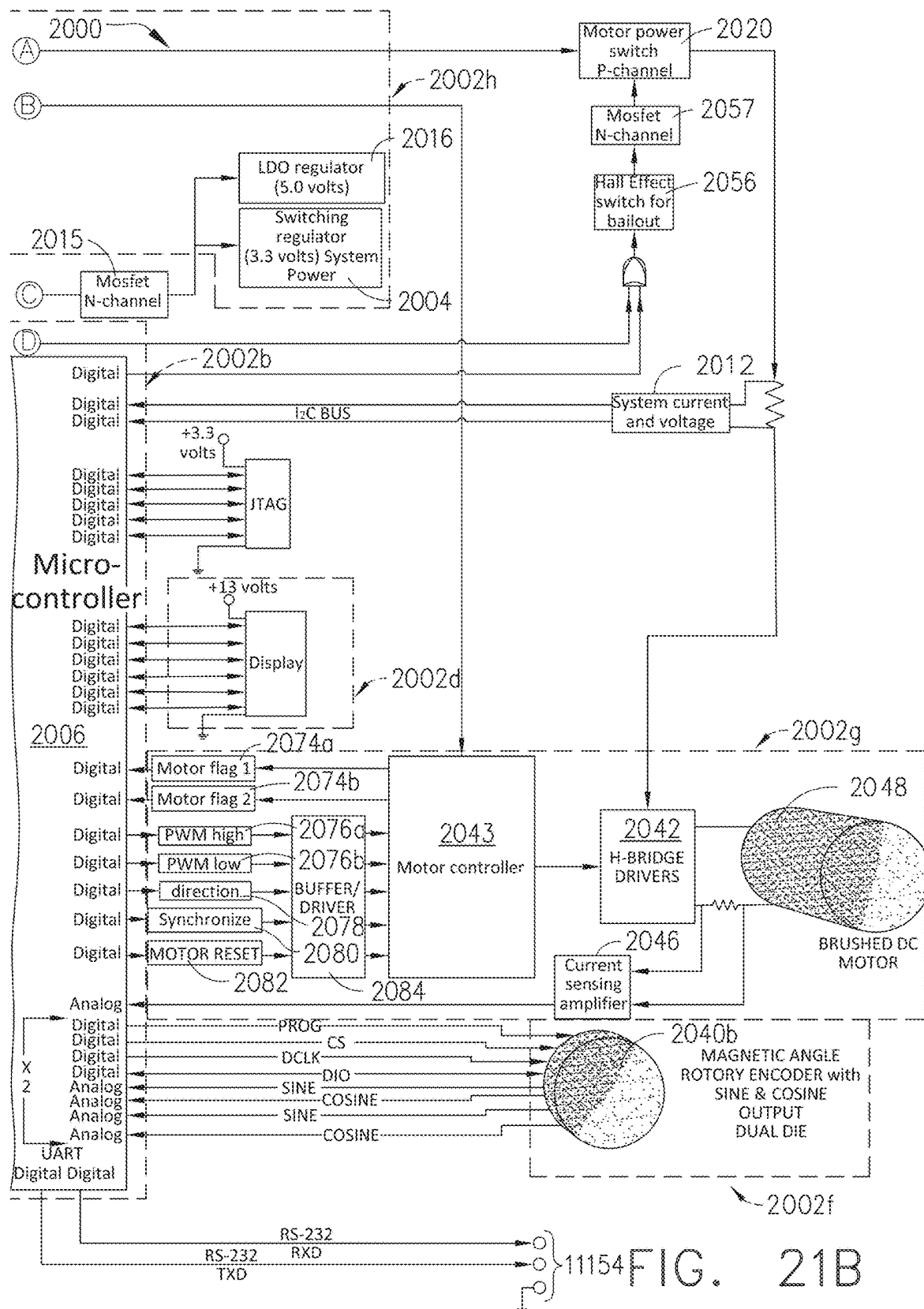


FIG. 20





11154 FIG. 21B

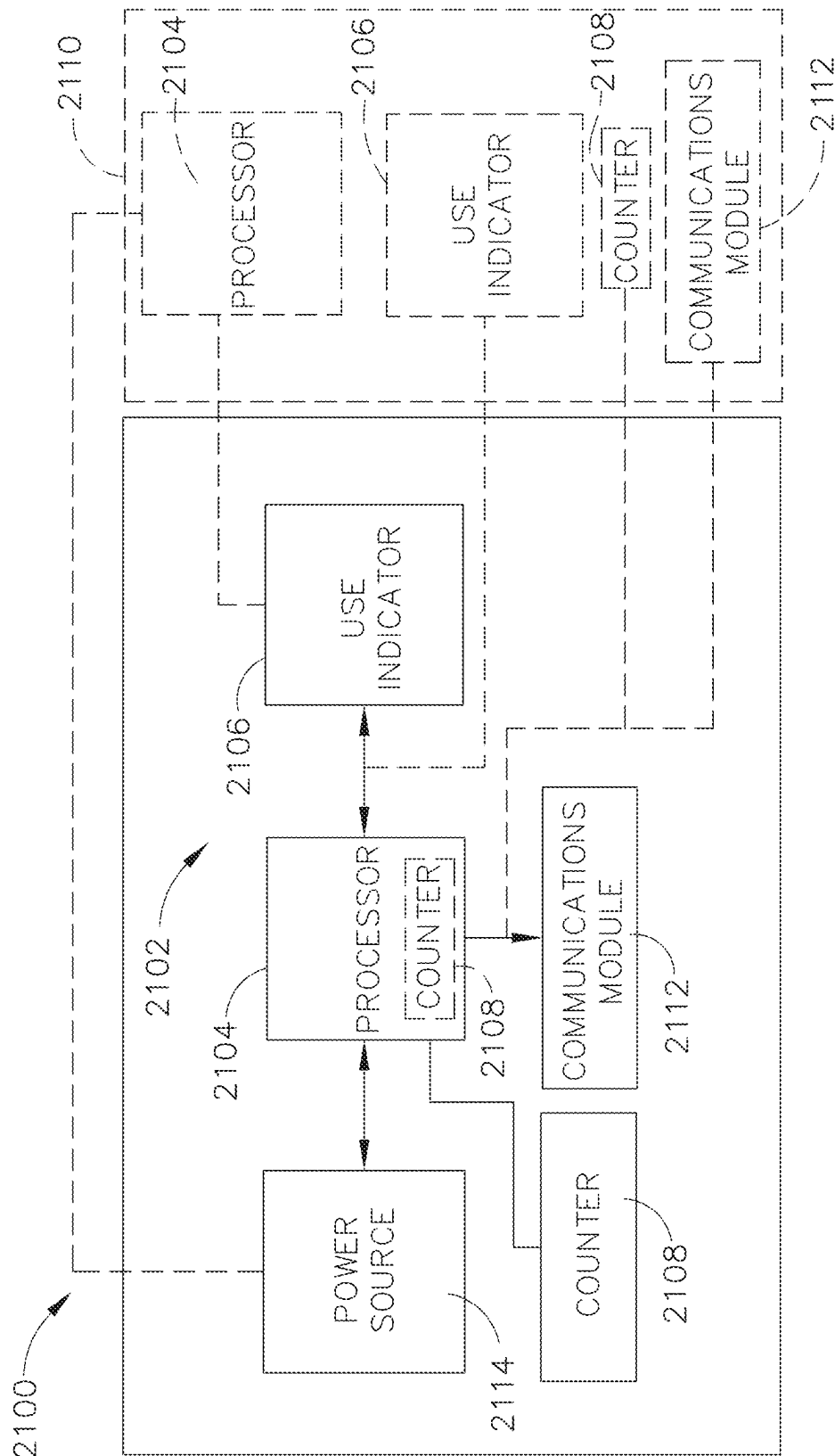


FIG. 22

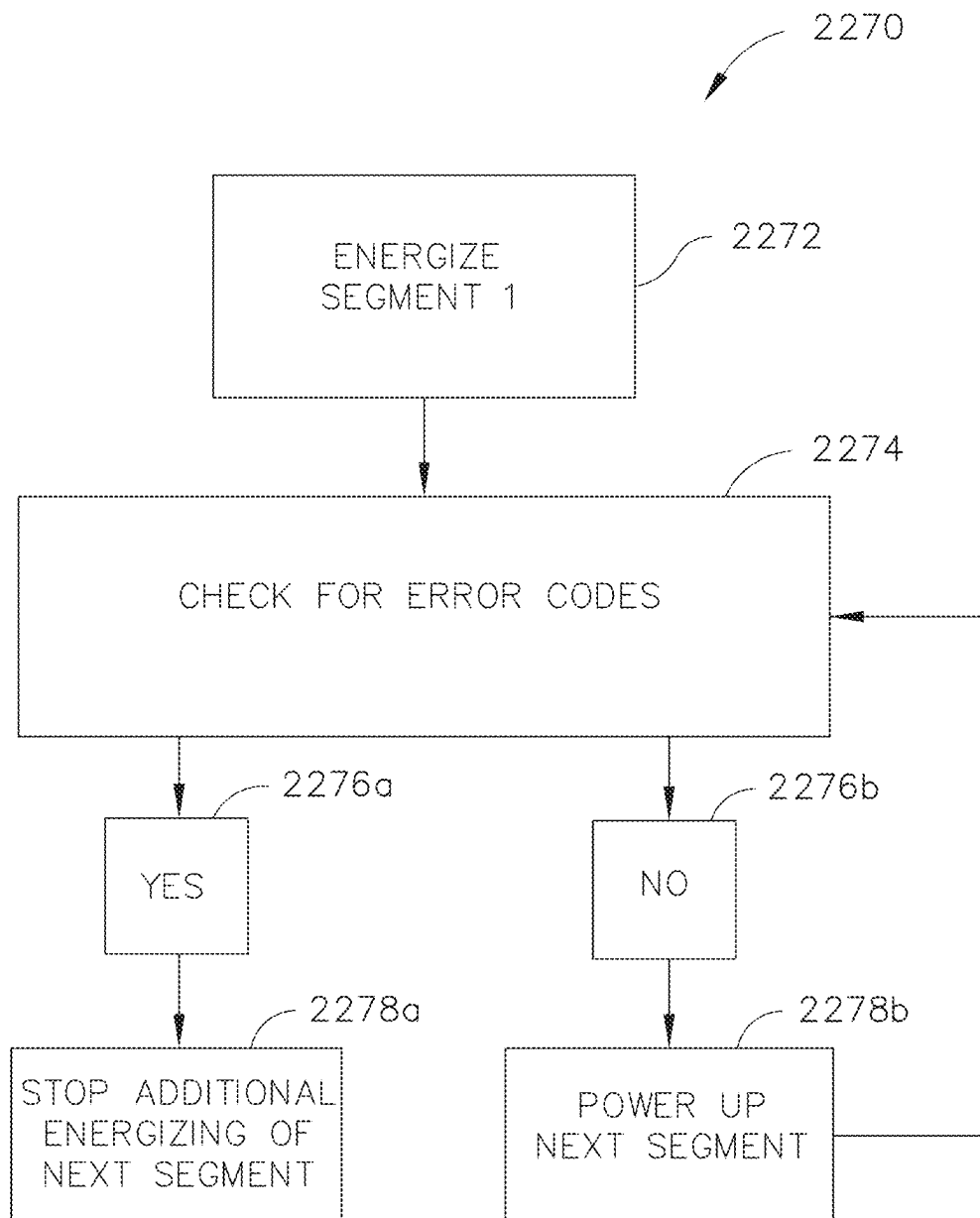


FIG. 23

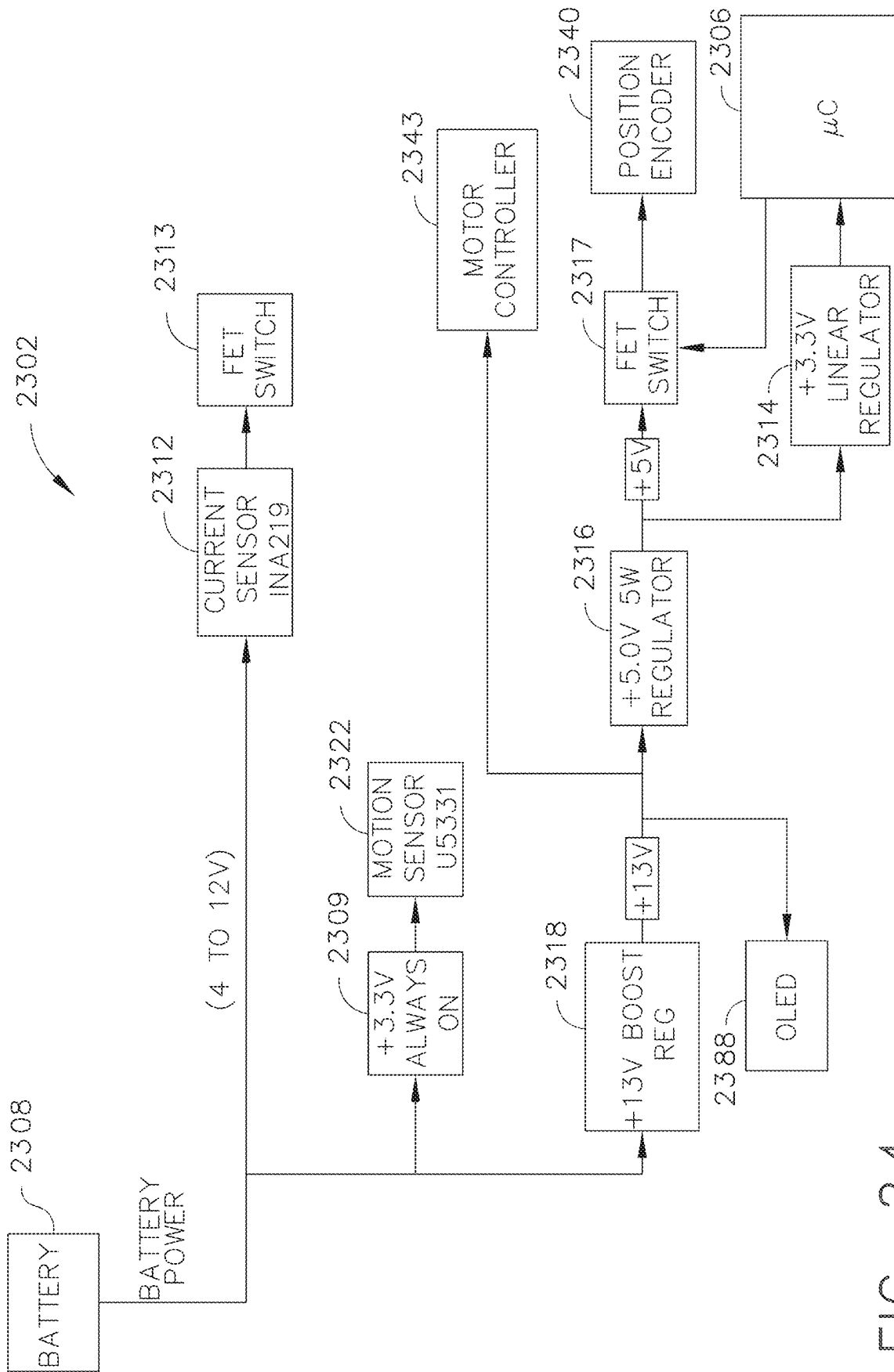
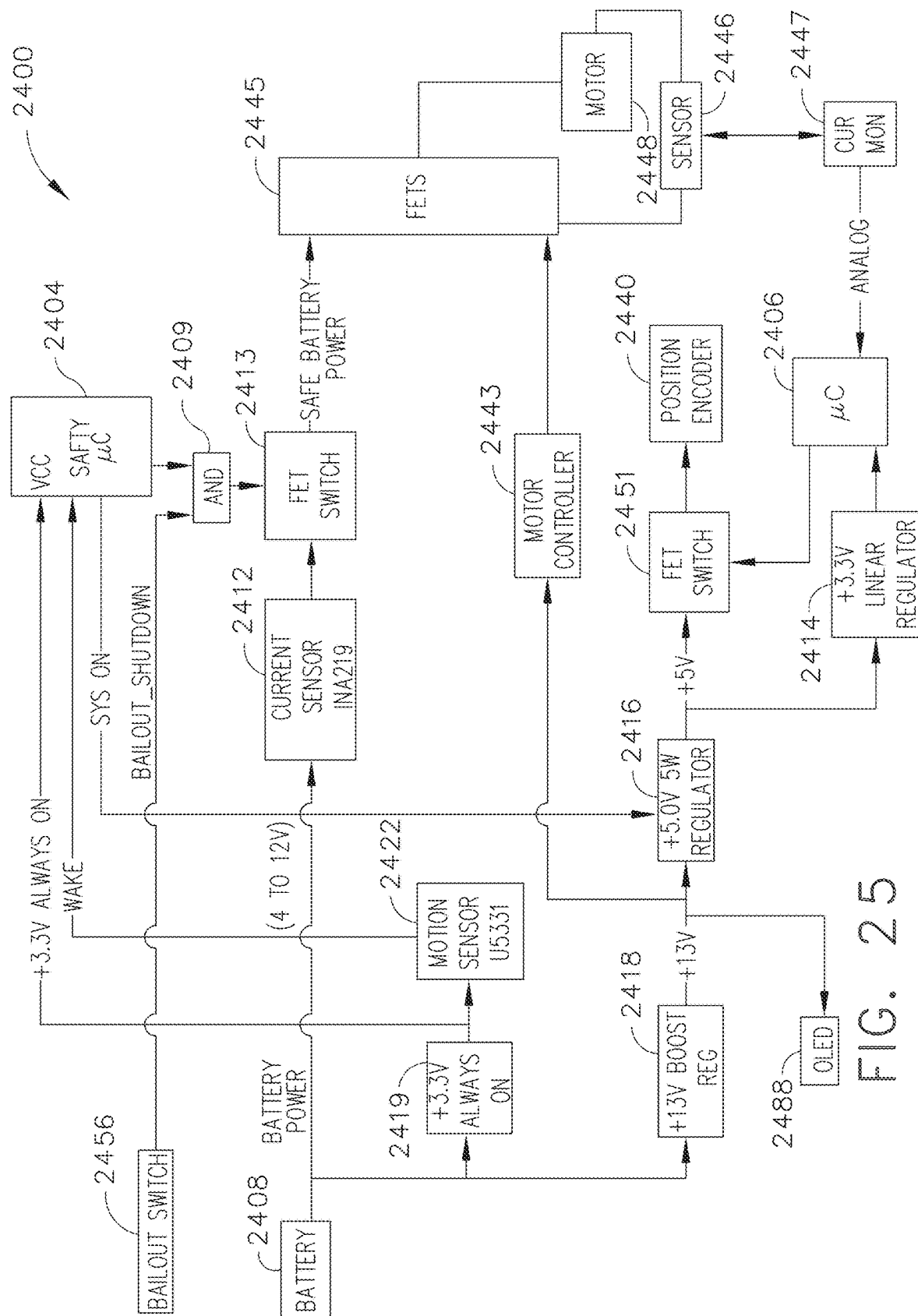


FIG. 24



520.1

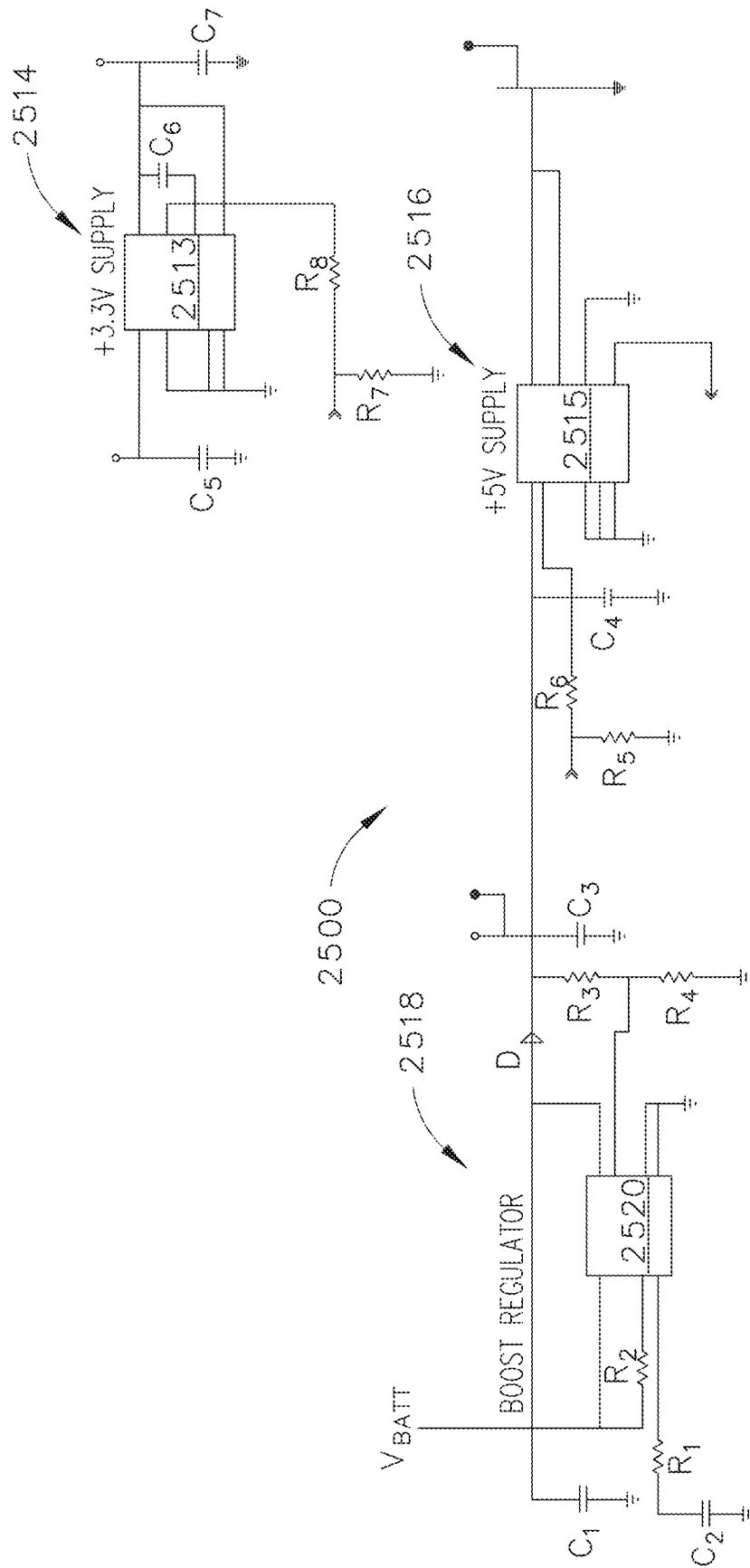


FIG. 26

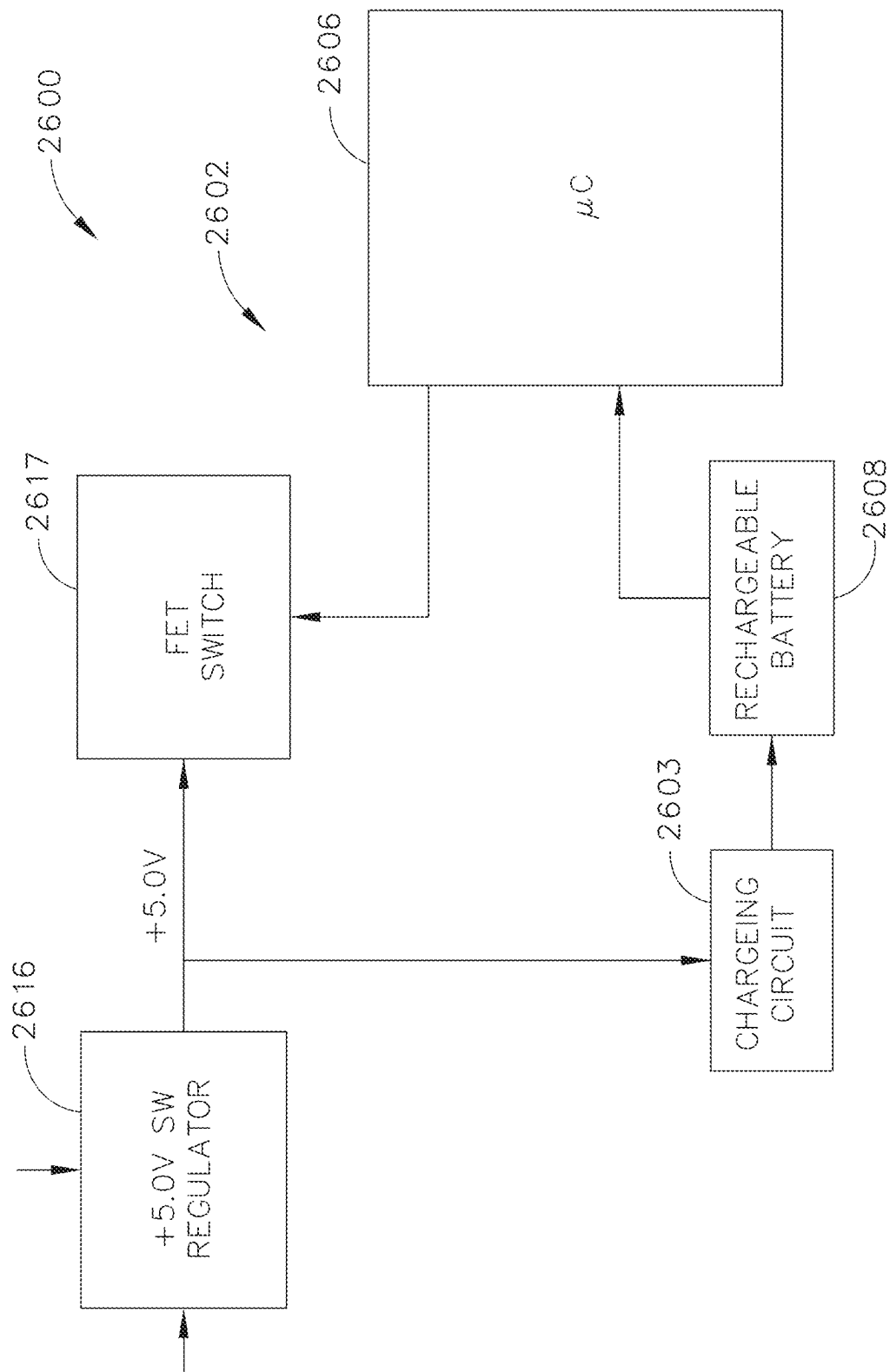


FIG. 27

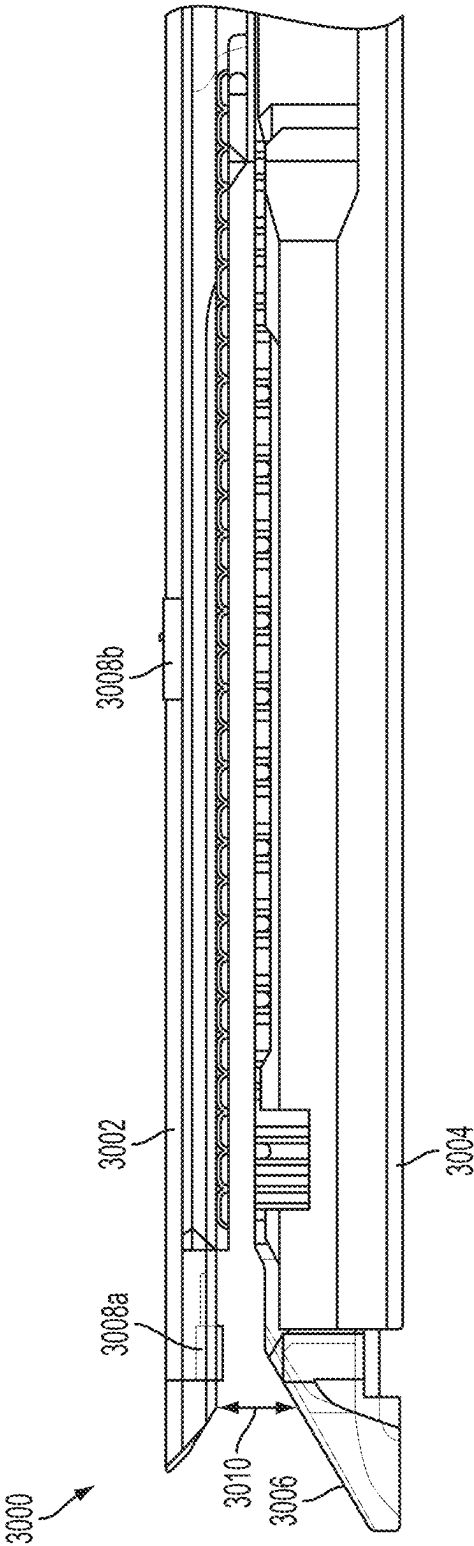


FIG. 28

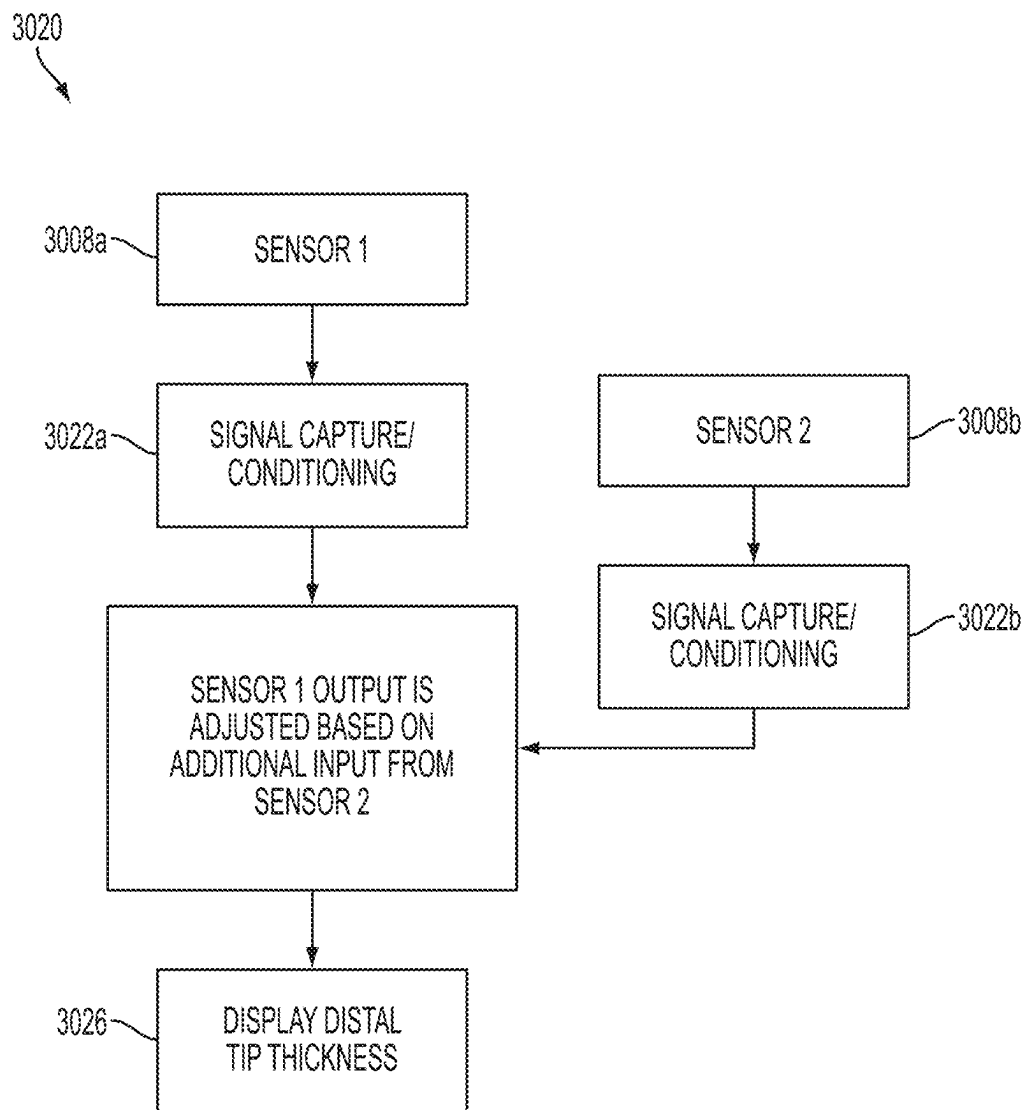


FIG. 29

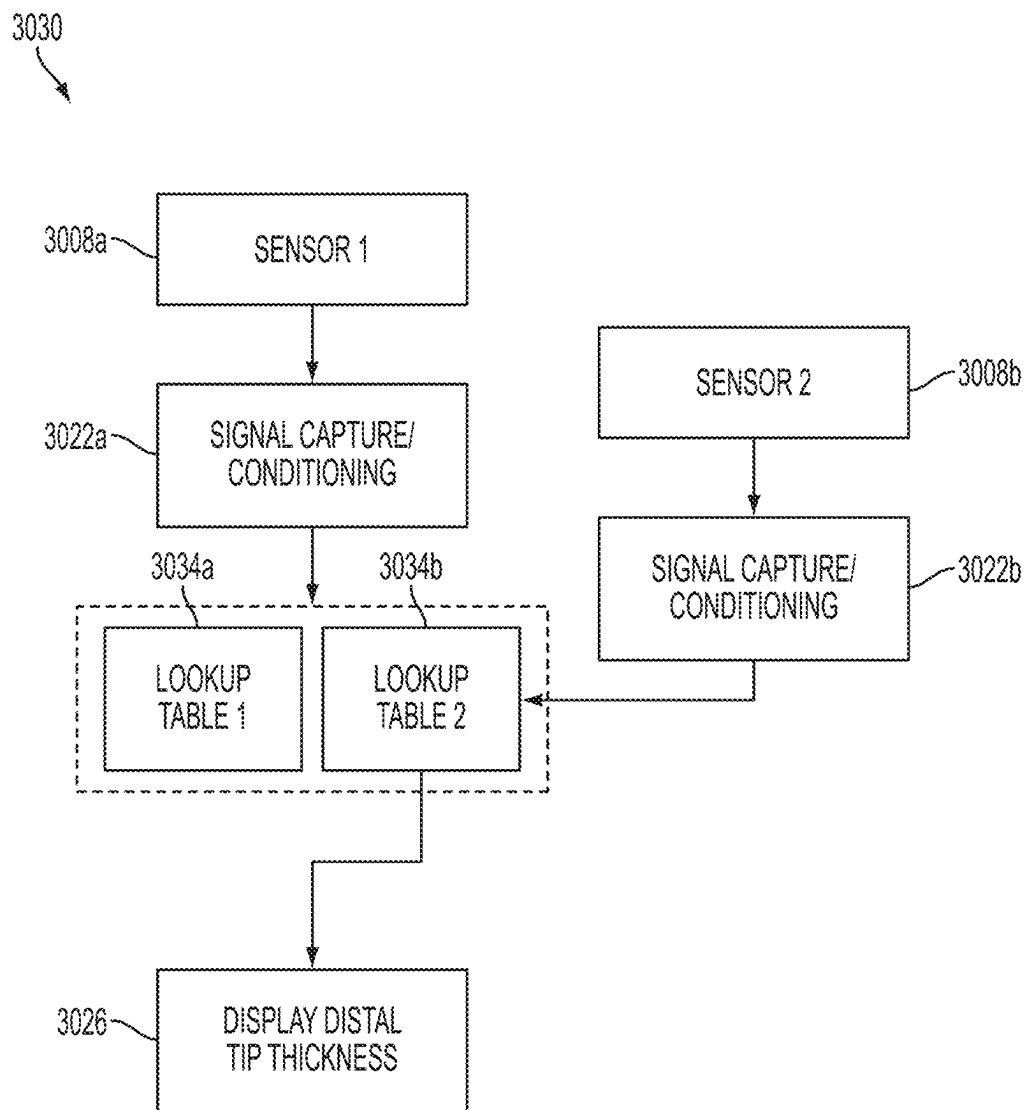


FIG. 30

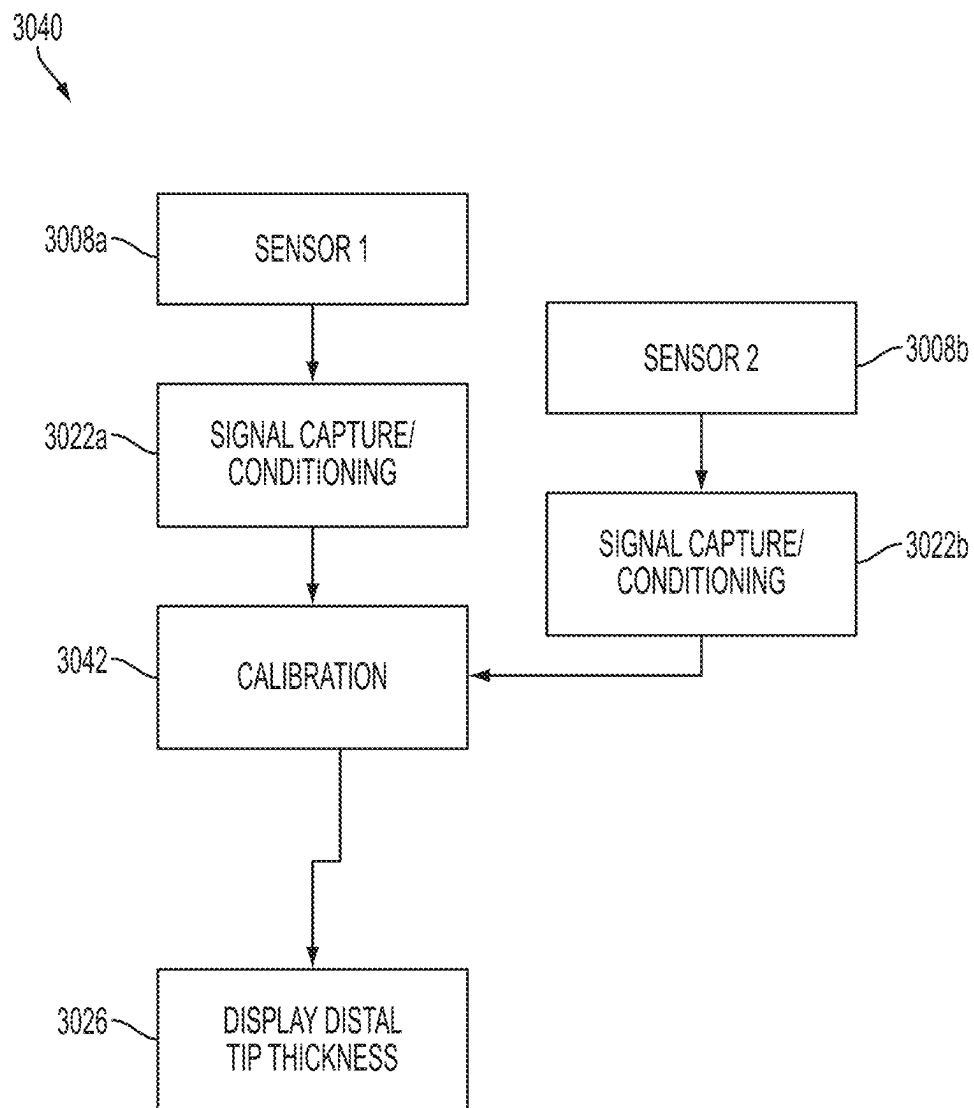


FIG. 31

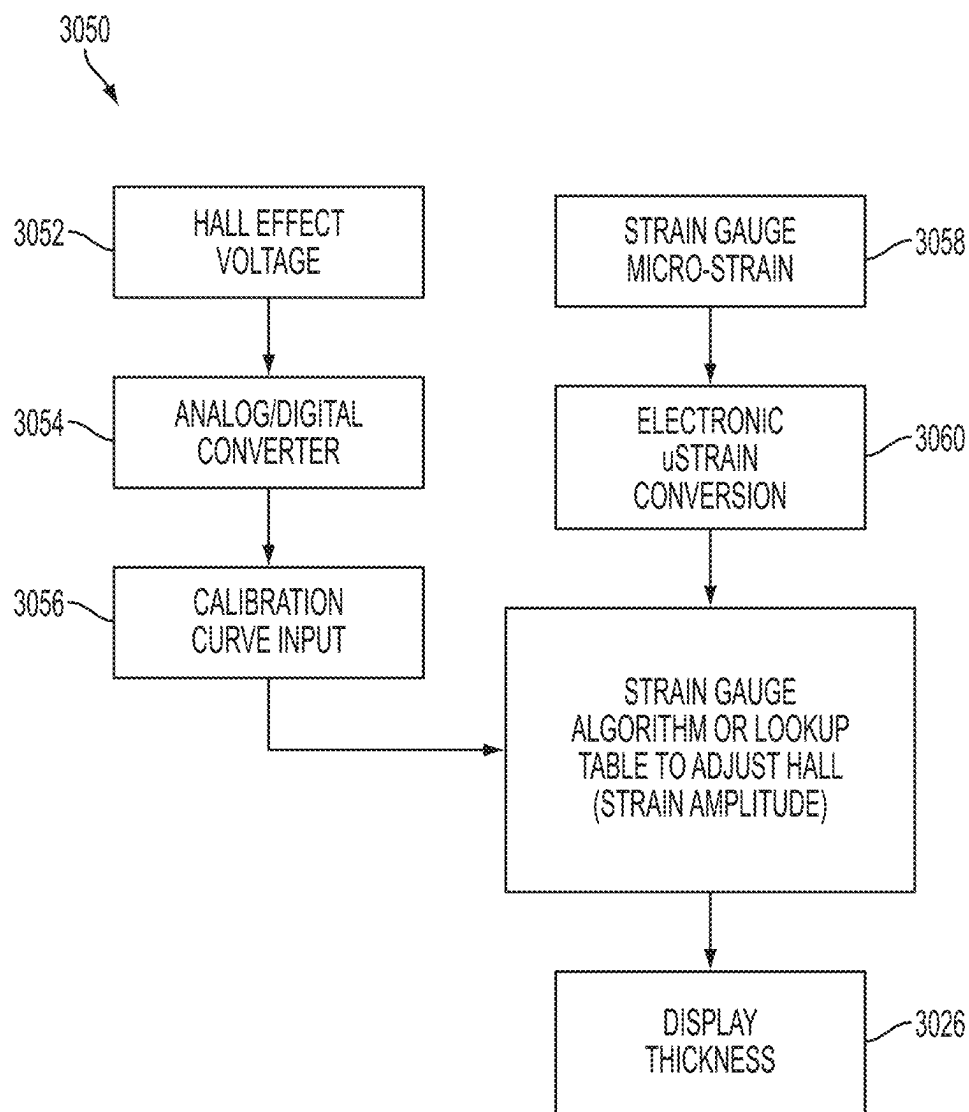


FIG. 32A

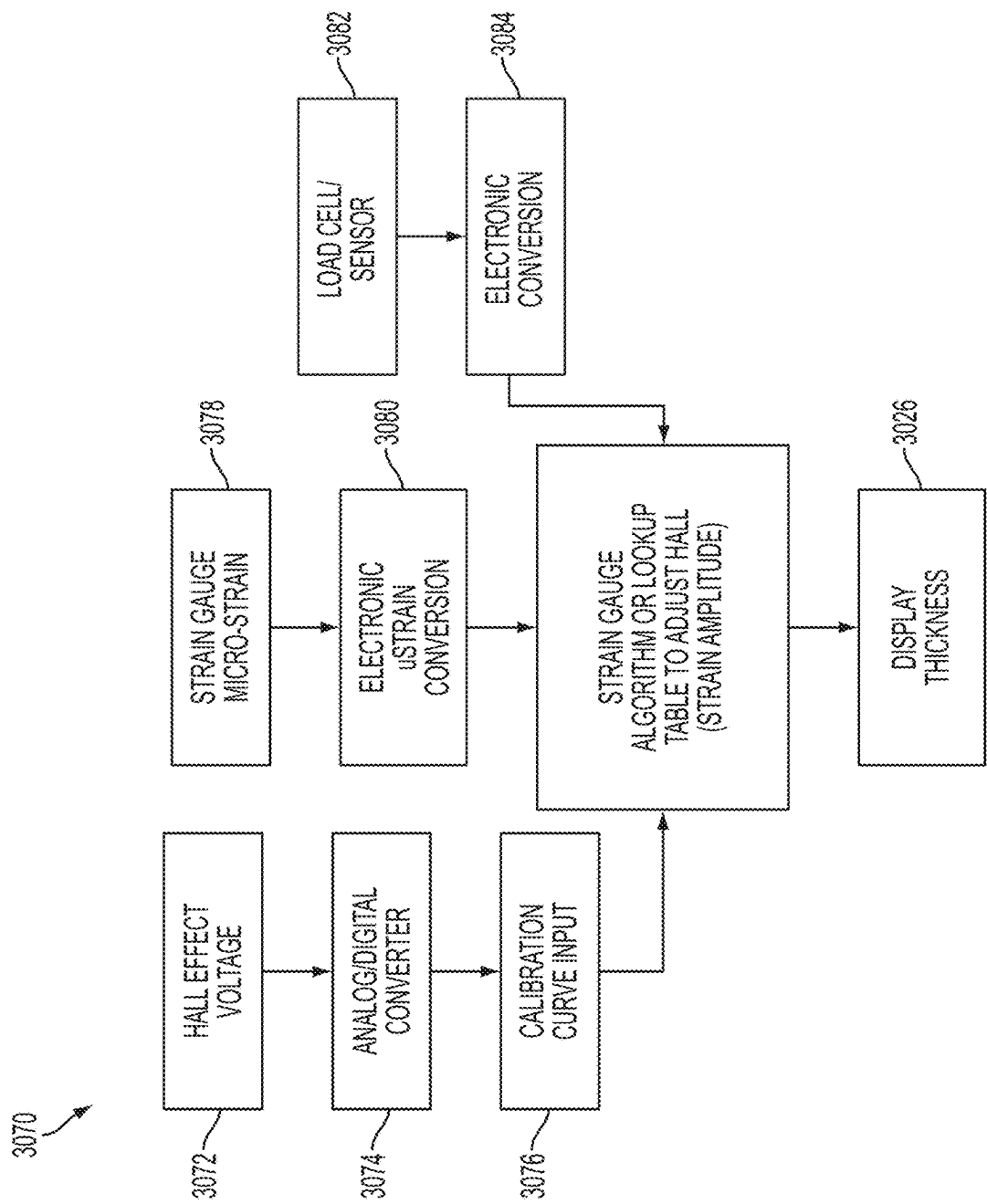


FIG. 32B

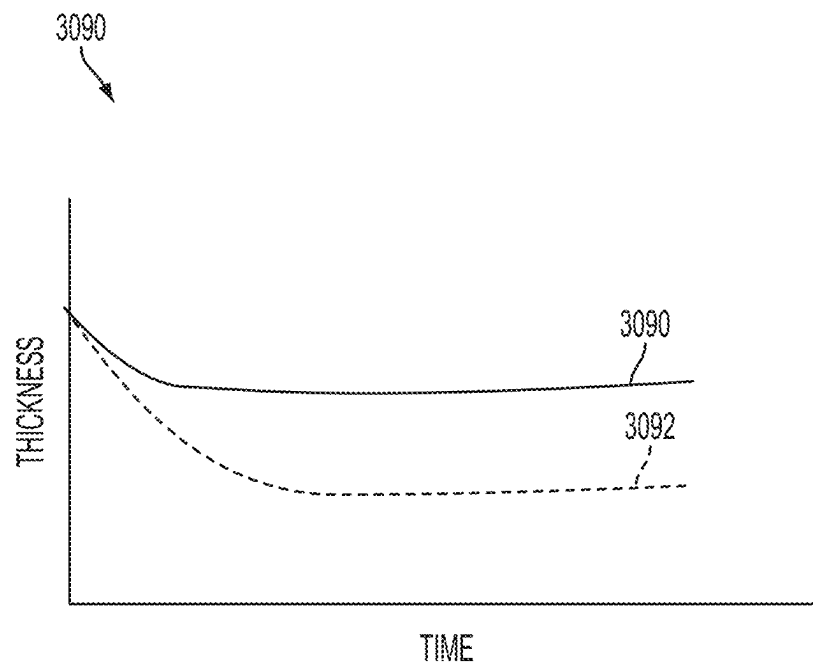


FIG. 33

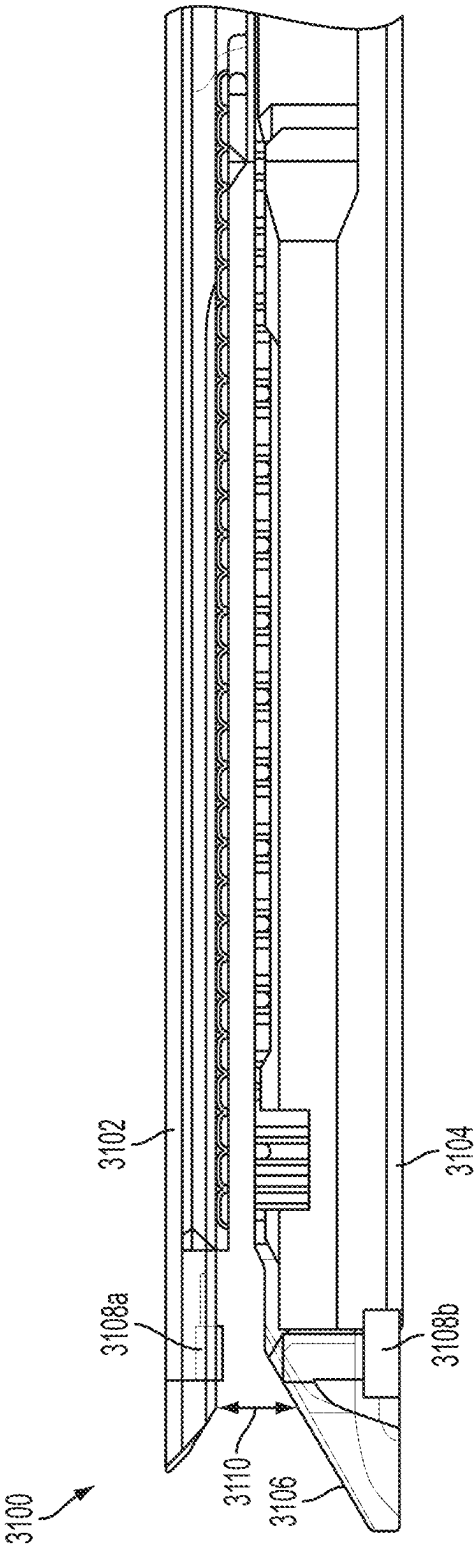


FIG. 34

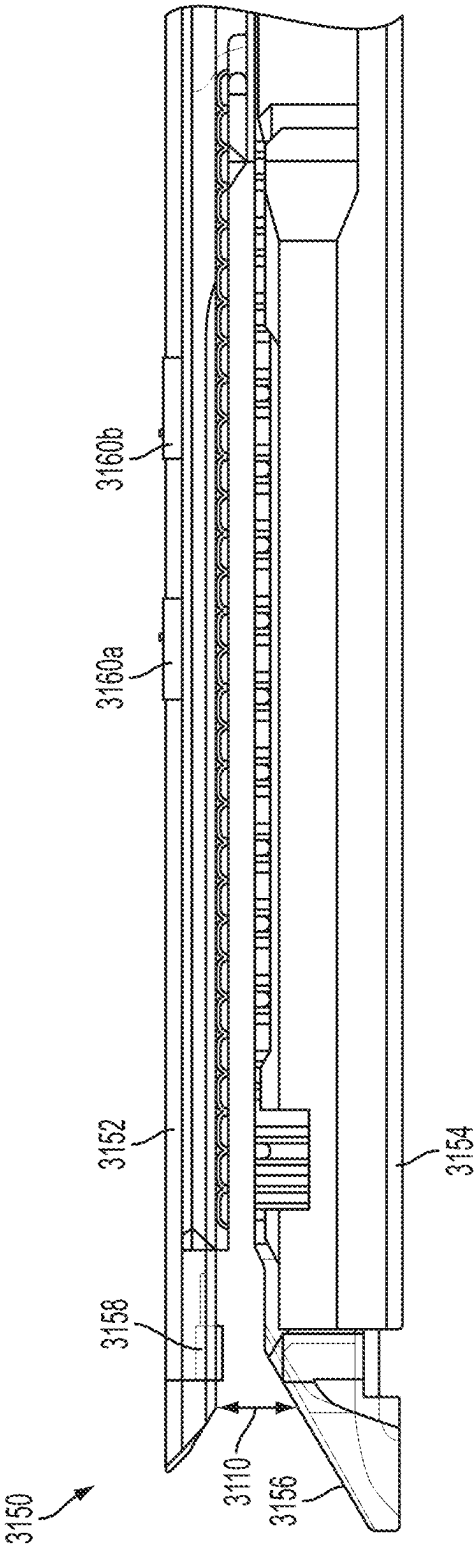


FIG. 35

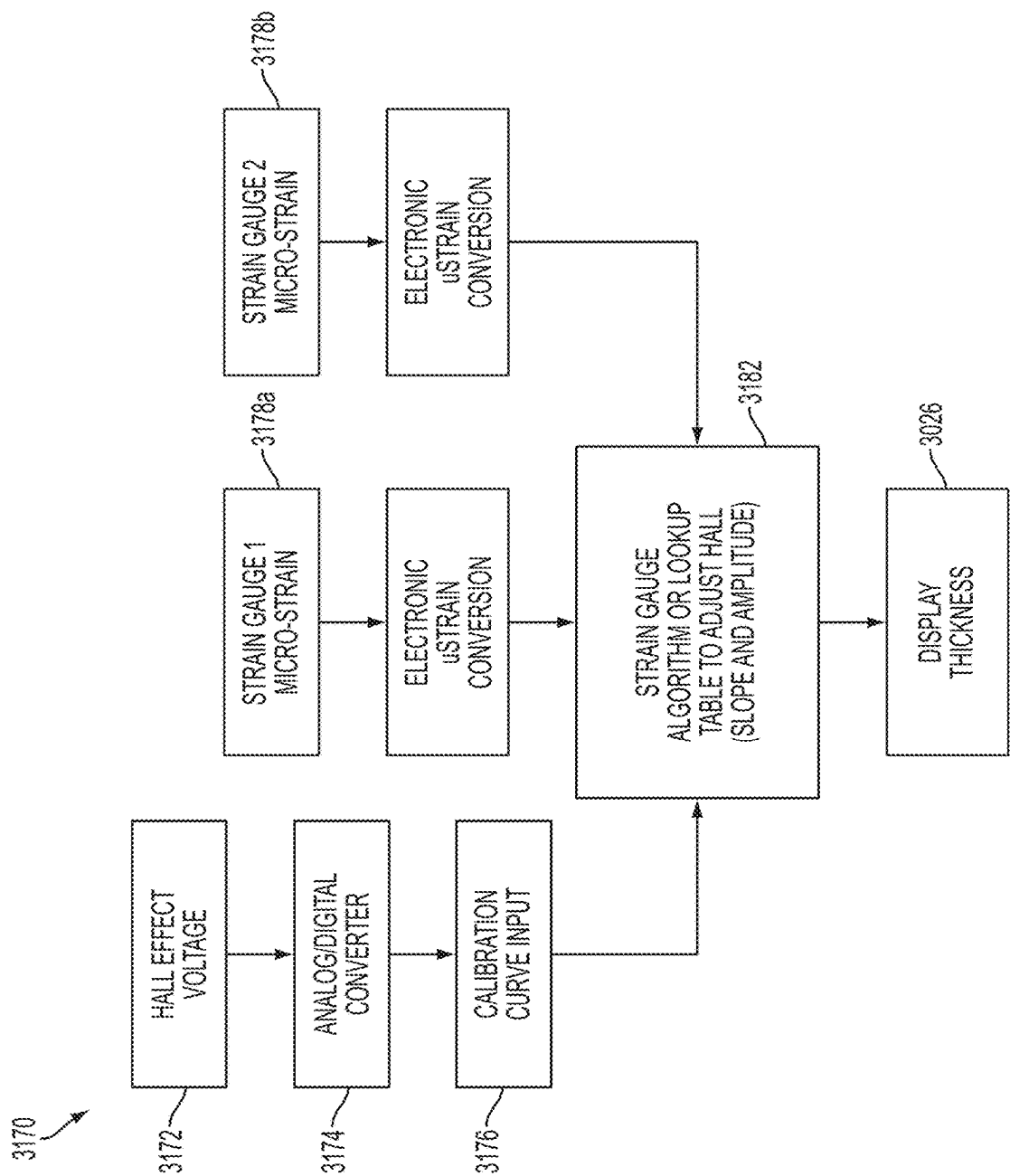


FIG. 36

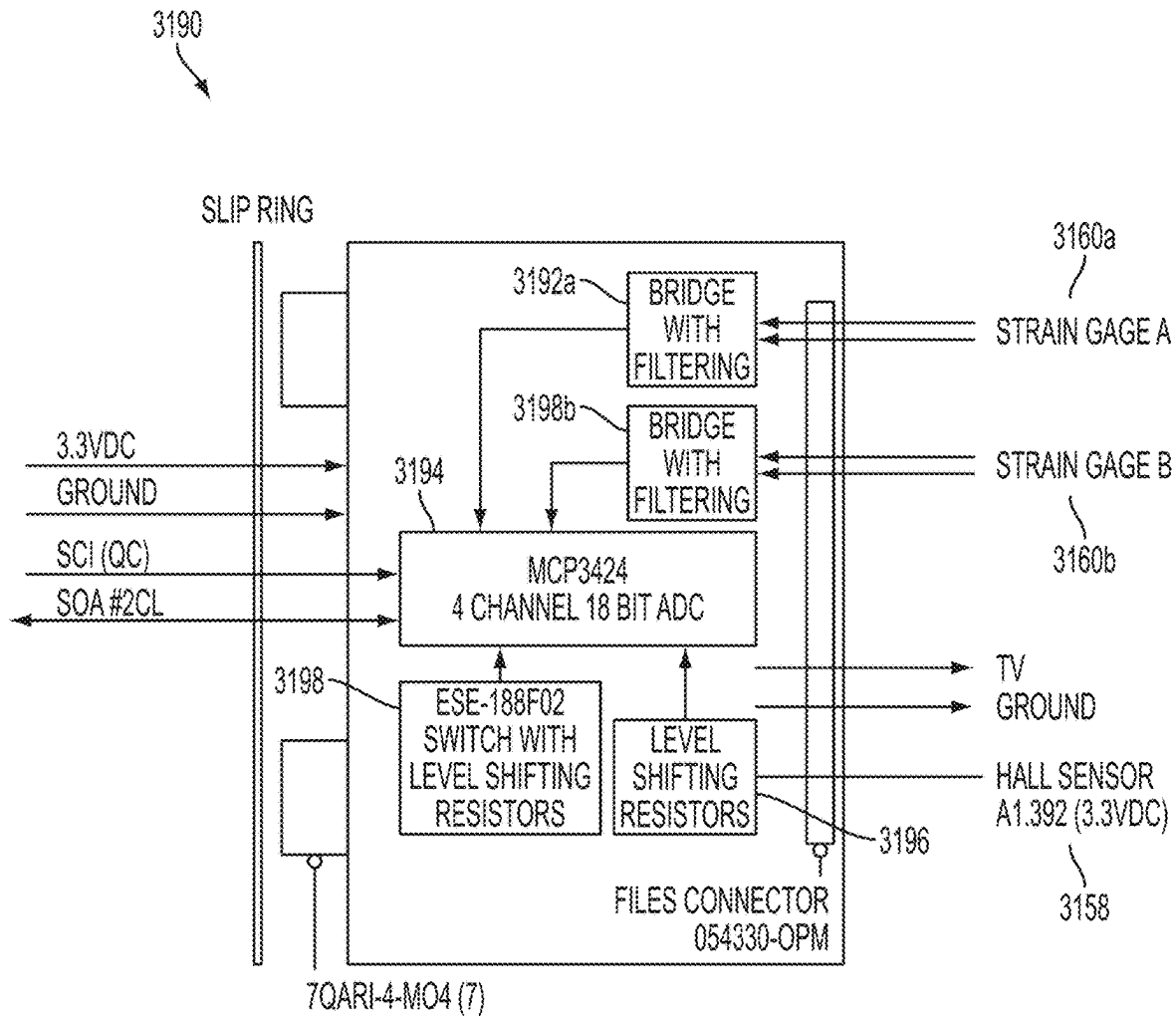


FIG. 37

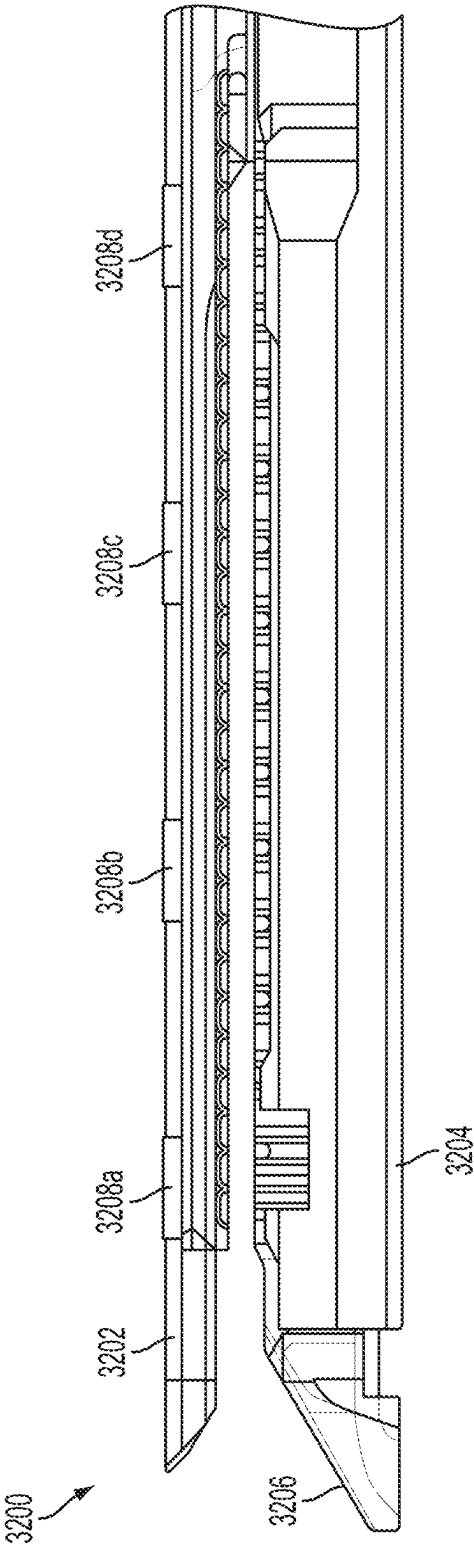


FIG. 38

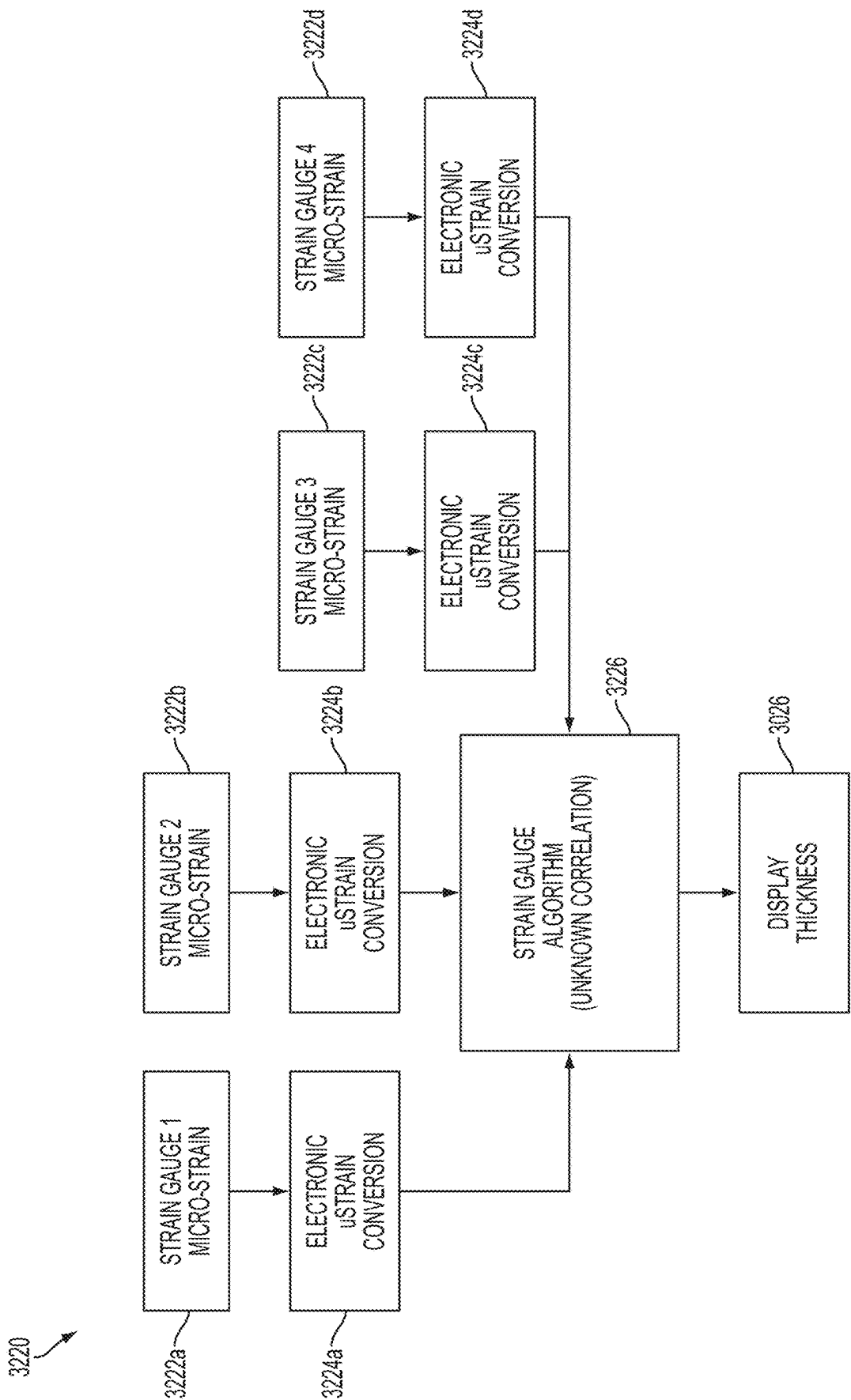


FIG. 39

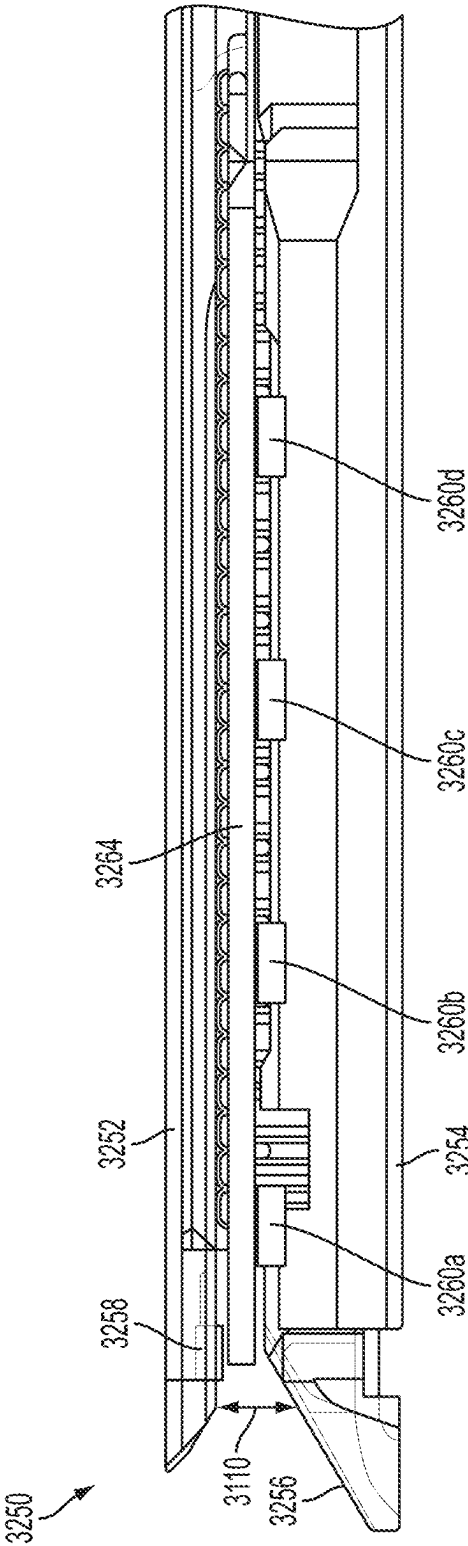


FIG. 40

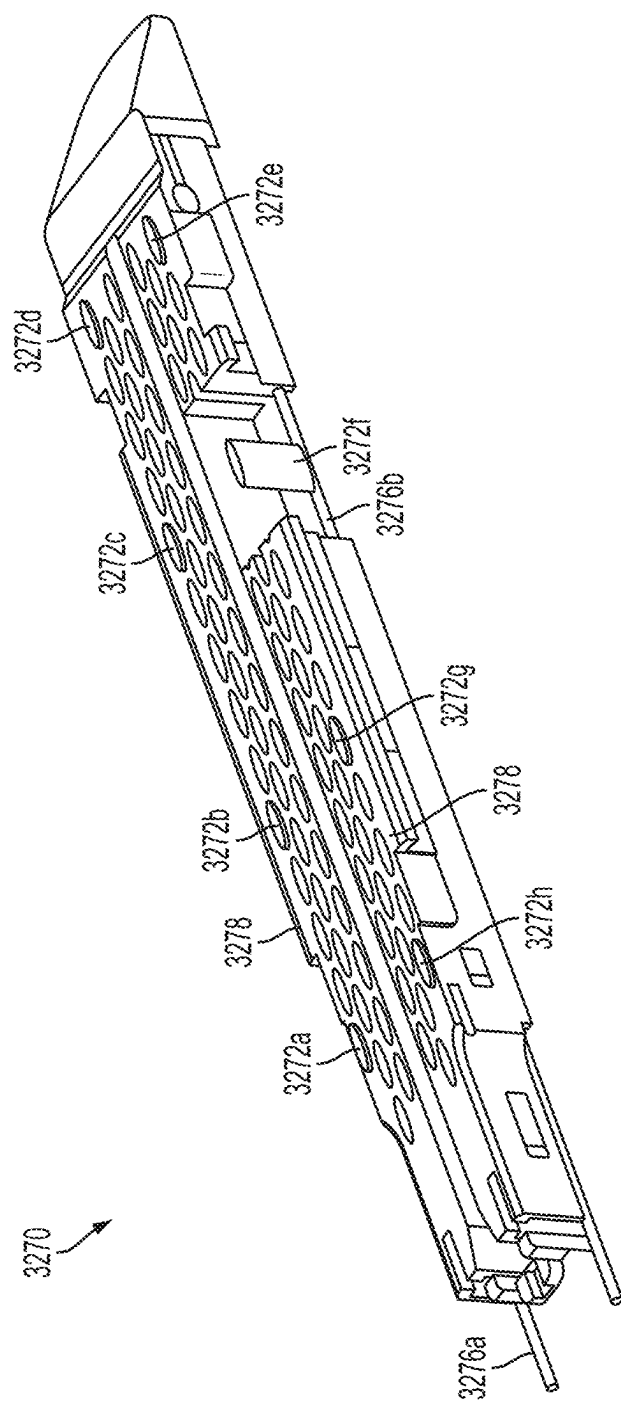


FIG. 41

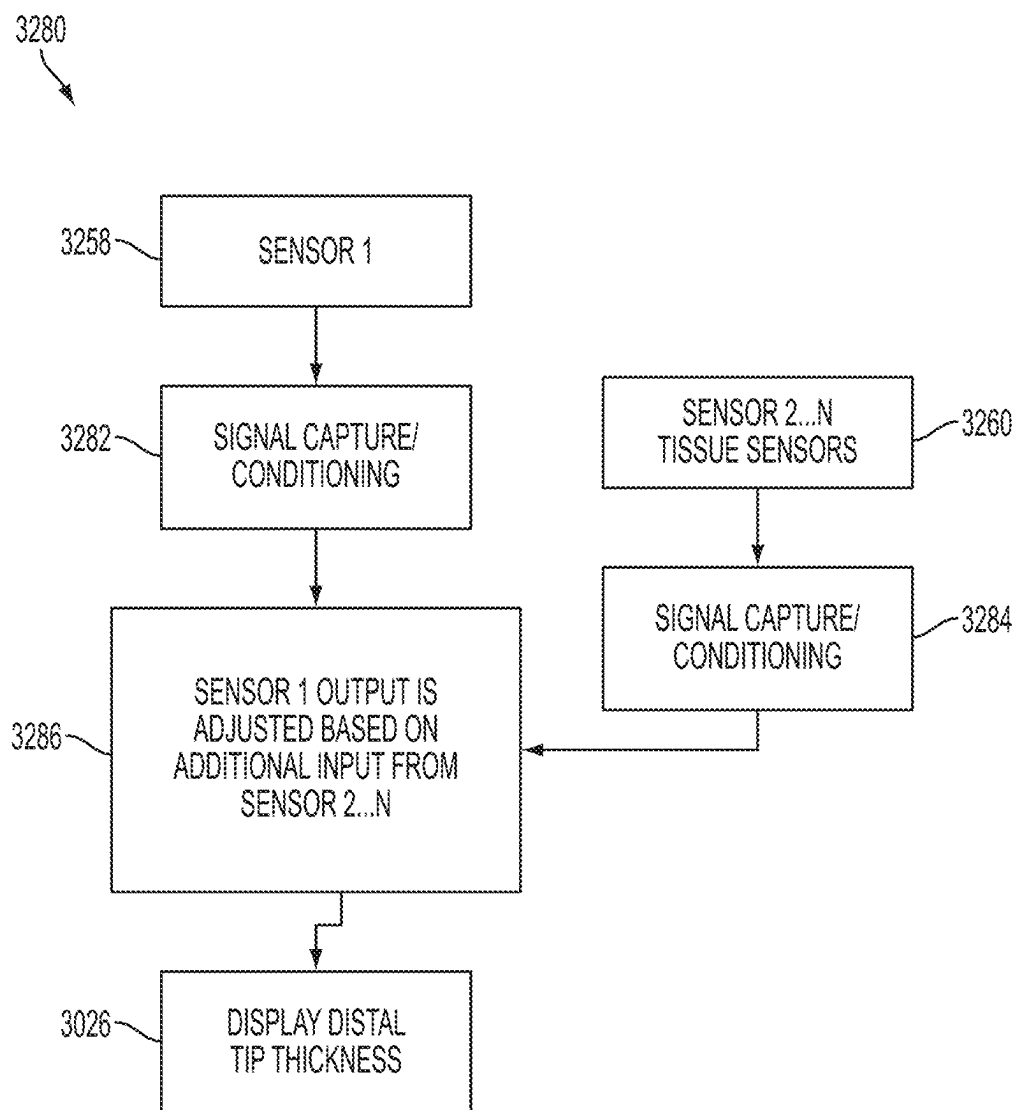


FIG. 42

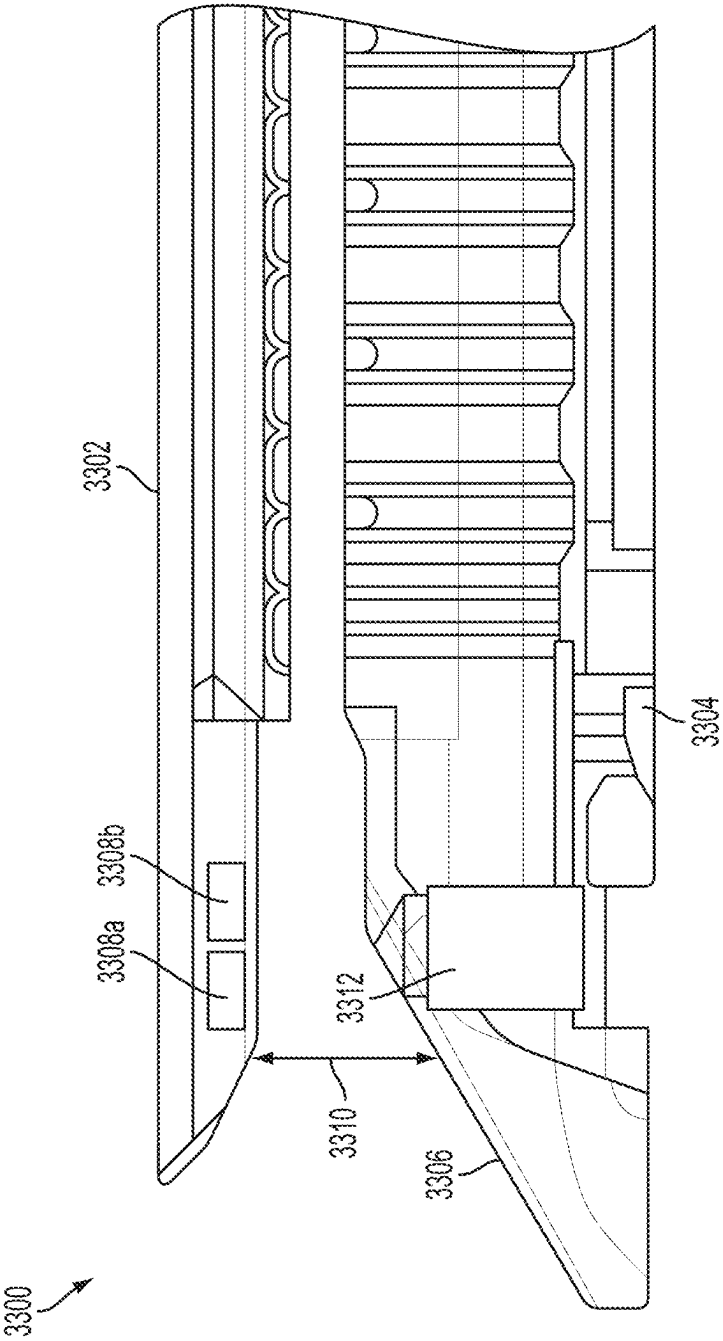


FIG. 43

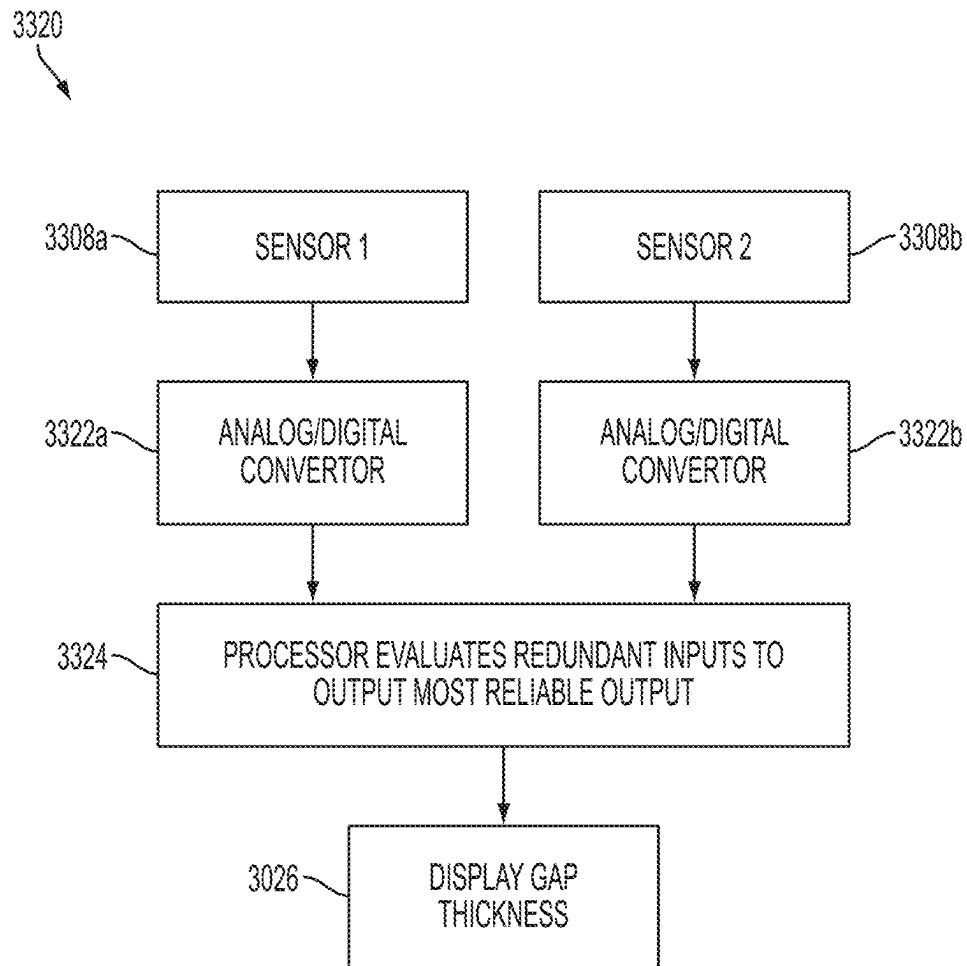


FIG. 44

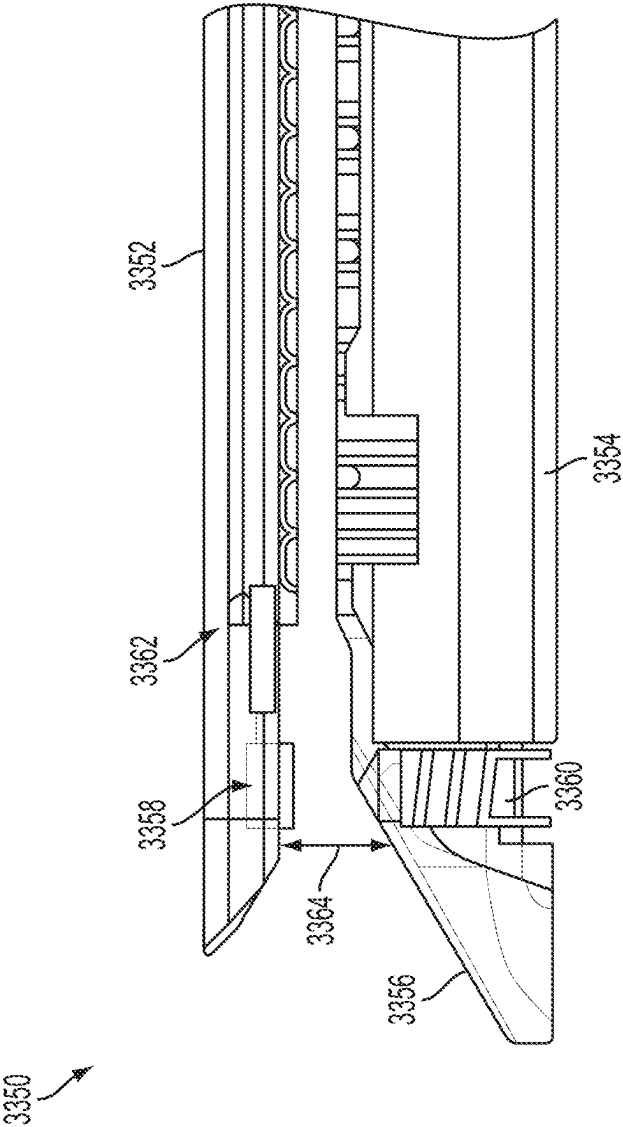


FIG. 45

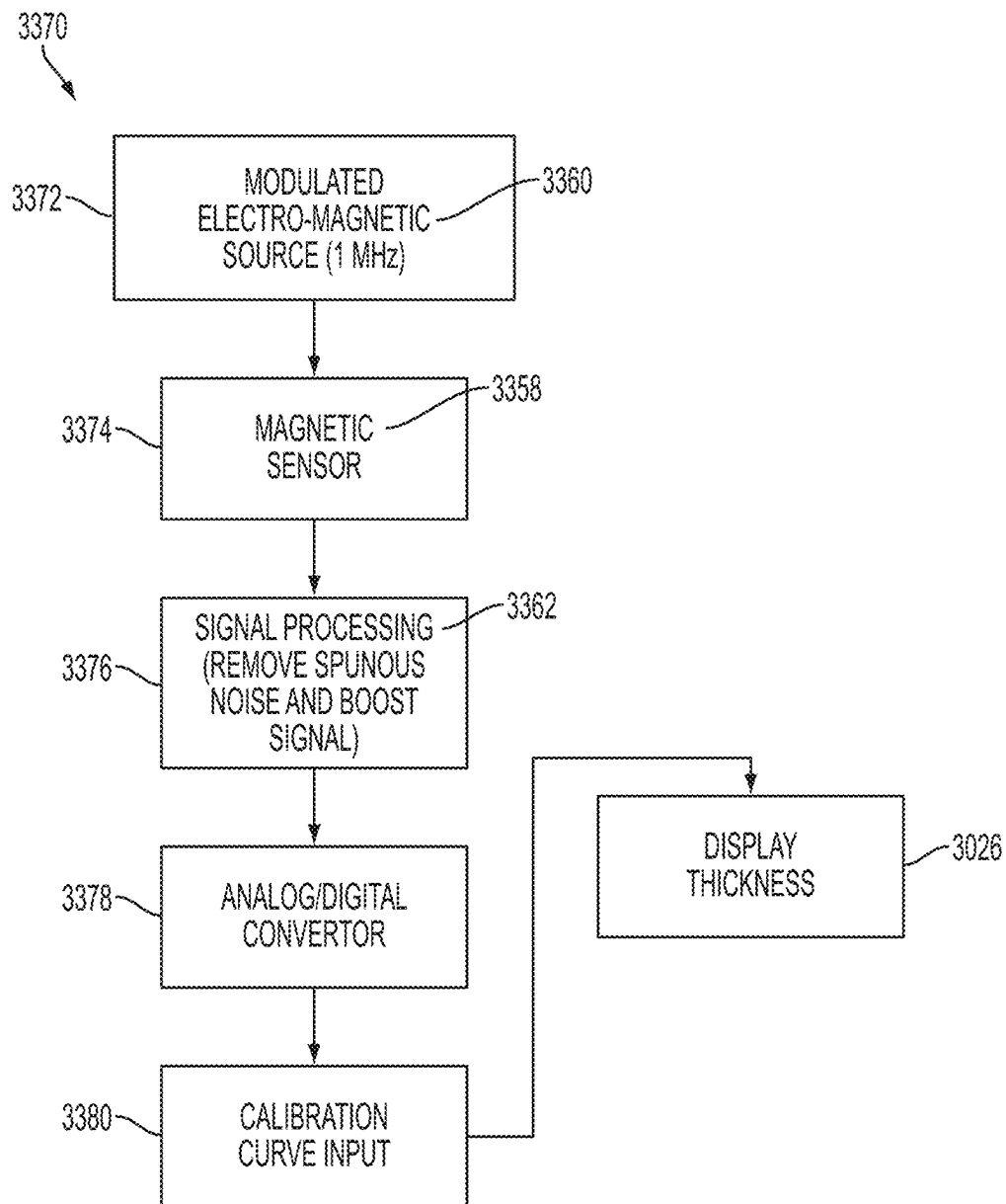


FIG. 46

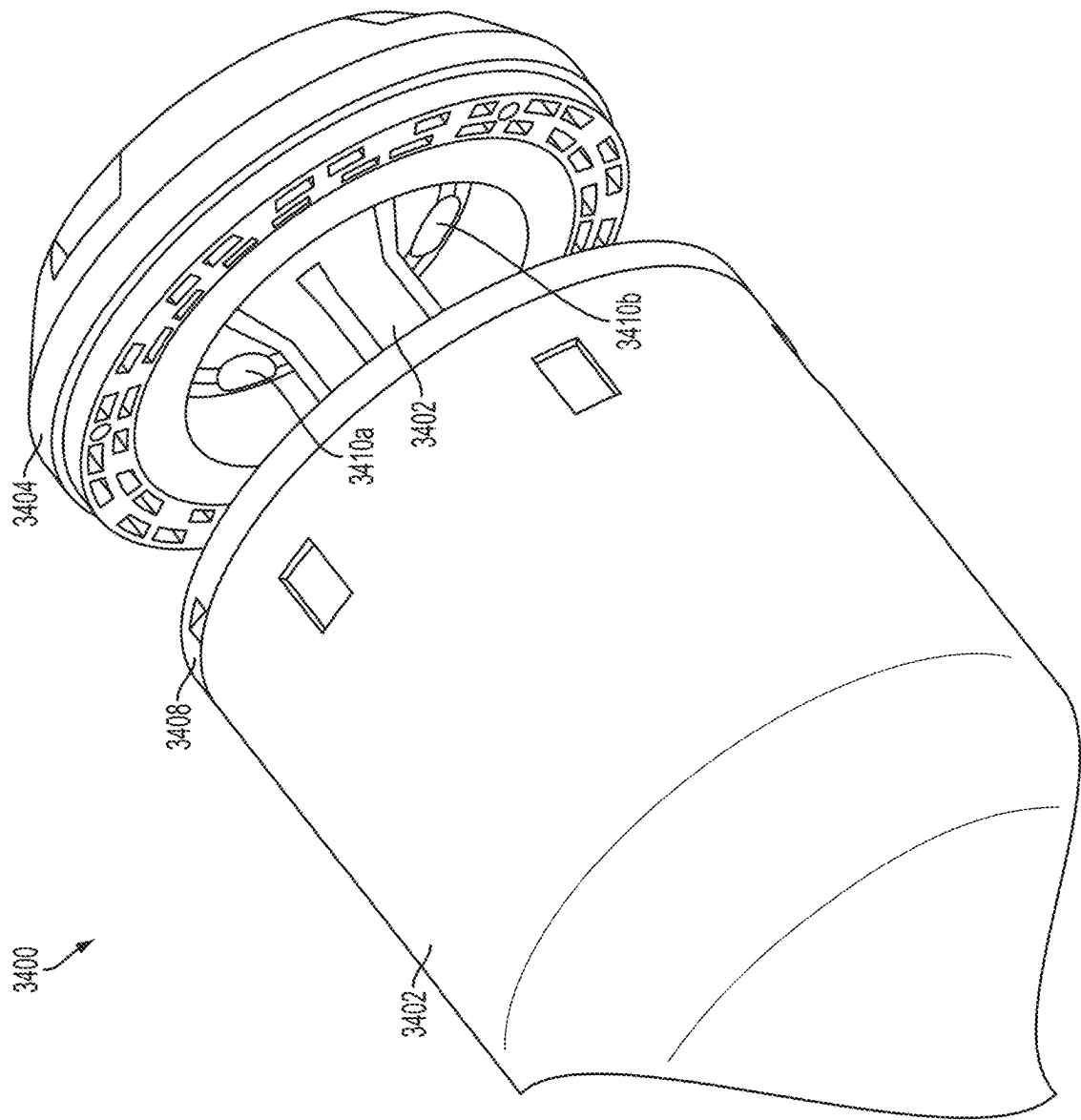


FIG. 47

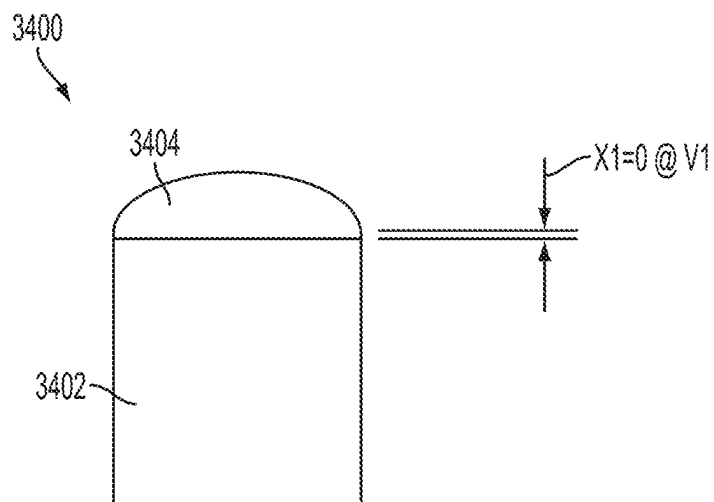


FIG. 48A

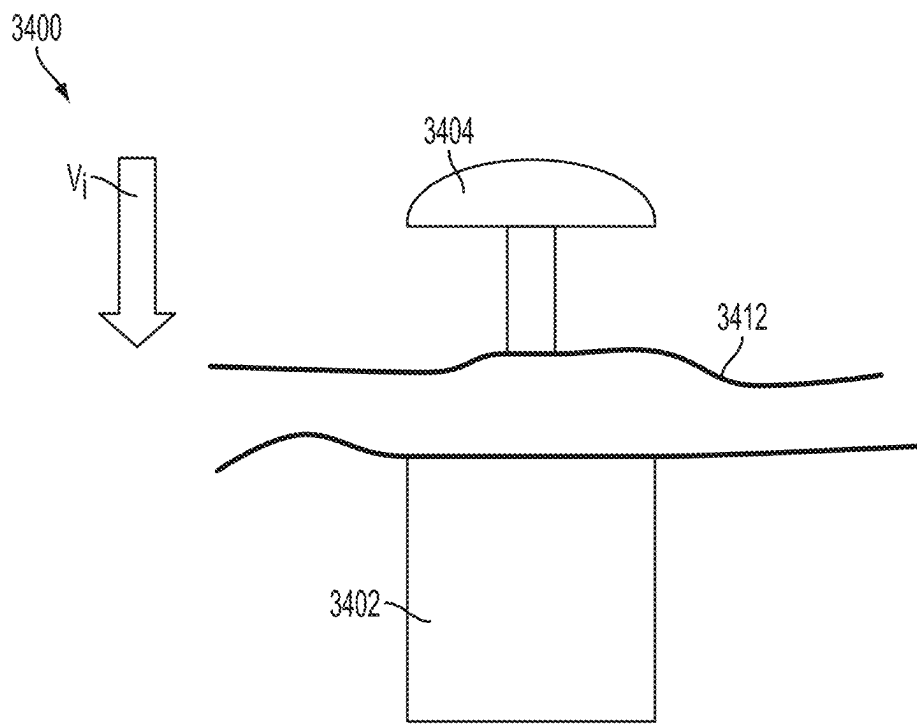


FIG. 48B

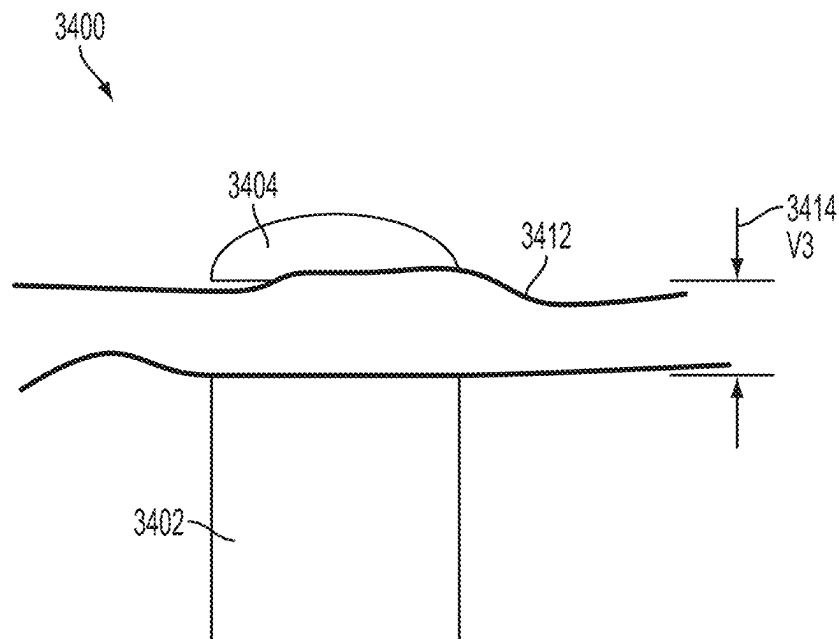


FIG. 48C

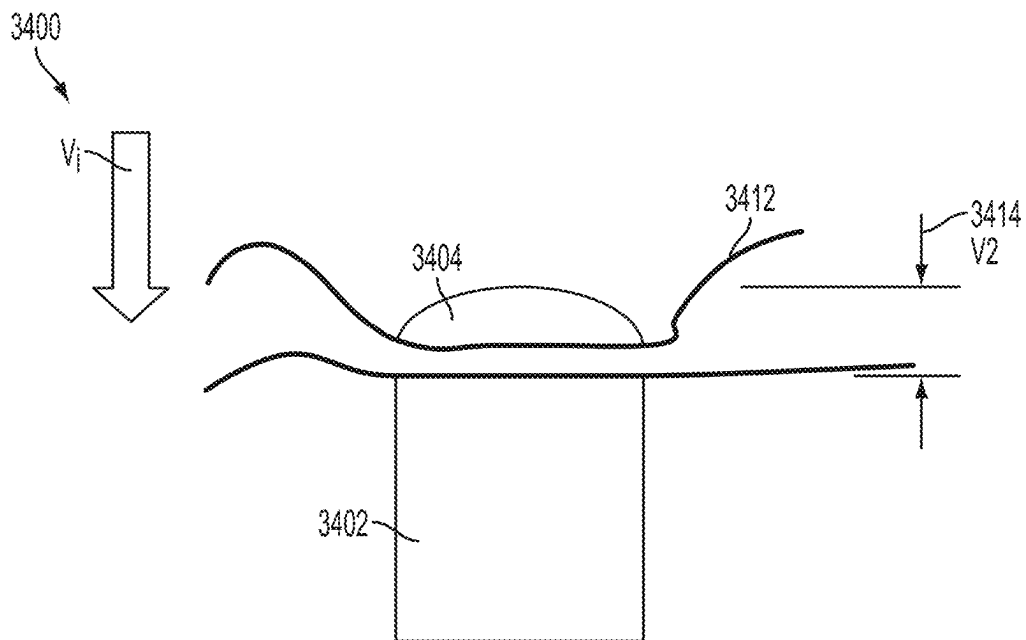


FIG. 48D

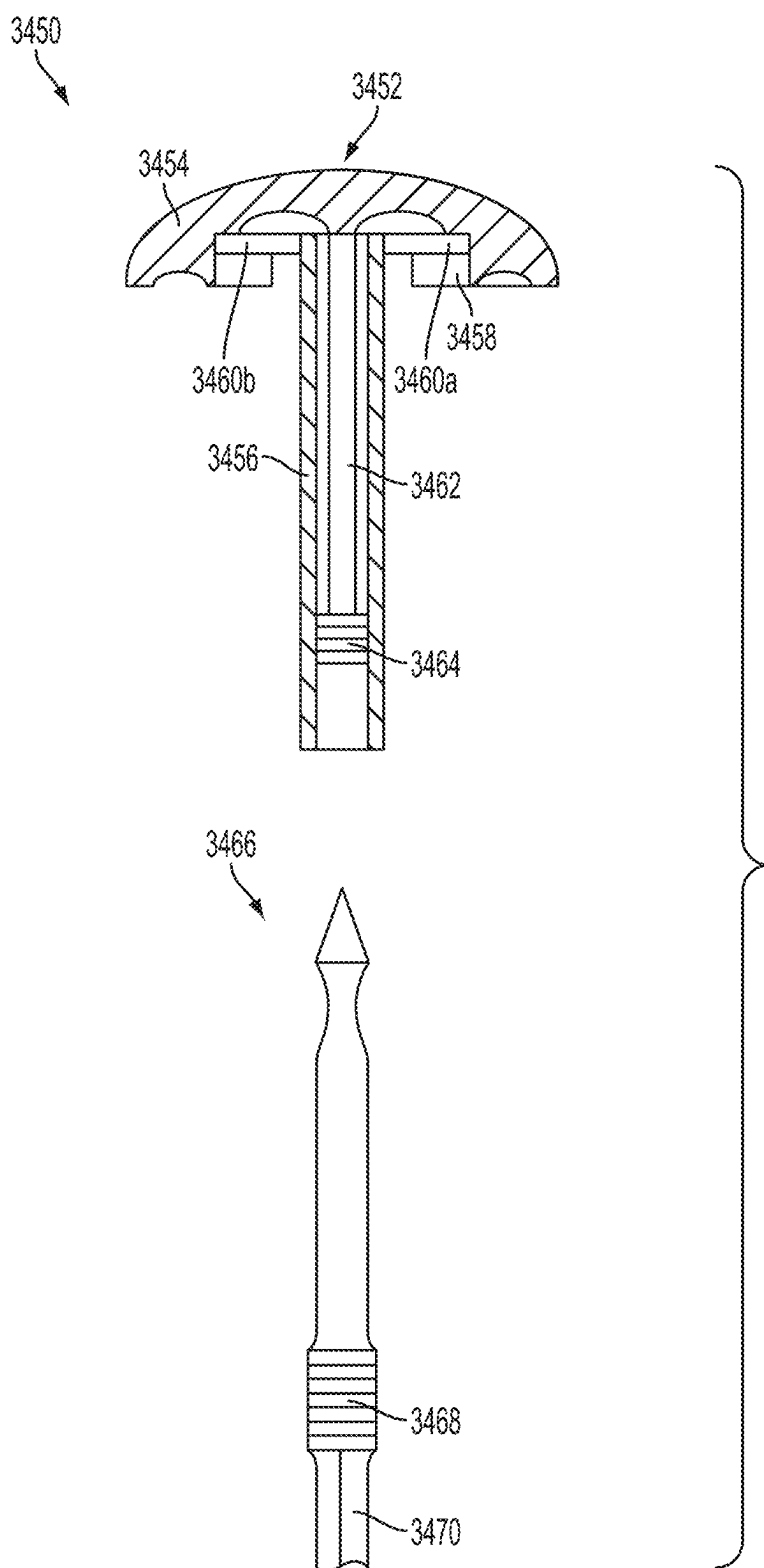


FIG. 49

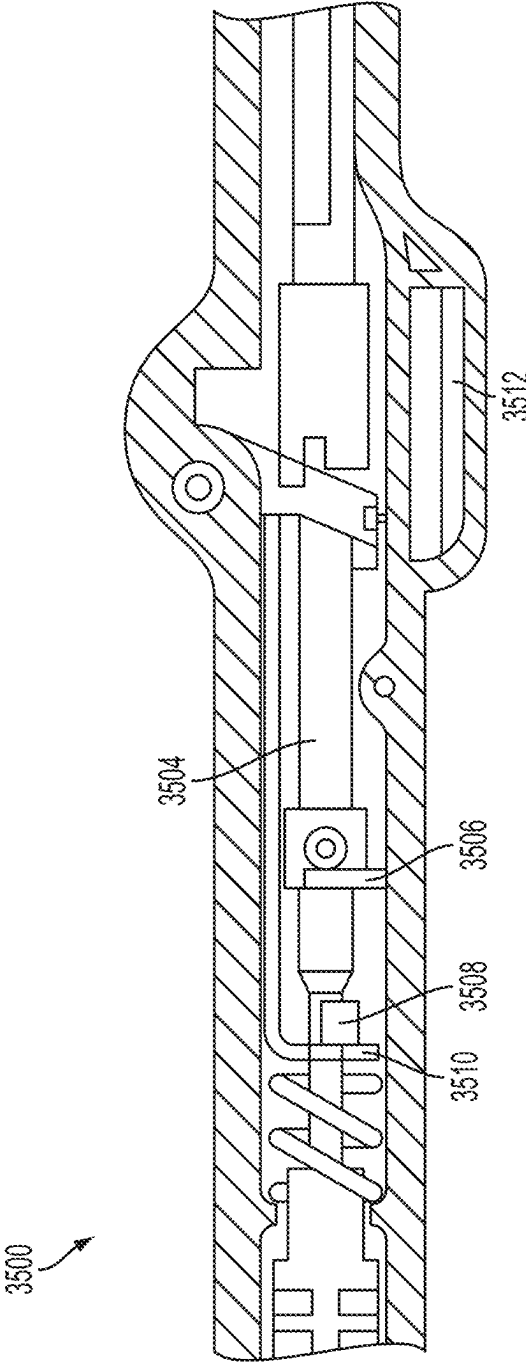


FIG. 50

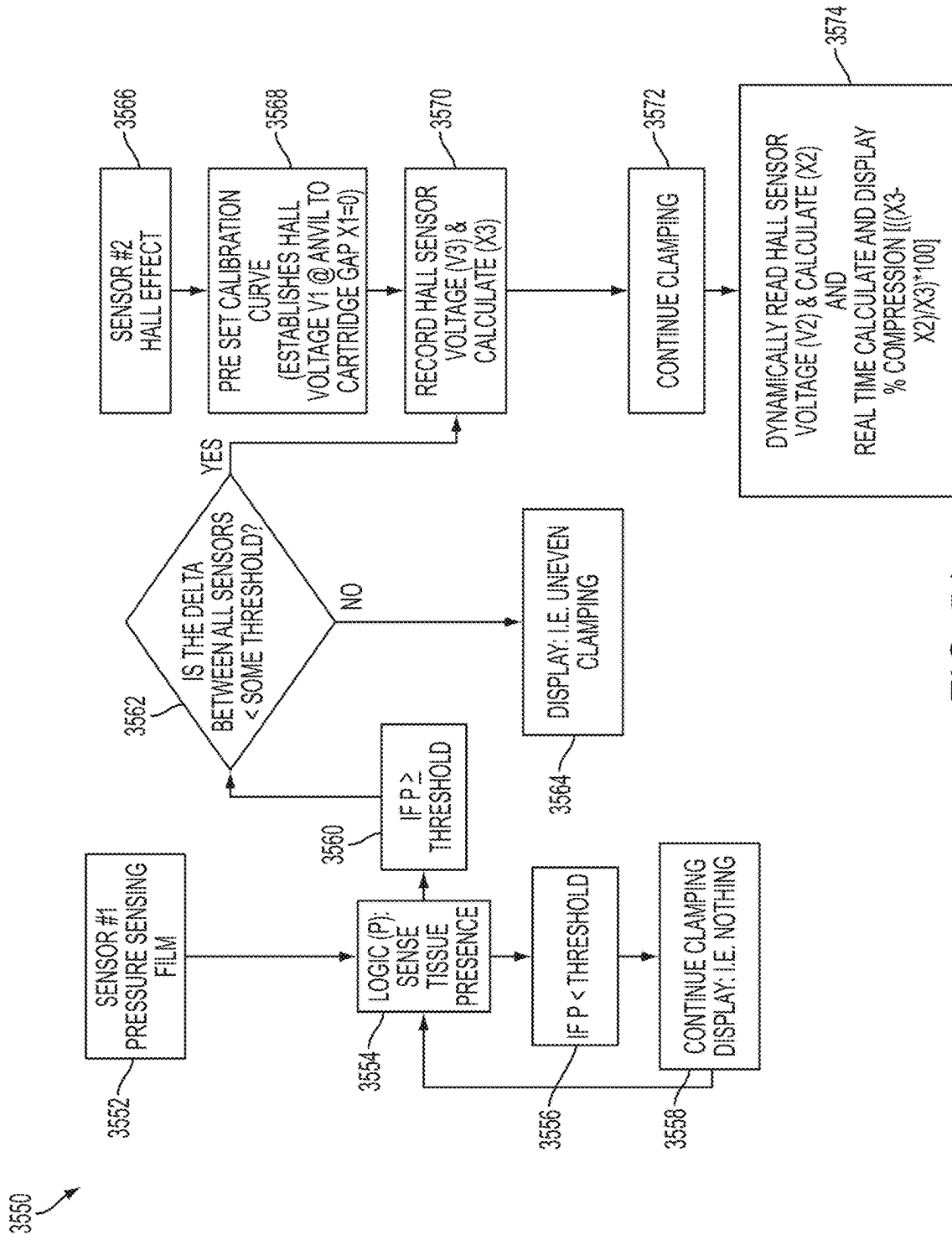
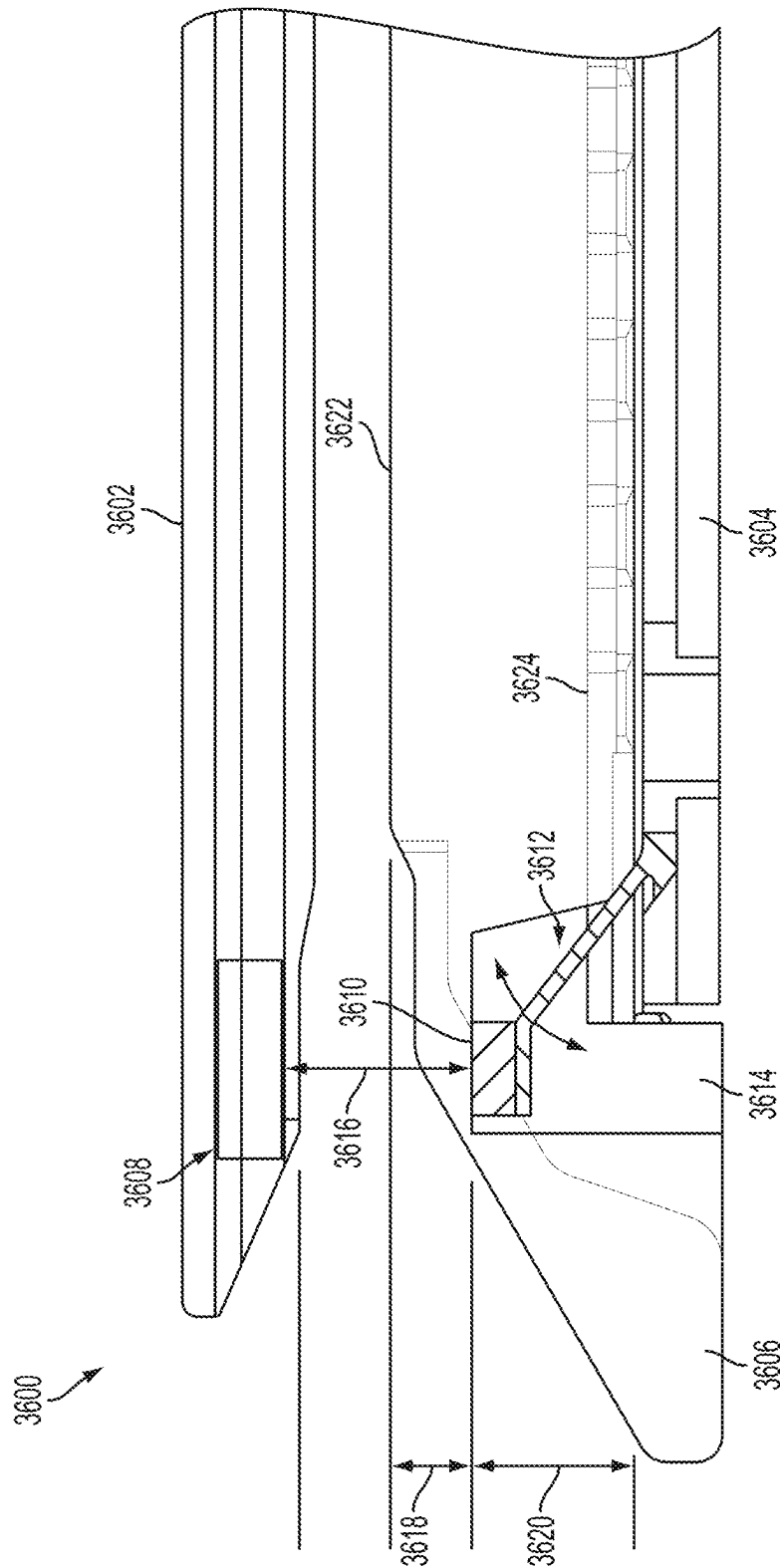


FIG. 51

52
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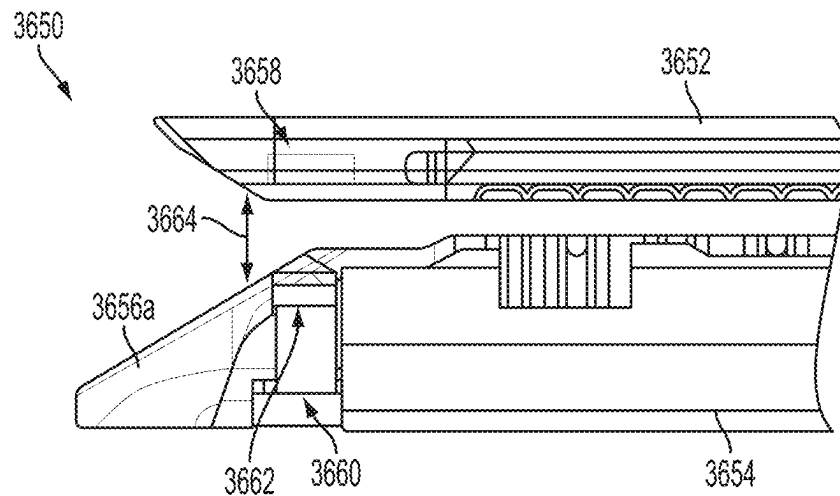


FIG. 53A

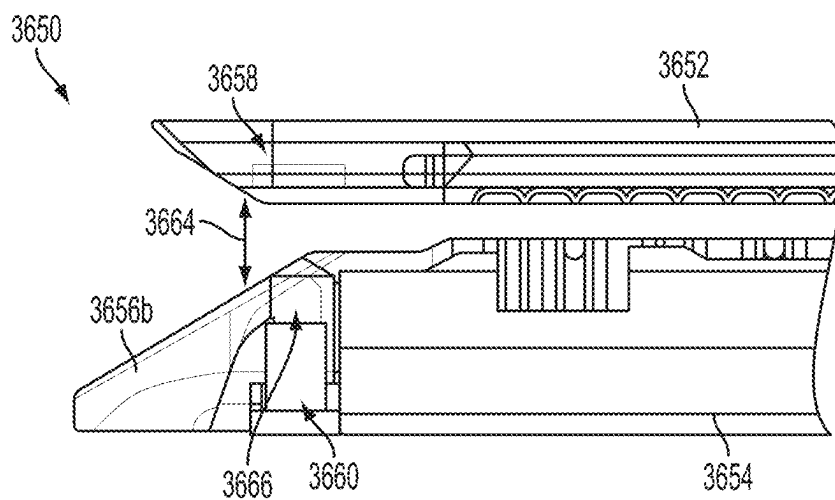


FIG. 53B

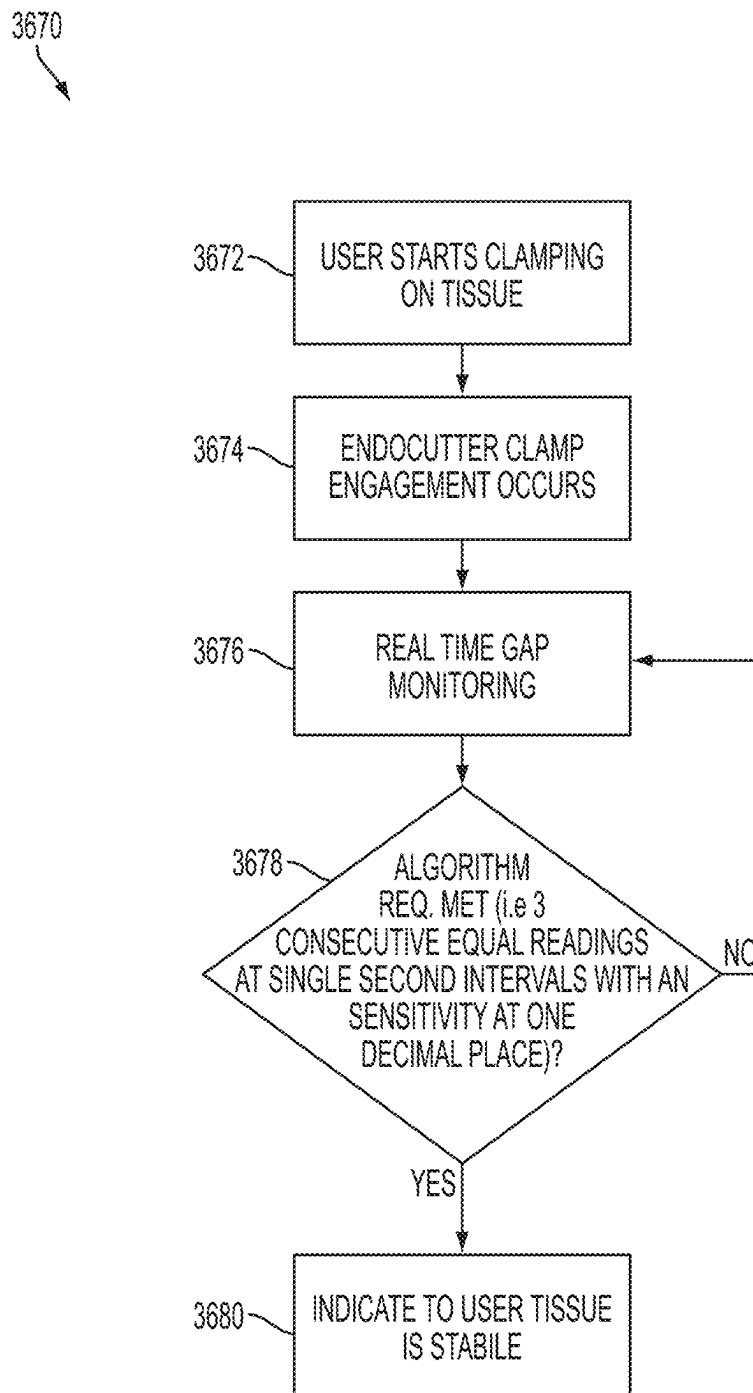


FIG. 54

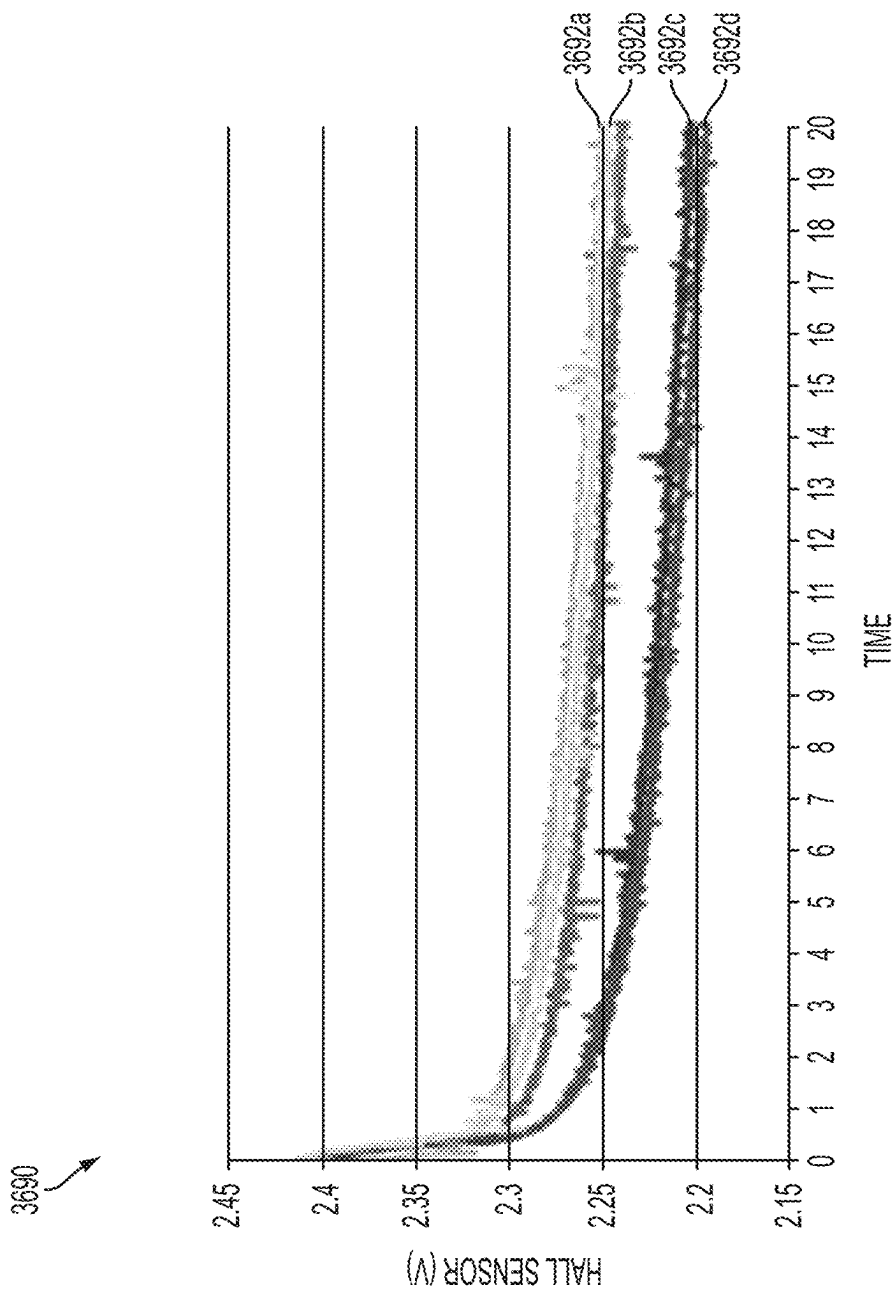


FIG. 55

3700

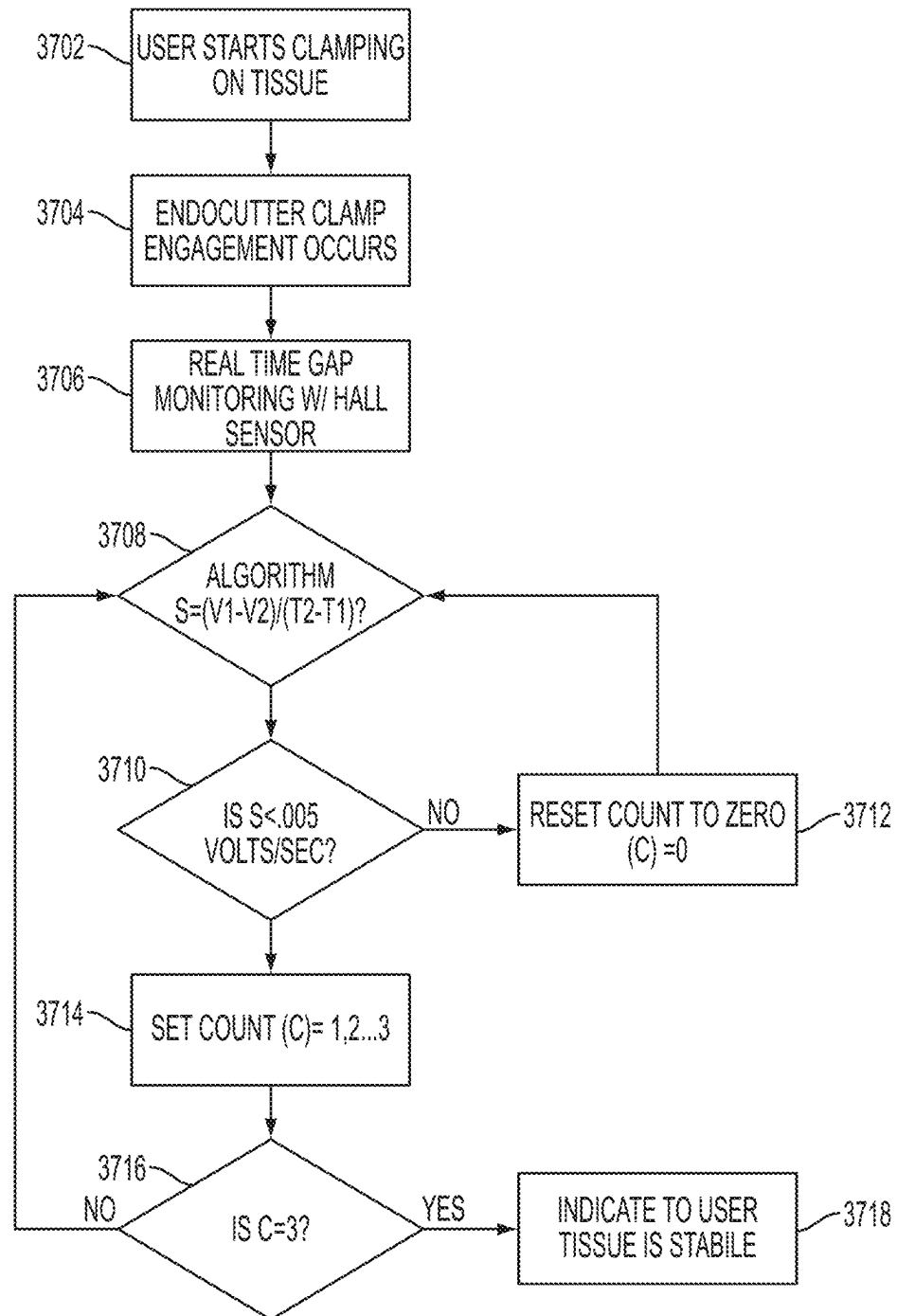


FIG. 56

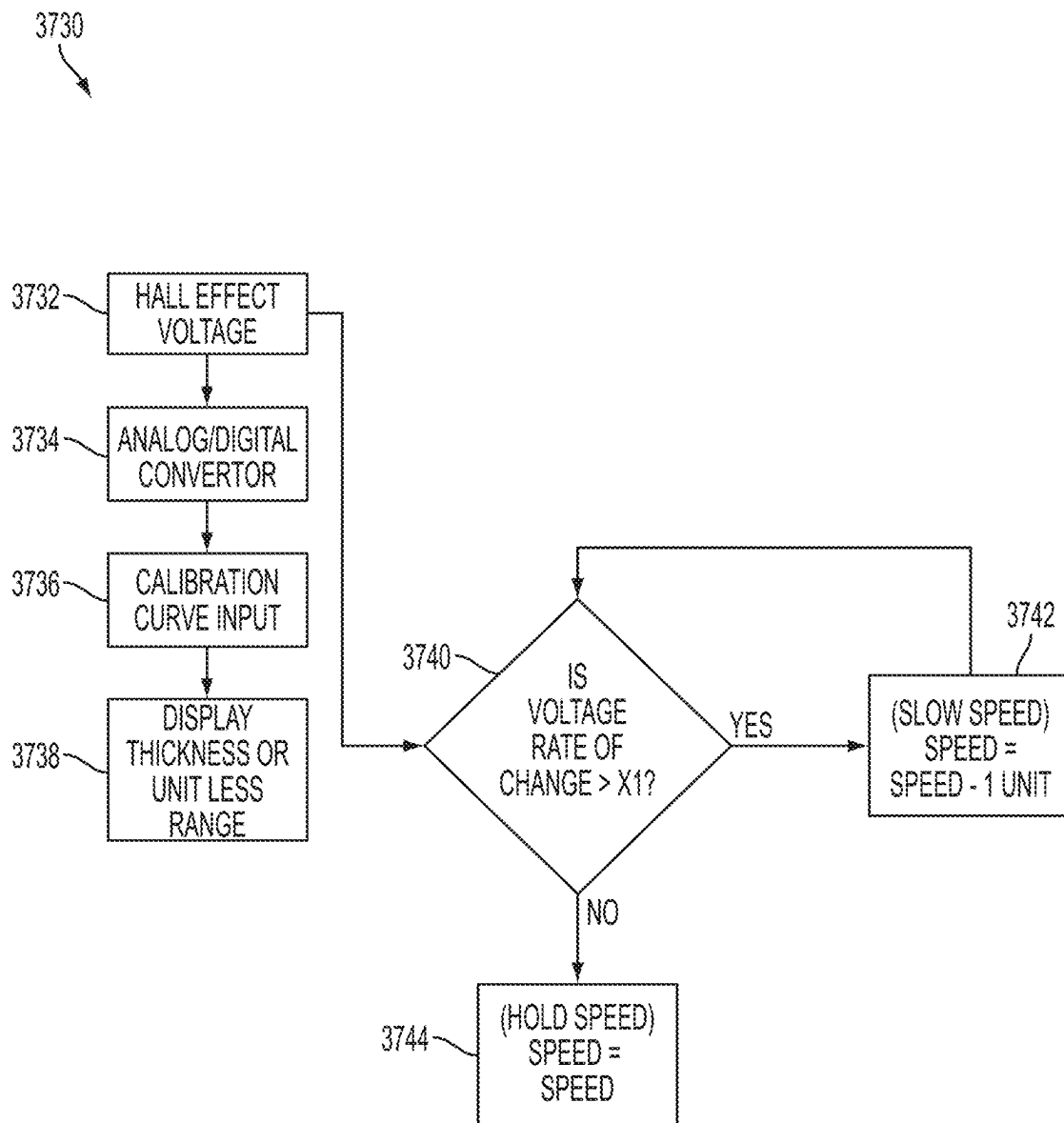


FIG. 57

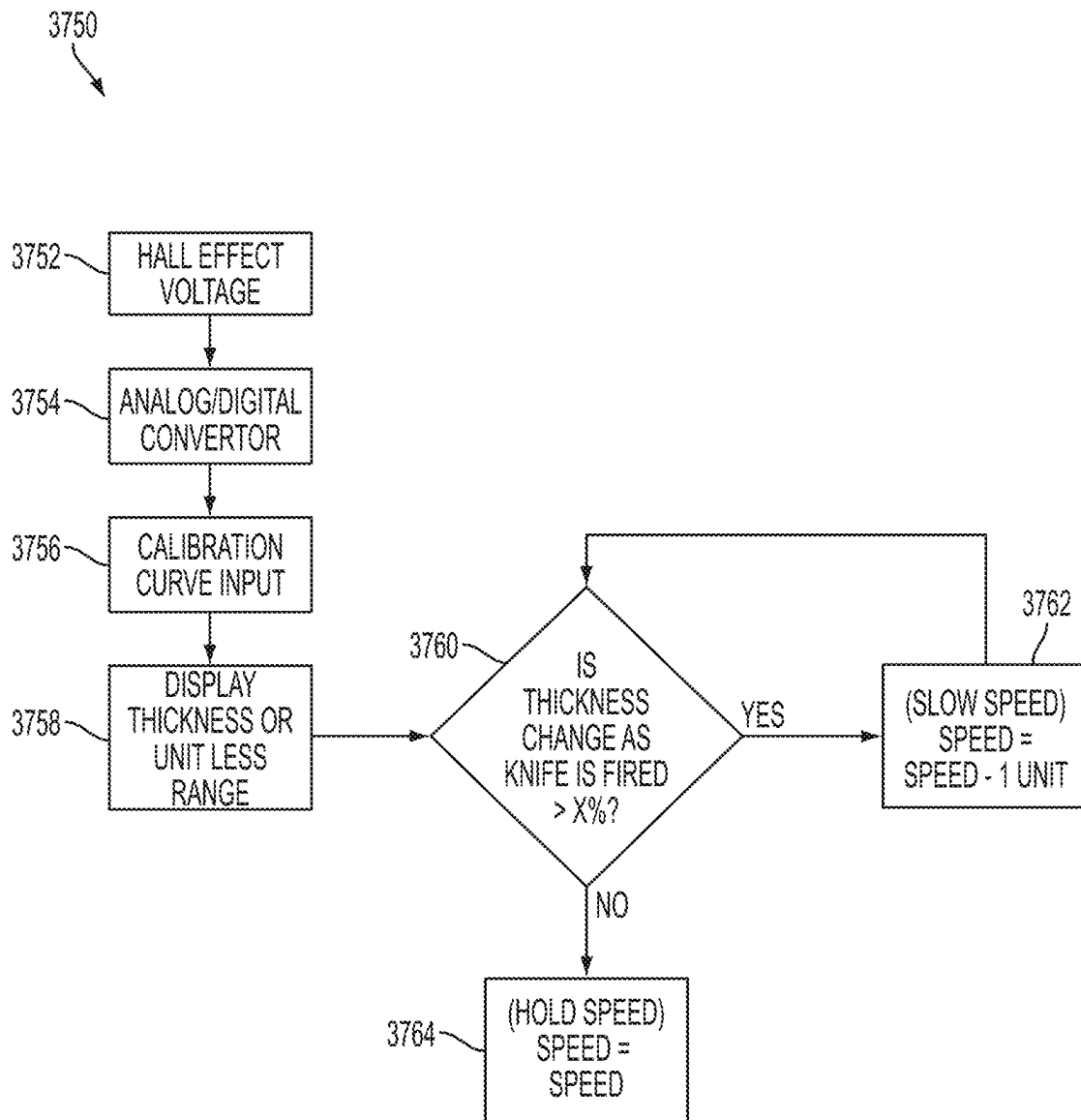


FIG. 58

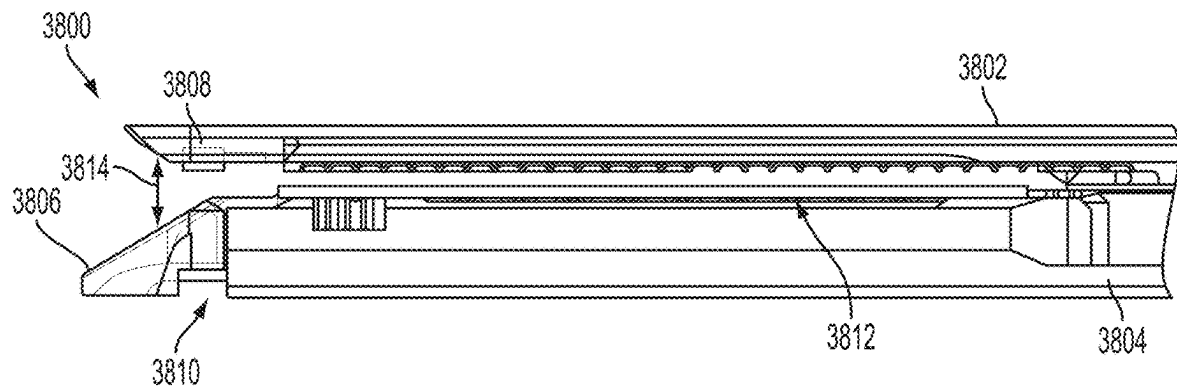


FIG. 59A

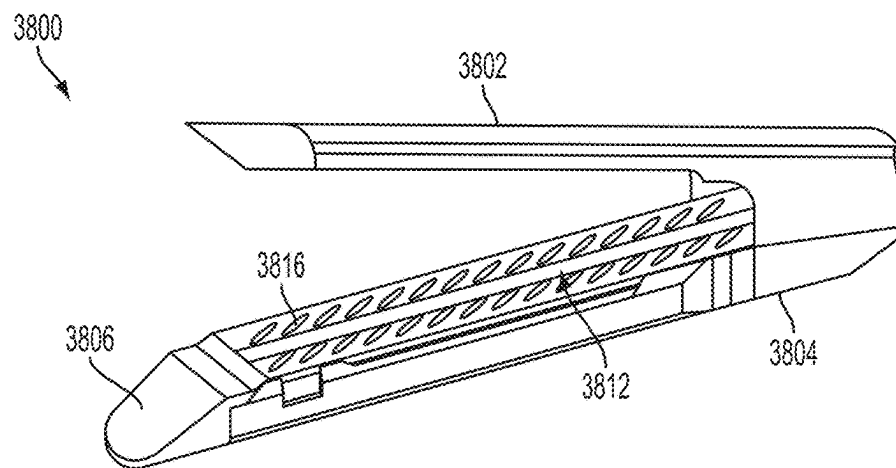


FIG. 59B

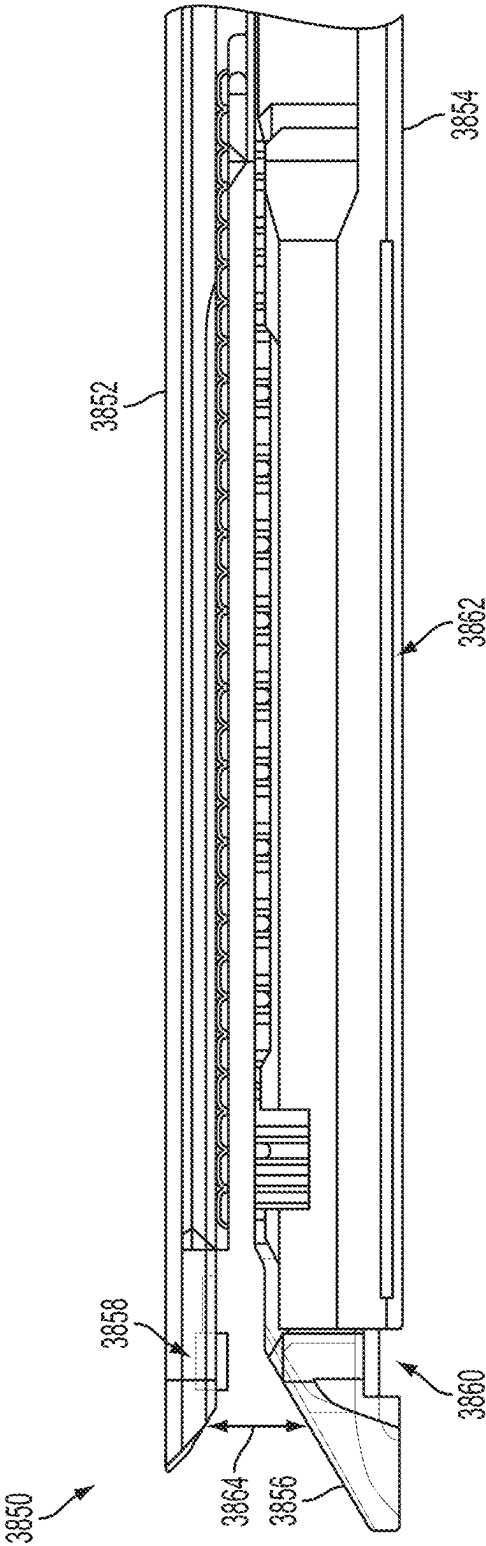


FIG. 60

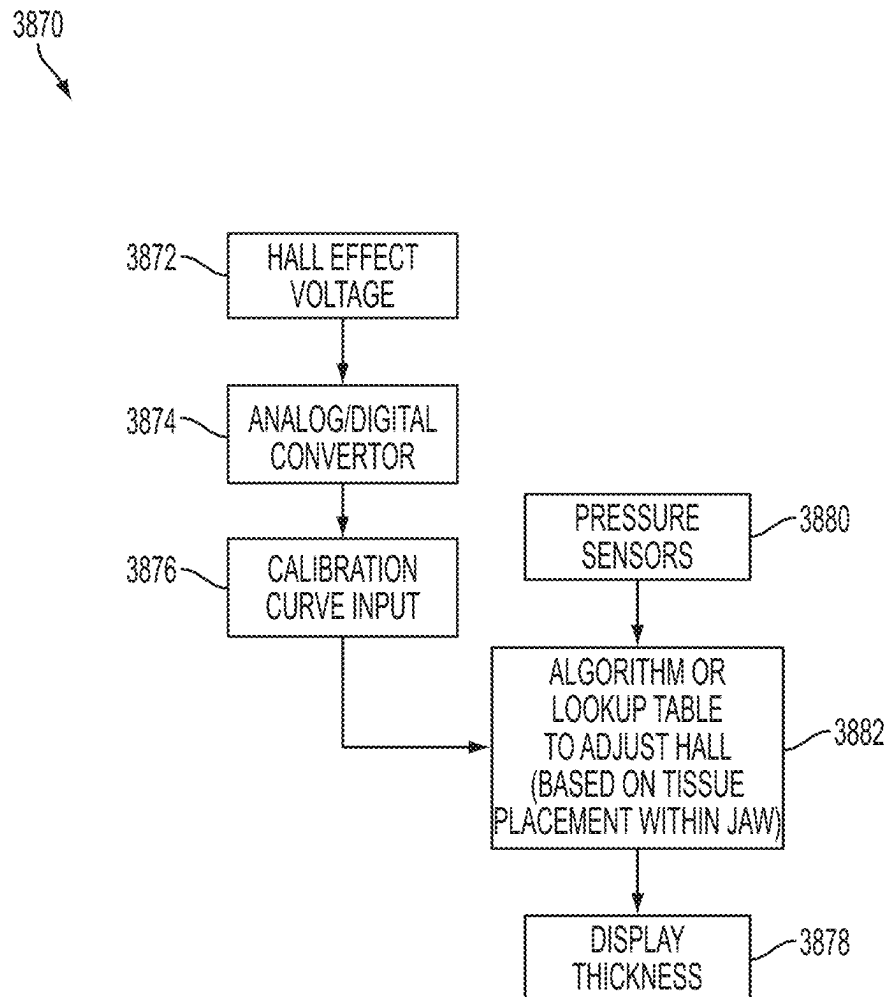


FIG. 61

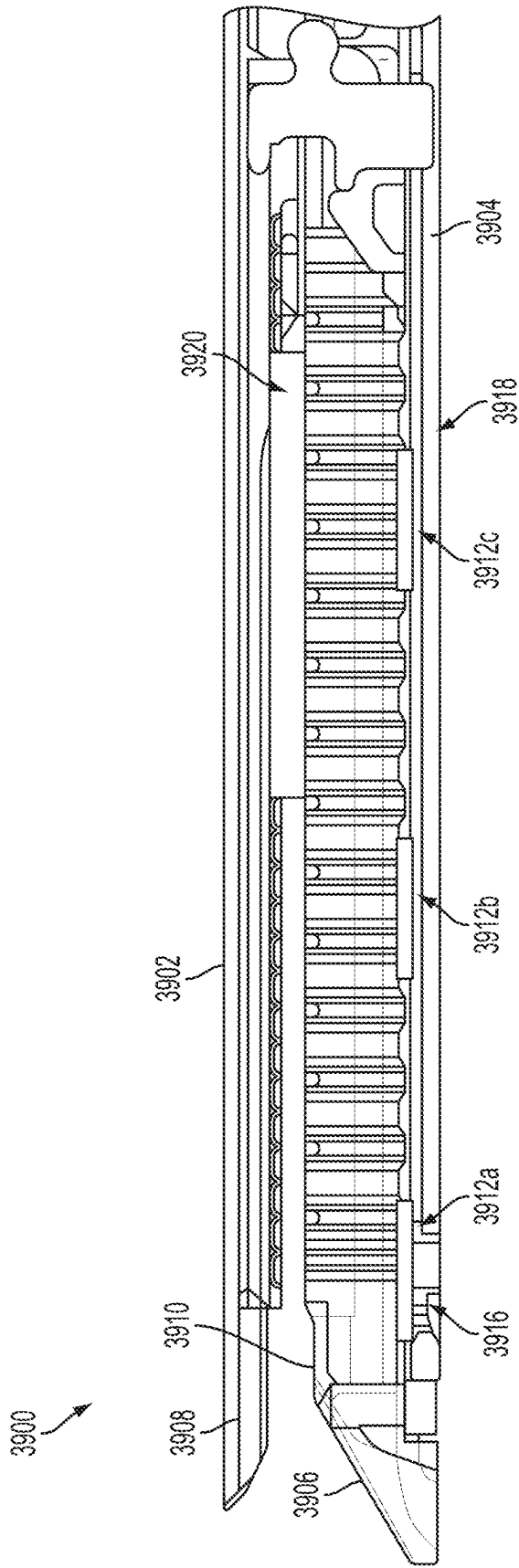


FIG. 62

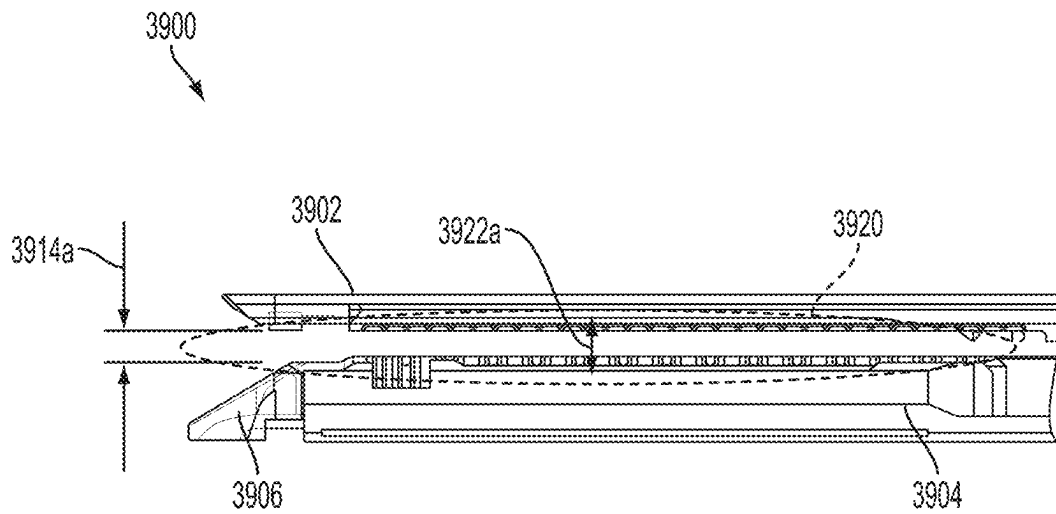


FIG. 63A

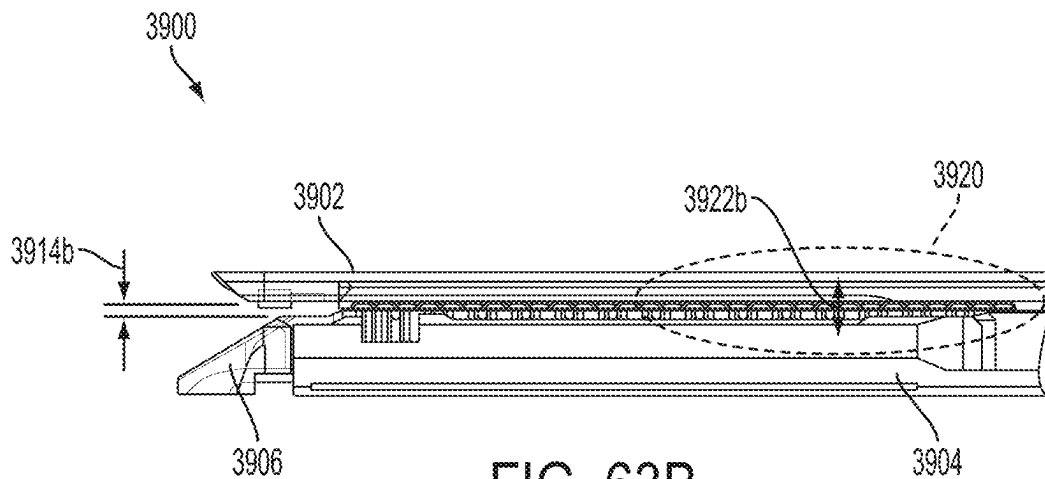


FIG. 63B

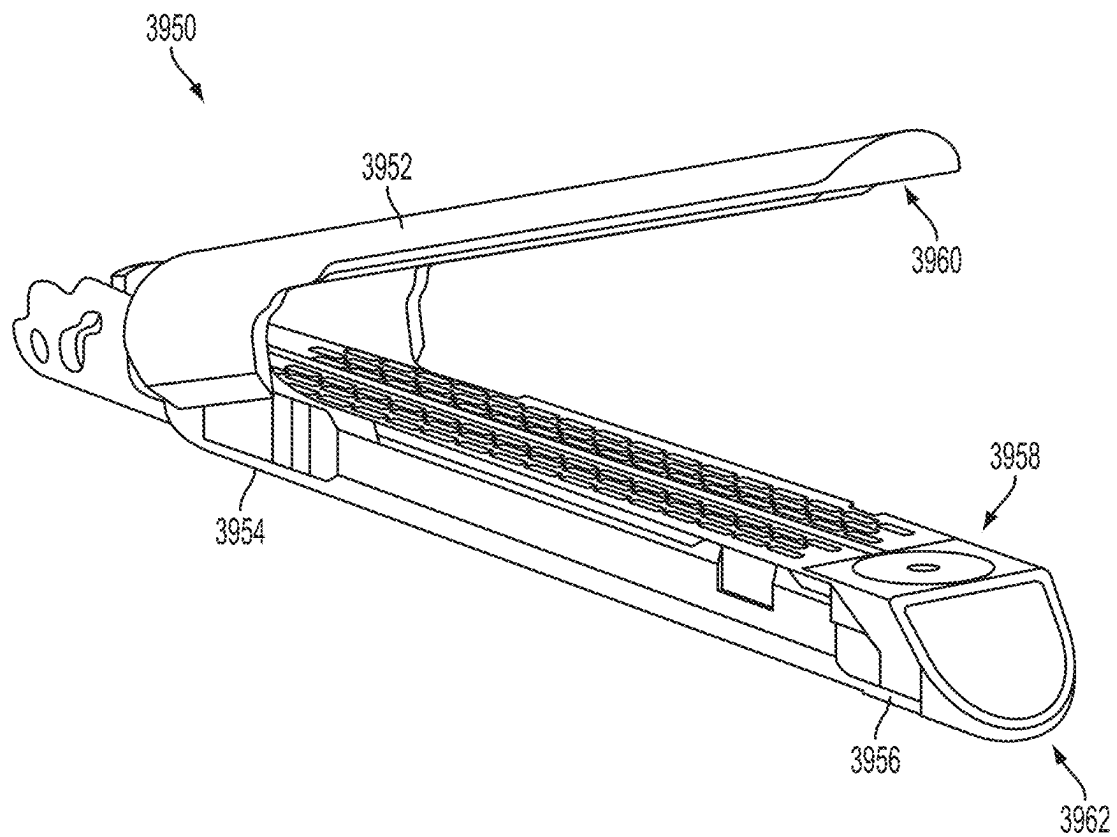


FIG. 64

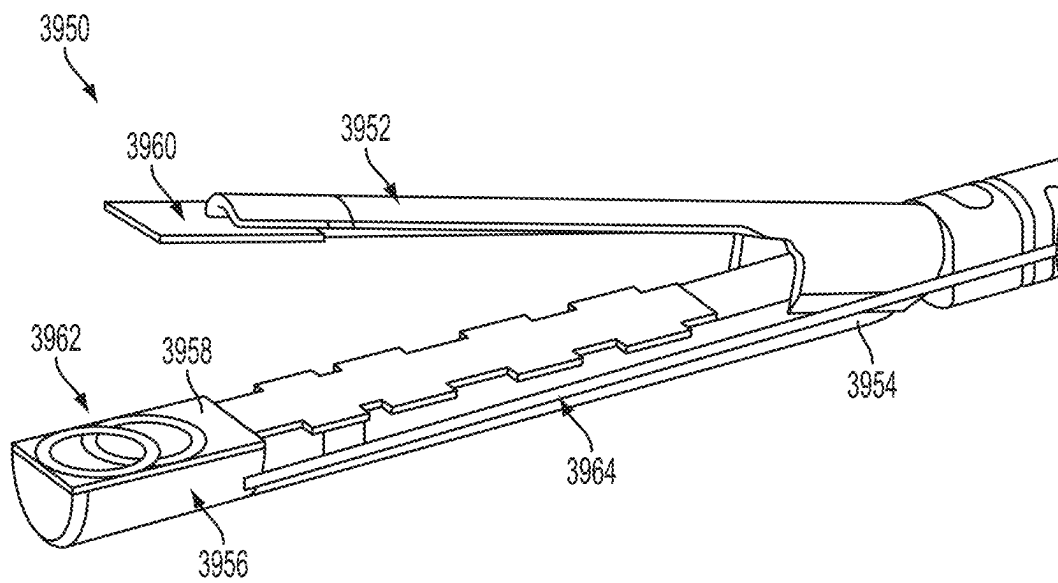


FIG. 65

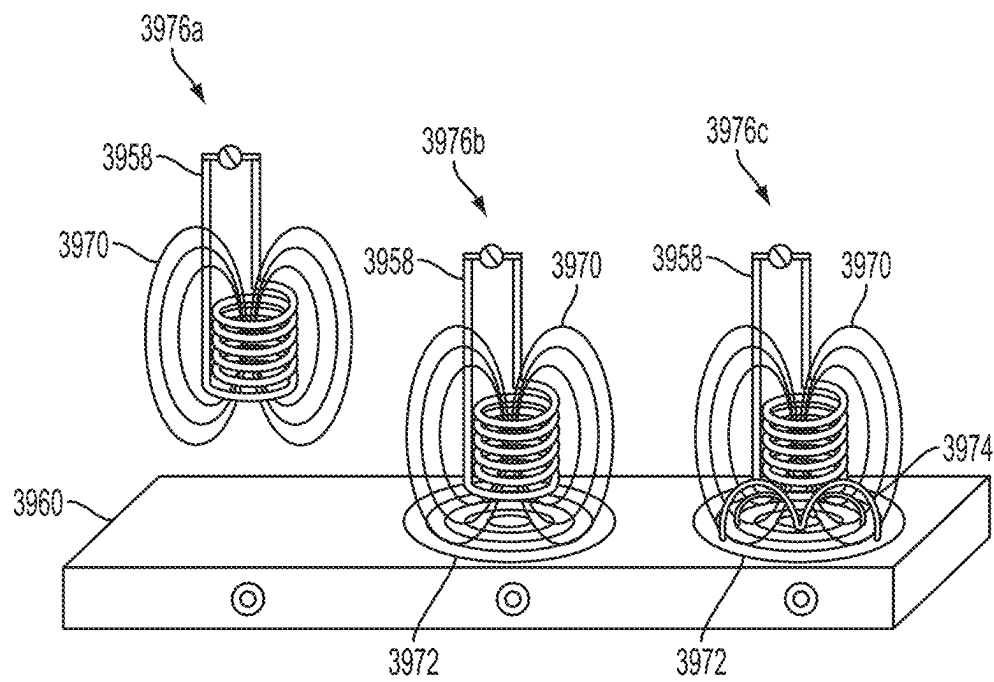


FIG. 66

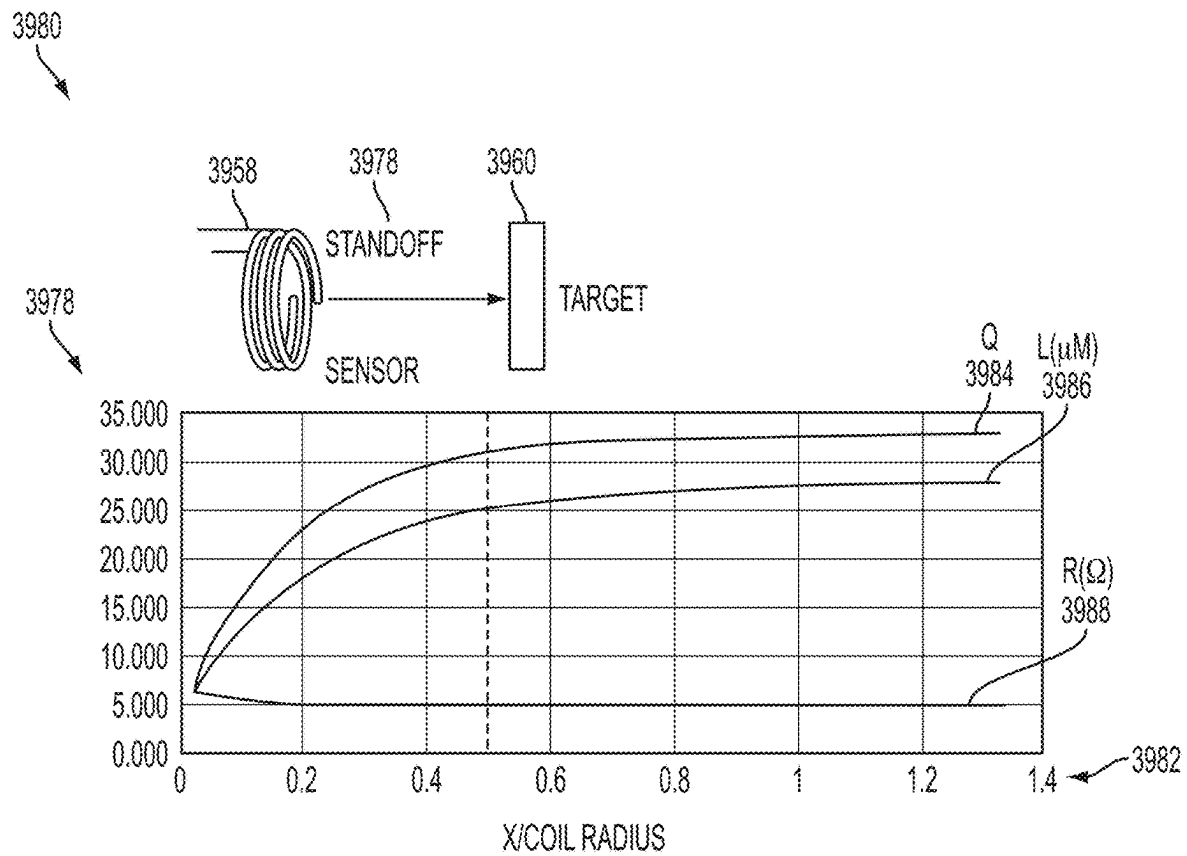


FIG. 67

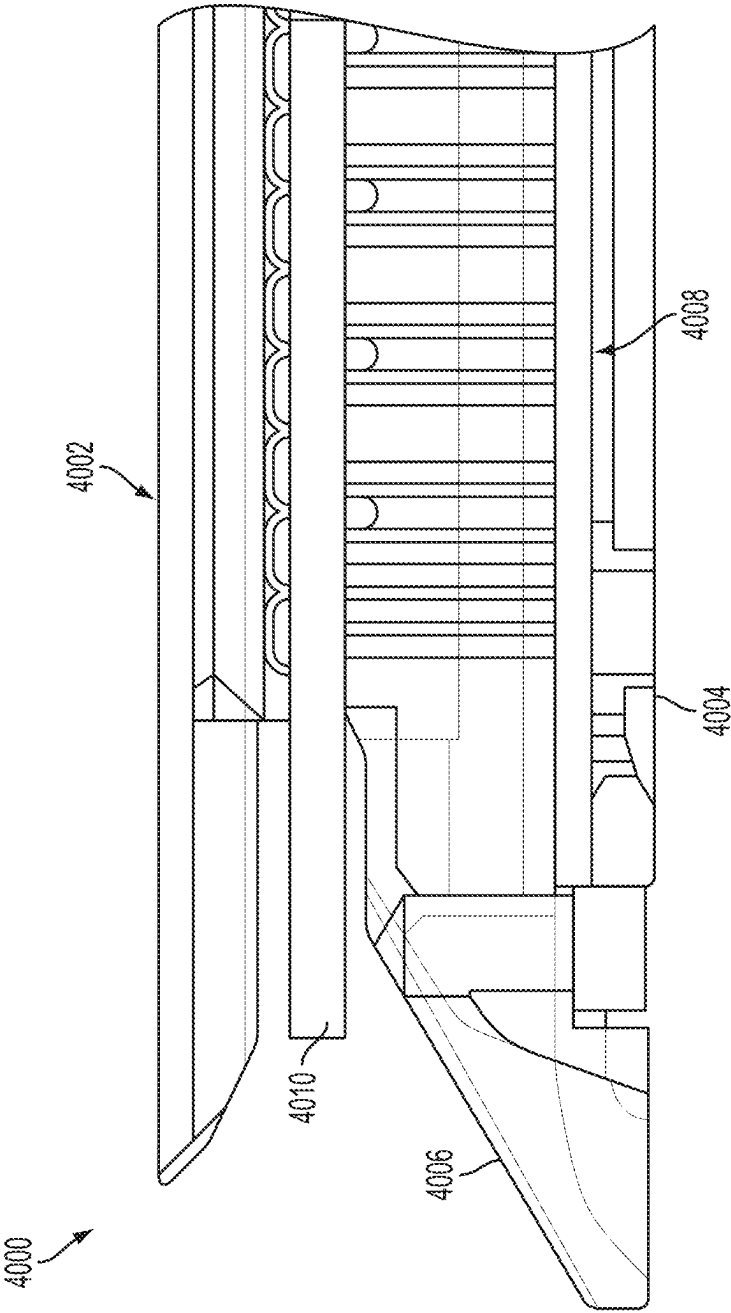


FIG. 68

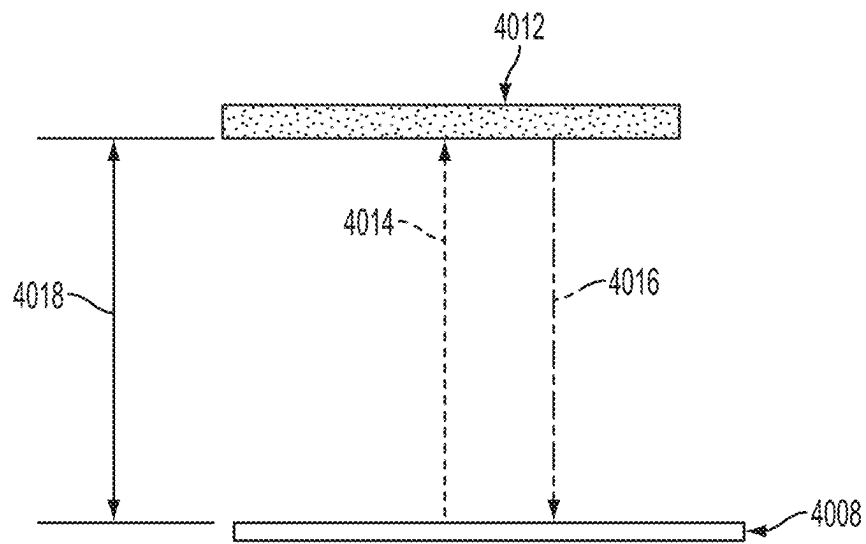


FIG. 69

4008

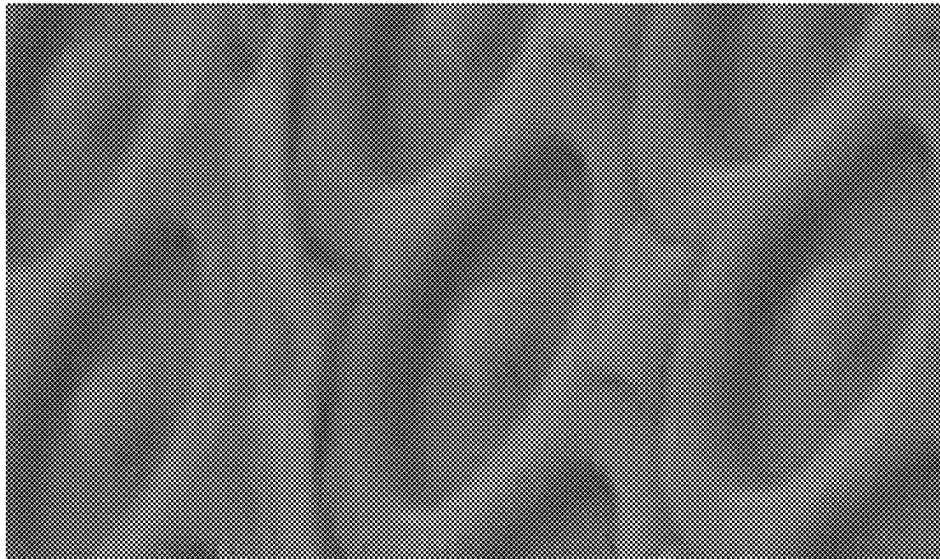


FIG. 70

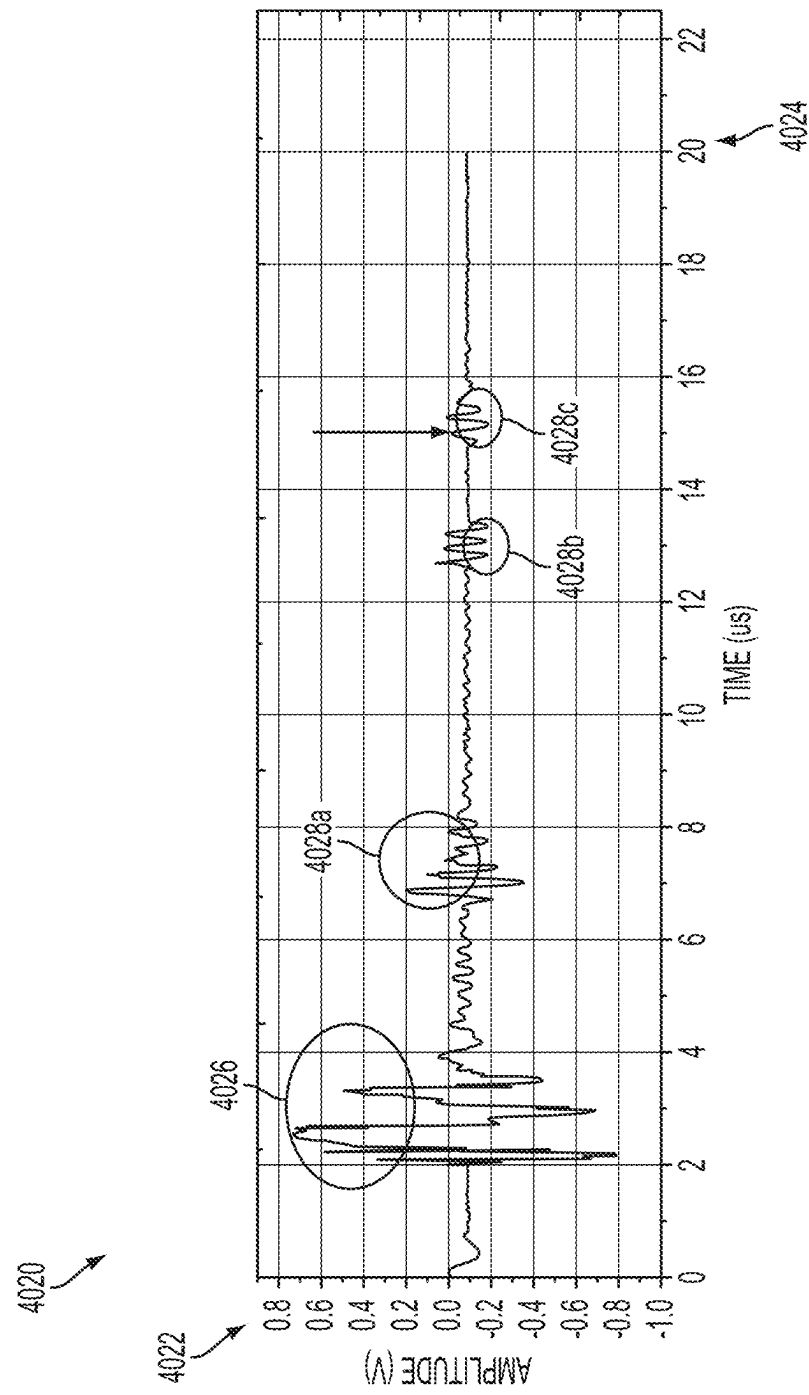


FIG. 71

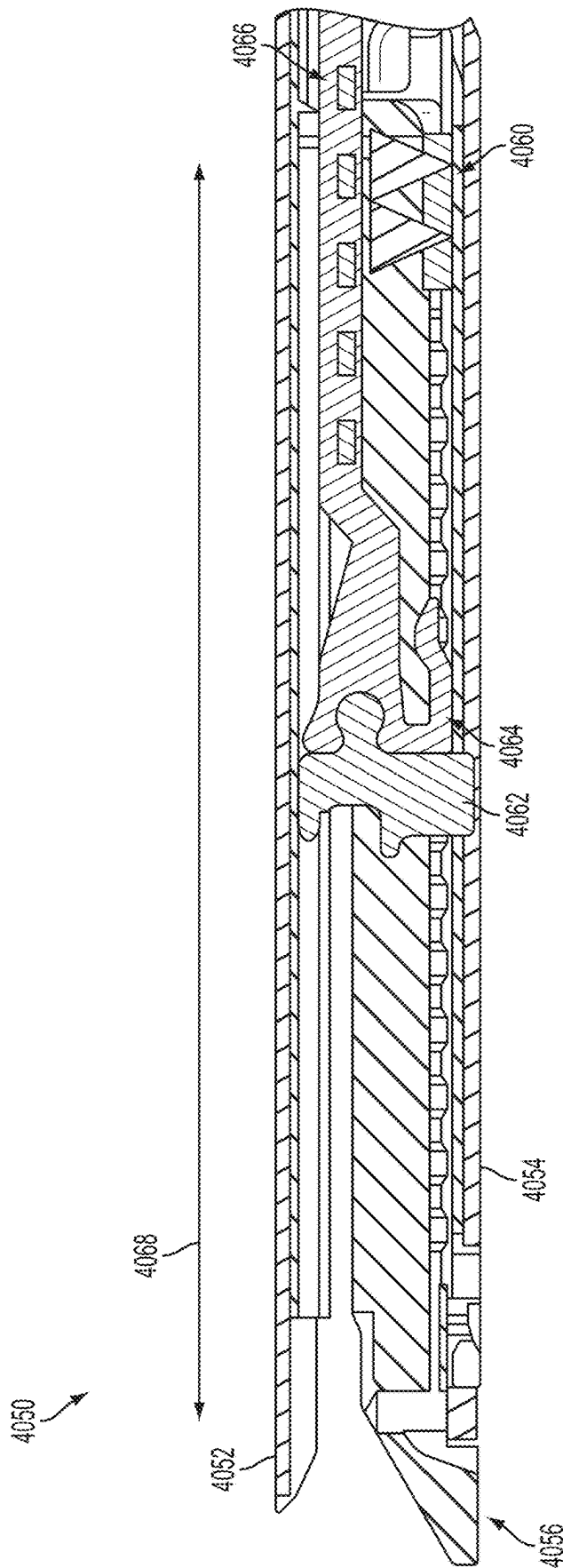


FIG. 72

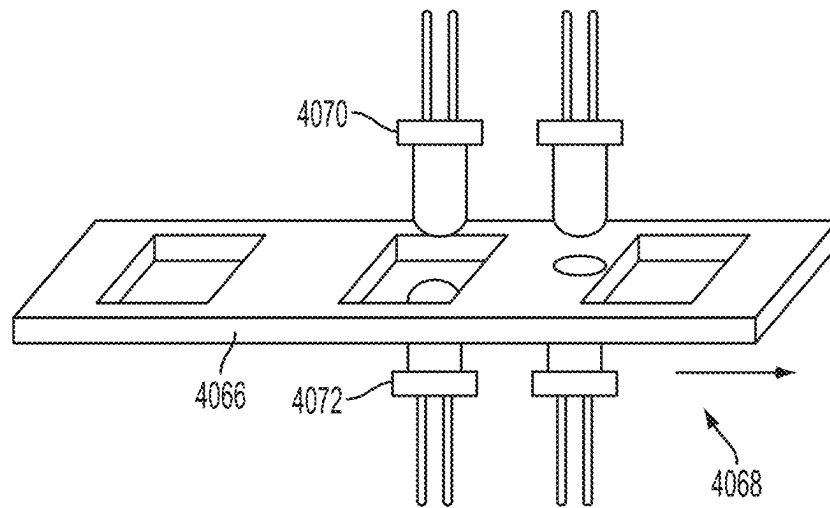


FIG. 73

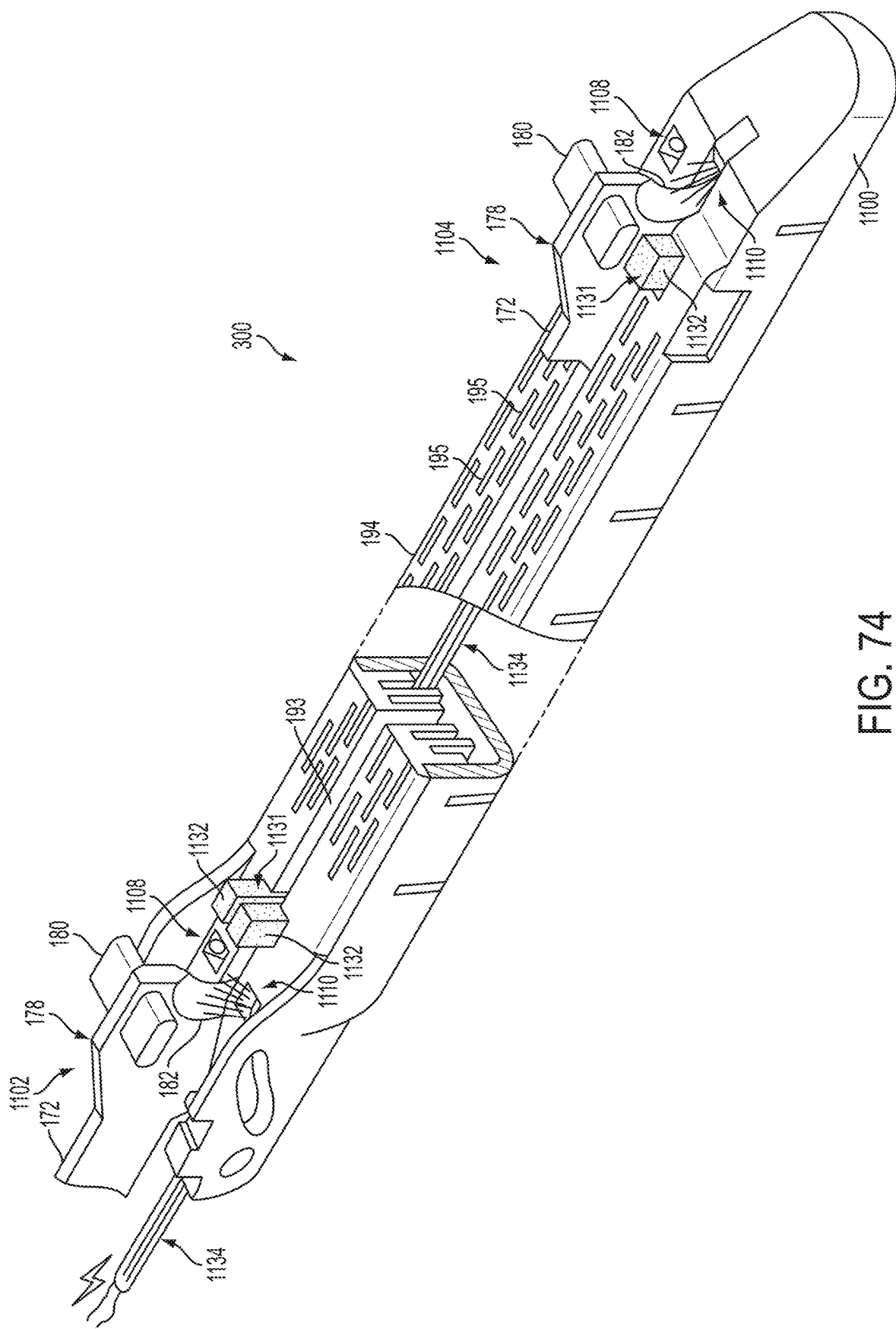


FIG. 74

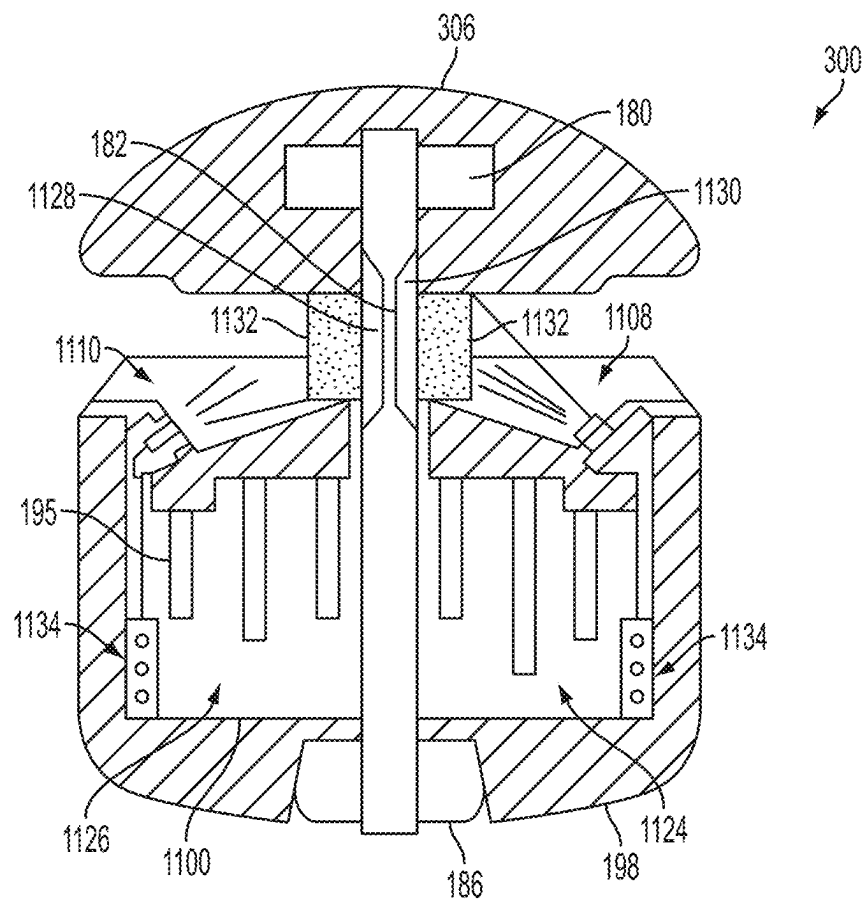


FIG. 75

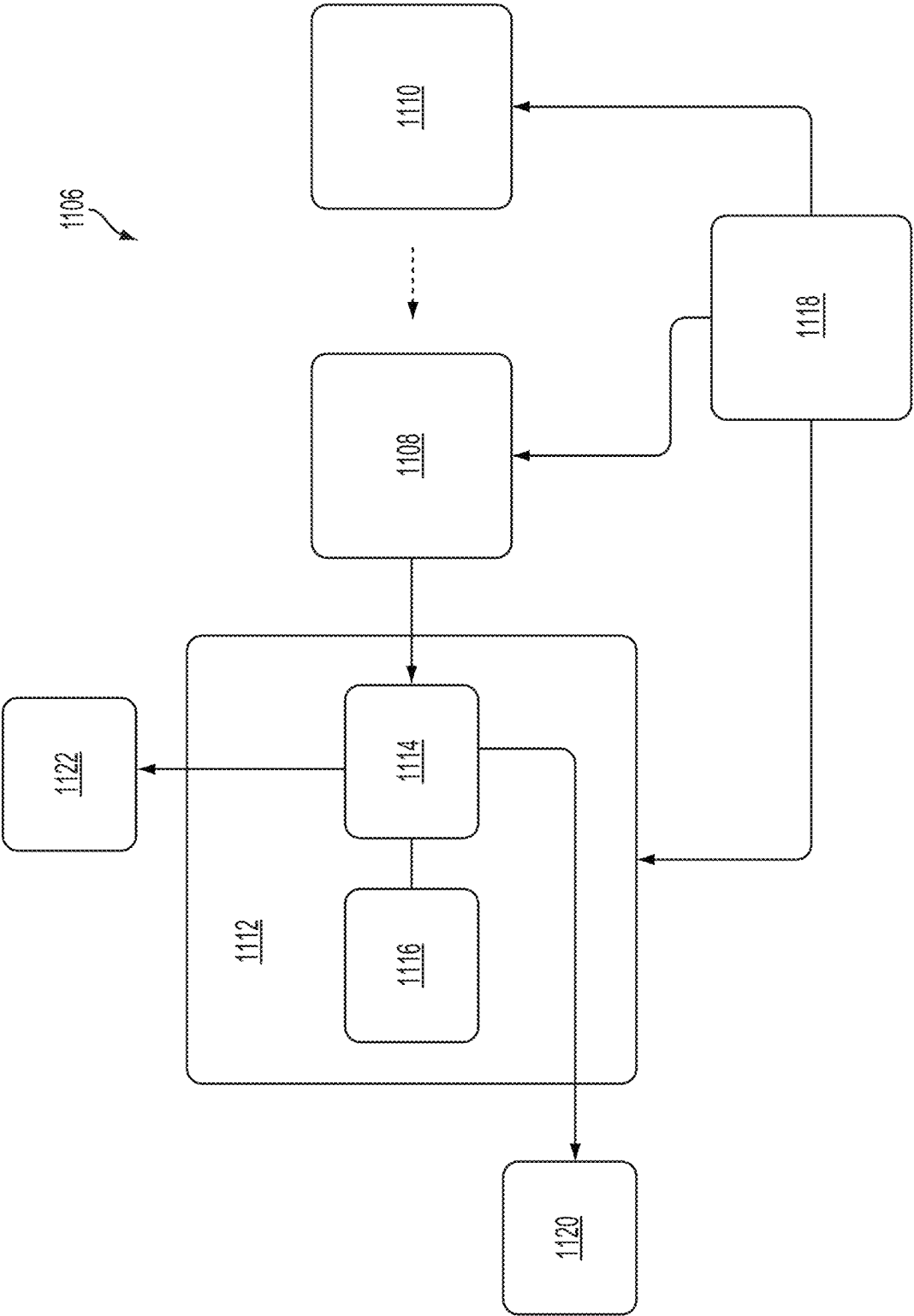


FIG. 76

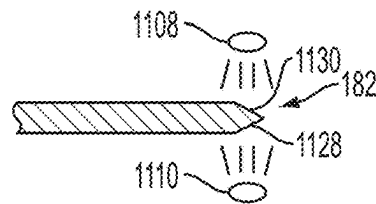


FIG. 77

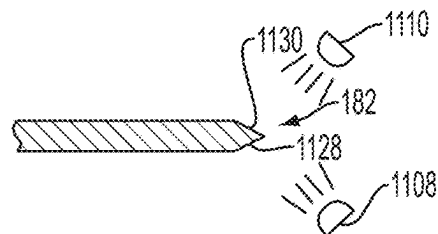


FIG. 78

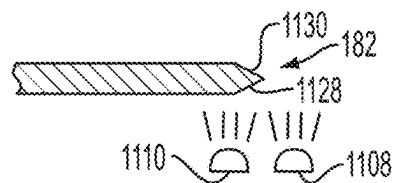


FIG. 79

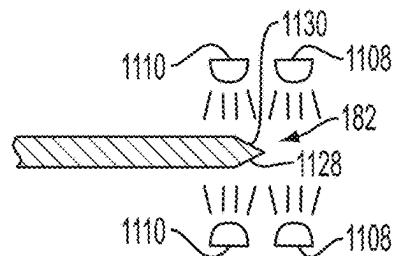


FIG. 80

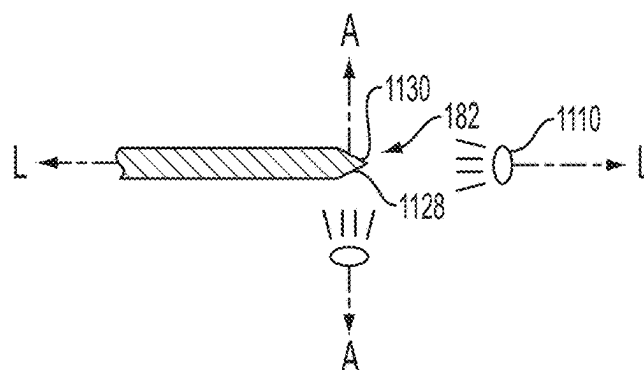


FIG. 81

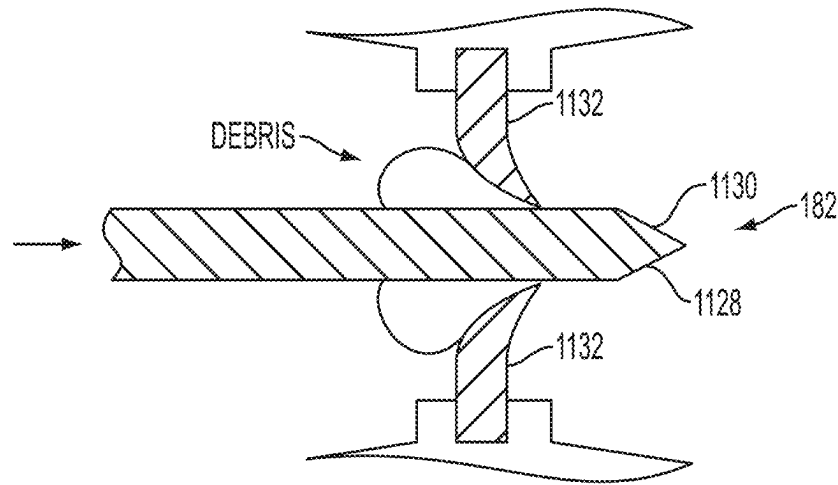


FIG. 82

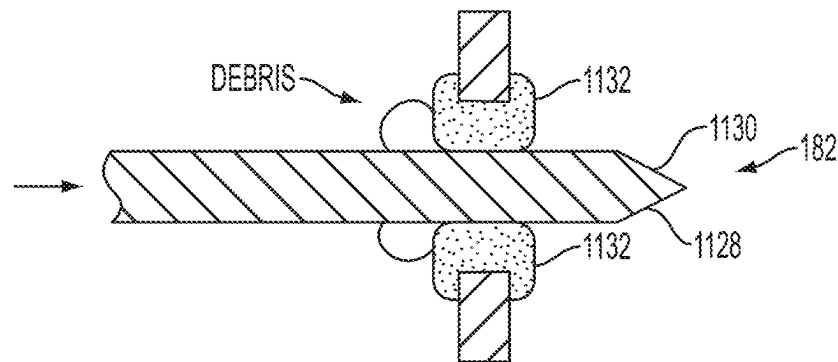


FIG. 83

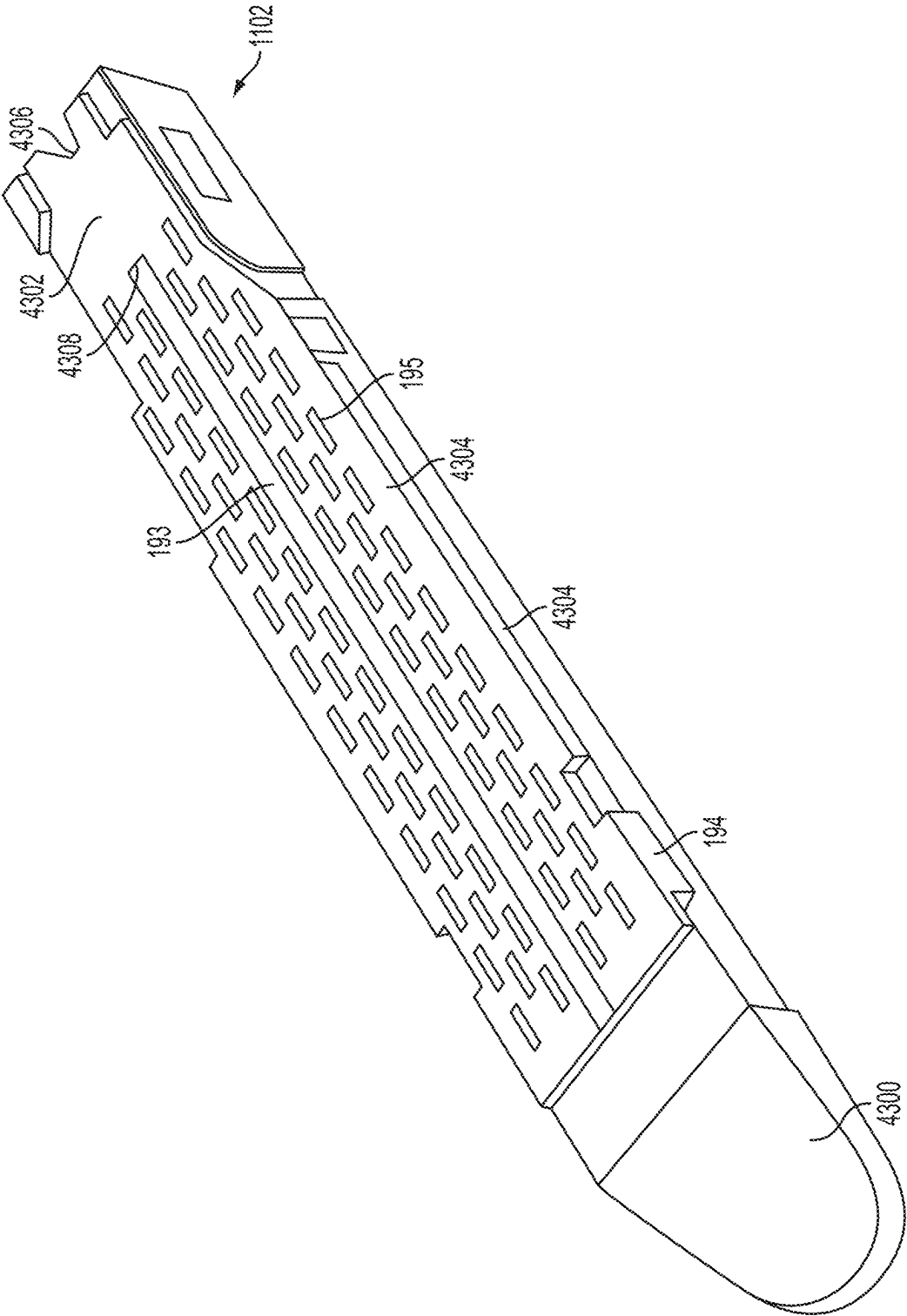


FIG. 84

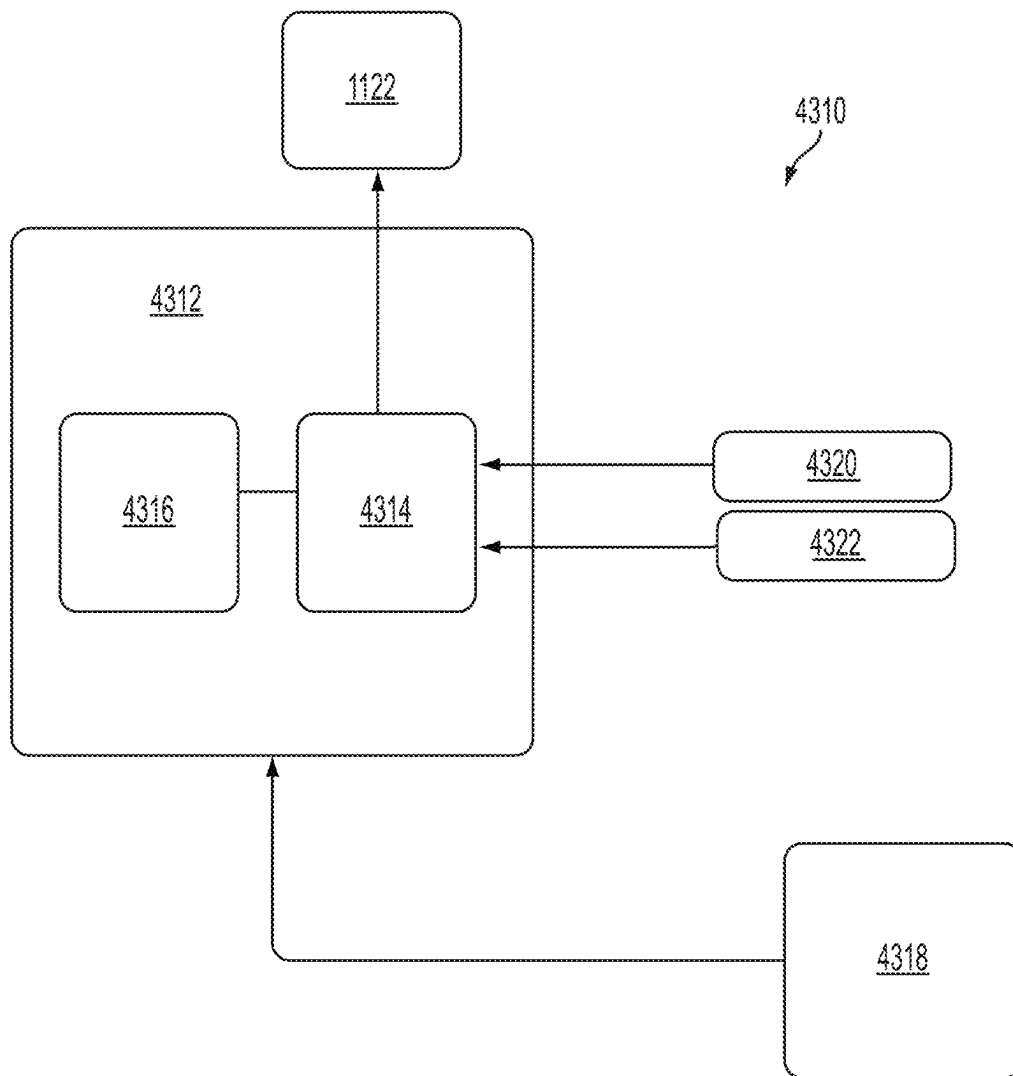


FIG. 85

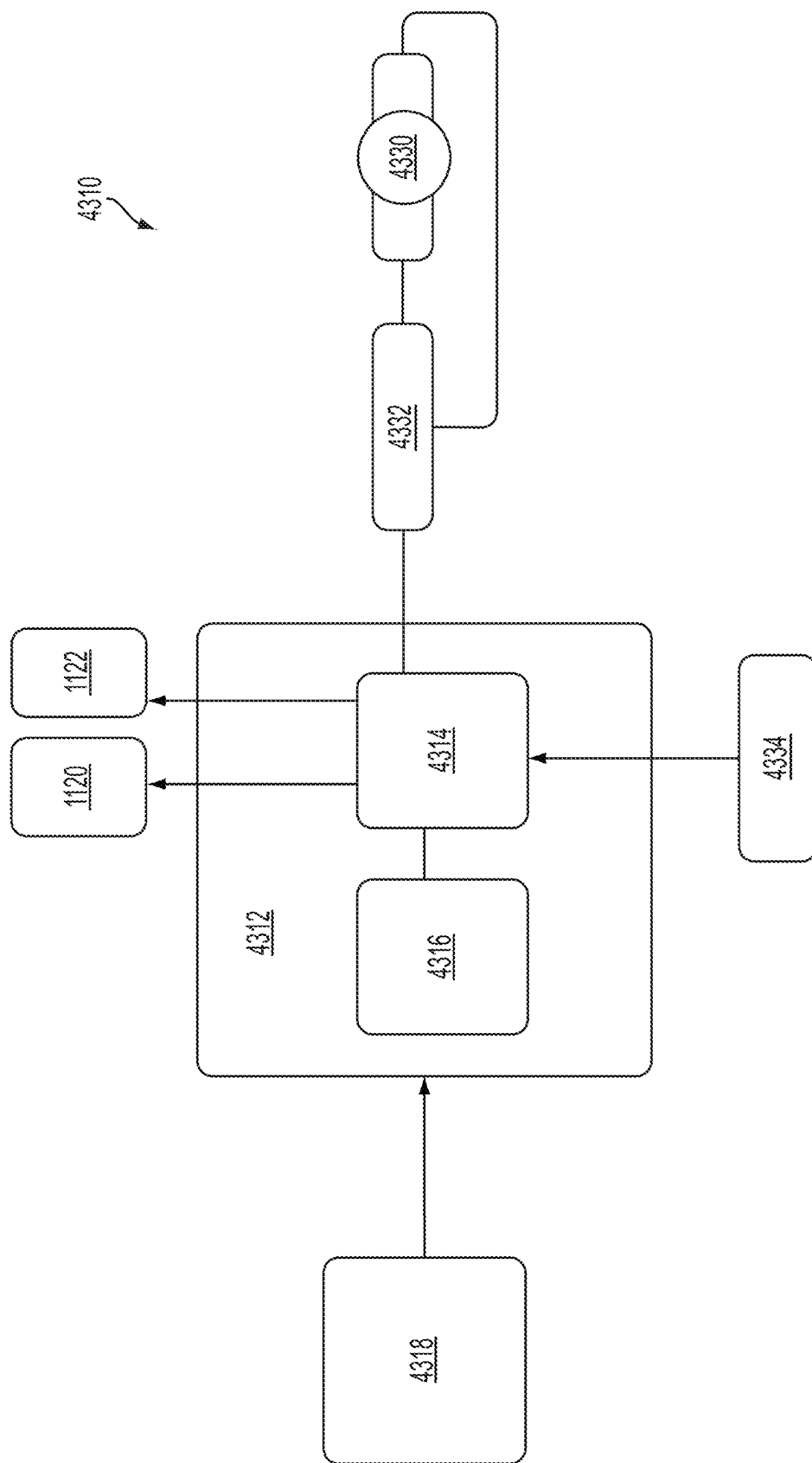


FIG. 86

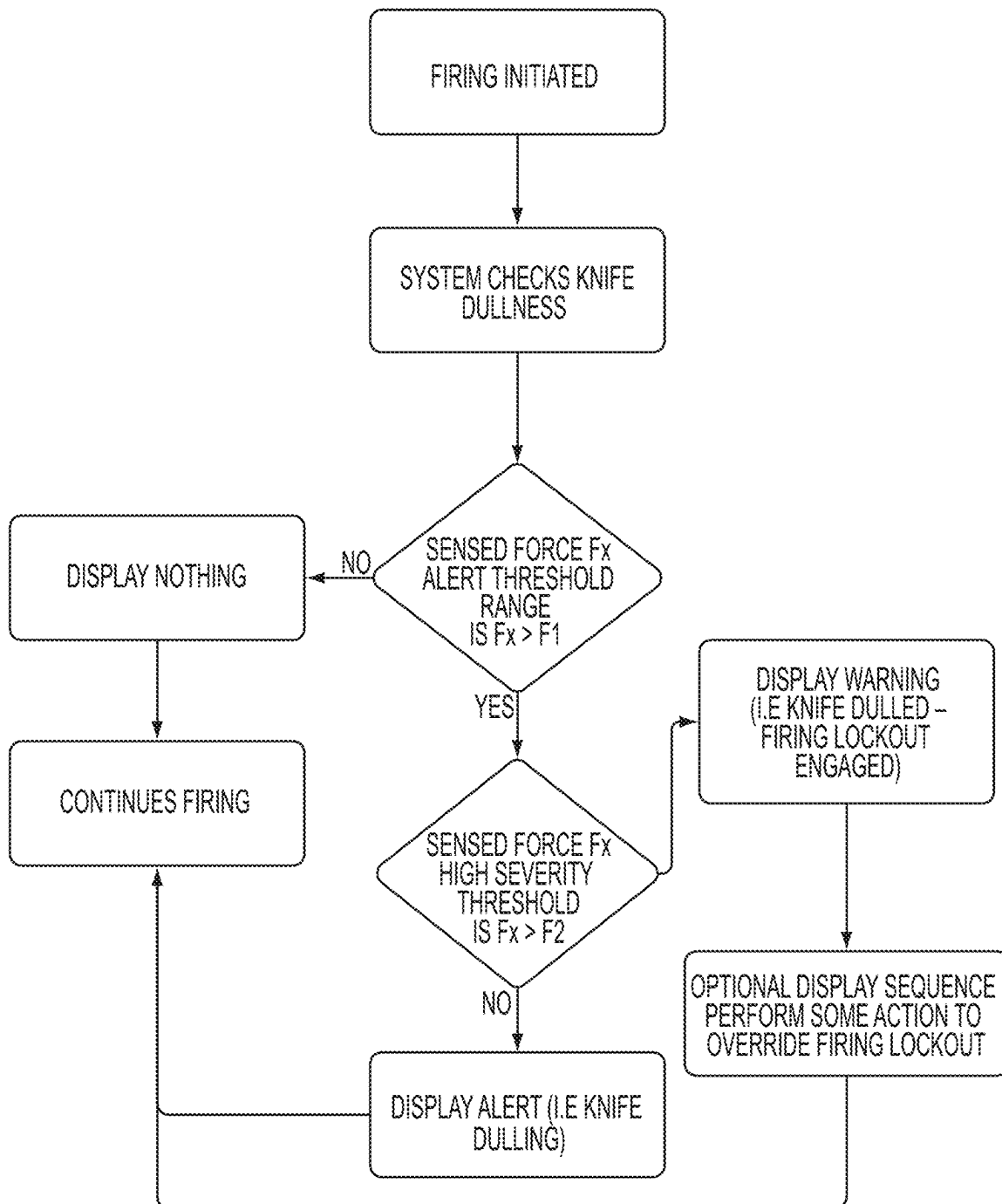


FIG. 87

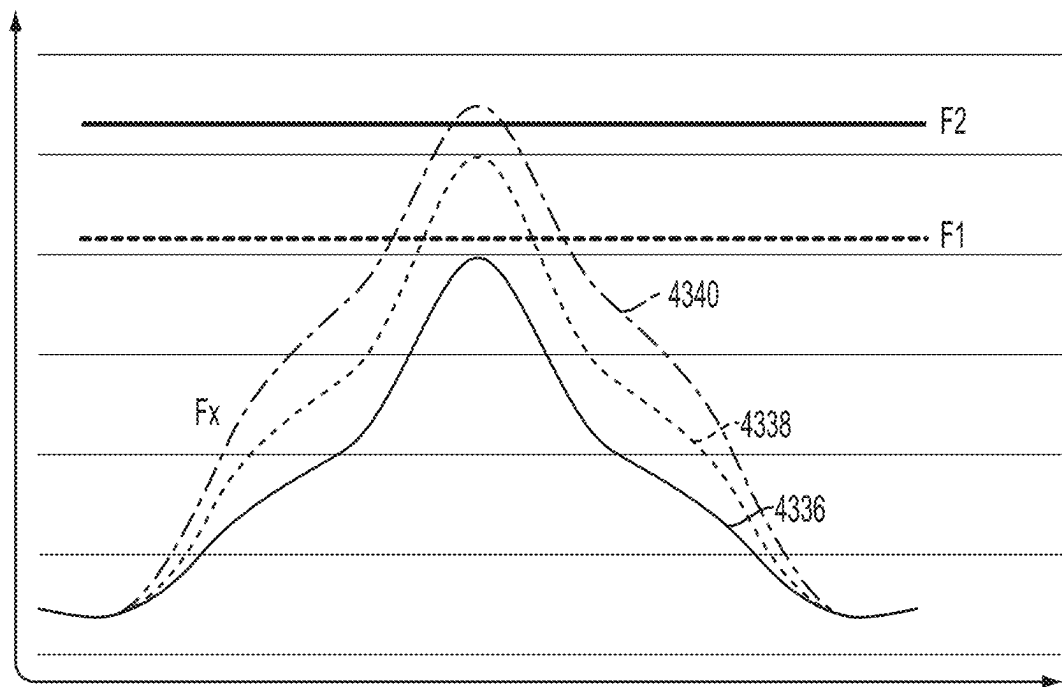


FIG. 88

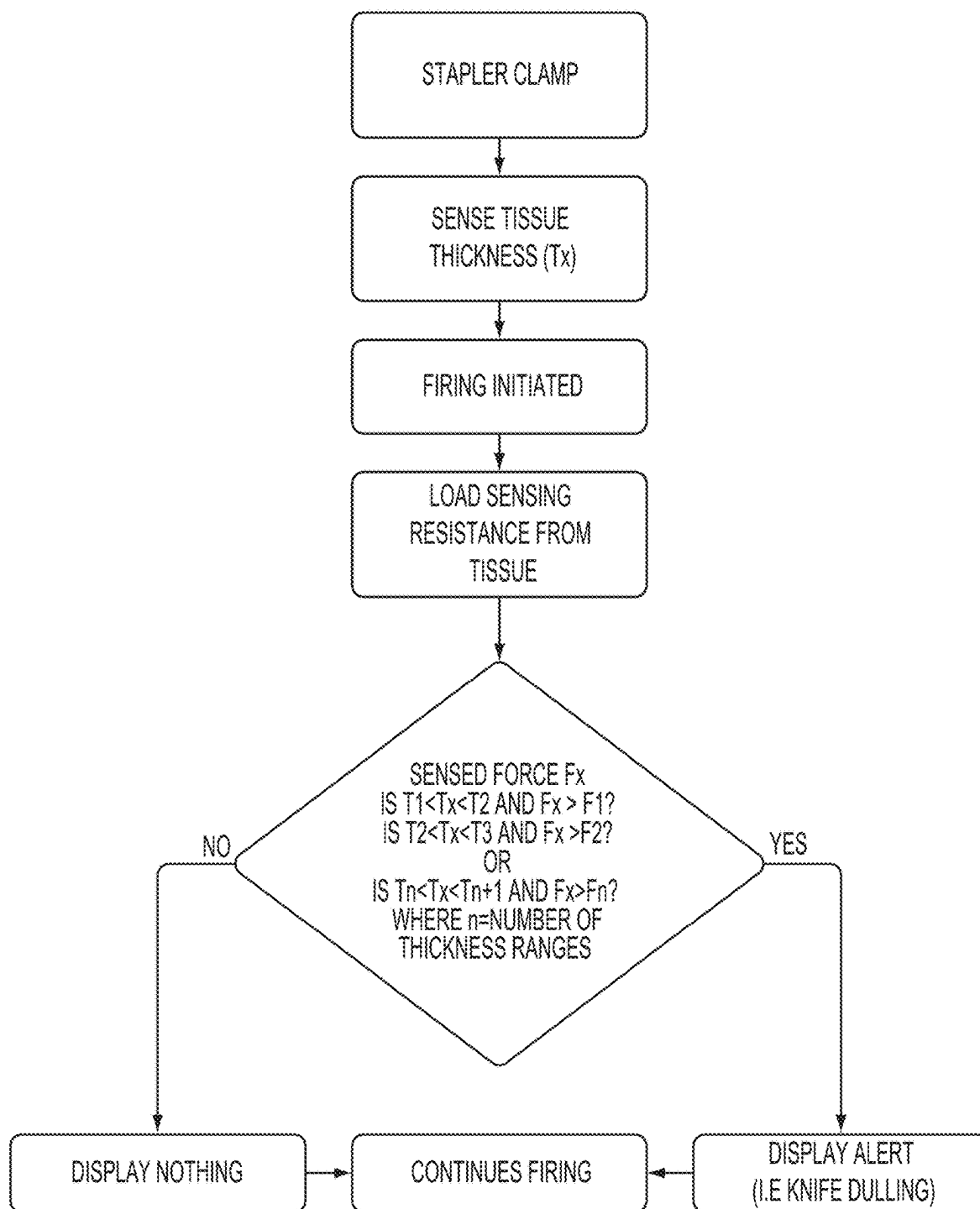


FIG. 89

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TISSUE THICKNESS RANGES (T) ▼	THRESHOLD FORCE (F) ▼
T1-T2	F1
T2-T3	F2
T3-T4	F3
-	-
-	-
-	-
Tn-1 - Tn	Fn

FIG. 90

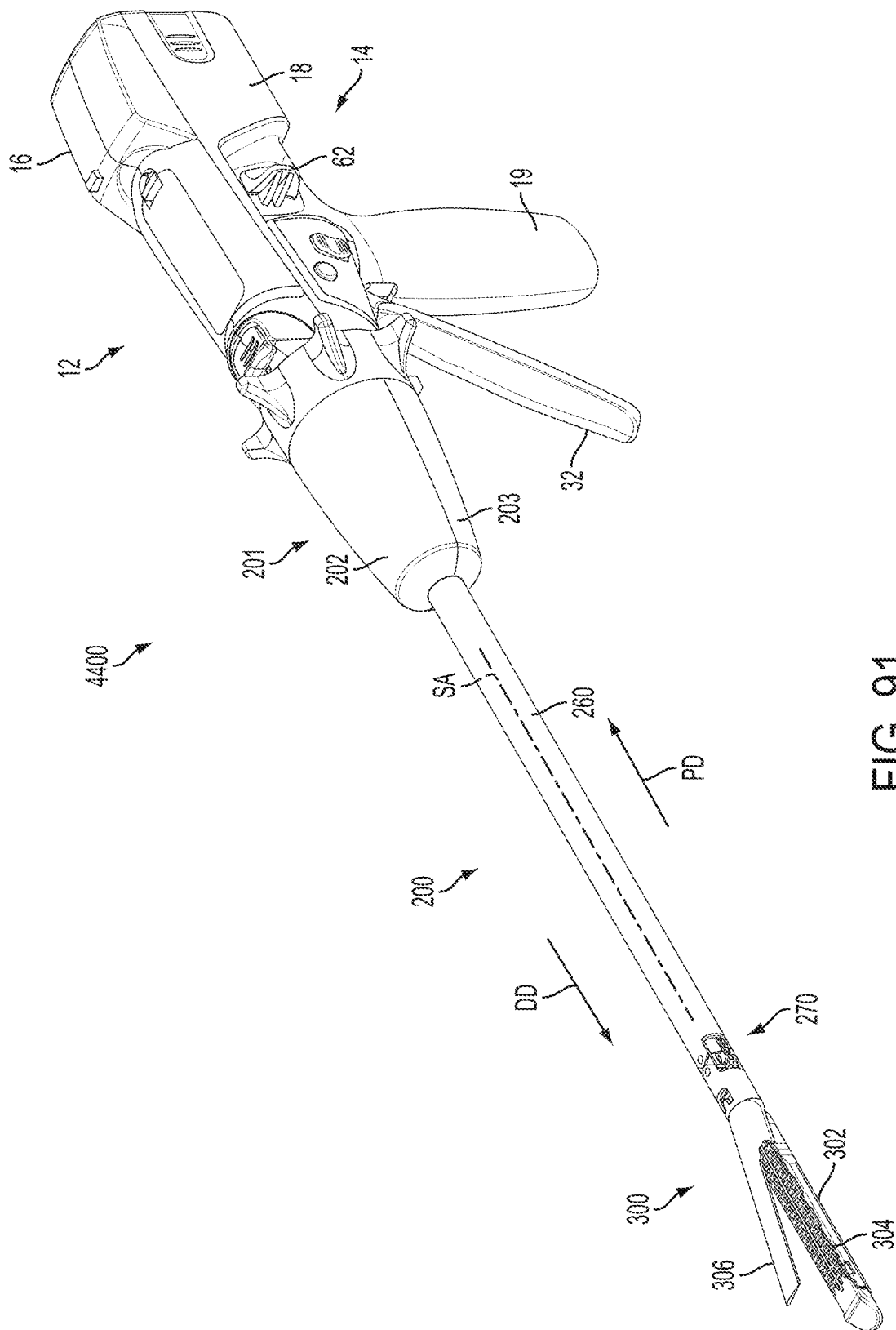


FIG. 91

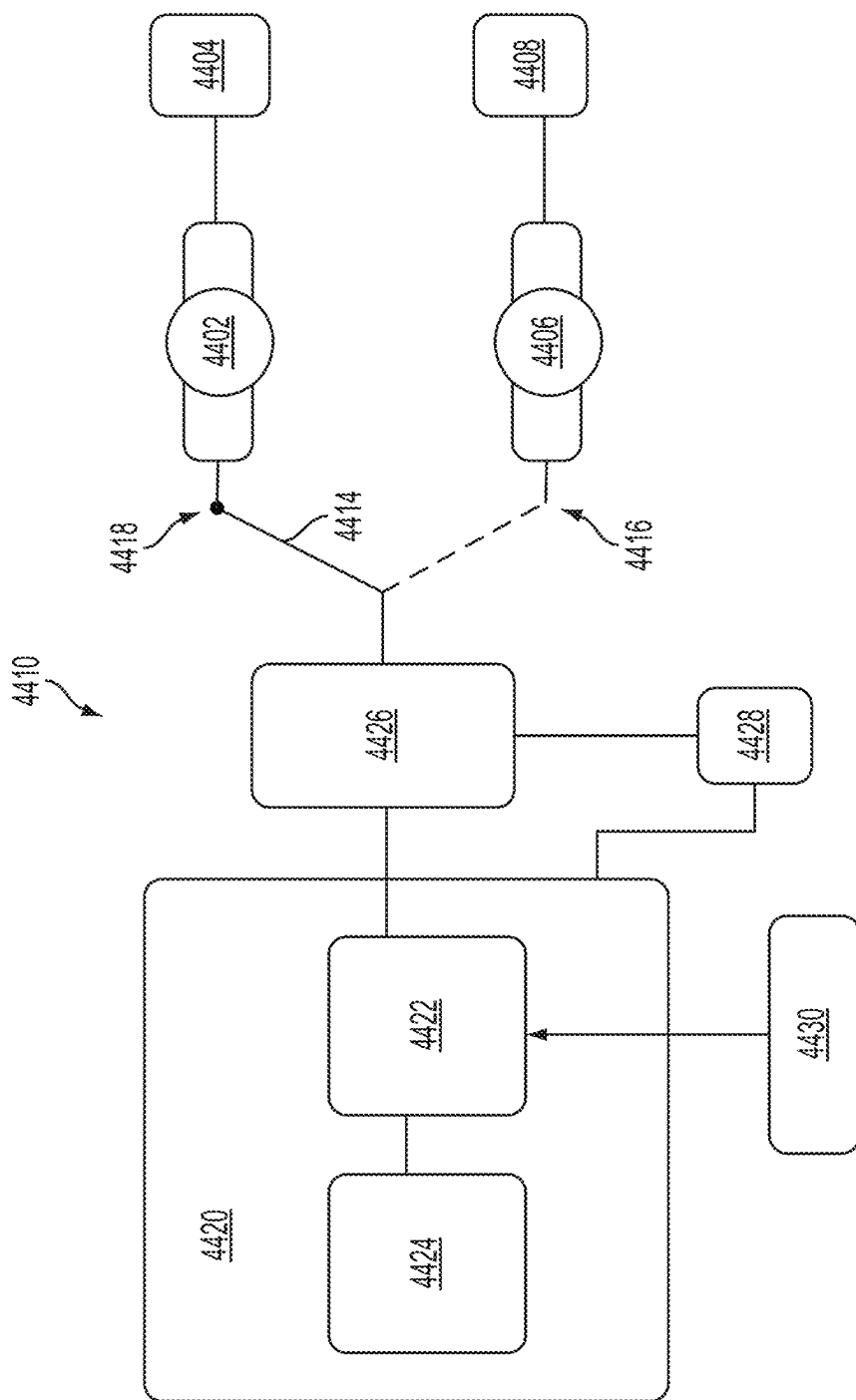


FIG. 92

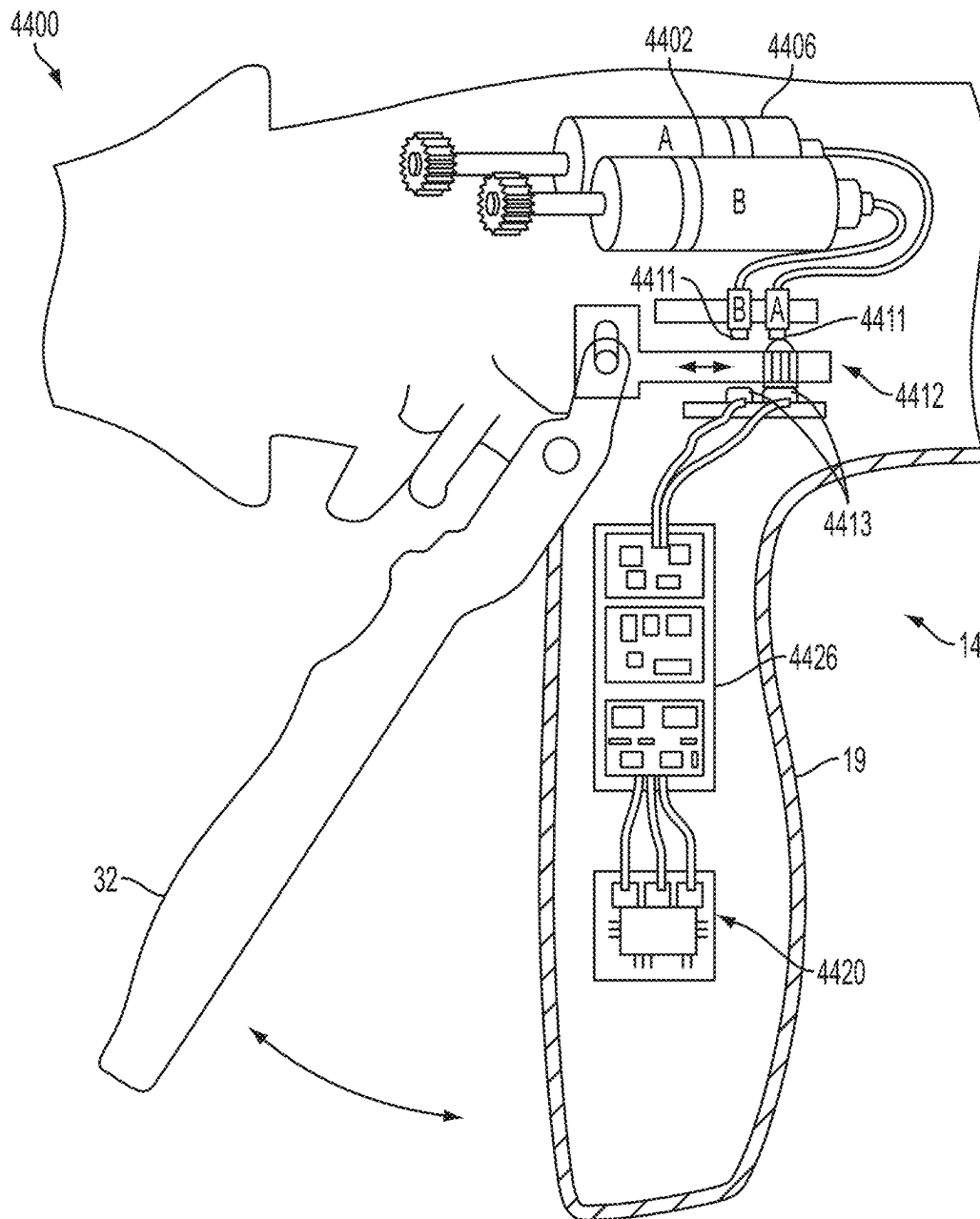


FIG. 93

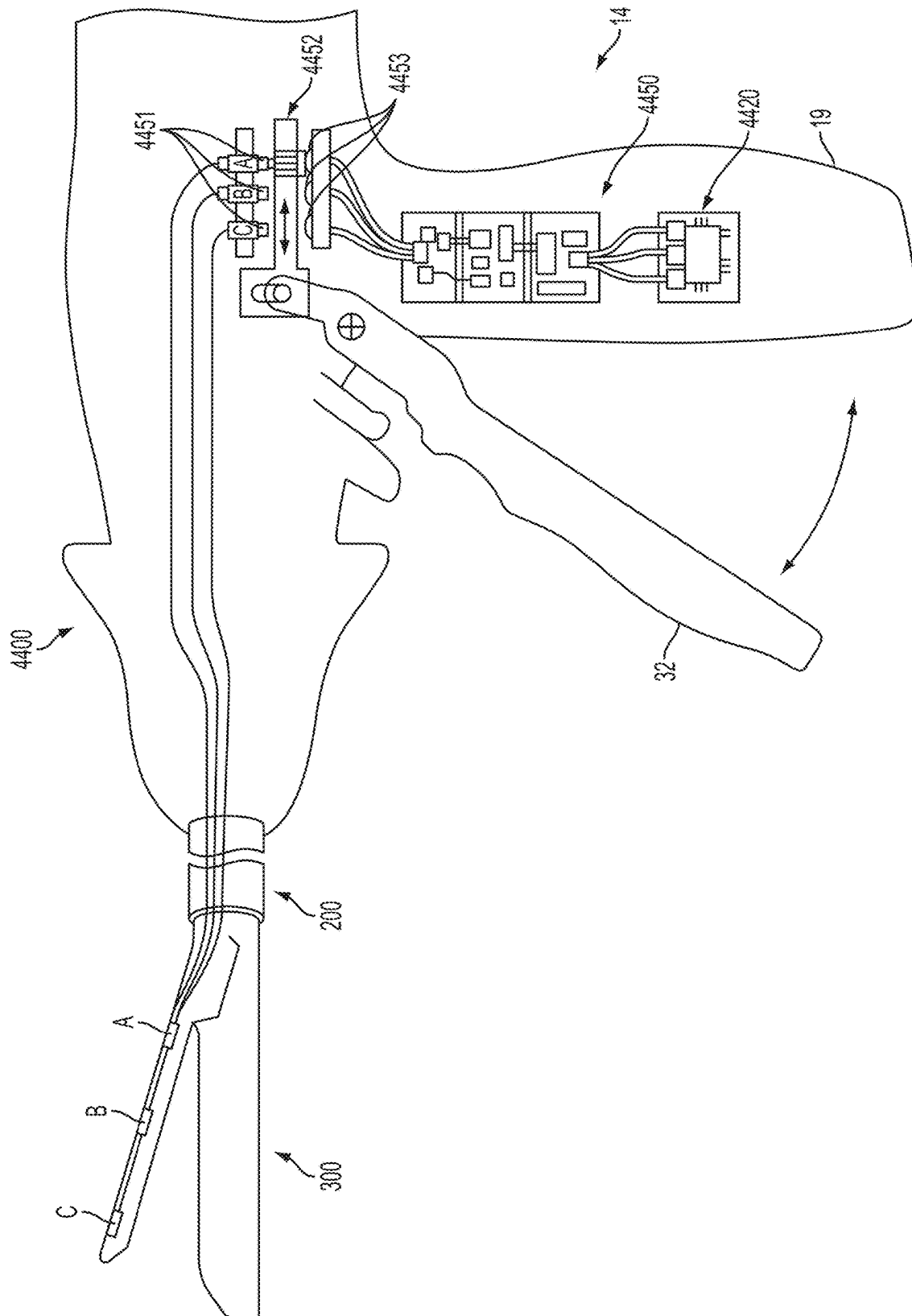


Fig. 4

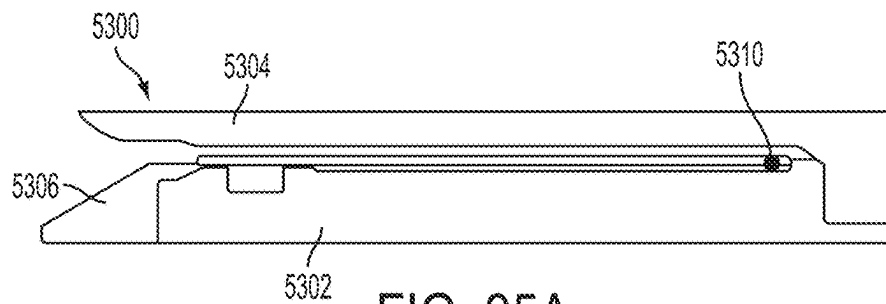


FIG. 95A

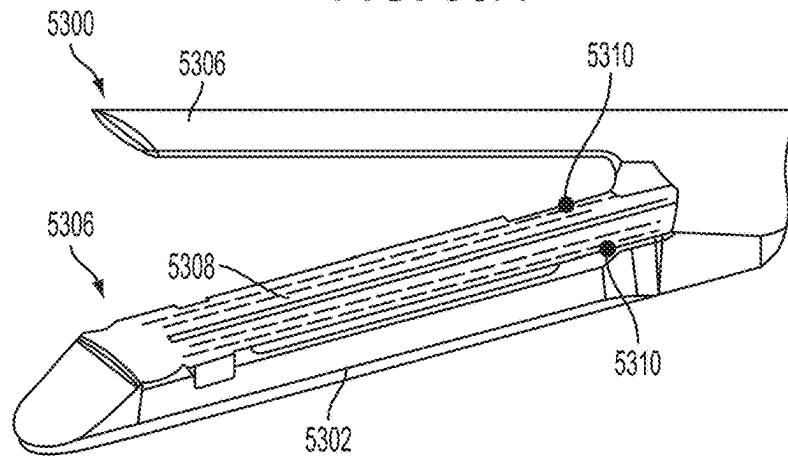


FIG. 95B

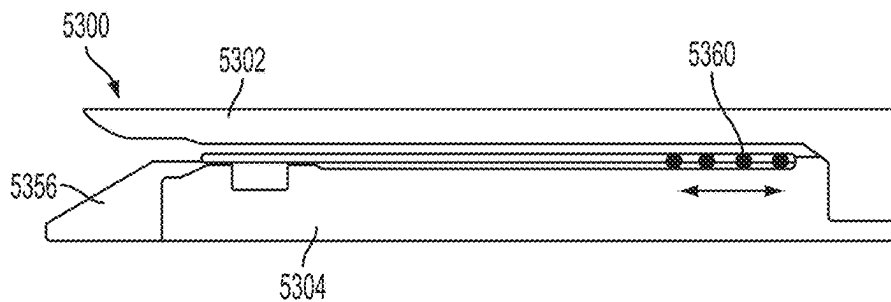


FIG. 96A

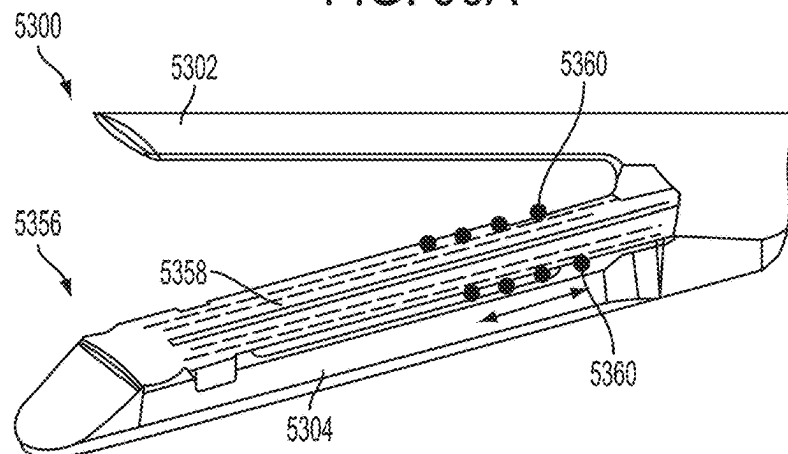


FIG. 96B

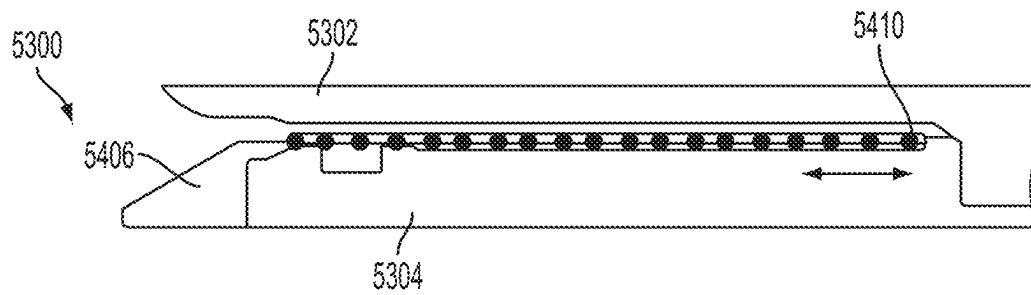


FIG. 97A

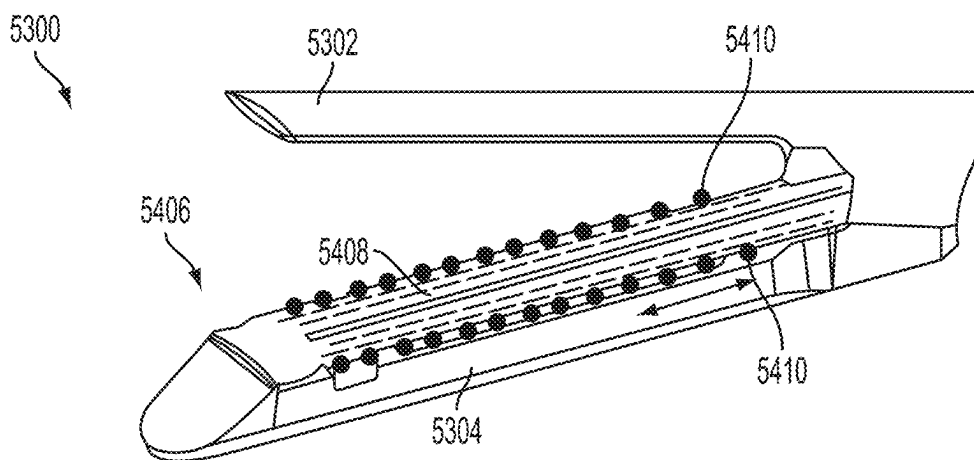


FIG. 97B

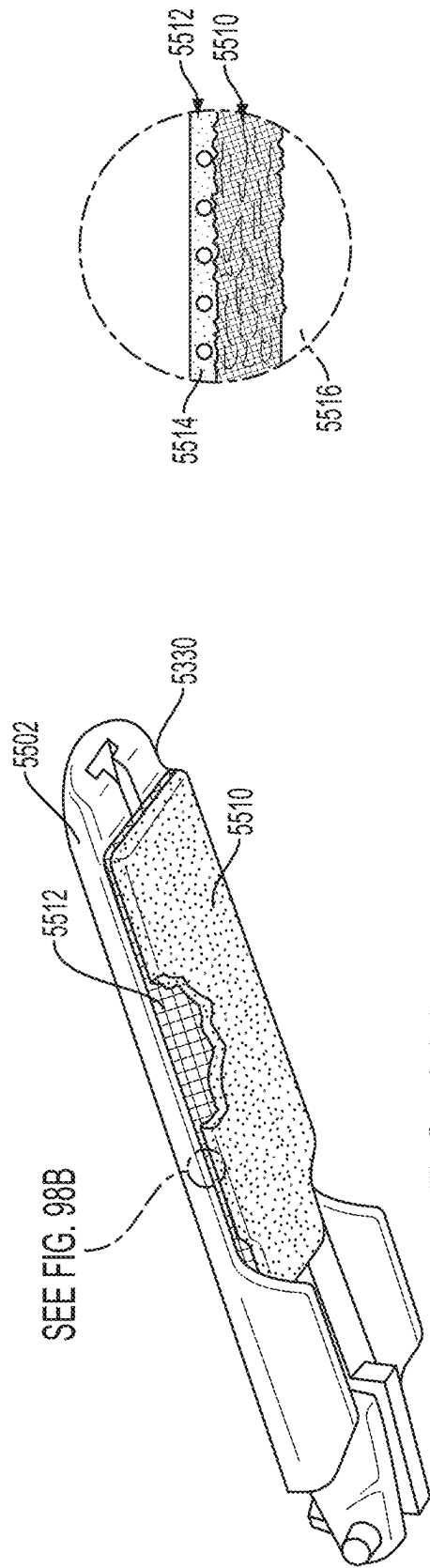


FIG. 98A

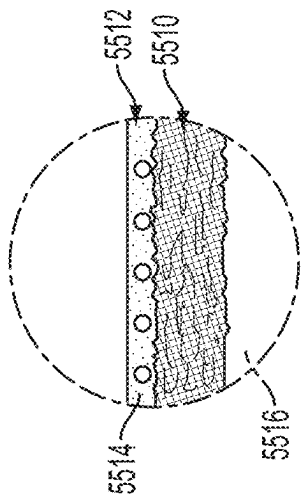


FIG. 98B

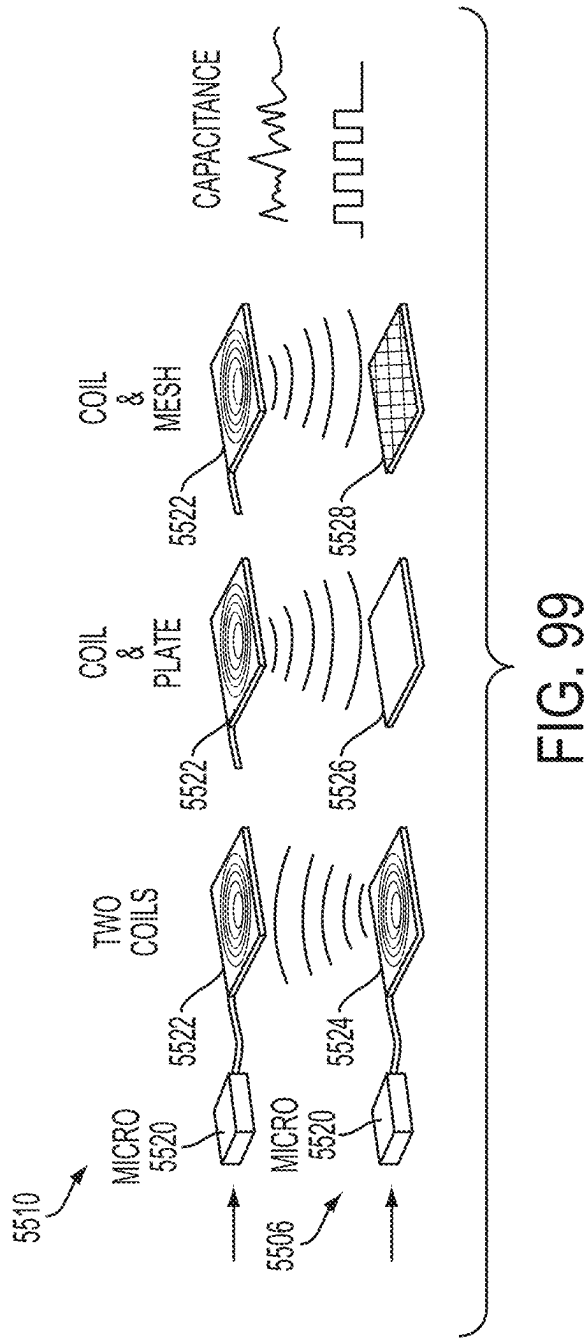


FIG. 99

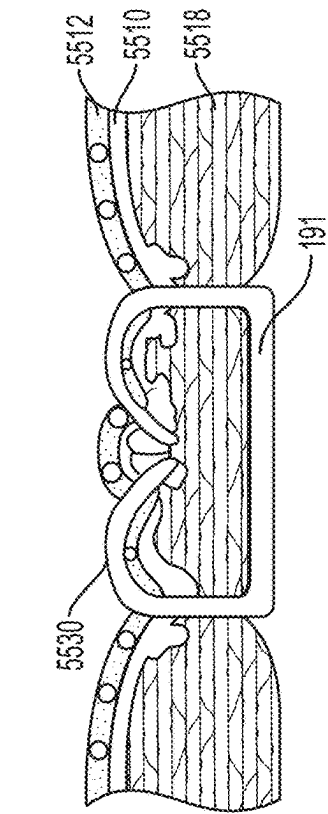


FIG. 100B

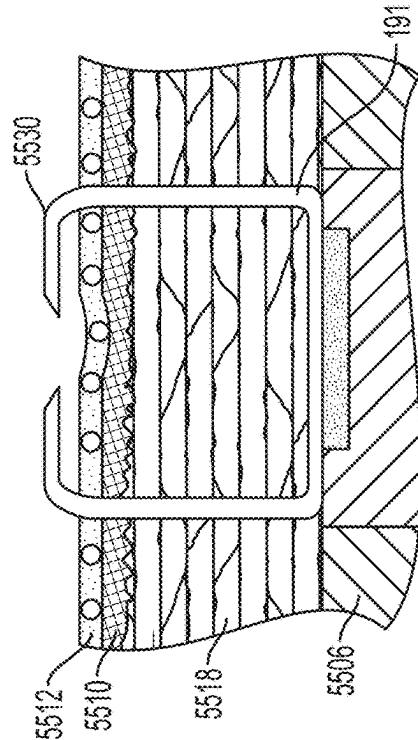


FIG. 100A

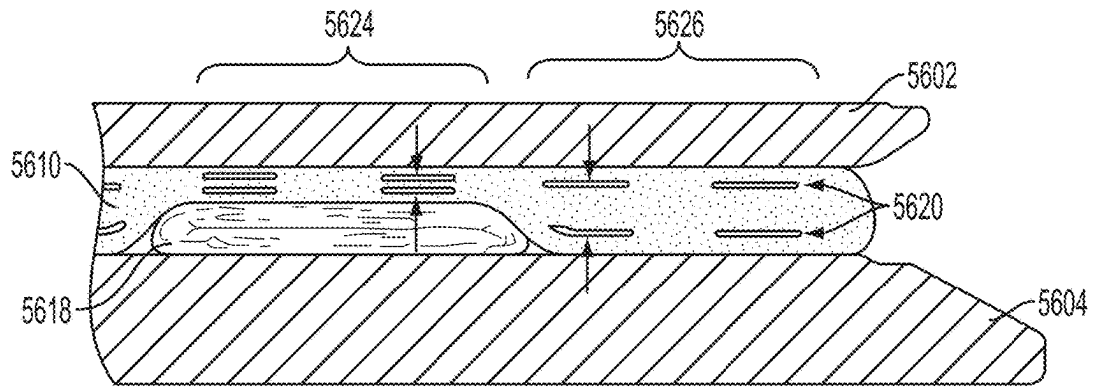


FIG. 101A

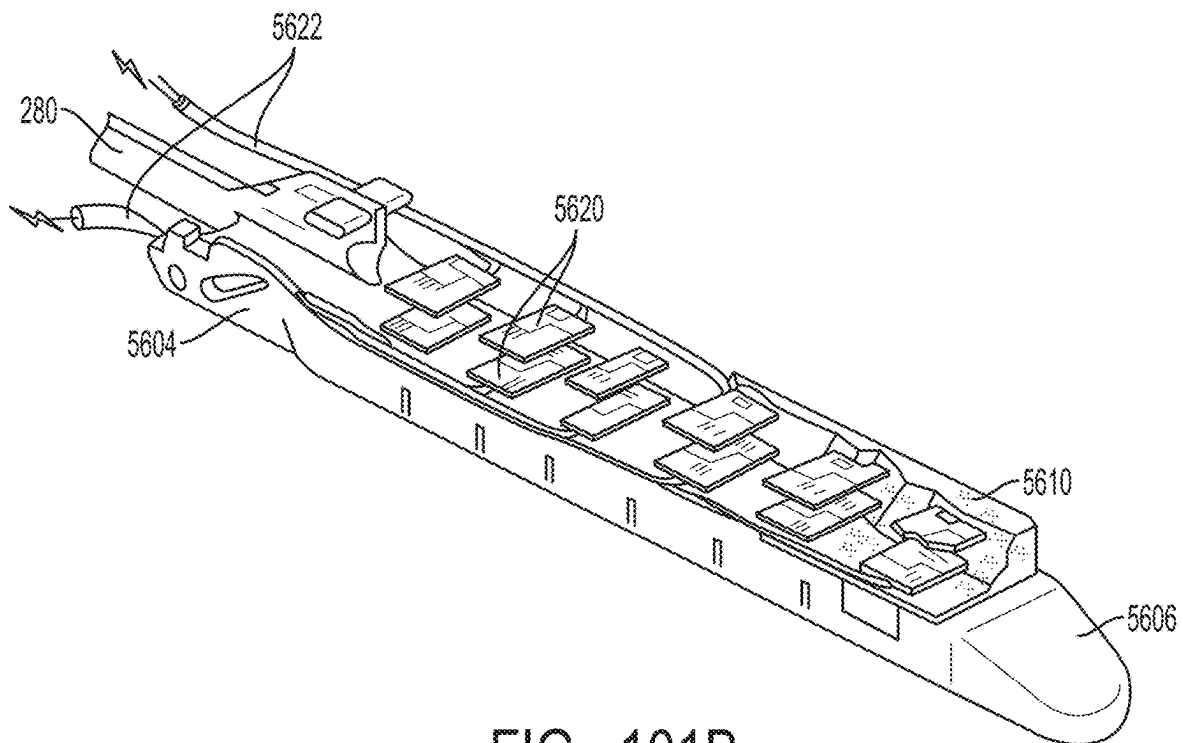
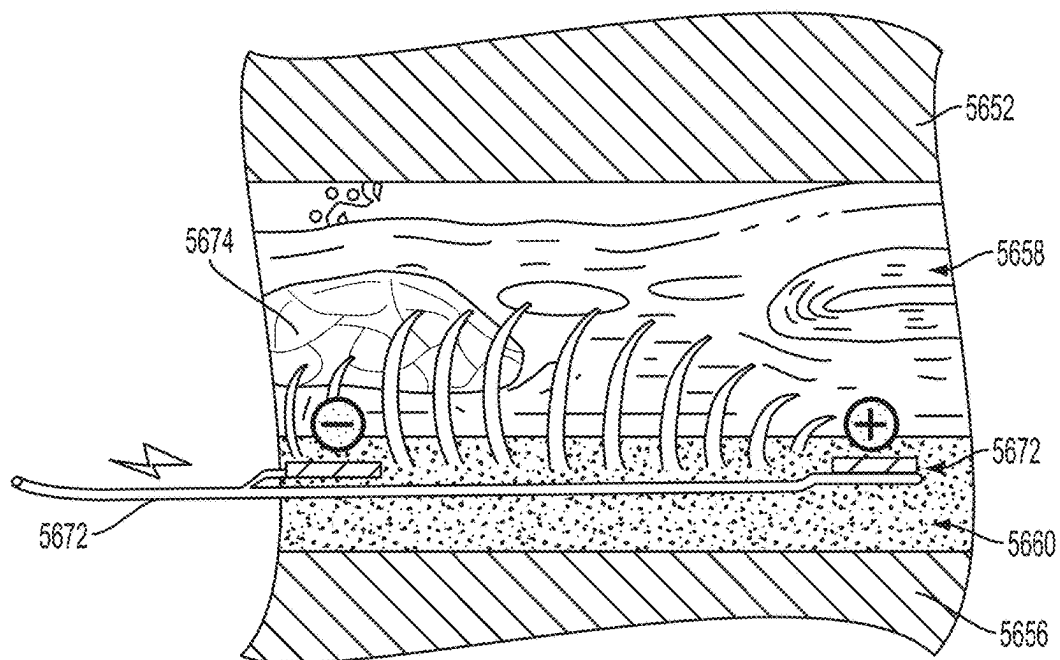
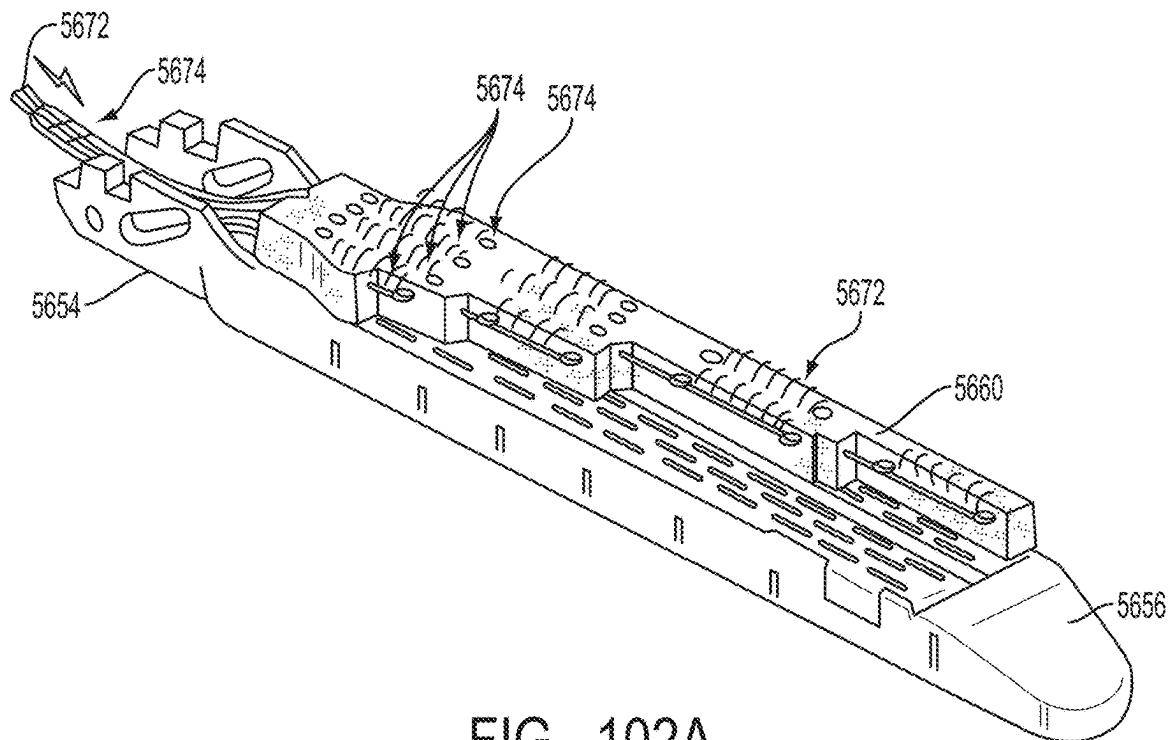


FIG. 101B



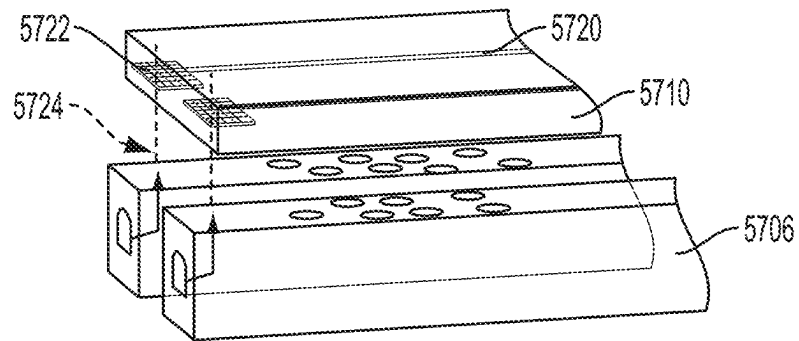


FIG. 103

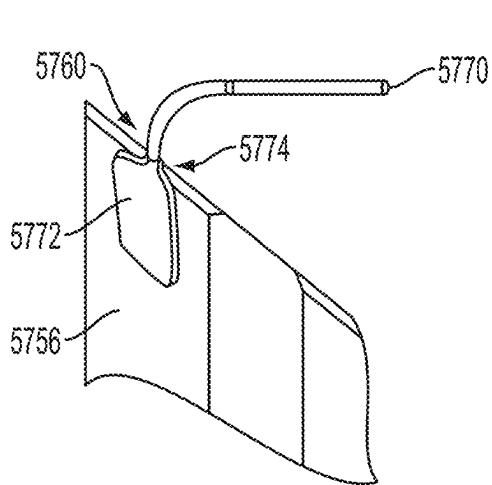


FIG. 104A

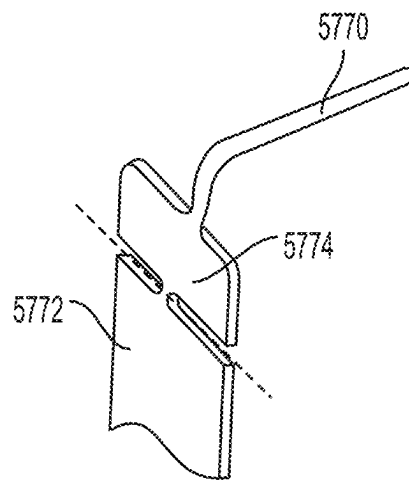


FIG. 104B

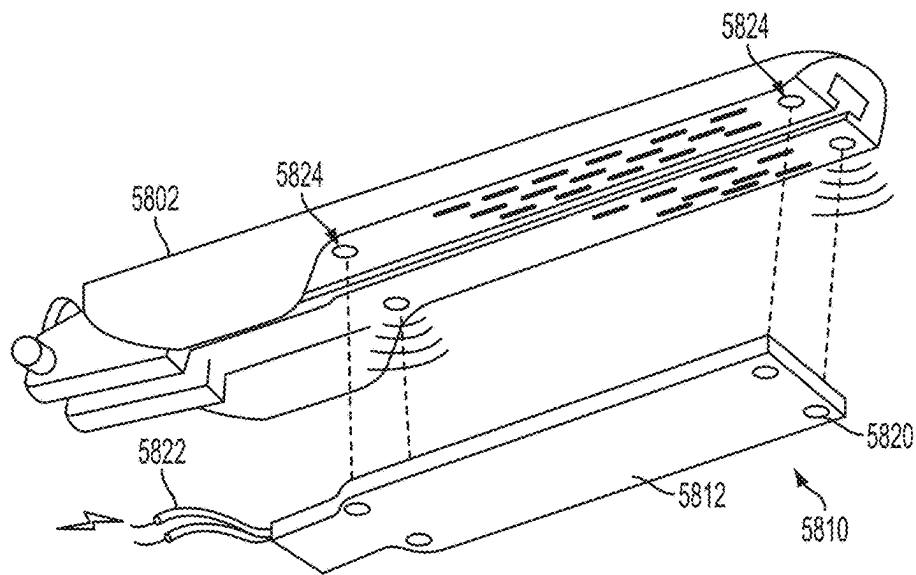


FIG. 105A

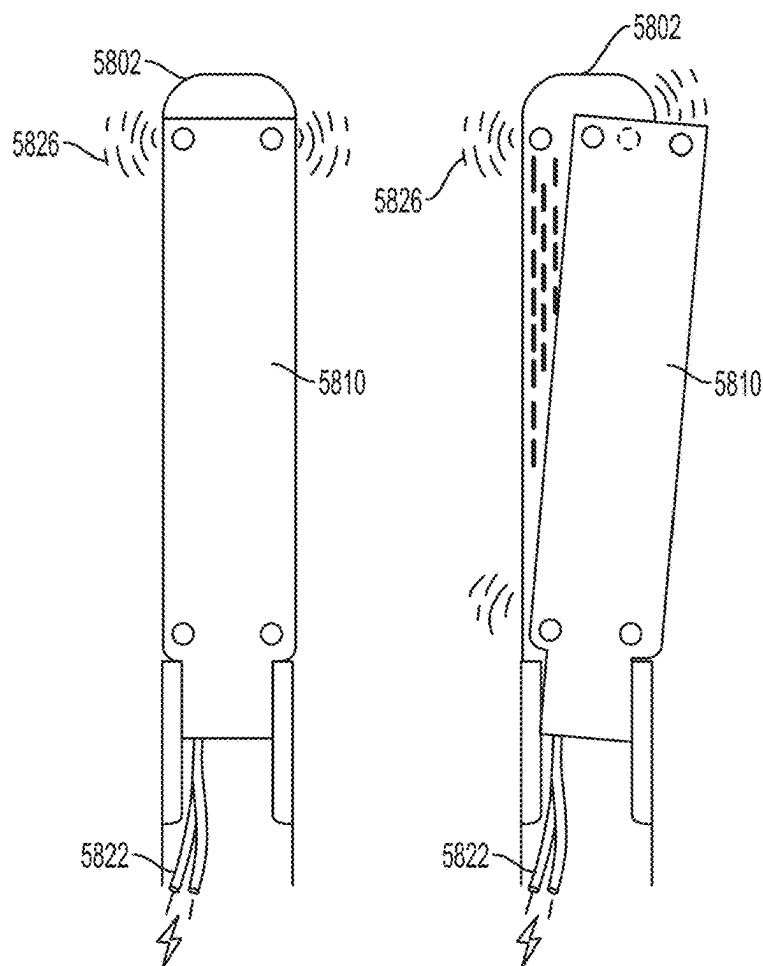


FIG. 105B

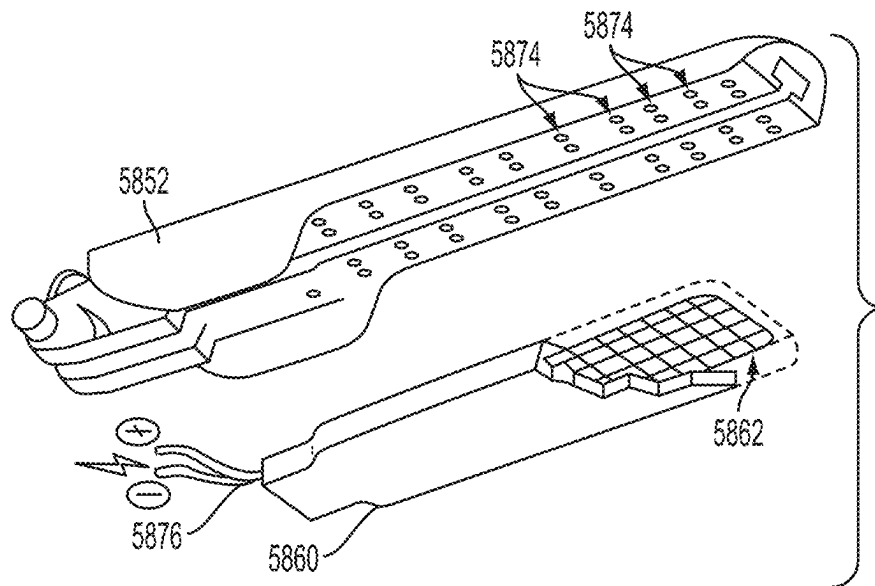


FIG . 106A

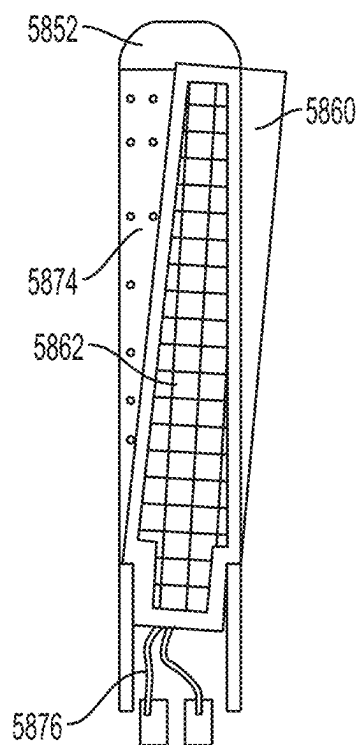


FIG . 106B

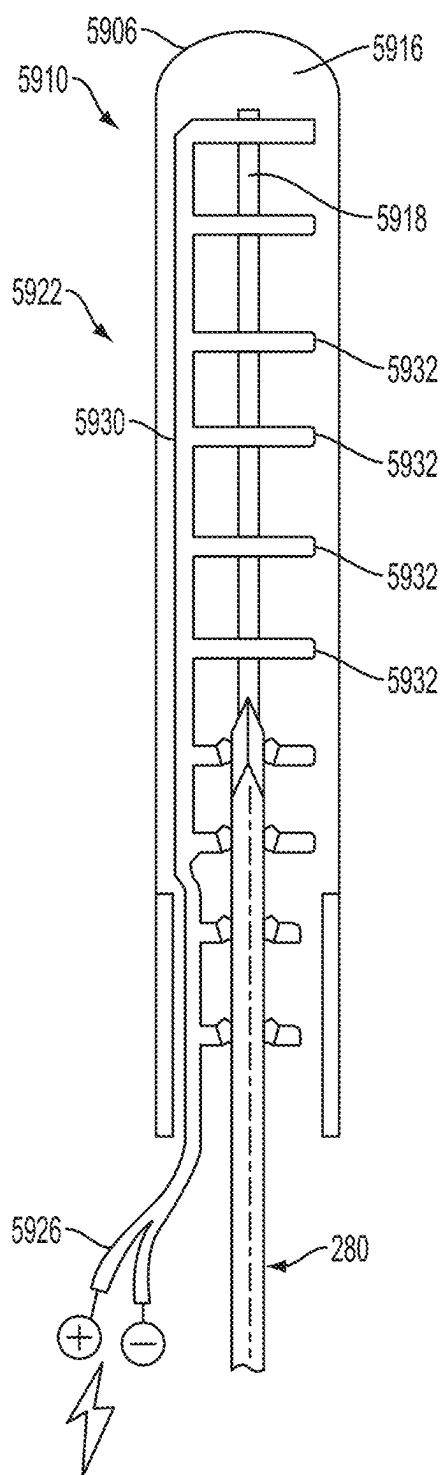


FIG . 107A

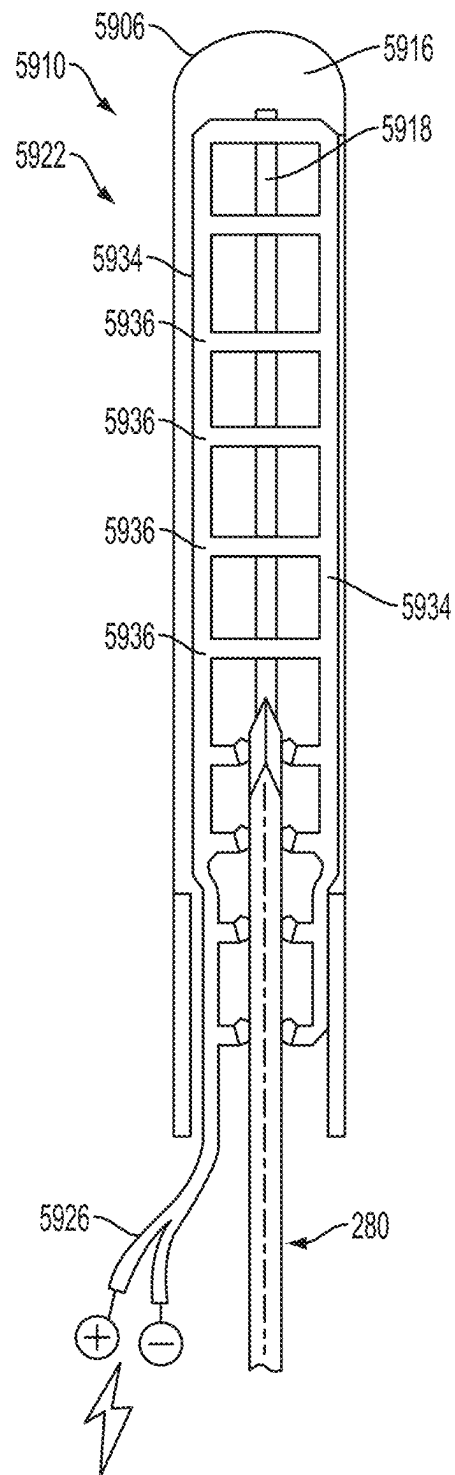


FIG . 107B

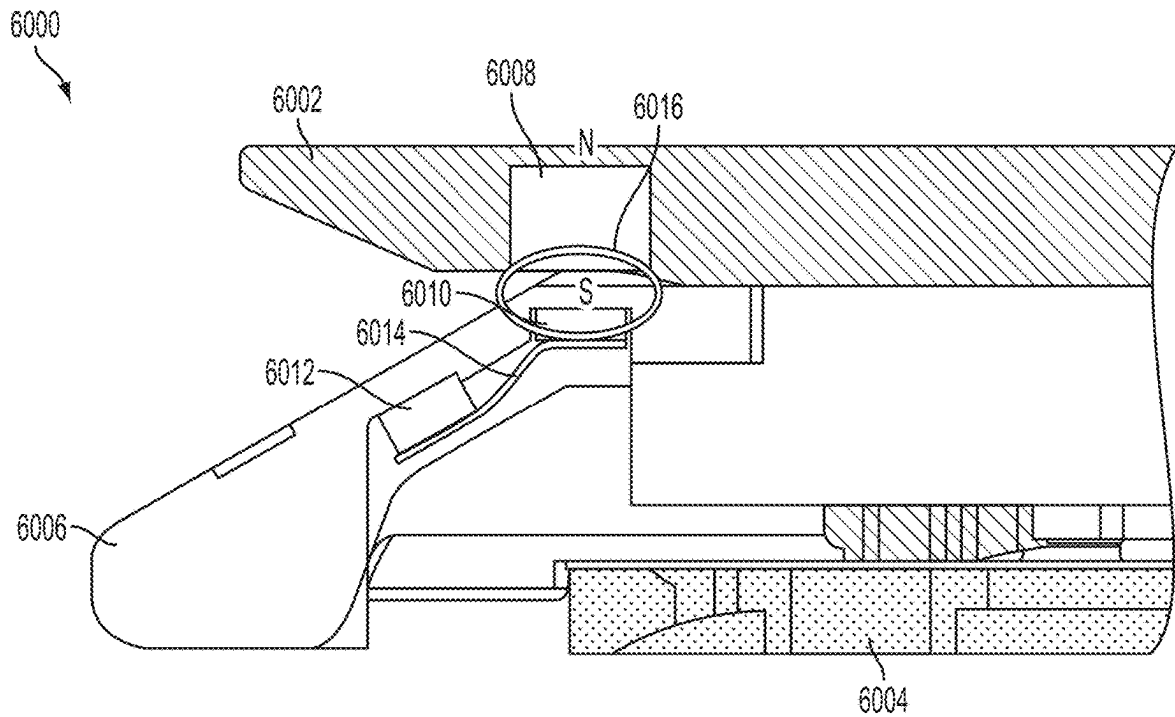


FIG. 108

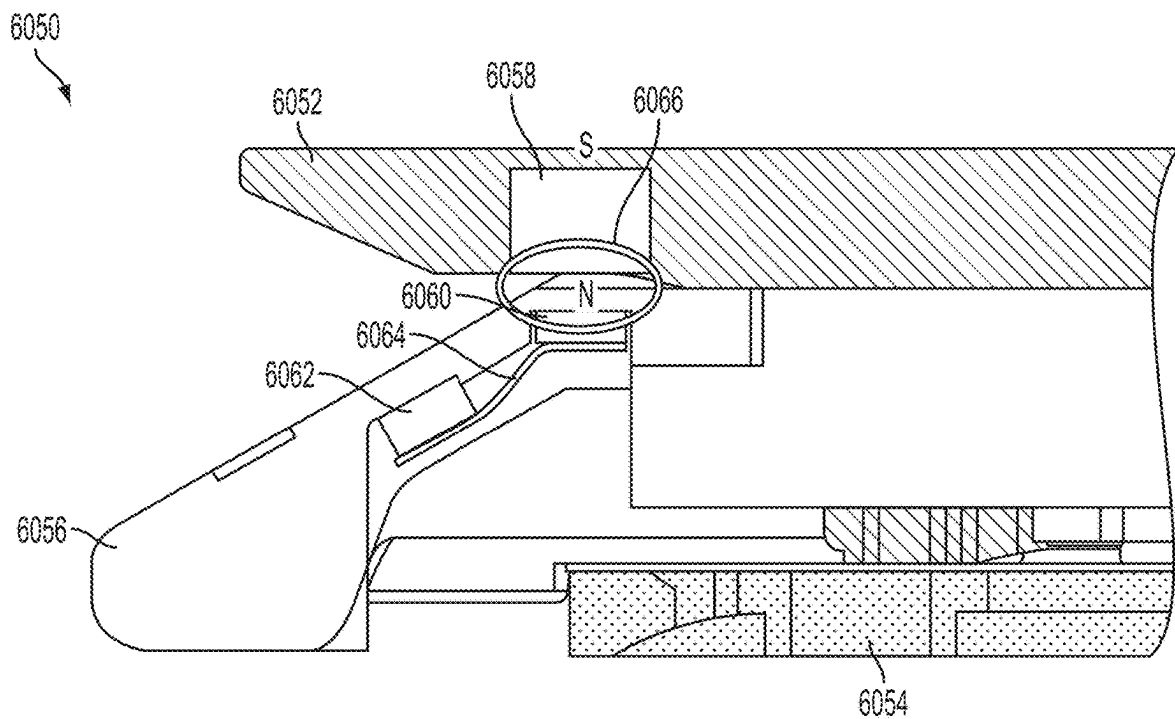


FIG. 109

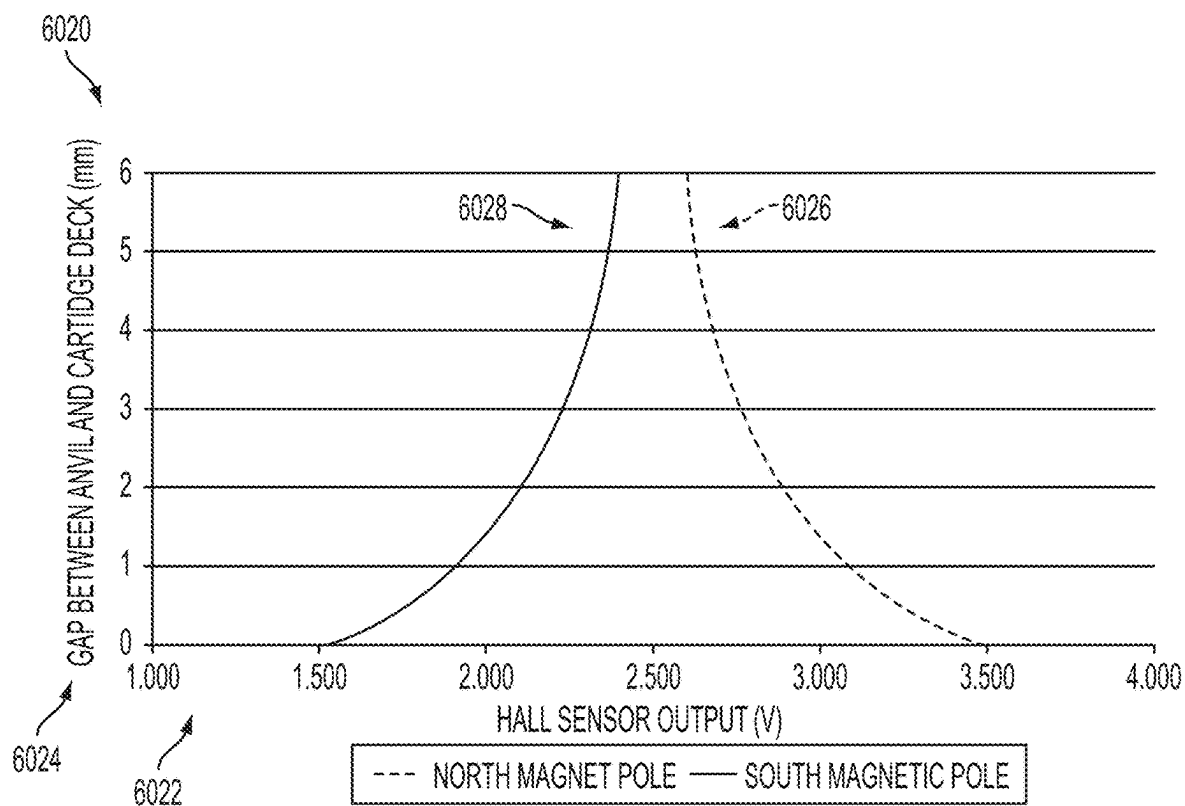


FIG. 110

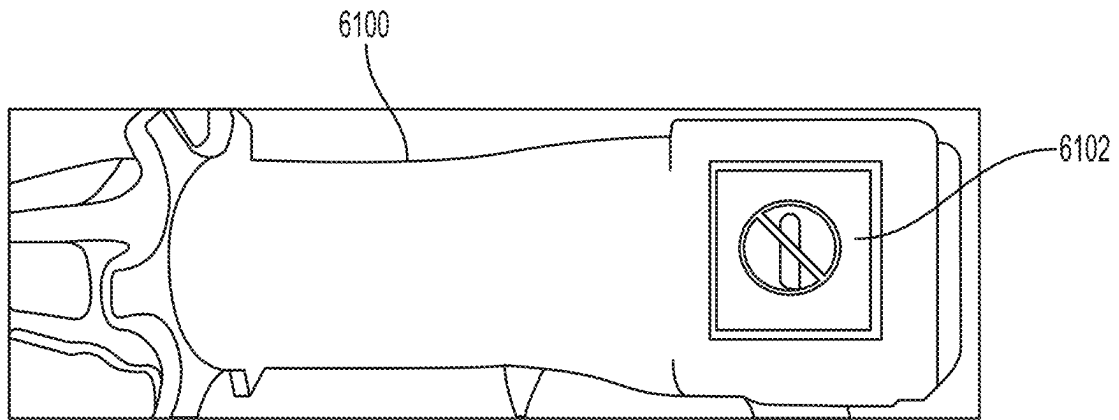


FIG. 111

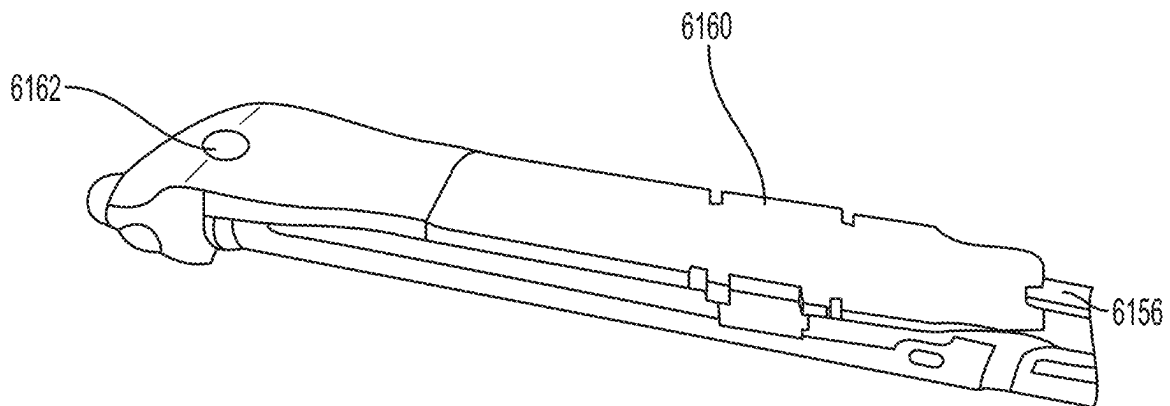


FIG. 112

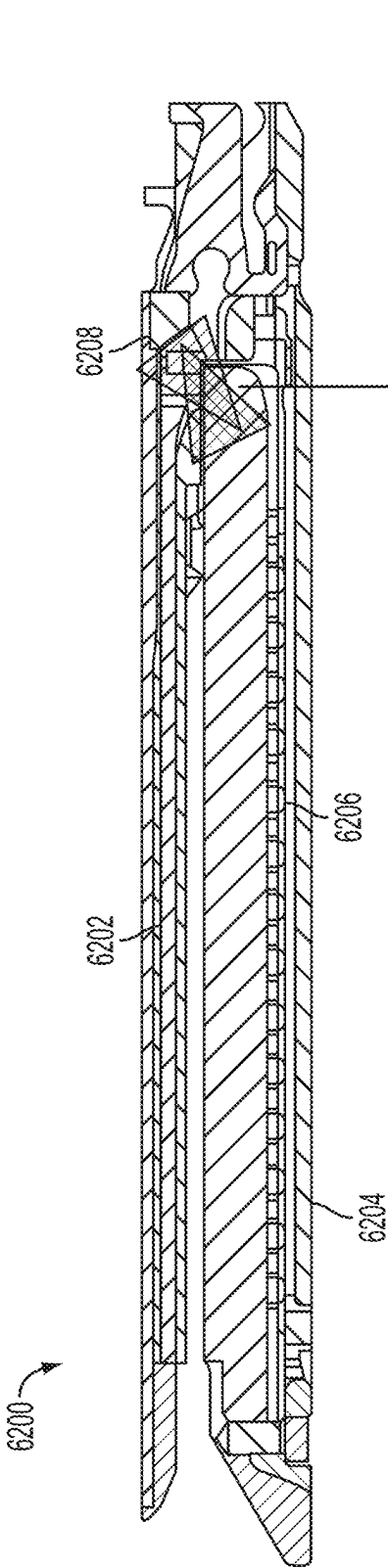


FIG. 113A

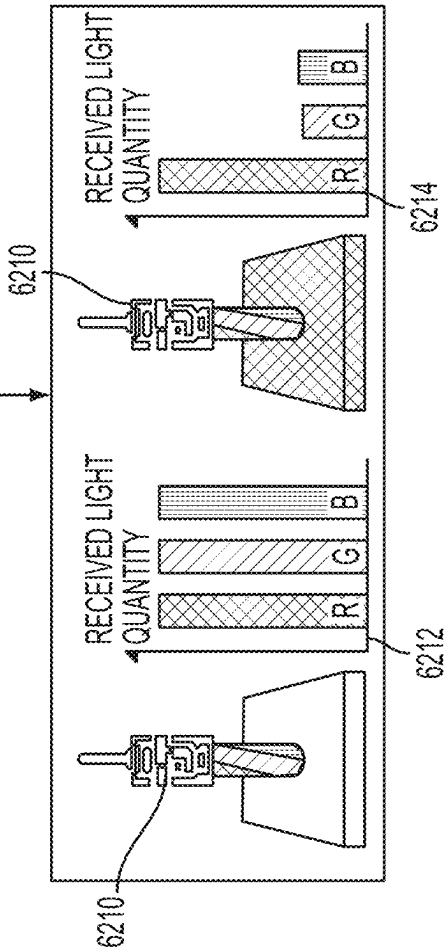


FIG. 113B

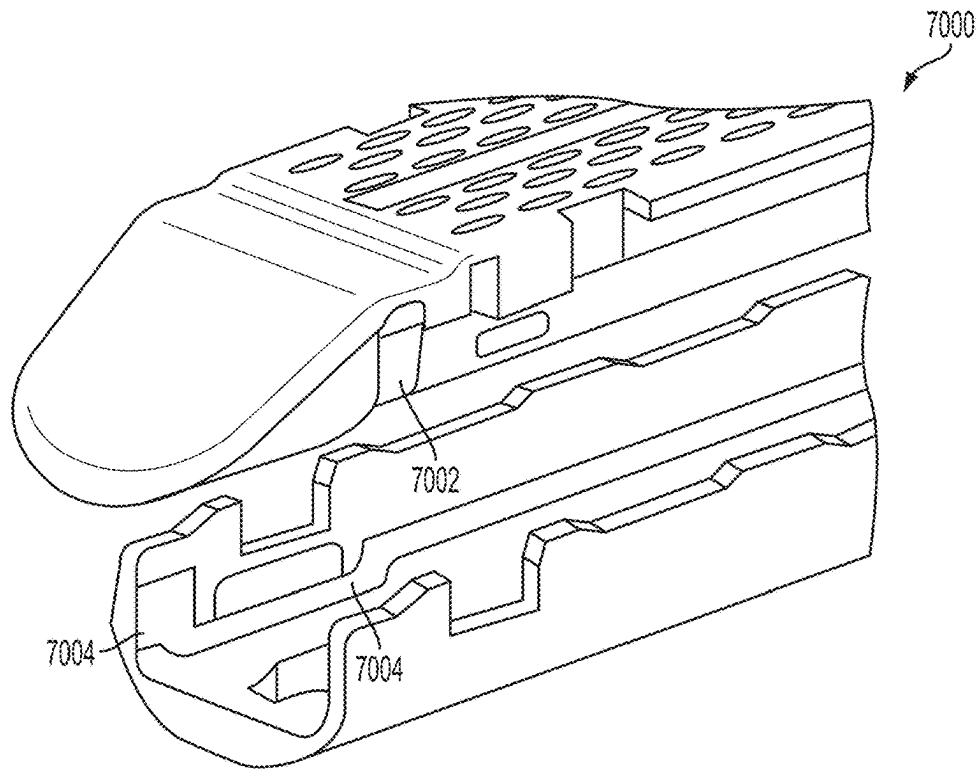


FIG. 114

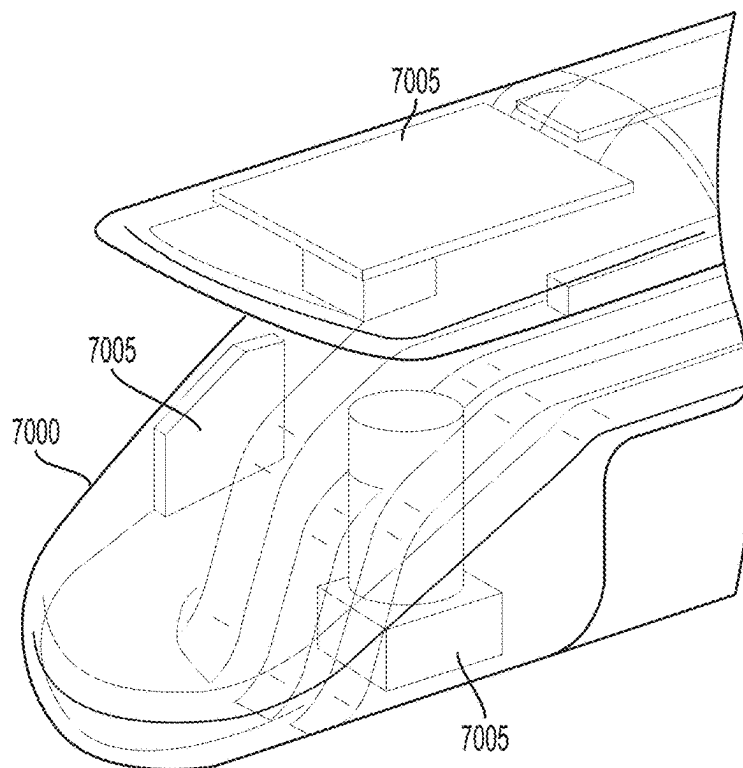


FIG. 115

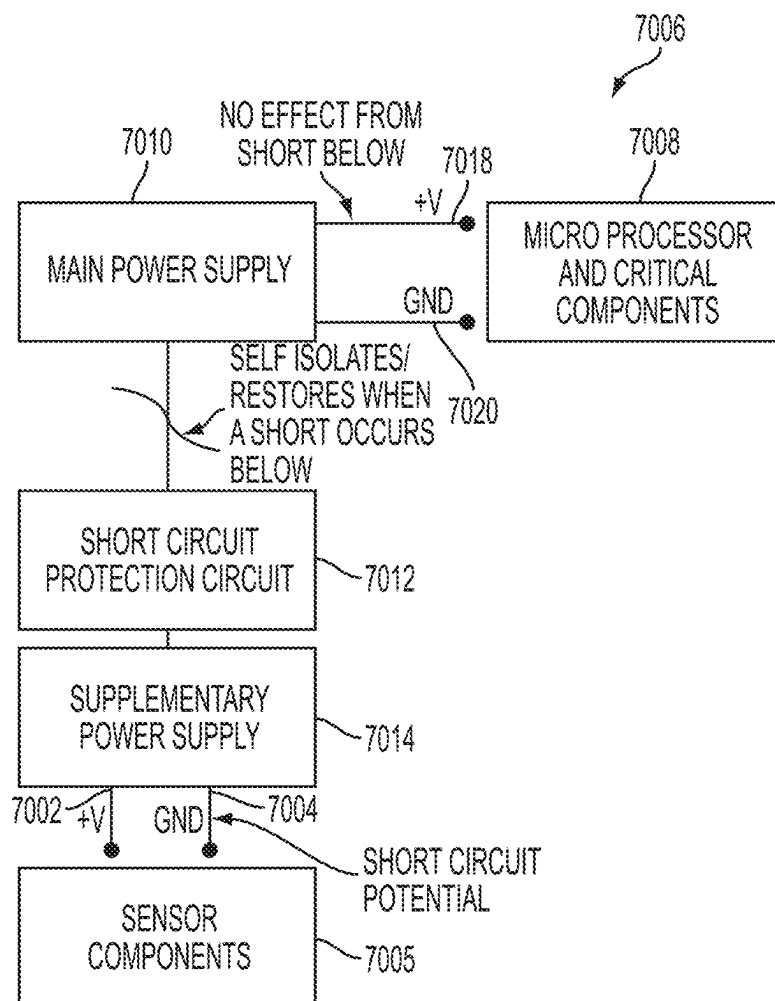


FIG. 116

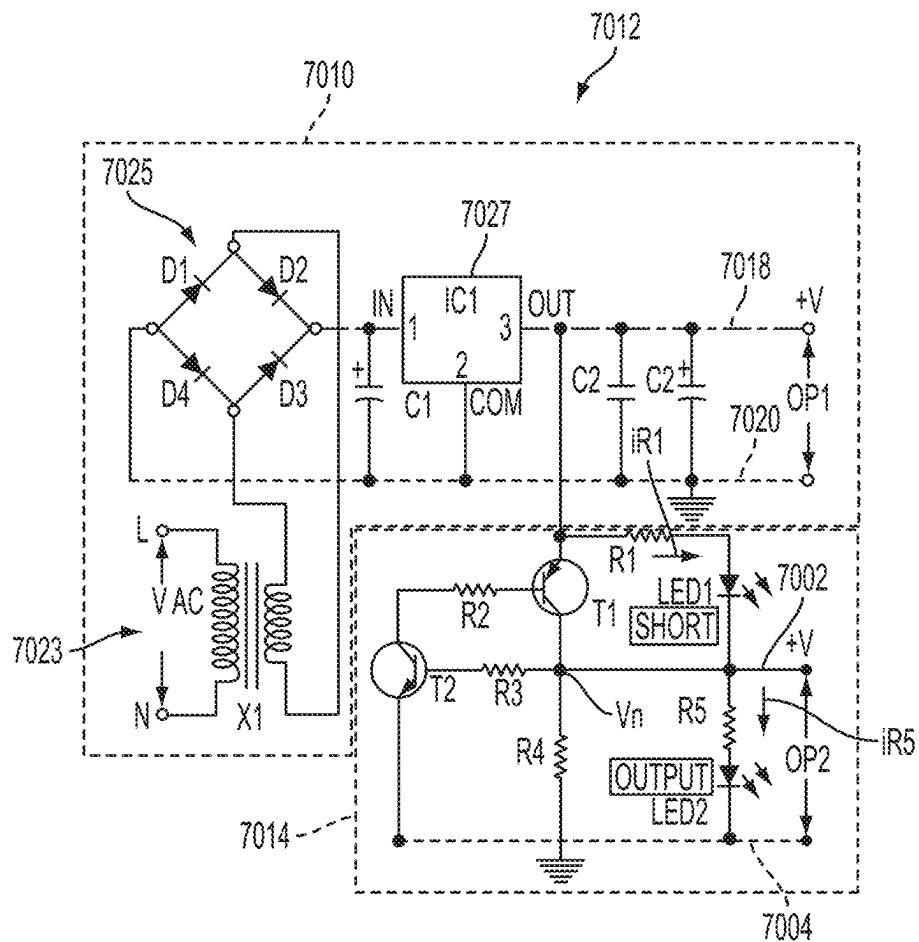


FIG. 117

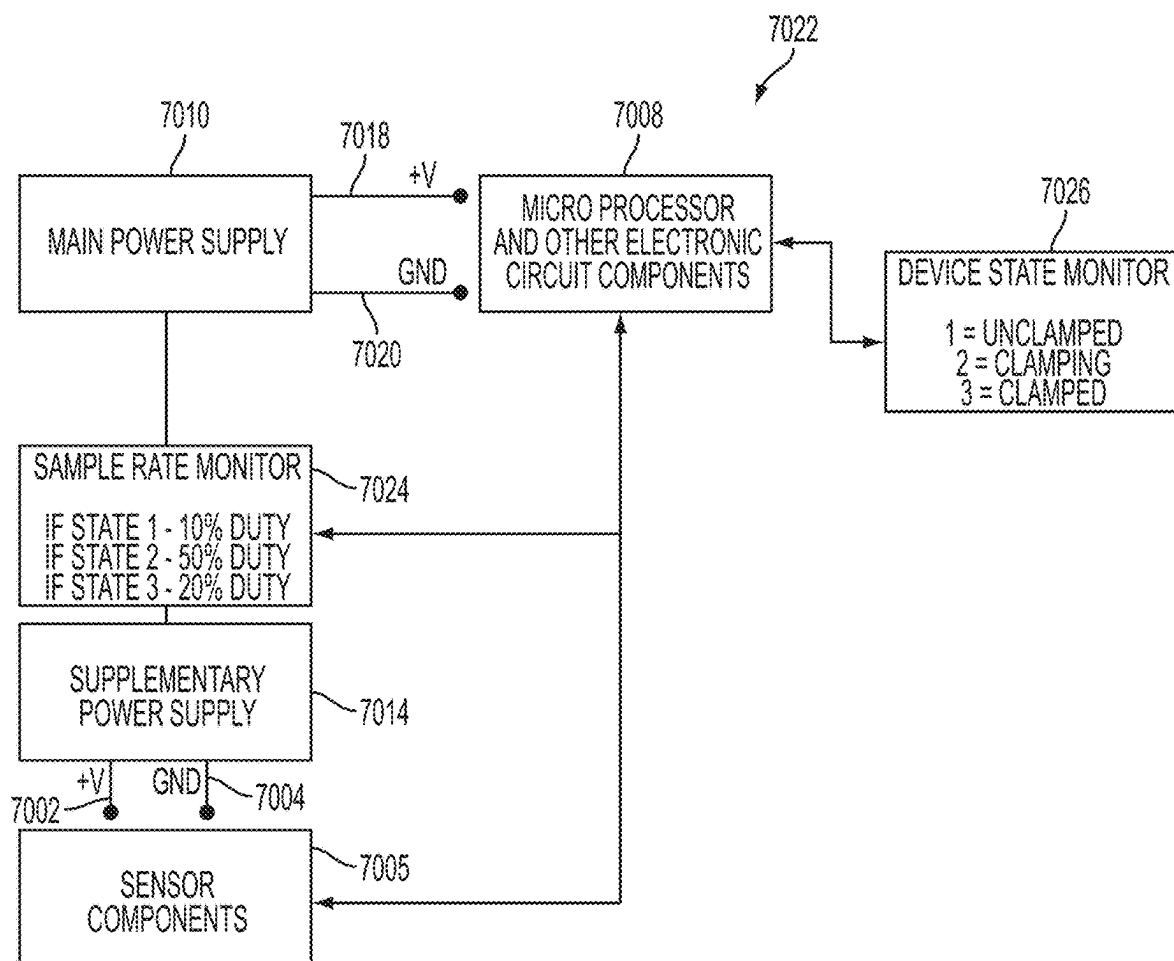


FIG. 118

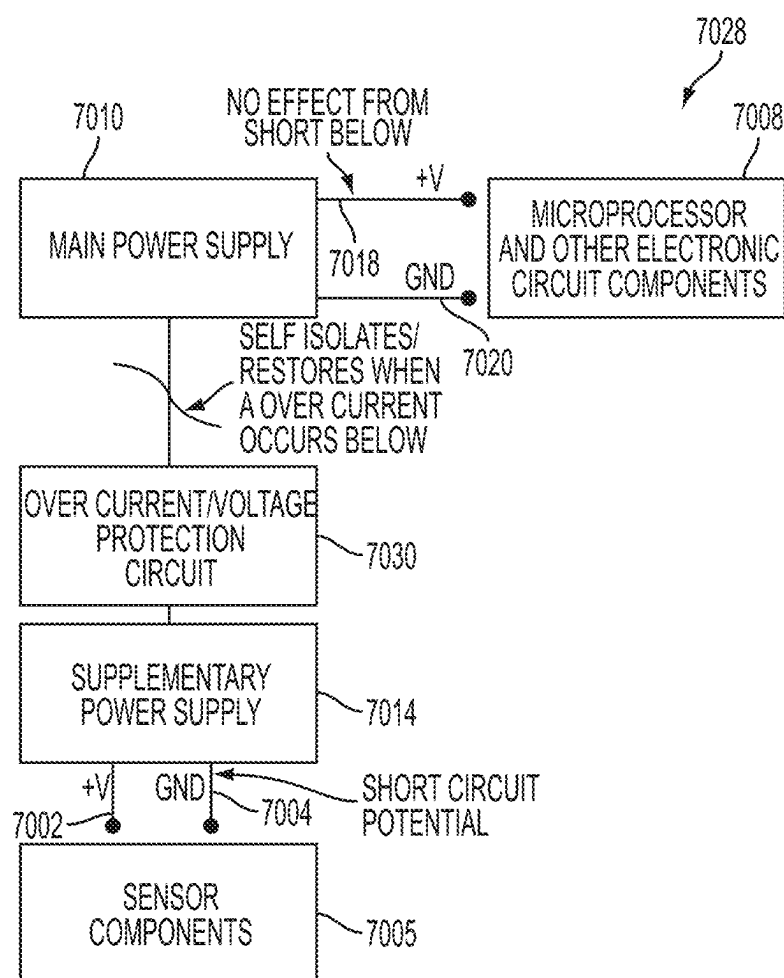


FIG. 119

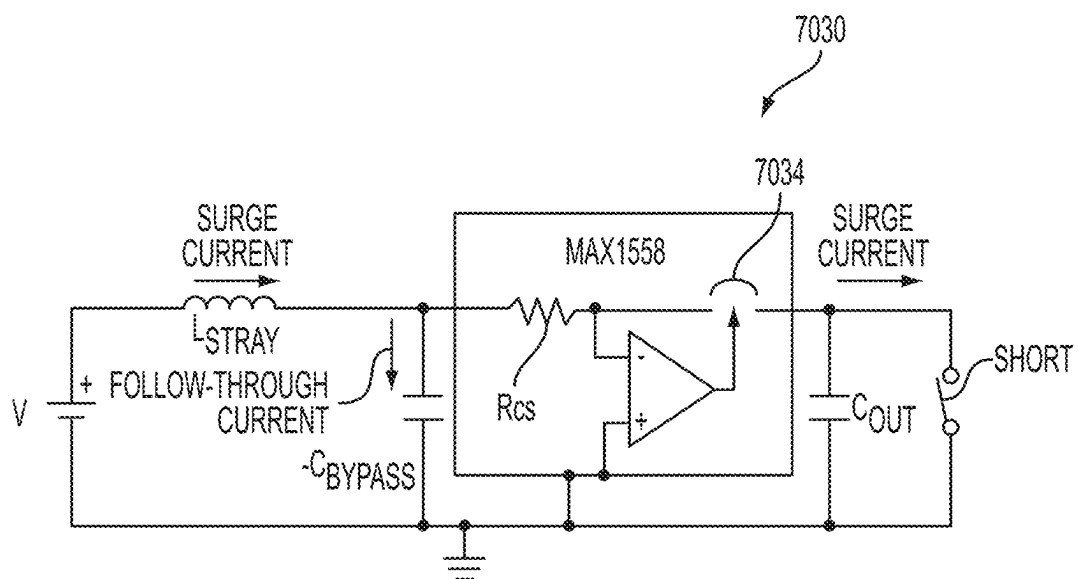


FIG. 120

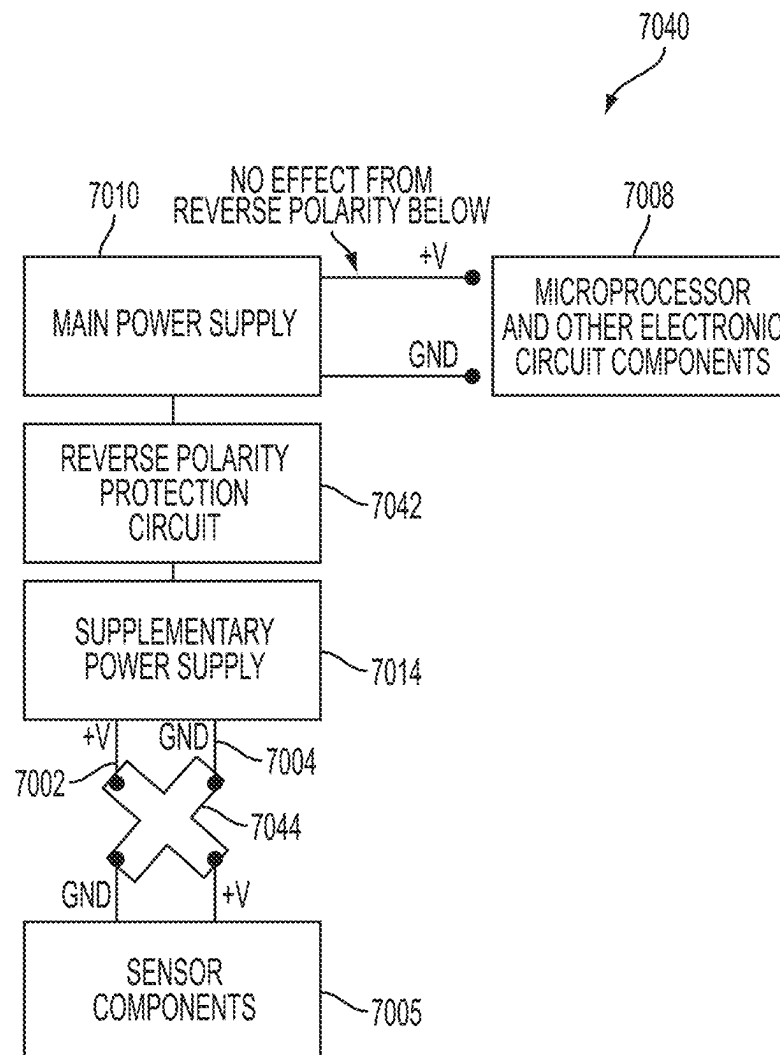


FIG. 121

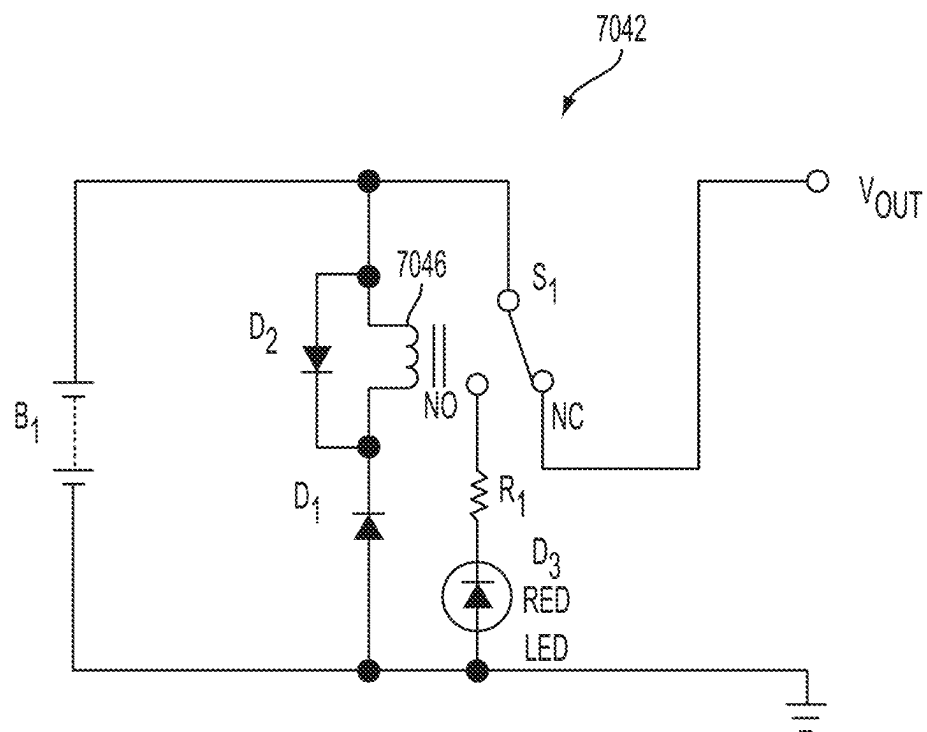


FIG. 122

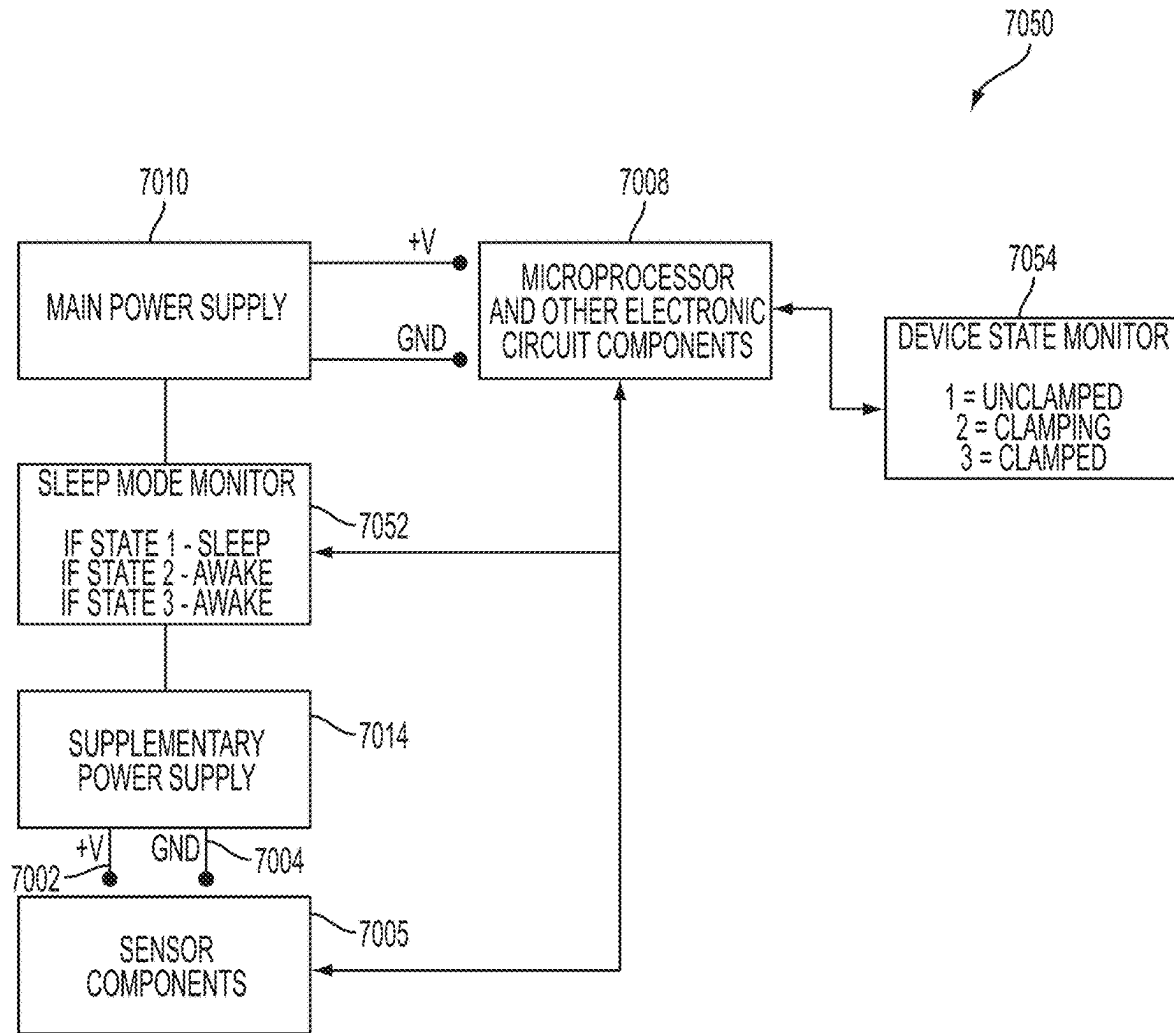


FIG. 123

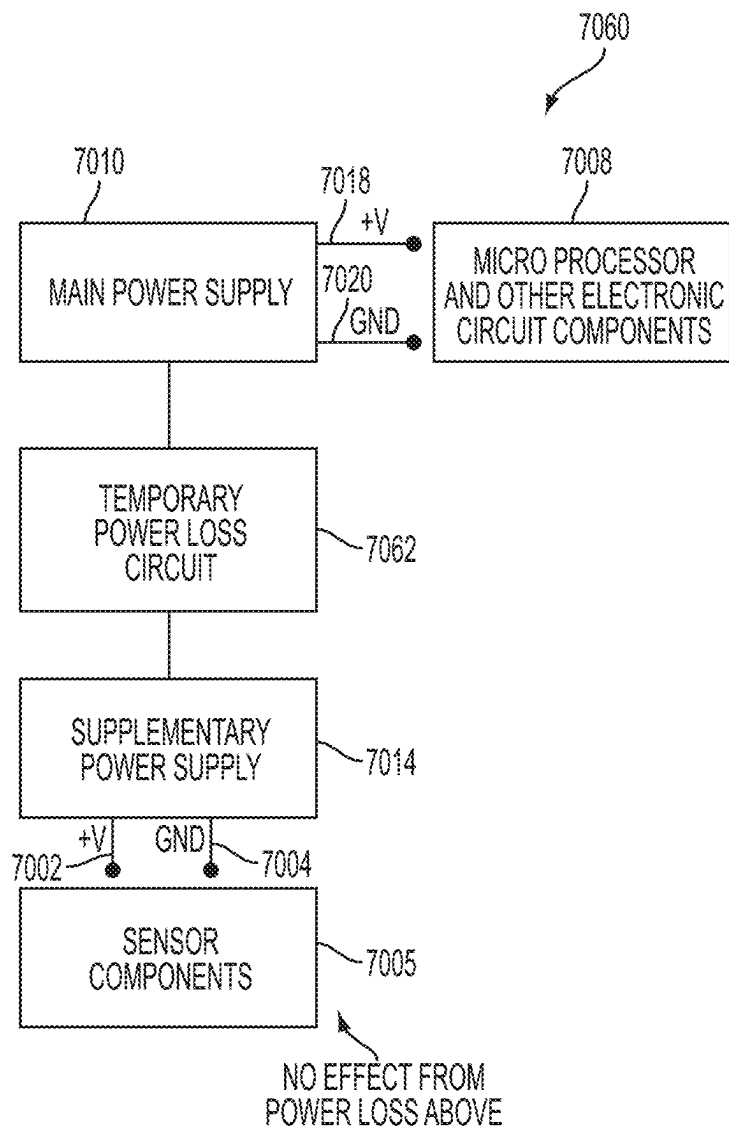


FIG. 124

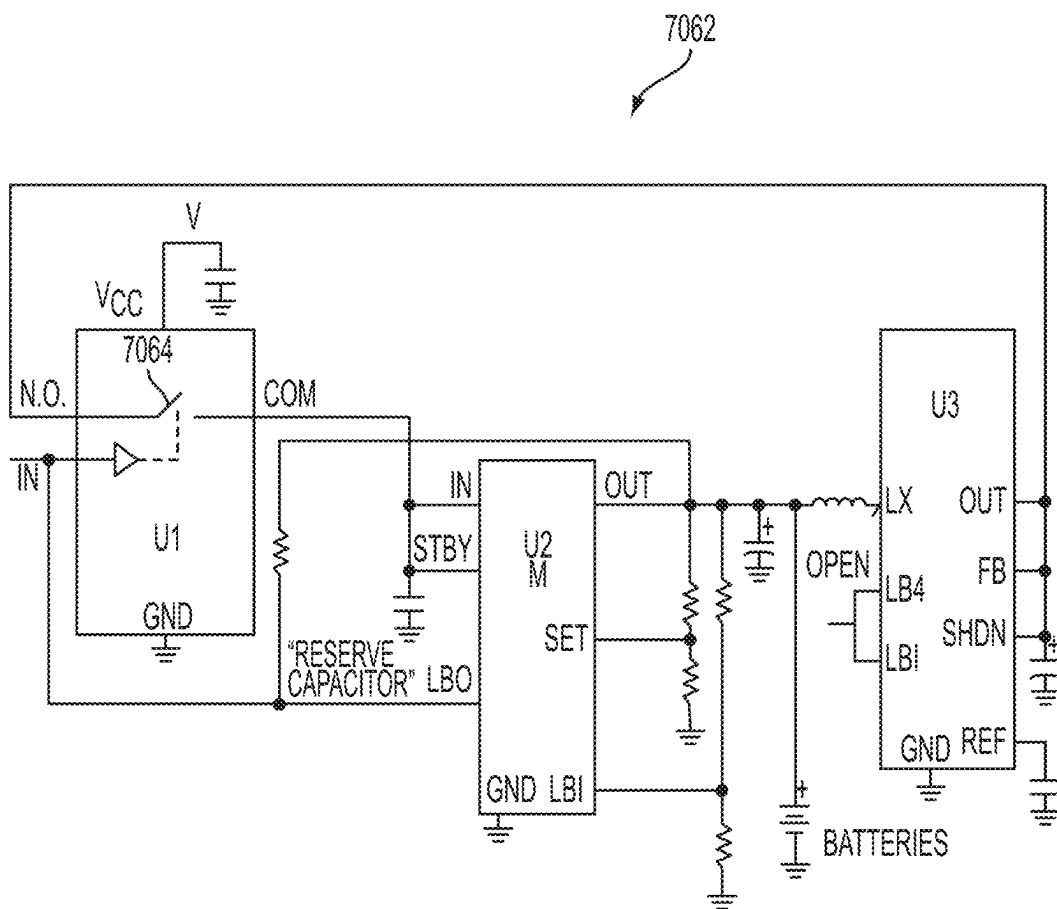


FIG. 125

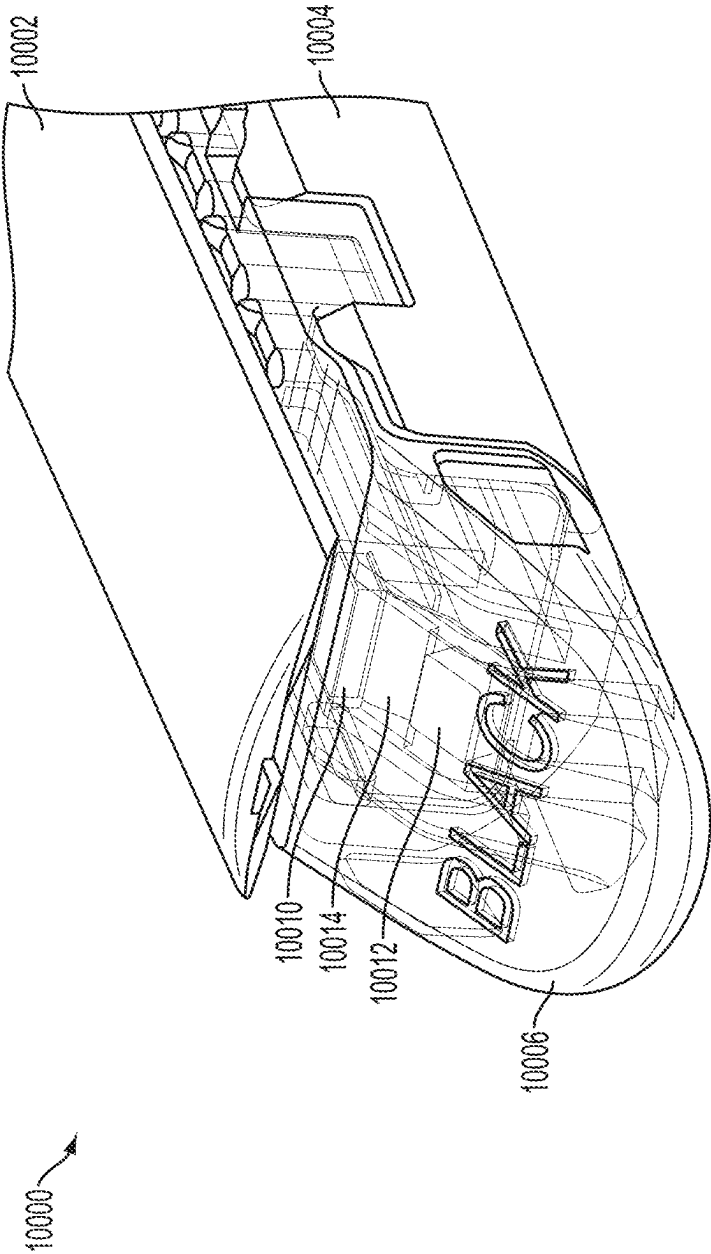


FIG. 126A

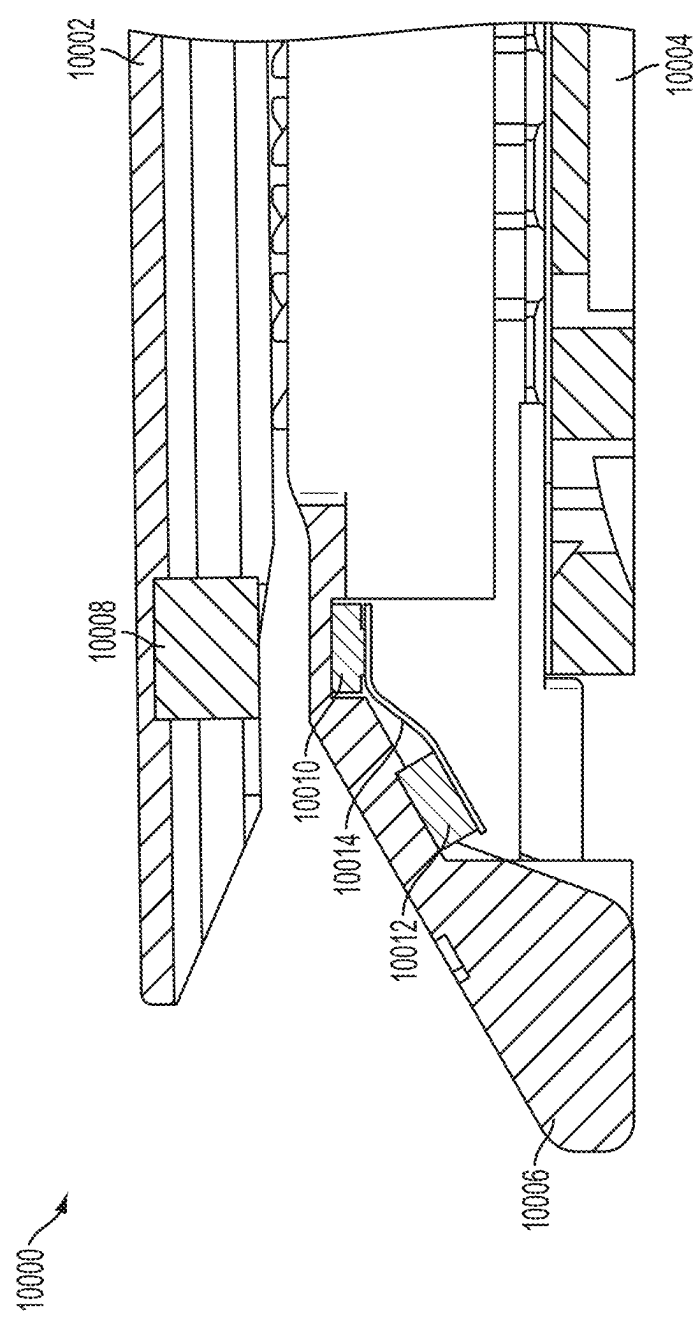


FIG. 126B

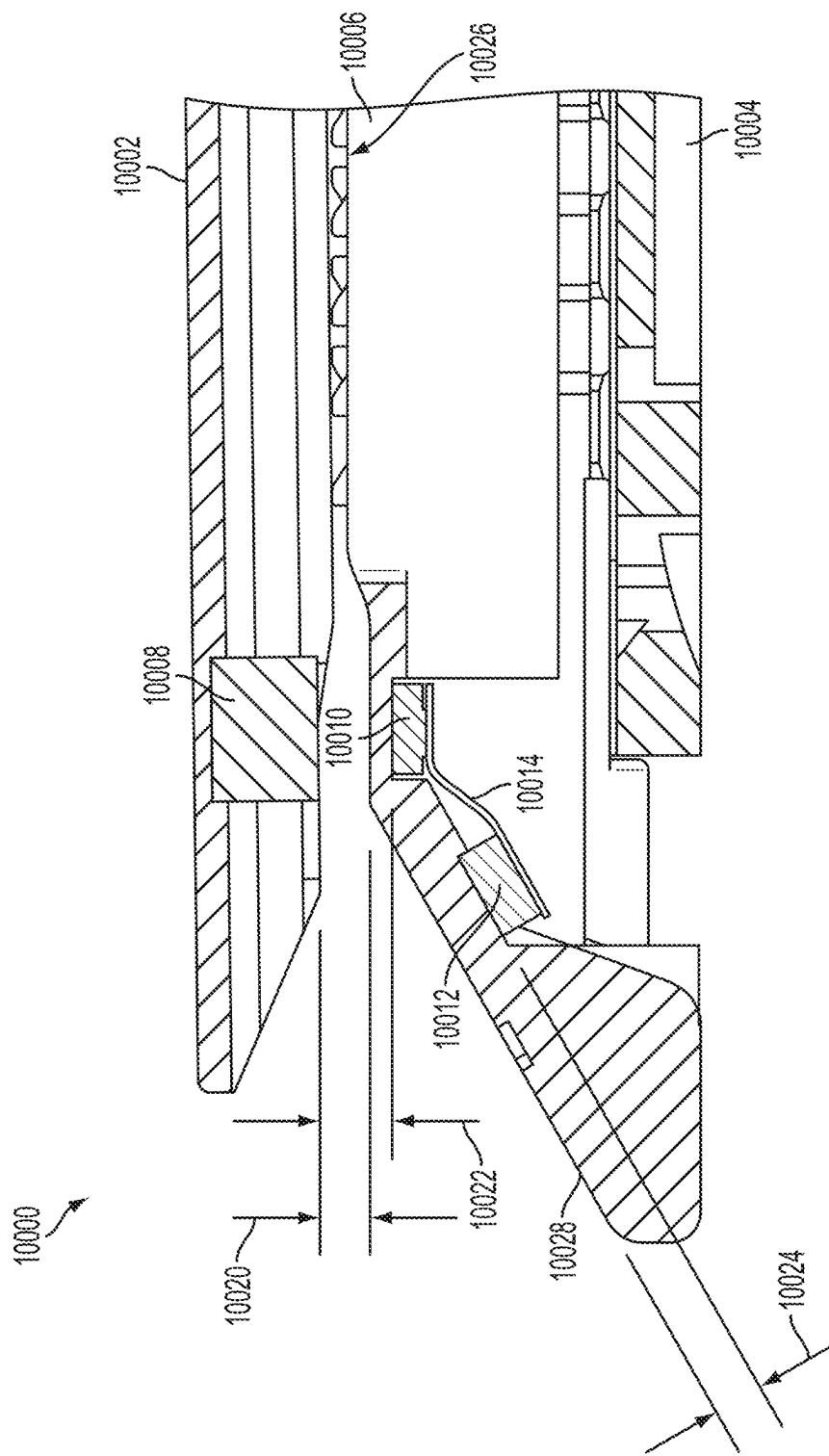


FIG. 127

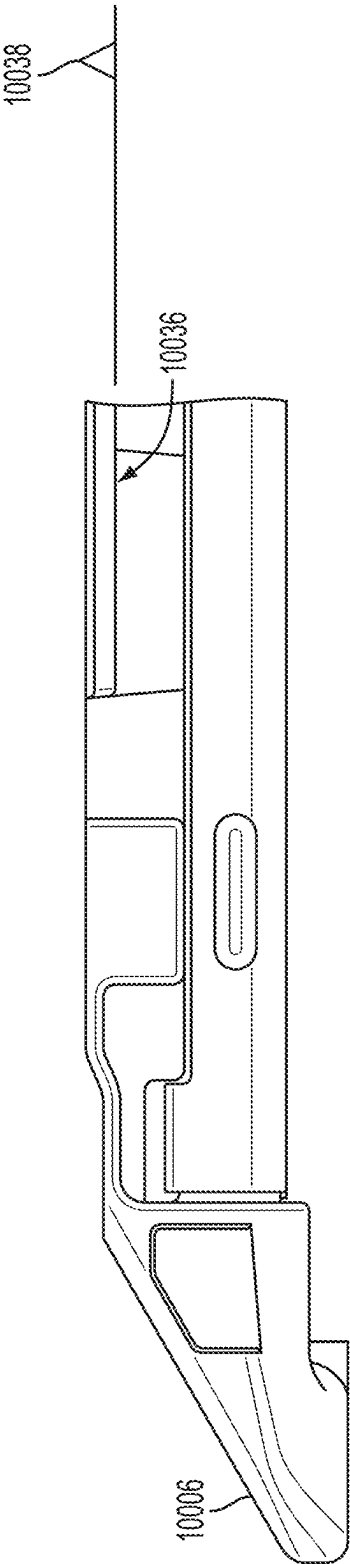


FIG. 128A

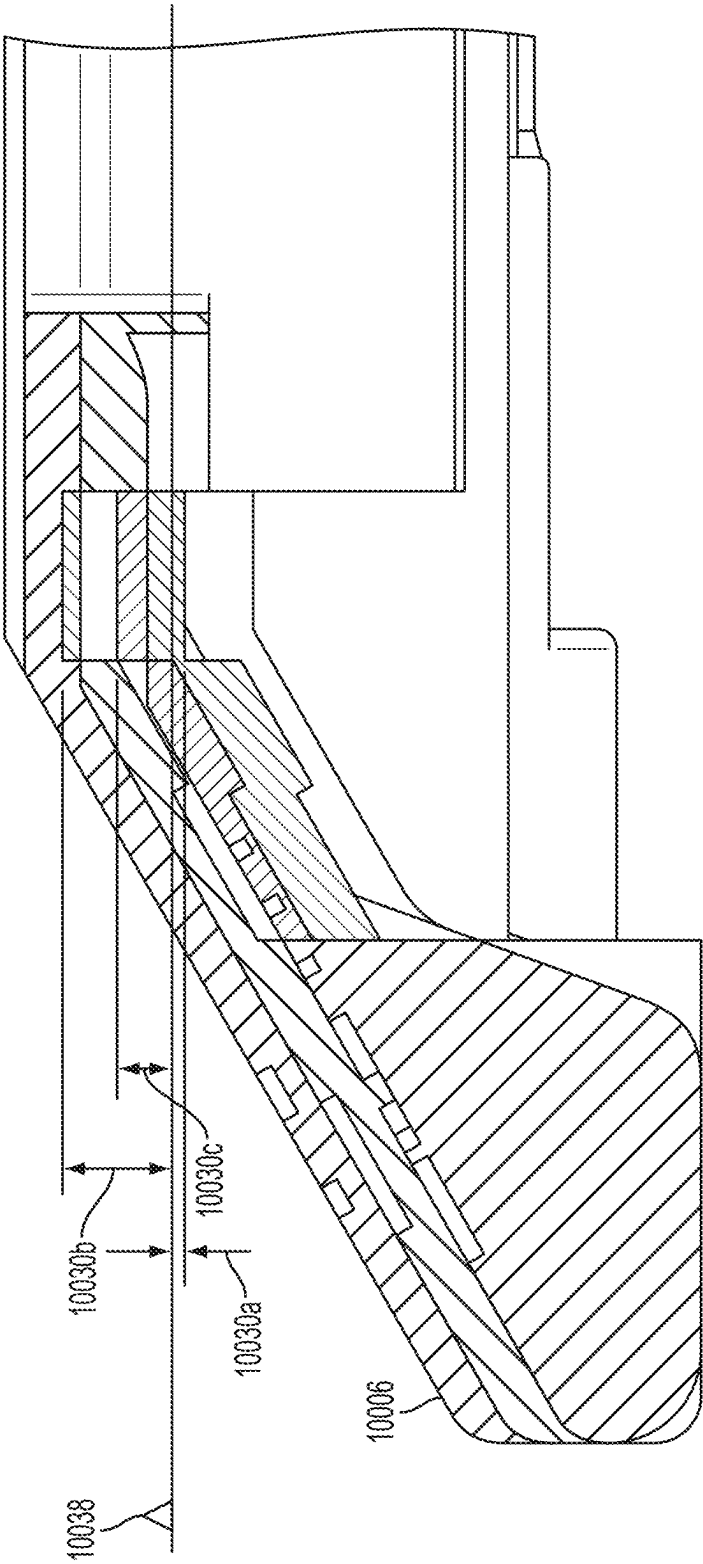


FIG. 128B

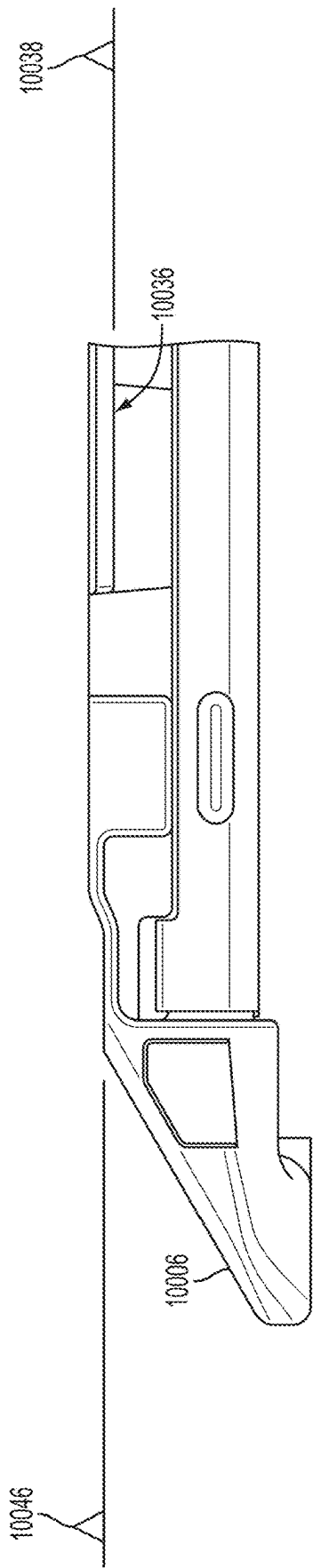


FIG. 128C

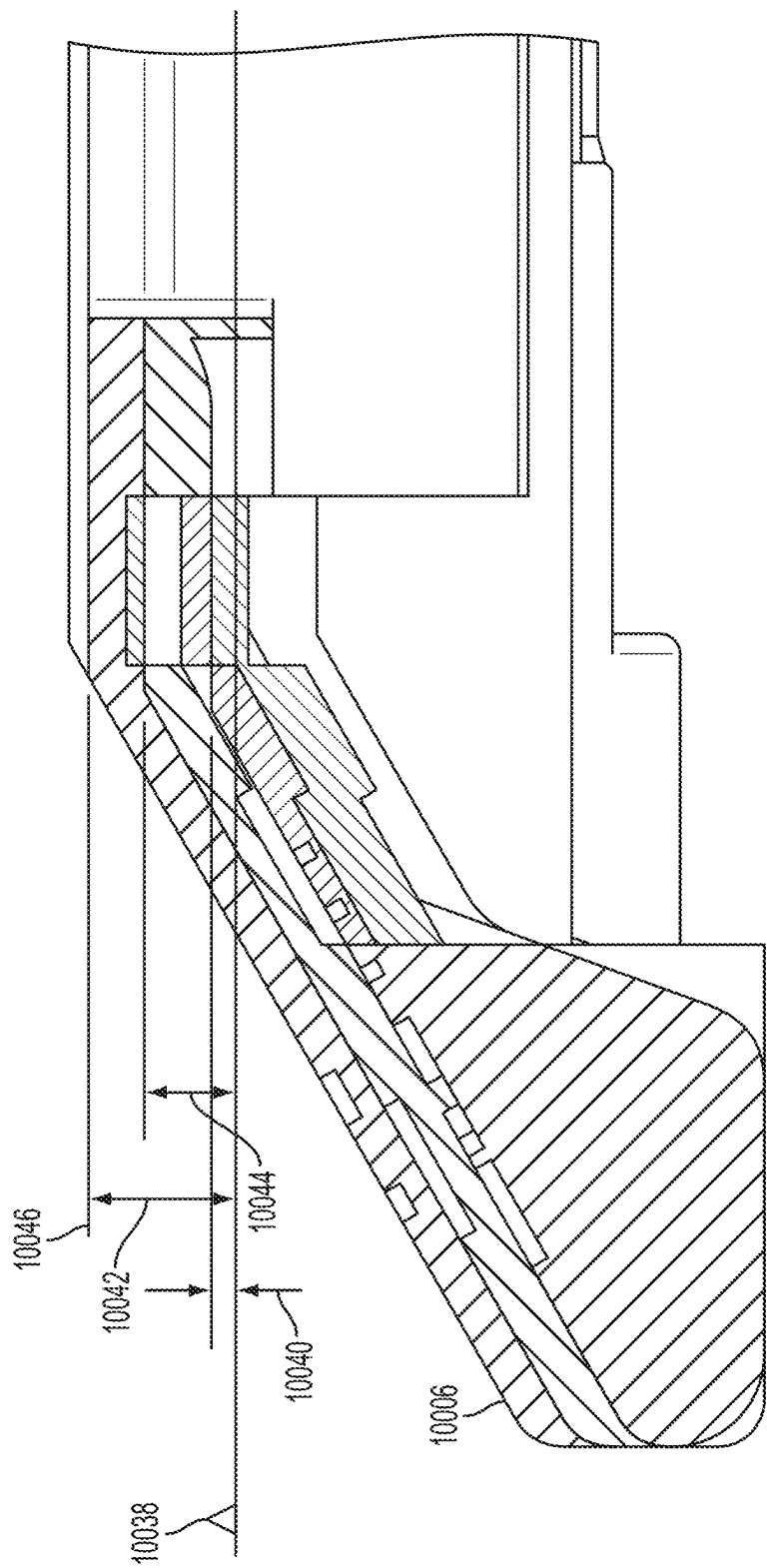
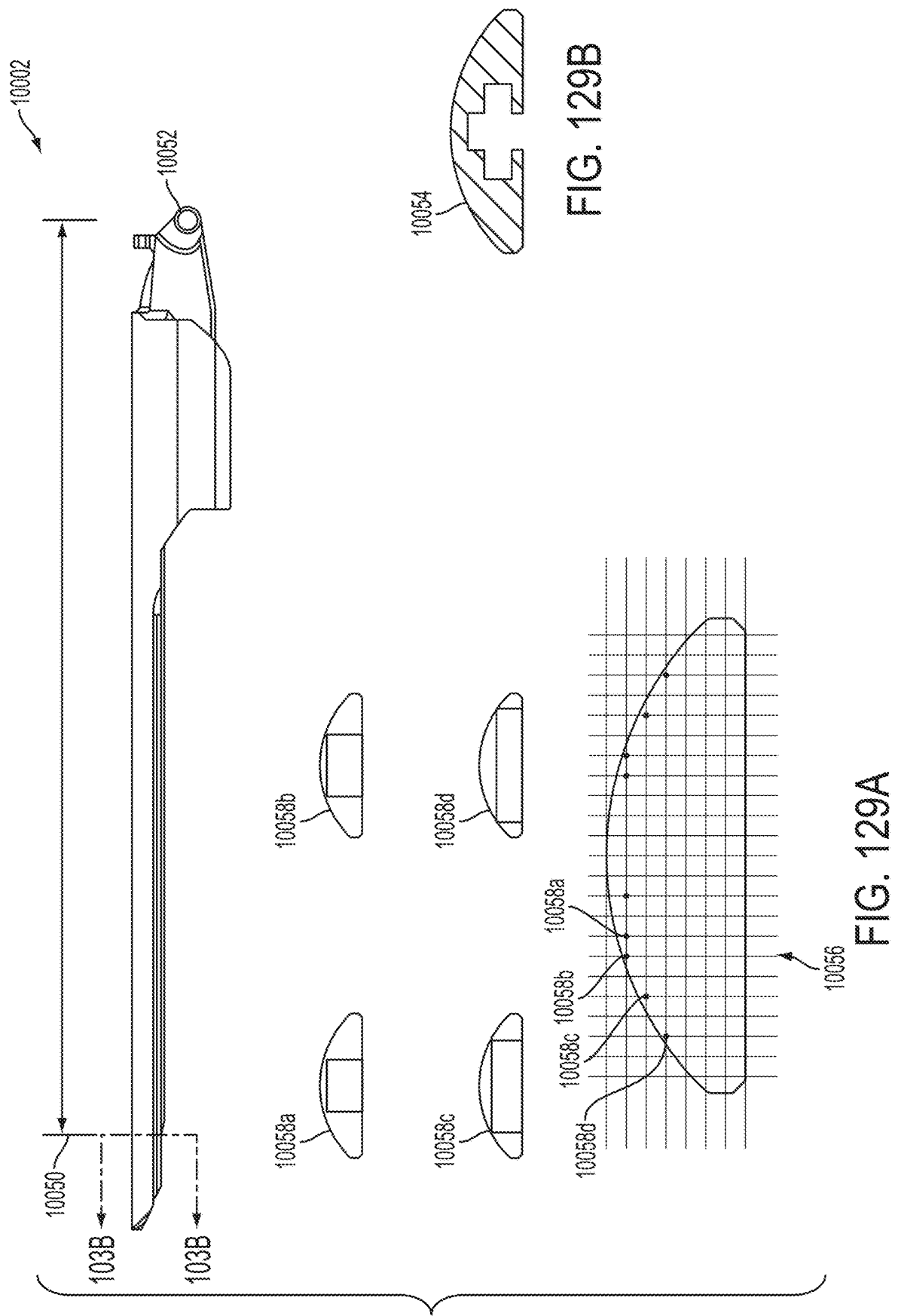


FIG. 128D



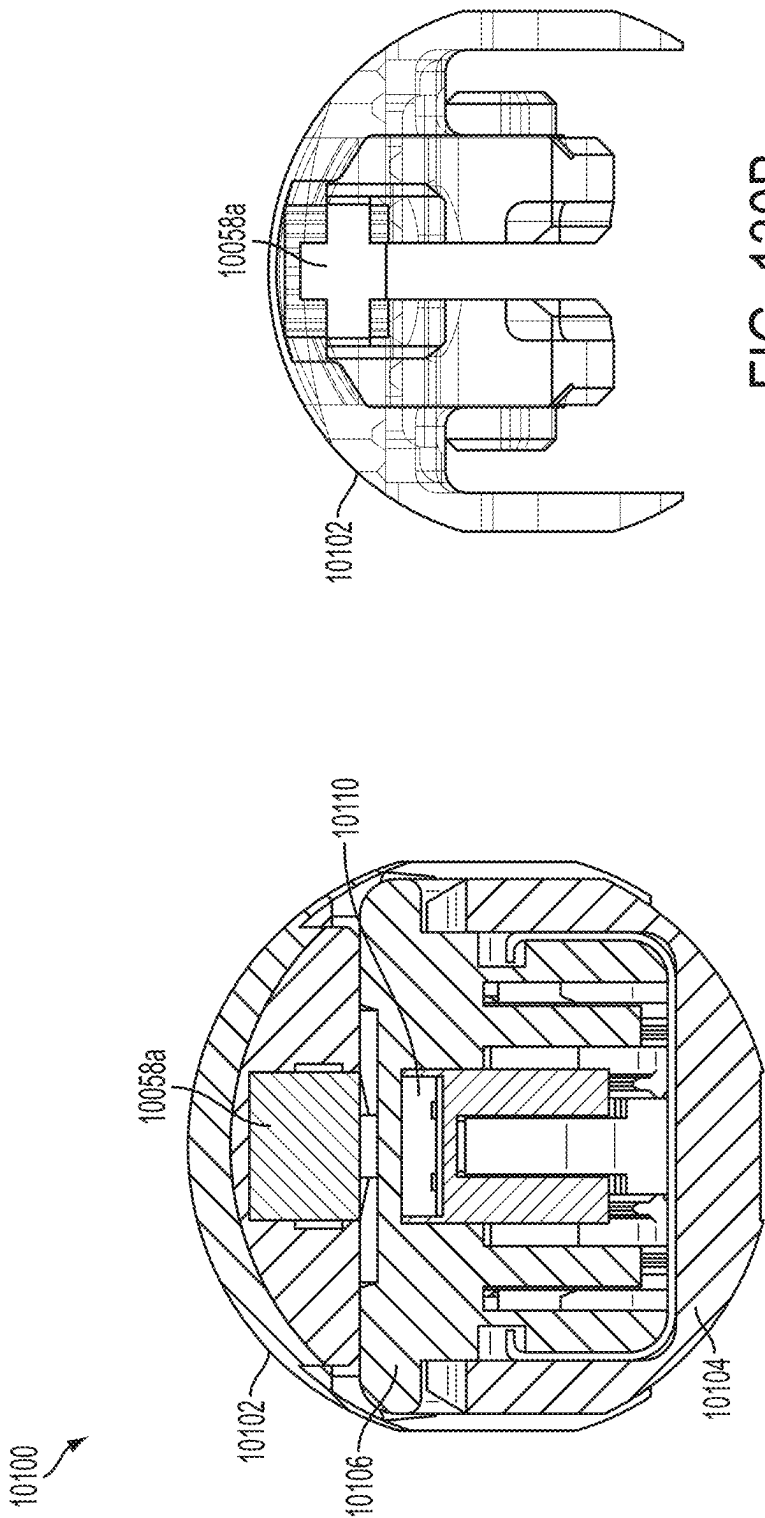


FIG. 130B

FIG. 130A

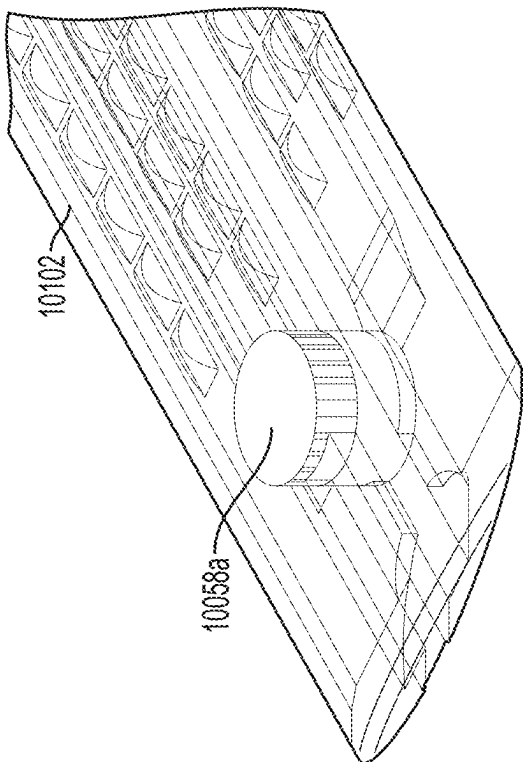


FIG. 130C

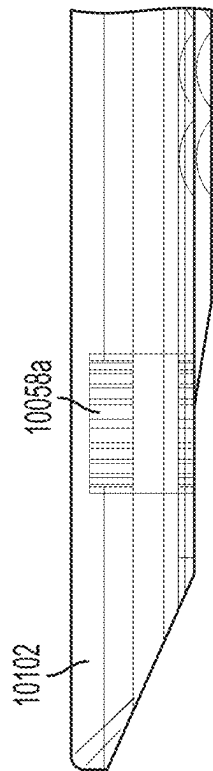


FIG. 130D

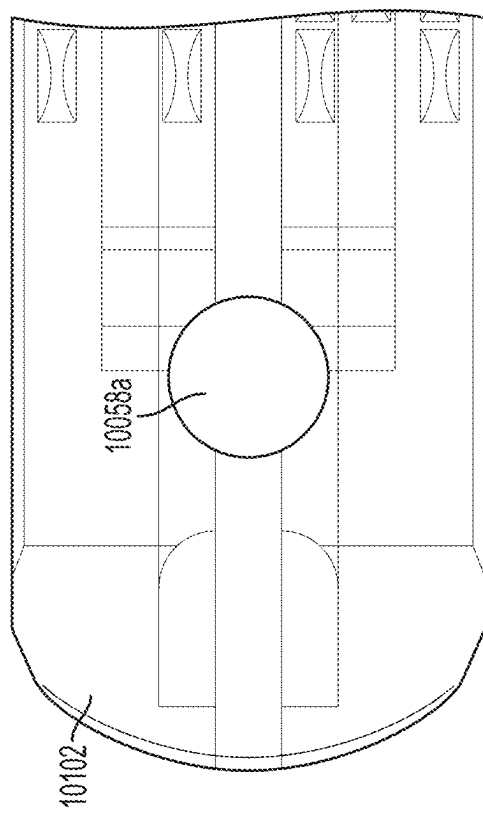


FIG. 130E

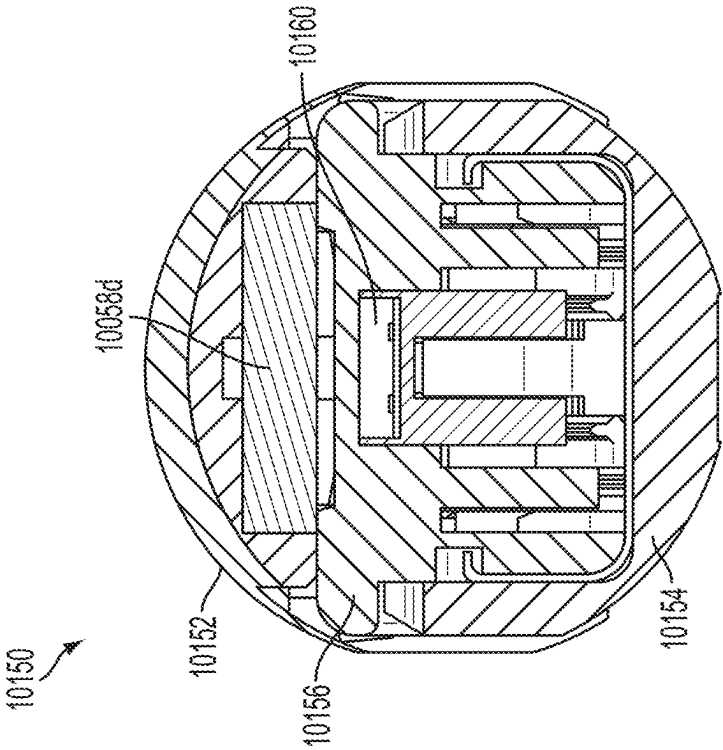


FIG. 131A

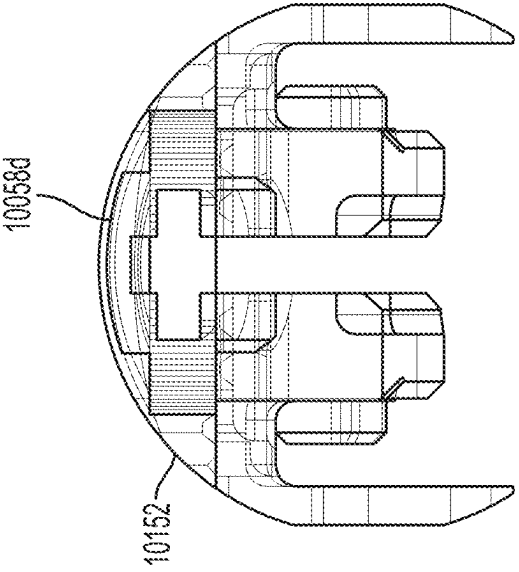


FIG. 131B

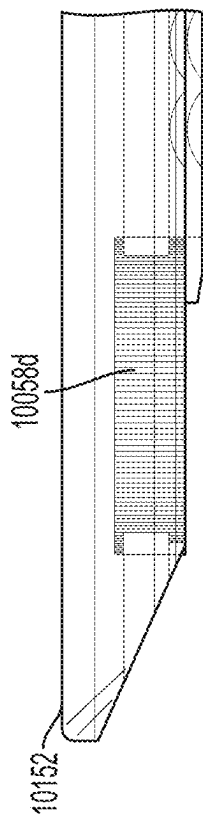


FIG. 131D

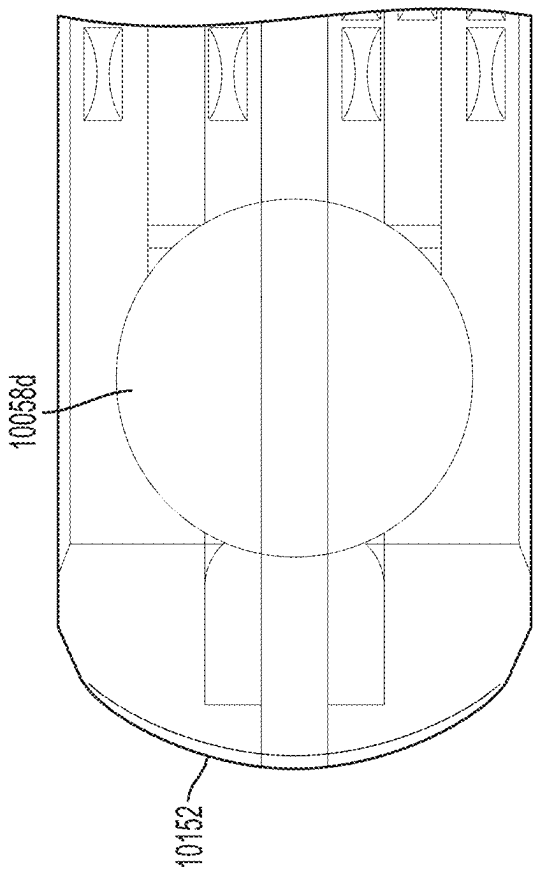


FIG. 131E

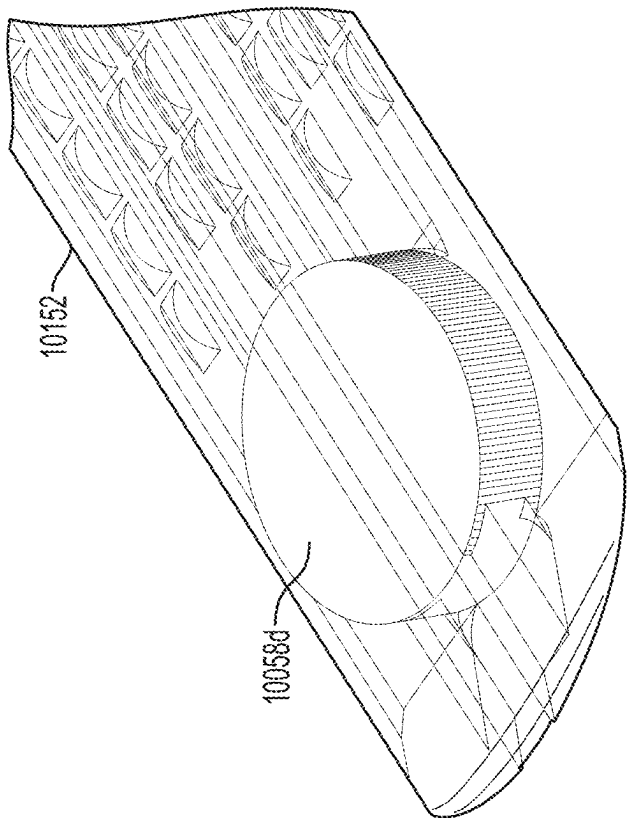


FIG. 131C

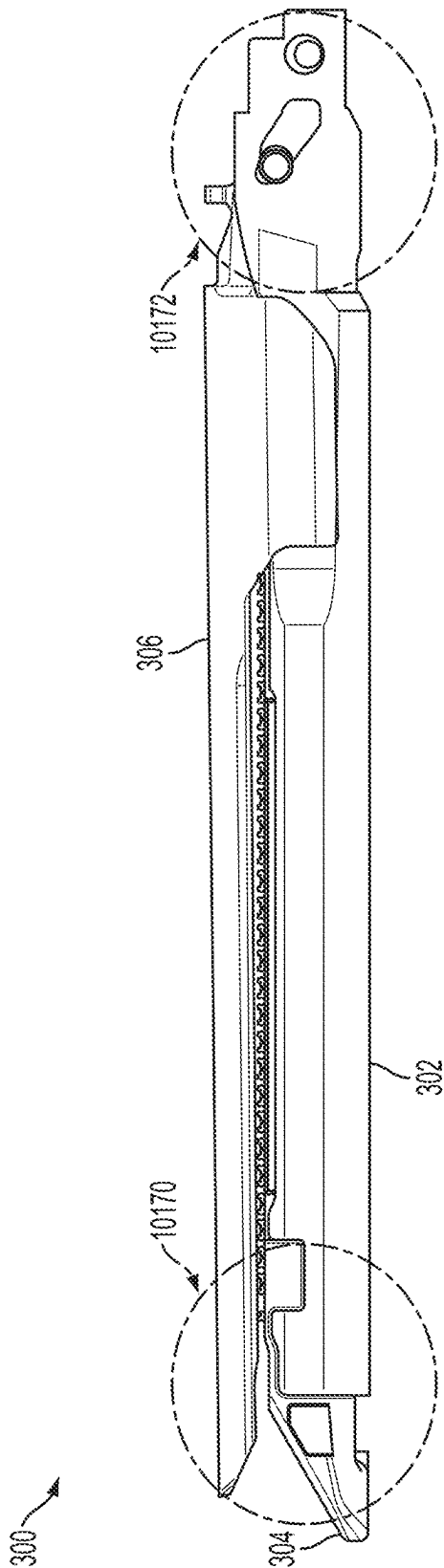


FIG. 132

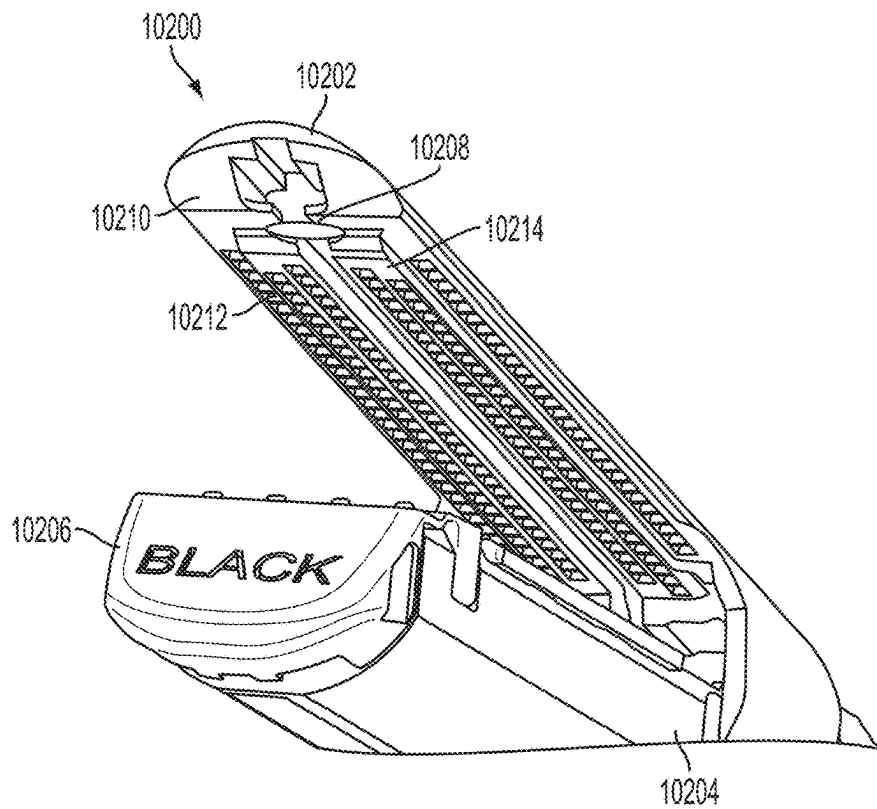


FIG. 133A

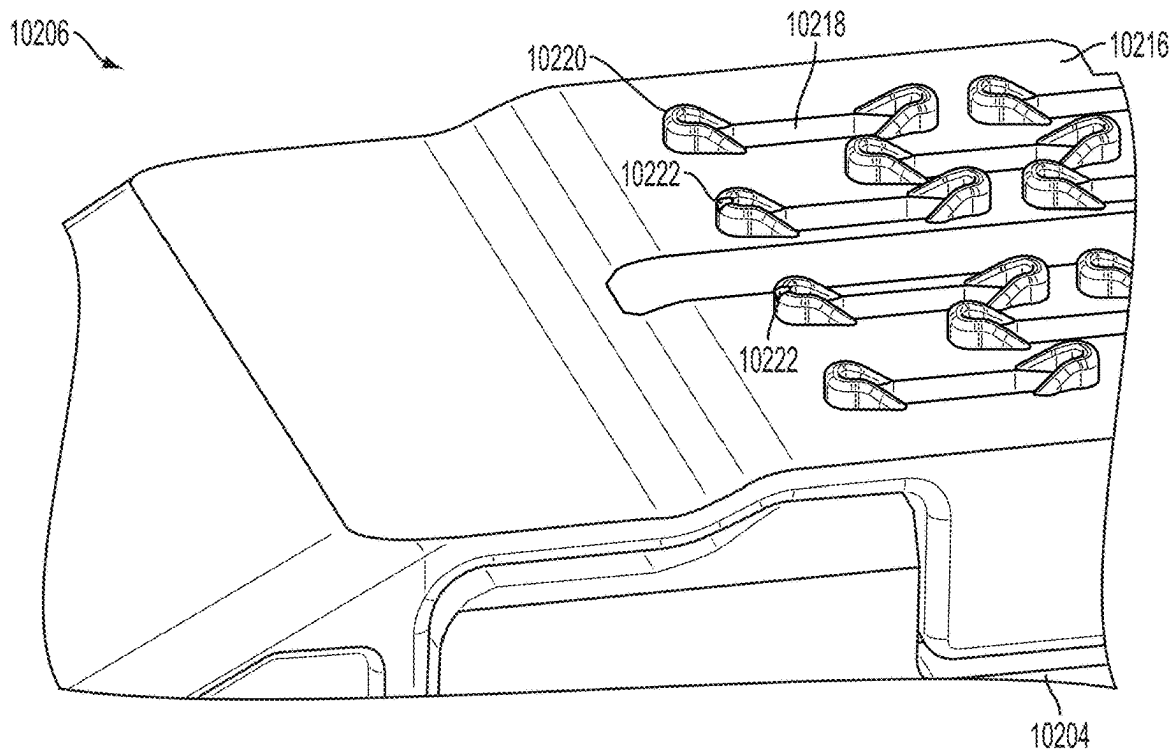


FIG. 133B

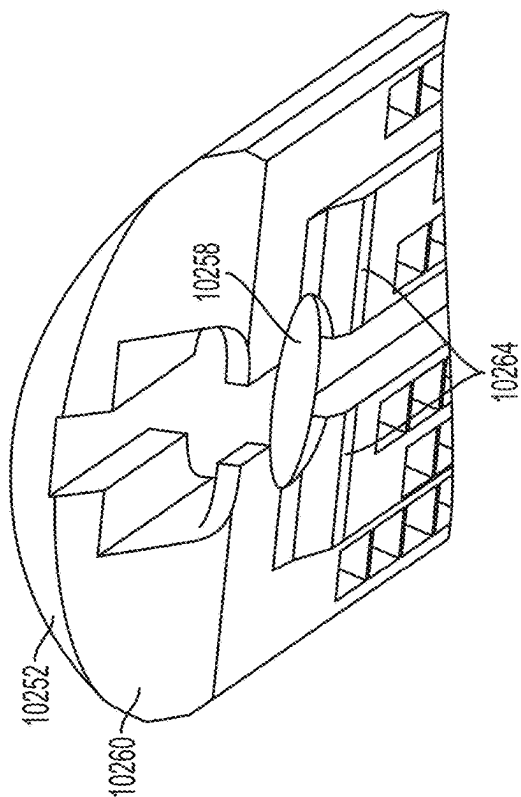


FIG. 134B

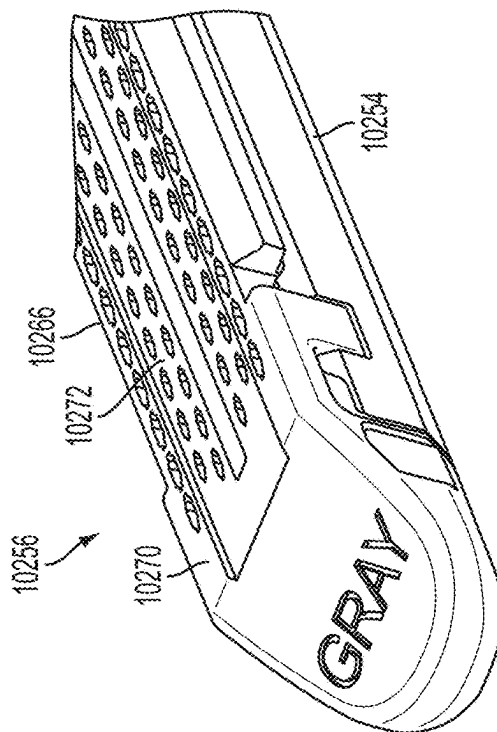


FIG. 134C

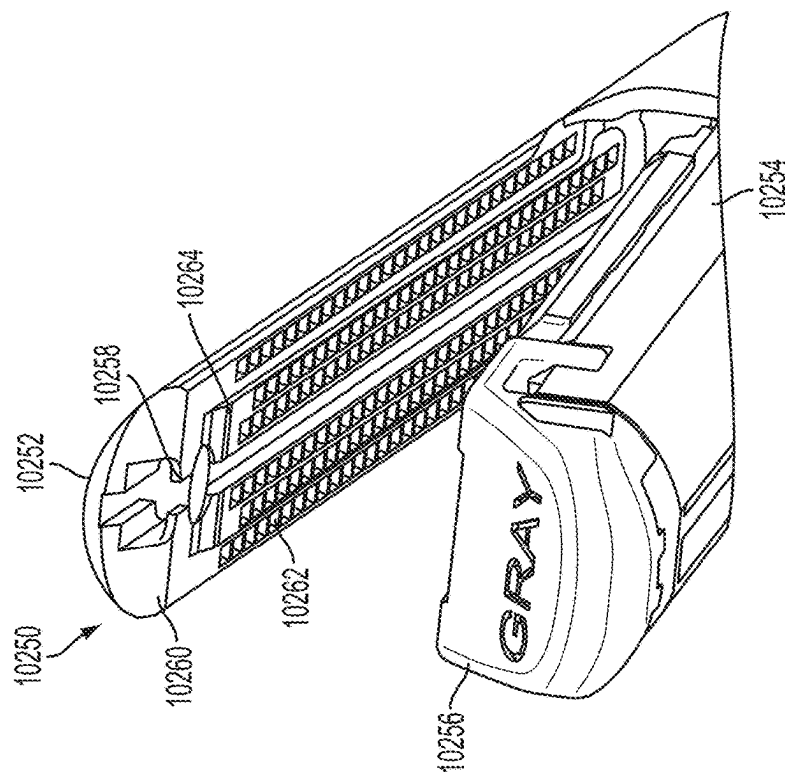


FIG. 134A

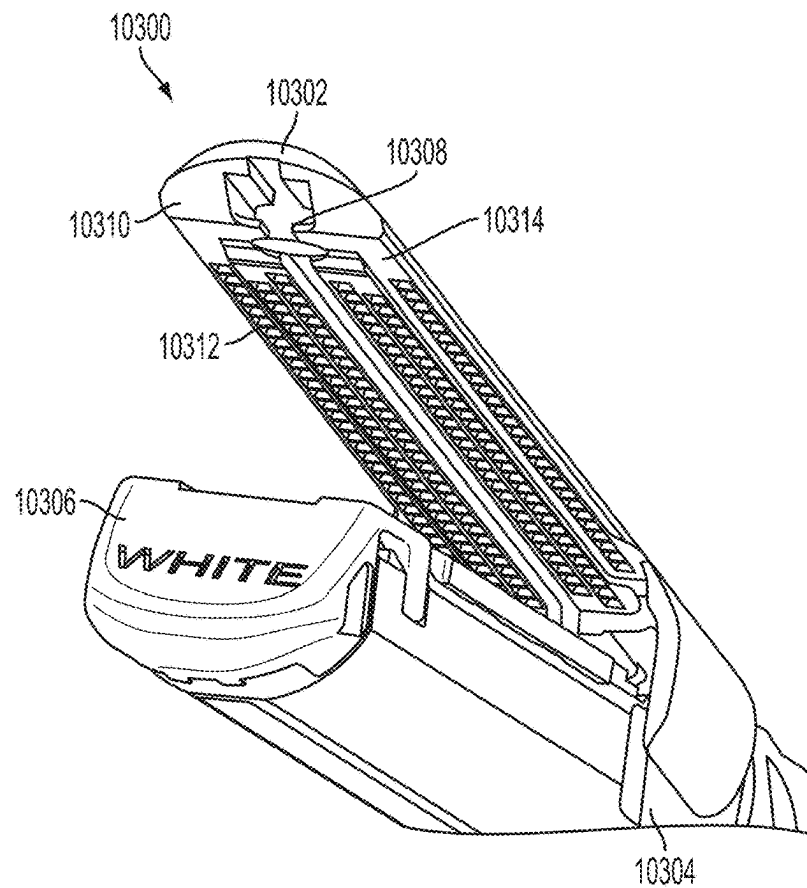


FIG. 135A

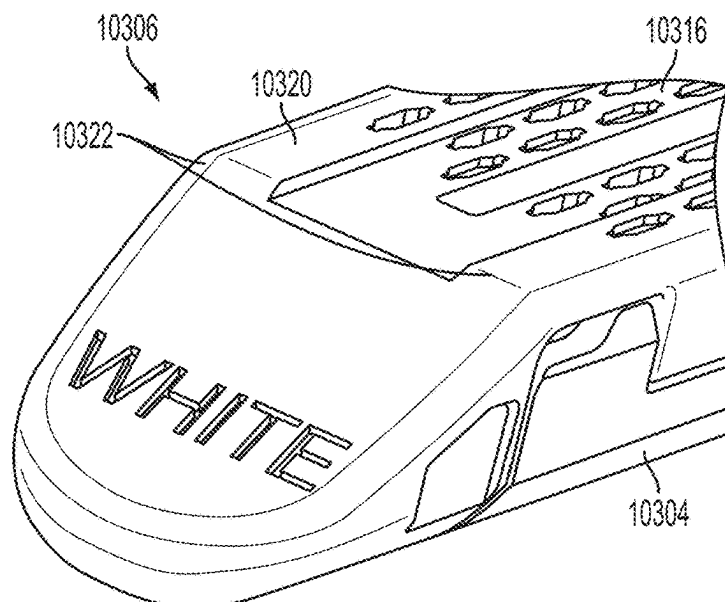


FIG. 135B

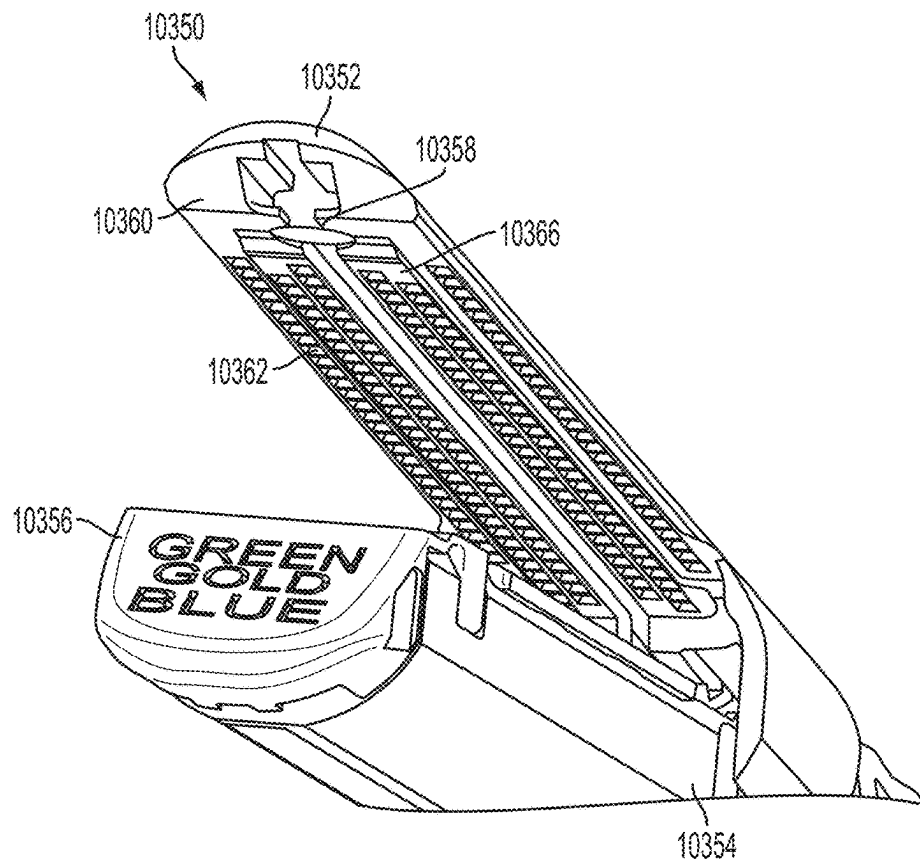


FIG. 136A

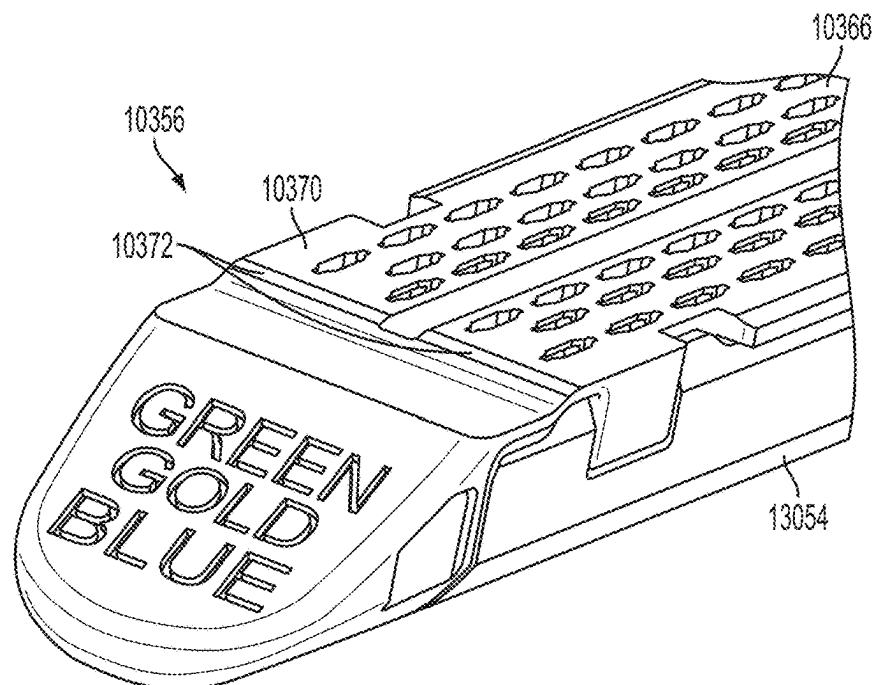
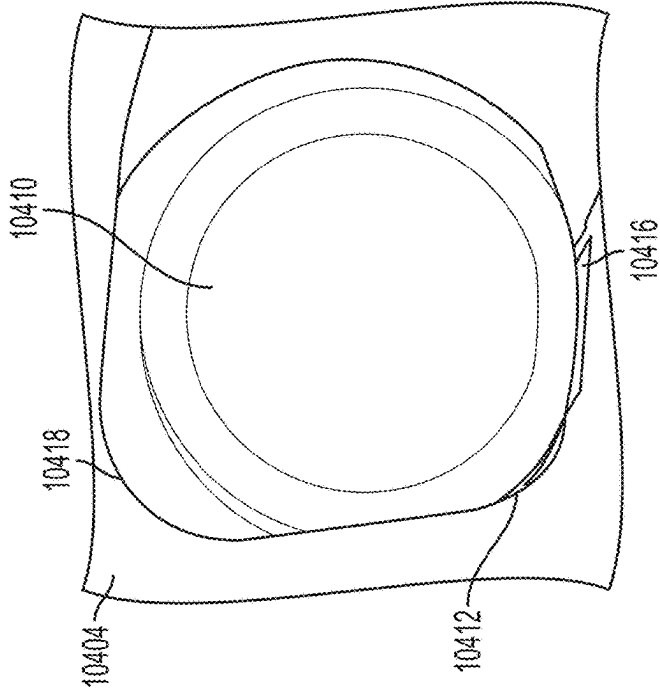
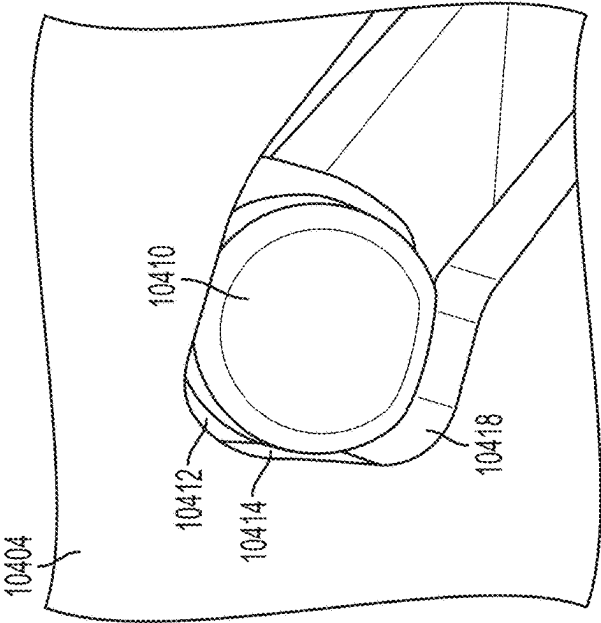
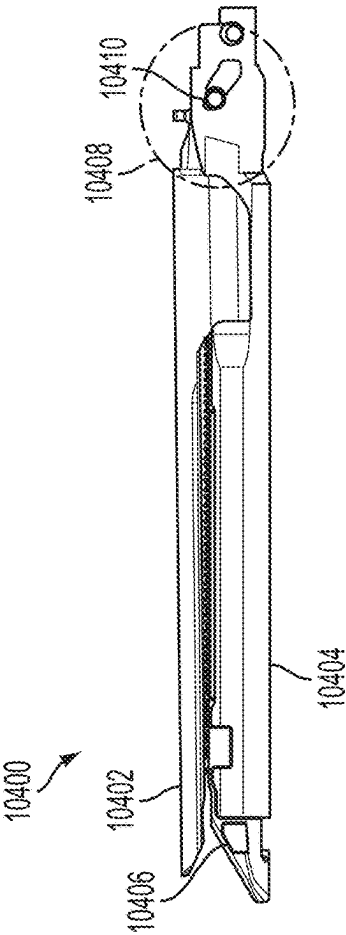


FIG. 136B



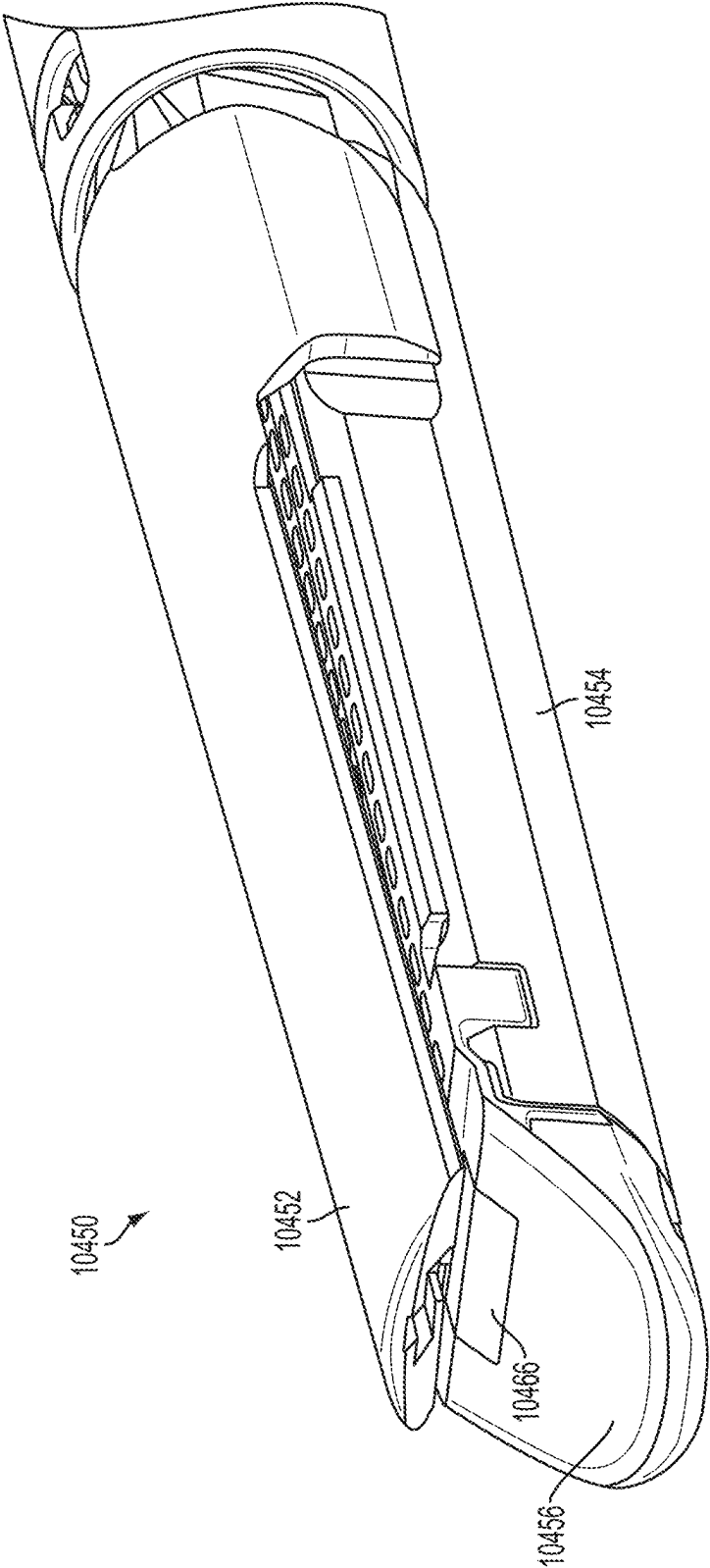
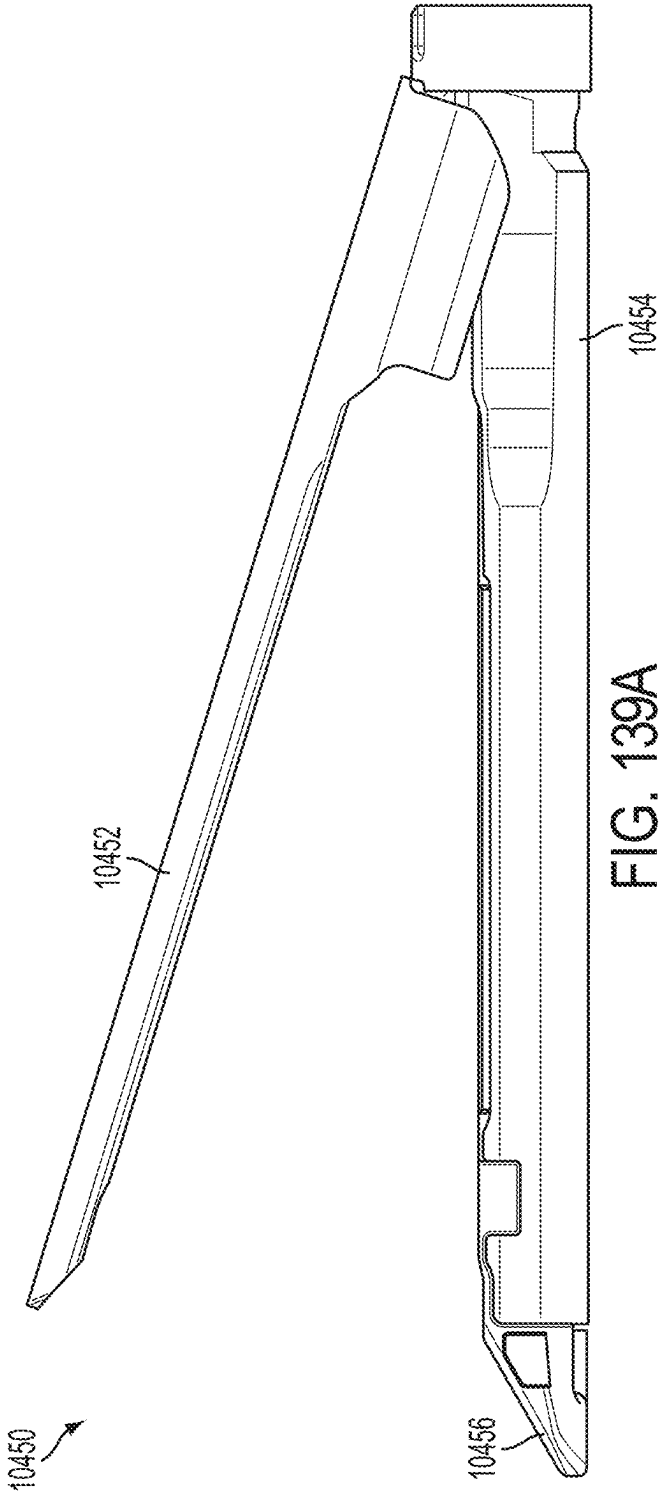
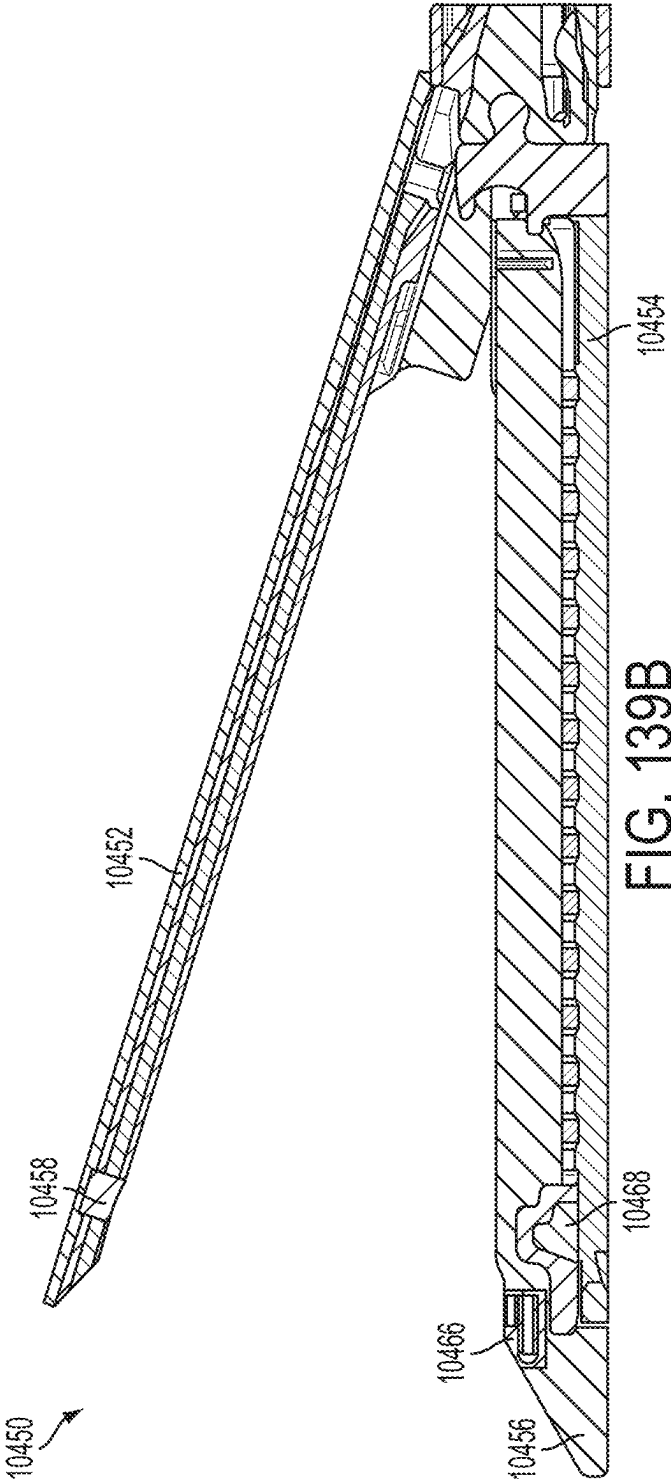


FIG. 138





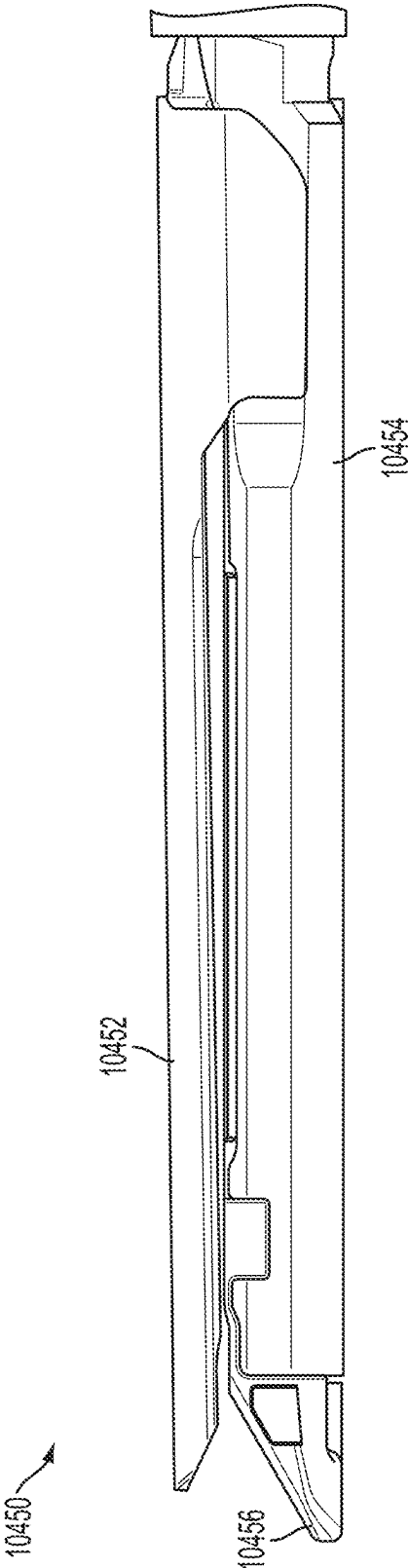


FIG. 139C

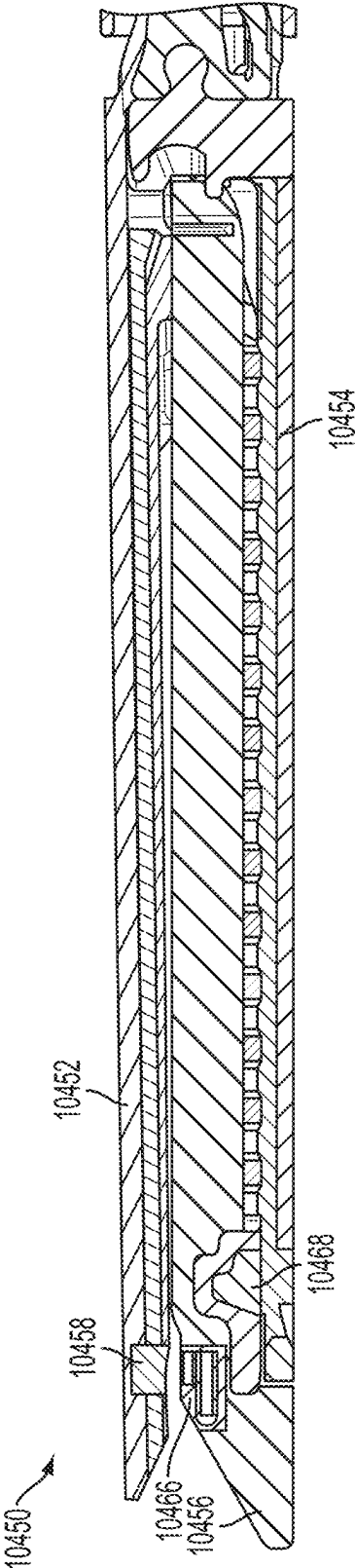


FIG. 139D

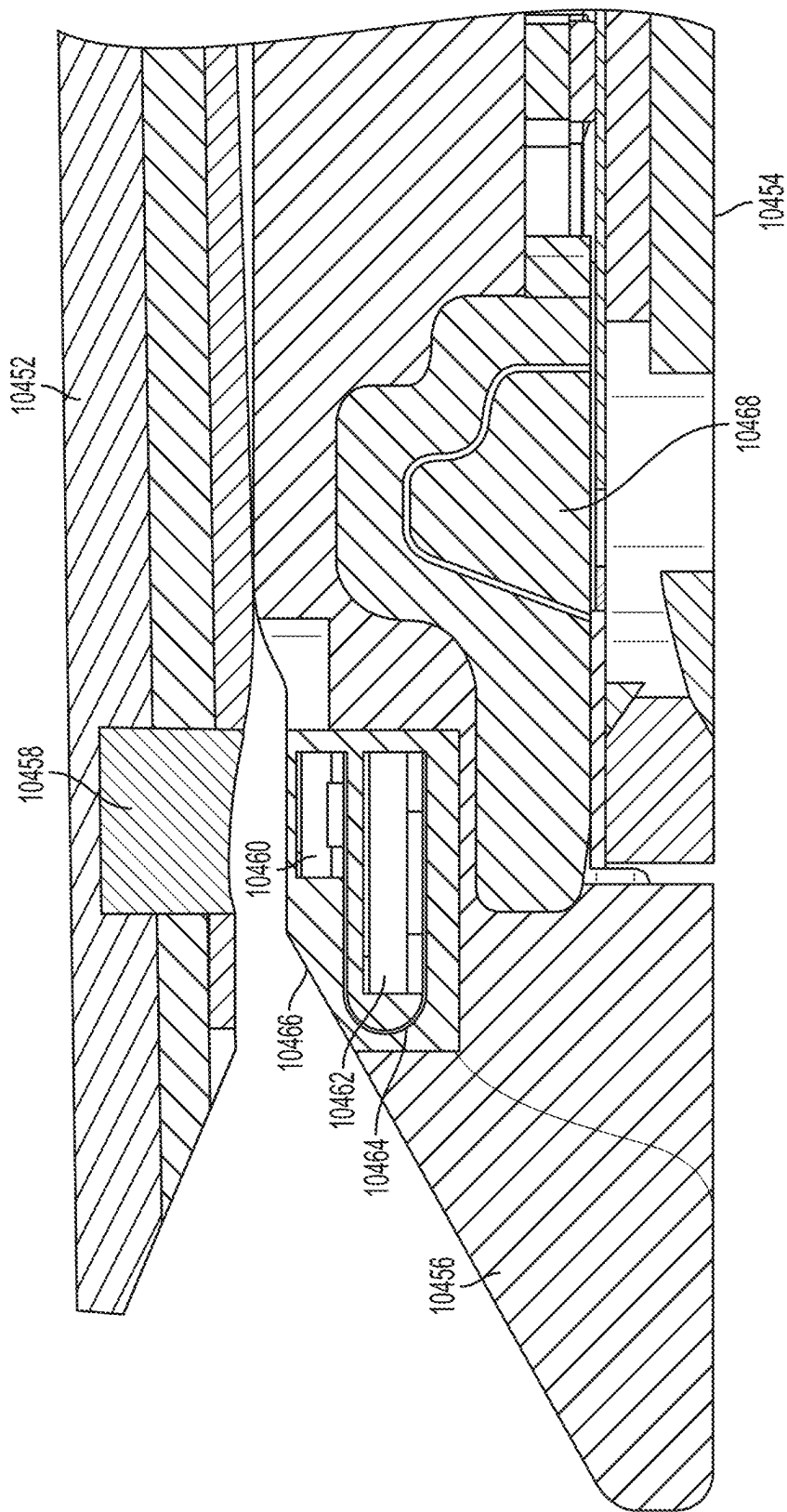


FIG. 140

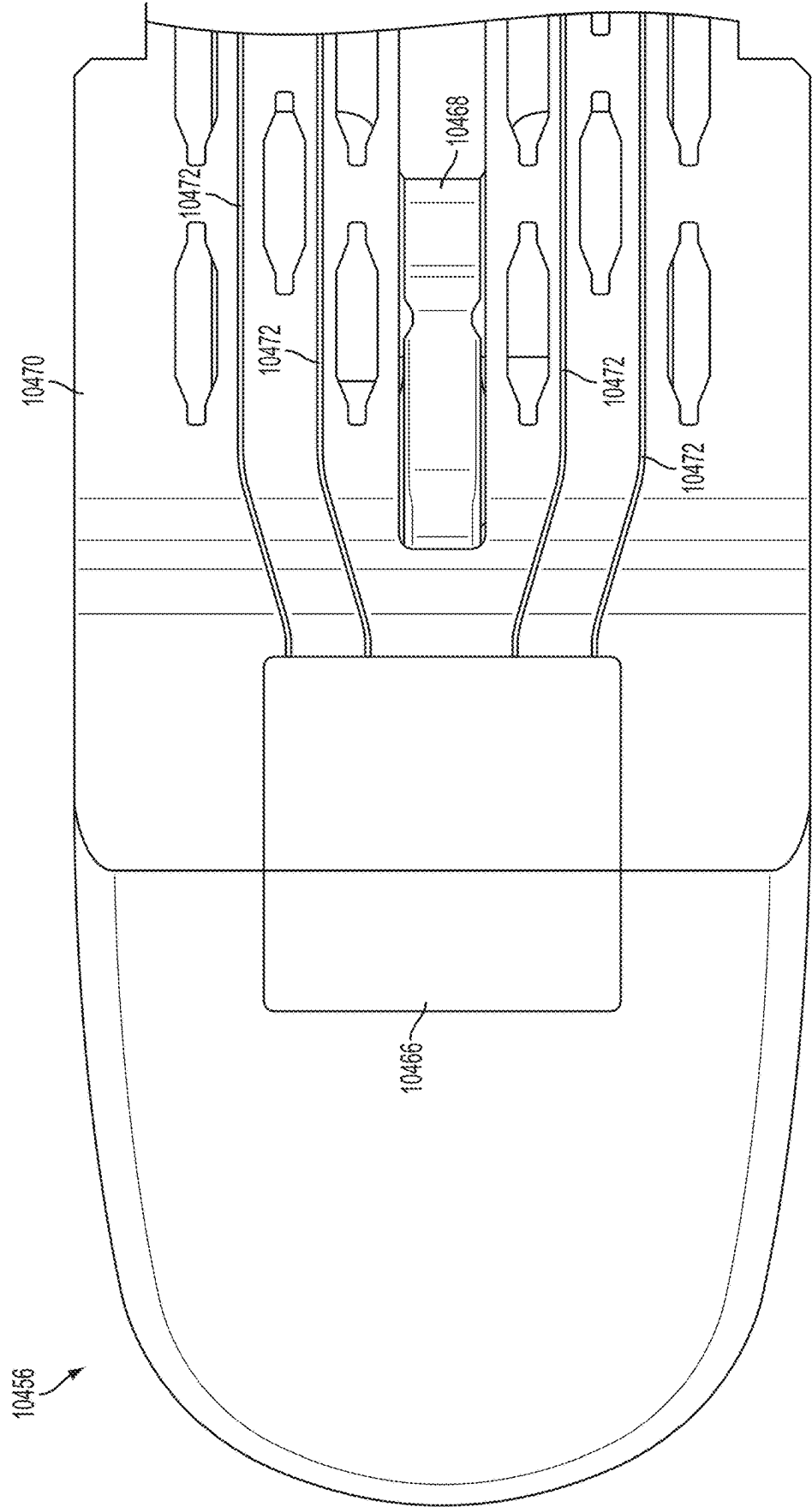


FIG. 141

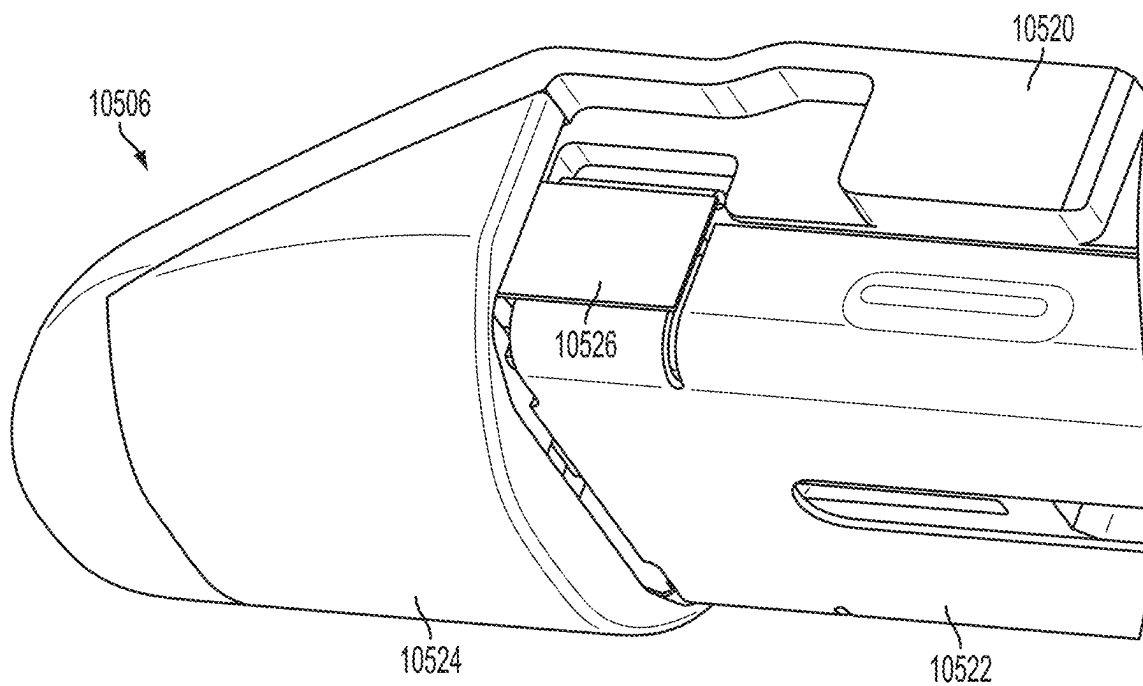


FIG. 142A

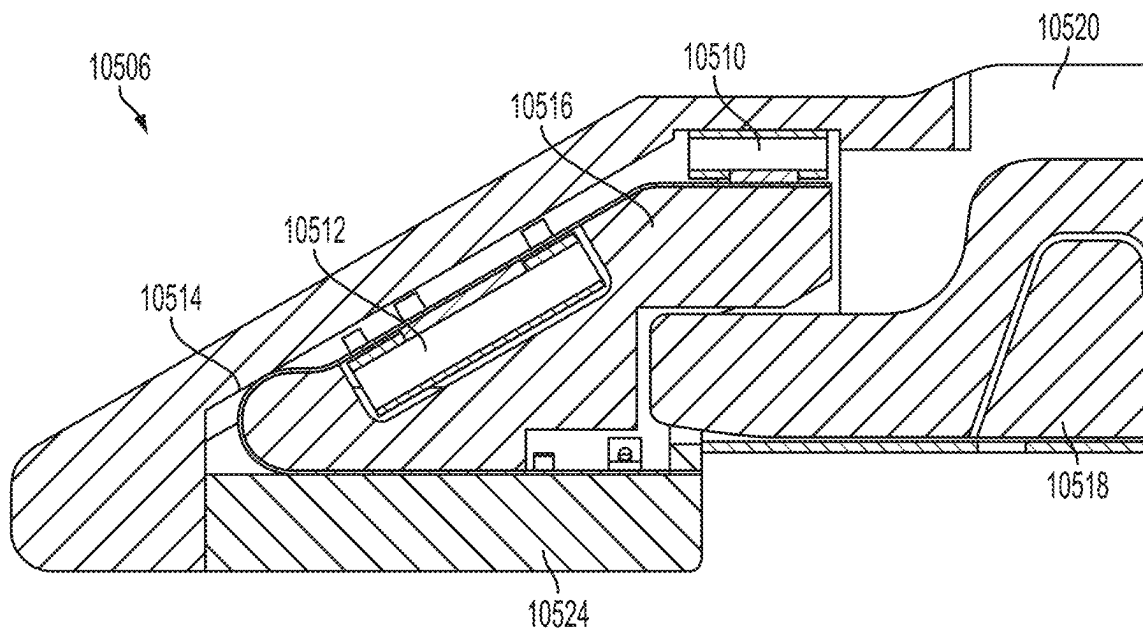
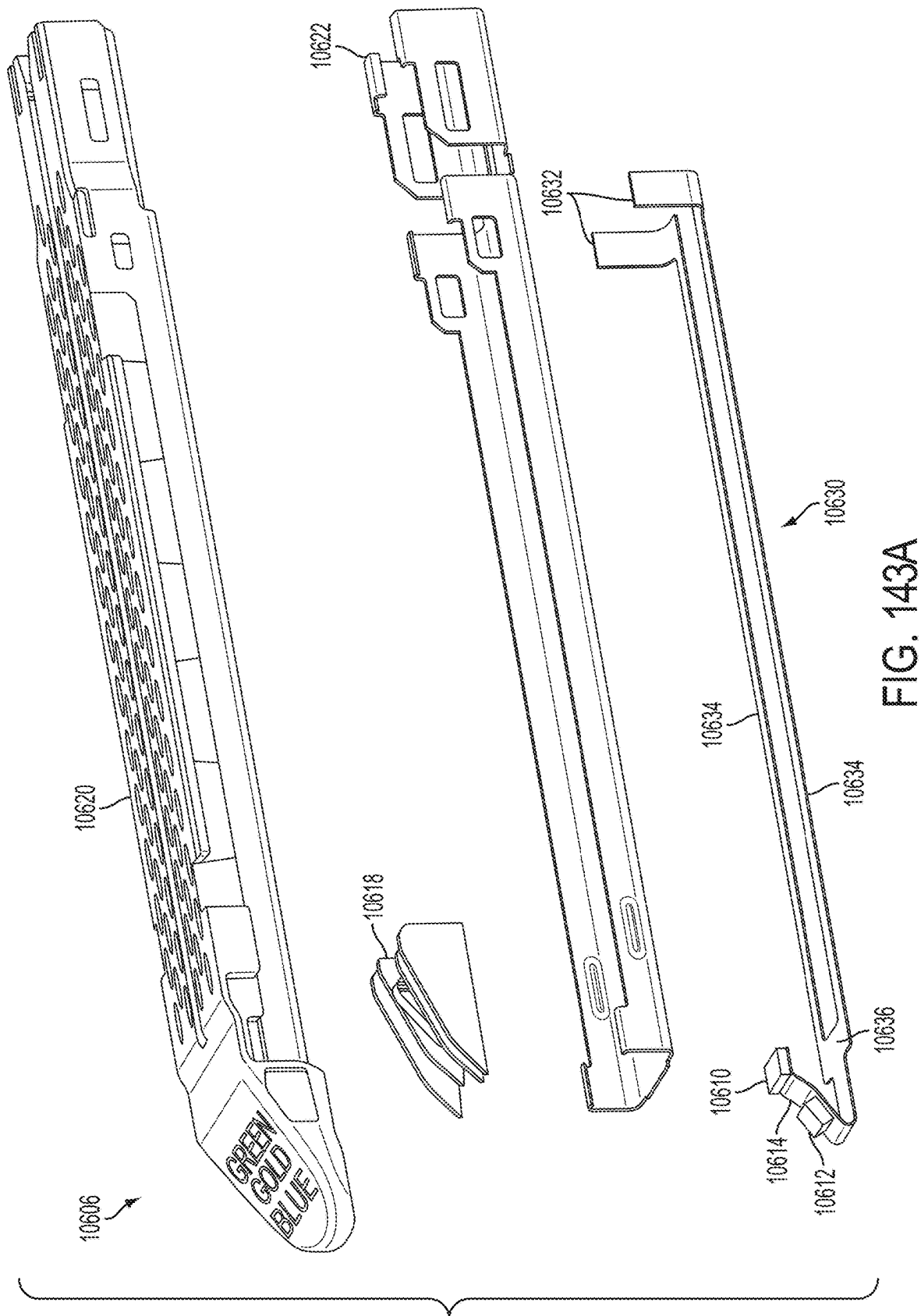


FIG. 142B



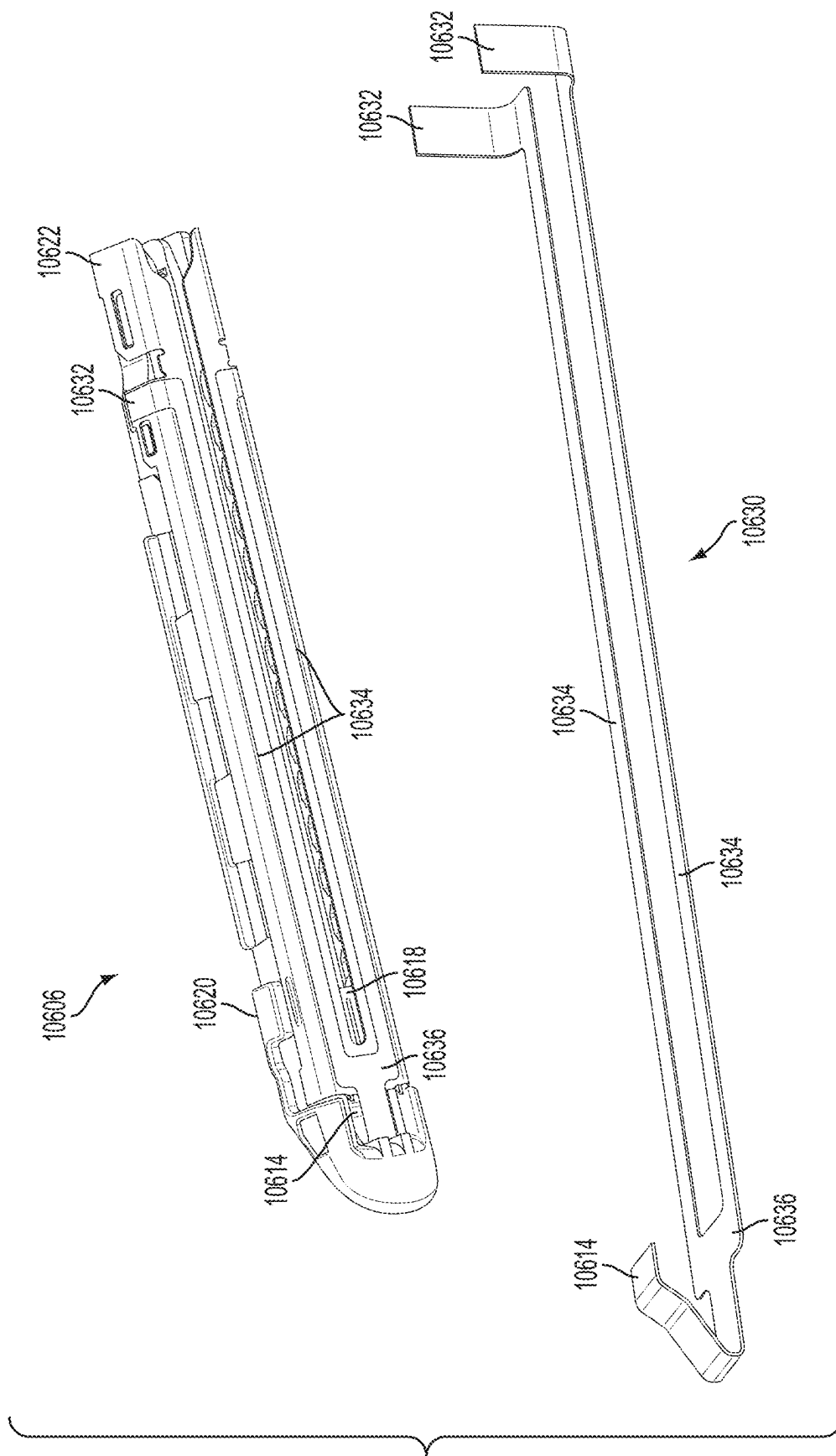


FIG. 143B

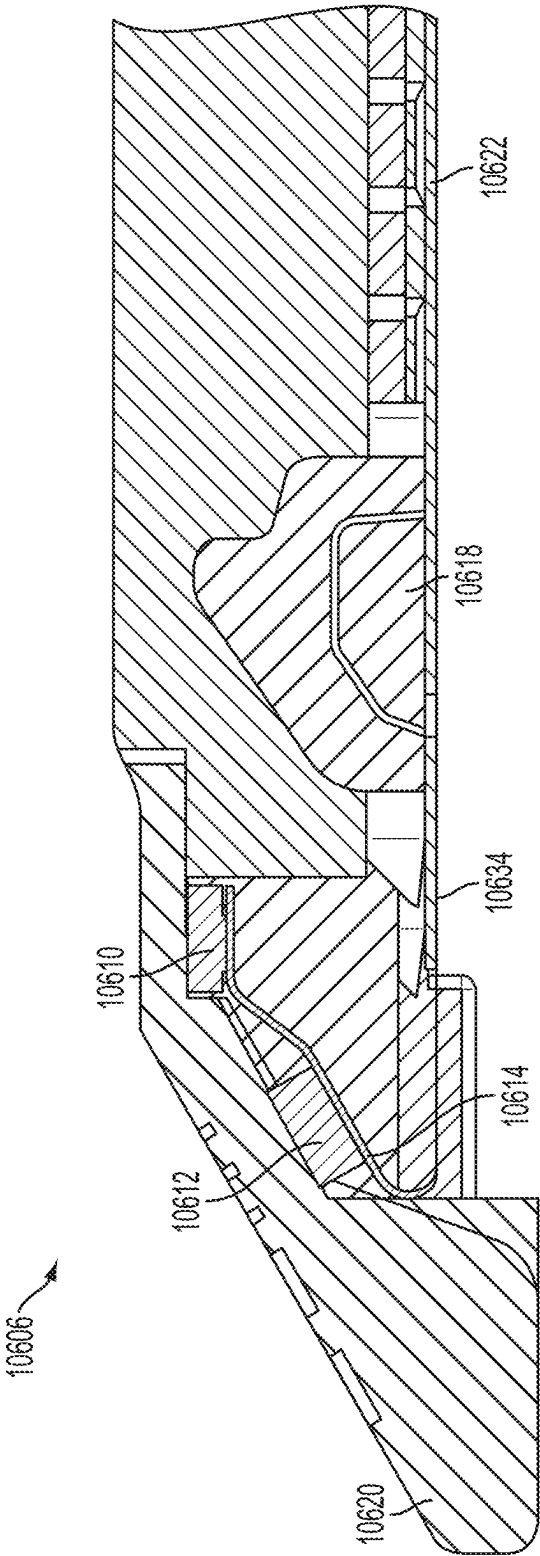
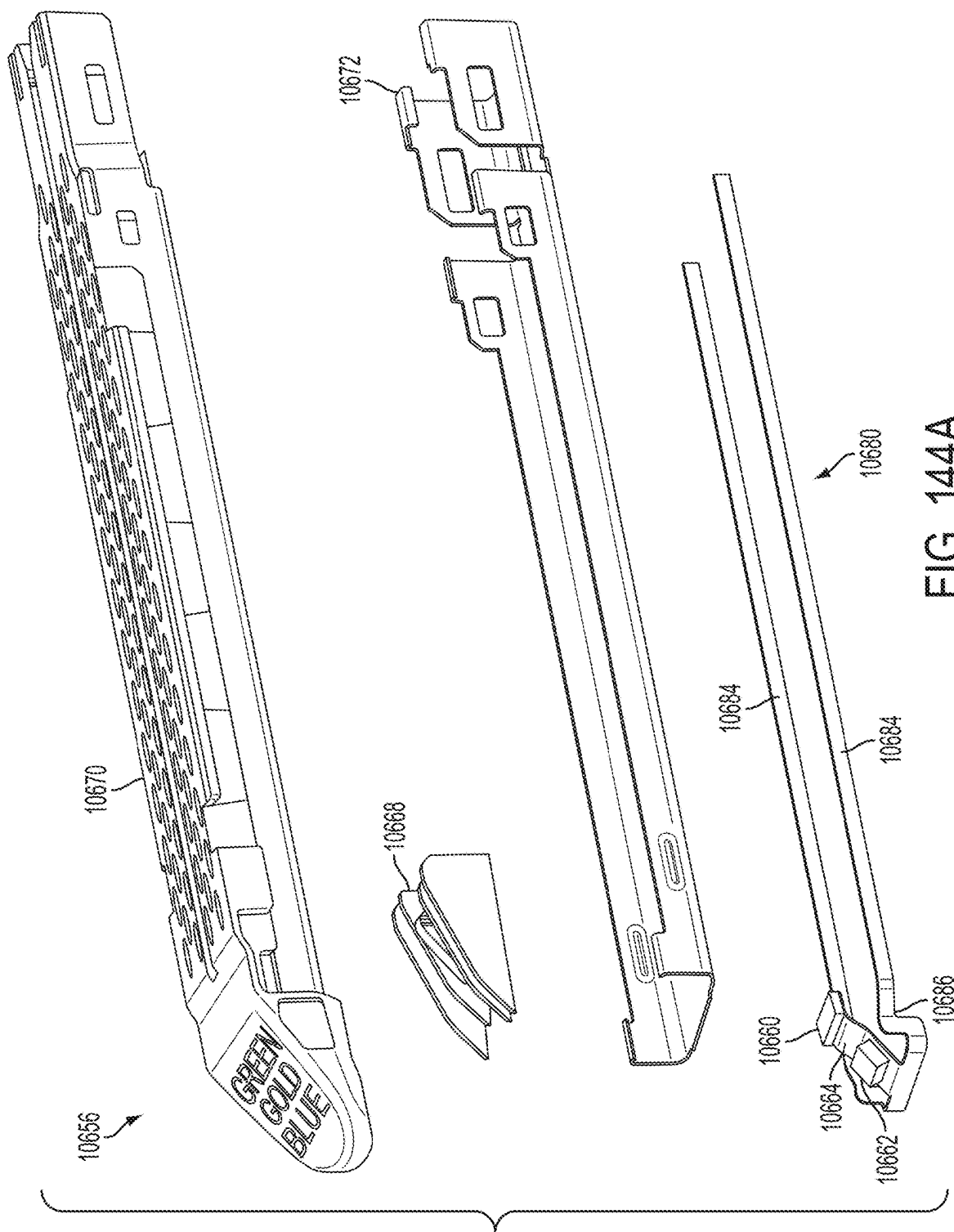
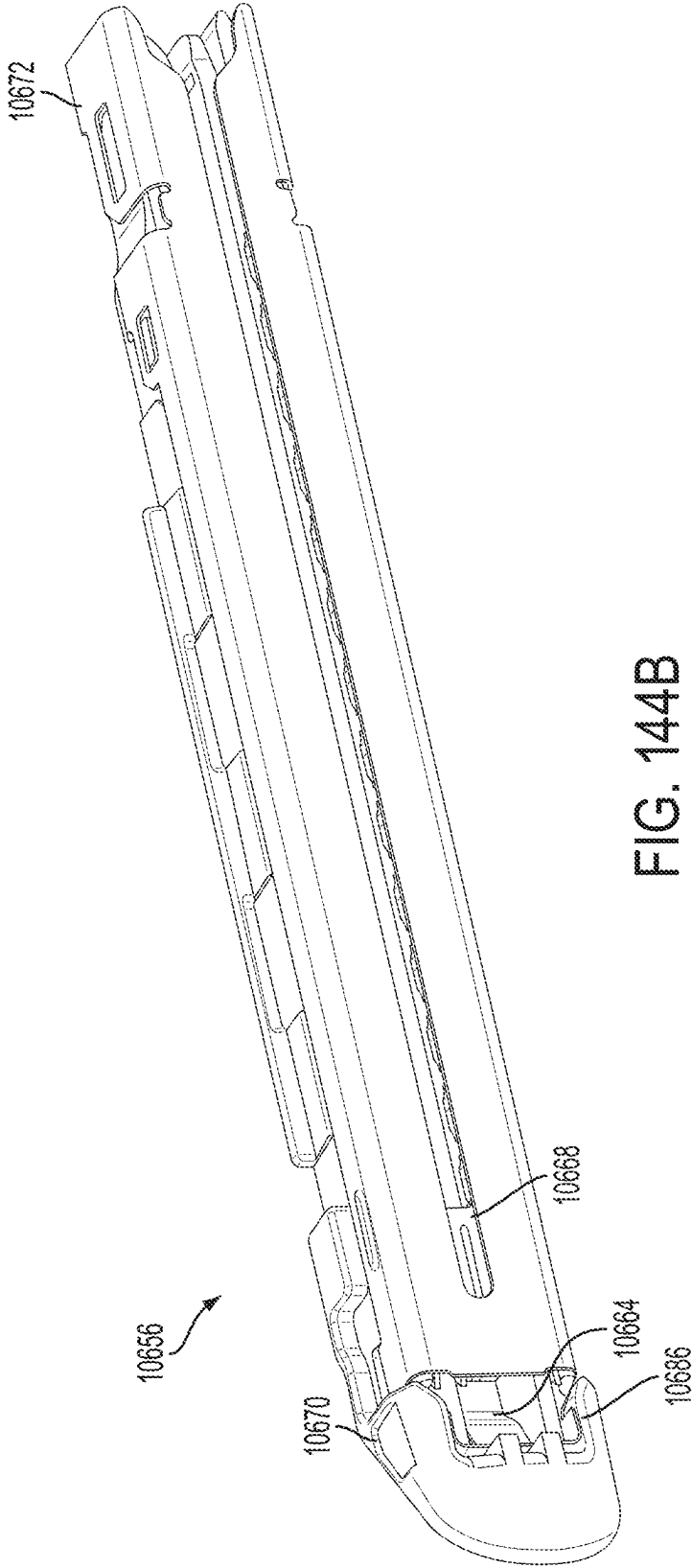
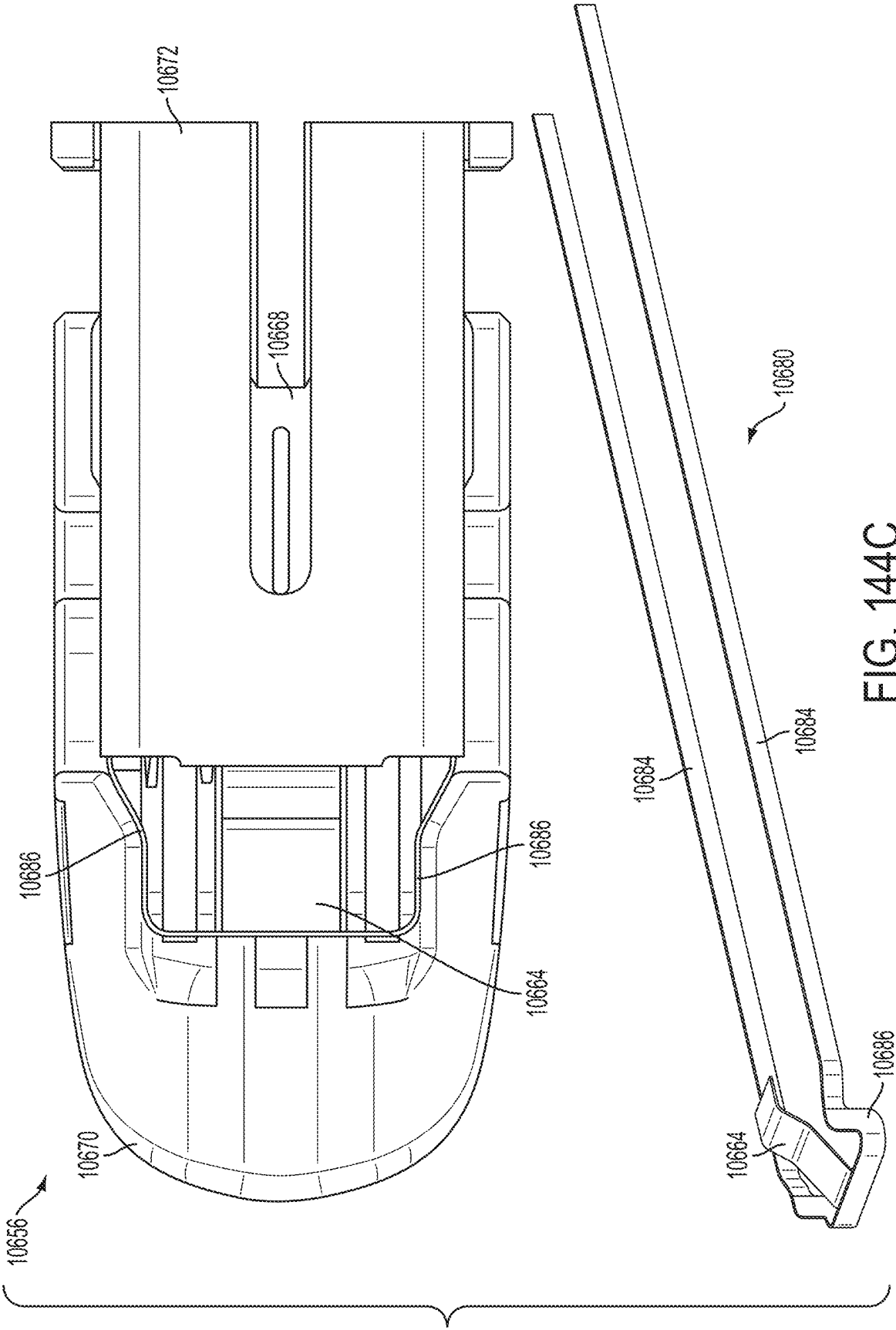


FIG. 143C







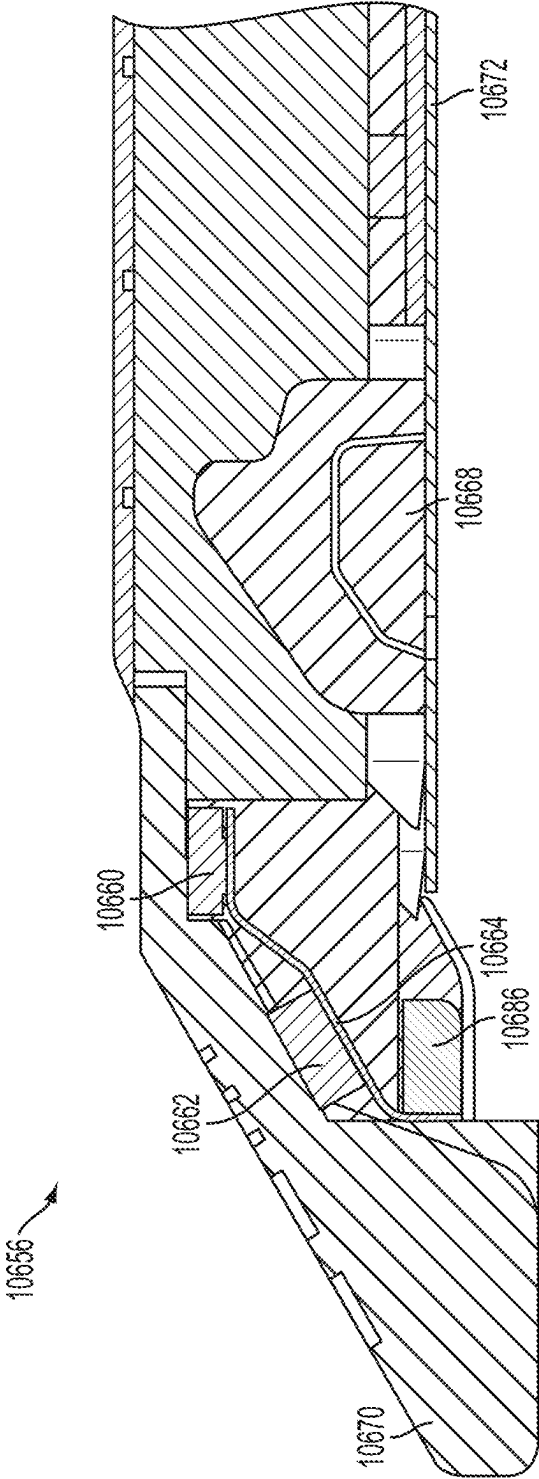


FIG. 144D

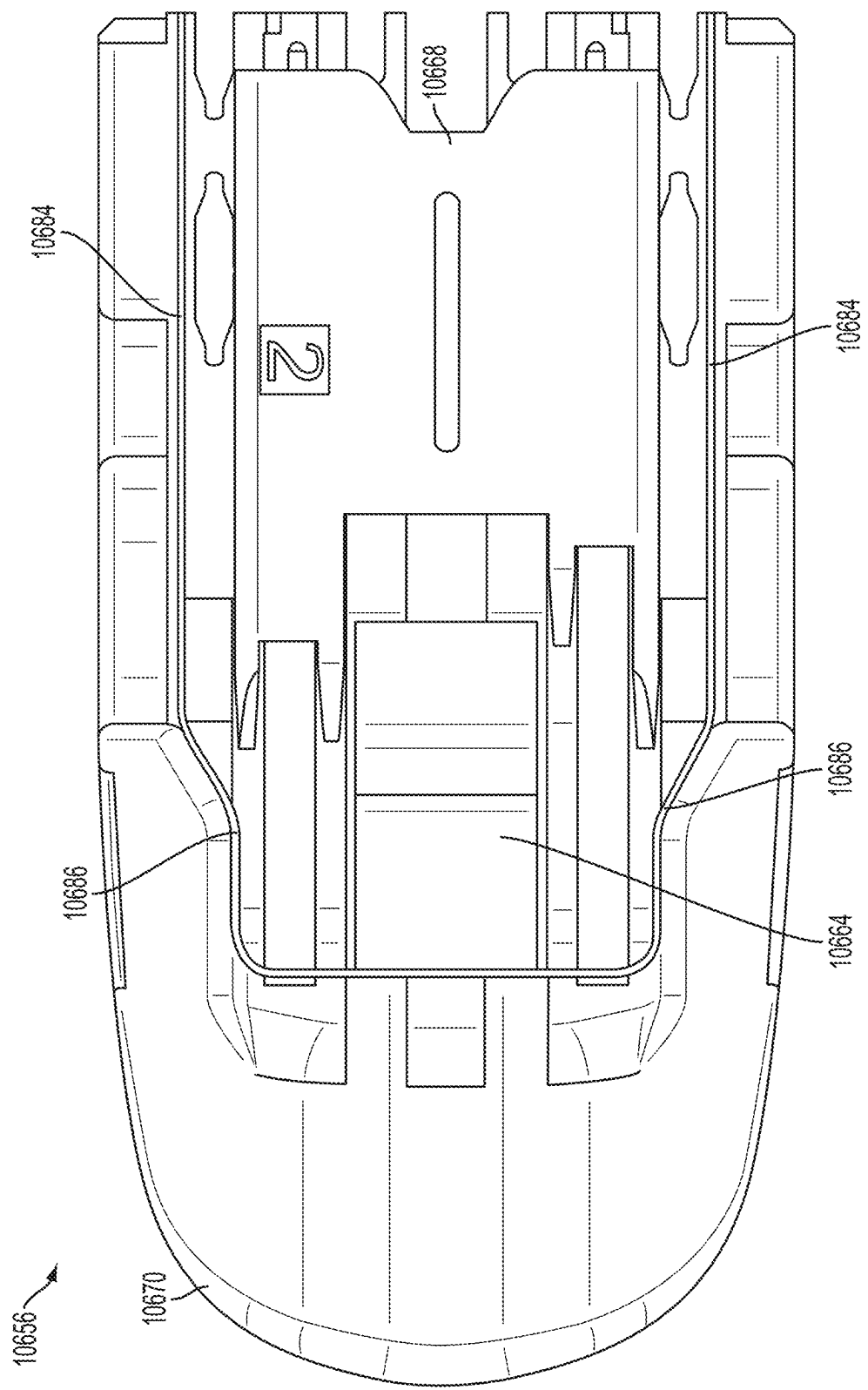
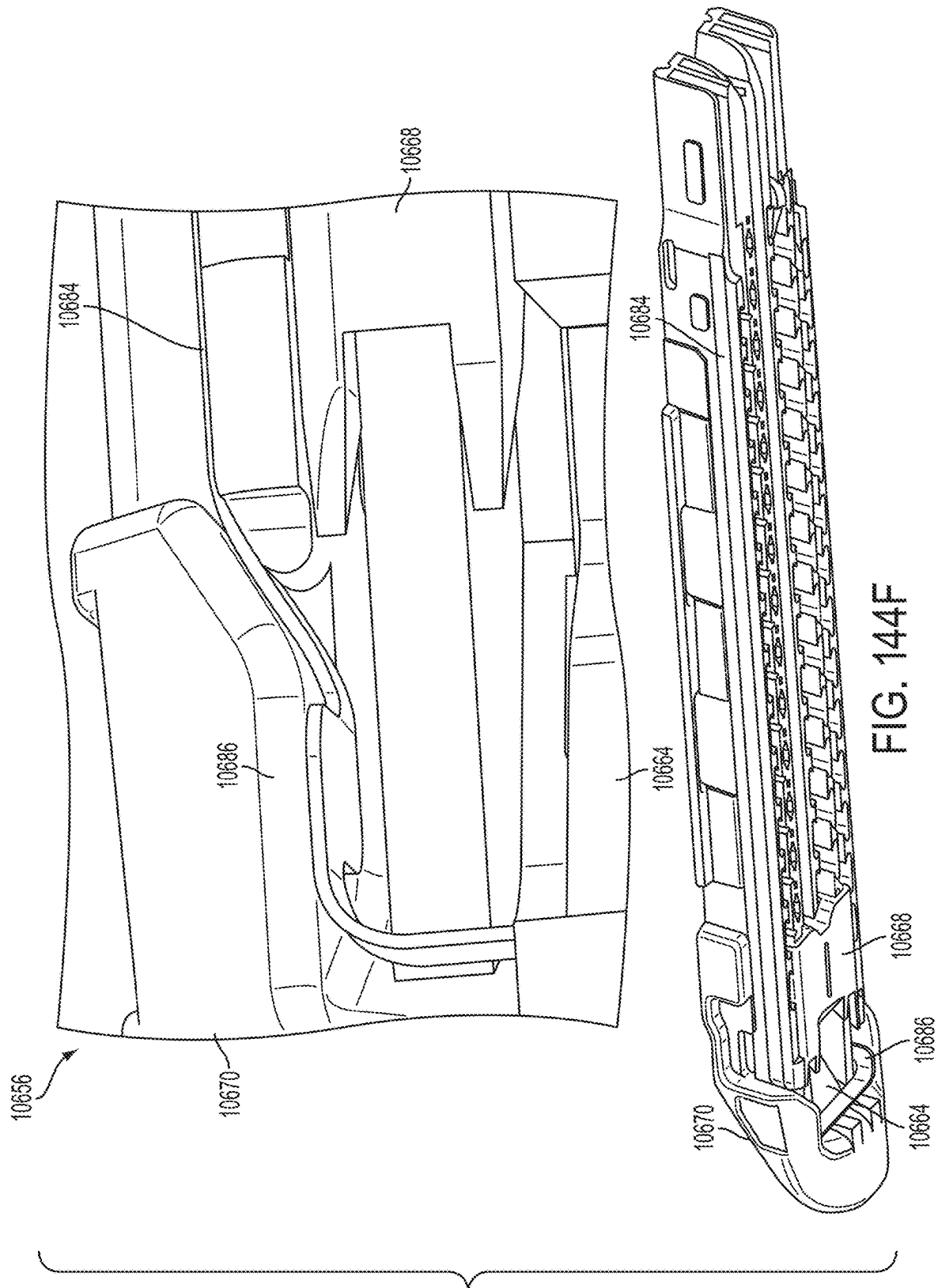
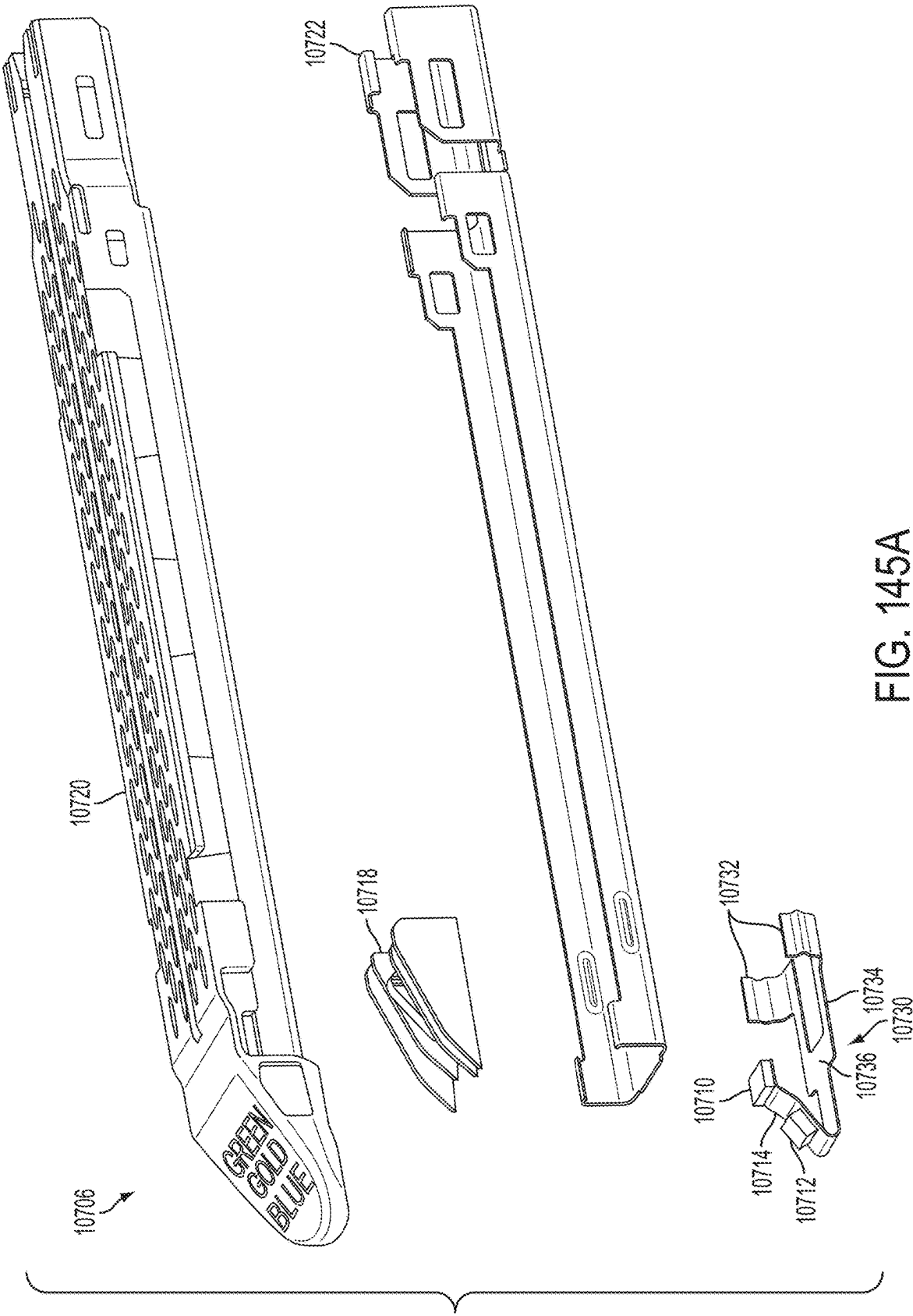


FIG. 144E





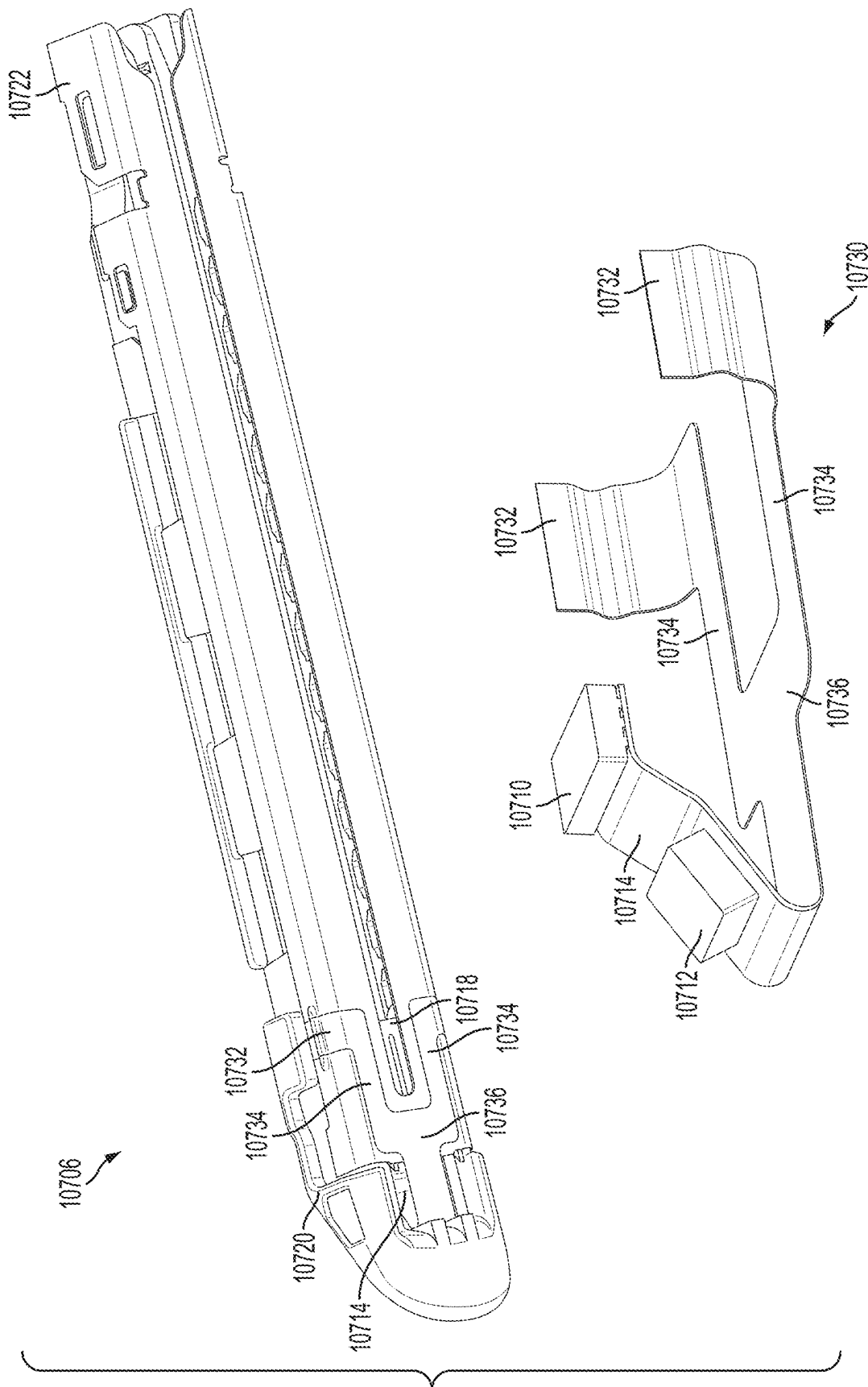


FIG. 145B

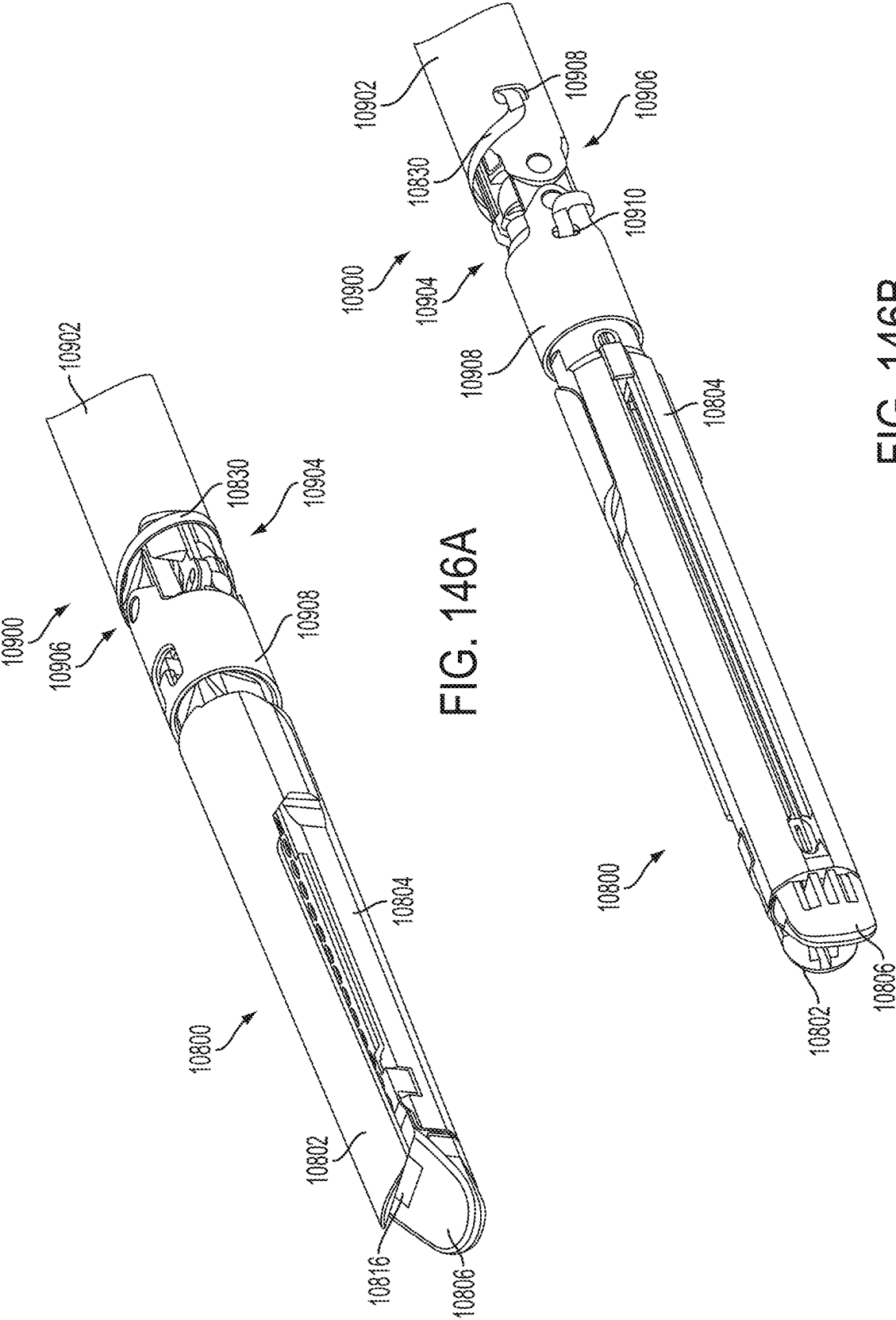


FIG. 146A

FIG. 146B

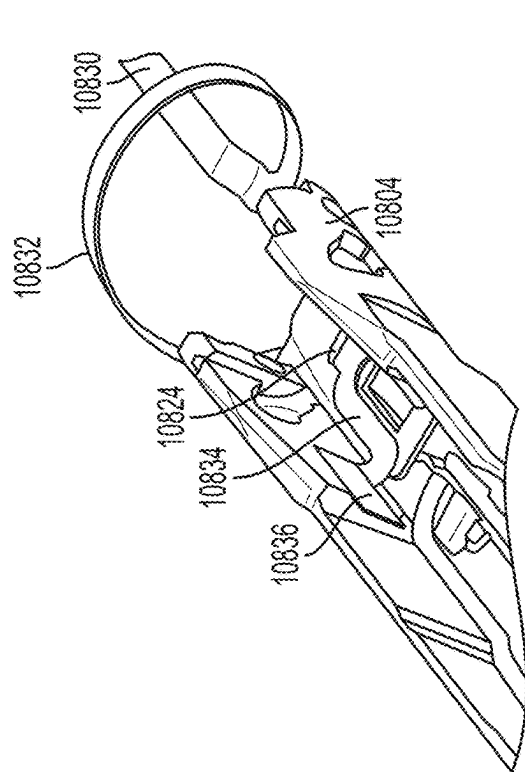
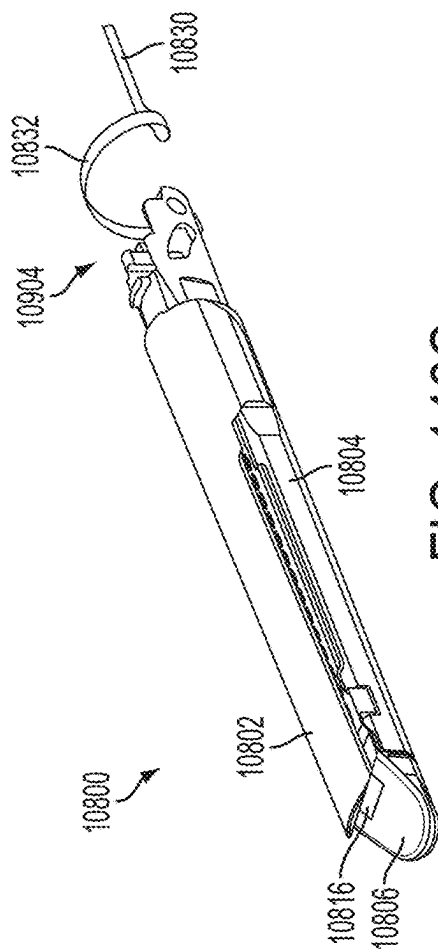


FIG. 146D

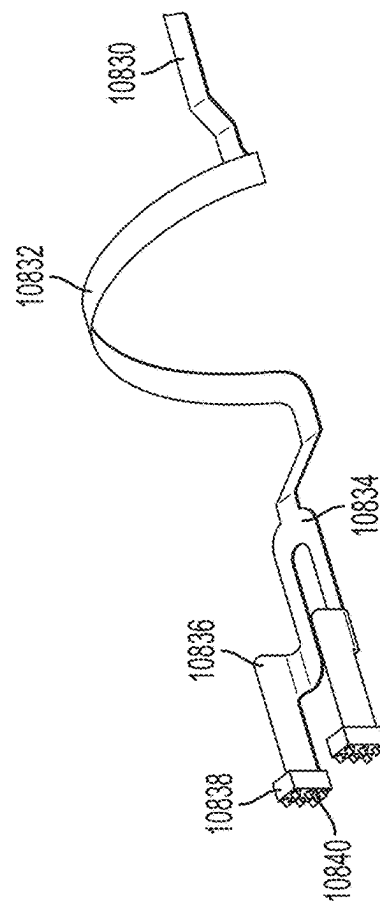
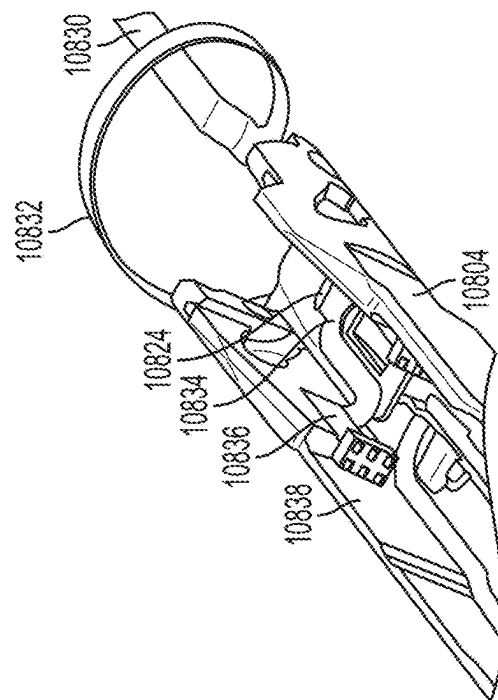


FIG. 146F

FIG. 146E

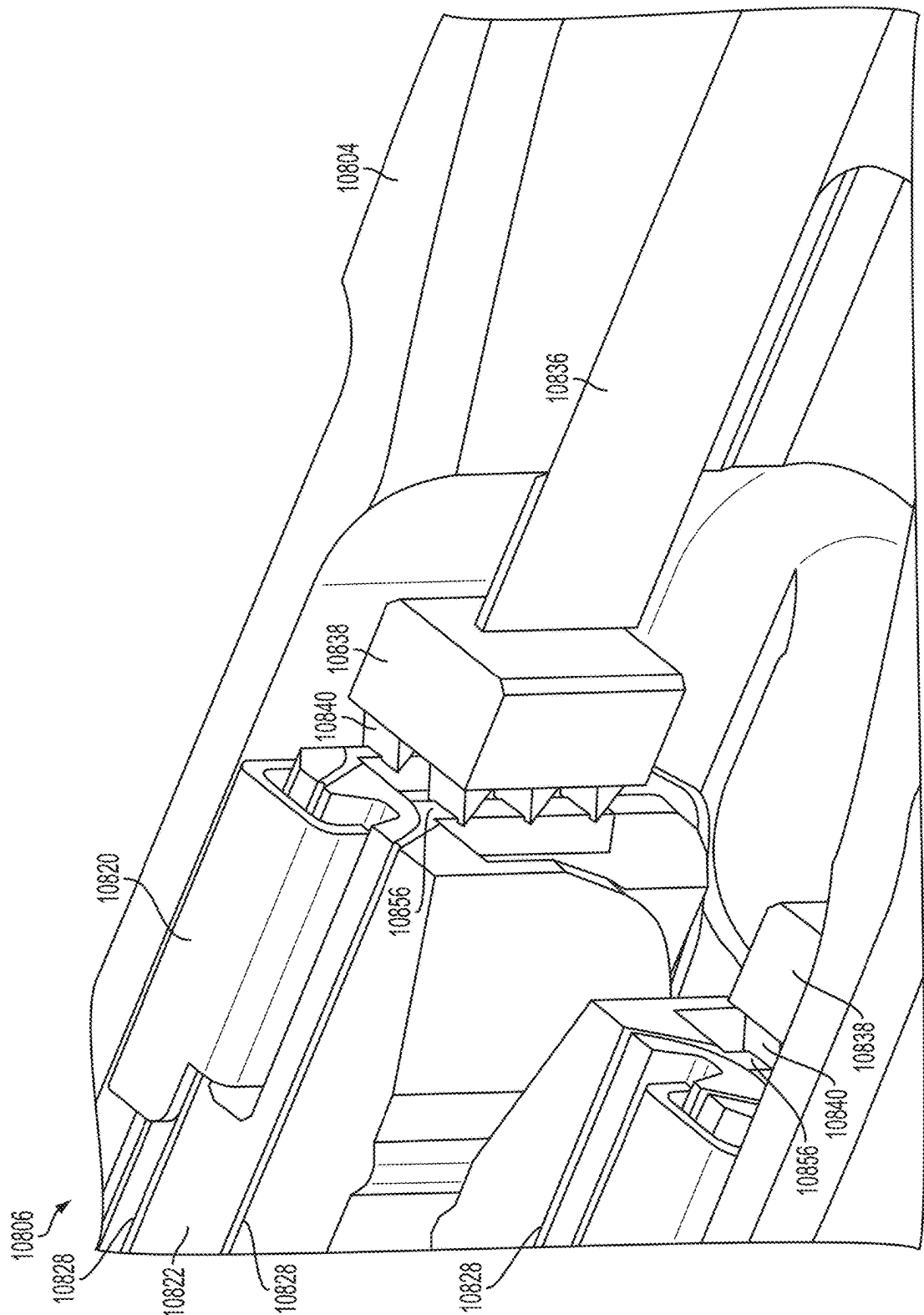
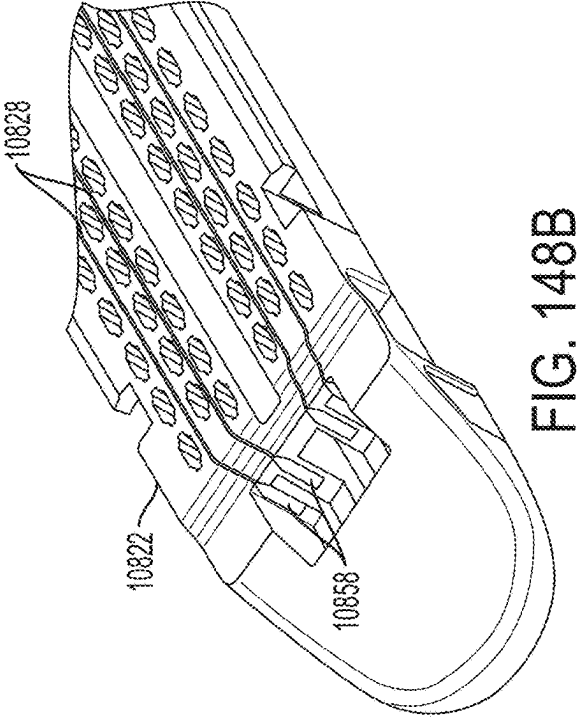
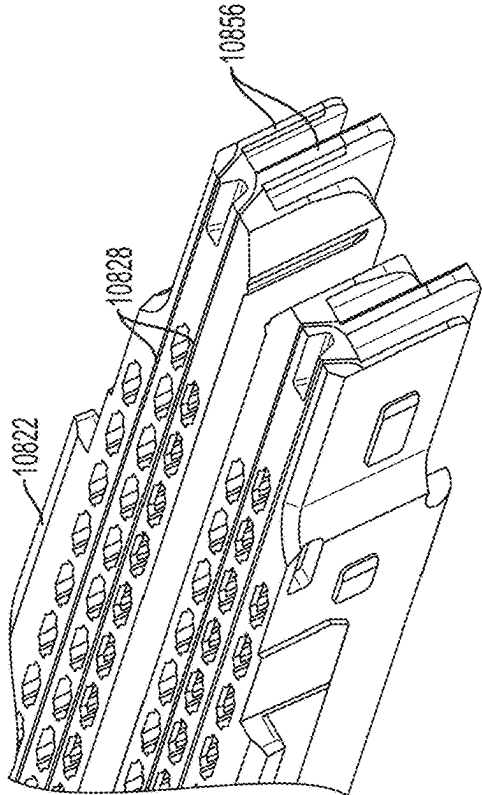
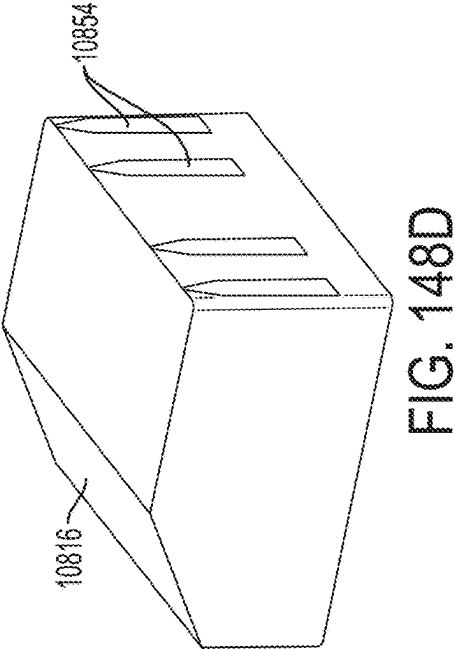
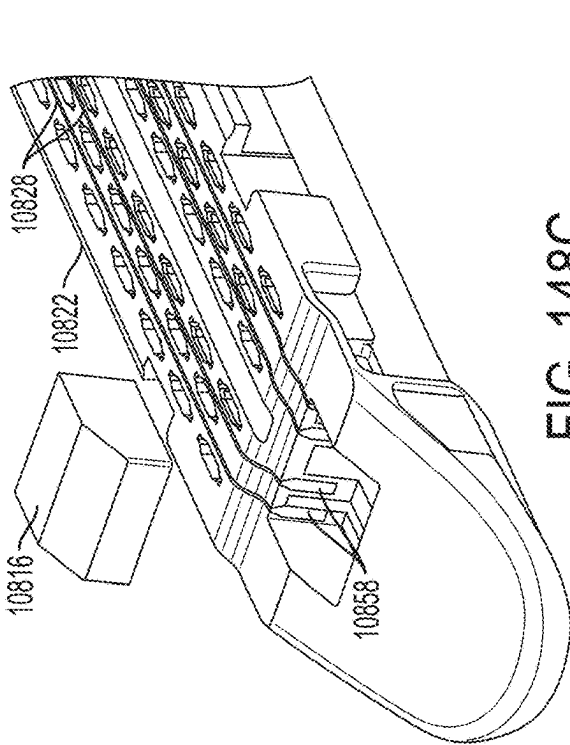


FIG. 147



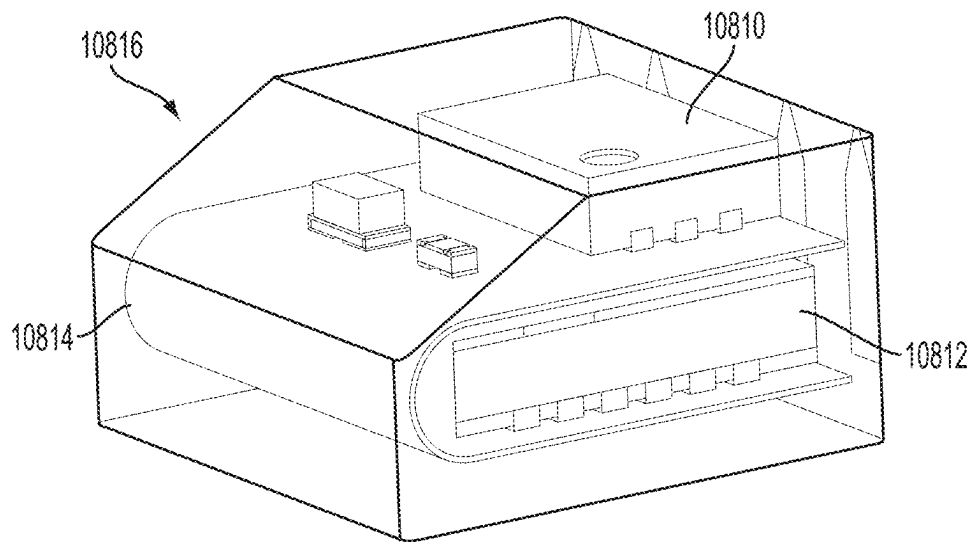


FIG. 149A

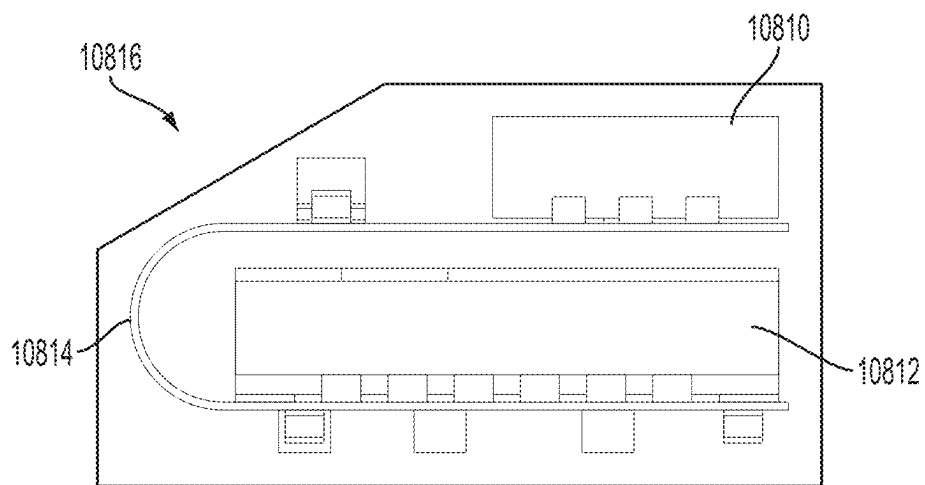


FIG. 149B

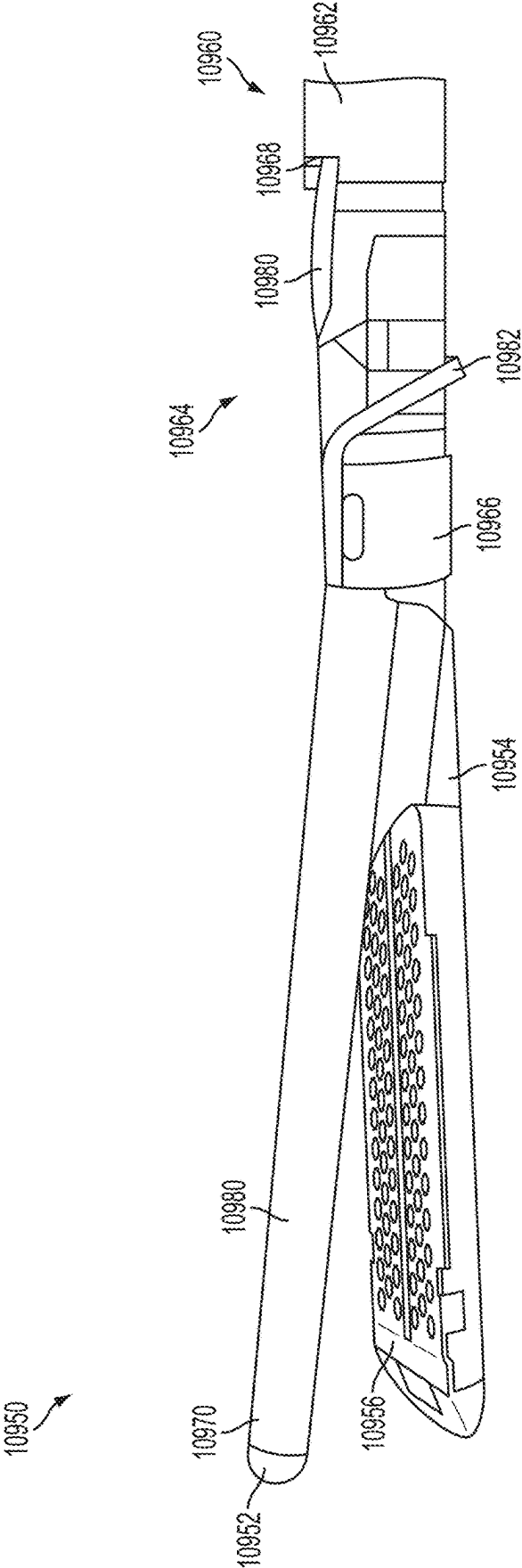


FIG. 150

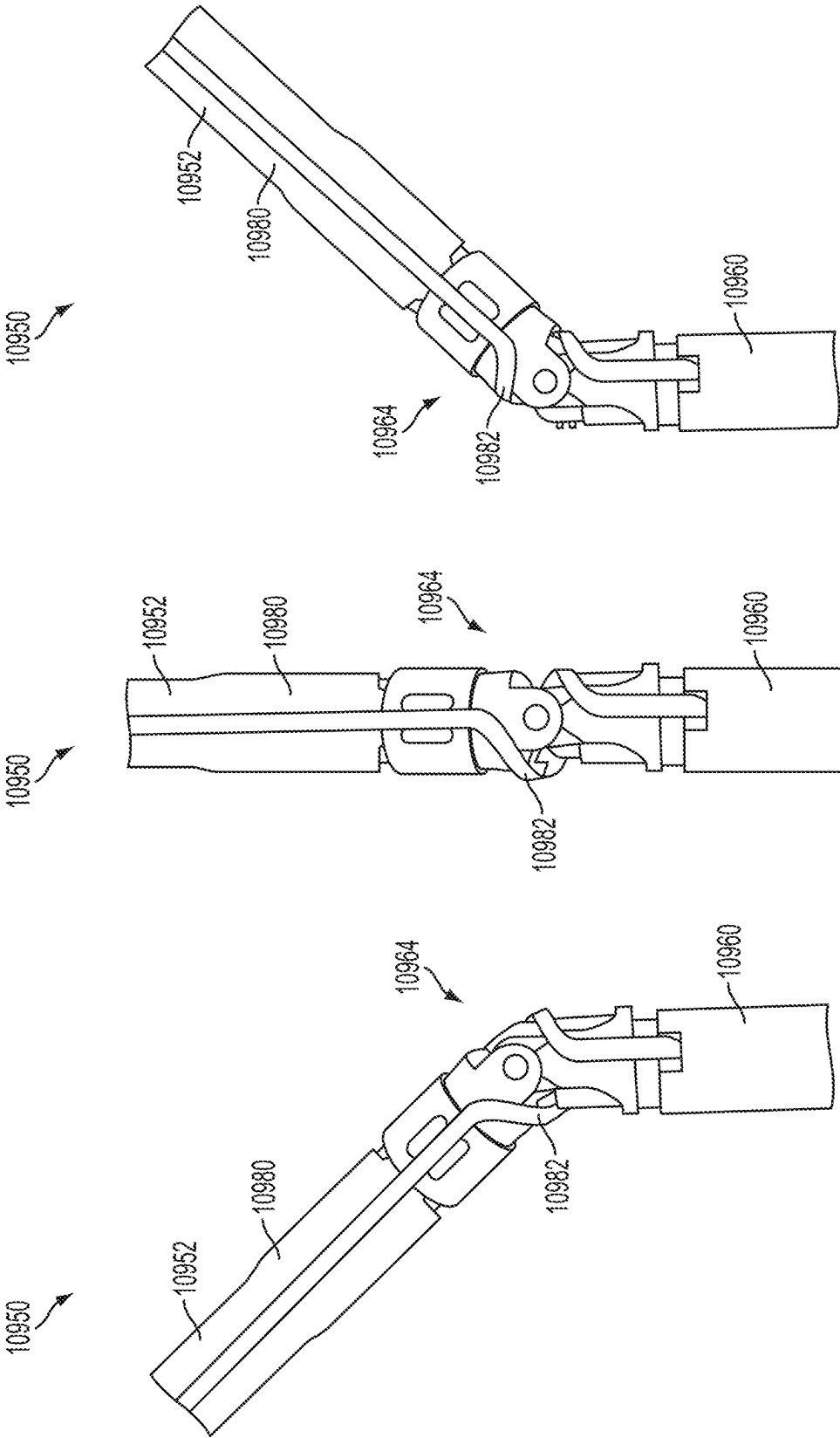


FIG. 151C

FIG. 151B

FIG. 151A

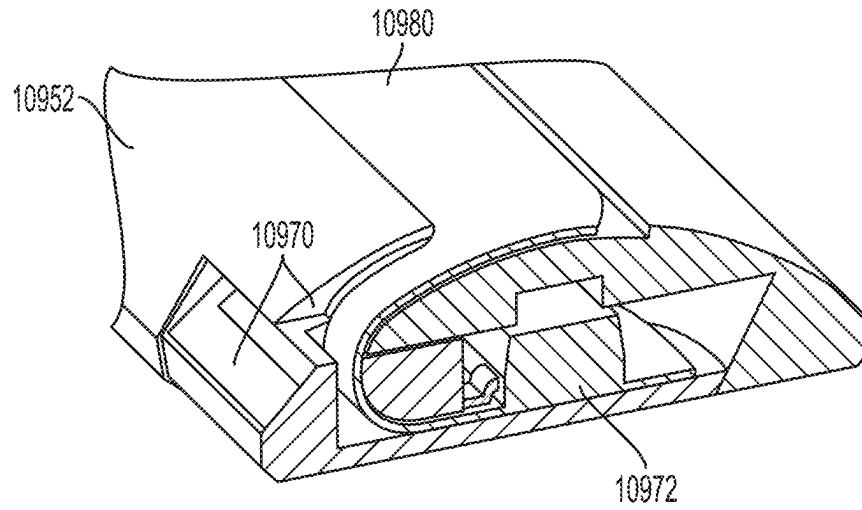


FIG. 152

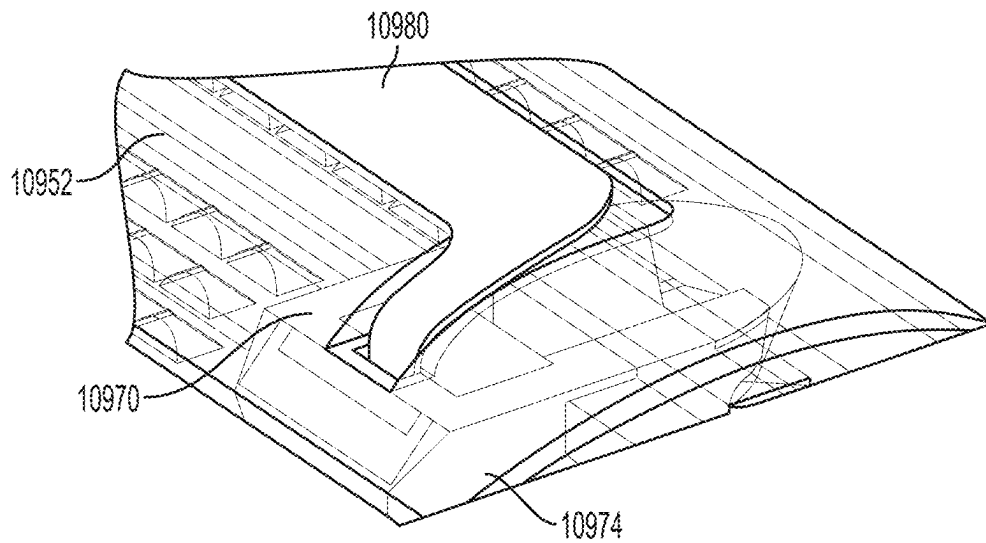


FIG. 153

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STAPLE CARTRIDGE ELECTRICAL CONTACTS**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/580,493, entitled LOCAL DISPLAY OF TISSUE PARAMETER STABILIZATION, filed Sep. 24, 2019, which issued on May 23, 2023 as U.S. Pat. No. 11,653,918, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 15/459,531, entitled POWERED MEDICAL DEVICE INCLUDING MEASUREMENT OF CLOSURE STATE OF JAWS, filed Mar. 15, 2017, which issued on Apr. 26, 2022 as U.S. Pat. No. 11,311,294, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 14/479,108, entitled LOCAL DISPLAY OF TISSUE PARAMETER STABILIZATION, filed Sep. 5, 2014, now U.S. Patent Application Publication No. 2016/0066913, the entire disclosures of which are hereby incorporated by reference herein.

This application is also related to U.S. patent application Ser. No. 14/479,103 entitled CIRCUITRY AND SENSORS FOR POWERED MEDICAL DEVICE, now U.S. Pat. No. 10,111,679, 14/479,119 entitled ADJUNCT WITH INTEGRATED SENSORS TO QUANTIFY TISSUE COMPRESSION, now U.S. Pat. No. 9,724,094, Ser. No. 14/478,908 entitled MONITORING DEVICE DEGRADATION BASED ON COMPONENT EVALUATION, now U.S. Pat. No. 9,737,301, Ser. No. 14/478,895 entitled MULTIPLE SENSORS WITH ONE SENSOR AFFECTING A SECOND SENSOR'S OUTPUT OR INTERPRETATION, now U.S. Pat. No. 9,757,128, Ser. No. 14/479,110 entitled POLARITY OF HALL MAGNET TO DETECT MIS-LOADED CARTRIDGE, now U.S. patent Ser. No. 14/479,098 entitled SMART CARTRIDGE WAKE UP OPERATION AND DATA RETENTION, now U.S. Pat. No. 10,135,242, and Ser. No. 14/479,115 entitled MULTIPLE MOTOR CONTROL FOR POWERED MEDICAL DEVICE, now U.S. Pat. No. 9,788,836, each of which is incorporated herein by reference in its entirety.

BACKGROUND

The present embodiments of the invention relate to surgical instruments and, in various circumstances, to surgical stapling and cutting instruments and staple cartridges therefor that are designed to staple and cut tissue.

SUMMARY

In one embodiment, a staple cartridge for use with a surgical stapler is provided. The staple cartridge comprises a cartridge body having a tissue-contacting surface; one or more LEDs positioned at the edges of the tissue-contacting surface; and a plurality of staple drivers within the cartridge body each supporting a staple. In one embodiment the one or more LEDs emit ultraviolet light or infrared light. In one embodiment, the staples are coated in a fluorescing dye.

In one embodiment, a surgical stapling system is provided. The surgical stapling system comprises an elongated shaft assembly configured to transmit actuation motions from an actuator; and an end effector for compressing and stapling tissue, the end effector operably coupled to the elongated shaft, the end effector comprising: an elongated

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channel; an anvil having a staple forming surface thereon, the anvil moveable relative to the elongated channel between an open position and a closed position; and a staple cartridge removably positioned within the elongated channel, the staple cartridge comprising: a cartridge body having a tissue-contacting surface in a confronting relationship with the anvil's staple forming surface when the anvil is in the closed position; one or more LEDs positioned to be visible when the anvil is in the closed position; and a plurality of staple drivers within the cartridge body each supporting a staple. In one embodiment, the one or more LEDs indicate that the compressed tissue is stable. In one embodiment, at least one LED is visible from either the left or the right side of the end effector, the at least one LED configured to flash at the rate of the tissue's stabilization and further configured to maintain a lit condition when the tissue is stable. In one embodiment, more than one LED is visible from either the left or the right side of the end effector. In one embodiment, the more than one LEDs light in sequence at the rate of the tissue's stabilization, such that when all the LEDs indicates that the tissue is stable. In one embodiment, the more than one LEDs light in sequence, the speed of the sequence indicating the rate of the tissue's stabilization, and wherein all the LEDs flash simultaneously when the tissue is stable. In one embodiment, the one or more LEDs indicate which portions of the end effector are in sufficient contact with the tissue.

In one embodiment, the system further comprises a processor, the processor configured to compare the tissue's parameters against the acceptable parameters for all staple cartridges. In one embodiment, the processor is further configured to compare the tissue's parameters against the acceptable parameters for the staple cartridge currently present in the elongated channel. In one embodiment, the LEDs indicate that the staple cartridge is appropriate for the tissue. In one embodiment, the one or more LEDs indicate that the staple cartridge is not compatible with the stapling system. In one embodiment, the one or more LEDs indicate that the end effector is enclosing more tissue than is suitable for the staple cartridge. In one embodiment, a surgical stapling system is provided. The surgical stapling system comprises: an elongated shaft assembly configured to transmit actuation motions from an actuator; and an end effector for compressing and stapling tissue, the end effector operably coupled to the elongated shaft, the end effector comprising: an elongated channel; an anvil having a staple forming surface thereon, the anvil moveable relative to the elongated channel between an open position and a closed position; and a staple cartridge removably positioned within the elongated channel, the staple cartridge comprising: a cartridge body having an upper surface in a confronting relationship with the anvil's staple forming surface when the anvil is in a closed position; one or more LEDs positioned to provide illumination to the area between the anvil and the staple cartridge when the anvil is in a closed position; and a plurality of staple drivers within the cartridge body each supporting a staple. In one embodiment, the one or more LEDs emit ultraviolet light or infrared light. In one embodiment, the staples are coated in a fluorescing dye.

BRIEF DESCRIPTION OF THE DRAWINGS

The features and advantages of the various embodiments of the invention, and the manner of attaining them, will become more apparent and the embodiment of the invention itself will be better understood by reference to the following

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description of embodiments of the embodiment of the invention taken in conjunction with the accompanying drawings, wherein:

FIG. 1 is a perspective view of a surgical instrument that has an interchangeable shaft assembly operably coupled thereto;

FIG. 2 is an exploded assembly view of the interchangeable shaft assembly and surgical instrument of FIG. 1;

FIG. 3 is another exploded assembly view showing portions of the interchangeable shaft assembly and surgical instrument of FIGS. 1 and 2;

FIG. 4 is an exploded assembly view of a portion of the surgical instrument of FIGS. 1-3;

FIG. 5 is a cross-sectional side view of a portion of the surgical instrument of FIG. 4 with the firing trigger in a fully actuated position;

FIG. 6 is another cross-sectional view of a portion of the surgical instrument of FIG. 5 with the firing trigger in an unactuated position;

FIG. 7 is an exploded assembly view of one form of an interchangeable shaft assembly;

FIG. 8 is another exploded assembly view of portions of the interchangeable shaft assembly of FIG. 7;

FIG. 9 is another exploded assembly view of portions of the interchangeable shaft assembly of FIGS. 7 and 8;

FIG. 10 is a cross-sectional view of a portion of the interchangeable shaft assembly of FIGS. 7-9;

FIG. 11 is a perspective view of a portion of the shaft assembly of FIGS. 7-10 with the switch drum omitted for clarity;

FIG. 12 is another perspective view of the portion of the interchangeable shaft assembly of FIG. 11 with the switch drum mounted thereon;

FIG. 13 is a perspective view of a portion of the interchangeable shaft assembly of FIG. 11 operably coupled to a portion of the surgical instrument of FIG. 1 illustrated with the closure trigger thereof in an unactuated position;

FIG. 14 is a right side elevational view of the interchangeable shaft assembly and surgical instrument of FIG. 13;

FIG. 15 is a left side elevational view of the interchangeable shaft assembly and surgical instrument of FIGS. 13 and 14;

FIG. 16 is a perspective view of a portion of the interchangeable shaft assembly of FIG. 11 operably coupled to a portion of the surgical instrument of FIG. 1 illustrated with the closure trigger thereof in an actuated position and a firing trigger thereof in an unactuated position;

FIG. 17 is a right side elevational view of the interchangeable shaft assembly and surgical instrument of FIG. 16;

FIG. 18 is a left side elevational view of the interchangeable shaft assembly and surgical instrument of FIGS. 16 and 17;

FIG. 18A is a right side elevational view of the interchangeable shaft assembly of FIG. 11 operably coupled to a portion of the surgical instrument of FIG. 1 illustrated with the closure trigger thereof in an actuated position and the firing trigger thereof in an actuated position;

FIG. 19 is a schematic of a system for powering down an electrical connector of a surgical instrument handle when a shaft assembly is not coupled thereto;

FIG. 20 is an exploded view of one embodiment of an end effector of the surgical instrument of FIG. 1;

FIGS. 21A-21B is a circuit diagram of the surgical instrument of FIG. 1 spanning two drawings sheets;

FIG. 22 illustrates one instance of a power assembly comprising a usage cycle circuit configured to generate a usage cycle count of the battery back;

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FIG. 23 illustrates one embodiment of a process for sequentially energizing a segmented circuit;

FIG. 24 illustrates one embodiment of a power segment comprising a plurality of daisy chained power converters;

FIG. 25 illustrates one embodiment of a segmented circuit configured to maximize power available for critical and/or power intense functions;

FIG. 26 illustrates one embodiment of a power system comprising a plurality of daisy chained power converters configured to be sequentially energized;

FIG. 27 illustrates one embodiment of a segmented circuit comprising an isolated control section;

FIG. 28 illustrates one embodiment of an end effector comprising a first sensor and a second sensor;

FIG. 29 is a logic diagram illustrating one embodiment of a process for adjusting the measurement of the first sensor based on input from the second sensor of the end effector illustrated in FIG. 28;

FIG. 30 is a logic diagram illustrating one embodiment of a process for determining a look-up table for a first sensor based on the input from a second sensor;

FIG. 31 is a logic diagram illustrating one embodiment of a process for calibrating a first sensor in response to an input from a second sensor;

FIG. 32A is a logic diagram illustrating one embodiment of a process for determining and displaying the thickness of a tissue section clamped between an anvil and a staple cartridge of an end effector;

FIG. 32B is a logic diagram illustrating one embodiment of a process for determining and displaying the thickness of a tissue section clamped between the anvil and the staple cartridge of the end effector;

FIG. 33 is a graph illustrating an adjusted Hall effect thickness measurement compared to an unmodified Hall effect thickness measurement;

FIG. 34 illustrates one embodiment of an end effector comprising a first sensor and a second sensor;

FIG. 35 illustrates one embodiment of an end effector comprising a first sensor and a plurality of second sensors;

FIG. 36 is a logic diagram illustrating one embodiment of a process for adjusting a measurement of a first sensor in response to a plurality of secondary sensors;

FIG. 37 illustrates one embodiment of a circuit configured to convert signals from a first sensor and a plurality of secondary sensors into digital signals receivable by a processor;

FIG. 38 illustrates one embodiment of an end effector comprising a plurality of sensors;

FIG. 39 is a logic diagram illustrating one embodiment of a process for determining one or more tissue properties based on a plurality of sensors;

FIG. 40 illustrates one embodiment of an end effector comprising a plurality of sensors coupled to a second jaw member;

FIG. 41 illustrates one embodiment of a staple cartridge comprising a plurality of sensors formed integrally therein;

FIG. 42 is a logic diagram illustrating one embodiment of a process for determining one or more parameters of a tissue section clamped within an end effector;

FIG. 43 illustrates one embodiment of an end effector comprising a plurality of redundant sensors;

FIG. 44 is a logic diagram illustrating one embodiment of a process for selecting the most reliable output from a plurality of redundant sensors;

FIG. 45 illustrates one embodiment of an end effector comprising a sensor comprising a specific sampling rate to limit or eliminate false signals;

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FIG. 46 is a logic diagram illustrating one embodiment of a process for generating a thickness measurement for a tissue section located between an anvil and a staple cartridge of an end effector;

FIG. 47 illustrates one embodiment of a circular stapler;

FIGS. 48A-48D illustrate a clamping process of the circular stapler illustrated in FIG. 47, where FIG. 48A illustrates the circular stapler in an initial position with the anvil and the body in a closed configuration, FIG. 48B illustrates that the anvil is moved distally to disengage with the body and create a gap configured to receive a tissue section therein, once the circular stapler 3400 is positioned, FIG. 48C illustrates the tissue section compressed to a predetermined compression between the anvil and the body, and FIG. 48D illustrates the circular stapler in position corresponding to staple deployment;

FIG. 49 illustrates one embodiment of a circular staple anvil and an electrical connector configured to interface therewith;

FIG. 50 illustrates one embodiment of a surgical instrument comprising a sensor coupled to a drive shaft of the surgical instrument;

FIG. 51 is a flow chart illustrating one embodiment of a process for determining uneven tissue loading in an end effector;

FIG. 52 illustrates one embodiment of an end effector configured to determine one or more parameters of a tissue section during a clamping operation;

FIGS. 53A and 53B illustrate an embodiment of an end effector configured to normalize a Hall effect voltage irrespective of a deck height of a staple cartridge;

FIG. 54 is a logic diagram illustrating one embodiment of a process for determining when the compression of tissue within an end effector, such as, for example, the end effector illustrated in FIGS. 53A-53B, has reached a steady state;

FIG. 55 is a graph illustrating various Hall effect sensor readings;

FIG. 56 is a logic diagram illustrating one embodiment of a process for determining when the compression of tissue within an end effector, such as, for example, the end effector illustrated in FIGS. 53A-53B, has reached a steady state;

FIG. 57 is a logic diagram illustrating one embodiment of a process for controlling an end effector to improve proper staple formation during deployment;

FIG. 58 is a logic diagram illustrating one embodiment of a process for controlling an end effector to allow for fluid evacuation and provide improved staple formation;

FIGS. 59A-59B illustrate one embodiment of an end effector comprising a pressure sensor;

FIG. 60 illustrates one embodiment of an end effector comprising a second sensor located between a staple cartridge and a second jaw member;

FIG. 61 is a logic diagram illustrating one embodiment of a process for determining and displaying the thickness of a tissue section clamped in an end effector, according to FIGS. 59A-59B or FIG. 60;

FIG. 62 illustrates one embodiment of an end effector comprising a plurality of second sensors located between a staple cartridge and an elongated channel;

FIGS. 63A and 63B further illustrate the effect of a full versus partial bite of tissue;

FIG. 64 illustrates one embodiment of an end effector comprising a coil and oscillator circuit for measuring the gap between the anvil and the staple cartridge;

FIG. 65 illustrates and alternate view of the end effector. As illustrated, in some embodiments external wiring may supply power to the oscillator circuit;

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FIG. 66 illustrates examples of the operation of a coil to detect eddy currents in a target;

FIG. 67 illustrates a graph of a measured quality factor, the measured inductance, and measure resistance of the radius of a coil as a function of the coil's standoff to a target;

FIG. 68 illustrates one embodiment of an end effector comprising an emitter and sensor placed between the staple cartridge and the elongated channel;

FIG. 69 illustrates an embodiment of an emitter and sensor in operation;

FIG. 70 illustrates the surface of an embodiment of an emitter and sensor comprising a MEMS transducer;

FIG. 71 illustrates a graph of an example of the reflected signal that may be measured by the emitter and sensor of FIG. 69;

FIG. 72 illustrates an embodiment of an end effector that is configured to determine the location of a cutting member or knife;

FIG. 73 illustrates an example of the code strip in operation with red LEDs and an infrared LEDs;

FIG. 74 illustrates a partial perspective view of an end effector of a surgical instrument comprising a staple cartridge according to various embodiments described herein;

FIG. 75 illustrates a elevational view of a portion of the end effector of FIG. 74 according to various embodiments described herein;

FIG. 76 illustrates a logic diagram of a module of the surgical instrument of FIG. 74 according to various embodiments described herein;

FIG. 77 illustrates a partial view of a cutting edge, an optical sensor, and a light source of the surgical instrument of FIG. 74 according to various embodiments described herein;

FIG. 78 illustrates a partial view of a cutting edge, an optical sensor, and a light source of the surgical instrument of FIG. 74 according to various embodiments described herein;

FIG. 79 illustrates a partial view of a cutting edge, an optical sensor, and a light source of the surgical instrument of FIG. 74 according to various embodiments described herein;

FIG. 80 illustrates a partial view of a cutting edge, optical sensors, and light sources of the surgical instrument of FIG. 74 according to various embodiments described herein;

FIG. 81 illustrates a partial view of a cutting edge, an optical sensor, and a light source of the surgical instrument of FIG. 74 according to various embodiments described herein;

FIG. 82 illustrates a partial view of a cutting edge between cleaning blades of the surgical instrument of FIG. 74 according to various embodiments described herein;

FIG. 83 illustrates a partial view of a cutting edge between cleaning sponges of the surgical instrument of FIG. 74 according to various embodiments described herein;

FIG. 84 illustrates a perspective view of a staple cartridge including a sharpness testing member according to various embodiments described herein;

FIG. 85 illustrates a logic diagram of a module of a surgical instrument according to various embodiments described herein;

FIG. 86 illustrates a logic diagram of a module of a surgical instrument according to various embodiments described herein;

FIG. 87 illustrates a logic diagram outlining a method for evaluating sharpness of a cutting edge of a surgical instrument according to various embodiments described herein;

FIG. 88 illustrates a chart of the forces applied against a cutting edge of a surgical instrument by the sharpness testing member of FIG. 84 at various sharpness levels according to various embodiments described herein;

FIG. 89 illustrates a flow chart outlining a method for determining whether a cutting edge of a surgical instrument is sufficiently sharp to transect tissue captured by the surgical instrument according to various embodiments described herein; and

FIG. 90 illustrates a table showing predefined tissue thicknesses and corresponding predefined threshold forces according to various embodiments described herein.

FIG. 91 illustrates a perspective view of a surgical instrument including a handle, a shaft assembly, and an end effector;

FIG. 92 illustrates a logic diagram of a common control module for use with a plurality of motors of the surgical instrument of FIG. 91;

FIG. 93 illustrates a partial elevational view of the handle of the surgical instrument of FIG. 91 with a removed outer casing;

FIG. 94 illustrates a partial elevational view of the surgical instrument of FIG. 91 with a removed outer casing.

FIG. 95A illustrates a side angle view of an end effector with the anvil in a closed position, illustrating one located on either side of the cartridge deck;

FIG. 95B illustrates a three-quarter angle view of the end effector with the anvil in an open position, and one LED located on either side of the cartridge deck;

FIG. 96A illustrates a side angle view of an end effector with the anvil in a closed position and a plurality of LEDs located on either side of the cartridge deck;

FIG. 96B illustrates a three-quarter angle view of the end effector with the anvil in an open position, and a plurality of LEDs located on either side of the cartridge deck;

FIG. 97A illustrates a side angle view of an end effector with the anvil in a closed position, and a plurality of LEDs from the proximal to the distal end of the staple cartridge, on either side of the cartridge deck; and

FIG. 97B illustrates a three-quarter angle view of the end effector with the anvil in an open position, illustrating a plurality of LEDs from the proximal to the distal end of the staple cartridge, and on either side of the cartridge deck.

FIG. 98A illustrates an embodiment wherein the tissue compensator is removably attached to the anvil portion of the end effector;

FIG. 98B illustrates a detail view of a portion of the tissue compensator shown in FIG. 98A;

FIG. 99 illustrates various example embodiments that use the layer of conductive elements and conductive elements in the staple cartridge to detect the distance between the anvil and the upper surface of the staple cartridge;

FIGS. 100A and 100B illustrate an embodiment of the tissue compensator comprising a layer of conductive elements in operation;

FIGS. 101A and 101B illustrate an embodiment of an end effector comprising a tissue compensator further comprising conductors embedded within;

FIGS. 102A and 102B illustrate an embodiment of an end effector comprising a tissue compensator further comprising conductors embedded therein;

FIG. 103 illustrates an embodiment of a staple cartridge and a tissue compensator wherein the staple cartridge provides power to the conductive elements that comprise the tissue compensator;

FIGS. 104A and 104B illustrate an embodiment of a staple cartridge and a tissue compensator wherein the staple

cartridge provides power to the conductive elements that comprise the tissue compensator;

FIGS. 105A and 105B illustrate an embodiment of an end effector comprising position sensing elements and a tissue compensator;

FIGS. 106A and 106B illustrate an embodiment of an end effector comprising position sensing elements and a tissue compensator;

FIGS. 107A and 107B illustrate an embodiment of a staple cartridge and a tissue compensator that is operable to indicate the position of a cutting member or knife bar;

FIG. 108 illustrates one embodiment of an end effector comprising a magnet and a Hall effect sensor wherein the detected magnetic field can be used to identify a staple cartridge;

FIG. 109 illustrates one embodiment of an end effector comprising a magnet and a Hall effect sensor wherein the detected magnetic field can be used to identify a staple cartridge;

FIG. 110 illustrates a graph of the voltage detected by a Hall effect sensor located in the distal tip of a staple cartridge, such as is illustrated in FIGS. 108 and 109, in response to the distance or gap between a magnet located in the anvil and the Hall effect sensor in the staple cartridge, such as illustrated in FIGS. 108 and 109;

FIG. 111 illustrates one embodiment of the housing of the surgical instrument, comprising a display;

FIG. 112 illustrates one embodiment of a staple retainer comprising a magnet;

FIGS. 113A and 113B illustrate one embodiment of an end effector comprising a sensor for identifying staple cartridges of different types;

FIG. 114 is a partial view of an end effector with sensor power conductors for transferring power and data signals between the connected components of the surgical instrument according to one embodiment.

FIG. 115 is a partial view of the end effector shown in FIG. 114 showing sensors and/or electronic components located in an end effector.

FIG. 116 is a block diagram of a surgical instrument electronic subsystem comprising a short circuit protection circuit for the sensors and/or electronic components according to one embodiment.

FIG. 117 is a short circuit protection circuit comprising a supplementary power supply circuit 7014 coupled to a main power supply circuit, according to one embodiment.

FIG. 118 is a block diagram of a surgical instrument electronic subsystem comprising a sample rate monitor to provide power reduction by limiting sample rates and/or duty cycle of the sensor components when the surgical instrument is in a non-sensing state, according to one embodiment.

FIG. 119 is a block diagram of a surgical instrument electronic subsystem comprising an over current/voltage protection circuit for sensors and/or electronic components of a surgical instrument, according to one embodiment.

FIG. 120 is an over current/voltage protection circuit for sensors and electronic components for a surgical instrument, according to one embodiment.

FIG. 121 is a block diagram of a surgical instrument electronic subsystem with a reverse polarity protection circuit for sensors and/or electronic components according to one embodiment.

FIG. 122 is a reverse polarity protection circuit for sensors and/or electronic components for a surgical instrument according to one embodiment.

FIG. 123 is a block diagram of a surgical instrument electronic subsystem with power reduction utilizing a sleep mode monitor for sensors and/or electronic components according to one embodiment.

FIG. 124 is a block diagram of a surgical instrument electronic subsystem comprising a temporary power loss circuit to provide protection against intermittent power loss for sensors and/or electronic components in modular surgical instruments.

FIG. 125 illustrates one embodiment of a temporary power loss circuit implemented as a hardware circuit.

FIG. 126A illustrates a perspective view of one embodiment of an end effector comprising a magnet and a Hall effect sensor in communication with a processor;

FIG. 126B illustrates a sideways cross-sectional view of one embodiment of an end effector comprising a magnet and a Hall effect sensor in communication with processor;

FIG. 127 illustrates one embodiment of the operable dimensions that relate to the operation of the Hall effect sensor;

FIG. 128A illustrates an external side view of an embodiment of a staple cartridge;

FIG. 128B illustrates various dimensions possible between the lower surface of the push-off lug and the top of the Hall effect sensor;

FIG. 128C illustrates an external side view of an embodiment of a staple cartridge;

FIG. 128D illustrates various dimensions possible between the lower surface of the push-off lug and the upper surface of the staple cartridge above the Hall effect sensor;

FIG. 129A further illustrates a front-end cross-sectional view 10054 of the anvil 10002 and the central axis point of the anvil;

FIG. 129B is a cross sectional view of a magnet shown in FIG. 129A;

FIGS. 130A-130E illustrate one embodiment of an end effector that comprises a magnet where FIG. 130A illustrates a front-end cross-sectional view of the end effector, FIG. 130B illustrates a front-end cutaway view of the anvil and the magnet in situ, FIG. 130C illustrates a perspective cutaway view of the anvil and the magnet, FIG. 130D illustrates a side cutaway view of the anvil and the magnet, and FIG. 130E illustrates a top cutaway view of the anvil and the magnet;

FIGS. 131A-131E illustrate another embodiment of an end effector that comprises a magnet where FIG. 131A illustrates a front-end cross-sectional view of the end effector, FIG. 131B illustrates a front-end cutaway view of the anvil and the magnet, in situ, FIG. 131C illustrates a perspective cutaway view of the anvil and the magnet, FIG. 131D illustrates a side cutaway view of the anvil and the magnet, and FIG. 131E illustrates a top cutaway view of the anvil and magnet;

FIG. 132 illustrates contact points between the anvil and either the staple cartridge and/or the elongated channel;

FIGS. 133A and 133B illustrate one embodiment of an end effector that is operable to use conductive surfaces at the distal contact point to create an electrical connection;

FIGS. 134A-134C illustrate one embodiment of an end effector that is operable to use conductive surfaces to form an electrical connection where FIG. 134A illustrates an end effector comprising an anvil, an elongated channel, and a staple cartridge, FIG. 134B illustrates the inside surface of the anvil further comprising first conductive surfaces located distally from the staple-forming indents, and FIG. 134C illustrates the staple cartridge comprising a cartridge body

and first conductive surfaces located such that they can come into contact with a second conductive surface located on the staple cartridge;

FIGS. 135A and 135B illustrate one embodiment of an end effector that is operable to use conductive surfaces to form an electrical connection where FIG. 135A illustrates an end effector comprising an anvil, an elongated channel, and a staple cartridge and FIGS. 109B 135B is a close-up view of the staple cartridge illustrating the first conductive surface located such that it can come into contact with second conductive surfaces;

FIGS. 136A and 136B illustrate one embodiment of an end effector that is operable to use conductive surfaces to form an electrical connection where FIG. 136A illustrates an end effector comprising an anvil, an elongated channel, and a staple cartridge and FIG. 136B is a close-up view of the staple cartridge illustrating the anvil further comprising a magnet and an inside surface, which further comprises a number of staple-forming indents;

FIGS. 137A-137C illustrate one embodiment of an end effector that is operable to use the proximal contact point to form an electrical connection where FIG. 137A illustrates the end effector, which comprises an anvil, an elongated channel, and a staple cartridge, FIG. 137B is a close-up view of a pin as it rests within an aperture defined in the elongated channel for that purpose, and FIG. 137C illustrates an alternate embodiment, with an alternate location for a second conductive surface on the surface of the aperture;

FIG. 138 illustrates one embodiment of an end effector with a distal sensor plug;

FIG. 139A illustrates the end effector shown in FIG. 138 with the anvil in an open position;

FIG. 139B illustrates a cross-sectional view of the end effector shown in FIG. 139A with the anvil in an open position;

FIG. 139C illustrates the end effector shown in FIG. 138 with the anvil in a closed position;

FIG. 139D illustrates a cross sectional view of the end effector shown in FIG. 139C with the anvil in a closed position;

FIG. 140 provides a close-up view of the cross section of the distal end of the end effector;

FIG. 141 illustrates a close-up top view of the staple cartridge that comprises a distal sensor plug;

FIG. 142A is a perspective view of the underside of a staple cartridge that comprises a distal sensor plug;

FIG. 142B illustrates a cross sectional view of the distal end of the staple cartridge;

FIGS. 143A-143C illustrate one embodiment of a staple cartridge that comprises a flex cable connected to a Hall effect sensor and processor where FIG. 143A is an exploded view of the staple cartridge, FIG. 143B illustrates the assembly of the staple cartridge and the flex cable in greater detail, and FIG. 143C illustrates a cross sectional view of the staple cartridge to illustrate the placement of the Hall effect sensor, processor, and conductive coupling within the distal end of the staple cartridge, in accordance with the present embodiment;

FIG. 144A-144F illustrate one embodiment of a staple cartridge that comprises a flex cable connected to a Hall effect sensor and a processor where FIG. 144A is an exploded view of the staple cartridge, FIG. 144B illustrates the assembly of the staple cartridge, FIG. 144C illustrates the underside of an assembled staple cartridge, and also illustrates the flex cable in greater detail, FIG. 144D illustrates a cross sectional view of the staple cartridge to illustrate the placement of the Hall effect sensor, processor,

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and conductive coupling, FIG. 144E illustrates the underside of the staple cartridge without the cartridge tray and including the wedge sled, in its most distal position, and FIG. 144F illustrates the staple cartridge without the cartridge tray in order to illustrate a possible placement for the cable traces;

FIGS. 145A and 145B illustrates one embodiment of a staple cartridge that comprises a flex cable, a Hall effect sensor, and a processor where FIG. 145A is an exploded view of the staple cartridge and FIG. 145B illustrates the assembly of the staple cartridge and the flex cable in greater detail;

FIG. 146A illustrates a perspective view of an end effector coupled to a shaft assembly;

FIG. 146B illustrates a perspective view of an underside of the end effector and shaft assembly shown in FIG. 146A;

FIG. 146C illustrates the end effector shown in FIGS. 146A and 146B with a flex cable and without the shaft assembly;

FIGS. 146D and 146E illustrate an elongated channel portion of the end effector shown in FIGS. 146A and 146B without the anvil or the staple cartridge, to illustrate how the flex cable shown in FIG. 146C can be seated within the elongated channel;

FIG. 146F illustrates the flex cable, shown in FIGS. 146C-120E 146C-146E, alone;

FIG. 147 illustrates a close up view of the elongated channel shown in FIGS. 146D and 146E with a staple cartridge coupled thereto;

FIGS. 148A-148D further illustrate one embodiment of a staple cartridge operative with the present embodiment of an end effector where FIG. 148A illustrates a close up view of the proximal end of the staple cartridge, FIG. 148B illustrates a close-up view of the distal end of the staple cartridge, with a space for a distal sensor plug, FIG. 148C further illustrates the distal sensor plug, and FIG. 148D illustrates the proximal-facing side of the distal sensor plug;

FIGS. 149A and 149B illustrate one embodiment of a distal sensor plug where FIG. 149A illustrates a cutaway view of the distal sensor plug and FIG. 149B further illustrates the Hall effect sensor and the processor operatively coupled to the flex board such that they are capable of communicating;

FIG. 150 illustrates an embodiment of an end effector with a flex cable operable to provide power to sensors and electronics in the distal tip of the anvil portion;

FIGS. 151A-151C illustrate the operation of the articulation joint and flex cable of the end effector where FIG. 151A illustrates a top view of the end effector with the end effector pivoted -45 degrees with respect to the shaft assembly, FIG. 151B illustrates a top view of the end effector, and FIG. 151C illustrates a top view of the end effector with the end effector pivoted +45 degrees with respect to the shaft assembly;

FIG. 152 illustrates cross-sectional view of the distal tip of an embodiment of an anvil with sensors and electronics; and

FIG. 153 illustrates a cutaway view of the distal tip of the anvil.

DESCRIPTION

Certain example embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those of ordinary skill in the art will understand that the

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devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting example embodiments. The features illustrated or described in connection with one example embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present embodiment of the invention.

Reference throughout the specification to “various embodiments,” “some embodiments,” “one embodiment,” or “an embodiment”, or the like, means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases “in various embodiments,” “in some embodiments,” “in one embodiment”, or “in an embodiment”, or the like, in places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment may be combined, in whole or in part, with the features structures, or characteristics of one or more other embodiments without limitation. Such modifications and variations are intended to be included within the scope of the present embodiment of the invention.

The terms “proximal” and “distal” are used herein with reference to a clinician manipulating the handle portion of the surgical instrument. The term “proximal” referring to the portion closest to the clinician and the term “distal” referring to the portion located away from the clinician. It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical,” “horizontal,” “up,” and “down” may be used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

Various example devices and methods are provided for performing laparoscopic and minimally invasive surgical procedures. However, the person of ordinary skill in the art will readily appreciate that the various methods and devices disclosed herein can be used in numerous surgical procedures and applications including, for example, in connection with open surgical procedures. As the present Detailed Description proceeds, those of ordinary skill in the art will further appreciate that the various instruments disclosed herein can be inserted into a body in any way, such as through a natural orifice, through an incision or puncture hole formed in tissue, etc. The working portions or end effector portions of the instruments can be inserted directly into a patient's body or can be inserted through an access device that has a working channel through which the end effector and elongated shaft of a surgical instrument can be advanced.

FIGS. 1-6 depict a motor-driven surgical cutting and fastening instrument 10 that may or may not be reused. In the illustrated embodiment, the instrument 10 includes a housing 12 that comprises a handle 14 that is configured to be grasped, manipulated and actuated by the clinician. The housing 12 is configured for operable attachment to an interchangeable shaft assembly 200 that has a surgical end effector 300 operably coupled thereto that is configured to perform one or more surgical tasks or procedures. As the present Detailed Description proceeds, it will be understood that the various unique and novel arrangements of the various forms of interchangeable shaft assemblies disclosed herein may also be effectively employed in connection with robotically-controlled surgical systems. Thus, the term

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“housing” may also encompass a housing or similar portion of a robotic system that houses or otherwise operably supports at least one drive system that is configured to generate and apply at least one control motion which could be used to actuate the interchangeable shaft assemblies disclosed herein and their respective equivalents. The term “frame” may refer to a portion of a handheld surgical instrument. The term “frame” may also represent a portion of a robotically controlled surgical instrument and/or a portion of the robotic system that may be used to operably control a surgical instrument. For example, the interchangeable shaft assemblies disclosed herein may be employed with various robotic systems, instruments, components and methods disclosed in U.S. patent application Ser. No. 13/118,241, entitled SURGICAL STAPLING INSTRUMENTS WITH ROTATABLE STAPLE DEPLOYMENT ARRANGEMENTS, now U.S. Pat. No. 9,072,535. U.S. patent application Ser. No. 13/118,241, entitled SURGICAL STAPLING INSTRUMENTS WITH ROTATABLE STAPLE DEPLOYMENT ARRANGEMENTS, now U.S. Pat. No. 9,072,535, is incorporated by reference herein in its entirety.

The housing 12 depicted in FIGS. 1-3 is shown in connection with an interchangeable shaft assembly 200 that includes an end effector 300 that comprises a surgical cutting and fastening device that is configured to operably support a surgical staple cartridge 304 therein. The housing 12 may be configured for use in connection with interchangeable shaft assemblies that include end effectors that are adapted to support different sizes and types of staple cartridges, have different shaft lengths, sizes, and types, etc. In addition, the housing 12 may also be effectively employed with a variety of other interchangeable shaft assemblies including those assemblies that are configured to apply other motions and forms of energy such as, for example, radio frequency (RF) energy, ultrasonic energy and/or motion to end effector arrangements adapted for use in connection with various surgical applications and procedures. Furthermore, the end effectors, shaft assemblies, handles, surgical instruments, and/or surgical instrument systems can utilize any suitable fastener, or fasteners, to fasten tissue. For instance, a fastener cartridge comprising a plurality of fasteners removably stored therein can be removably inserted into and/or attached to the end effector of a shaft assembly.

FIG. 1 illustrates the surgical instrument 10 with an interchangeable shaft assembly 200 operably coupled thereto. FIGS. 2 and 3 illustrate attachment of the interchangeable shaft assembly 200 to the housing 12 or handle 14. As shown in FIG. 4, the handle 14 may comprise a pair of interconnectable handle housing segments 16 and 18 that may be interconnected by screws, snap features, adhesive, etc. In the illustrated arrangement, the handle housing segments 16, 18 cooperate to form a pistol grip portion 19 that can be gripped and manipulated by the clinician. As will be discussed in further detail below, the handle 14 operably supports a plurality of drive systems therein that are configured to generate and apply various control motions to corresponding portions of the interchangeable shaft assembly that is operably attached thereto.

Referring now to FIG. 4, the handle 14 may further include a frame 20 that operably supports a plurality of drive systems. For example, the frame 20 can operably support a “first” or closure drive system, generally designated as 30, which may be employed to apply closing and opening motions to the interchangeable shaft assembly 200 that is operably attached or coupled thereto. In at least one form, the closure drive system 30 may include an actuator in the

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form of a closure trigger 32 that is pivotally supported by the frame 20. More specifically, as illustrated in FIG. 4, the closure trigger 32 is pivotally coupled to the housing 14 by a pin 33. Such arrangement enables the closure trigger 32 to be manipulated by a clinician such that when the clinician grips the pistol grip portion 19 of the handle 14, the closure trigger 32 may be easily pivoted from a starting or “unactuated” position to an “actuated” position and more particularly to a fully compressed or fully actuated position. The closure trigger 32 may be biased into the unactuated position by spring or other biasing arrangement (not shown). In various forms, the closure drive system 30 further includes a closure linkage assembly 34 that is pivotally coupled to the closure trigger 32. As shown in FIG. 4, the closure linkage assembly 34 may include a first closure link 36 and a second closure link 38 that are pivotally coupled to the closure trigger 32 by a pin 35. The second closure link 38 may also be referred to herein as an “attachment member” and include a transverse attachment pin 37.

Still referring to FIG. 4, it can be observed that the first closure link 36 may have a locking wall or end 39 thereon that is configured to cooperate with a closure release assembly 60 that is pivotally coupled to the frame 20. In at least one form, the closure release assembly 60 may comprise a release button assembly 62 that has a distally protruding locking pawl 64 formed thereon. The release button assembly 62 may be pivoted in a counterclockwise direction by a release spring (not shown). As the clinician depresses the closure trigger 32 from its unactuated position towards the pistol grip portion 19 of the handle 14, the first closure link 36 pivots upward to a point wherein the locking pawl 64 drops into retaining engagement with the locking wall 39 on the first closure link 36 thereby preventing the closure trigger 32 from returning to the unactuated position. See FIG. 18. Thus, the closure release assembly 60 serves to lock the closure trigger 32 in the fully actuated position. When the clinician desires to unlock the closure trigger 32 to permit it to be biased to the unactuated position, the clinician simply pivots the closure release button assembly 62 such that the locking pawl 64 is moved out of engagement with the locking wall 39 on the first closure link 36. When the locking pawl 64 has been moved out of engagement with the first closure link 36, the closure trigger 32 may pivot back to the unactuated position. Other closure trigger locking and release arrangements may also be employed.

Further to the above, FIGS. 13-15 illustrate the closure trigger 32 in its unactuated position which is associated with an open, or unclamped, configuration of the shaft assembly 200 in which tissue can be positioned between the jaws of the shaft assembly 200. FIGS. 16-18 illustrate the closure trigger 32 in its actuated position which is associated with a closed, or clamped, configuration of the shaft assembly 200 in which tissue is clamped between the jaws of the shaft assembly 200. Upon comparing FIGS. 14 and 17, the reader will appreciate that, when the closure trigger 32 is moved from its unactuated position (FIG. 14) to its actuated position (FIG. 17), the closure release button 62 is pivoted between a first position (FIG. 14) and a second position (FIG. 17). The rotation of the closure release button 62 can be referred to as being an upward rotation; however, at least a portion of the closure release button 62 is being rotated toward the circuit board 100. Referring to FIG. 4, the closure release button 62 can include an arm 61 extending therefrom and a magnetic element 63, such as a permanent magnet, for example, mounted to the arm 61. When the closure release button 62 is rotated from its first position to its second position, the magnetic element 63 can move toward the

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circuit board **100**. The circuit board **100** can include at least one sensor configured to detect the movement of the magnetic element **63**. In at least one embodiment, a Hall effect sensor **65**, for example, can be mounted to the bottom surface of the circuit board **100**. The Hall effect sensor **65** can be configured to detect changes in a magnetic field surrounding the Hall effect sensor **65** caused by the movement of the magnetic element **63**. The Hall effect sensor **65** can be in signal communication with a microcontroller **1500** (FIG. **19**), for example, which can determine whether the closure release button **62** is in its first position, which is associated with the unactuated position of the closure trigger **32** and the open configuration of the end effector, its second position, which is associated with the actuated position of the closure trigger **32** and the closed configuration of the end effector, and/or any position between the first position and the second position.

In at least one form, the handle **14** and the frame **20** may operably support another drive system referred to herein as a firing drive system **80** that is configured to apply firing motions to corresponding portions of the interchangeable shaft assembly attached thereto. The firing drive system may **80** also be referred to herein as a “second drive system”. The firing drive system **80** may employ an electric motor **82**, located in the pistol grip portion **19** of the handle **14**. In various forms, the motor **82** may be a DC brushed driving motor having a maximum rotation of, approximately, 25,000 RPM, for example. In other arrangements, the motor may include a brushless motor, a cordless motor, a synchronous motor, a stepper motor, or any other suitable electric motor. The motor **82** may be powered by a power source **90** that in one form may comprise a removable power pack **92**. As shown in FIG. **4**, for example, the power pack **92** may comprise a proximal housing portion **94** that is configured for attachment to a distal housing portion **96**. The proximal housing portion **94** and the distal housing portion **96** are configured to operably support a plurality of batteries **98** therein. Batteries **98** may each comprise, for example, a Lithium Ion (“LI”) or other suitable battery. The distal housing portion **96** is configured for removable operable attachment to a control circuit board assembly **100** which is also operably coupled to the motor **82**. A number of batteries **98** may be connected in series may be used as the power source for the surgical instrument **10**. In addition, the power source **90** may be replaceable and/or rechargeable.

As outlined above with respect to other various forms, the electric motor **82** can include a rotatable shaft (not shown) that operably interfaces with a gear reducer assembly **84** that is mounted in meshing engagement with a with a set, or rack, of drive teeth **122** on a longitudinally-movable drive member **120**. In use, a voltage polarity provided by the power source **90** can operate the electric motor **82** in a clockwise direction wherein the voltage polarity applied to the electric motor by the battery can be reversed in order to operate the electric motor **82** in a counter-clockwise direction. When the electric motor **82** is rotated in one direction, the drive member **120** will be axially driven in the distal direction “DD”. When the motor **82** is driven in the opposite rotary direction, the drive member **120** will be axially driven in a proximal direction “PD”. The handle **14** can include a switch which can be configured to reverse the polarity applied to the electric motor **82** by the power source **90**. As with the other forms described herein, the handle **14** can also include a sensor that is configured to detect the position of the drive member **120** and/or the direction in which the drive member **120** is being moved.

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Actuation of the motor **82** can be controlled by a firing trigger **130** that is pivotally supported on the handle **14**. The firing trigger **130** may be pivoted between an unactuated position and an actuated position. The firing trigger **130** may be biased into the unactuated position by a spring **132** or other biasing arrangement such that when the clinician releases the firing trigger **130**, it may be pivoted or otherwise returned to the unactuated position by the spring **132** or biasing arrangement. In at least one form, the firing trigger **130** can be positioned “outboard” of the closure trigger **32** as was discussed above. In at least one form, a firing trigger safety button **134** may be pivotally mounted to the closure trigger **32** by pin **35**. The safety button **134** may be positioned between the firing trigger **130** and the closure trigger **32** and have a pivot arm **136** protruding therefrom. See FIG. **4**. When the closure trigger **32** is in the unactuated position, the safety button **134** is contained in the handle **14** where the clinician cannot readily access it and move it between a safety position preventing actuation of the firing trigger **130** and a firing position wherein the firing trigger **130** may be fired. As the clinician depresses the closure trigger **32**, the safety button **134** and the firing trigger **130** pivot down wherein they can then be manipulated by the clinician.

As discussed above, the handle **14** can include a closure trigger **32** and a firing trigger **130**. Referring to FIGS. **14-18A**, the firing trigger **130** can be pivotally mounted to the closure trigger **32**. The closure trigger **32** can include an arm **31** extending therefrom and the firing trigger **130** can be pivotally mounted to the arm **31** about a pivot pin **33**. When the closure trigger **32** is moved from its unactuated position (FIG. **14**) to its actuated position (FIG. **17**), the firing trigger **130** can descend downwardly, as outlined above. After the safety button **134** has been moved to its firing position, referring primarily to FIG. **18A**, the firing trigger **130** can be depressed to operate the motor of the surgical instrument firing system. In various instances, the handle **14** can include a tracking system, such as system **800**, for example, configured to determine the position of the closure trigger **32** and/or the position of the firing trigger **130**. With primary reference to FIGS. **14**, **17**, and **18A**, the tracking system **800** can include a magnetic element, such as permanent magnet **802**, for example, which is mounted to an arm **801** extending from the firing trigger **130**. The tracking system **800** can comprise one or more sensors, such as a first Hall effect sensor **803** and a second Hall effect sensor **804**, for example, which can be configured to track the position of the magnet **802**. Upon comparing FIGS. **14** and **17**, the reader will appreciate that, when the closure trigger **32** is moved from its unactuated position to its actuated position, the magnet **802** can move between a first position adjacent the first Hall effect sensor **803** and a second position adjacent the second Hall effect sensor **804**. Upon comparing FIGS. **17** and **18A**, the reader will further appreciate that, when the firing trigger **130** is moved from an unfired position (FIG. **17**) to a fired position (FIG. **18A**), the magnet **802** can move relative to the second Hall effect sensor **804**. The sensors **803** and **804** can track the movement of the magnet **802** and can be in signal communication with a microcontroller on the circuit board **100**. With data from the first sensor **803** and/or the second sensor **804**, the microcontroller can determine the position of the magnet **802** along a predefined path and, based on that position, the microcontroller can determine whether the closure trigger **32** is in its unactuated position, its actuated position, or a position therebetween. Similarly, with data from the first sensor **803** and/or the second sensor **804**, the microcontroller can determine the position of the magnet **802** along a predefined path and, based on that position, the

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microcontroller can determine whether the firing trigger **130** is in its unfired position, its fully fired position, or a position therebetween.

As indicated above, in at least one form, the longitudinally movable drive member **120** has a rack of teeth **122** formed thereon for meshing engagement with a corresponding drive gear **86** of the gear reducer assembly **84**. At least one form also includes a manually-actuatable “bailout” assembly **140** that is configured to enable the clinician to manually retract the longitudinally movable drive member **120** should the motor **82** become disabled. The bailout assembly **140** may include a lever or bailout handle assembly **142** that is configured to be manually pivoted into ratcheting engagement with teeth **124** also provided in the drive member **120**. Thus, the clinician can manually retract the drive member **120** by using the bailout handle assembly **142** to ratchet the drive member **120** in the proximal direction “PD”. U.S. Pat. No. 8,608,045 discloses bailout arrangements and other components, arrangements and systems that may also be employed with the various instruments disclosed herein. U.S. patent application Ser. No. 12/249,117, entitled POWERED SURGICAL CUTTING AND STAPLING APPARATUS WITH MANUALLY RETRACTABLE FIRING SYSTEM, now U.S. Pat. No. 8,608,045, is hereby incorporated by reference in its entirety.

Turning now to FIGS. **1** and **7**, the interchangeable shaft assembly **200** includes a surgical end effector **300** that comprises an elongated channel **302** that is configured to operably support a staple cartridge **304** therein. The end effector **300** may further include an anvil **306** that is pivotally supported relative to the elongated channel **302**. The interchangeable shaft assembly **200** may further include an articulation joint **270** and an articulation lock **350** (FIG. **8**) which can be configured to releasably hold the end effector **300** in a desired position relative to a shaft axis SA-SA. Details regarding the construction and operation of the end effector **300**, the articulation joint **270** and the articulation lock **350** are set forth in U.S. patent application Ser. No. 13/803,086, filed Mar. 14, 2013, entitled ARTICULATABLE SURGICAL INSTRUMENT COMPRISING AN ARTICULATION LOCK, now U.S. Patent Application Publication No. 2014/0263541. The entire disclosure of U.S. patent application Ser. No. 13/803,086, filed Mar. 14, 2013, entitled ARTICULATABLE SURGICAL INSTRUMENT COMPRISING AN ARTICULATION LOCK, now U.S. Patent Application Publication No. 2014/0263541 is hereby incorporated by reference herein. As shown in FIGS. **7** and **8**, the interchangeable shaft assembly **200** can further include a proximal housing or nozzle **201** comprised of nozzle portions **202** and **203**. The interchangeable shaft assembly **200** can further include a closure tube **260** which can be utilized to close and/or open the anvil **306** of the end effector **300**. Primarily referring now to FIGS. **8** and **9**, the shaft assembly **200** can include a spine **210** which can be configured to fixably support a shaft frame portion **212** of the articulation lock **350**. See FIG. **8**. The spine **210** can be configured to, one, slidably support a firing member **220** therein and, two, slidably support the closure tube **260** which extends around the spine **210**. The spine **210** can also be configured to slidably support a proximal articulation driver **230**. The articulation driver **230** has a distal end **231** that is configured to operably engage the articulation lock **350**. The articulation lock **350** interfaces with an articulation frame **352** that is adapted to operably engage a drive pin (not shown) on the end effector frame (not shown). As indicated above, further details regarding the operation of the articulation lock **350** and the articulation frame may be found in

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U.S. patent application Ser. No. 13/803,086. In various circumstances, the spine **210** can comprise a proximal end **211** which is rotatably supported in a chassis **240**. In one arrangement, for example, the proximal end **211** of the spine **210** has a thread **214** formed thereon for threaded attachment to a spine bearing **216** configured to be supported within the chassis **240**. See FIG. **7**. Such an arrangement facilitates rotatable attachment of the spine **210** to the chassis **240** such that the spine **210** may be selectively rotated about a shaft axis SA-SA relative to the chassis **240**.

Referring primarily to FIG. **7**, the interchangeable shaft assembly **200** includes a closure shuttle **250** that is slidably supported within the chassis **240** such that it may be axially moved relative thereto. As shown in FIGS. **3** and **7**, the closure shuttle **250** includes a pair of proximally-protruding hooks **252** that are configured for attachment to the attachment pin **37** that is attached to the second closure link **38** as will be discussed in further detail below. A proximal end **261** of the closure tube **260** is coupled to the closure shuttle **250** for relative rotation thereto. For example, a U shaped connector **263** is inserted into an annular slot **262** in the proximal end **261** of the closure tube **260** and is retained within vertical slots **253** in the closure shuttle **250**. See FIG. **7**. Such an arrangement serves to attach the closure tube **260** to the closure shuttle **250** for axial travel therewith while enabling the closure tube **260** to rotate relative to the closure shuttle **250** about the shaft axis SA-SA. A closure spring **268** is journaled on the closure tube **260** and serves to bias the closure tube **260** in the proximal direction “PD” which can serve to pivot the closure trigger into the unactuated position when the shaft assembly is operably coupled to the handle **14**.

In at least one form, the interchangeable shaft assembly **200** may further include an articulation joint **270**. Other interchangeable shaft assemblies, however, may not be capable of articulation. As shown in FIG. **7**, for example, the articulation joint **270** includes a double pivot closure sleeve assembly **271**. According to various forms, the double pivot closure sleeve assembly **271** includes an end effector closure sleeve assembly **272** having upper and lower distally projecting tangs **273**, **274**. An end effector closure sleeve assembly **272** includes a horseshoe aperture **275** and a tab **276** for engaging an opening tab on the anvil **306** in the various manners described in U.S. patent application Ser. No. 13/803,086, filed Mar. 14, 2013, entitled ARTICULATABLE SURGICAL INSTRUMENT COMPRISING AN ARTICULATION LOCK, now U.S. Patent Application Publication No. 2014/0263541, which has been incorporated by reference herein. As described in further detail therein, the horseshoe aperture **275** and tab **276** engage a tab on the anvil when the anvil **306** is opened. An upper double pivot link **277** includes upwardly projecting distal and proximal pivot pins that engage respectively an upper distal pin hole in the upper proximally projecting tang **273** and an upper proximal pin hole in an upper distally projecting tang **264** on the closure tube **260**. A lower double pivot link **278** includes upwardly projecting distal and proximal pivot pins that engage respectively a lower distal pin hole in the lower proximally projecting tang **274** and a lower proximal pin hole in the lower distally projecting tang **265**. See also FIG. **8**.

In use, the closure tube **260** is translated distally (direction “DD”) to close the anvil **306**, for example, in response to the actuation of the closure trigger **32**. The anvil **306** is closed by distally translating the closure tube **260** and thus the shaft closure sleeve assembly **272**, causing it to strike a proximal surface on the anvil **306** in the manner described in the

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aforementioned reference U.S. patent application Ser. No. 13/803,086. As was also described in detail in that reference, the anvil **306** is opened by proximally translating the closure tube **260** and the shaft closure sleeve assembly **272**, causing tab **276** and the horseshoe aperture **275** to contact and push against the anvil tab to lift the anvil **306**. In the anvil-open position, the shaft closure tube **260** is moved to its proximal position.

As indicated above, the surgical instrument **10** may further include an articulation lock **350** of the types and construction described in further detail in U.S. patent application Ser. No. 13/803,086 which can be configured and operated to selectively lock the end effector **300** in position. Such arrangement enables the end effector **300** to be rotated, or articulated, relative to the shaft closure tube **260** when the articulation lock **350** is in its unlocked state. In such an unlocked state, the end effector **300** can be positioned and pushed against soft tissue and/or bone, for example, surrounding the surgical site within the patient in order to cause the end effector **300** to articulate relative to the closure tube **260**. The end effector **300** may also be articulated relative to the closure tube **260** by an articulation driver **230**.

As was also indicated above, the interchangeable shaft assembly **200** further includes a firing member **220** that is supported for axial travel within the shaft spine **210**. The firing member **220** includes an intermediate firing shaft portion **222** that is configured for attachment to a distal cutting portion or knife bar **280**. The firing member **220** may also be referred to herein as a "second shaft" and/or a "second shaft assembly". As shown in FIGS. **8** and **9**, the intermediate firing shaft portion **222** may include a longitudinal slot **223** in the distal end thereof which can be configured to receive a tab **284** on the proximal end **282** of the distal knife bar **280**. The longitudinal slot **223** and the proximal end **282** can be sized and configured to permit relative movement therebetween and can comprise a slip joint **286**. The slip joint **286** can permit the intermediate firing shaft portion **222** of the firing drive **220** to be moved to articulate the end effector **300** without moving, or at least substantially moving, the knife bar **280**. Once the end effector **300** has been suitably oriented, the intermediate firing shaft portion **222** can be advanced distally until a proximal sidewall of the longitudinal slot **223** comes into contact with the tab **284** in order to advance the knife bar **280** and fire the staple cartridge positioned within the channel **302**. As can be further seen in FIGS. **8** and **9**, the shaft spine **210** has an elongate opening or window **213** therein to facilitate assembly and insertion of the intermediate firing shaft portion **222** into the shaft frame **210**. Once the intermediate firing shaft portion **222** has been inserted therein, a top frame segment **215** may be engaged with the shaft frame **212** to enclose the intermediate firing shaft portion **222** and knife bar **280** therein. Further description of the operation of the firing member **220** may be found in U.S. patent application Ser. No. 13/803,086.

Further to the above, the shaft assembly **200** can include a clutch assembly **400** which can be configured to selectively and releasably couple the articulation driver **230** to the firing member **220**. In one form, the clutch assembly **400** includes a lock collar, or sleeve **402**, positioned around the firing member **220** wherein the lock sleeve **402** can be rotated between an engaged position in which the lock sleeve **402** couples the articulation driver **360** to the firing member **220** and a disengaged position in which the articulation driver **360** is not operably coupled to the firing member **200**. When lock sleeve **402** is in its engaged position, distal movement of the firing member **220** can

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move the articulation driver **360** distally and, correspondingly, proximal movement of the firing member **220** can move the articulation driver **230** proximally. When lock sleeve **402** is in its disengaged position, movement of the firing member **220** is not transmitted to the articulation driver **230** and, as a result, the firing member **220** can move independently of the articulation driver **230**. In various circumstances, the articulation driver **230** can be held in position by the articulation lock **350** when the articulation driver **230** is not being moved in the proximal or distal directions by the firing member **220**.

Referring primarily to FIG. **9**, the lock sleeve **402** can comprise a cylindrical, or an at least substantially cylindrical, body including a longitudinal aperture **403** defined therein configured to receive the firing member **220**. The lock sleeve **402** can comprise diametrically-opposed, inwardly-facing lock protrusions **404** and an outwardly-facing lock member **406**. The lock protrusions **404** can be configured to be selectively engaged with the firing member **220**. More particularly, when the lock sleeve **402** is in its engaged position, the lock protrusions **404** are positioned within a drive notch **224** defined in the firing member **220** such that a distal pushing force and/or a proximal pulling force can be transmitted from the firing member **220** to the lock sleeve **402**. When the lock sleeve **402** is in its engaged position, the second lock member **406** is received within a drive notch **232** defined in the articulation driver **230** such that the distal pushing force and/or the proximal pulling force applied to the lock sleeve **402** can be transmitted to the articulation driver **230**. In effect, the firing member **220**, the lock sleeve **402**, and the articulation driver **230** will move together when the lock sleeve **402** is in its engaged position. On the other hand, when the lock sleeve **402** is in its disengaged position, the lock protrusions **404** may not be positioned within the drive notch **224** of the firing member **220** and, as a result, a distal pushing force and/or a proximal pulling force may not be transmitted from the firing member **220** to the lock sleeve **402**. Correspondingly, the distal pushing force and/or the proximal pulling force may not be transmitted to the articulation driver **230**. In such circumstances, the firing member **220** can be slid proximally and/or distally relative to the lock sleeve **402** and the proximal articulation driver **230**.

As shown in FIGS. **8-12**, the shaft assembly **200** further includes a switch drum **500** that is rotatably received on the closure tube **260**. The switch drum **500** comprises a hollow shaft segment **502** that has a shaft boss **504** formed thereon for receive an outwardly protruding actuation pin **410** therein. In various circumstances, the actuation pin **410** extends through a slot **267** into a longitudinal slot **408** provided in the lock sleeve **402** to facilitate axial movement of the lock sleeve **402** when it is engaged with the articulation driver **230**. A rotary torsion spring **420** is configured to engage the boss **504** on the switch drum **500** and a portion of the nozzle housing **203** as shown in FIG. **10** to apply a biasing force to the switch drum **500**. The switch drum **500** can further comprise at least partially circumferential openings **506** defined therein which, referring to FIGS. **5** and **6**, can be configured to receive circumferential mounts **204**, **205** extending from the nozzle halves **202**, **203** and permit relative rotation, but not translation, between the switch drum **500** and the proximal nozzle **201**. As shown in those Figures, the mounts **204** and **205** also extend through openings **266** in the closure tube **260** to be seated in recesses **211** in the shaft spine **210**. However, rotation of the nozzle **201** to a point where the mounts **204**, **205** reach the end of their respective slots **506** in the switch drum **500** will result

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in rotation of the switch drum **500** about the shaft axis SA-SA. Rotation of the switch drum **500** will ultimately result in the rotation of the actuation pin **410** and the lock sleeve **402** between its engaged and disengaged positions. Thus, in essence, the nozzle **201** may be employed to operably engage and disengage the articulation drive system with the firing drive system in the various manners described in further detail in U.S. patent application Ser. No. 13/803,086.

As also illustrated in FIGS. 8-12, the shaft assembly **200** can comprise a slip ring assembly **600** which can be configured to conduct electrical power to and/or from the end effector **300** and/or communicate signals to and/or from the end effector **300**, for example. The slip ring assembly **600** can comprise a proximal connector flange **604** mounted to a chassis flange **242** extending from the chassis **240** and a distal connector flange **601** positioned within a slot defined in the shaft housings **202**, **203**. The proximal connector flange **604** can comprise a first face and the distal connector flange **601** can comprise a second face which is positioned adjacent to and movable relative to the first face. The distal connector flange **601** can rotate relative to the proximal connector flange **604** about the shaft axis SA-SA. The proximal connector flange **604** can comprise a plurality of concentric, or at least substantially concentric, conductors **602** defined in the first face thereof. A connector **607** can be mounted on the proximal side of the connector flange **601** and may have a plurality of contacts (not shown) wherein each contact corresponds to and is in electrical contact with one of the conductors **602**. Such an arrangement permits relative rotation between the proximal connector flange **604** and the distal connector flange **601** while maintaining electrical contact therebetween. The proximal connector flange **604** can include an electrical connector **606** which can place the conductors **602** in signal communication with a shaft circuit board **610** mounted to the shaft chassis **240**, for example. In at least one instance, a wiring harness comprising a plurality of conductors can extend between the electrical connector **606** and the shaft circuit board **610**. The electrical connector **606** may extend proximally through a connector opening **243** defined in the chassis mounting flange **242**. See FIG. 7. U.S. patent application Ser. No. 13/800,067, entitled STAPLE CARTRIDGE TISSUE THICKNESS SENSOR SYSTEM, filed on Mar. 13, 2013, now U.S. Patent Application Publication No. 2014/0263552, is incorporated by reference in its entirety. U.S. patent application Ser. No. 13/800,025, entitled STAPLE CARTRIDGE TISSUE THICKNESS SENSOR SYSTEM, filed on Mar. 13, 2013, now U.S. Patent Application Publication No. 2014/0263552, is incorporated by reference in its entirety. Further details regarding slip ring assembly **600** may be found in U.S. patent application Ser. No. 13/803,086.

As discussed above, the shaft assembly **200** can include a proximal portion which is fixably mounted to the handle **14** and a distal portion which is rotatable about a longitudinal axis. The rotatable distal shaft portion can be rotated relative to the proximal portion about the slip ring assembly **600**, as discussed above. The distal connector flange **601** of the slip ring assembly **600** can be positioned within the rotatable distal shaft portion. Moreover, further to the above, the switch drum **500** can also be positioned within the rotatable distal shaft portion. When the rotatable distal shaft portion is rotated, the distal connector flange **601** and the switch drum **500** can be rotated synchronously with one another. In addition, the switch drum **500** can be rotated between a first position and a second position relative to the distal connector

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tor flange **601**. When the switch drum **500** is in its first position, the articulation drive system may be operably disengaged from the firing drive system and, thus, the operation of the firing drive system may not articulate the end effector **300** of the shaft assembly **200**. When the switch drum **500** is in its second position, the articulation drive system may be operably engaged with the firing drive system and, thus, the operation of the firing drive system may articulate the end effector **300** of the shaft assembly **200**. When the switch drum **500** is moved between its first position and its second position, the switch drum **500** is moved relative to distal connector flange **601**. In various instances, the shaft assembly **200** can comprise at least one sensor configured to detect the position of the switch drum **500**. Turning now to FIGS. 11 and 12, the distal connector flange **601** can comprise a Hall effect sensor **605**, for example, and the switch drum **500** can comprise a magnetic element, such as permanent magnet **505**, for example. The Hall effect sensor **605** can be configured to detect the position of the permanent magnet **505**. When the switch drum **500** is rotated between its first position and its second position, the permanent magnet **505** can move relative to the Hall effect sensor **605**. In various instances, Hall effect sensor **605** can detect changes in a magnetic field created when the permanent magnet **505** is moved. The Hall effect sensor **605** can be in signal communication with the shaft circuit board **610** and/or the handle circuit board **100**, for example. Based on the signal from the Hall effect sensor **605**, a microcontroller on the shaft circuit board **610** and/or the handle circuit board **100** can determine whether the articulation drive system is engaged with or disengaged from the firing drive system.

Referring again to FIGS. 3 and 7, the chassis **240** includes at least one, and preferably two, tapered attachment portions **244** formed thereon that are adapted to be received within corresponding dovetail slots **702** formed within a distal attachment flange portion **700** of the frame **20**. Each dovetail slot **702** may be tapered or, stated another way, be somewhat V-shaped to seatingly receive the attachment portions **244** therein. As can be further seen in FIGS. 3 and 7, a shaft attachment lug **226** is formed on the proximal end of the intermediate firing shaft **222**. As will be discussed in further detail below, when the interchangeable shaft assembly **200** is coupled to the handle **14**, the shaft attachment lug **226** is received in a firing shaft attachment cradle **126** formed in the distal end **125** of the longitudinal drive member **120**. See FIGS. 3 and 6.

Various shaft assembly embodiments employ a latch system **710** for removably coupling the shaft assembly **200** to the housing **12** and more specifically to the frame **20**. As shown in FIG. 7, for example, in at least one form, the latch system **710** includes a lock member or lock yoke **712** that is movably coupled to the chassis **240**. In the illustrated embodiment, for example, the lock yoke **712** has a U-shape with two spaced downwardly extending legs **714**. The legs **714** each have a pivot lug **716** formed thereon that are adapted to be received in corresponding holes **245** formed in the chassis **240**. Such arrangement facilitates pivotal attachment of the lock yoke **712** to the chassis **240**. The lock yoke **712** may include two proximally protruding lock lugs **714** that are configured for releasable engagement with corresponding lock detents or grooves **704** in the distal attachment flange **700** of the frame **20**. See FIG. 3. In various forms, the lock yoke **712** is biased in the proximal direction by spring or biasing member (not shown). Actuation of the lock yoke **712** may be accomplished by a latch button **722** that is slidably mounted on a latch actuator assembly **720**

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that is mounted to the chassis **240**. The latch button **722** may be biased in a proximal direction relative to the lock yoke **712**. As will be discussed in further detail below, the lock yoke **712** may be moved to an unlocked position by biasing the latch button in the distal direction which also causes the lock yoke **712** to pivot out of retaining engagement with the distal attachment flange **700** of the frame **20**. When the lock yoke **712** is in "retaining engagement" with the distal attachment flange **700** of the frame **20**, the lock lugs **716** are retainingly seated within the corresponding lock detents or grooves **704** in the distal attachment flange **700**.

When employing an interchangeable shaft assembly that includes an end effector of the type described herein that is adapted to cut and fasten tissue, as well as other types of end effectors, it may be desirable to prevent inadvertent detachment of the interchangeable shaft assembly from the housing during actuation of the end effector. For example, in use the clinician may actuate the closure trigger **32** to grasp and manipulate the target tissue into a desired position. Once the target tissue is positioned within the end effector **300** in a desired orientation, the clinician may then fully actuate the closure trigger **32** to close the anvil **306** and clamp the target tissue in position for cutting and stapling. In that instance, the first drive system **30** has been fully actuated. After the target tissue has been clamped in the end effector **300**, it may be desirable to prevent the inadvertent detachment of the shaft assembly **200** from the housing **12**. One form of the latch system **710** is configured to prevent such inadvertent detachment.

As can be most particularly seen in FIG. 7, the lock yoke **712** includes at least one and preferably two lock hooks **718** that are adapted to contact corresponding lock lug portions **256** that are formed on the closure shuttle **250**. Referring to FIGS. 13-15, when the closure shuttle **250** is in an unactuated position (i.e., the first drive system **30** is unactuated and the anvil **306** is open), the lock yoke **712** may be pivoted in a distal direction to unlock the interchangeable shaft assembly **200** from the housing **12**. When in that position, the lock hooks **718** do not contact the lock lug portions **256** on the closure shuttle **250**. However, when the closure shuttle **250** is moved to an actuated position (i.e., the first drive system **30** is actuated and the anvil **306** is in the closed position), the lock yoke **712** is prevented from being pivoted to an unlocked position. See FIGS. 16-18. Stated another way, if the clinician were to attempt to pivot the lock yoke **712** to an unlocked position or, for example, the lock yoke **712** was inadvertently bumped or contacted in a manner that might otherwise cause it to pivot distally, the lock hooks **718** on the lock yoke **712** will contact the lock lugs **256** on the closure shuttle **250** and prevent movement of the lock yoke **712** to an unlocked position.

Attachment of the interchangeable shaft assembly **200** to the handle **14** will now be described with reference to FIG. 3. To commence the coupling process, the clinician may position the chassis **240** of the interchangeable shaft assembly **200** above or adjacent to the distal attachment flange **700** of the frame **20** such that the tapered attachment portions **244** formed on the chassis **240** are aligned with the dovetail slots **702** in the frame **20**. The clinician may then move the shaft assembly **200** along an installation axis **IA** that is perpendicular to the shaft axis **SA-SA** to seat the attachment portions **244** in "operable engagement" with the corresponding dovetail receiving slots **702**. In doing so, the shaft attachment lug **226** on the intermediate firing shaft **222** will also be seated in the cradle **126** in the longitudinally movable drive member **120** and the portions of pin **37** on the second closure link **38** will be seated in the corresponding

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hooks **252** in the closure yoke **250**. As used herein, the term "operable engagement" in the context of two components means that the two components are sufficiently engaged with each other so that upon application of an actuation motion thereto, the components may carry out their intended action, function and/or procedure.

As discussed above, at least five systems of the interchangeable shaft assembly **200** can be operably coupled with at least five corresponding systems of the handle **14**. A first system can comprise a frame system which couples and/or aligns the frame or spine of the shaft assembly **200** with the frame **20** of the handle **14**. Another system can comprise a closure drive system **30** which can operably connect the closure trigger **32** of the handle **14** and the closure tube **260** and the anvil **306** of the shaft assembly **200**. As outlined above, the closure tube attachment yoke **250** of the shaft assembly **200** can be engaged with the pin **37** on the second closure link **38**. Another system can comprise the firing drive system **80** which can operably connect the firing trigger **130** of the handle **14** with the intermediate firing shaft **222** of the shaft assembly **200**.

As outlined above, the shaft attachment lug **226** can be operably connected with the cradle **126** of the longitudinal drive member **120**. Another system can comprise an electrical system which can signal to a controller in the handle **14**, such as microcontroller, for example, that a shaft assembly, such as shaft assembly **200**, for example, has been operably engaged with the handle **14** and/or, two, conduct power and/or communication signals between the shaft assembly **200** and the handle **14**. For instance, the shaft assembly **200** can include an electrical connector **1410** that is operably mounted to the shaft circuit board **610**. The electrical connector **1410** is configured for mating engagement with a corresponding electrical connector **1400** on the handle control board **100**. Further details regarding the circuitry and control systems may be found in U.S. patent application Ser. No. 13/803,086, the entire disclosure of which was previously incorporated by reference herein. The fifth system may consist of the latching system for releasably locking the shaft assembly **200** to the handle **14**.

Referring again to FIGS. 2 and 3, the handle **14** can include an electrical connector **1400** comprising a plurality of electrical contacts. Turning now to FIG. 19, the electrical connector **1400** can comprise a first contact **1401a**, a second contact **1401b**, a third contact **1401c**, a fourth contact **1401d**, a fifth contact **1401e**, and a sixth contact **1401f**, for example. While the illustrated embodiment utilizes six contacts, other embodiments are envisioned which may utilize more than six contacts or less than six contacts.

As illustrated in FIG. 19, the first contact **1401a** can be in electrical communication with a transistor **1408**, contacts **1401b-1401e** can be in electrical communication with a microcontroller **1500**, and the sixth contact **1401f** can be in electrical communication with a ground. In certain circumstances, one or more of the electrical contacts **1401b-1401e** may be in electrical communication with one or more output channels of the microcontroller **1500** and can be energized, or have a voltage potential applied thereto, when the handle **1042** is in a powered state. In some circumstances, one or more of the electrical contacts **1401b-1401e** may be in electrical communication with one or more input channels of the microcontroller **1500** and, when the handle **14** is in a powered state, the microcontroller **1500** can be configured to detect when a voltage potential is applied to such electrical contacts. When a shaft assembly, such as shaft assembly **200**, for example, is assembled to the handle **14**, the electrical contacts **1401a-1401f** may not communicate with each

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other. When a shaft assembly is not assembled to the handle 14, however, the electrical contacts 1401a-1401f of the electrical connector 1400 may be exposed and, in some circumstances, one or more of the contacts 1401a-1401f may be accidentally placed in electrical communication with each other. Such circumstances can arise when one or more of the contacts 1401a-1401f come into contact with an electrically conductive material, for example. When this occurs, the microcontroller 1500 can receive an erroneous input and/or the shaft assembly 200 can receive an erroneous output, for example. To address this issue, in various circumstances, the handle 14 may be unpowered when a shaft assembly, such as shaft assembly 200, for example, is not attached to the handle 14.

In other circumstances, the handle 1042 can be powered when a shaft assembly, such as shaft assembly 200, for example, is not attached thereto. In such circumstances, the microcontroller 1500 can be configured to ignore inputs, or voltage potentials, applied to the contacts in electrical communication with the microcontroller 1500, i.e., contacts 1401b-1401e, for example, until a shaft assembly is attached to the handle 14. Even though the microcontroller 1500 may be supplied with power to operate other functionalities of the handle 14 in such circumstances, the handle 14 may be in a powered-down state. In a way, the electrical connector 1400 may be in a powered-down state as voltage potentials applied to the electrical contacts 1401b-1401e may not affect the operation of the handle 14. The reader will appreciate that, even though contacts 1401b-1401e may be in a powered-down state, the electrical contacts 1401a and 1401f, which are not in electrical communication with the microcontroller 1500, may or may not be in a powered-down state. For instance, sixth contact 1401f may remain in electrical communication with a ground regardless of whether the handle 14 is in a powered-up or a powered-down state.

Furthermore, the transistor 1408, and/or any other suitable arrangement of transistors, such as transistor 1410, for example, and/or switches may be configured to control the supply of power from a power source 1404, such as a battery 90 within the handle 14, for example, to the first electrical contact 1401a regardless of whether the handle 14 is in a powered-up or a powered-down state. In various circumstances, the shaft assembly 200, for example, can be configured to change the state of the transistor 1408 when the shaft assembly 200 is engaged with the handle 14. In certain circumstances, further to the below, a Hall effect sensor 1402 can be configured to switch the state of transistor 1410 which, as a result, can switch the state of transistor 1408 and ultimately supply power from power source 1404 to first contact 1401a. In this way, both the power circuits and the signal circuits to the connector 1400 can be powered down when a shaft assembly is not installed to the handle 14 and powered up when a shaft assembly is installed to the handle 14.

In various circumstances, referring again to FIG. 19, the handle 14 can include the Hall effect sensor 1402, for example, which can be configured to detect a detectable element, such as a magnetic element 1407 (FIG. 3), for example, on a shaft assembly, such as shaft assembly 200, for example, when the shaft assembly is coupled to the handle 14. The Hall effect sensor 1402 can be powered by a power source 1406, such as a battery, for example, which can, in effect, amplify the detection signal of the Hall effect sensor 1402 and communicate with an input channel of the microcontroller 1500 via the circuit illustrated in FIG. 19. Once the microcontroller 1500 has received an input indicating that a shaft assembly has been at least partially

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coupled to the handle 14, and that, as a result, the electrical contacts 1401a-1401f are no longer exposed, the microcontroller 1500 can enter into its normal, or powered-up, operating state. In such an operating state, the microcontroller 1500 will evaluate the signals transmitted to one or more of the contacts 1401b-1401e from the shaft assembly and/or transmit signals to the shaft assembly through one or more of the contacts 1401b-1401e in normal use thereof. In various circumstances, the shaft assembly 200 may have to be fully seated before the Hall effect sensor 1402 can detect the magnetic element 1407. While a Hall effect sensor 1402 can be utilized to detect the presence of the shaft assembly 200, any suitable system of sensors and/or switches can be utilized to detect whether a shaft assembly has been assembled to the handle 14, for example. In this way, further to the above, both the power circuits and the signal circuits to the connector 1400 can be powered down when a shaft assembly is not installed to the handle 14 and powered up when a shaft assembly is installed to the handle 14.

In various embodiments, any number of magnetic sensing elements may be employed to detect whether a shaft assembly has been assembled to the handle 14, for example. For example, the technologies used for magnetic field sensing include search coil, fluxgate, optically pumped, nuclear precession, SQUID, Hall-effect, anisotropic magnetoresistance, giant magnetoresistance, magnetic tunnel junctions, giant magnetoimpedance, magnetostrictive/piezoelectric composites, magnetodiode, magnetotransistor, fiber optic, magneto optic, and microelectromechanical systems-based magnetic sensors, among others.

Referring to FIG. 19, the microcontroller 1500 may generally comprise a microprocessor ("processor") and one or more memory units operationally coupled to the processor. By executing instruction code stored in the memory, the processor may control various components of the surgical instrument, such as the motor, various drive systems, and/or a user display, for example. The microcontroller 1500 may be implemented using integrated and/or discrete hardware elements, software elements, and/or a combination of both. Examples of integrated hardware elements may include processors, microprocessors, microcontrollers, integrated circuits, application specific integrated circuits (ASIC), programmable logic devices (PLD), digital signal processors (DSP), field programmable gate arrays (FPGA), logic gates, registers, semiconductor devices, chips, microchips, chip sets, microcontrollers, system-on-chip (SoC), and/or system-in-package (SIP). Examples of discrete hardware elements may include circuits and/or circuit elements such as logic gates, field effect transistors, bipolar transistors, resistors, capacitors, inductors, and/or relays. In certain instances, the microcontroller 1500 may include a hybrid circuit comprising discrete and integrated circuit elements or components on one or more substrates, for example.

Referring to FIG. 19, the microcontroller 1500 may be an LM 4F230H5QR, available from Texas Instruments, for example. In certain instances, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), internal read-only memory (ROM) loaded with StellarisWare® software, 2 KB electrically erasable programmable read-only memory (EEPROM), one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QEI) analog, one or more 12-bit Analog-to-Digital Converters (ADC) with 12 analog input channels,

among other features that are readily available. Other micro-controllers may be readily substituted for use with the present disclosure. Accordingly, the present disclosure should not be limited in this context.

As discussed above, the handle **14** and/or the shaft assembly **200** can include systems and configurations configured to prevent, or at least reduce the possibility of, the contacts of the handle electrical connector **1400** and/or the contacts of the shaft electrical connector **1410** from becoming shorted out when the shaft assembly **200** is not assembled, or completely assembled, to the handle **14**. Referring to FIG. **3**, the handle electrical connector **1400** can be at least partially recessed within a cavity **1409** defined in the handle frame **20**. The six contacts **1401a-1401f** of the electrical connector **1400** can be completely recessed within the cavity **1409**. Such arrangements can reduce the possibility of an object accidentally contacting one or more of the contacts **1401a-1401f**. Similarly, the shaft electrical connector **1410** can be positioned within a recess defined in the shaft chassis **240** which can reduce the possibility of an object accidentally contacting one or more of the contacts **1411a-1411f** of the shaft electrical connector **1410**. With regard to the particular embodiment depicted in FIG. **3**, the shaft contacts **1411a-1411f** can comprise male contacts. In at least one embodiment, each shaft contact **1411a-1411f** can comprise a flexible projection extending therefrom which can be configured to engage a corresponding handle contact **1401a-1401f**, for example. The handle contacts **1401a-1401f** can comprise female contacts. In at least one embodiment, each handle contact **1401a-1401f** can comprise a flat surface, for example, against which the male shaft contacts **1401a-1401f** can wipe, or slide, against and maintain an electrically conductive interface therebetween. In various instances, the direction in which the shaft assembly **200** is assembled to the handle **14** can be parallel to, or at least substantially parallel to, the handle contacts **1401a-1401f** such that the shaft contacts **1411a-1411f** slide against the handle contacts **1401a-1401f** when the shaft assembly **200** is assembled to the handle **14**. In various alternative embodiments, the handle contacts **1401a-1401f** can comprise male contacts and the shaft contacts **1411a-1411f** can comprise female contacts. In certain alternative embodiments, the handle contacts **1401a-1401f** and the shaft contacts **1411a-1411f** can comprise any suitable arrangement of contacts.

In various instances, the handle **14** can comprise a connector guard configured to at least partially cover the handle electrical connector **1400** and/or a connector guard configured to at least partially cover the shaft electrical connector **1410**. A connector guard can prevent, or at least reduce the possibility of, an object accidentally touching the contacts of an electrical connector when the shaft assembly is not assembled to, or only partially assembled to, the handle. A connector guard can be movable. For instance, the connector guard can be moved between a guarded position in which it at least partially guards a connector and an unguarded position in which it does not guard, or at least guards less of, the connector. In at least one embodiment, a connector guard can be displaced as the shaft assembly is being assembled to the handle. For instance, if the handle comprises a handle connector guard, the shaft assembly can contact and displace the handle connector guard as the shaft assembly is being assembled to the handle. Similarly, if the shaft assembly comprises a shaft connector guard, the handle can contact and displace the shaft connector guard as the shaft assembly is being assembled to the handle. In various instances, a connector guard can comprise a door, for example. In at least one instance, the door can comprise a beveled surface

which, when contacted by the handle or shaft, can facilitate the displacement of the door in a certain direction. In various instances, the connector guard can be translated and/or rotated, for example. In certain instances, a connector guard can comprise at least one film which covers the contacts of an electrical connector. When the shaft assembly is assembled to the handle, the film can become ruptured. In at least one instance, the male contacts of a connector can penetrate the film before engaging the corresponding contacts positioned underneath the film.

As described above, the surgical instrument can include a system which can selectively power-up, or activate, the contacts of an electrical connector, such as the electrical connector **1400**, for example. In various instances, the contacts can be transitioned between an unactivated condition and an activated condition. In certain instances, the contacts can be transitioned between a monitored condition, a deactivated condition, and an activated condition. For instance, the microcontroller **1500**, for example, can monitor the contacts **1401a-1401f** when a shaft assembly has not been assembled to the handle **14** to determine whether one or more of the contacts **1401a-1401f** may have been shorted. The microcontroller **1500** can be configured to apply a low voltage potential to each of the contacts **1401a-1401f** and assess whether only a minimal resistance is present at each of the contacts. Such an operating state can comprise the monitored condition. In the event that the resistance detected at a contact is high, or above a threshold resistance, the microcontroller **1500** can deactivate that contact, more than one contact, or, alternatively, all of the contacts. Such an operating state can comprise the deactivated condition. If a shaft assembly is assembled to the handle **14** and it is detected by the microcontroller **1500**, as discussed above, the microcontroller **1500** can increase the voltage potential to the contacts **1401a-1401f**. Such an operating state can comprise the activated condition.

The various shaft assemblies disclosed herein may employ sensors and various other components that require electrical communication with the controller in the housing. These shaft assemblies generally are configured to be able to rotate relative to the housing necessitating a connection that facilitates such electrical communication between two or more components that may rotate relative to each other. When employing end effectors of the types disclosed herein, the connector arrangements must be relatively robust in nature while also being somewhat compact to fit into the shaft assembly connector portion.

Referring to FIG. **20**, a non-limiting form of the end effector **300** is illustrated. As described above, the end effector **300** may include the anvil **306** and the staple cartridge **304**. In this non-limiting embodiment, the anvil **306** is coupled to an elongate channel **198**. For example, apertures **199** can be defined in the elongate channel **198** which can receive pins **152** extending from the anvil **306** and allow the anvil **306** to pivot from an open position to a closed position relative to the elongate channel **198** and staple cartridge **304**. In addition, FIG. **20** shows a firing bar **172**, configured to longitudinally translate into the end effector **300**. The firing bar **172** may be constructed from one solid section, or in various embodiments, may include a laminate material comprising, for example, a stack of steel plates. A distally projecting end of the firing bar **172** can be attached to an E-beam **178** that can, among other things, assist in spacing the anvil **306** from a staple cartridge **304** positioned in the elongate channel **198** when the anvil **306** is in a closed position. The E-beam **178** can also include a sharpened cutting edge **182** which can be used to sever tissue as the

E-beam **178** is advanced distally by the firing bar **172**. In operation, the E-beam **178** can also actuate, or fire, the staple cartridge **304**. The staple cartridge **304** can include a molded cartridge body **194** that holds a plurality of staples **191** resting upon staple drivers **192** within respective upwardly open staple cavities **195**. A wedge sled **190** is driven distally by the E-beam **178**, sliding upon a cartridge tray **196** that holds together the various components of the replaceable staple cartridge **304**. The wedge sled **190** upwardly cams the staple drivers **192** to force out the staples **191** into deforming contact with the anvil **306** while a cutting surface **182** of the E-beam **178** severs clamped tissue.

Further to the above, the E-beam **178** can include upper pins **180** which engage the anvil **306** during firing. The E-beam **178** can further include middle pins **184** and a bottom foot **186** which can engage various portions of the cartridge body **194**, cartridge tray **196** and elongate channel **198**. When a staple cartridge **304** is positioned within the elongate channel **198**, a slot **193** defined in the cartridge body **194** can be aligned with a slot **197** defined in the cartridge tray **196** and a slot **189** defined in the elongate channel **198**. In use, the E-beam **178** can slide through the aligned slots **193**, **197**, and **189** wherein, as indicated in FIG. **20**, the bottom foot **186** of the E-beam **178** can engage a groove running along the bottom surface of channel **198** along the length of slot **189**, the middle pins **184** can engage the top surfaces of cartridge tray **196** along the length of longitudinal slot **197**, and the upper pins **180** can engage the anvil **306**. In such circumstances, the E-beam **178** can space, or limit the relative movement between, the anvil **306** and the staple cartridge **304** as the firing bar **172** is moved distally to fire the staples from the staple cartridge **304** and/or incise the tissue captured between the anvil **306** and the staple cartridge **304**. Thereafter, the firing bar **172** and the E-beam **178** can be retracted proximally allowing the anvil **306** to be opened to release the two stapled and severed tissue portions (not shown).

Having described a surgical instrument **10** in general terms, the description now turns to a detailed description of various electrical/electronic component of the surgical instrument **10**. Turning now to FIGS. **21A-21B**, where one embodiment of a segmented circuit **2000** comprising a plurality of circuit segments **2002a-2002g** is illustrated. The segmented circuit **2000** comprising the plurality of circuit segments **2002a-2002g** is configured to control a powered surgical instrument, such as, for example, the surgical instrument **10** illustrated in FIGS. **1-18A**, without limitation. The plurality of circuit segments **2002a-2002g** is configured to control one or more operations of the powered surgical instrument **10**. A safety processor segment **2002a** (Segment 1) comprises a safety processor **2004**. A primary processor segment **2002b** (Segment 2) comprises a primary processor **2006**. The safety processor **2004** and/or the primary processor **2006** are configured to interact with one or more additional circuit segments **2002c-2002g** to control operation of the powered surgical instrument **10**. The primary processor **2006** comprises a plurality of inputs coupled to, for example, one or more circuit segments **2002c-2002g**, a battery **2008**, and/or a plurality of switches **2058a-2070**. The segmented circuit **2000** may be implemented by any suitable circuit, such as, for example, a printed circuit board assembly (PCBA) within the powered surgical instrument **10**. It should be understood that the term processor as used herein includes any microprocessor, microcontroller, or other basic computing device that incorporates the functions of a computer's central processing unit (CPU) on an integrated circuit or at most a few integrated circuits. The processor is

a multipurpose, programmable device that accepts digital data as input, processes it according to instructions stored in its memory, and provides results as output. It is an example of sequential digital logic, as it has internal memory. Processors operate on numbers and symbols represented in the binary numeral system.

In one embodiment, the main processor **2006** may be any single core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one embodiment, the safety processor **2004** may be a safety microcontroller platform comprising two microcontroller-based families such as TMS570 and RM4x known under the trade name Hercules ARM Cortex R4, also by Texas Instruments. Nevertheless, other suitable substitutes for microcontrollers and safety processor may be employed, without limitation. In one embodiment, the safety processor **2004** may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

In certain instances, the main processor **2006** may be an LM 4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, internal ROM loaded with StellarisWare® software, 2 KB EEPROM, one or more PWM modules, one or more QEI analog, one or more 12-bit ADC with 12 analog input channels, among other features that are readily available for the product datasheet. Other processors may be readily substituted and, accordingly, the present disclosure should not be limited in this context.

In one embodiment, the segmented circuit **2000** comprises an acceleration segment **2002c** (Segment 3). The acceleration segment **2002c** comprises an acceleration sensor **2022**. The acceleration sensor **2022** may comprise, for example, an accelerometer. The acceleration sensor **2022** is configured to detect movement or acceleration of the powered surgical instrument **10**. In some embodiments, input from the acceleration sensor **2022** is used, for example, to transition to and from a sleep mode, identify an orientation of the powered surgical instrument, and/or identify when the surgical instrument has been dropped. In some embodiments, the acceleration segment **2002c** is coupled to the safety processor **2004** and/or the primary processor **2006**.

In one embodiment, the segmented circuit **2000** comprises a display segment **2002d** (Segment 4). The display segment **2002d** comprises a display connector **2024** coupled to the primary processor **2006**. The display connector **2024** couples the primary processor **2006** to a display **2028** through one or more display driver integrated circuits **2026**. The display driver integrated circuits **2026** may be integrated with the display **2028** and/or may be located separately from the display **2028**. The display **2028** may comprise any suitable display, such as, for example, an organic light-emitting diode (OLED) display, a liquid-crystal display (LCD), and/or any other suitable display. In some embodiments, the display segment **2002d** is coupled to the safety processor **2004**.

In some embodiments, the segmented circuit **2000** comprises a shaft segment **2002e** (Segment 5). The shaft segment **2002e** comprises one or more controls for a shaft **2004** coupled to the surgical instrument **10** and/or one or more controls for an end effector **2006** coupled to the shaft **2004**. The shaft segment **2002e** comprises a shaft connector **2030**

configured to couple the primary processor **2006** to a shaft PCBA **2031**. The shaft PCBA **2031** comprises a first articulation switch **2036**, a second articulation switch **2032**, and a shaft PCBA EEPROM **2034**. In some embodiments, the shaft PCBA EEPROM **2034** comprises one or more parameters, routines, and/or programs specific to the shaft **2004** and/or the shaft PCBA **2031**. The shaft PCBA **2031** may be coupled to the shaft **2004** and/or integral with the surgical instrument **10**. In some embodiments, the shaft segment **2002e** comprises a second shaft EEPROM **2038**. The second shaft EEPROM **2038** comprises a plurality of algorithms, routines, parameters, and/or other data corresponding to one or more shafts **2004** and/or end effectors **2006** which may be interfaced with the powered surgical instrument **10**.

In some embodiments, the segmented circuit **2000** comprises a position encoder segment **2002f** (Segment 6). The position encoder segment **2002f** comprises one or more magnetic rotary position encoders **2040a-2040b**. The one or more magnetic rotary position encoders **2040a-2040b** are configured to identify the rotational position of a motor **2048**, a shaft **2004**, and/or an end effector **2006** of the surgical instrument **10**. In some embodiments, the magnetic rotary position encoders **2040a-2040b** may be coupled to the safety processor **2004** and/or the primary processor **2006**.

In some embodiments, the segmented circuit **2000** comprises a motor segment **2002g** (Segment 7). The motor segment **2002g** comprises a motor **2048** configured to control one or more movements of the powered surgical instrument **10**. The motor **2048** is coupled to the primary processor **2006** by an H-Bridge driver **2042** and one or more H-bridge field-effect transistors (FETs) **2044**. The H-bridge FETs **2044** are coupled to the safety processor **2004**. A motor current sensor **2046** is coupled in series with the motor **2048** to measure the current draw of the motor **2048**. The motor current sensor **2046** is in signal communication with the primary processor **2006** and/or the safety processor **2004**. In some embodiments, the motor **2048** is coupled to a motor electromagnetic interference (EMI) filter **2050**.

The segmented circuit **2000** comprises a power segment **2002h** (Segment 8). A battery **2008** is coupled to the safety processor **2004**, the primary processor **2006**, and one or more of the additional circuit segments **2002c-2002g**. The battery **2008** is coupled to the segmented circuit **2000** by a battery connector **2010** and a current sensor **2012**. The current sensor **2012** is configured to measure the total current draw of the segmented circuit **2000**. In some embodiments, one or more voltage converters **2014a, 2014b, 2016** are configured to provide predetermined voltage values to one or more circuit segments **2002a-2002g**. For example, in some embodiments, the segmented circuit **2000** may comprise 3.3V voltage converters **2014a-2014b** and/or 5V voltage converters **2016**. A boost converter **2018** is configured to provide a boost voltage up to a predetermined amount, such as, for example, up to 13V. The boost converter **2018** is configured to provide additional voltage and/or current during power intensive operations and prevent brownout or low-power conditions.

In some embodiments, the safety segment **2002a** comprises a motor power interrupt **2020**. The motor power interrupt **2020** is coupled between the power segment **2002h** and the motor segment **2002g**. The safety segment **2002a** is configured to interrupt power to the motor segment **2002g** when an error or fault condition is detected by the safety processor **2004** and/or the primary processor **2006** as discussed in more detail herein. Although the circuit segments **2002a-2002g** are illustrated with all components of the circuit segments **2002a-2002h** located in physical proximity,

one skilled in the art will recognize that a circuit segment **2002a-2002h** may comprise components physically and/or electrically separate from other components of the same circuit segment **2002a-2002g**. In some embodiments, one or more components may be shared between two or more circuit segments **2002a-2002g**.

In some embodiments, a plurality of switches **2056-2070** are coupled to the safety processor **2004** and/or the primary processor **2006**. The plurality of switches **2056-2070** may be configured to control one or more operations of the surgical instrument **10**, control one or more operations of the segmented circuit **2000**, and/or indicate a status of the surgical instrument **10**. For example, a bail-out door switch **2056** is configured to indicate the status of a bail-out door. A plurality of articulation switches, such as, for example, a left side articulation left switch **2058a**, a left side articulation right switch **2060a**, a left side articulation center switch **2062a**, a right side articulation left switch **2058b**, a right side articulation right switch **2060b**, and a right side articulation center switch **2062b** are configured to control articulation of a shaft **2004** and/or an end effector **2006**. A left side reverse switch **2064a** and a right side reverse switch **2064b** are coupled to the primary processor **2006**. In some embodiments, the left side switches comprising the left side articulation left switch **2058a**, the left side articulation right switch **2060a**, the left side articulation center switch **2062a**, and the left side reverse switch **2064a** are coupled to the primary processor **2006** by a left flex connector **2072a**. The right side switches comprising the right side articulation left switch **2058b**, the right side articulation right switch **2060b**, the right side articulation center switch **2062b**, and the right side reverse switch **2064b** are coupled to the primary processor **2006** by a right flex connector **2072b**. In some embodiments, a firing switch **2066**, a clamp release switch **2068**, and a shaft engaged switch **2070** are coupled to the primary processor **2006**.

The plurality of switches **2056-2070** may comprise, for example, a plurality of handle controls mounted to a handle of the surgical instrument **10**, a plurality of indicator switches, and/or any combination thereof. In various embodiments, the plurality of switches **2056-2070** allow a surgeon to manipulate the surgical instrument, provide feedback to the segmented circuit **2000** regarding the position and/or operation of the surgical instrument, and/or indicate unsafe operation of the surgical instrument **10**. In some embodiments, additional or fewer switches may be coupled to the segmented circuit **2000**, one or more of the switches **2056-2070** may be combined into a single switch, and/or expanded to multiple switches. For example, in one embodiment, one or more of the left side and/or right side articulation switches **2058a-2064b** may be combined into a single multi-position switch.

In one embodiment, the safety processor **2004** is configured to implement a watchdog function, among other safety operations. The safety processor **2004** and the primary processor **2006** of the segmented circuit **2000** are in signal communication. A microprocessor alive heartbeat signal is provided at output **2096**. The acceleration segment **2002c** comprises an accelerometer **2022** configured to monitor movement of the surgical instrument **10**. In various embodiments, the accelerometer **2022** may be a single, double, or triple axis accelerometer. The accelerometer **2022** may be employed to measure proper acceleration that is not necessarily the coordinate acceleration (rate of change of velocity). Instead, the accelerometer sees the acceleration associated with the phenomenon of weight experienced by a test mass at rest in the frame of reference of the accelerometer

2022. For example, the accelerometer **2022** at rest on the surface of the earth will measure an acceleration $g=9.8 \text{ m/s}^2$ (gravity) straight upwards, due to its weight. Another type of acceleration that accelerometer **2022** can measure is g-force acceleration. In various other embodiments, the accelerometer **2022** may comprise a single, double, or triple axis accelerometer. Further, the acceleration segment **2002c** may comprise one or more inertial sensors to detect and measure acceleration, tilt, shock, vibration, rotation, and multiple degrees-of-freedom (DoF). A suitable inertial sensor may comprise an accelerometer (single, double, or triple axis), a magnetometer to measure a magnetic field in space such as the earth's magnetic field, and/or a gyroscope to measure angular velocity.

In one embodiment, the safety processor **2004** is configured to implement a watchdog function with respect to one or more circuit segments **2002c-2002h**, such as, for example, the motor segment **2002g**. In this regard, the safety processor **2004** employs the watchdog function to detect and recover from malfunctions of the primary processor **2006**. During normal operation, the safety processor **2004** monitors for hardware faults or program errors of the primary processor **2004** and to initiate corrective action or actions. The corrective actions may include placing the primary processor **2006** in a safe state and restoring normal system operation. In one embodiment, the safety processor **2004** is coupled to at least a first sensor. The first sensor measures a first property of the surgical instrument **10**. In some embodiments, the safety processor **2004** is configured to compare the measured property of the surgical instrument **10** to a predetermined value. For example, in one embodiment, a motor sensor **2040a** is coupled to the safety processor **2004**. The motor sensor **2040a** provides motor speed and position information to the safety processor **2004**. The safety processor **2004** monitors the motor sensor **2040a** and compares the value to a maximum speed and/or position value and prevents operation of the motor **2048** above the predetermined values. In some embodiments, the predetermined values are calculated based on real-time speed and/or position of the motor **2048**, calculated from values supplied by a second motor sensor **2040b** in communication with the primary processor **2006**, and/or provided to the safety processor **2004** from, for example, a memory module coupled to the safety processor **2004**.

In some embodiments, a second sensor is coupled to the primary processor **2006**. The second sensor is configured to measure the first physical property. The safety processor **2004** and the primary processor **2006** are configured to provide a signal indicative of the value of the first sensor and the second sensor respectively. When either the safety processor **2004** or the primary processor **2006** indicates a value outside of an acceptable range, the segmented circuit **2000** prevents operation of at least one of the circuit segments **2002c-2002h**, such as, for example, the motor segment **2002g**. For example, in the embodiment illustrated in FIGS. 21A-21B, the safety processor **2004** is coupled to a first motor position sensor **2040a** and the primary processor **2006** is coupled to a second motor position sensor **2040b**. The motor position sensors **2040a**, **2040b** may comprise any suitable motor position sensor, such as, for example, a magnetic angle rotary input comprising a sine and cosine output. The motor position sensors **2040a**, **2040b** provide respective signals to the safety processor **2004** and the primary processor **2006** indicative of the position of the motor **2048**.

The safety processor **2004** and the primary processor **2006** generate an activation signal when the values of the

first motor sensor **2040a** and the second motor sensor **2040b** are within a predetermined range. When either the primary processor **2006** or the safety processor **2004** to detect a value outside of the predetermined range, the activation signal is terminated and operation of at least one circuit segment **2002c-2002h**, such as, for example, the motor segment **2002g**, is interrupted and/or prevented. For example, in some embodiments, the activation signal from the primary processor **2006** and the activation signal from the safety processor **2004** are coupled to an AND gate. The AND gate is coupled to a motor power switch **2020**. The AND gate maintains the motor power switch **2020** in a closed, or on, position when the activation signal from both the safety processor **2004** and the primary processor **2006** are high, indicating a value of the motor sensors **2040a**, **2040b** within the predetermined range. When either of the motor sensors **2040a**, **2040b** detect a value outside of the predetermined range, the activation signal from that motor sensor **2040a**, **2040b** is set low, and the output of the AND gate is set low, opening the motor power switch **2020**. In some embodiments, the value of the first sensor **2040a** and the second sensor **2040b** is compared, for example, by the safety processor **2004** and/or the primary processor **2006**. When the values of the first sensor and the second sensor are different, the safety processor **2004** and/or the primary processor **2006** may prevent operation of the motor segment **2002g**.

In some embodiments, the safety processor **2004** receives a signal indicative of the value of the second sensor **2040b** and compares the second sensor value to the first sensor value. For example, in one embodiment, the safety processor **2004** is coupled directly to a first motor sensor **2040a**. A second motor sensor **2040b** is coupled to a primary processor **2006**, which provides the second motor sensor **2040b** value to the safety processor **2004**, and/or coupled directly to the safety processor **2004**. The safety processor **2004** compares the value of the first motor sensor **2040a** to the value of the second motor sensor **2040b**. When the safety processor **2004** detects a mismatch between the first motor sensor **2040a** and the second motor sensor **2040b**, the safety processor **2004** may interrupt operation of the motor segment **2002g**, for example, by cutting power to the motor segment **2002g**.

In some embodiments, the safety processor **2004** and/or the primary processor **2006** is coupled to a first sensor **2040a** configured to measure a first property of a surgical instrument and a second sensor **2040b** configured to measure a second property of the surgical instrument. The first property and the second property comprise a predetermined relationship when the surgical instrument is operating normally. The safety processor **2004** monitors the first property and the second property. When a value of the first property and/or the second property inconsistent with the predetermined relationship is detected, a fault occurs. When a fault occurs, the safety processor **2004** takes at least one action, such as, for example, preventing operation of at least one of the circuit segments, executing a predetermined operation, and/or resetting the primary processor **2006**. For example, the safety processor **2004** may open the motor power switch **2020** to cut power to the motor circuit segment **2002g** when a fault is detected.

In one embodiment, the safety processor **2004** is configured to execute an independent control algorithm. In operation, the safety processor **2004** monitors the segmented circuit **2000** and is configured to control and/or override signals from other circuit components, such as, for example, the primary processor **2006**, independently. The safety processor

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cessor **2004** may execute a preprogrammed algorithm and/or may be updated or programmed on the fly during operation based on one or more actions and/or positions of the surgical instrument **10**. For example, in one embodiment, the safety processor **2004** is reprogrammed with new parameters and/or safety algorithms each time a new shaft and/or end effector is coupled to the surgical instrument **10**. In some embodiments, one or more safety values stored by the safety processor **2004** are duplicated by the primary processor **2006**. Two-way error detection is performed to ensure values and/or parameters stored by either of the processors **2004**, **2006** are correct.

In some embodiments, the safety processor **2004** and the primary processor **2006** implement a redundant safety check. The safety processor **2004** and the primary processor **2006** provide periodic signals indicating normal operation. For example, during operation, the safety processor **2004** may indicate to the primary processor **2006** that the safety processor **2004** is executing code and operating normally. The primary processor **2006** may, likewise, indicate to the safety processor **2004** that the primary processor **2006** is executing code and operating normally. In some embodiments, communication between the safety processor **2004** and the primary processor **2006** occurs at a predetermined interval. The predetermined interval may be constant or may be variable based on the circuit state and/or operation of the surgical instrument **10**.

FIG. 22 illustrates one example of a power assembly **2100** comprising a usage cycle circuit **2102** configured to monitor a usage cycle count of the power assembly **2100**. The power assembly **2100** may be coupled to a surgical instrument **2110**. The usage cycle circuit **2102** comprises a processor **2104** and a use indicator **2106**. The use indicator **2106** is configured to provide a signal to the processor **2104** to indicate a use of the battery back **2100** and/or a surgical instrument **2110** coupled to the power assembly **2100**. A “use” may comprise any suitable action, condition, and/or parameter such as, for example, changing a modular component of a surgical instrument **2110**, deploying or firing a disposable component coupled to the surgical instrument **2110**, delivering electrosurgical energy from the surgical instrument **2110**, reconditioning the surgical instrument **2110** and/or the power assembly **2100**, exchanging the power assembly **2100**, recharging the power assembly **2100**, and/or exceeding a safety limitation of the surgical instrument **2110** and/or the battery back **2100**.

In some instances, a usage cycle, or use, is defined by one or more power assembly **2100** parameters. For example, in one instance, a usage cycle comprises using more than 5% of the total energy available from the power assembly **2100** when the power assembly **2100** is at a full charge level. In another instance, a usage cycle comprises a continuous energy drain from the power assembly **2100** exceeding a predetermined time limit. For example, a usage cycle may correspond to five minutes of continuous and/or total energy draw from the power assembly **2100**. In some instances, the power assembly **2100** comprises a usage cycle circuit **2102** having a continuous power draw to maintain one or more components of the usage cycle circuit **2102**, such as, for example, the use indicator **2106** and/or a counter **2108**, in an active state.

The processor **2104** maintains a usage cycle count. The usage cycle count indicates the number of uses detected by the use indicator **2106** for the power assembly **2100** and/or the surgical instrument **2110**. The processor **2104** may increment and/or decrement the usage cycle count based on input from the use indicator **2106**. The usage cycle count is

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used to control one or more operations of the power assembly **2100** and/or the surgical instrument **2110**. For example, in some instances, a power assembly **2100** is disabled when the usage cycle count exceeds a predetermined usage limit. Although the instances discussed herein are discussed with respect to incrementing the usage cycle count above a predetermined usage limit, those skilled in the art will recognize that the usage cycle count may start at a predetermined amount and may be decremented by the processor **2104**. In this instance, the processor **2104** initiates and/or prevents one or more operations of the power assembly **2100** when the usage cycle count falls below a predetermined usage limit.

The usage cycle count is maintained by a counter **2108**. The counter **2108** comprises any suitable circuit, such as, for example, a memory module, an analog counter, and/or any circuit configured to maintain a usage cycle count. In some instances, the counter **2108** is formed integrally with the processor **2104**. In other instances, the counter **2108** comprises a separate component, such as, for example, a solid state memory module. In some instances, the usage cycle count is provided to a remote system, such as, for example, a central database. The usage cycle count is transmitted by a communications module **2112** to the remote system. The communications module **2112** is configured to use any suitable communications medium, such as, for example, wired and/or wireless communication. In some instances, the communications module **2112** is configured to receive one or more instructions from the remote system, such as, for example, a control signal when the usage cycle count exceeds the predetermined usage limit.

In some instances, the use indicator **2106** is configured to monitor the number of modular components used with a surgical instrument **2110** coupled to the power assembly **2100**. A modular component may comprise, for example, a modular shaft, a modular end effector, and/or any other modular component. In some instances, the use indicator **2106** monitors the use of one or more disposable components, such as, for example, insertion and/or deployment of a staple cartridge within an end effector coupled to the surgical instrument **2110**. The use indicator **2106** comprises one or more sensors for detecting the exchange of one or more modular and/or disposable components of the surgical instrument **2110**.

In some instances, the use indicator **2106** is configured to monitor single patient surgical procedures performed while the power assembly **2100** is installed. For example, the use indicator **2106** may be configured to monitor firings of the surgical instrument **2110** while the power assembly **2100** is coupled to the surgical instrument **2110**. A firing may correspond to deployment of a staple cartridge, application of electrosurgical energy, and/or any other suitable surgical event. The use indicator **2106** may comprise one or more circuits for measuring the number of firings while the power assembly **2100** is installed. The use indicator **2106** provides a signal to the processor **2104** when a single patient procedure is performed and the processor **2104** increments the usage cycle count.

In some instances, the use indicator **2106** comprises a circuit configured to monitor one or more parameters of the power source **2114**, such as, for example, a current draw from the power source **2114**. The one or more parameters of the power source **2114** correspond to one or more operations performable by the surgical instrument **2110**, such as, for example, a cutting and sealing operation. The use indicator **2106** provides the one or more parameters to the processor

2104, which increments the usage cycle count when the one or more parameters indicate that a procedure has been performed.

In some instances, the use indicator **2106** comprises a timing circuit configured to increment a usage cycle count after a predetermined time period. The predetermined time period corresponds to a single patient procedure time, which is the time required for an operator to perform a procedure, such as, for example, a cutting and sealing procedure. When the power assembly **2100** is coupled to the surgical instrument **2110**, the processor **2104** polls the use indicator **2106** to determine when the single patient procedure time has expired. When the predetermined time period has elapsed, the processor **2104** increments the usage cycle count. After incrementing the usage cycle count, the processor **2104** resets the timing circuit of the use indicator **2106**.

In some instances, the use indicator **2106** comprises a time constant that approximates the single patient procedure time. In one embodiment, the usage cycle circuit **2102** comprises a resistor-capacitor (RC) timing circuit **2506**. The RC timing circuit comprises a time constant defined by a resistor-capacitor pair. The time constant is defined by the values of the resistor and the capacitor. In one embodiment, the usage cycle circuit **2552** comprises a rechargeable battery and a clock. When the power assembly **2100** is installed in a surgical instrument, the rechargeable battery is charged by the power source. The rechargeable battery comprises enough power to run the clock for at least the single patient procedure time. The clock may comprise a real time clock, a processor configured to implement a time function, or any other suitable timing circuit.

Referring back to FIG. 2, in some instances, the use indicator **2106** comprises a sensor configured to monitor one or more environmental conditions experienced by the power assembly **2100**. For example, the use indicator **2106** may comprise an accelerometer. The accelerometer is configured to monitor acceleration of the power assembly **2100**. The power assembly **2100** comprises a maximum acceleration tolerance. Acceleration above a predetermined threshold indicates, for example, that the power assembly **2100** has been dropped. When the use indicator **2106** detects acceleration above the maximum acceleration tolerance, the processor **2104** increments a usage cycle count. In some instances, the use indicator **2106** comprises a moisture sensor. The moisture sensor is configured to indicate when the power assembly **2100** has been exposed to moisture. The moisture sensor may comprise, for example, an immersion sensor configured to indicate when the power assembly **2100** has been fully immersed in a cleaning fluid, a moisture sensor configured to indicate when moisture is in contact with the power assembly **2100** during use, and/or any other suitable moisture sensor.

In some instances, the use indicator **2106** comprises a chemical exposure sensor. The chemical exposure sensor is configured to indicate when the power assembly **2100** has come into contact with harmful and/or dangerous chemicals. For example, during a sterilization procedure, an inappropriate chemical may be used that leads to degradation of the power assembly **2100**. The processor **2104** increments the usage cycle count when the use indicator **2106** detects an inappropriate chemical.

In some instances, the usage cycle circuit **2102** is configured to monitor the number of reconditioning cycles experienced by the power assembly **2100**. A reconditioning cycle may comprise, for example, a cleaning cycle, a sterilization cycle, a charging cycle, routine and/or preventative maintenance, and/or any other suitable reconditioning cycle.

The use indicator **2106** is configured to detect a reconditioning cycle. For example, the use indicator **2106** may comprise a moisture sensor to detect a cleaning and/or sterilization cycle. In some instances, the usage cycle circuit **2102** monitors the number of reconditioning cycles experienced by the power assembly **2100** and disables the power assembly **2100** after the number of reconditioning cycles exceeds a predetermined threshold.

The usage cycle circuit **2102** may be configured to monitor the number of power assembly **2100** exchanges. The usage cycle circuit **2102** increments the usage cycle count each time the power assembly **2100** is exchanged. When the maximum number of exchanges is exceeded the usage cycle circuit **2102** locks out the power assembly **2100** and/or the surgical instrument **2110**. In some instances, when the power assembly **2100** is coupled the surgical instrument **2110**, the usage cycle circuit **2102** identifies the serial number of the power assembly **2100** and locks the power assembly **2100** such that the power assembly **2100** is usable only with the surgical instrument **2110**. In some instances, the usage cycle circuit **2102** increments the usage cycle each time the power assembly **2100** is removed from and/or coupled to the surgical instrument **2110**.

In some instances, the usage cycle count corresponds to sterilization of the power assembly **2100**. The use indicator **2106** comprises a sensor configured to detect one or more parameters of a sterilization cycle, such as, for example, a temperature parameter, a chemical parameter, a moisture parameter, and/or any other suitable parameter. The processor **2104** increments the usage cycle count when a sterilization parameter is detected. The usage cycle circuit **2102** disables the power assembly **2100** after a predetermined number of sterilizations. In some instances, the usage cycle circuit **2102** is reset during a sterilization cycle, a voltage sensor to detect a recharge cycle, and/or any suitable sensor. The processor **2104** increments the usage cycle count when a reconditioning cycle is detected. The usage cycle circuit **2102** is disabled when a sterilization cycle is detected. The usage cycle circuit **2102** is reactivated and/or reset when the power assembly **2100** is coupled to the surgical instrument **2110**. In some instances, the use indicator comprises a zero power indicator. The zero power indicator changes state during a sterilization cycle and is checked by the processor **2104** when the power assembly **2100** is coupled to a surgical instrument **2110**. When the zero power indicator indicates that a sterilization cycle has occurred, the processor **2104** increments the usage cycle count.

A counter **2108** maintains the usage cycle count. In some instances, the counter **2108** comprises a non-volatile memory module. The processor **2104** increments the usage cycle count stored in the non-volatile memory module each time a usage cycle is detected. The memory module may be accessed by the processor **2104** and/or a control circuit, such as, for example, the control circuit **200**. When the usage cycle count exceeds a predetermined threshold, the processor **2104** disables the power assembly **2100**. In some instances, the usage cycle count is maintained by a plurality of circuit components. For example, in one instance, the counter **2108** comprises a resistor (or fuse) pack. After each use of the power assembly **2100**, a resistor (or fuse) is burned to an open position, changing the resistance of the resistor pack. The power assembly **2100** and/or the surgical instrument **2110** reads the remaining resistance. When the last resistor of the resistor pack is burned out, the resistor pack has a predetermined resistance, such as, for example, an infinite resistance corresponding to an open circuit, which indicates that the power assembly **2100** has reached its usage

limit. In some instances, the resistance of the resistor pack is used to derive the number of uses remaining.

In some instances, the usage cycle circuit **2102** prevents further use of the power assembly **2100** and/or the surgical instrument **2110** when the usage cycle count exceeds a predetermined usage limit. In one instance, the usage cycle count associated with the power assembly **2100** is provided to an operator, for example, utilizing a screen formed integrally with the surgical instrument **2110**. The surgical instrument **2110** provides an indication to the operator that the usage cycle count has exceeded a predetermined limit for the power assembly **2100**, and prevents further operation of the surgical instrument **2110**.

In some instances, the usage cycle circuit **2102** is configured to physically prevent operation when the predetermined usage limit is reached. For example, the power assembly **2100** may comprise a shield configured to deploy over contacts of the power assembly **2100** when the usage cycle count exceeds the predetermined usage limit. The shield prevents recharge and use of the power assembly **2100** by covering the electrical connections of the power assembly **2100**.

In some instances, the usage cycle circuit **2102** is located at least partially within the surgical instrument **2110** and is configured to maintain a usage cycle count for the surgical instrument **2110**. FIG. 22 illustrates one or more components of the usage cycle circuit **2102** within the surgical instrument **2110** in phantom, illustrating the alternative positioning of the usage cycle circuit **2102**. When a predetermined usage limit of the surgical instrument **2110** is exceeded, the usage cycle circuit **2102** disables and/or prevents operation of the surgical instrument **2110**. The usage cycle count is incremented by the usage cycle circuit **2102** when the use indicator **2106** detects a specific event and/or requirement, such as, for example, firing of the surgical instrument **2110**, a predetermined time period corresponding to a single patient procedure time, based on one or more motor parameters of the surgical instrument **2110**, in response to a system diagnostic indicating that one or more predetermined thresholds are met, and/or any other suitable requirement. As discussed above, in some instances, the use indicator **2106** comprises a timing circuit corresponding to a single patient procedure time. In other instances, the use indicator **2106** comprises one or more sensors configured to detect a specific event and/or condition of the surgical instrument **2110**.

In some instances, the usage cycle circuit **2102** is configured to prevent operation of the surgical instrument **2110** after the predetermined usage limit is reached. In some instances, the surgical instrument **2110** comprises a visible indicator to indicate when the predetermined usage limit has been reached and/or exceeded. For example, a flag, such as a red flag, may pop-up from the surgical instrument **2110**, such as from the handle, to provide a visual indication to the operator that the surgical instrument **2110** has exceeded the predetermined usage limit. As another example, the usage cycle circuit **2102** may be coupled to a display formed integrally with the surgical instrument **2110**. The usage cycle circuit **2102** displays a message indicating that the predetermined usage limit has been exceeded. The surgical instrument **2110** may provide an audible indication to the operator that the predetermined usage limit has been exceeded. For example, in one instance, the surgical instrument **2110** emits an audible tone when the predetermined usage limit is exceeded and the power assembly **2100** is removed from the surgical instrument **2110**. The audible tone indicates the last

use of the surgical instrument **2110** and indicates that the surgical instrument **2110** should be disposed or reconditioned.

In some instances, the usage cycle circuit **2102** is configured to transmit the usage cycle count of the surgical instrument **2110** to a remote location, such as, for example, a central database. The usage cycle circuit **2102** comprises a communications module **2112** configured to transmit the usage cycle count to the remote location. The communications module **2112** may utilize any suitable communications system, such as, for example, wired or wireless communications system. The remote location may comprise a central database configured to maintain usage information. In some instances, when the power assembly **2100** is coupled to the surgical instrument **2110**, the power assembly **2100** records a serial number of the surgical instrument **2110**. The serial number is transmitted to the central database, for example, when the power assembly **2100** is coupled to a charger. In some instances, the central database maintains a count corresponding to each use of the surgical instrument **2110**. For example, a bar code associated with the surgical instrument **2110** may be scanned each time the surgical instrument **2110** is used. When the use count exceeds a predetermined usage limit, the central database provides a signal to the surgical instrument **2110** indicating that the surgical instrument **2110** should be discarded.

The surgical instrument **2110** may be configured to lock and/or prevent operation of the surgical instrument **2110** when the usage cycle count exceeds a predetermined usage limit. In some instances, the surgical instrument **2110** comprises a disposable instrument and is discarded after the usage cycle count exceeds the predetermined usage limit. In other instances, the surgical instrument **2110** comprises a reusable surgical instrument which may be reconditioned after the usage cycle count exceeds the predetermined usage limit. The surgical instrument **2110** initiates a reversible lockout after the predetermined usage limit is met. A technician reconditions the surgical instrument **2110** and releases the lockout, for example, utilizing a specialized technician key configured to reset the usage cycle circuit **2102**.

In some embodiments, the segmented circuit **2000** is configured for sequential start-up. An error check is performed by each circuit segment **2002a-2002g** prior to energizing the next sequential circuit segment **2002a-2002g**. FIG. 23 illustrates one embodiment of a process for sequentially energizing a segmented circuit **2270**, such as, for example, the segmented circuit **2000**. When a battery **2008** is coupled to the segmented circuit **2000**, the safety processor **2004** is energized **2272**. The safety processor **2004** performs a self-error check **2274**. When an error is detected **2276a**, the safety processor stops energizing the segmented circuit **2000** and generates an error code **2278a**. When no errors are detected **2276b**, the safety processor **2004** initiates **2278b** power-up of the primary processor **2006**. The primary processor **2006** performs a self-error check. When no errors are detected, the primary processor **2006** begins sequential power-up of each of the remaining circuit segments **2278b**. Each circuit segment is energized and error checked by the primary processor **2006**. When no errors are detected, the next circuit segment is energized **2278b**. When an error is detected, the safety processor **2004** and/or the primary process stops energizing the current segment and generates an error **2278a**. The sequential start-up continues until all of the circuit segments **2002a-2002g** have been energized. In some embodiments, the segmented circuit **2000** transitions from sleep mode following a similar sequential power-up process **11250**.

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FIG. 24 illustrates one embodiment of a power segment 2302 comprising a plurality of daisy chained power converters 2314, 2316, 2318. The power segment 2302 comprises a battery 2308. The battery 2308 is configured to provide a source voltage, such as, for example, 12V. A current sensor 2312 is coupled to the battery 2308 to monitor the current draw of a segmented circuit and/or one or more circuit segments. The current sensor 2312 is coupled to an FET switch 2313. The battery 2308 is coupled to one or more voltage converters 2309, 2314, 2316. An always on converter 2309 provides a constant voltage to one or more circuit components, such as, for example, a motion sensor 2322. The always on converter 2309 comprises, for example, a 3.3V converter. The always on converter 2309 may provide a constant voltage to additional circuit components, such as, for example, a safety processor (not shown). The battery 2308 is coupled to a boost converter 2318. The boost converter 2318 is configured to provide a boosted voltage above the voltage provided by the battery 2308. For example, in the illustrated embodiment, the battery 2308 provides a voltage of 12V. The boost converter 2318 is configured to boost the voltage to 13V. The boost converter 2318 is configured to maintain a minimum voltage during operation of a surgical instrument, for example, the surgical instrument 10 illustrated in FIGS. 69-71. Operation of a motor can result in the power provided to the primary processor 2306 dropping below a minimum threshold and creating a brownout or reset condition in the primary processor 2306. The boost converter 2318 ensures that sufficient power is available to the primary processor 2306 and/or other circuit components, such as the motor controller 2343, during operation of the surgical instrument 10. In some embodiments, the boost converter 2318 is coupled directly one or more circuit components, such as, for example, an OLED display 2388.

The boost converter 2318 is coupled to one or more step-down converters to provide voltages below the boosted voltage level. A first voltage converter 2316 is coupled to the boost converter 2318 and provides a first stepped-down voltage to one or more circuit components. In the illustrated embodiment, the first voltage converter 2316 provides a voltage of 5V. The first voltage converter 2316 is coupled to a rotary position encoder 2340. A FET switch 2317 is coupled between the first voltage converter 2316 and the rotary position encoder 2340. The FET switch 2317 is controlled by the processor 2306. The processor 2306 opens the FET switch 2317 to deactivate the position encoder 2340, for example, during power intensive operations. The first voltage converter 2316 is coupled to a second voltage converter 2314 configured to provide a second stepped-down voltage. The second stepped-down voltage comprises, for example, 3.3V. The second voltage converter 2314 is coupled to a processor 2306. In some embodiments, the boost converter 2318, the first voltage converter 2316, and the second voltage converter 2314 are coupled in a daisy chain configuration. The daisy chain configuration allows the use of smaller, more efficient converters for generating voltage levels below the boosted voltage level. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

FIG. 25 illustrates one embodiment of a segmented circuit 2400 configured to maximize power available for critical and/or power intense functions. The segmented circuit 2400 comprises a battery 2408. The battery 2408 is configured to provide a source voltage such as, for example, 12V. The source voltage is provided to a plurality of voltage converters 2409, 2418. An always-on voltage converter 2409 pro-

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vides a constant voltage to one or more circuit components, for example, a motion sensor 2422 and a safety processor 2404. The always-on voltage converter 2409 is directly coupled to the battery 2408. The always-on converter 2409 provides a voltage of 3.3V, for example. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

The segmented circuit 2400 comprises a boost converter 2418. The boost converter 2418 provides a boosted voltage above the source voltage provided by the battery 2408, such as, for example, 13V. The boost converter 2418 provides a boosted voltage directly to one or more circuit components, such as, for example, an OLED display 2488 and a motor controller 2443. By coupling the OLED display 2488 directly to the boost converter 2418, the segmented circuit 2400 eliminates the need for a power converter dedicated to the OLED display 2488. The boost converter 2418 provides a boosted voltage to the motor controller 2443 and the motor 2448 during one or more power intensive operations of the motor 2448, such as, for example, a cutting operation. The boost converter 2418 is coupled to a step-down converter 2416. The step-down converter 2416 is configured to provide a voltage below the boosted voltage to one or more circuit components, such as, for example, 5V. The step-down converter 2416 is coupled to, for example, a FET switch 2451 and a position encoder 2440. The FET switch 2451 is coupled to the primary processor 2406. The primary processor 2406 opens the FET switch 2451 when transitioning the segmented circuit 2400 to sleep mode and/or during power intensive functions requiring additional voltage delivered to the motor 2448. Opening the FET switch 2451 deactivates the position encoder 2440 and eliminates the power draw of the position encoder 2440. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

The step-down converter 2416 is coupled to a linear converter 2414. The linear converter 2414 is configured to provide a voltage of, for example, 3.3V. The linear converter 2414 is coupled to the primary processor 2406. The linear converter 2414 provides an operating voltage to the primary processor 2406. The linear converter 2414 may be coupled to one or more additional circuit components. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

The segmented circuit 2400 comprises a bailout switch 2456. The bailout switch 2456 is coupled to a bailout door on the surgical instrument 10. The bailout switch 2456 and the safety processor 2404 are coupled to an AND gate 2419. The AND gate 2419 provides an input to a FET switch 2413. When the bailout switch 2456 detects a bailout condition, the bailout switch 2456 provides a bailout shutdown signal to the AND gate 2419. When the safety processor 2404 detects an unsafe condition, such as, for example, due to a sensor mismatch, the safety processor 2404 provides a shutdown signal to the AND gate 2419. In some embodiments, both the bailout shutdown signal and the shutdown signal are high during normal operation and are low when a bailout condition or an unsafe condition is detected. When the output of the AND gate 2419 is low, the FET switch 2413 is opened and operation of the motor 2448 is prevented. In some embodiments, the safety processor 2404 utilizes the shutdown signal to transition the motor 2448 to an off state in sleep mode. A third input to the FET switch 2413 is provided by a current sensor 2412 coupled to the battery 2408. The current sensor 2412 monitors the current drawn by the circuit 2400 and opens the FET switch 2413 to shut-off power to the motor 2448 when an electrical current

above a predetermined threshold is detected. The FET switch **2413** and the motor controller **2443** are coupled to a bank of FET switches **2445** configured to control operation of the motor **2448**.

A motor current sensor **2446** is coupled in series with the motor **2448** to provide a motor current sensor reading to a current monitor **2447**. The current monitor **2447** is coupled to the primary processor **2406**. The current monitor **2447** provides a signal indicative of the current draw of the motor **2448**. The primary processor **2406** may utilize the signal from the motor current **2447** to control operation of the motor, for example, to ensure the current draw of the motor **2448** is within an acceptable range, to compare the current draw of the motor **2448** to one or more other parameters of the circuit **2400** such as, for example, the position encoder **2440**, and/or to determine one or more parameters of a treatment site. In some embodiments, the current monitor **2447** may be coupled to the safety processor **2404**.

In some embodiments, actuation of one or more handle controls, such as, for example, a firing trigger, causes the primary processor **2406** to decrease power to one or more components while the handle control is actuated. For example, in one embodiment, a firing trigger controls a firing stroke of a cutting member. The cutting member is driven by the motor **2448**. Actuation of the firing trigger results in forward operation of the motor **2448** and advancement of the cutting member. During firing, the primary processor **2406** closes the FET switch **2451** to remove power from the position encoder **2440**. The deactivation of one or more circuit components allows higher power to be delivered to the motor **2448**. When the firing trigger is released, full power is restored to the deactivated components, for example, by closing the FET switch **2451** and reactivating the position encoder **2440**.

In some embodiments, the safety processor **2404** controls operation of the segmented circuit **2400**. For example, the safety processor **2404** may initiate a sequential power-up of the segmented circuit **2400**, transition of the segmented circuit **2400** to and from sleep mode, and/or may override one or more control signals from the primary processor **2406**. For example, in the illustrated embodiment, the safety processor **2404** is coupled to the step-down converter **2416**. The safety processor **2404** controls operation of the segmented circuit **2400** by activating or deactivating the step-down converter **2416** to provide power to the remainder of the segmented circuit **2400**.

FIG. 26 illustrates one embodiment of a power system **2500** comprising a plurality of daisy chained power converters **2514**, **2516**, **2518** configured to be sequentially energized. The plurality of daisy chained power converters **2514**, **2516**, **2518** may be sequentially activated by, for example, a safety processor during initial power-up and/or transition from sleep mode. The safety processor may be powered by an independent power converter (not shown). For example, in one embodiment, when a battery voltage V_{BATT} is coupled to the power system **2500** and/or an accelerometer detects movement in sleep mode, the safety processor initiates a sequential start-up of the daisy chained power converters **2514**, **2516**, **2518**. The safety processor activates the 13V boost section **2518**. The boost section **2518** is energized and performs a self-check. In some embodiments, the boost section **2518** comprises an integrated circuit **2520** configured to boost the source voltage and to perform a self check. A diode D prevents power-up of a 5V supply section **2516** until the boost section **2518** has completed a self-check and provided a signal to the diode D indicating that the boost section **2518** did not identify any

errors. In some embodiments, this signal is provided by the safety processor. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

The 5V supply section **2516** is sequentially powered-up after the boost section **2518**. The supply section **2516** performs a self-check during power-up to identify any errors in the 5V supply section **2516**. The 5V supply section **2516** comprises an integrated circuit **2515** configured to provide a step-down voltage from the boost voltage and to perform an error check. When no errors are detected, the supply section **2516** completes sequential power-up and provides an activation signal to the 3.3V supply section **2514**. In some embodiments, the safety processor provides an activation signal to the 3.3V supply section **2514**. The 3.3V supply section comprises an integrated circuit **2513** configured to provide a step-down voltage from the 5V supply section **2516** and perform a self-error check during power-up. When no errors are detected during the self-check, the 3.3V supply section **2514** provides power to the primary processor. The primary processor is configured to sequentially energize each of the remaining circuit segments. By sequentially energizing the power system **2500** and/or the remainder of a segmented circuit, the power system **2500** reduces error risks, allows for stabilization of voltage levels before loads are applied, and prevents large current draws from all hardware being turned on simultaneously in an uncontrolled manner. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

In one embodiment, the power system **2500** comprises an over voltage identification and mitigation circuit. The over voltage identification and mitigation circuit is configured to detect a monopolar return current in the surgical instrument and interrupt power from the power segment when the monopolar return current is detected. The over voltage identification and mitigation circuit is configured to identify ground floatation of the power system. The over voltage identification and mitigation circuit comprises a metal oxide varistor. The over voltage identification and mitigation circuit comprises at least one transient voltage suppression diode.

FIG. 27 illustrates one embodiment of a segmented circuit **2600** comprising an isolated control section **2602**. The isolated control section **2602** isolates control hardware of the segmented circuit **2600** from a power section (not shown) of the segmented circuit **2600**. The control section **2602** comprises, for example, a primary processor **2606**, a safety processor (not shown), and/or additional control hardware, for example, a FET Switch **2617**. The power section comprises, for example, a motor, a motor driver, and/or a plurality of motor MOSFETS. The isolated control section **2602** comprises a charging circuit **2603** and a rechargeable battery **2608** coupled to a 5V power converter **2616**. The charging circuit **2603** and the rechargeable battery **2608** isolate the primary processor **2606** from the power section. In some embodiments, the rechargeable battery **2608** is coupled to a safety processor and any additional support hardware. Isolating the control section **2602** from the power section allows the control section **2602**, for example, the primary processor **2606**, to remain active even when main power is removed, provides a filter, through the rechargeable battery **2608**, to keep noise out of the control section **2602**, isolates the control section **2602** from heavy swings in the battery voltage to ensure proper operation even during heavy motor loads, and/or allows for real-time operating system (RTOS) to be used by the segmented circuit **2600**. In some

embodiments, the rechargeable battery **2608** provides a stepped-down voltage to the primary processor, such as, for example, 3.3V. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

Use of Multiple Sensors with One Sensor Affecting a Second Sensor's Output or Interpretation

FIG. **28** illustrates one embodiment of an end effector **3000** comprising a first sensor **3008a** and a second sensor **3008b**. The end effector **3000** is similar to the end effector **300** described above. The end effector **3000** comprises a first jaw member, or anvil, **3002** pivotally coupled to a second jaw member **3004**. The second jaw member **3004** is configured to receive a staple cartridge **3006** therein. The staple cartridge **3006** comprises a plurality of staples (not shown). The plurality of staples is deployable from the staple cartridge **3006** during a surgical operation. The end effector **3000** comprises a first sensor **3008a**. The first sensor **3008a** is configured to measure one or more parameters of the end effector **3000**. For example, in one embodiment, the first sensor **3008a** is configured to measure the gap **3010** between the anvil **3002** and the second jaw member **3004**. The first sensor **3008a** may comprise, for example, a Hall effect sensor configured to detect a magnetic field generated by a magnet **3012** embedded in the second jaw member **3004** and/or the staple cartridge **3006**. As another example, in one embodiment, the first sensor **3008a** is configured to measure one or more forces exerted on the anvil **3002** by the second jaw member **3004** and/or tissue clamped between the anvil **3002** and the second jaw member **3004**.

The end effector **3000** comprises a second sensor **3008b**. The second sensor **3008b** is configured to measure one or more parameters of the end effector **3000**. For example, in various embodiments, the second sensor **3008b** may comprise a strain gauge configured to measure the magnitude of the strain in the anvil **3002** during a clamped condition. The strain gauge provides an electrical signal whose amplitude varies with the magnitude of the strain. In various embodiments, the first sensor **3008a** and/or the second sensor **3008b** may comprise, for example, a magnetic sensor such as, for example, a Hall effect sensor, a strain gauge, a pressure sensor, a force sensor, an inductive sensor such as, for example, an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor for measuring one or more parameters of the end effector **3000**. The first sensor **3008a** and the second sensor **3008b** may be arranged in a series configuration and/or a parallel configuration. In a series configuration, the second sensor **3008b** may be configured to directly affect the output of the first sensor **3008a**. In a parallel configuration, the second sensor **3008b** may be configured to indirectly affect the output of the first sensor **3008a**.

In one embodiment, the one or more parameters measured by the first sensor **3008a** are related to the one or more parameters measured by the second sensor **3008b**. For example, in one embodiment, the first sensor **3008a** is configured to measure the gap **3010** between the anvil **3002** and the second jaw member **3004**. The gap **3010** is representative of the thickness and/or compressibility of a tissue section clamped between the anvil **3002** and the staple cartridge **3006**. The first sensor **3008a** may comprise, for example, a Hall effect sensor configured to detect a magnetic field generated by a magnet **3012** coupled to the second jaw member **3004** and/or the staple cartridge **3006**. Measuring at a single location accurately describes the compressed tissue thickness for a calibrated full bit of tissue, but may provide inaccurate results when a partial bite of tissue is placed

between the anvil **3002** and the second jaw member **3004**. A partial bite of tissue, either a proximal partial bite or a distal partial bite, changes the clamping geometry of the anvil **3002**.

In some embodiments, the second sensor **3008b** is configured to detect one or more parameters indicative of a type of tissue bite, for example, a full bite, a partial proximal bite, and/or a partial distal bite. The measurement of the second sensor **3008b** may be used to adjust the measurement of the first sensor **3008a** to accurately represent a proximal or distal positioned partial bite's true compressed tissue thickness. For example, in one embodiment, the second sensor **3008b** comprises a strain gauge, such as, for example, a micro-strain gauge, configured to monitor the amplitude of the strain in the anvil during a clamped condition. The amplitude of the strain of the anvil **3002** is used to modify the output of the first sensor **3008a**, for example, a Hall effect sensor, to accurately represent a proximal or distal positioned partial bite's true compressed tissue thickness. The first sensor **3008a** and the second sensor **3008b** may be measured in real-time during a clamping operation. Real-time measurement allows time based information to be analyzed, for example, by the primary processor **2006**, and used to select one or more algorithms and/or look-up tables to recognize tissue characteristics and clamping positioning to dynamically adjust tissue thickness measurements.

In some embodiments, the thickness measurement of the first sensor **3008a** may be provided to an output device of a surgical instrument **10** coupled to the end effector **3000**. For example, in one embodiment, the end effector **3000** is coupled to the surgical instrument **10** comprising a display **2028**. The measurement of the first sensor **3008a** is provided to a processor, for example, the primary processor **2006**. The primary processor **2006** adjusts the measurement of the first sensor **3008a** based on the measurement of the second sensor **3008b** to reflect the true tissue thickness of a tissue section clamped between the anvil **3002** and the staple cartridge **3006**. The primary processor **2006** outputs the adjusted tissue thickness measurement and an indication of full or partial bite to the display **2028**. An operator may determine whether or not to deploy the staples in the staple cartridge **3006** based on the displayed values.

In some embodiments, the first sensor **3008a** and the second sensor **3008b** may be located in different environments, such as, for example, the first sensor **3008a** being located within a patient at a treatment site and the second sensor **3008b** being located externally to the patient. The second sensor **3008b** may be configured to calibrate and/or modify the output of the first sensor **3008a**. The first sensor **3008a** and/or the second sensor **3008b** may comprise, for example, an environmental sensor. Environmental sensors may comprise, for example, temperature sensors, humidity sensors, pressure sensors, and/or any other suitable environmental sensor.

FIG. **29** is a logic diagram illustrating one embodiment of a process **3020** for adjusting the measurement of a first sensor **3008a** based on input from a second sensor **3008b**. A first signal is captured **3022a** by the first sensor **3008a**. The first signal **3022a** may be conditioned based on one or more predetermined parameters, such as, for example, a smoothing function, a look-up table, and/or any other suitable conditioning parameters. A second signal is captured **3022b** by the second sensor **3008b**. The second signal **3022b** may be conditioned based on one or more predetermined conditioning parameters. The first signal **3022a** and the second signal **3022b** are provided to a processor, such as, for example, the primary processor **2006**. The processor **2006**

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adjusts the measurement of the first sensor **3022a**, as represented by the first signal **3022a**, based on the second signal **3022b** from the second sensor. For example, in one embodiment, the first sensor **3022a** comprises a Hall effect sensor and the second sensor **3022b** comprises a strain gauge. The distance measurement of the first sensor **3022a** is adjusted by the amplitude of the strain measured by the second sensor **3022b** to determine the fullness of the bite of tissue in the end effector **3000**. The adjusted measurement is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

FIG. **30** is a logic diagram illustrating one embodiment of a process **3030** for determining a look-up table for a first sensor **3008a** based on the input from a second sensor **3008b**. The first sensor **3008a** captures **3022a** a signal indicative of one or more parameters of the end effector **3000**. The first signal **3022a** may be conditioned based on one or more predetermined parameters, such as, for example, a smoothing function, a look-up table, and/or any other suitable conditioning parameters. A second signal is captured **3022b** by the second sensor **3008b**. The second signal **3022b** may be conditioned based on one or more predetermined conditioning parameters. The first signal **3022a** and the second signal **3022b** are provided to a processor, such as, for example, the primary processor **2006**. The processor **2006** selects a look-up table from one or more available look-up tables **3034a**, **3034b** based on the value of the second signal. The selected look-up table is used to convert the first signal into a thickness measurement of the tissue located between the anvil **3002** and the staple cartridge **3006**. The adjusted measurement is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

FIG. **31** is a logic diagram illustrating one embodiment of a process **3040** for calibrating a first sensor **3008a** in response to an input from a second sensor **3008b**. The first sensor **3008a** is configured to capture **3022a** a signal indicative of one or more parameters of the end effector **3000**. The first signal **3022a** may be conditioned based on one or more predetermined parameters, such as, for example, a smoothing function, a look-up table, and/or any other suitable conditioning parameters. A second signal is captured **3022b** by the second sensor **3008b**. The second signal **3022b** may be conditioned based on one or more predetermined conditioning parameters. The first signal **3022a** and the second signal **3022b** are provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** calibrates **3042** the first signal **3022a** in response to the second signal **3022b**. The first signal **3022a** is calibrated **3042** to reflect the fullness of the bite of tissue in the end effector **3000**. The calibrated signal is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

FIG. **32A** is a logic diagram illustrating one embodiment of a process **3050** for determining and displaying the thickness of a tissue section clamped between the anvil **3002** and the staple cartridge **3006** of the end effector **3000**. The process **3050** comprises obtaining a Hall effect voltage **3052**, for example, through a Hall effect sensor located at the distal tip of the anvil **3002**. The Hall effect voltage **3052** is provided to an analog to digital convertor **3054** and converted into a digital signal. The digital signal is provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** calibrates **3056** the curve input of the Hall effect voltage **3052** signal. A strain gauge **3058**, such as, for example, a micro-strain gauge, is configured to measure one or more parameters of the end effector

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3000, such as, for example, the amplitude of the strain exerted on the anvil **3002** during a clamping operation. The measured strain is converted **3060** to a digital signal and provided to the processor, such as, for example, the primary processor **2006**. The primary processor **2006** uses one or more algorithms and/or lookup tables to adjust the Hall effect voltage **3052** in response to the strain measured by the strain gauge **3058** to reflect the true thickness and fullness of the bite of tissue clamped by the anvil **3002** and the staple cartridge **3006**. The adjusted thickness is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

In some embodiments, the surgical instrument can further comprise a load cell or sensor **3082**. The load sensor **3082** can be located, for instance, in the shaft assembly **200**, described above, or in the housing **12**, also described above. FIG. **32B** is a logic diagram illustrating one embodiment of a process **3070** for determining and displaying the thickness of a tissue section clamped between the anvil **3002** and the staple cartridge **3006** of the end effector **3000**. The process comprises obtaining a Hall effect voltage **3072**, for example, through a Hall effect sensor located at the distal tip of the anvil **3002**. The Hall effect voltage **3072** is provided to an analog to digital convertor **3074** and converted into a digital signal. The digital signal is provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** applies calibrates **3076** the curve input of the Hall effect voltage **3072** signal. A strain gauge **3078**, such as, for example, a micro-strain gauge, is configured to measure one or more parameters of the end effector **3000**, such as, for example, the amplitude of the strain exerted on the anvil **3002** during a clamping operation. The measured strain is converted **3080** to a digital signal and provided to the processor, such as, for example, the primary processor **2006**. The load sensor **3082** measures the clamping force of the anvil **3002** against the staple cartridge **3006**. The measured clamping force is converted **3084** to a digital signal and provided to the processor, such as for example, the primary processor **2006**. The primary processor **2006** uses one or more algorithms and/or lookup tables to adjust the Hall effect voltage **3072** in response to the strain measured by the strain gauge **3078** and the clamping force measured by the load sensor **3082** to reflect the true thickness and fullness of the bite of tissue clamped by the anvil **3002** and the staple cartridge **3006**. The adjusted thickness is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

FIG. **33** is a graph **3090** illustrating an adjusted Hall effect thickness measurement **3094** compared to an unmodified Hall effect thickness measurement **3092**. As shown in FIG. **33**, the unmodified Hall effect thickness measurement **3092** indicates a thicker tissue measurement, as the single sensor is unable to compensate for partial distal/proximal bites that result in incorrect thickness measurements. The adjusted thickness measurement **3094** is generated by, for example, the process **3050** illustrated in FIG. **32A**. The Hall effect thickness measurement **3092** is calibrated based on input from one or more additional sensors, such as, for example, a strain gauge. The adjusted Hall effect thickness **3094** reflects the true thickness of the tissue located between an anvil **3002** and a staple cartridge **3006**.

FIG. **34** illustrates one embodiment of an end effector **3100** comprising a first sensor **3108a** and a second sensor **3108b**. The end effector **3100** is similar to the end effector **3000** illustrated in FIG. **28**. The end effector **3100** comprises a first jaw member, or anvil, **3102** pivotally coupled to a second jaw member **3104**. The second jaw member **3104** is

configured to receive a staple cartridge **3106** therein. The end effector **3100** comprises a first sensor **3108a** coupled to the anvil **3102**. The first sensor **3108a** is configured to measure one or more parameters of the end effector **3100**, such as, for example, the gap **3110** between the anvil **3102** and the staple cartridge **3106**. The gap **3110** may correspond to, for example, a thickness of tissue clamped between the anvil **3102** and the staple cartridge **3106**. The first sensor **3108a** may comprise any suitable sensor for measuring one or more parameters of the end effector. For example, in various embodiments, the first sensor **3108a** may comprise a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

In some embodiments, the end effector **3100** comprises a second sensor **3108b**. The second sensor **3108b** is coupled to second jaw member **3104** and/or the staple cartridge **3106**. The second sensor **3108b** is configured to detect one or more parameters of the end effector **3100**. For example, in some embodiments, the second sensor **3108b** is configured to detect one or more instrument conditions such as, for example, a color of the staple cartridge **3106** coupled to the second jaw member **3104**, a length of the staple cartridge **3106**, a clamping condition of the end effector **3100**, the number of uses/number of remaining uses of the end effector **3100** and/or the staple cartridge **3106**, and/or any other suitable instrument condition. The second sensor **3108b** may comprise any suitable sensor for detecting one or more instrument conditions, such as, for example, a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

The end effector **3100** may be used in conjunction with any of the processes shown in FIGS. 29-33. For example, in one embodiment, input from the second sensor **3108b** may be used to calibrate the input of the first sensor **3108a**. The second sensor **3108b** may be configured to detect one or more parameters of the staple cartridge **3106**, such as, for example, the color and/or length of the staple cartridge **3106**. The detected parameters, such as the color and/or the length of the staple cartridge **3106**, may correspond to one or more properties of the cartridge, such as, for example, the height of the cartridge deck, the thickness of tissue useable/optimal for the staple cartridge, and/or the pattern of the staples in the staple cartridge **3106**. The known parameters of the staple cartridge **3106** may be used to adjust the thickness measurement provided by the first sensor **3108a**. For example, if the staple cartridge **3106** has a higher deck height, the thickness measurement provided by the first sensor **3108a** may be reduced to compensate for the added deck height. The adjusted thickness may be displayed to an operator, for example, through a display **2026** coupled to the surgical instrument **10**.

FIG. 35 illustrates one embodiment of an end effector **3150** comprising a first sensor **3158** and a plurality of second sensors **3160a**, **3160b**. The end effector **3150** comprises a first jaw member, or anvil, **3152** and a second jaw member **3154**. The second jaw member **3154** is configured to receive a staple cartridge **3156**. The anvil **3152** is pivotally moveable with respect to the second jaw member **3154** to clamp tissue between the anvil **3152** and the staple cartridge **3156**. The anvil comprises a first sensor **3158**. The first sensor **3158** is configured to detect one or more parameters of the end effector **3150**, such as, for example, the gap **3110** between the anvil **3152** and the staple cartridge **3156**. The

gap **3110** may correspond to, for example, a thickness of tissue clamped between the anvil **3152** and the staple cartridge **3156**. The first sensor **3158** may comprise any suitable sensor for measuring one or more parameters of the end effector. For example, in various embodiments, the first sensor **3158** may comprise a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

In some embodiments, the end effector **3150** comprises a plurality of secondary sensors **3160a**, **3160b**. The secondary sensors **3160a**, **3160b** are configured to detect one or more parameters of the end effector **3150**. For example, in some embodiments, the secondary sensors **3160a**, **3160b** are configured to measure an amplitude of strain exerted on the anvil **3152** during a clamping procedure. In various embodiments, the secondary sensors **3160a**, **3160b** may comprise a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor. The secondary sensors **3160a**, **3160b** may be configured to measure one or more identical parameters at different locations of the anvil **3152**, different parameters at identical locations on the anvil **3152**, and/or different parameters at different locations on the anvil **3152**.

FIG. 36 is a logic diagram illustrating one embodiment of a process **3170** for adjusting a measurement of a first sensor **3158** in response to a plurality of secondary sensors **3160a**, **3160b**. In one embodiment, a Hall effect voltage is obtained **3172**, for example, by a Hall effect sensor. The Hall effect voltage is converted **3174** by an analog to digital convertor. The converted Hall effect voltage signal is calibrated **3176**. The calibrated curve represents the thickness of a tissue section located between the anvil **3152** and the staple cartridge **3156**. A plurality of secondary measurements are obtained **3178a**, **3178b** by a plurality of secondary sensors, such as, for example, a plurality of strain gauges. The input of the strain gauges is converted **3180a**, **3180b** into one or more digital signals, for example, by a plurality of electronic μ Strain conversion circuits. The calibrated Hall effect voltage and the plurality of secondary measurements are provided to a processor, such as, for example, the primary processor **2006**. The primary processor utilizes the secondary measurements to adjust **3182** the Hall effect voltage, for example, by applying an algorithm and/or utilizing one or more look-up tables. The adjusted Hall effect voltage represents the true thickness and fullness of the bite of tissue clamped by the anvil **3152** and the staple cartridge **3156**. The adjusted thickness is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

FIG. 37 illustrates one embodiment of a circuit **3190** configured to convert signals from the first sensor **3158** and the plurality of secondary sensors **3160a**, **3160b** into digital signals receivable by a processor, such as, for example, the primary processor **2006**. The circuit **3190** comprises an analog-to-digital convertor **3194**. In some embodiments, the analog-to-digital convertor **3194** comprises a 4-channel, 18-bit analog to digital convertor. Those skilled in the art will recognize that the analog-to-digital convertor **3194** may comprise any suitable number of channels and/or bits to convert one or more inputs from analog to digital signals. The circuit **3190** comprises one or more level shifting resistors **3196** configured to receive an input from the first sensor **3158**, such as, for example, a Hall effect sensor. The

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level shifting resistors **3196** adjust the input from the first sensor, shifting the value to a higher or lower voltage depending on the input. The level shifting resistors **3196** provide the level-shifted input from the first sensor **3158** to the analog-to-digital convertor.

In some embodiments, a plurality of secondary sensors **3160a**, **3160b** are coupled to a plurality of bridges **3192a**, **3192b** within the circuit **3190**. The plurality of bridges **3192a**, **3192b** may provide filtering of the input from the plurality of secondary sensors **3160a**, **3160b**. After filtering the input signals, the plurality of bridges **3192a**, **3192b** provide the inputs from the plurality of secondary sensors **3160a**, **3160b** to the analog-to-digital convertor **3194**. In some embodiments, a switch **3198** coupled to one or more level shifting resistors may be coupled to the analog-to-digital convertor **3194**. The switch **3198** is configured to calibrate one or more of the input signals, such as, for example, an input from a Hall effect sensor. The switch **3198** may be engaged to provide one or more level shifting signals to adjust the input of one or more of the sensors, such as, for example, to calibrate the input of a Hall effect sensor. In some embodiments, the adjustment is not necessary, and the switch **3198** is left in the open position to decouple the level shifting resistors. The switch **3198** is coupled to the analog-to-digital convertor **3194**. The analog-to-digital convertor **3194** provides an output to one or more processors, such as, for example, the primary processor **2006**. The primary processor **2006** calculates one or more parameters of the end effector **3150** based on the input from the analog-to-digital convertor **3194**. For example, in one embodiment, the primary processor **2006** calculates a thickness of tissue located between the anvil **3152** and the staple cartridge **3156** based on input from one or more sensors **3158**, **3160a**, **3160b**.

FIG. **38** illustrates one embodiment of an end effector **3200** comprising a plurality of sensors **3208a-3208d**. The end effector **3200** comprises an anvil **3202** pivotally coupled to a second jaw member **3204**. The second jaw member **3204** is configured to receive a staple cartridge **3206** therein. The anvil **3202** comprises a plurality of sensors **3208a-3208d** thereon. The plurality of sensors **3208a-3208d** is configured to detect one or more parameters of the end effector **3200**, such as, for example, the anvil **3202**. The plurality of sensors **3208a-3208d** may comprise one or more identical sensors and/or different sensors. The plurality of sensors **3208a-3208d** may comprise, for example, magnetic sensors, such as a Hall effect sensor, strain gauges, pressure sensors, inductive sensors, such as an eddy current sensor, resistive sensors, capacitive sensors, optical sensors, and/or any other suitable sensors or combination thereof. For example, in one embodiment, the plurality of sensors **3208a-3208d** may comprise a plurality of strain gauges.

In one embodiment, the plurality of sensors **3208a-3208d** allows a robust tissue thickness sensing process to be implemented. By detecting various parameters along the length of the anvil **3202**, the plurality of sensors **3208a-3208d** allow a surgical instrument, such as, for example, the surgical instrument **10**, to calculate the tissue thickness in the jaws regardless of the bite, for example, a partial or full bite. In some embodiments, the plurality of sensors **3208a-3208d** comprises a plurality of strain gauges. The plurality of strain gauges is configured to measure the strain at various points on the anvil **3202**. The amplitude and/or the slope of the strain at each of the various points on the anvil **3202** can be used to determine the thickness of tissue in between the anvil **3202** and the staple cartridge **3206**. The plurality of strain gauges may be configured to optimize maximum amplitude and/or slope differences based on clamping

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dynamics to determine thickness, tissue placement, and/or material properties of the tissue. Time based monitoring of the plurality of sensors **3208a-3208d** during clamping allows a processor, such as, for example, the primary processor **2006**, to utilize algorithms and look-up tables to recognize tissue characteristics and clamping positions and dynamically adjust the end effector **3200** and/or tissue clamped between the anvil **3202** and the staple cartridge **3206**.

FIG. **39** is a logic diagram illustrating one embodiment of a process **3220** for determining one or more tissue properties based on a plurality of sensors **3208a-3208d**. In one embodiment, a plurality of sensors **3208a-3208d** generate **3222a-3222d** a plurality of signals indicative of one or more parameters of the end effector **3200**. The plurality of generated signals is converted **3224a-3224d** to digital signals and provided to a processor. For example, in one embodiment comprising a plurality of strain gauges, a plurality of electronic μ Strain (micro-strain) conversion circuits convert **3224a-3224d** the strain gauge signals to digital signals. The digital signals are provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** determines **3226** one or more tissue characteristics based on the plurality of signals. The processor **2006** may determine the one or more tissue characteristics by applying an algorithm and/or a look-up table. The one or more tissue characteristics are displayed **3026** to an operator, for example, by a display **2026** embedded in the surgical instrument **10**.

FIG. **40** illustrates one embodiment of an end effector **3250** comprising a plurality of sensors **3260a-3260d** coupled to a second jaw member **3254**. The end effector **3250** comprises an anvil **3252** pivotally coupled to a second jaw member **3254**. The anvil **3252** is moveable relative to the second jaw member **3254** to clamp one or more materials, such as, for example, a tissue section **3264**, therebetween. The second jaw member **3254** is configured to receive a staple cartridge **3256**. A first sensor **3258** is coupled to the anvil **3252**. The first sensor is configured to detect one or more parameters of the end effector **3150**, such as, for example, the gap **3110** between the anvil **3252** and the staple cartridge **3256**. The gap **3110** may correspond to, for example, a thickness of tissue clamped between the anvil **3252** and the staple cartridge **3256**. The first sensor **3258** may comprise any suitable sensor for measuring one or more parameters of the end effector. For example, in various embodiments, the first sensor **3258** may comprise a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

A plurality of secondary sensors **3260a-3260d** is coupled to the second jaw member **3254**. The plurality of secondary sensors **3260a-3260d** may be formed integrally with the second jaw member **3254** and/or the staple cartridge **3256**. For example, in one embodiment, the plurality of secondary sensors **3260a-3260d** is disposed on an outer row of the staple cartridge **3256** (see FIG. **41**). The plurality of secondary sensors **3260a-3260d** are configured to detect one or more parameters of the end effector **3250** and/or a tissue section **3264** clamped between the anvil **3252** and the staple cartridge **3256**. The plurality of secondary sensors **3260a-3260d** may comprise any suitable sensors for detecting one or more parameters of the end effector **3250** and/or the tissue section **3264**, such as, for example, magnetic sensors, such as a Hall effect sensor, strain gauges, pressure sensors, inductive sensors, such as an eddy current sensor, resistive

sensors, capacitive sensors, optical sensors, and/or any other suitable sensors or combination thereof. The plurality of secondary sensors **3260a-3260d** may comprise identical sensors and/or different sensors.

In some embodiments, the plurality of secondary sensors **3260a-3260d** comprises dual purpose sensors and tissue stabilizing elements. The plurality of secondary sensors **3260a-3260d** comprise electrodes and/or sensing geometries configured to create a stabilized tissue condition when the plurality of secondary sensors **3260a-3260d** are engaged with a tissue section **3264**, such as, for example, during a clamping operation. In some embodiments, one or more of the plurality of secondary sensors **3260a-3260d** may be replaced with non-sensing tissue stabilizing elements. The secondary sensors **3260a-3260d** create a stabilized tissue condition by controlling tissue flow, staple formation, and/or other tissue conditions during a clamping, stapling, and/or other treatment process.

FIG. **41** illustrates one embodiment of a staple cartridge **3270** comprising a plurality of sensors **3272a-3272h** formed integrally therein. The staple cartridge **3270** comprises a plurality of rows containing a plurality of holes for storing staples therein. One or more of the holes in the outer row **3278** are replaced with one of the plurality of sensors **3272a-3272h**. A cut-away section **3274** is shown to illustrate a sensor **3272f** coupled to a sensor wire **3276b**. The sensor wires **3276a**, **3276b** may comprise a plurality of wires for coupling the plurality of sensors **3272a-3272h** to one or more circuits of a surgical instrument, such as, for example, the surgical instrument **10**. In some embodiments, one or more of the plurality of sensors **3272a-3272h** comprise dual purpose sensor and tissue stabilizing elements having electrodes and/or sensing geometries configured to provide tissue stabilization. In some embodiments, the plurality of sensors **3272a-3272h** may be replaced with and/or copopulated with a plurality of tissue stabilizing elements. Tissue stabilization may be provided by, for example, controlling tissue flow and/or staple formation during a clamping and/or stapling process. The plurality of sensors **3272a-3272h** provide signals to one or more circuits of the surgical instrument **10** to enhance feedback of stapling performance and/or tissue thickness sensing.

FIG. **42** is a logic diagram illustrating one embodiment of a process **3280** for determining one or more parameters of a tissue section **3264** clamped within an end effector, such as, for example, the end effector **3250** illustrated in FIG. **40**. In one embodiment, a first sensor **3258** is configured to detect one or more parameters of the end effector **3250** and/or a tissue section **3264** located between the anvil **3252** and the staple cartridge **3256**. A first signal is generated **3282** by the first sensors **3258**. The first signal is indicative of the one or more parameters detected by the first sensor **3258**. One or more secondary sensors **3260** are configured to detect one or more parameters of the end effector **3250** and/or the tissue section **3264**. The secondary sensors **3260** may be configured to detect the same parameters, additional parameters, or different parameters as the first sensor **3258**. Secondary signals **3284** are generated by the secondary sensors **3260**. The secondary signals **3284** are indicative of the one or more parameters detected by the secondary sensors **3260**. The first signal and the secondary signals are provided to a processor, such as, for example, a primary processor **2006**. The processor **2006** adjusts **3286** the first signal generated by the first sensor **3258** based on input generated by the secondary sensors **3260**. The adjusted signal may be indicative of, for example, the true thickness of a tissue section **3264** and the

fullness of the bite. The adjusted signal is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

FIG. **43** illustrates one embodiment of an end effector **3300** comprising a plurality of redundant sensors **3308a**, **3308b**. The end effector **3300** comprises a first jaw member, or anvil, **3302** pivotally coupled to a second jaw member **3304**. The second jaw member **3304** is configured to receive a staple cartridge **3306** therein. The anvil **3302** is moveable with respect to the staple cartridge **3306** to grasp a material, such as, for example, a tissue section, between the anvil **3302** and the staple cartridge **3306**. A plurality of sensors **3308a**, **3308b** is coupled to the anvil. The plurality of sensors **3308a**, **3308b** are configured to detect one or more parameters of the end effector **3300** and/or a tissue section located between the anvil **3302** and the staple cartridge **3306**. In some embodiments, the plurality of sensors **3308a**, **3308b** are configured to detect a gap **3310** between the anvil **3302** and the staple cartridge **3306**. The gap **3310** may correspond to, for example, the thickness of tissue located between the anvil **3302** and the staple cartridge **3306**. The plurality of sensors **3308a**, **3308b** may detect the gap **3310** by, for example, detecting a magnetic field generated by a magnet **3312** coupled to the second jaw member **3304**.

In some embodiments, the plurality of sensors **3308a**, **3308b** comprise redundant sensors. The redundant sensors are configured to detect the same properties of the end effector **3300** and/or a tissue section located between the anvil **3302** and the staple cartridge **3306**. The redundant sensors may comprise, for example, Hall effect sensors configured to detect the gap **3310** between the anvil **3302** and the staple cartridge **3306**. The redundant sensors provide signals representative of one or more parameters allowing a processor, such as, for example, the primary processor **2006**, to evaluate the multiple inputs and determine the most reliable input. In some embodiments, the redundant sensors are used to reduce noise, false signals, and/or drift. Each of the redundant sensors may be measured in real-time during clamping, allowing time-based information to be analyzed and algorithms and/or look-up tables to recognize tissue characteristics and clamping positioning dynamically. The input of one or more of the redundant sensors may be adjusted and/or selected to identify the true tissue thickness and bite of a tissue section located between the anvil **3302** and the staple cartridge **3306**.

FIG. **44** is a logic diagram illustrating one embodiment of a process **3320** for selecting the most reliable output from a plurality of redundant sensors, such as, for example, the plurality of sensors **3308a**, **3308b** illustrated in FIG. **43**. In one embodiment, a first signal is generated by a first sensor **3308a**. The first signal is converted **3322a** by an analog-to-digital convertor. One or more additional signals are generated by one or more redundant sensors **3308b**. The one or more additional signals are converted **3322b** by an analog-to-digital convertor. The converted signals are provided to a processor, such as, for example, the primary processor **2006**. The primary processor evaluates **3324** the redundant inputs to determine the most reliable output. The most reliable output may be selected based on one or more parameters, such as, for example, algorithms, look-up tables, input from additional sensors, and/or instrument conditions. After selecting the most reliable output, the processor may adjust the output based on one or more additional sensors to reflect, for example, the true thickness and bite of a tissue section located between the anvil **3302** and the staple cartridge

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3306. The adjusted most reliable output is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

FIG. **45** illustrates one embodiment of an end effector **3350** comprising a sensor **3358** comprising a specific sampling rate to limit or eliminate false signals. The end effector **3350** comprises a first jaw member, or anvil, **3352** pivotably coupled to a second jaw member **3354**. The second jaw member **3354** is configured to receive a staple cartridge **3356** therein. The staple cartridge **3356** contains a plurality of staples that may be delivered to a tissue section located between the anvil **3352** and the staple cartridge **3356**. A sensor **3358** is coupled to the anvil **3352**. The sensor **3358** is configured to detect one or more parameters of the end effector **3350**, such as, for example, the gap **3364** between the anvil **3352** and the staple cartridge **3356**. The gap **3364** may correspond to the thickness of a material, such as, for example, a tissue section, and/or the fullness of a bite of material located between the anvil **3352** and the staple cartridge **3356**. The sensor **3358** may comprise any suitable sensor for detecting one or more parameters of the end effector **3350**, such as, for example, a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

In one embodiment, the sensor **3358** comprises a magnetic sensor configured to detect a magnetic field generated by an electromagnetic source **3360** coupled to the second jaw member **3354** and/or the staple cartridge **3356**. The electromagnetic source **3360** generates a magnetic field detected by the sensor **3358**. The strength of the detected magnetic field may correspond to, for example, the thickness and/or fullness of a bite of tissue located between the anvil **3352** and the staple cartridge **3356**. In some embodiments, the electromagnetic source **3360** generates a signal at a known frequency, such as, for example, 1 MHz. In other embodiments, the signal generated by the electromagnetic source **3360** may be adjustable based on, for example, the type of staple cartridge **3356** installed in the second jaw member **3354**, one or more additional sensor, an algorithm, and/or one or more parameters.

In one embodiment, a signal processor **3362** is coupled to the end effector **3350**, such as, for example, the anvil **3352**. The signal processor **3362** is configured to process the signal generated by the sensor **3358** to eliminate false signals and to boost the input from the sensor **3358**. In some embodiments, the signal processor **3362** may be located separately from the end effector **3350**, such as, for example, in the handle **14** of a surgical instrument **10**. In some embodiments, the signal processor **3362** is formed integrally with and/or comprises an algorithm executed by a general processor, such as, for example, the primary processor **2006**. The signal processor **3362** is configured to process the signal from the sensor **3358** at a frequency substantially equal to the frequency of the signal generated by the electromagnetic source **3360**. For example, in one embodiment, the electromagnetic source **3360** generates a signal at a frequency of 1 MHz. The signal is detected by the sensor **3358**. The sensor **3358** generates a signal indicative of the detected magnetic field which is provided to the signal processor **3362**. The signal is processed by the signal processor **3362** at a frequency of 1 MHz to eliminate false signals. The processed signal is provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** correlates the received signal to one or more parameters of the

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end effector **3350**, such as, for example, the gap **3364** between the anvil **3352** and the staple cartridge **3356**.

FIG. **46** is a logic diagram illustrating one embodiment of a process **3370** for generating a thickness measurement for a tissue section located between an anvil and a staple cartridge of an end effector, such as, for example, the end effector **3350** illustrated in FIG. **45**. In one embodiment of the process **3370**, a signal is generated **3372** by a modulated electromagnetic source **3360**. The generated signal may comprise, for example, a 1 MHz signal. A magnetic sensor **3358** is configured to detect **3374** the signal generated by the electromagnetic source **3360**. The magnetic sensor **3358** generates a signal indicative of the detected magnetic field and provides the signal to a signal processor **3362**. The signal processor **3362** processes **3376** the signal to remove noise, false signals, and/or to boost the signal. The processed signal is provided to an analog-to-digital converter for conversion **3378** to a digital signal. The digital signal may be calibrated **3380**, for example, by application of a calibration curve input algorithm and/or look-up table. The signal processing **3376**, conversion **3378**, and calibration **3380** may be performed by one or more circuits. The calibrated signal is displayed **3026** to a user by, for example, a display **2026** formed integrally with a surgical instrument **10**.

Although the various embodiments so far described comprise an end effector having first and second jaw members pivotally coupled, the described embodiments are not so limited. For example, in one embodiment, the end effector may comprise a circular stapler end effector. FIG. **47** illustrates one embodiment of a circular stapler **3400** configured to implement one or more of the processes described in FIGS. **28-46**. The circular stapler **3400** comprises a body **3402**. The body **3402** may be coupled to a shaft, such as, for example, the shaft assembly **200** of the surgical instrument **10**. The body **3402** is configured to receive a staple cartridge and/or one or more staples therein (not shown). An anvil **3404** is moveably coupled to the body **3402**. The anvil **3404** may be coupled to the body **3402** by, for example, a shaft **3406**. The shaft **3406** is receivable within a cavity within the body (not shown). In some embodiments, a breakaway washer **3408** is coupled to the anvil **3404**. The breakaway washer **3408** may comprise a buttress or reinforcing material during stapling.

In some embodiments, the circular stapler **3400** comprises a plurality of sensors **3410a**, **3410b**. The plurality of sensor **3410a**, **3410b** is configured to detect one or more parameters of the circular stapler **3400** and/or a tissue section located between the body **3402** and the anvil **3404**. The plurality of sensors **3410a**, **3410b** may be coupled to any suitable portion of the anvil **3404**, such as, for example, being positioned under the breakaway washer **3408**. The plurality of sensors **3410a**, **3410b** may be arranged in any suitable arrangement, such as, for example, being equally spaced about the perimeter of the anvil **3404**. The plurality of sensors **3410a**, **3410b** may comprise any suitable sensors for detecting one or more parameters of the end effector **3400** and/or a tissue section located between the body **3402** and the anvil **3404**. For example, the plurality of sensors **3410a**, **3410b** may comprise magnetic sensors, such as a Hall effect sensor, strain gauges, pressure sensors, inductive sensors, such as an eddy current sensor, resistive sensors, capacitive sensors, optical sensors, any combination thereof, and/or any other suitable sensor.

In one embodiment, the plurality of sensors **3410a**, **3410b** comprise a plurality of pressure sensors positioned under the breakaway washer **3408**. Each of the sensors **3410a**, **3410b** is configured to detect a pressure generated by the presence

of compressed tissue between the body **3402** and the anvil **3404**. In some embodiments the plurality of sensors **3410a**, **3410b** are configured to detect the impedance of a tissue section located between the anvil **3404** and the body **3402**. The detected impedance may be indicative of the thickness and/or fullness of tissue located between the anvil **3404** and the body **3402**. The plurality of sensors **3410a**, **3410b** generate a plurality of signals indicative of the detected pressure. The plurality of generated signals is provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** applies one or more algorithms and/or look-up tables based on the input from the plurality of sensors **3410a**, **3410b** to determine one or more parameters of the end effector **3400** and/or a tissue section located between the body **3402** and the anvil **3404**. For example, in one embodiment comprising a plurality of pressure sensors, the processor **2006** is configured to apply an algorithm to quantitatively compare the output of the plurality of sensors **3410a**, **3410b** with respect to each other and with respect to a predetermined threshold. In one embodiment, if the delta, or difference, between the outputs of the plurality of sensors **3410a**, **3410b** is greater than a predetermined threshold, feedback is provided to the operator indicating a potential uneven loading condition. In some embodiments, the end effector **3400** may be coupled to a shaft comprising one or more additional sensors, such as, for example, the drive shaft **3504** described in connection to FIG. **50** below.

FIGS. **48A-48D** illustrate a clamping process of the circular stapler **3400** illustrated in FIG. **47**. FIG. **48A** illustrates the circular stapler **3400** in an initial position with the anvil **3404** and the body **3402** in a closed configuration. The circular stapler **3400** is positioned at a treatment site in the closed configuration. Once the circular stapler **3400** is positioned, the anvil **3404** is moved distally to disengage with the body **3402** and create a gap configured to receive a tissue section **3412** therein, as illustrated in FIG. **48B**. The tissue section **3412** is compressed to a predetermined compression **3414** between the anvil **3404** and the body **3402**, as shown in FIG. **48C**. The tissue section **3412** is further compressed between the anvil **3404** and the body **3402**. The additional compression deploys one or more staples from the body **3402** into the tissue section **3412**. The staples are shaped by the anvil **3404**. FIG. **48D** illustrates the circular stapler **3400** in position corresponding to staple deployment. Proper staple deployment is dependent on obtaining a proper bite of tissue between the body **3402** and the anvil **3404**. The plurality of sensors **3410a**, **3410b** disposed on the anvil **3404** allow a processor to determine that a proper bite of tissue is located between the anvil **3404** and the body **3402** prior to deployment of the staples.

FIG. **49** illustrates one embodiment of a circular staple anvil **3452** and an electrical connector **3466** configured to interface therewith. The anvil **3452** comprises an anvil head **3454** coupled to an anvil shaft **3456**. A breakaway washer **3458** is coupled to the anvil head **3452**. A plurality of pressure sensors **3460a**, **3460b** are coupled to the anvil head **3452** between the anvil head **3452** and the breakaway washer **3458**. A flex circuit **3462** is formed on the shaft **3456**. The flex circuit **3462** is coupled to the plurality of pressure sensors **3460a**, **3460b**. One or more contacts **3464** are formed on the shaft **3456** to couple the flex circuit **3462** to one or more circuits, such as, for example, the control circuit **2000** of the surgical instrument **10**. The flex circuit **3462** may be coupled to the one or more circuits by an electrical connector **3466**. The electrical connector **3466** is coupled to the anvil **3454**. For example, in one embodiment, the shaft **3456** is hollow and configured to receive the electrical

connector **3466** therein. The electrical connector **3466** comprises a plurality of contacts **3468** configured to interface with the contacts **3464** formed on the anvil shaft **3456**. The plurality of contacts **3468** on the electrical connector **3466** are coupled to a flex circuit **3470** which is coupled the one or more circuits, such as, for example, a control circuit **2000**.

FIG. **50** illustrates one embodiment of a surgical instrument **3500** comprising a sensor **3506** coupled to a drive shaft **3504** of the surgical instrument **3500**. The surgical instrument **3500** may be similar to the surgical instrument **10** described above. The surgical instrument **3500** comprises a handle **3502** and a drive shaft **3504** coupled to a distal end of the handle. The drive shaft **3504** is configured to receive an end effector (not shown) at the distal end. A sensor **3506** is fixedly mounted in the drive shaft **3504**. The sensor **3506** is configured to detect one or more parameters of the drive shaft **3504**. The sensor **3506** may comprise any suitable sensor, such as, for example, a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

In some embodiments, the sensor **3506** comprises a magnetic Hall effect sensor. A magnet **3508** is located within the drive shaft **3504**. The sensor **3506** is configured to detect a magnetic field generated by the magnet **3508**. The magnet **3508** is coupled to a spring-backed bracket **3510**. The spring-backed bracket **3510** is coupled to the end effector. The spring-backed bracket **3510** is moveable in response to an action of the end effector, for example, compression of an anvil towards a body and/or second jaw member. The spring-backed bracket **3510** moves the magnet **3508** in response to the movement of the end effector. The sensor **3506** detects the change in the magnetic field generated by the magnet **3508** and generates a signal indicative of the movement of the magnet **3508**. The movement of the magnet **3508** may correspond to, for example, the thickness of tissue clamped by the end effector. The thickness of the tissue may be displayed to an operator by, for example, a display **3512** embedded in the handle **3502** of the surgical instrument **3500**. In some embodiments, the Hall effect sensor **3508** may be combined with one or more additional sensors, such as, for example, the pressure sensors illustrated in FIG. **47**.

FIG. **51** is a flow chart illustrating one embodiment of a process **3550** for determining uneven tissue loading in an end effector, for example, the end effector **3400** illustrated in FIG. **47** coupled to the surgical instrument **3500** illustrated in FIG. **50**. In one embodiment, the process **3550** comprises utilizing one or more first sensors **3552**, such as, for example, a plurality of pressure sensors, to detect **3554** the presence of tissue within an end effector. During a clamping operation of the end effector **3400**, the input from the pressure sensors, **P**, is analyzed to determine the value of **P**. If **P** is less **3556** than a predetermined threshold, the end effector **3400** continues **3558** the clamping operation. If **P** is greater than or equal to **3560** the predetermined threshold, clamping is stopped. The delta (difference) between the plurality of sensors **3552** is compared **3562**. If the delta is greater than a predetermined delta, the surgical instrument **3500** displays **3564** a warning to the user. The warning may comprise, for example, a message indicating that there is uneven clamping in the end effector. If the delta is less than or equal to the predetermined delta, the input of the one or more sensors **3552** is compared to an input from an additional sensor **3566**.

In some embodiments, a second sensor **3566** is configured to detect one or more parameters of the surgical instrument **3500**. For example, in one some embodiments, a magnetic sensor, such as, for example, a Hall effect sensor, is located in a shaft **3504** of the surgical instrument **3500**. The second sensor **3566** generates a signal indicative of the one or more parameters of the surgical instrument **3500**. A preset calibration curve is applied **3568** to the input from the second sensor **3566**. The preset calibration curve may adjust **3568** a signal generated by the second sensor **3566**, such as, for example, a Hall voltage generated by a Hall effect sensor. For example, in one embodiment, the Hall effect voltage is adjusted such that the generated Hall effect voltage is set at a predetermined value when the gap between the anvil **3404** and the body **3402**, $X1$, is equal to zero. The adjusted sensor **3566** input is used to calculate **3570** a distance, $X3$, between the anvil **3404** and the body **3402** when the pressure threshold P is met. The clamping process is continued **3572** to deploy a plurality of staples into the tissue section clamped in the end effector **3400**. The input from the second sensor **3566** changes dynamically during the clamping procedure and is used to calculate the distance, $X2$, between the anvil **3404** and the body **3402** in real-time. A real-time percent compression is calculated **3574** and displayed to an operator. In one embodiment, the percent compression is calculated as: $[(X3-X2)/X3]*100$.

In some embodiments, one or more of the sensors illustrated in FIGS. **28-50** are used to indicate: whether the anvil is attached to the body of the surgical device; the compressed tissue gap; and/or whether the anvil is in a proper position for removing the device, or any combination of these indicators.

In some embodiments, one or more of the sensors illustrated in FIGS. **28-50** are used to affect device performance. One or more control parameters of a surgical device **10** may be adjusted by at least one sensor output. For example, in some embodiments, the speed control of a firing operation may be adjusted by the output of one or more sensors, such as, for example, a Hall effect sensor. In some embodiments, one or more the sensors may adjust a closure and/or clamping operation based on load and/or tissue type. In some embodiments, multiple stage compression sensors allow the surgical instrument **10** to stop closure at a predetermined load and/or a predetermined displacement. The control circuit **2000** may apply one or more predetermined algorithms to apply varying compression to a tissue section to determine a tissue type, for example, based on a tissue response. The algorithms may be varied based on closure rate and/or predetermined tissue parameters. In some embodiments, one or more sensors are configured to detect a tissue property and one or more sensors are configured to detect a device property and/or configuration parameter. For example, in one embodiment, capacitive blocks may be formed integrally with a staple cartridge to measure skew.

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FIG. **52** illustrates one embodiment of an end effector **3600** configured to determine one or more parameters of a tissue section during a clamping operation. The end effector **3600** comprises a first jaw member, or anvil, **3602** pivotally coupled to a second jaw member **3604**. The second jaw member **3604** is configured to receive a staple cartridge **3606** therein. The staple cartridge **3606** contains a plurality of staples (not shown) configured to be deployed into a tissue section during a clamping and stapling operation. The staple cartridge **3606** comprises a staple cartridge deck **3622** having a predetermined height. The staple cartridge **3606** further comprises a slot **3624** defined within the body of the

staple cartridge, similar to slot **193** described above. A Hall effect sensor **3608** is configured to detect the distance **3616** between the Hall effect sensor **3608** and a magnet **3610** coupled to the second jaw member **3604**. The distance **3616** between the Hall effect sensor **3608** and the magnet **3610** is indicative of a thickness of tissue located between the anvil **3602** and the staple cartridge deck **3622**.

The second jaw member **3604** is configured to receive a plurality of staple cartridge **3606** types. The types of staple cartridge **3606** may vary by, for example, containing different length staples, comprising a buttress material, and/or containing different types of staples. In some embodiments, the height **3618** of the staple cartridge deck **3622** may vary based on the type of staple cartridge **3606** coupled to the second jaw member **3604**. The varying cartridge height **3618** may result in an inaccurate thickness measurement by the Hall effect sensor **3608**. For example, in one embodiment, a first cartridge comprises a first cartridge deck height X and a second cartridge comprises a second cartridge deck height Y , where $Y > X$. A fixed Hall effect sensor **3608** and fixed magnet will produce an accurate thickness measurement only for one of the two cartridge deck heights. In some embodiments, an adjustable magnet is used to compensate for various deck heights.

In some embodiments, the second jaw member **3604** and the staple cartridge **3606** comprise a magnet cavity **3614**. The magnet cavity **3614** is configured to receive the magnet **3610** therein. The magnet is coupled to a spring-arm **3612**. The spring-arm **3612** is configured to bias the magnet towards the upper surface of the magnet cavity **3614**. A depth **3620** of the magnet cavity **3614** varies depending on the deck height **3618** of the staple cartridge **3606**. For example, each staple cartridge **3606** may define a cavity depth **3620** such that the upper surface of the cavity **3614** is a set distance from the plane of the deck **3622**. The magnet **3610** is biased against the upper surface of the cavity **3614**. The magnetic reference of the magnet **3610**, as viewed by the Hall effect sensor **3608**, is consistent relative to all cartridge decks but variable relative to the slot **3624**. For example, in some embodiments, the upper-biased magnet **3610** and the cavity **3614** provide a set distance **3616** from the Hall effect sensor **3608** to the magnet **3610**, regardless of the staple cartridge **3606** inserted into the second jaw member **3604**. The set distance **3616** allows the Hall effect sensor **3608** to generate an accurate thickness measurement irrespective of the staple cartridge **3606** type. In some embodiments, the depth **3620** of the cavity **3614** may be adjusted to calibrate the Hall effect sensor **3608** for one or more surgical procedures.

FIGS. **53A** and **53B** illustrate an embodiment of an end effector **3650** configured to normalize a Hall effect voltage irrespective of a deck height of a staple cartridge **3656**. FIG. **53A** illustrates one embodiment of the end effector **3650** comprising a first cartridge **3656a** inserted therein. The end effector **3650** comprises a first jaw member, or anvil, **3652** pivotally coupled to a second jaw member **3654** to grasp tissue therebetween. The second jaw member **3654** is configured to receive a staple cartridge **3656a**. The staple cartridge **3656a** may comprise a variety of staple lengths, buttress materials, and/or deck heights. A magnetic sensor **3658**, such as, for example, a Hall effect sensor, is coupled to the anvil **3652**. The magnetic sensor **3658** is configured to detect a magnetic field generated by a magnet **3660**. The detected magnetic field strength is indicative of the distance **3664** between the magnetic sensor **3658** and the magnet **3660**, which may be indicative of, for example, a thickness of a tissue section grasped between the anvil **3652** and the

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staple cartridge **3656**. As noted above, various staple cartridges **3656a** may comprise varying deck heights which create differences in the calibrated compression gap **3664**.

In some embodiments, a magnetic attenuator **3662** is coupled to the staple cartridge **3656a**. The magnetic attenuator **3662** is configured to attenuate the magnetic flux generated to by the magnet **3660**. The magnetic attenuator **3662** is calibrated to produce a magnetic flux based on the height of the staple cartridge **3656a**. By attenuating the magnet **3660** based on the staple cartridge **3656** type, the magnetic attenuator **3662** normalizes the magnetic sensor **3658** signal to the same calibration level for various deck heights. The magnetic attenuator **3662** may comprise any suitable magnet attenuator, such as, for example, a ferrous metallic cap. The magnetic attenuator **3662** is molded into the staple cartridge **3656a** such that the magnetic attenuator **3662** is positioned above the magnet **3660** when the staple cartridge **3656** is inserted into the second jaw member **3654**.

In some embodiments, attenuation of the magnet **3660** is not required for the deck height of the staple cartridge. FIG. **53B** illustrates one embodiment of the end effector **3650** comprising a second staple cartridge **3656b** coupled to the second jaw member **3654**. The second staple cartridge **3656b** comprises a deck height matching the calibration of the magnet **3660** and the Hall effect sensor **3658**, and therefore does not require attenuation. As shown in FIG. **53B**, the second staple cartridge **3656b** comprises a cavity **3666** in place of the magnetic attenuator **3662** of the first staple cartridge **3656a**. In some embodiments, larger and/or smaller attenuation members are provided depending on the height of the cartridge deck. The design of the attenuation member **3662** shape may be optimized to create features in the response signal generated by the Hall effect sensor **3658** that allow for the distinction of one or more additional cartridge attributes.

FIG. **54** is a logic diagram illustrating one embodiment of a process **3670** for determining when the compression of tissue within an end effector, such as, for example, the end effector **3650** illustrated in FIGS. **53A-53B**, has reached a steady state. In some embodiments, a clinician initiates **3672** a clamping procedure to clamp tissue within the end effector, for example, between an anvil **3652** and staple cartridge **3656**. The end effector engages **3674** with tissue during the clamping procedure. Once the tissue has been engaged **3674**, the end effector begins **3676** real time gap monitoring. The real time gap monitoring monitors the gap between, for example, the anvil **3652** and the staple cartridge **3656** of the end effector **3650**. The gap may be monitored by, for example, a sensor **3658**, such as a Hall effect sensor, coupled to the end effector **3650**. The sensor **3658** may be coupled to a processor, such as, for example, the primary processor **2006**. The processor determines **3678** when tissue clamping requirements of the end effector **3650** and/or the staple cartridge **3656** have been met. Once the processor determines that the tissue has stabilized, the process indicates **3680** to the user that the tissue has stabilized. The indication may be provided by, for example, a display embedded within a surgical instrument **10**.

In some embodiments, the gap measurement is provided by a Hall effect sensor. The Hall effect sensor may be located, for example, at the distal tip of an anvil **3652**. The Hall effect sensor is configured to measure the gap between the anvil **3652** and a staple cartridge **3656** deck at the distal tip. The measured gap may be used to calculate a jaw closure gap and/or to monitor a change in tissue compression of a tissue section clamped in the end effector **3650**. In one embodiment, the Hall effect sensor is coupled to a processor,

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such as, for example, the primary processor **2006**. The processor is configured to receive real time measurements from the Hall effect sensor and compare the received signal to a predetermined set of criteria. For example, in one embodiment, a logic equation at equally spaced intervals, such as one second, is used to indicate stabilization of a tissue section to the user when a gap reading remains unchanged for a predetermined interval, such as, for example, 3.0 seconds. Tissue stabilization may also be indicated after a predetermined time period, such as, for example, 15.0 seconds. As another example, tissue stabilization may be indicated when $y_n = y_{n+1} = y_{n+2}$, where y equals a gap measurement of the Hall effect sensor and n is a predetermined measurement interval. A surgical instrument **10** may display an indication to a user, such as, for example, a graphical and/or numerical representation, when stabilization has occurred.

FIG. **55** is a graph **3690** illustrating various Hall effect sensor readings **3692a-3692d**. As shown in graph **3690**, a thickness, or compression, of a tissue section stabilizes after a predetermined time period. A processor, such as, for example, the primary processor **2006**, may be configured to indicate when the calculated thickness from a sensor, such as a Hall effect sensor, is relatively consistent or constant over a predetermined time period. The processor **2006** may indicate to a user, for example, through a number display, that the tissue has stabilized.

FIG. **56** is a logic diagram illustrating one embodiment of a process **3700** for determining when the compression of tissue within an end effector, such as, for example, the end effector **3650** illustrated in FIGS. **53A-53B**, has reached a steady state. In some embodiments, a clinician initiates **3702** a clamping procedure to clamp tissue within the end effector, for example, between an anvil **3652** and staple cartridge **3656**. The end effector engages **3704** with tissue during the clamping procedure. Once the tissue has been engaged **3704**, the end effector begins **3706** real time gap monitoring. The real time gap monitoring technique monitors **3706** the gap between, for example, the anvil **3652** and the staple cartridge **3656** of the end effector **3650**. The gap may be monitored **3706** by, for example, a sensor **3658**, such as a Hall effect sensor, coupled to the end effector **3650**. The sensor **3658** may be coupled to a processor, such as, for example, the primary processor **2006**. The processor is configured to execute one or more algorithms determine when tissue section compressed by the end effector **3650** has stabilized.

For example, in the embodiment illustrated in FIG. **56**, the process **3700** is configured to utilize a slope calculation to determine stabilization of tissue. The processor calculates **3708** the slope, S , of an input from a sensor, such as a Hall effect sensor. The slope may be calculated **3708** by, for example, the equation $S = ((V_1 - V_2)) / ((T_1 - T_2))$. The processor compares **3710** the calculated slope to a predetermined value, such as, for example, 0.005 volts/sec. If the value of the calculated slope is greater than the predetermined value, the processor resets **3712** a count, C , to zero. If the calculated slope is less than or equal to the predetermined value, the processor increments **3714** the value of the count C . The count, C , is compared **3716** to a predetermined threshold value, such as, for example, 3. If the value of the count C is greater than or equal to the predetermined threshold value, the processor indicates **3718** to the user that the tissue section has stabilized. If the value of the count C is less than the predetermined threshold value, the processor continues monitoring the sensor **3658**. In various embodi-

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ments, the slope of the sensor input, the change in the slope, and/or any other suitable change in the input signal may be monitored.

In some embodiments, an end effector, such as for example, the end effectors **3600**, **3650** illustrated in FIGS. **52**, **53A**, and **53B** may comprise a cutting member deployable therein. The cutting member may comprise, for example, an I-Beam configured to simultaneously cut a tissue section located between an anvil **3602** and a staple cartridge **3608** and to deploy staples from the staple cartridge **3608**. In some embodiments, the I-Beam may comprise only a cutting member and/or may only deploy one or more staples. Tissue flow during firing may affect the proper formation of staples. For example, during I-Beam deployment, fluid in the tissue may cause the thickness of tissue to temporarily increase, causing improper deployment of staples.

FIG. **57** is a logic diagram illustrating one embodiment of a process **3730** for controlling an end effector to improve proper staple formation during deployment. The control process **3730** comprises generating **3732** a sensor measurement indicative of the thickness of a tissue section within the end effector **3650**, such as for example, a Hall effect voltage generated by a Hall effect sensor. The sensor measurement is converted **3734** to a digital signal by an analog-to-digital converter. The digital signal is calibrated **3736**. The calibration **3736** may be performed by, for example, a processor and/or a dedicated calibration circuit. The digital signal is calibrated **3736** based on one or more calibration curve inputs. The calibrated digital signal is displayed **3738** to an operator by, for example, a display **2026** embedded in a surgical instrument **10**. The calibrated signal may be displayed **3738** as a thickness measurement of a tissue section grasped between the anvil **3652** and the staple cartridge **3656** and/or as a unit-less range.

In some embodiments, the generated **3732** Hall effect voltage is used to control an I-beam. For example, in the illustrated embodiment, the Hall effect voltage is provided to a processor configured to control deployment of an I-Beam within an end effector, such as, for example, the primary processor **2006**. The processor receives the Hall effect voltage and calculates the voltage rate of change over a predetermined time period. The processor compares **3740** the calculated rate of change to a predetermined value, $x1$. If the calculated rate of change is greater than the predetermined value, $x1$, the processor slows **3742** the speed of the I-Beam. The speed may be reduced by, for example, decrementing a speed variable by a predetermined unit. If the calculated voltage rate of change is less than or equal to the predetermined value, $x1$, the processor maintains **3744** the current speed of the I-Beam.

In some embodiments, the processor may temporarily reduce the speed of the I-Beam to compensate, for example, for thicker tissue, uneven loading, and/or any other tissue characteristic. For example, in one embodiment, the processor is configured to monitor **3740** the rate of voltage change of a Hall effect sensor. If the rate of change monitored **3740** by the processor exceeds a first predetermined value, $x1$, the processor slows down or stops deployment of the I-Beam until the rate of change is less than a second predetermined value, $x2$. When the rate of change is less than the second predetermined value, $x2$, the processor may return the I-beam to normal speed. In some embodiments, the sensor input may be generated by for example, a pressure sensor, a strain gauge, a Hall effect sensor, and/or any other suitable sensor. In some embodiments, the processor may implement one or more pause points during deployment of an I-Beam.

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For example, in some embodiments, the processor may implement three predetermined pause points, at which the processor pauses deployment of the I-Beam for a predetermined time period. The pause points are configured to provide optimized tissue flow control.

FIG. **58** is a logic diagram illustrating one embodiment of a process **3750** for controlling an end effector to allow for fluid evacuation and provide improved staple formation. The process **3750** comprises generating **3752** a sensor measurement, such as, for example, a Hall effect voltage. The sensor measurement may be indicative of, for example, the thickness of a tissue section grasped between an anvil **3652** and a staple cartridge **3656** of an end effector **3650**. The generated signal is provided to an analog-to-digital converter for conversion **3754** to a digital signal. The converted signal is calibrated **3756** based on one or more inputs, such as, for example, a second sensor input and/or a predetermined calibration curve. The calibrated signal is representative of one or more parameters of the end effector **3650**, such as, for example, the thickness of a tissue section grasped therein. The calibrated thickness measurement may be displayed to a user as a thickness and/or as a unit-less range. The calibrated thickness may be displayed by, for example, a display **2026** embedded in a surgical instrument **10** coupled to the end effector **3650**.

In some embodiments, the calibrated thickness measurement is used to control deployment of an I-Beam and/or other firing member within the end effector **3650**. The calibrated thickness measurement is provided to a processor. The processor compares **3760** the change in the calibrated thickness measurement to a predetermined threshold percentage, x . If the rate of change of the thickness measurement is greater than x , the processor slows **3762** the speed, or rate of deployment, of the I-Beam within the end effector. The processor may slow **3762** the speed of the I-Beam by, for example, decrementing a speed variable by a predetermined unit. If the rate of change of the thickness measurement is less than or equal to x , the processor maintains **3764** the speed of the I-Beam within the end effector **3650**. The real time feedback of tissue thickness and/or compression allows the surgical instrument **10** to affect the firing speed to allow for fluid evacuation and/or provide improved staple form.

In some embodiments, the sensor reading generated **3752** by the sensor, for example, a Hall effect voltage, may be adjusted by one or more additional sensor inputs. For example, in one embodiment, a generated **3752** Hall effect voltage may be adjusted by an input from a micro-strain gauge sensor on the anvil **3652**. The micro-strain gauge may be configured to monitor the strain amplitude of the anvil **3652**. The generated **3752** Hall effect voltage may be adjusted by the monitored strain amplitude to indicate, for example, partial proximal or distal tissue bites. Time based monitoring of the micro-strain and Hall effect sensor output during clamping allows one or more algorithms and/or look-up tables to recognize tissue characteristics and clamping positioning and dynamically adjust tissue thickness measurements to control firing speed of, for example, an I-Beam. In some embodiments, the processor may implement one or more pause points during deployment of an I-Beam. For example, in some embodiments, the processor may implement three predetermined pause points, at which the processor pauses deployment of the I-Beam for a predetermined time period. The pause points are configured to provide optimized tissue flow control.

FIGS. **59A-59B** illustrate one embodiment of an end effector **3800** comprising a pressure sensor. The end effector

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3800 comprises a first jaw member, or anvil, **3802** pivotally coupled to a second jaw member **3804**. The second jaw member **3804** is configured to receive a staple cartridge **3806** therein. The staple cartridge **3806** comprises a plurality of staples. A first sensor **3808** is coupled to the anvil **3802** at a distal tip. The first sensor **3808** is configured to detect one or more parameters of the end effector, such as, for example, the distance, or gap **3814**, between the anvil **3802** and the staple cartridge **3806**. The first sensor **3808** may comprise any suitable sensor, such as, for example, a magnetic sensor. A magnet **3810** may be coupled to the second jaw member **3804** and/or the staple cartridge **3806** to provide a magnetic signal to the magnetic sensor.

In some embodiments, the end effector **3800** comprises a second sensor **3812**. The second sensor **3812** is configured to detect one or more parameters of the end effector **3800** and/or a tissue section located therebetween. The second sensor **3812** may comprise any suitable sensor, such as, for example, one or more pressure sensors. The second sensor **3812** may be coupled to the anvil **3802**, the second jaw member **3804**, and/or the staple cartridge **3806**. A signal from the second sensor **3812** may be used to adjust the measurement of the first sensor **3808** to adjust the reading of the first sensor to accurately represent proximal and/or distal positioned partial bites true compressed tissue thickness. In some embodiments, the second sensor **3812** may be surrogate with respect to the first sensor **3808**.

In some embodiments, the second sensor **3812** may comprise, for example, a single continuous pressure sensing film and/or an array of pressure sensing films. The second sensor **3812** is coupled to the deck of the staple cartridge **3806** along the central axis covering, for example, a slot **3816** configured to receive a cutting and/or staple deployment member. The second sensor **3812** provides signals indicate of the amplitude of pressure applied by the tissue during a clamping procedure. During firing of the cutting and/or deployment member, the signal from the second sensor **3812** may be severed, for example, by cutting electrical connections between the second sensor **3812** and one or more circuits. In some embodiments, a severed circuit of the second sensor **3812** may be indicative of a spent staple cartridge **3806**. In other embodiments, the second sensor **3812** may be positioned such that deployment of a cutting and/or deployment member does not sever the connection to the second sensor **3812**.

FIG. 60 illustrates one embodiment of an end effector **3850** comprising a second sensor **3862** located between a staple cartridge **3806** and a second jaw member **3804**. The end effector **3850** comprises a first jaw member, or anvil, **3852** pivotally coupled to a second jaw member **3854**. The second jaw member **3854** is configured to receive a staple cartridge **3856** therein. A first sensor **3858** is coupled to the anvil **3852** at a distal tip. The first sensor **3858** is configured to detect one or more parameters of the end effector **3850**, such as, for example, the distance, or gap **3864**, between the anvil **3852** and the staple cartridge **3856**. The first sensor **3858** may comprise any suitable sensor, such as, for example, a magnetic sensor. A magnet **3860** may be coupled to the second jaw member **3854** and/or the staple cartridge **3856** to provide a magnetic signal to the magnetic sensor. In some embodiments, the end effector **3850** comprises a second sensor **3862** similar in all respect to the second sensor **3812** of FIGS. 59A-59B, except that it is located between the staple cartridge **3856** and the second jaw member **3864**.

FIG. 61 is a logic diagram illustrating one embodiment of a process **3870** for determining and displaying the thickness

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of a tissue section clamped in an end effector **3800** or **3850**, according to FIGS. B59A-59B or FIG. 60. The process comprises obtaining a Hall effect voltage **3872**, for example, through a Hall effect sensor located at the distal tip of the anvil **3802**. The Hall effect voltage **3872** is proved to an analog to digital converter **3874** and converted into a digital signal. The digital signal is provided to a process, such as for example the primary processor **2006**. The primary processor **2006** calibrates **3874** the curve input of the Hall effect voltage **3872** signal. Pressure sensors, such as for example second sensor **3812**, is configured to measure **3880** one or more parameters of, for example, the end effector **3800**, such as for example the amount of pressure being exerted by the anvil **3802** on the tissue clamped in the end effector **3800**. In some embodiments the pressure sensors may comprise a single continuous pressure sensing film and/or array of pressure sensing films. The pressure sensors may thus be operable determine variations in the measure pressure at different locations between the proximal and distal ends of the end effector **3800**. The measured pressure is provided to the processor, such as for example the primary processor **2006**. The primary processor **2006** uses one or more algorithms and/or lookup tables to adjust **3882** the Hall effect voltage **3872** in response to the pressure measured by the pressure sensors **3880** to more accurately reflect the thickness of the tissue clamped between, for example, the anvil **3802** and the staple cartridge **3806**. The adjusted thickness is displayed **3878** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

FIG. 62 illustrates one embodiment of an end effector **3900** comprising a plurality of second sensors **3912a-3912b** located between a staple cartridge **3906** and an elongated channel **3916**. The end effector **3900** comprises a first jaw member or anvil **3902** pivotally coupled to a second jaw member or elongated channel **3904**. The elongated channel **3904** is configured to receive a staple cartridge **3906** therein. The anvil **3902** further comprises a first sensor **3908** located in the distal tip. The first sensor **3908** is configured to detect one or more parameters of the end effector **3900**, such as, for example, the distance, or gap, between the anvil **3902** and the staple cartridge **3906**. The first sensor **3908** may comprise any suitable sensor, such as, for example, a magnetic sensor. A magnet **3910** may be coupled to the elongated channel **3904** and/or the staple cartridge **3906** to provide a magnetic signal to the first sensor **3908**. In some embodiments, the end effector **3900** comprises a plurality of second sensors **3912a-3912c** located between the staple cartridge **3906** and the elongated channel **3904**. The second sensors **3912a-3912c** may comprise any suitable sensors, such as for instance piezo-resistive pressure film strips. In some embodiments, the second sensors **3912a-3912c** may be uniformly distributed between the distal and proximal ends of the end effector **3900**.

In some embodiments, signals from the second sensors **3912a-3912c** may be used to adjust the measurement of the first sensor **3908**. For instance, the signals from the second sensors **3912a-3912c** may be used to adjust the reading of the first sensor **3908** to accurately represent the gap between the anvil **3908** and the staple cartridge **3906**, which may vary between the distal and proximal ends of the end effector **3900**, depending on the location and/or density of tissue **3920** between the anvil **3902** and the staple cartridge **3906**. FIG. 11 illustrates an example of a partial bite of tissue **3920**. As illustrated for purposes of this example, the tissue is located only in the proximal area of the end effector **3900**, creating a high pressure **3918** area near the proximal area of

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the end effector **3900** and a corresponding low pressure **3916** area near the distal end of the end effector.

FIGS. **63A** and **63B** further illustrate the effect of a full versus partial bite of tissue **3920**. FIG. **63A** illustrates the end effector **3900** with a full bite of tissue **3920**, where the tissue **3920** is of uniform density. With a full bite of tissue **3920** of uniform density, the measured first gap **3914a** at the distal tip of the end effector **3900** may be approximately the same as the measured second gap **3922a** in the middle or proximal end of the end effector **3900**. For example, the first gap **3914a** may measure 2.4 mm, and the second gap may measure 2.3 mm. FIG. **63B** illustrates an end effector **3900** with a partial bite of tissue **3920**, or alternatively a full bit of tissue **3920** of non-uniform density. In this case, the first gap **3914b** will measure less than the second gap **3922b** measured at the thickest or densest portion of the tissue **3920**. For example, the first gap may measure 1.0 mm, while the second gap may measure 1.9 mm. In the conditions illustrated in FIGS. **63A** and **63B**, signals from the second sensors **3912a-3912c**, such as for instance measured pressure at different points along the length of the end effector **3900**, may be employed by the instrument to determine tissue **3920** placement and/or material properties of the tissue **3920**. The instrument may further be operable to use measured pressure over time to recognize tissue characteristics and tissue position, and dynamically adjust tissue thickness measurements.

FIG. **64** illustrates one embodiment of an end effector **3950** comprising a coil **3958** and oscillator circuit **3962** for measuring the gap between the anvil and the staple cartridge **3956**. The end effector **3950** comprises a first jaw member or anvil **3952** pivotally coupled to a second jaw member or elongated channel **3954**. The elongated channel **3954** is configured to receive a staple cartridge **3956** therein. In some embodiments the staple cartridge **3954** further comprises a coil **3958** and an oscillator circuit **3962** located at the distal end. The coil **3958** and oscillator circuit **3962** are operable as eddy current sensors and/or inductive sensors. The coil **3958** and oscillator circuit **3962** can detect eddy currents and/or induction as a target **3960**, such as for instance the distal tip of the anvil **3952**, approaches the coil **3958**. The eddy current and/or induction detected by the coil **3958** and oscillator circuit **3962** can be used to detect the distance or gap between the anvil **3952** and staple cartridge **3956**.

FIG. **65** illustrates and alternate view of the end effector **3950**. As illustrated, in some embodiments external wiring **3964** may supply power to the oscillator circuit **3962**. The external wiring **3964** may be placed along the outside of the elongated channel **3954**.

FIG. **66** illustrates examples of the operation of a coil **3958** to detect eddy currents **3972** in a target **3960**. Alternating current flowing through the coil **3958** at a chosen frequency generates a magnetic field **3970** around the coil **3958**. When the coil **3958** is at its position **3976a** a certain distance away from the target **3960**, the coil **3958** will not induce an eddy current **3972**. When the coil **3958** is at a position **3976b** close to an electrically conductive target **3960** and eddy current **3972** is produced in the target **3960**. When the coil **3958** is at a position **3976c** near a flaw in the target **3960**, the flaw may disrupt the eddy current circulation; in this case, the magnetic coupling with the coil **3958** is changed and a defect signal **3974** can be read by measuring the coil impedance variation.

FIG. **67** illustrates a graph **3980** of a measured quality factor **3984**, the measured inductance **3986**, and measured resistance **3988** of the radius of a coil **3958** as a function of

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the coil's **3958** standoff **3978** to a target **3960**. The quality factor **3984** depends on the standoff **3978**, while both the inductance **3986** and resistance **3988** are functions of displacement. A higher quality factor **3984** results in a more purely reactive sensor. The specific value of the inductance **3986** is constrained only by the need for a manufacturable coil **3958** and a practical circuit design that burns a reasonable amount of energy at a reasonable frequency. Resistance **3988** is a parasitic effect.

The graph **3980** illustrates how inductance **3986**, resistance **3988**, and the quality factor **3984** depend on the target standoff **3978**. As the standoff **3978** increases, the inductance **3986** increases by a factor of four, the resistance **3988** decreases slightly and as a consequence the quality factor **3984** increases. The change in all three parameters is highly nonlinear and each curve tends to decay roughly exponentially as standoff **3978** increases. The rapid loss of sensitivity with distance strictly limits the range of an eddy current sensor to approximately $\frac{1}{2}$ the coil diameter.

FIG. **68** illustrates one embodiment of an end effector **4000** comprising an emitter and sensor **4008** placed between the staple cartridge **4006** and the elongated channel **4004**. The end effector **4000** comprises a first jaw member or anvil **4002** pivotally coupled to a second jaw member or elongated channel **4004**. The elongated channel **4004** is configured to receive a staple cartridge **4006** therein. In some embodiments, the end effector **4000** further comprises an emitter and sensor **4008** located between the staple cartridge **4006** and the elongated channel **4004**. The emitter and sensor **4008** can be any suitable device, such as for instance a MEMS ultrasonic transducer. In some embodiments, the emitter and sensor may be placed along the length of the end effector **4000**.

FIG. **69** illustrates an embodiment of an emitter and sensor **4008** in operation. The emitter and sensor **4008** may be operable to emit a pulse **4014** and sense the reflected signal **4016** of the pulse **4014**. The emitter and sensor **4008** may further be operable to measure the time of flight **4018** between the issuance of the pulse **4014** and the reception of the reflected signal **4016**. The measured time of flight **4018** can be used to determine the thickness of tissue compressed in the end effector **4000** along the entire length of the end effector **4000**. In some embodiments, the emitter and sensor **4008** may be coupled to a processor, such as for instance the primary processor **2006**. The processor **2006** may be operable to use the time of flight **4018** to determine additional information about the tissue, such as for instance whether the tissue was diseased, bunched, or damaged. The surgical instrument can further be operable to convey this information to the operator of the instrument.

FIG. **70** illustrates the surface of an embodiment of an emitter and sensor **4008** comprising a MEMS transducer.

FIG. **71** illustrates a graph **4020** of an example of the reflected signal **4016** that may be measured by the emitter and sensor **4008** of FIG. **69**. FIG. **71** illustrates the amplitude **4022** of the reflected signal **4016** as a function of time **4024**. As illustrated, the amplitude of the transmitted pulse **4026** is greater than the amplitude of the reflected pulses **4028a-4028c**. The amplitude of the transmitted pulse **4026** may be of a known or expected value. The first reflected pulse **4028a** may be, for example, from the tissue enclosed by the end effector **4000**. The second reflected pulse **4028b** may be, for example, from the lower surface of the anvil **4002**. The third reflected pulse **4028c** may be, for example, from the upper surface of the anvil **4002**.

FIG. **72** illustrates an embodiment of an end effector **4050** that is configured to determine the location of a cutting

member or knife **4058**. The end effector **4050** comprises a first jaw member or anvil **4052** pivotally coupled to a second jaw member or elongated channel **4054**. The elongated channel **4054** is configured to receive a staple cartridge **4056** therein. The staple cartridge **4056** further comprises a slot **4058** (not shown) and a cutting member or knife **4062** located therein. The knife **4062** is operably coupled to a knife bar **4064**. The knife bar **4064** is operable to move the knife **4062** from the proximal end of the slot **4058** to the distal end. The end effector **4050** may further comprise an optical sensor **4060** located near the proximal end of the slot **4058**. The optical sensor may be coupled to a processor, such as for instance the primary processor **2006**. The optical sensor **4060** may be operable to emit an optical signal towards the knife bar **4064**. The knife bar **4064** may further comprise a code strip **4066** along its length. The code strip **4066** may comprise cut-outs, notches, reflective pieces, or any other configuration that is optically readable. The code strip **4066** is placed such that the optical signal from the optical sensor **4060** will reflect off or through the code strip **4066**. As the knife **4062** and knife bar **4064** move **4068** along the slot **4058**, the optical sensor **4060** will detect the reflection of the emitted optical signal coupled to the code strip **4066**. The optical sensor **4060** may be operable to communicate the detected signal to the processor **2006**. The processor **2006** may be configured to use the detected signal to determine the position of the knife **4062**. The position of the knife **4062** may be sensed more precisely by designing the code strip **4066** such that the detected optical signal has a gradual rise and fall.

FIG. **73** illustrates an example of the code strip **4066** in operation with red LEDs **4070** and infrared LEDs **4072**. For purposes of this example only, the code strip **4066** comprises cut-outs. As the code strip **4066** moves **4068**, the light emitted by the red LEDs **4070** will be interrupted as the cut-outs passed before it. The infrared LEDs **4072** will therefore detect the motion **4068** of the code strip **4066**, and therefore, by extension, the motion of the knife **4062**. Monitoring Device Degradation Based on Component Evaluation

FIG. **74** depicts a partial view of the end effector **300** of the surgical instrument **10**. In the example form depicted in FIG. **74**, the end effector **300** comprises a staple cartridge **1100** which is similar in many respects to the staple cartridge **304** (FIG. **20**). Several parts of the end effector **300** are omitted to enable a clearer understanding of the present disclosure. In certain instances, the end effector **300** may include a first jaw such as, for example, the anvil **306** (FIG. **20**) and a second jaw such as, for example, the channel **198** (FIG. **20**). In certain instances, as described above, the channel **198** may accommodate a staple cartridge such as, for example, the staple cartridge **304** or the staple cartridge **1100**, for example. At least one of the channel **198** and the anvil **306** may be movable relative to the other one of the channel **198** and the anvil **306** to capture tissue between the staple cartridge **1100** and the anvil **306**. Various actuation assemblies are described herein to facilitation motion of the channel **198** and/or the anvil **306** between an open configuration (FIG. **1**) and a closed configuration (FIG. **75**), for example

In certain instances, as described above, the E-beam **178** can be advanced distally to deploy the staples **191** into the captured tissue and/or advance the cutting edge **182** between a plurality of positions to engage and cut the captured tissue. As illustrated in FIG. **74**, the cutting edge **182** can be advanced distally along a path defined by the slot **193**, for example. In certain instances, the cutting edge **182** can be

advanced from a proximal portion **1102** of the staple cartridge **1100** to a distal portion **1104** of the staple cartridge **1100** to cut the captured tissue, as illustrated in FIG. **74**. In certain instances, the cutting edge **182** can be retracted proximally from the distal portion **1104** to the proximal portion **1102** by retraction of the E-beam **178** proximally, for example.

In certain instances, the cutting edge **182** can be employed to cut tissue captured by the end effector **300** in multiple procedures. The reader will appreciate that repetitive use of the cutting edge **182** may affect the sharpness of the cutting edge **182**. The reader will also appreciate that as the sharpness of the cutting edge **182** decreases, the force required to cut the captured tissue with the cutting edge **182** may increase. Referring to FIGS. **74-76**, in certain instances, the surgical instrument **10** may comprise a module **1106** (FIG. **76**) for monitoring the sharpness of the cutting edge **182** during, before, and/or after operation of the surgical instrument **10** in a surgical procedure, for example. In certain instances, the module **1106** can be employed to test the sharpness of the cutting edge **182** prior to utilizing the cutting edge **182** to cut the captured tissue. In certain instances, the module **1106** can be employed to test the sharpness of the cutting edge **182** after the cutting edge **182** has been used to cut the captured tissue. In certain instances, the module **1106** can be employed to test the sharpness of the cutting edge **182** prior to and after the cutting edge **182** is used to cut the captured tissue. In certain instances, the module **1106** can be employed to test the sharpness of the cutting edge **1106** at the proximal portion **1102** and/or at the distal portion **1104**.

Referring to FIGS. **74-76**, the module **1106** may include one or more sensors such as, for example, an optical sensor **1108**; the optical sensor **1108** of the module **1106** can be employed to test the reflective ability of the cutting edge **182**, for example. In certain instances, the ability of the cutting edge **182** to reflect light may correlate with the sharpness of the cutting edge **182**. In other words, a decrease in the sharpness of the cutting edge **182** may result in a decrease in the ability of the cutting edge **182** to reflect the light. Accordingly, in certain instances, the dullness of the cutting edge **182** can be evaluated by monitoring the intensity of the light reflected from the cutting edge **182**, for example. In certain instances, the optical sensor **1108** may define a light sensing region. The optical sensor **1108** can be oriented such that the optical sensing region is disposed in the path of the cutting edge **182**, for example. The optical sensor **1108** may be employed to sense the light reflected from the cutting edge **182** while the cutting edge **182** is in the optical sensing region, for example. A decrease in intensity of the reflected light beyond a threshold can indicate that the sharpness of the cutting edge **182** has decreased beyond an acceptable level.

Referring again to FIGS. **74-76**, the module **1106** may include one or more lights sources such as, for example, a light source **1110**. In certain instances, the module **1106** may include a microcontroller **1112** ("controller") which may be operably coupled to the optical sensor **1108**, as illustrated in FIG. **76**. In certain instances, the controller **1112** may include a microprocessor **1114** ("processor") and one or more computer readable mediums or memory units **1116** ("memory"). In certain instances, the memory **1116** may store various program instructions, which when executed may cause the processor **1114** to perform a plurality of functions and/or calculations described herein. In certain instances, the memory **1116** may be coupled to the processor **1114**, for example. A power source **1118** can be configured

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to supply power to the controller **1112**, the optical sensors **1108**, and/or the light sources **1110**, for example. In certain instances, the power source **1118** may comprise a battery (or “battery pack” or “power pack”), such as a Li ion battery, for example. In certain instances, the battery pack may be configured to be releasably mounted to the handle **14** for supplying power to the surgical instrument **10**. A number of battery cells connected in series may be used as the power source **4428**. In certain instances, the power source **1118** may be replaceable and/or rechargeable, for example.

The controller **1112** and/or other controllers of the present disclosure may be implemented using integrated and/or discrete hardware elements, software elements, and/or a combination of both. Examples of integrated hardware elements may include processors, microprocessors, microcontrollers, integrated circuits, ASICs, PLDs, DSPs, FPGAs, logic gates, registers, semiconductor devices, chips, microchips, chip sets, microcontrollers, SoC, and/or SIP. Examples of discrete hardware elements may include circuits and/or circuit elements such as logic gates, field effect transistors, bipolar transistors, resistors, capacitors, inductors, and/or relays. In certain instances, the controller **1112** may include a hybrid circuit comprising discrete and integrated circuit elements or components on one or more substrates, for example.

In certain instances, the controller **1112** and/or other controllers of the present disclosure may be an LM4F230H5QR, available from Texas Instruments, for example. In certain instances, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, internal ROM loaded with StellarisWare® software, 2 KB EEPROM, one or more PWM modules, one or more QEI analog, one or more 12-bit ADC with 12 analog input channels, among other features that are readily available. Other microcontrollers may be readily substituted for use with the present disclosure. Accordingly, the present disclosure should not be limited in this context.

In certain instances, the light source **1110** can be employed to emit light which can be directed at the cutting edge **182** in the optical sensing region, for example. The optical sensor **1108** may be employed to measure the intensity of the light reflected from the cutting edge **182** while in the optical sensing region in response to exposure to the light emitted by the light source **1110**. In certain instances, the processor **1114** may receive one or more values of the measured intensity of the reflected light and may store the one or more values of the measured intensity of the reflected light on the memory **1116**, for example. The stored values can be detected and/or recorded before, after, and/or during a plurality of surgical procedures performed by the surgical instrument **10**, for example.

In certain instances, the processor **1114** may compare the measured intensity of the reflected light to a predefined threshold values that may be stored on the memory **1116**, for example. In certain instances, the controller **1112** may conclude that the sharpness of the cutting edge **182** has dropped below an acceptable level if the measured light intensity exceeds the predefined threshold value by 1%, 5%, 10%, 25%, 50%, 100% and/or more than 100%, for example. In certain instances, the processor **1114** can be employed to detect a decreasing trend in the stored values of the measured intensity of the light reflected from the cutting edge **182** while in the optical sensing region.

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In certain instances, the surgical instrument **10** may include one or more feedback systems such as, for example, the feedback system **1120**. In certain instances, the processor **1114** can employ the feedback system **1120** to alert a user if the measured light intensity of the light reflected from cutting edge **182** while in the optical sensing region is beyond the stored threshold value, for example. In certain instances, the feedback system **1120** may comprise one or more visual feedback systems such as display screens, backlights, and/or LEDs, for example. In certain instances, the feedback system **1120** may comprise one or more audio feedback systems such as speakers and/or buzzers, for example. In certain instances, the feedback system **1120** may comprise one or more haptic feedback systems, for example. In certain instances, the feedback system **1120** may comprise combinations of visual, audio, and/or tactile feedback systems, for example.

In certain instances, the surgical instrument **10** may comprise a firing lockout mechanism **1122** which can be employed to prevent advancement of the cutting edge **182**. Various suitable firing lockout mechanisms are described in greater detail in U.S. Patent Publication No. 2014/0001231, entitled FIRING SYSTEM LOCKOUT ARRANGEMENTS FOR SURGICAL INSTRUMENTS, and filed Jun. 28, 2012, which is hereby incorporated by reference herein in its entirety. In certain instances, as illustrated in FIG. 76, the processor **1114** can be operably coupled to the lockout mechanism **1122**; the processor **1114** may employ the lockout mechanism **1122** to prevent advancement of the cutting edge **182** in the event it is determined that the measured intensity of the light reflected from the cutting edge **182** is beyond the stored threshold, for example. In other words, the processor **1114** may activate the lockout mechanism **1122** if the cutting edge is not sufficiently sharp to cut the tissue captured by the end effector **300**.

In certain instances, the optical sensor **1108** and the light source **1110** can be housed at a distal portion of the shaft assembly **200**. In certain instances, the sharpness of cutting edge **182** can be evaluated by the optical sensor **1108**, as described above, prior to transitioning the cutting edge **182** into the end effector **300**. The firing bar **172** (FIG. 20) may advance the cutting edge **182** through the optical sensing region defined by the optical sensor **1108** while the cutting edge **182** is in the shaft assembly **182** and prior to entering the end effector **300**, for example. In certain instances, the sharpness of cutting edge **182** can be evaluated by the optical sensor **1108** after retracting the cutting edge **182** proximally from the end effector **300**. The firing bar **172** (FIG. 20) may retract the cutting edge **182** through the optical sensing region defined by the optical sensor **1108** after retracting the cutting edge **182** from the end effector **300** into the shaft assembly **200**, for example.

In certain instances, the optical sensor **1108** and the light source **1110** can be housed at a proximal portion of the end effector **300** which can be proximal to the staple cartridge **1100**, for example. The sharpness of cutting edge **182** can be evaluated by the optical sensor **1108** after transitioning the cutting edge **182** into the end effector **300** but prior to engaging the staple cartridge **1100**, for example. In certain instances, the firing bar **172** (FIG. 20) may advance the cutting edge **182** through the optical sensing region defined by the optical sensor **1108** while the cutting edge **182** is in the end effector **300** but prior to engaging the staple cartridge **1100**, for example.

In various instances, the sharpness of cutting edge **182** can be evaluated by the optical sensor **1108** as the cutting edge **182** is advanced by the firing bar **172** through the slot **193**.

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As illustrated in FIG. 74, the optical sensor 1108 and the light source 1110 can be housed at the proximal portion 1102 of the staple cartridge 1100, for example; and the sharpness of cutting edge 182 can be evaluated by the optical sensor 1108 at the proximal portion 1102, for example. The firing bar 172 (FIG. 20) may advance the cutting edge 182 through the optical sensing region defined by the optical sensor 1108 at the proximal portion 1102 before the cutting edge 182 engages tissue captured between the staple cartridge 1100 and the anvil 306, for example. In certain instances, as illustrated in FIG. 74, the optical sensor 1108 and the light source 1110 can be housed at the distal portion 1104 of the staple cartridge 1100, for example. The sharpness of cutting edge 182 can be evaluated by the optical sensor 1108 at the distal portion 1104. In certain instances, the firing bar 172 (FIG. 20) may advance the cutting edge 182 through the optical sensing region defined by the optical sensor 1108 at the distal portion 1104 after the cutting edge 182 has passed through the tissue captured between the staple cartridge 1100 and the anvil 306, for example.

Referring again to FIG. 74, the staple cartridge 1100 may comprise a plurality of optical sensors 1108 and a plurality of corresponding light sources 1110, for example. In certain instances, a pair of the optical sensor 1108 and the light source 1110 can be housed at the proximal portion 1102 of the staple cartridge 1100, for example; and a pair of the optical sensor 1108 and the light source 1110 can be housed at the distal portion 1104 of the staple cartridge 1100, for example. In such instances, the sharpness of the cutting edge 182 can be evaluated a first time at the proximal portion 1102 prior to engaging the tissue, for example, and a second time at the distal portion 1104 after passing through the captured tissue, for example.

The reader will appreciate that an optical sensor 1108 may evaluate the sharpness of the cutting edge 182 a plurality of times during a surgical procedure. For example, the sharpness of the cutting edge can be evaluated a first time during advancement of the cutting edge 182 through the slot 193 in a firing stroke, and a second time during retraction of the cutting edge 182 through the slot 193 in a return stroke, for example. In other words, the light reflected from the cutting edge 182 can be measured by the same optical sensor 1108 once as the cutting edge is advanced through the optical sensing region, and once as the cutting edge 182 is retracted through the optical sensing region, for example.

The reader will appreciate that the processor 1114 may receive a plurality of readings of the intensity of the light reflected from the cutting edge 182 from one or more of the optical sensors 1108. In certain instances, the processor 1114 may be configured to discard outliers and calculate an average reading from the plurality of readings, for example. In certain instances, the average reading can be compared to a threshold stored in the memory 1116, for example. In certain instances, the processor 1114 may be configured to alert a user through the feedback system 1120 and/or activate the lockout mechanism 1122 if it is determined that the calculated average reading is beyond the threshold stored in the memory 1116, for example.

In certain instances, as illustrated in FIGS. 75, 77, and 78, a pair of the optical sensor 1108 and the light source 1110 can be positioned on opposite sides of the staple cartridge 1100. In other words, the optical sensor 1108 can be positioned on a first side 1124 of the slot 193, for example, and the light source 1110 can be positioned on a second side 1126, opposite the first side 1124, of the slot 193, for example. In certain instances, the pair of the optical sensor 1108 and the light source 1110 can be substantially disposed

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in a plane transecting the staple cartridge 1100, as illustrated in FIG. 75. The pair of the optical sensor 1108 and the light source 1110 can be oriented to define an optical sensing region that is positioned, or at least substantially positioned, on the plane transecting the staple cartridge 1100, for example. Alternatively, the pair of the optical sensor 1108 and the light source 1110 can be oriented to define an optical sensing region that is positioned proximal to the plane transecting the staple cartridge 1100, for example, as illustrated in FIG. 78.

In certain instances, a pair of the optical sensor 1108 and the light source 1110 can be positioned on a same side of the staple cartridge 1100. In other words, as illustrated in FIG. 79, the pair of the optical sensor 1108 and the light source 1110 can be positioned on a first side of the cutting edge 182, e.g. the side 1128, as the cutting edge 182 is advanced through the slot 193. In such instances, the light source 1110 can be oriented to direct light at the side 1128 of the cutting edge 182; and the intensity of the light reflected from the side 1128, as measured by the optical sensor 1108, may represent the sharpness of the side 1128.

In certain instances, as illustrated in FIG. 80, a second pair of the optical sensor 1108 and the light source 1110 can be positioned on a second side of the cutting edge 182, e.g. the side 1130, for example. The second pair can be employed to evaluate the sharpness of the side 1130. For example, the light source 1110 of the second pair can be oriented to direct light at the side 1130 of the cutting edge 182; and the intensity of the light reflected from the side 1130, as measured by the optical sensor 1108 of the second pair, may represent the sharpness of the side 1130. In certain instances, the processor can be configured to assess the sharpness of the cutting edge 182 based upon the measured intensities of the light reflected from the sides 1128 and 1130 of the cutting edge 182, for example.

In certain instances, as illustrated in FIG. 75, a pair of the optical sensor 1108 and the light source 1110 can be housed at the distal portion 1104 of the staple cartridge 1100. As illustrated in FIG. 81, the light source 1108 can be positioned, or at least substantially positioned, on an axis LL which extends longitudinally along the path of the cutting edge 182 through the slot 193, for example. In addition, the light source 1110 can be positioned distal to the cutting edge 182 and oriented to direct light at the cutting edge 182 as the cutting edge is advanced toward the light source 1110, for example. Furthermore, the optical sensor 1108 can be positioned, or at least substantially positioned, along an axis AA that intersects the axis LL, as illustrated in FIG. 81. In certain instances, the axis AA may be perpendicular to the axis LL, for example. In any event, the optical sensor 1108 can be oriented to define an optical sensing region at the intersection of the axis LL and the axis AA, for example.

The reader will appreciate that the position, orientation and/or number of optical sensors and corresponding light sources described herein in connection with the surgical instrument 10 are example embodiments intended for illustration purposes. Various other arrangements of optical sensors and light sources can be employed by the present disclosure to evaluate the sharpness of the cutting edge 182.

The reader will appreciate that advancement of the cutting edge 182 through the tissue captured by the end effector 300 may cause the cutting edge to collect tissue debris and/or bodily fluids during each firing of the surgical instrument 10. Such debris may interfere with the ability of the module 1106 to accurately evaluate the sharpness of the cutting edge 182. In certain instances, the surgical instrument 10 can be equipped with one or more cleaning mechanisms which can

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be employed to clean the cutting edge **182** prior to evaluating the sharpness of the cutting edge **182**, for example. In certain instances, as illustrated in FIG. **82**, a cleaning mechanism **1131** may comprise one or more cleaning members **1132**, for example. In certain instances, the cleaning members **1132** can be disposed on opposite sides of the slot **193** to receive the cutting edge **182** therebetween (See FIG. **82**) as the cutting edge **182** is advanced through the slot **193**, for example. In certain instances, as illustrated in FIG. **82**, the cleaning members **1132** may comprise wiper blades, for example. In certain instances, as illustrated in FIG. **830**, the cleaning members **1132** may comprise sponges, for example. The reader will appreciate that various other cleaning members can be employed to clean the cutting edge **182**, for example.

Referring to FIG. **74**, in certain instances, the staple cartridge **1100** may include a first pair of the optical sensor **1108** and the light source **1110**, which can be housed in the proximal portion **1102** of the staple cartridge **1100**, for example. Furthermore, as illustrated in FIG. **74**, the staple cartridge **1100** may include a first pair of the cleaning members **1132**, which can be housed in the proximal portion **1102** on opposite sides of the slot **193**. The first pair of the cleaning members **1132** can be positioned distal to the first pair of the optical sensor **1108** and the light source **1110**, for example. As illustrated in FIG. **74**, the staple cartridge **1100** may include a second pair of the optical sensor **1108** and the light source **1110**, which can be housed in the distal portion **1104** of the staple cartridge **1100**, for example. As illustrated in FIG. **74**, the staple cartridge **1100** may include a second pair of the cleaning members **1132**, which can be housed in the distal portion **1104** on opposite sides of the slot **193**. The second pair of the cleaning members **1132** can be positioned proximal to the second pair of the optical sensor **1108** and the light source **1110**.

Further to the above, as illustrated in FIG. **74**, the cutting edge **182** may be advanced distally in a firing stroke to cut tissue captured by the end effector **300**. As the cutting edge is advanced, a first evaluation of the sharpness of the cutting edge **182** can be performed by the first pair of the optical sensor **1108** and the light source **1110** prior to tissue engagement by the cutting edge **182**, for example. A second evaluation of the sharpness of the cutting edge **182** can be performed by the second pair of the optical sensor **1108** and the light source **1110** after the cutting edge **182** has transected the captured tissue, for example. The cutting edge **182** may be advanced through the second pair of the cleaning members **1132** prior to the second evaluation of the sharpness of the cutting edge **182** to remove any debris collected by the cutting edge **182** during the transection of the captured tissue.

Further to the above, as illustrated in FIG. **74**, the cutting edge **182** may be retracted proximally in a return stroke. As the cutting edge is retracted, a third evaluation of the sharpness of the cutting edge **182** can be performed by the first pair of the optical sensor **1108** and the light source **1110** during the return stroke. The cutting edge **182** may be retracted through the first pair of the cleaning members **1132** prior to the third evaluation of the sharpness of the cutting edge **182** to remove any debris collected by the cutting edge **182** during the transection of the captured tissue, for example.

In certain instances, one or more of the light sources **1110** may comprise one or more optical fiber cables. In certain instances, one or more flex circuits **1134** can be employed to transmit energy from the power source **1118** to the optical sensors **1108** and/or the light sources **1110**. In certain

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instances, the flex circuits **1134** may be configured to transmit one or more of the readings of the optical sensors **1108** to the controller **1112**, for example.

Referring now to FIG. **84**, a staple cartridge **4300** is depicted; the staple cartridge **4300** is similar in many respects to the staple cartridge **304** (FIG. **20**). For example, the staple cartridge **4300** can be employed with the end effector **300**. In certain instances, as illustrated in FIG. **84**, the staple cartridge **4300** may comprise a sharpness testing member **4302** which can be employed to test the sharpness of the cutting edge **182**. In certain instances, the sharpness testing member **4302** can be attached to and/or integrated with the cartridge body **194** of the staple cartridge **4300**, for example. In certain instances, the sharpness testing member **4302** can be disposed in the proximal portion **1102** of the staple cartridge **4300**, for example. In certain instances, as illustrated in FIG. **84**, the sharpness testing member **4302** can be disposed onto a cartridge deck **4304** of the staple cartridge **4300**, for example.

In certain instances, as illustrated in FIG. **84**, the sharpness testing member **4302** can extend across the slot **193** of the staple cartridge **4300** to bridge, or at least partially bridge, the gap defined by the slot **193**, for example. In certain instances, the sharpness testing member **4302** may interrupt, or at least partially interrupt, the path of the cutting edge **182**. The cutting edge **182** may engage, cut, and/or pass through the sharpness testing member **4302** as the cutting edge **182** is advanced during a firing stroke, for example. In certain instances, the cutting edge **182** may be configured to engage, cut, and/or pass through the sharpness testing member **4302** prior to engaging tissue captured by the end effector **300** in a firing stroke, for example. In certain instances, the cutting edge **182** may be configured to engage the sharpness testing member **4302** at a proximal end **4306** of the sharpness testing member **4302**, and exit and/or disengage the sharpness testing member **4302** at a distal end **4308** of the sharpness testing member **4302**, for example. In certain instances, the cutting edge **182** can travel and/or cut through the sharpness testing member **4302** a distance (D) between the proximal end **4306** and the distal end **4308**, for example, as the cutting edge **182** is advanced during a firing stroke.

Referring primarily to FIGS. **84** and **85**, the surgical instrument **10** may comprise a sharpness testing module **4310** for testing the sharpness of the cutting edge **182**, for example. In certain instances, the module **4310** can evaluate the sharpness of the cutting edge **182** by testing the ability of the cutting edge **182** to be advanced through the sharpness testing member **4302**. For example, the module **4310** can be configured to observe the time period the cutting edge **182** takes to fully transect and/or completely pass through at least a predetermined portion of the sharpness testing member **4302**. If the observed time period exceeds a predetermined threshold, the module **4310** may conclude that the sharpness of the cutting edge **182** has dropped below an acceptable level, for example.

In certain instances, the module **4310** may include a microcontroller **4312** ("controller") which may include a microprocessor **4314** ("processor") and one or more computer readable mediums or memory units **4316** ("memory"). In certain instances, the memory **4316** may store various program instructions, which when executed may cause the processor **4314** to perform a plurality of functions and/or calculations described herein. In certain instances, the memory **4316** may be coupled to the processor **4314**, for example. A power source **4318** can be configured to supply power to the controller **4312**, for example. In certain

instances, the power source **4318** may comprise a battery (or “battery pack” or “power pack”), such as a Li ion battery, for example. In certain instances, the battery pack may be configured to be releasably mounted to the handle **14**. A number of battery cells connected in series may be used as the power source **4318**. In certain instances, the power source **4318** may be replaceable and/or rechargeable, for example.

In certain instances, the processor **4313** can be operably coupled to the feedback system **1120** and/or the lockout mechanism **1122**, for example.

Referring to FIGS. **84** and **85**, the module **4310** may comprise one or more position sensors. Example position sensors and positioning systems suitable for use with the present disclosure are described in U.S. patent application Ser. No. 13/803,210, entitled SENSOR ARRANGEMENTS FOR ABSOLUTE POSITIONING SYSTEM FOR SURGICAL INSTRUMENTS, and filed Mar. 14, 2013, now U.S. Pat. No. 9,808,244, the disclosure of which is hereby incorporated by reference herein in its entirety. In certain instances, the module **4310** may include a first position sensor **4320** and a second position sensor **4322**. In certain instances, the first position sensor **4320** can be employed to detect a first position of the cutting edge **182** at the proximal end **4306** of the sharpness testing member **4302**, for example; and the second position sensor **4322** can be employed to detect a second position of the cutting edge **182** at the distal end **4308** of the sharpness testing member **4302**, for example.

In certain instances, the position sensors **4320** and **4322** can be employed to provide first and second position signals, respectively, to the microcontroller **4312**. It will be appreciated that the position signals may be analog signals or digital values based on the interface between the microcontroller **4312** and the position sensors **4320** and **4322**. In one embodiment, the interface between the microcontroller **4312** and the position sensors **4320** and **4322** can be a standard serial peripheral interface (SPI), and the position signals can be digital values representing the first and second positions of the cutting edge **182**, as described above.

Further to the above, the processor **4314** may determine the time period between receiving the first position signal and receiving the second position signal. The determined time period may correspond to the time it takes the cutting edge **182** to advance through the sharpness testing member **4302** from the first position at the proximal end **4306** of the sharpness testing member **4302**, for example, to the second position at the distal end **4308** of the sharpness testing member **4302**, for example. In at least one example, the controller **4312** may include a time element which can be activated by the processor **4314** upon receipt of the first position signal, and deactivated upon receipt of the second position signal. The time period between the activation and deactivation of the time element may correspond to the time it takes the cutting edge **182** to advance from the first position to the second position, for example. The time element may comprise a real time clock, a processor configured to implement a time function, or any other suitable timing circuit.

In various instances, the controller **4312** can compare the time period it takes the cutting edge **182** to advance from the first position to the second position to a predefined threshold value to assess whether the sharpness of the cutting edge **182** has dropped below an acceptable level, for example. In certain instances, the controller **4312** may conclude that the sharpness of the cutting edge **182** has dropped below an acceptable level if the measured time period exceeds the

predefined threshold value by 1%, 5%, 10%, 25%, 50%, 100% and/or more than 100%, for example.

Referring to FIG. **86**, in various instances, an electric motor **4330** can drive the firing bar **172** (FIG. **20**) to advance the cutting edge **182** during a firing stroke and/or to retract the cutting edge **182** during a return stroke, for example. A motor driver **4332** can control the electric motor **4330**; and a microcontroller such as, for example, the microcontroller **4312** can be in signal communication with the motor driver **4332**. As the electric motor **4330** advances the cutting edge **182**, the microcontroller **4312** can determine the current drawn by the electric motor **4330**, for example. In such instances, the force required to advance the cutting edge **182** can correspond to the current drawn by the electric motor **4330**, for example. Referring still to FIG. **86**, the microcontroller **4312** of the surgical instrument **10** can determine if the current drawn by the electric motor **4330** increases during advancement of the cutting edge **182** and, if so, can calculate the percentage increase of the current.

In certain instances, the current drawn by the electric motor **4330** may increase significantly while the cutting edge **182** is in contact with the sharpness testing member **4302** due to the resistance of the sharpness testing member **4302** to the cutting edge **182**. For example, the current drawn by the electric motor **4330** may increase significantly as the cutting edge **182** engages, passes and/or cuts through the sharpness testing member **4302**. The reader will appreciate that the resistance of the sharpness testing member **4302** to the cutting edge **182** depends, in part, on the sharpness of the cutting edge **182**; and as the sharpness of the cutting edge **182** decreases from repetitive use, the resistance of the sharpness testing member **4302** to the cutting edge **182** will increase. Accordingly, the value of the percentage increase of the current drawn by the motor **4330** while the cutting edge is in contact with the sharpness testing member **4302** can increase as the sharpness of the cutting edge **182** decreases from repetitive use, for example.

In certain instances, the determined value of the percentage increase of the current drawn by the motor **4330** can be the maximum detected percentage increase of the current drawn by the motor **4330**. In various instances, the microcontroller **4312** can compare the determined value of the percentage increase of the current drawn by the motor **4330** to a predefined threshold value of the percentage increase of the current drawn by the motor **4330**. If the determined value exceeds the predefined threshold value, the microcontroller **4312** may conclude that the sharpness of the cutting edge **182** has dropped below an acceptable level, for example.

In certain instances, as illustrated in FIG. **86**, the processor **4314** can be in communication with the feedback system **1120** and/or the lockout mechanism **1122**, for example. In certain instances, the processor **4314** can employ the feedback system **1120** to alert a user if the determined value of the percentage increase of the current drawn by the motor **4330** exceeds the predefined threshold value, for example. In certain instances, the processor **4314** may employ the lockout mechanism **1122** to prevent advancement of the cutting edge **182** if the determined value of the percentage increase of the current drawn by the motor **4330** exceeds the predefined threshold value, for example.

In various instances, the microcontroller **43312** can utilize an algorithm to determine the change in current drawn by the electric motor **4330**. For example, a current sensor can detect the current drawn by the electric motor **4330** during the firing stroke. The current sensor can continually detect the current drawn by the electric motor and/or can intermittently detect the current draw by the electric motor. In

various instances, the algorithm can compare the most recent current reading to the immediately preceding current reading, for example. Additionally or alternatively, the algorithm can compare a sample reading within a time period X to a previous current reading. For example, the algorithm can compare the sample reading to a previous sample reading within a previous time period X, such as the immediately preceding time period X, for example. In other instances, the algorithm can calculate the trending average of current drawn by the motor. The algorithm can calculate the average current draw during a time period X that includes the most recent current reading, for example, and can compare that average current draw to the average current draw during an immediately preceding time period time X, for example.

Referring to FIG. 87, a method is depicted for evaluating the sharpness of the cutting edge 182 of the surgical instrument 10; and various responses are outlined in the event the sharpness of the cutting edge 182 drops to and/or below an alert threshold and/or a high severity threshold, for example. In various instances, a microcontroller such as, for example, the microcontroller 4312 can be configured to implement the method depicted in FIG. 87. In certain instances, the surgical instrument 10 may include a load cell 4334 (FIG. 86); as illustrated in FIG. 86, the microcontroller 4312 may be in communication with the load cell 4334. In certain instances, the load cell 4334 may include a force sensor such as, for example, a strain gauge, which can be operably coupled to the firing bar 172, for example. In certain instances, the microcontroller 4312 may employ the load cell 4334 to monitor the force (Fx) applied to the cutting edge 182 as the cutting edge 182 is advanced during a firing stroke.

In certain instances, as illustrated in FIG. 88, the load cell 4334 can be configured to monitor the force (Fx) applied to the cutting edge 182 while the cutting edge 182 is engaged and/or in contact with the sharpness testing member 4302, for example. The reader will appreciate that the force (Fx) applied by the sharpness testing member 4302 to the cutting edge 182 while the cutting edge 182 is engaged and/or in contact with the sharpness testing member 4302 may depend, at least in part, on the sharpness of the cutting edge 182. In certain instances, a decrease in the sharpness of the cutting edge 182 can result in an increase in the force (Fx) required for the cutting edge 182 to cut or pass through the sharpness testing member 4302. For example, as illustrated in FIG. 88, graphs 4336, 4338, and 4340 represent the force (Fx) applied to the cutting edge 182 while the cutting edge 182 travels a predefined distance (D) through three identical, or at least substantially identical, sharpness testing members 4302. The graph 4336 corresponds to a first sharpness of the cutting edge 182; the graph 4338 corresponds to a second sharpness of the cutting edge 182; and the graph 4340 corresponds to a third sharpness of the cutting edge 182. The first sharpness is greater than the second sharpness, and the second sharpness is greater than the third sharpness.

In certain instances, the microcontroller 4312 may compare a maximum value of the monitored force (Fx) applied to the cutting edge 182 to one or more predefined threshold values. In certain instances, as illustrated in FIG. 88, the predefined threshold values may include an alert threshold (F1) and/or a high severity threshold (F2). In certain instances, as illustrated in the graph 4336 of FIG. 88, the monitored force (Fx) can be less than the alert threshold (F1), for example. In such instances, as illustrated in FIG. 87, the sharpness of the cutting edge 182 is at a good level and the microcontroller 4312 may take no action to alert a

user as to the status of the cutting edge 182 or may inform the user that the sharpness of the cutting edge 182 is within an acceptable range.

In certain instances, as illustrated in the graph 4338 of FIG. 88, the monitored force (Fx) can be more than the alert threshold (F1) but less than the high severity threshold (F2), for example. In such instances, as illustrated in FIG. 87, the sharpness of the cutting edge 182 can be dulling but still within an acceptable level. The microcontroller 4312 may take no action to alert a user as to the status of the cutting edge 182. Alternatively, the microcontroller 4312 may inform the user that the sharpness of the cutting edge 182 is within an acceptable range. Alternatively or additionally, the microcontroller 4312 may determine or estimate the number of cutting cycles remaining in the lifecycle of the cutting edge 182 and may alert the user accordingly.

In certain instances, the memory 4316 may include a database or a table that correlates the number of cutting cycles remaining in the lifecycle of the cutting edge 182 to predetermined values of the monitored force (Fx). The processor 4314 may access the memory 4316 to determine the number of cutting cycles remaining in the lifecycle of the cutting edge 182 which correspond to a particular measured value of the monitored force (Fx) and may alert the user to the number of cutting cycles remaining in the lifecycle of the cutting edge 182, for example.

In certain instances, as illustrated in the graph 4340 of FIG. 88, the monitored force (Fx) can be more than the high severity threshold (F2), for example. In such instances, as illustrated in FIG. 87, the sharpness of the cutting edge 182 can be below an acceptable level. In response, the microcontroller 4312 may employ the feedback system 1120 to warn the user that the cutting edge 182 is too dull for safe use, for example. In certain instances, the microcontroller 4312 may employ the lockout mechanism 1122 to prevent advancement of the cutting edge 182 upon detection that the monitored force (Fx) exceeds the high severity threshold (F2), for example. In certain instances, the microcontroller 4312 may employ the feedback system 1122 to provide instructions to the user for overriding the lockout mechanism 1122, for example.

Referring to FIG. 89, a method is depicted for determining whether a cutting edge such as, for example, the cutting edge 182 is sufficiently sharp to be employed in transecting a tissue of a particular tissue thickness that is captured by the end effector 300, for example. In certain instances, the microcontroller 4312 can be implemented to perform the method depicted in FIG. 16, for example. As described above, repetitive use of the cutting edge 182 may dull or reduce the sharpness of the cutting edge 182 which may increase the force required for the cutting edge 182 to transect the captured tissue. In other words, the sharpness level of the cutting edge 182 can be defined by the force required for the cutting edge 182 to transect the captured tissue, for example. The reader will appreciate that the force required for the cutting edge 182 to transect a captured tissue may also depend on the thickness of the captured tissue. In certain instances, the greater the thickness of the captured tissue, the greater the force required for the cutting edge 182 to transect the captured tissue at the same sharpness level, for example.

In certain instances, the cutting edge 182 may be sufficiently sharp for transecting a captured tissue comprising a first thickness but may not be sufficiently sharp for transecting a captured tissue comprising a second thickness greater than the first thickness, for example. In certain instances, a sharpness level of the cutting edge 182, as defined by the

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force required for the cutting edge 182 to transect a captured tissue, may be adequate for transecting the captured tissue if the captured tissue comprises a tissue thickness that is in a particular range of tissue thicknesses, for example. In certain instances, as illustrated in FIG. 90, the memory 4316 can store one or more predefined ranges of tissue thicknesses of tissue captured by the end effector 300; and predefined threshold forces associated with the predefined ranges of tissue thicknesses. In certain instances, each predefined threshold force may represent a minimum sharpness level of the cutting edge 182 that is suitable for transecting a captured tissue comprising a tissue thickness (Tx) encompassed by the range of tissue thicknesses that is associated with the predefined threshold force. In certain instances, if the force (Fx) required for the cutting edge 182 to transect the captured tissue, comprising the tissue thickness (Tx), exceeds the predefined threshold force associated with the predefined range of tissue thicknesses that encompasses the tissue thickness (Tx), the cutting edge 182 may not be sufficiently sharp to transect the captured tissue, for example.

In certain instances, the predefined threshold forces and their corresponding predefined ranges of tissue thicknesses can be stored in a database and/or a table on the memory 4316 such as, for example, a table 4342, as illustrated in FIG. 90. In certain instances, the processor 4314 can be configured to receive a measured value of the force (Fx) required for the cutting edge 182 to transect a captured tissue and a measured value of the tissue thickness (Tx) of the captured tissue. The processor 4314 may access the table 4342 to determine the predefined range of tissue thicknesses that encompasses the measured tissue thickness (Tx). In addition, the processor 4314 may compare the measured force (Fx) to the predefined threshold force associated with the predefined range of tissue thicknesses that encompasses the tissue thickness (Tx). In certain instances, if the measured force (Fx) exceeds the predefined threshold force, the processor 4314 may conclude that the cutting edge 182 may not be sufficiently sharp to transect the captured tissue, for example.

Further to the above, the processor 4314 may employ one or more tissue thickness sensing modules such as, for example, a tissue thickness sensing module 4336 to determine the thickness of the captured tissue. Various suitable tissue thickness sensing modules are described in the present disclosure. In addition, various tissue thickness sensing devices and methods, which are suitable for use with the present disclosure, are disclosed in U.S. Publication No. US 2011/0155781, entitled SURGICAL CUTTING INSTRUMENT THAT ANALYZES TISSUE THICKNESS, and filed Dec. 24, 2009, now U.S. Pat. No. 8,851,354, the entire disclosure of which is hereby incorporated by reference herein.

In certain instances, the processor 4314 may employ the load cell 4334 to measure the force (Fx) required for the cutting edge 182 to transect a captured tissue comprising a tissue thickness (Tx). The reader will appreciate that the force applied to the cutting edge 182 by the captured tissue, while the cutting edge 182 is engaged and/or in contact with the captured tissue, may increase as the cutting edge 182 is advanced against the captured tissue up to the force (Fx) at which the cutting edge 182 may transect the captured tissue. In certain instances, the processor 4314 may employ the load cell 4334 to continually monitor the force applied by the captured tissue against the cutting edge 182 as the cutting edge 182 is advanced against the captured tissue. The processor 4314 may continually compare the monitored

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force to the predefined threshold force associated with the predefined tissue thickness range encompassing the tissue thickness (Tx) of the captured tissue. In certain instances, if the monitored force exceeds the predefined threshold force, the processor 4314 may conclude that the cutting edge is not sufficiently sharp to safely transect the captured tissue, for example.

The method described in FIG. 89 outline various example actions that can be taken by the processor 4313 in the event it is determined that the cutting edge 182 is not be sufficiently sharp to safely transect the captured tissue, for example. In certain instances, the microcontroller 4312 may warn the user that the cutting edge 182 is too dull for safe use, for example, through the feedback system 1120, for example. In certain instances, the microcontroller 4312 may employ the lockout mechanism 1122 to prevent advancement of the cutting edge 182 upon concluding that the cutting edge 182 is not sufficiently sharp to safely transect the captured tissue, for example. In certain instances, the microcontroller 4312 may employ the feedback system 1122 to provide instructions to the user for overriding the lockout mechanism 1122, for example.

Multiple Motor Control for Powered Medical Device

FIGS. 91-93 illustrate various embodiments of an apparatus, system, and method for employing a common control module with a plurality of motors in connection with a surgical instrument such as, for example, a surgical instrument 4400. The surgical instrument 4400 is similar in many respects to other surgical instruments described by the present disclosure such as, for example, the surgical instrument 10 of FIG. 1 which is described in greater detail above. For example, as illustrated in FIG. 91, the surgical instrument 4400 includes the housing 12, the handle 14, the closure trigger 32, the shaft assembly 200, and the surgical end effector 300. Accordingly, for conciseness and clarity of disclosure, a detailed description of certain features of the surgical instrument 4400, which are common with the surgical instrument 10, will not be repeated here.

Referring primarily to FIG. 92, the surgical instrument 4400 may include a plurality of motors which can be activated to perform various functions in connection with the operation of the surgical instrument 4400. In certain instances, a first motor can be activated to perform a first function; a second motor can be activated to perform a second function; and a third motor can be activated to perform a third function. In certain instances, the plurality of motors of the surgical instrument 4400 can be individually activated to cause articulation, closure, and/or firing motions in the end effector 300. The articulation, closure, and/or firing motions can be transmitted to the end effector 300 through the shaft assembly 200, for example.

In certain instances, as illustrated in FIG. 92, the surgical instrument 4400 may include a firing motor 4402. The firing motor 4402 may be operably coupled to a firing drive assembly 4404 which can be configured to transmit firing motions generated by the motor 4402 to the end effector 300. In certain instances, the firing motions generated by the motor 4402 may cause the staples 191 to be deployed from the staple cartridge 304 into tissue captured by the end effector 300 and/or the cutting edge 182 to be advanced to cut the captured tissue, for example.

In certain instances, as illustrated in FIG. 92, the surgical instrument 4400 may include an articulation motor 4406, for example. The motor 4406 may be operably coupled to an articulation drive assembly 4408 which can be configured to transmit articulation motions generated by the motor 4406 to the end effector 300. In certain instances, the articulation

motions may cause the end effector **300** to articulate relative to the shaft assembly **200**, for example. In certain instances, the surgical instrument **4400** may include a closure motor, for example. The closure motor may be operably coupled to a closure drive assembly which can be configured to transmit closure motions to the end effector **300**. In certain instances, the closure motions may cause the end effector **300** to transition from an open configuration to an approximated configuration to capture tissue, for example. The reader will appreciate that the motors described herein and their corresponding drive assemblies are intended as examples of the types of motors and/or driving assemblies that can be employed in connection with the present disclosure. The surgical instrument **4400** may include various other motors which can be utilized to perform various other functions in connection with the operation of the surgical instrument **4400**.

As described above, the surgical instrument **4400** may include a plurality of motors which may be configured to perform various independent functions. In certain instances, the plurality of motors of the surgical instrument **4400** can be individually or separately activated to perform one or more functions while the other motors remain inactive. For example, the articulation motor **4406** can be activated to cause the end effector **300** to be articulated while the firing motor **4402** remains inactive. Alternatively, the firing motor **4402** can be activated to fire the plurality of staples **191** and/or advance the cutting edge **182** while the articulation motor **4406** remains inactive.

In certain instances, the surgical instrument **4400** may include a common control module **4410** which can be employed with a plurality of motors of the surgical instrument **4400**. In certain instances, the common control module **4410** may accommodate one of the plurality of motors at a time. For example, the common control module **4410** can be separably couplable to the plurality of motors of the surgical instrument **4400** individually. In certain instances, a plurality of the motors of the surgical instrument **4400** may share one or more common control modules such as the module **4410**. In certain instances, a plurality of motors of the surgical instrument **4400** can be individually and selectively engaged the common control module **4410**. In certain instances, the module **4410** can be selectively switched from interfacing with one of a plurality of motors of the surgical instrument **4400** to interfacing with another one of the plurality of motors of the surgical instrument **4400**.

In at least one example, the module **4410** can be selectively switched between operable engagement with the articulation motor **4406** and operable engagement with the firing motor **4402**. In at least one example, as illustrated in FIG. 92, a switch **4414** can be moved or transitioned between a plurality of positions and/or states such as a first position **4416** and a second position **4418**, for example. In the first position **4416**, the switch **4414** may electrically couple the module **4410** to the articulation motor **4406**; and in the second position **4418**, the switch **4414** may electrically couple the module **4410** to the firing motor **4402**, for example. In certain instances, the module **4410** can be electrically coupled to the articulation motor **4406**, while the switch **4414** is in the first position **4416**, to control the operation of the motor **4406** to articulate the end effector **300** to a desired position. In certain instances, the module **4410** can be electrically coupled to the firing motor **4402**, while the switch **4414** is in the second position **4418**, to control the operation of the motor **4402** to fire the plurality of staples **191** and/or advance the cutting edge **182**, for example. In certain instances, the switch **4414** may be a mechanical

switch, an electromechanical switch, a solid state switch, or any suitable switching mechanism.

Referring now to FIG. 93, an outer casing of the handle **14** of the surgical instrument **4400** is removed and several features and elements of the surgical instrument **4400** are also removed for clarity of disclosure. In certain instances, as illustrated in FIG. 93, the surgical instrument **4400** may include an interface **4412** which can be selectively transitioned between a plurality of positions and/or states. In a first position and/or state, the interface **4412** may couple the module **4410** to a first motor such as, for example, the articulation motor **4406**; and in a second position and/or state, the interface **4412** may couple the module **4410** to a second motor such as, for example, the firing motor **4402**. Additional positions and/or states of the interface **4412** are contemplated by the present disclosure.

In certain instances, the interface **4412** is movable between a first position and a second position, wherein the module **4410** is coupled to a first motor in the first position and a second motor in the second position. In certain instances, the module **4410** is decoupled from first motor as the interface **4412** is moved from the first position; and the module **4410** is decoupled from second motor as the interface **4412** is moved from the second position. In certain instances, a switch or a trigger can be configured to transition the interface **4412** between the plurality of positions and/or states. In certain instances, a trigger can be movable to simultaneously effectuate the end effector and transition the control module **4410** from operable engagement with one of the motors of the surgical instrument **4400** to operable engagement with another one of the motors of the surgical instrument **4400**.

In at least one example, as illustrated in FIG. 93, the closure trigger **32** can be operably coupled to the interface **4412** and can be configured to transition the interface **4412** between a plurality of positions and/or states. As illustrated in FIG. 93, the closure trigger **32** can be movable, for example during a closure stroke, to transition the interface **4412** from a first position and/or state to a second position and/or state while transitioning the end effector **300** to an approximated configuration to capture tissue by the end effector, for example.

In certain instances, in the first position and/or state, the module **4410** can be electrically coupled to a first motor such as, for example, the articulation motor **4406**, and in the second position and/or state, the module **4410** can be electrically coupled to a second motor such as, for example, the firing motor **4402**. In the first position and/or state, the module **4410** may be engaged with the articulation motor **4406** to allow the user to articulate the end effector **300** to a desired position; and the module **4410** may remain engaged with the articulation motor **4406** until the trigger **32** is actuated. As the user actuates the closure trigger **32** to capture tissue by the end effector **300** at the desired position, the interface **4412** can be transitioned or shifted to transition the module **4410** from operable engagement with the articulation motor **4406**, for example, to operable engagement with the firing motor **4402**, for example. Once operable engagement with the firing motor **4402** is established, the module **4410** may take control of the firing motor **4402**; and the module **4410** may activate the motor **4402**, in response to user input, to fire the plurality of staples **191** and/or advance the cutting edge **182**, for example.

In certain instances, as illustrated in FIG. 93, the module **4410** may include a plurality of electrical and/or mechanical contacts **4411** adapted for coupling engagement with the interface **4412**. The plurality of motors of the surgical

instrument **4400**, which share the module **4410**, may each comprise one or more corresponding electrical and/or mechanical contacts **4413** adapted for coupling engagement with the interface **4412**, for example.

In various instances, the motors of the surgical instrument **4400** can be electrical motors. In certain instances, one or more of the motors of the surgical instrument **4400** can be a DC brushed driving motor having a maximum rotation of, approximately, 25,000 RPM, for example. In other arrangements, the motors of the surgical instrument **4400** may include one or more motors selected from a group of motors comprising a brushless motor, a cordless motor, a synchronous motor, a stepper motor, or any other suitable electric motor.

In various instances, as illustrated in FIG. **92**, the common control module **4410** may comprise a motor driver **4426** which may comprise one or more H-Bridge field-effect transistors (FETs). The motor driver **4426** may modulate the power transmitted from a power source **4428** to a motor coupled to the module **4410** based on input from a microcontroller **4420** ("controller"), for example. In certain instances, the controller **4420** can be employed to determine the current drawn by the motor, for example, while the motor is coupled to the module **4410**, as described above.

In certain instances, the controller **4420** may include a microprocessor **4422** ("processor") and one or more computer readable mediums or memory units **4424** ("memory"). In certain instances, the memory **4424** may store various program instructions, which when executed may cause the processor **4422** to perform a plurality of functions and/or calculations described herein. In certain instances, one or more of the memory units **4424** may be coupled to the processor **4422**, for example.

In certain instances, the power source **4428** can be employed to supply power to the controller **4420**, for example. In certain instances, the power source **4428** may comprise a battery (or "battery pack" or "power pack"), such as a Li ion battery, for example. In certain instances, the battery pack may be configured to be releasably mounted to the handle **14** for supplying power to the surgical instrument **4400**. A number of battery cells connected in series may be used as the power source **4428**. In certain instances, the power source **4428** may be replaceable and/or rechargeable, for example.

In various instances, the processor **4422** may control the motor driver **4426** to control the position, direction of rotation, and/or velocity of a motor that is coupled to the module **4410**. In certain instances, the processor **4422** can signal the motor driver **4426** to stop and/or disable a motor that is coupled to the module **4410**. It should be understood that the term processor as used herein includes any suitable microprocessor, microcontroller, or other basic computing device that incorporates the functions of a computer's central processing unit (CPU) on an integrated circuit or at most a few integrated circuits. The processor is a multipurpose, programmable device that accepts digital data as input, processes it according to instructions stored in its memory, and provides results as output. It is an example of sequential digital logic, as it has internal memory. Processors operate on numbers and symbols represented in the binary numeral system.

In one instance, the processor **4422** may be any single core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In certain instances, the microcontroller **4420** may be an LM 4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments

LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, internal ROM loaded with StellarisWare® software, 2 KB EEPROM, one or more PWM modules, one or more QEI analog, one or more 12-bit ADC with 12 analog input channels, among other features that are readily available for the product datasheet. Other microcontrollers may be readily substituted for use with the module **4410**. Accordingly, the present disclosure should not be limited in this context.

In certain instances, the memory **4424** may include program instructions for controlling each of the motors of the surgical instrument **4400** that are couplable to the module **4410**. For example, the memory **4424** may include program instructions for controlling the articulation motor **4406**. Such program instructions may cause the processor **4422** to control the articulation motor **4406** to articulate the end effector **300** in accordance with user input while the articulation motor **4406** is coupled to the module **4410**. In another example, the memory **4424** may include program instructions for controlling the firing motor **4402**. Such program instructions may cause the processor **4422** to control the firing motor **4402** to fire the plurality of staples **191** and/or advance the cutting edge **182** in accordance with user input while the firing motor **4402** is coupled to the module **4410**.

In certain instances, one or more mechanisms and/or sensors such as, for example, sensors **4430** can be employed to alert the processor **4422** to the program instructions that should be used in a particular setting. For example, the sensors **4430** may alert the processor **4422** to use the program instructions associated with articulation of the end effector **300** while the module **4410** is coupled to the articulation motor **4406**; and the sensors **4430** may alert the processor **4422** to use the program instructions associated with firing the surgical instrument **4400** while the module **4410** is coupled to the firing motor **4402**. In certain instances, the sensors **4430** may comprise position sensors which can be employed to sense the position of the switch **4414**, for example. Accordingly, the processor **4422** may use the program instructions associated with articulation of the end effector **300** upon detecting, through the sensors **4430** for example, that the switch **4414** is in the first position **4416**; and the processor **4422** may use the program instructions associated with firing the surgical instrument **4400** upon detecting, through the sensors **4430** for example, that the switch **4414** is in the second position **4418**.

Referring now to FIG. **94**, an outer casing of the surgical instrument **4400** is removed and several features and elements of the surgical instrument **4400** are also removed for clarity of disclosure. As illustrated in FIG. **94**, the surgical instrument **4400** may include a plurality of sensors which can be employed to perform various functions in connection with the operation of the surgical instrument **4400**. For example, as illustrated in FIG. **94**, the surgical instrument **4400** may include sensors A, B, and/or C. In certain instances, the sensor A can be employed to perform a first function, for example; the sensor B can be employed to perform a second function, for example; and the sensor C can be employed to perform a third function, for example. In certain instances, the sensor A can be employed to sense a thickness of the tissue captured by the end effector **300** during a first segment of a closure stroke; the sensor B can be employed to sense the tissue thickness during a second segment of the closure stroke following the first segment; and the sensor C can be employed to sense the tissue

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thickness during a third segment of the closure stroke following the second segment, for example. In certain instances, the sensors A, B, and C can be disposed along the end effector 300, for example.

In certain instances, the sensors A, B, and C can be arranged, as illustrated in FIG. 94, such that the sensor A is disposed proximal to the sensor B, and the sensor C is disposed proximal to the sensor B, for example. In certain instances, as illustrated in FIG. 94, the sensor A can sense the tissue thickness of the tissue captured by the end effector 300 at a first position; the sensor B can sense the tissue thickness of the tissue captured by the end effector 300 at a second position distal to the first position; and the sensor C can sense the tissue thickness of the tissue captured by the end effector 300 at a third position distal to the second position, for example. The reader will appreciate that the sensors described herein are intended as examples of the types of sensors which can be employed in connection with the present disclosure. Other suitable sensors and sensing arrangements can be employed by the present disclosure.

In certain instances, the surgical instrument 4400 may include a common control module 4450 which can be similar in many respects to the module 4410. For example, the module 4450, like the module 4410, may comprise the controller 4420, the processor 4422, and/or the memory 4424. In certain instances, the power source 4428 can supply power to the module 4450, for example. In certain instances, the surgical instrument 4400 may include a plurality of sensors such as the sensors A, B, and C, for example, which can be activated to perform various functions in connection with the operation of the surgical instrument 4400. In certain instances, one of the sensors A, B, and C, for example, can be individually or separately activated to perform one or more functions while the other sensors remain inactive. In certain instances, a plurality of sensors of the surgical instrument 4400 such as, for example, the sensors A, B, and C may share the module 4450. In certain instances, only one of the sensors A, B, and C can be coupled to the module 4450 at a time. In certain instances, the plurality of sensors of the surgical instrument 4400 can be individually and separately couplable to the module 4450, for example. In at least one example, the module 4450 can be selectively switched between operable engagement with sensor A, Sensor B, and/or Sensor C.

In certain instances, as illustrated in FIG. 94, the module 4450 can be disposed in the handle 14, for example, and the sensors that share the module 4450 can be disposed in the end effector 300, for example. The reader will appreciate that the module 4450 and/or the sensors that share the module 4450 are not limited to the above identified positions. In certain instances, the module 4450 and the sensors that share the module 4450 can be disposed in the end effector 300, for example. Other arrangements for the positions of the module 4450 and/or the sensors that share the module 4450 are contemplated by the present disclosure.

In certain instances, as illustrated in FIG. 94, an interface 4452 can be employed to manage the coupling and/or decoupling of the sensors of the surgical instrument 4400 to the module 4450. In certain instances, the interface 4452 can be selectively transitioned between a plurality of positions and/or states. In a first position and/or state, the interface 4452 may couple the module 4450 to the sensor A, for example; in a second position and/or state, the interface 4452 may couple the module 4450 to the sensor B, for example; and in a third position and/or state, the interface 4452 may couple the module 4450 to the sensor C, for

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example. Additional positions and/or states of the interface 4452 are contemplated by the present disclosure.

In certain instances, the interface 4452 is movable between a first position, a second position, and/or a third position, for example, wherein the module 4450 is coupled to a first sensor in the first position, a second sensor in the second position, and a third sensor in the third position. In certain instances, the module 4450 is decoupled from first sensor as the interface 4452 is moved from the first position; the module 4450 is decoupled from second sensor as the interface 4452 is moved from the second position; and the module 4450 is decoupled from third sensor as the interface 4452 is moved from the third position. In certain instances, a switch or a trigger can be configured to transition the interface 4452 between the plurality of positions and/or states. In certain instances, a trigger can be movable to simultaneously effectuate the end effector and transition the control module 4450 from operable engagement with one of the sensors that share the module 4450 to operable engagement with another one of the sensors that share the module 4450, for example.

In at least one example, as illustrated in FIG. 94, the closure trigger 32 can be operably coupled to the interface 4450 and can be configured to transition the interface 4450 between a plurality of positions and/or states. As illustrated in FIG. 94, the closure trigger 32 can be moveable between a plurality of positions, for example during a closure stroke, to transition the interface 4450 between a first position and/or state wherein the module 4450 is electrically coupled to the sensor A, for example, a second position and/or state wherein the module 4450 is electrically coupled to the sensor B, for example, and/or a third position and/or state wherein the module 4450 is electrically coupled to the sensor C, for example.

In certain instances, a user may actuate the closure trigger 32 to capture tissue by the end effector 300. Actuation of the closure trigger may cause the interface 4452 to be transitioned or shifted to transition the module 4450 from operable engagement with the sensor A, for example, to operable engagement with the sensor B, for example, and/or from operable engagement with sensor B, for example, to operable engagement with sensor C, for example.

In certain instances, the module 4450 may be coupled to the sensor A while the trigger 32 is in a first actuated position. As the trigger 32 is actuated past the first actuated position and toward a second actuated position, the module 4450 may be decoupled from the sensor A. Alternatively, the module 4450 may be coupled to the sensor A while the trigger 32 is in an unactuated position. As the trigger 32 is actuated past the unactuated position and toward a second actuated position, the module 4450 may be decoupled from the sensor A. In certain instances, the module 4450 may be coupled to the sensor B while the trigger 32 is in the second actuated position. As the trigger 32 is actuated past the second actuated position and toward a third actuated position, the module 4450 may be decoupled from the sensor B. In certain instances, the module 4450 may be coupled to the sensor C while the trigger 32 is in the third actuated position.

In certain instances, as illustrated in FIG. 94, the module 4450 may include a plurality of electrical and/or mechanical contacts 4451 adapted for coupling engagement with the interface 4452. The plurality of sensors of the surgical instrument 4400, which share the module 4450, may each comprise one or more corresponding electrical and/or mechanical contacts 4453 adapted for coupling engagement with the interface 4452, for example.

In certain instances, the processor **4422** may receive input from the plurality of sensors that share the module **4450** while the sensors are coupled to the module **4452**. For example, the processor **4422** may receive input from the sensor A while the sensor A is coupled to the module **4450**; the processor **4422** may receive input from the sensor B while the sensor B is coupled to the module **4450**; and the processor **4422** may receive input from the sensor C while the sensor C is coupled to the module **4450**. In certain instances, the input can be a measurement value such as, for example, a measurement value of a tissue thickness of tissue captured by the end effector **300**. In certain instances, the processor **4422** may store the input from one or more of the sensors A, B, and C on the memory **4426**. In certain instances, the processor **4422** may perform various calculations based on the input provided by the sensors A, B, and C, for example.

Local Display of Tissue Parameter Stabilization

FIGS. **95A** and **1B** illustrate one embodiment of an end effector **5300** comprising a staple cartridge **5306** that further comprises two light-emitting diodes (LEDs) **5310**. The end effector **5300** is similar to the end effector **300** described above. The end effector comprises a first jaw member or anvil **5302**, pivotally coupled to a second jaw member or elongated channel **5304**. The elongated channel **5304** is configured to receive the staple cartridge **5306** therein. The staple cartridge **5306** comprises a plurality of staples (not shown). The plurality of staples are deployable from the staple cartridge **5306** during a surgical operation. The staple cartridge **5306** further comprises two LEDs **5310** mounted on the upper surface, or cartridge deck **5308** of the staple cartridge **5306**. The LEDs **5310** are mounted such that they will be visible when the anvil **5304** is in a closed position. Furthermore, the LEDs **5310** can be sufficiently bright to be visible through any tissue that may be obscuring a direct view of the LEDs **5310**. Additionally, one LED **5310** can be mounted on either side of the staple cartridge **5306** such that at least one LED **5310** is visible from either side of the end effector **5300**. The LED **5310** can be mounted near the proximal end of the staple cartridge **530**, as illustrated, or may be mounted at the distal end of the staple cartridge **5306**.

The LEDs **5310** may be in communication with a processor or microcontroller, such as for instance microcontroller **1500** of FIG. **19**. The microcontroller **1500** can be configured to detect a property of tissue compressed by the anvil **5304** against the cartridge deck **5308**. Tissue that is enclosed by the end effector **5300** may change height as fluid within the tissue is exuded from the tissue's layers. Stapling the tissue before it has sufficiently stabilized may affect the effectiveness of the staples. Tissue stabilization is typically communicated as a rate of change, where the rate of change indicates how rapidly the tissue enclosed by the end effector is changing height.

The LEDs **5310** mounted to the staple cartridge **5306**, in the view of the operator of the instrument, can be used to indicate rate at which the enclosed tissue is stabilizing and/or whether the tissue has reached a stable state. The LEDs **5310** can, for example, be configured to flash at a rate that directly correlates to the rate of stabilization of the tissue, that is, can flash quickly initially, flash slower as the tissue stabilizes, and remain steady when the tissue is stable. Alternatively, the LEDs **5310** can flash slowly initially, flash more quickly as the tissue stabilizes, and turn off when the tissue is stable.

The LEDs **5310** mounted on the staple cartridge **5306** can be used additionally or optionally to indicate other informa-

tion. Examples of other information include, but are not limited to: whether the end effector **5300** is enclosing a sufficient amount of tissue, whether the staple cartridge **5306** is appropriate for the enclosed tissue, whether there is more tissue enclosed than is appropriate for the staple cartridge **5306**, whether the staple cartridge **5306** is not compatible with the surgical instrument, or any other indicator that would be useful to the operator of the instrument. The LEDs **5310** can indicate information by either flashing at a particular rate, turning on or off at a particular instance, lighting in different colors for different information. The LEDs **5310** can alternatively or additionally be used to illuminate the area of operation. In some embodiments the LEDs **5310** can be selected to emit ultraviolet or infrared light to illuminate information not visible under normal light, where that information is printed on the staple cartridge **5300** or on a tissue compensator (not illustrated). Alternatively or additionally, the staples can be coated with a fluorescing dye and the wavelength of the LEDs **5310** chosen so that the LEDs **5310** cause the fluorescing dye to glow. By illuminating the staples with the LEDs **5310** allows the operator of the instrument to see the staples after they have been driven.

Returning to FIGS. **95A** and **95B**, FIG. **95A** illustrates a side angle view of the end effector **5300** with the anvil **5304** in a closed position. The illustrated embodiment comprises, by way of example, one LED **5310** located on either side of the cartridge deck **5308**. FIG. **95B** illustrates a three-quarter angle view of the end effector **5300** with the anvil **5304** in an open position, and one LED **5310** located on either side of the cartridge deck **5308**.

FIGS. **96A** and **96B** illustrate one embodiment of the end effector **5300** comprising a staple cartridge **5356** that further comprises a plurality of LEDs **5360**. The staple cartridge **5356** comprises a plurality of LEDs **5360** mounted on the cartridge deck **5358** of the staple cartridge **5356**. The LEDs **5360** are mounted such that they will be visible when the anvil **5304** is in a closed position. Furthermore, the LEDs **5360** can be sufficiently bright to be visible through any tissue that may be obscuring a direct view of the LEDs **5360**. Additionally, the same number of LEDs **5360** can be mounted on either side of the staple cartridge **5356** such that the same number of LEDs **5360** is visible from either side of the end effector **5300**. The LEDs **5360** can be mounted near the proximal end of the staple cartridge **5356**, as illustrated, or may be mounted at the distal end of the staple cartridge **5356**.

The LEDs **5360** may be in communication with a processor or microcontroller, such as for instance microcontroller **1500** of FIG. **15**. The microcontroller **1500** can be configured to detect a property of tissue compressed by the anvil **5304** against the cartridge deck **5358**, such as the rate of stabilization of the tissue, as described above. The LEDs **5360** can be used to indicate the rate at which the enclosed tissue is stabilizing and/or whether the tissue has reached a stable state. The LEDs **5360** can be configured, for instance, to light in sequence starting at the proximal end of the staple cartridge **5356** with each subsequent LED **5360** lighting at the rate at which the enclosed tissue is stabilizing; when the tissue is stable, all the LEDs **5360** can be lit. Alternatively, the LEDs **5360** can light in sequence beginning at the distal end of the staple cartridge **5356**. Yet another alternative is for the LEDs **5360** to light in a sequential, repeating sequence, with the sequence starting at either the proximal or distal end of the LEDs **5360**. The rate at which the LEDs **5360** light and/or the speed of the repeat can indicate the rate at which the enclosed tissue is stabilizing. It is understood that these are only examples of how the LEDs **5360** can indicate

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information about the tissue, and that other combinations of the sequence in which the LEDs 5360 light, the rate at which they light, and or their on or off state are possible. It is also understood that the LEDs 5360 can be used to communicate some other information to the operator of the surgical instrument, or to light the work area, as described above.

Returning to FIGS. 96A and 96B, FIG. 96A illustrates a side angle view of the end effector 5300 with the anvil 5304 in a closed position. The illustrated embodiment comprises, by way of example, a plurality of LEDs 5360 located on either side of the cartridge deck 5358. FIG. 96B illustrates a three-quarter angle view of the end effector 5300 with the anvil 5304 in an open position, illustrating a plurality of LEDs 5360 located on either side of the cartridge deck 5358.

FIGS. 97A and 97B illustrate one embodiment of the end effector 5300 comprising a staple cartridge 5406 that further comprises a plurality of LEDs 5410. The staple cartridge 5406 comprises a plurality of LEDs 5410 mounted on the cartridge deck 5408 of the staple cartridge 5406, with the LEDs 5410 placed continuously from the proximal to the distal end of the staple cartridge 5406. The LEDs 5410 are mounted such that they will be visible when the anvil 5302 is in a closed position. The same number of LEDs 5410 can be mounted on either side of the staple cartridge 5406 such that the same number of LEDs 5410 is visible from either side of the end effector 5300.

The LEDs 5410 can be in communication with a processor or microcontroller, such as for instance microcontroller 1500 of FIG. 15. The microcontroller 1500 can be configured to detect a property of tissue compressed by the anvil 5304 against the cartridge deck 5408, such as the rate of stabilization of the tissue, as described above. The LEDs 5410 can be configured to be turned on or off in sequences or groups as desired to indicate the rate of stabilization of the tissue and/or that the tissue is stable. The LEDs 5410 can further be configured communicate some other information to the operator of the surgical instrument, or to light the work area, as described above. Additionally or alternatively, the LEDs 5410 can be configured to indicate which areas of the end effector 5300 contain stable tissue, and or what areas of the end effector 5300 are enclosing tissue, and/or if those areas are enclosing sufficient tissue. The LEDs 5410 can further be configured to indicate if any portion of the enclosed tissue is unsuitable for the staple cartridge 5406.

Returning to FIGS. 97A and 97B, FIG. 97A illustrates a side angle view of the end effector 5300 with the anvil 5304 in a closed position. The illustrated embodiment comprises, by way of example, a plurality of LEDs 5410 from the proximal to the distal end of the staple cartridge 5406, on either side of the cartridge deck 5408. FIG. 97B illustrates a three-quarter angle view of the end effector 5300 with the anvil 5304 in an open position, illustrating a plurality of LEDs 5410 from the proximal to the distal end of the staple cartridge 5406, and on either side of the cartridge deck 5408. Adjunct with Integrated Sensors to Quantify Tissue Compression

FIG. 98A illustrates an embodiment of an end effector 5500 comprising a tissue compensator 5510 that further comprises a layer of conductive elements 5512. The end effector 5500 is similar to the end effector 300 described above. The end effector 5500 comprises a first jaw member, or anvil, 5502 pivotally coupled to a second jaw member 5504 (not shown). The second jaw member 5504 is configured to receive a staple cartridge 5506 therein (not shown). The staple cartridge 5506 comprises a plurality of staples (not shown). The plurality of staples 191 is deployable from the staple cartridge 3006 during a surgical operation. In

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some embodiments, the end effector 5500 further comprises a tissue compensator 5510 removably positioned on the anvil 5502 or on the staple cartridge 5506. FIG. 98B illustrates a detail view of a portion of the tissue compensator 5510 shown in FIG. 98A.

As described above, the plurality of staples 191 can be deployed between an unfired position and a fired position, such that staple legs 5530 move through and penetrate tissue 5518 compressed between the anvil 5502 and the staple cartridge 5506, and contact the anvil's 5502 staple-forming surface. In embodiments that include a tissue compensator 5510, the staple legs 5530 also penetrate and puncture the tissue compensator 5510. As the staple legs 5530 are deformed against the anvil's staple-forming surface, each staple 191 can capture a portion of the tissue 5518 and the tissue compensator 5510 and apply a compressive force to the tissue 5518. The tissue compensator 5510 thus remains in place with the staples 191 after the surgical instrument 10 is withdrawn from the patient's body. Because they are to be retained by the patient's body, the tissue compensators 5510 are composed of biodegradable and/or biodegradable materials. The tissue compensators 5510 are described in further detail in U.S. Pat. No. 8,657,176, entitled TISSUE THICKNESS COMPENSATOR FOR SURGICAL STAPLER, which is incorporated herein by reference in its entirety.

Returning to FIG. 98A, in some embodiments, the tissue compensator 5510 comprises a layer of conductive elements 5512. The conductive elements 5512 can comprise any combination of conductive materials in any number of configurations, such as for instance coils of wire, a mesh or grid of wires, conductive strips, conductive plates, electrical circuits, microprocessors, or any combination thereof. The layer containing conductive elements 5512 can be located on the anvil-facing surface 5514 of the tissue compensator 5510. Alternatively or additionally, the layer of conductive elements 5512 can be located on the staple cartridge-facing surface 5516 of the tissue compensator 5510. Alternatively or additionally, the layer of conductive elements 5512 can be embedded within the tissue compensator 5510. Alternatively, the layer of conductive elements 5512 can comprise all of the tissue compensator 5510, such as when a conductive material is uniformly or non-uniformly distributed in the material comprising the tissue compensator 5510.

FIG. 98A illustrates an embodiment wherein the tissue compensator 5510 is removably attached to the anvil 5502 portion of the end effector 5500. The tissue compensator 5510 would be so attached before the end effector 5500 would be inserted into a patient's body. Additionally or alternatively, a tissue compensator 5610 can be attached to a staple cartridge 5506 (not illustrated) after or before the staple cartridge 5506 is applied to the end effector 6600 and before the device is inserted into a patient's body.

FIG. 99 illustrates various example embodiments that use the layer of conductive elements 5512 and conductive elements 5524, 5526, and 5528 in the staple cartridge 5506 to detect the distance between the anvil 5502 and the upper surface of the staple cartridge 5506. The distance between the anvil 5502 and the staple cartridge 5506 indicates the amount and/or density of tissue 5518 compressed therebetween. This distance can additionally or alternatively indicate which areas of the end effector 5500 contain tissue. The tissue 5518 thickness, density, and/or location can be communicated to the operator of the surgical instrument 10.

In the illustrated example embodiments, the layer of conductive elements 5512 is located on the anvil-facing surface 5514 of the tissue compensator 5510, and comprises one or more coils of wire 5522 in communication with a

microprocessor 5520. The microprocessor 5500 can be located in the end effector 5500 or any component thereof, or can be located in the housing 12 of the instrument, or can comprise any microprocessor or microcontroller previously described. In the illustrated example embodiments, the staple cartridge 5506 also includes conductive elements, which can be any one of: one or more coils of wire 5524, one or more conductive plates 5526, a mesh of wires 5528, or any other convenient configuration, or any combination thereof. The staple cartridge's 5506 conductive elements can be in communication with the same microprocessor 5520 or some other microprocessor in the instrument.

When the anvil 5502 is in a closed position and thus is compressing tissue 5518 against staple cartridge 5506, the layer of conductive elements 5512 of the tissue compensator 5510 can capacitively couple with the conductors in staple cartridge 5506. The strength of the capacitive field between the layer of conductive elements 5512 and the conductive elements of the staple cartridge 5506 can be used to determine the amount of tissue 5518 being compressed. Alternatively, the staple cartridge 5506 can comprise eddy current sensors in communication with a microprocessor 5520, wherein the eddy current sensors are operable to sense the distance between the anvil 5502 and the upper surface of the staple cartridge 5506 using eddy currents.

It is understood that other configurations of conductive elements are possible, and that the embodiments of FIG. 99 are by way of example only, and not limitation. For example, in some embodiments the layer of conductive elements 5512 can be located on the staple cartridge-facing surface 5516 of the tissue compensator 5510. Also, in some embodiments the conductive elements 5524, 5526, and/or 5528 can be located on or within the anvil 5502. Thus in some embodiments, the layer of conductive elements 5512 can capacitively couple with conductive elements in the anvil 5502 and thereby sense properties of tissue 5518 enclosed within the end effector.

It can also be recognized that tissue compensator 5512 can comprise a layer of conductive elements 5512 on both the anvil-facing surface 5514 and the cartridge-facing surface 5516. A system to detect the amount, density, and/or location of tissue 5518 compressed by the anvil 5502 against the staple cartridge 5506 can comprise conductors or sensors either in the anvil 5502, the staple cartridge 5506, or both. Embodiments that include conductors or sensors in both the anvil 5502 and the staple cartridge 5506 can optionally achieve enhanced results by allowing differential analysis of the signals that can be achieved by this configuration.

FIGS. 100A and 100B illustrate an embodiment of the tissue compensator 5510 comprising a layer of conductive elements 5512 in operation. FIG. 100A illustrates one of the plurality of staples 191 after it has been deployed. As illustrated, the staple 191 has penetrated both the tissue 5518 and the tissue compensator 5510. The layer of conductive elements 5512 may comprise, for example, mesh wires. Upon penetrating the layer of conductive elements 5512, the staple legs 5530 may puncture the mesh of wires, thus altering the conductivity of the layer of conductive elements 5512. This change in the conductivity can be used to indicate the locations of each of the plurality of staples 191. The location of the staples 191 can be compared against the expected location of the staples, and this comparison can be used to determine if any staples did not fire or if any staples are not where they are expected to be.

FIG. 100A also illustrates staple legs 5530 that failed to completely deform. FIG. 100B illustrates staple legs 5530 that have properly and completely deformed. As illustrated

in FIG. 100B, the layer of conductive elements 5512 can be punctured by the staple legs 5530 a second time, such as when the staple legs 5530 deform against the staple-forming surface of the anvil 5502 and turn back towards the tissue 5518. The secondary breaks in the layer of conductive elements 5512 can be used to indicate complete staple 191 formation, as illustrated in FIG. 100B, or incomplete staple 191 formation, as in FIG. 100A.

FIGS. 101A and 101B illustrate an embodiment of an end effector 5600 comprising a tissue compensator 5610 further comprising conductors 5620 embedded within. The end effector 5600 comprises a first jaw member, or anvil, 5602 pivotally coupled to a second jaw member 5604. The second jaw member 5604 is configured to receive a staple cartridge 5606 therein. In some embodiments, the end effector 5600 further comprises a tissue compensator 5610 removably positioned on the anvil 5602 or the staple cartridge 5606.

Turning first to FIG. 4B, FIG. 4B illustrates a cutaway view of the tissue compensator 5610 removably positioned on the staple cartridge 5606. The cutaway view illustrates an array of conductors 5620 embedded within the material that comprises the tissue compensator 5610. The array of conductors 5620 can be arranged in an opposing configuration, and the opposing elements can be separated by insulating material. The array of conductors 5620 are each coupled to one or more conductive wires 5622. The conductive wires 5622 allow the array of conductors 5620 to communicate with a microprocessor, such as for instance microprocessor 1500. The array of conductors 5620 may span the width of the tissue compensator 5610 such that they will be in the path of a cutting member or knife bar 280. As the knife bar 280 advances, it will sever, destroy, or otherwise disable the conductors 5620, and thereby indicate its position within the end effector 5600. The array of conductors 5610 can comprise conductive elements, electric circuits, microprocessors, or any combination thereof.

Turning now to FIG. 101A, FIG. 101A illustrates a close-up cutaway view of the end effector 5600 with the anvil 5602 in a closed position. In a closed position, the anvil 5602 can compress tissue 5618 and the tissue compensator 5610 against the staple cartridge 5606. In some cases, only a part of the end effector 5600 may be enclosing the tissue 5618. In areas of the end effector 5600 that are enclosing tissue 5618, the tissue compensator 5610 may be compressed 5624 a greater amount than areas that do not enclose tissue 5618, where the tissue compensator 5618 may remain uncompressed 5626 or be less compressed. In areas of greater compression 5624, the array of conductors 5620 will also be compressed, while in uncompressed 5626 areas, the array of conductors 5620 will be further apart. Hence, the conductivity, resistance, capacitance, and/or some other electrical property between the array of conductors 5620 can indicate which areas of the end effector 5600 contain tissue.

FIGS. 102A and 102B illustrate an embodiment of an end effector 5650 comprising a tissue compensator 5660 further comprising conductors 5662 embedded therein. The end effector 5650 comprises a first jaw member, or anvil, 5652 pivotally coupled to a second jaw member 5654. The second jaw member 5654 is configured to receive a staple cartridge 5656 therein. In some embodiments, the end effector 5650 further comprises a tissue compensator 5660 removably positioned on the anvil 5652 or the staple cartridge 5656.

FIG. 102A illustrates a cutaway view of the tissue compensator 5660 removably positioned on the staple cartridge 5656. The cutaway view illustrates conductors 5670 embedded within the material that comprises the tissue compensator 5660. Each of the conductors 5672 is coupled to a

conductive wire **5672**. The conductive wires **5672** allow the array of conductors **5672** to communicate with a microprocessor, such as for instance microprocessor **1500**. The conductors **5672** may comprise conductive elements, electric circuits, microprocessors, or any combination thereof.

FIG. **102A** illustrates a close-up side view of the end effector **5650** with the anvil **5652** in a closed position. In a closed position, the anvil **5652** can compress tissue **5658** and the tissue compensator **5660** against the staple cartridge **5656**. The conductors **5672** embedded within the tissue compensator **5660** can be operable to apply pulses of electrical current **5674**, at predetermined frequencies, to the tissue **5658**. The same or additional conductors **5672** can detect the response of the tissue **5658** and transmit this response to a microprocessor or microcontroller located in the instrument. The response of the tissue **5658** to the electrical pulses **5674** can be used to determine a property of the tissue **5658**. For example, the galvanic response of the tissue **5658** indicates the tissue's **5658** moisture content. As another example, measurement of the electrical impedance through the tissue **5658** could be used to determine the conductivity of the tissue **5648**, which is an indicator of the tissue type. Other properties that can be determined include by way of example and not limitation: oxygen content, salinity, density, and/or the presence of certain chemicals. By combining data from several sensors, other properties could be determined, such as blood flow, blood type, the presence of antibodies, etc.

FIG. **103** illustrates an embodiment of a staple cartridge **5706** and a tissue compensator **5710** wherein the staple cartridge **5706** provides power to the conductive elements **5720** that comprise the tissue compensator **5710**. As illustrated, the staple cartridge **5706** comprises electrical contacts **5724** in the form of patches, spokes, bumps, or some other raised configuration. The tissue compensator **5710** comprises mesh or solid contact points **5722** that can electrically couple to the contacts **5724** on the staple cartridge **5706**.

FIGS. **104A** and **104B** illustrate an embodiment of a staple cartridge **5756** and a tissue compensator **5760** wherein the staple cartridge provides power to the conductive elements **5770** that comprise the tissue compensator **5710**. As illustrated in FIG. **104A**, the tissue compensator **5760** comprises an extension or tab **5772** configured to come into contact with the staple cartridge **5756**. The tab **5772** may contact and adhere to an electrical contact (not shown) on the staple cartridge **5756**. The tab **5772** further comprises a break point **5774** located in a wire comprising the conductive elements **5770** of the tissue compensator **5760**. When the tissue compensator **5760** is compressed, such as when an anvil is in a closed position towards the staple cartridge **5756**, the break point **5774** will break, thus allowing the tissue compensator **5756** to become free from the staple cartridge **5756**. FIG. **104B** illustrates another embodiment employing a break point **5774** positioned in the tab **5772**.

FIGS. **105A** and **F8B** illustrate an embodiment of an end effector **5800** comprising position sensing elements **5824** and a tissue compensator **5810**. The end effector **5800** comprises a first jaw member, or anvil, **5802** pivotally coupled to a second jaw member **5804** (not shown). The second jaw member **5804** is configured to receive a staple cartridge **5806** (not shown) therein. In some embodiments, the end effector **5800** further comprises a tissue compensator **5810** removably positioned on the anvil **5802** or the staple cartridge **5806**.

FIG. **105A** illustrates the anvil **5804** portion of the end effector **5800**. In some embodiments the anvil **5804** com-

prises position sensing elements **5824**. The position sensing elements **5824** can comprise, for example, electrical contacts, magnets, RF sensors, etc. The position sensing elements **5824** can be located in key locations, such as for instance the corner points where the tissue compensator **5810** will be attached, or along the exterior edges of the anvil's **5802** tissue-facing surface. In some embodiments, the tissue compensator **5810** can comprise position indicating elements **5820**. The position indicating elements **5820** can be located in corresponding locations to the position sensing elements **5824** on the anvil **5802**, or in proximal locations, or in overlapping locations. The tissue compensator **5810** optionally further comprises a layer of conductive elements **5812**. The layer of conductive elements **5812** and/or the position indicating elements **5820** can be electrically coupled to conductive wires **5822**. The conductive wires **5822** can provide communication with a microprocessor, such as for instance microprocessor **1500**.

FIG. **F8B** illustrates an embodiment the position sensing elements **5824** and position indicating elements **5820** in operation. When the tissue compensator **5810** is positioned, the anvil **5802** can sense **5826** that the tissue compensator **5810** is properly position. When the tissue compensator **5810** is misaligned or missing entirely, the anvil **5802** (or some other component) can sense **5826** that the tissue compensator **5810** is misaligned. If the misalignment is above a threshold magnitude, a warning can be signaled to the operator of the instrument, and/or a function of the instrument can be disabled to prevent the staples from being fired.

In FIGS. **105A** and **105B** the position sensing elements **5824** are illustrated as a part of the anvil **5804** by way of example only. It is understood that the position sensing elements **5824** can be located instead or additionally on the staple cartridge **5806**. It is also understood that the location of the position sensing elements **5824** and the position indicating elements **5820** can be reversed, such that the tissue compensator **5810** is operable to indicate whether it is properly aligned.

FIGS. **106A** and **F9B** illustrate an embodiment of an end effector **5850** comprising position sensing elements **5874** and a tissue compensator **5860**. The end effector **5850** comprises a first jaw member, or anvil, **5852** pivotally coupled to a second jaw member **5854** (not shown). The second jaw member **5854** is configured to receive a staple cartridge **5856** (not show) therein. In some embodiments, the end effector **5850** further comprises a tissue compensator **5860** removably positioned on the anvil **5852** or the staple cartridge **5856**.

FIG. **106A** illustrates the anvil **5852** portion of the end effector **5850**. In some embodiments, the anvil **5854** comprises an array of conductive elements **5474**. The array of conductive elements **5474** can comprise, for example, electrical contacts, magnets, RF sensors, etc. The array of conductive elements **5474** are arrayed along the length of the tissue-facing surface of the anvil **5852**. In some embodiments, the tissue compensator **5860** can comprise a layer of conductive elements **5862**, wherein the conductive elements comprise a grid or mesh of wires. The layer of conductive elements **5862** may be coupled to conductive wires **5876**. The conductive wires **5862** can provide communication with a microprocessor, such as for instance microprocessor **1500**.

FIG. **106A** illustrates an embodiment wherein of the conductive elements **5474** of the anvil **5852** and the layer of conductive elements **5862** are operable to indicate whether the tissue compensator **5860** is misaligned or missing. As illustrated, the array of conductive elements **5874** is oper-

able to electrically couple with the layer of conductive elements **5862**. When the tissue compensator **5860** is misaligned or missing, the electrical coupling will be incomplete. If the misalignment is above a threshold magnitude, a warning can be signaled to the operator of the instrument, and/or a function of the instrument can be disabled to prevent the staples from being fired.

It is understood that the array of conductive elements **5874** may additionally or alternatively be located on the staple cartridge **5856**. It is also understood that the any of the anvil **5852**, staple cartridge **5856**, and/or tissue compensator **5860** may be operable to indicate misalignment of the tissue compensator **5860**.

FIGS. **107A** and **107B** illustrate an embodiment of a staple cartridge **5906** and a tissue compensator **5910** that is operable to indicate the position of a cutting member or knife bar **280**. FIG. **107A** is a top-down view of the staple cartridge **5906** that has a tissue compensator **5920** placed on its upper surface **5916**. The staple cartridge **5906** further comprises a cartridge channel **5918** operable to accept a cutting member or knife bar **280**. FIG. **107A** illustrates only the layer of conductive elements **5922** of the tissue compensator **5910**, for clarity. As illustrated, the layer of conductive elements **5922** comprises a lengthwise segment **5930** that is located off-center. The lengthwise segment **5930** is coupled to conductive wires **5926**. The conductive wires **5926** allow the layer of conductive elements **5922** to communicate with a microprocessor, such as for instance microprocessor **1500**. The layer of conductive elements **5922** further comprises horizontal elements **5932** coupled to the lengthwise segment **5930** and spanning the width of the tissue compensator **5910**, and thus crossing the path of the knife bar **280**. As the knife bar **280** advances, it will sever the horizontal elements **5932** and thereby alter an electrical property of the layer of conductive elements **5922**. For example, the advancing of the knife bar **280** may alter the resistance, capacitance, conductivity, or some other electrical property of the layer of conductive elements **5922**. As each horizontal element **5932** is severed by the knife bar **280**, the change in the electrical properties of the layer of conductive elements **5922** will indicate the position of the knife bar **280**.

FIG. **107B** illustrates an alternate configuration for the layer of conductive elements **5922**. As illustrated, the layer of conductive elements **5922** comprises a lengthwise segment **5934** on either side of the cartridge channel **5918**. The layer of conductive elements **5922** further comprises horizontal elements **5936** coupled to both of the lengthwise segments **5934**, thus spanning the path of the knife bar **280**. As the knife bar **280**, the resistance, for example between the knife bar and the horizontal elements **5936** can be measured and used to determine the location of the knife bar **280**. Other configurations of the layer of conductive elements **5922** can be used to accomplish the same result, such as for instance any of the arrangements illustrated in FIGS. **98A** through **102B**. For example, the layer of conductive elements **5922** can comprise a wire mesh or grid, such that as the knife bar **280** advances it can sever the wire mesh and thereby change the conductivity in the wire mesh. This change in conductivity can be used to indicate the position of the knife bar **280**.

Other uses for the layer of conductive elements **5922** can be imagined. For example, a specific resistance can be created in the layer of conductive elements **592**, or a binary ladder of resistors or conductors can be implemented, such that simple data can be stored in the tissue compensator **5910**. This data can be extracted from the tissue compen-

sator **5910** by conductive elements in the anvil and/or staple cartridge when either electrically couple with the layer of conductive elements **5922**. The data can represent, for example, a serial number, a "use by" date, etc.

5 Polarity of Hall Magnet to Detect Misloaded Cartridge

FIG. **108** illustrates one embodiment of an end effector **6000** comprising a magnet **6008** and a Hall effect sensor **6010** wherein the detected magnetic field **6016** can be used to identify a staple cartridge **6006**. The end effector **6000** is similar to the end effector **300** described above. The end effector **6000** comprises a first jaw member or anvil **6002**, pivotally coupled to a second jaw member or elongated channel **6004**. The elongated channel **6004** is configured to operably support a staple cartridge **6006** therein. The staple cartridge **6006** is similar to the staple cartridge **304** described above. The anvil **6002** further comprises a magnet **6008**. The staple cartridge **6006** further comprises a Hall effect sensor **6010** and a processor **6012**. The Hall effect sensor **6010** is operable to communicate with the processor **6012** through a conductive coupling **6014**. The Hall effect sensor **6010** is positioned within the staple cartridge **6006** to operatively couple with the magnet **6008** when the anvil **6002** is in a closed position. The Hall effect sensor **6010** can be operable to detect the magnetic field **6016** produced by the magnet **6008**. The polarity of the magnetic field **6016** can be one of north or south depending on the orientation of the magnet **6008** within the anvil **6002**. In the illustrated embodiment of FIG. **108**, the magnet **6008** is oriented such that its south pole is directed towards the staple cartridge **6006**. The Hall effect sensor **6010** can be operable to detect the magnetic field **6016** produced by a south pole. If the Hall effect sensor **6010** detects a magnetic south pole, then the staple cartridge **6006** can be identified as of a first type.

FIG. **109** illustrates an embodiment of an end effector **6050** comprising a magnet **6058** and a Hall effect sensor **6060** wherein the detected magnetic field **6066** can be used to identify a staple cartridge **6056**. The end effector **6050** comprises a first jaw member or anvil **6052**, pivotally coupled to a second jaw member or elongated channel **6054**. The elongated channel **6054** is configured to operably support a staple cartridge **6056** therein. The anvil **6052** further comprises a magnet **6058**. The staple cartridge **6056** further comprises a Hall effect sensor **6060** in communication with a processor **6062** over a conductive coupling **6064**. The Hall effect sensor **6060** is positioned such that it will operatively couple with the magnet **6058** when the anvil **6052** is in a closed position. The Hall effect sensor **6060** can be operable to detect the magnetic field **6066** produced by the magnet **6058**. In the illustrated embodiment, the magnet **6058** is oriented such that its north magnetic pole is directed towards the staple cartridge **6056**. The Hall effect sensor **6060** can be operable to detect the magnetic field **6066** produced by a north pole. If the Hall effect sensor **6060** detects a north magnetic pole, then the staple cartridge **6056** can be identified as a second type.

It can be recognized that the second type staple cartridge **6056** of FIG. **109** can be substituted for the first type staple cartridge **6006** of FIG. **108**, and vice versa. In FIG. **108**, the second type staple cartridge **6056** would be operable to detect a magnetic north pole, but will detect a magnetic south pole instead. In this case, end effector **6000** will identify the staple cartridge **6056** as being of the second type. If the end effector **6000** did not expect a staple cartridge **6056** of the second type, the operator of the instrument can be alerted, and/or a function of the instrument can be disabled. The type of the staple cartridge **6056** can additionally or alternatively be used to identify some

parameter of the staple cartridge **6056**, such as for instance the length of the cartridge and/or the height and length of the staples.

Similarly, as shown in FIG. **109**, the first type staple cartridge **6006** can be substituted for the second staple cartridge **6056**. The first type staple cartridge **6006** would be operable to detect a south magnetic pole, but will instead detect a north magnetic pole. In this case, the end effector **6050** will identify the staple cartridge **6006** as being of the first type.

FIG. **110** illustrates a graph **6020** of the voltage **6022** detected by a Hall effect sensor located in the distal tip of a staple cartridge, such as is illustrated in FIGS. **108** and **109**, in response to the distance or gap **6024** between a magnet located in the anvil and the Hall effect sensor in the staple cartridge, such as illustrated in FIGS. **108** and **109**. As illustrated FIG. **110**, when the magnet in the anvil is oriented such that its north pole is towards the staple cartridge, the voltage will tend towards a first value as the magnet comes in proximity to the Hall effect sensor; when the magnet is oriented with its south pole towards the staple cartridge, the voltage will tend towards a second, different value. The measured voltage can be used by the instrument to identify the staple cartridge.

FIG. **111** illustrates one embodiment of the housing **6100** of the surgical instrument, comprising a display **6102**. The housing **6100** is similar to the housing **12** described above. The display **6102** can be operable to convey information to the operator of the instrument, such as for instance, that the staple cartridge coupled to the end effector is inappropriate for the present application. Additionally or alternatively, the display **6102** can display the parameters of the staple cartridge, such as the length of the cartridge and/or the height and length of the staples.

FIG. **112** illustrates one embodiment of a staple retainer **6160** comprising a magnet **6162**. The staple retainer **6160** can be operably coupled to a staple cartridge **6156** and functions to prevent staples from exiting of the staple cartridge **6156**. The staple retainer **6160** can be left in place when the staple cartridge **6156** is applied to an end effector. In some embodiments, the staple retainer **6160** comprises a magnet **6162** located in the distal area of the staple retainer **6160**. The anvil of the end effector can comprise a Hall effect sensor operable to couple with the magnet **6162** in the staple retainer **6160**. The Hall effect sensor can be operable to detect the properties of the magnet **6162**, such as for instance the magnetic field strength and magnetic polarity. The magnetic field strength can be varied by, for example, placing the magnet **6162** in different locations and/or depths on or in the staple retainer **6160**, or by selecting magnets **6162** of different compositions. The different properties of the magnet **6162** can be used to identify staple cartridges of different types.

FIGS. **113A** and **113B** illustrate one embodiment of an end effector **6200** comprising a sensor **6208** for identifying staple cartridges **6206** of different types. The end effector **6200** comprises a first jaw member or anvil **6202**, pivotally coupled to a second jaw member or elongated channel **6204**. The elongated channel **6204** is configured to operably support a staple cartridge **6206** therein. The end effector **6200** further comprises a sensor **6208** located in the proximal area. The sensor **6208** can be any of an optical sensor, a magnetic sensor, an electrical sensor, or any other suitable sensor.

The sensor **6208** can be operable to detect a property of the staple cartridge **6206** and thereby identify the staple cartridge **6206** type. FIG. **113B** illustrates an example where the sensor **6208** is an optical emitter and detector **6210**. The

body of the staple cartridge **6206** can be different colors, such that the color identifies the staple cartridge **6206** type. An optical emitter and detector **6210** can be operable to interrogate the color of the staple cartridge **6206** body. In the illustrated example, the optical emitter and detector **6210** can detect white **6212** by receiving reflected light in the red, green, and blue spectrums in equal intensity. The optical emitter and detector **6210** can detect red **6214** by receiving very little reflected light in the green and blue spectrums while receiving light in the red spectrum in greater intensity.

Alternately or additionally, the optical emitter and detector **6210**, or another suitable sensor **6208**, can interrogate and identify some other symbol or marking on the staple cartridge **6206**. The symbol or marking can be any one of a barcode, a shape or character, a color-coded emblem, or any other suitable marking. The information read by the sensor **6208** can be communicated to a microcontroller in the surgical device **10**, such as for instance microcontroller **1500**. The microcontroller **1500** can be configured to communicate information about the staple cartridge **6206** to the operator of the instrument. For instance, the identified staple cartridge **6206** may not be appropriate for a given application; in such case, the operator of the instrument can be informed, and/or a function of the instrument is inappropriate. In such instance, microcontroller **1500** can optionally be configured to disable a function of surgical instrument can be disabled. Alternatively or additionally, microcontroller **1500** can be configured to inform the operator of the surgical instrument **10** of the parameters of the identified staple cartridge **6206** type, such as for instance the length of the staple cartridge **6206**, or information about the staples, such as the height and length.

Smart Cartridge Wake Up Operation and Data Retention

In one embodiment the surgical instrument described herein comprises short circuit protection techniques for sensors and/or electronic components. To enable such sensors and other electronic technology both power and data signals are transferred between modular components of the surgical instrument. During assembly of modular sensor components electrical conductors that when connected are used to transfer power and data signals between the connected components are typically exposed.

FIG. **114** is a partial view of an end effector **7000** with electrical conductors **7002**, **7004** for transferring power and data signals between the connected components of the surgical instrument according to one embodiment. There is potential for these electrical conductors **7002**, **7004** to become shorted and thus damage critical system electronic components. FIG. **115** is a partial view of the end effector **7000** shown in FIG. **114** showing sensors and/or electronic components **7005** located in the end effector. With reference now to both FIGS. **114** and **115**, in various embodiments the surgical instruments disclosed throughout the present disclosure provide real time feedback about the compressibility and thickness of tissue using electronic sensors. Modular architectures will enable the configuration of custom modular shafts to employ job specific technologies. To enable sensors and other electronic circuit components in surgical instruments it is necessary to transfer both power and data signals between a secondary circuit comprising the modular sensor and/or electronic circuit components **7005**. During the assembly of the modular sensors and/or electronic components **7005** the electrical conductors **7002**, **7004** are exposed such that when connected, they are used to transfer power and data signals between the connected sensors and/or electronic components **7005**. Because there is a potential for these electrical conductors **7002**, **7004** to

become short circuited during the assembly process and thus damage other system electronic circuits, various embodiments of the surgical instruments described herein comprise short circuit protection techniques for the sensors and/or electronic components **7005**

In one embodiment, the present disclosure provides a short circuit protection circuit **7012** for the sensors and/or electronic components **7005** of the secondary circuits of the surgical instrument. FIG. **116** is a block diagram of a surgical instrument electronic subsystem **7006** comprising a short circuit protection circuit **7012** for the sensors and/or electronic components **7005** according to one embodiment. A main power supply circuit **7010** is connected to a primary circuit comprising a microprocessor and other electronic components **7008** (processor **7008** hereinafter) through main power supply terminals **7018**, **7020**. The main power supply circuit **7010** also is connected to a short circuit protection circuit **7012**. The short circuit protection circuit **7012** is coupled to a supplementary power supply circuit **7014**, which supplies power to the sensors and/or electronic components **7005** via the electrical conductors **7002**, **7004**.

To reduce damage to the processor **7008** connected to the main power supply terminals **7018**, **7020**, during a short circuit between the electrical conductors **7002**, **7004** of the power supply terminals feeding the sensors and/or electronic components **7005**, a self isolating/restoring short circuit protection circuit **7012** is provided. In one embodiment, the short circuit protection circuit **7012** may be implemented by coupling a supplementary power supply circuit **7014** to the main power supply circuit **7010**. In circumstances when the supplementary power supply circuit **7014** power conductors **7002**, **7004** are shorted, the supplementary power supply circuit **7014** isolates itself from the main power supply circuit **7010** to prevent damage to the processor **7008** of the surgical instrument. Thus, there is virtually no effect to the processor **7008** and other electronic circuit components coupled to the main power supply terminals **7018**, **7020** when a short circuit occurs in the electrical conductors **7002**, **7004** of the supplementary power supply circuit **7014**. Accordingly, in the event that a short circuit occurs between the electrical conductors **7002**, **7004** of the supplementary power supply circuit **7014**, the main power supply circuit **7010** is unaffected and remains active to supply power to the protected processor **7008** such that the processor **7008** can monitor the short circuit condition. When the short circuit between the electrical conductors **7002**, **7004** of the supplementary power supply circuit **7014** is remedied, the supplementary power supply circuit **7014** rejoins the main power supply circuit **7010** and is available once again to supply power to the sensor components **7005**. The short circuit protection circuit **7012** also may be monitored to indicate one or more short circuit conditions to the end user of the surgical instrument. The short circuit protection circuit **7012** also may be monitored to lockout the firing of the surgical instrument when a short circuit event is indicated. Many supplementary protection circuits may be networked together to isolate, detect, or protect other circuit functions.

Accordingly, in one aspect, the present disclosure provides a short circuit protection circuit **7012** for electrical conductors **7002**, **7004** in the end effector **7000** (FIGS. **114** and **115**) or other elements of the surgical instrument. In one embodiment, the short circuit protection circuit **7012** employs a supplementary self-isolating/restoring power supply circuit **7014** coupled to the main power supply circuit **7010**. The short circuit protection circuit **7012** may be monitored to indicate one or more short circuit conditions to the end user of the surgical instrument. In the event of a short

circuit, the short circuit protection circuit **7012** may be employed to lock-out the surgical instrument from being fired or other device operations. Many other supplementary protection circuits may be networked together to isolate, detect, or protect other circuit functions.

FIG. **117** is a short circuit protection circuit **7012** comprising a supplementary power supply circuit **7014** coupled to a main power supply circuit **7010**, according to one embodiment. The main power supply circuit **7010** comprises a transformer **7023** (X1) coupled to a full wave rectifier **7025** implemented with diodes **91-94**. The full wave rectifier **7025** is coupled to the voltage regulator **7027**. The output (OUT) of the voltage regulator **7027** is coupled to both the output terminals **7018**, **7020** of the main power supply circuit **7010** (OP1) and the supplementary power supply circuit **7014**. An input capacitor C1 filters the input voltage in the voltage regulator **7027** and one or more capacitors C2 filter the output of the voltage regulator **7027**.

In the embodiment illustrated in FIG. **117**, the supplementary power supply circuit **7014** comprises a pair of transistors T1, T2 configured to control the power supply output OP2 between the electrical conductors **7002**, **7004**. During normal operation when the electrical conductors **7002**, **7004** are not shorted, the output OP2 supplies power to the sensor components **7005**. Once the transistors T1 and T2 are turned ON (activated) and begin conducting current, the current from the output of the voltage regulator **7027** is shunted by the first transistor T1 such that no current flows through R1 and $i_{R1}=0$. The output voltage of the regulator +V is applied at the node such the $V_n \sim +V$, which is then the output voltage OP2 of the supplementary power supply circuit **7014** and the first transistor T1 drives the current to the sensor components **7005** through the output terminal **7002**, where output terminal **7004** is the current return path. A portion of the output current i_{RS} is diverted through R5 to drive the output indicator LED2. The current through the LED2 is i_{RS} . As long as the node voltage V_n is above the threshold necessary to turn ON (activate) the second transistor T2, the supplementary power supply circuit **7014** operates as a power supply circuit to feed the sensors and/or electronic components **7005**.

When the electrical conductors **7002**, **7004** of the secondary circuit are shorted, the node voltage V_n drops to ground or zero and the second transistor T2 turns OFF and stops conducting, which turns OFF the first transistor T1. When the first transistor T1 is cut-OFF, the output voltage +V of the voltage regulator **7027** causes current i_{R1} to flow through the short circuit indicator LED1 and through to ground via the short circuit between the electrical conductors **7002**, **7004**. Thus, no current flows through R5 and $i_{RS}=0A$ and $+V_{OP2}=0V$. The supplementary power supply circuit **7014** isolates itself from the main power supply circuit **7010** until the short circuit is removed. During the short circuit only the short circuit indicator LED1 is energized while the output indicator LED2 is not. When the short circuit between the electrical conductors **7002**, **7004** is removed, the node voltage V_n rises until T2 turns ON and subsequently turning T1 ON. When T1 and T2 are turned ON (are biased in a conducting state such as saturation), until the node voltage V_n reaches $+V_{OP2}$ and the supplementary power supply circuit **7014** resumes its power supply function for the sensor components **7005**. Once the supplementary power supply circuit **7014** restores its power supply function, the short circuit indicator LED1 turns OFF and the output indicator LED2 turns ON. The cycle is repeated in the event of another short circuit between the supplementary power supply circuit **7014** electrical conductors **7002**, **7004**.

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In one embodiment, a sample rate monitor is provided to enable power reduction by limiting sample rates and/or duty cycle of the sensor components when the surgical instrument is in a non-sensing state. FIG. 118 is a block diagram of a surgical instrument electronic subsystem 7022 comprising a sample rate monitor 7024 to provide power reduction by limiting sample rates and/or duty cycle of the sensors and/or electronic components 7005 of the secondary circuit when the surgical instrument is in a non-sensing state, according to one embodiment. As shown in FIG. 118, the surgical instrument electronic subsystem 7022 comprises a processor 7008 coupled to a main power supply circuit 7010. The main power supply circuit 7010 is coupled to a sample rate monitor circuit 7024. A supplementary power supply circuit 7014 is coupled to the sample rate 7024 as powers the sensors and/or electronic components 7005 via the electrical conductors 7002, 7004. The primary circuit comprising the processor 7008 is coupled to a device state monitor 7026. In various embodiments, the surgical instrument electronic subsystem 7022 provides real time feedback about the compressibility and thickness of tissue using the sensors and/or electronic components 7005 as previously described herein. The modular architecture of the surgical instrument enables the configuration of custom modular shafts to employ function job specific technologies. To enable such additional functionality, electronic connection points and components are employed to transfer both power and signal between modular components of the surgical instrument. An increase in the number of sensors and/or electronic components 7005 increases the power consumption of the surgical instrument system 7022 and creates the need for various techniques for reducing power consumption of the surgical instrument system 7022.

In one embodiment, to reduce power consumption, a surgical instrument configured with sensors and/or electronic components 7005 (secondary circuit) comprises a sample rate monitor 7024, which can be implemented as a hardware circuit or software technique to reduce the sample rate and/or duty cycle for the sensors and/or electronic components 7005. The sample rate monitor 7024 operates in conjunction with the device state monitor 7026. The device state monitor 7026 senses the state of various electrical/mechanical subsystems of the surgical instrument. In the embodiment illustrated in FIG. 118, the device state monitor 7026 whether the state of the end effector is in an unclamped (State 1), a clamping (State 2), or a clamped (State 3) state of operation.

The sample rate monitor 7024 sets the sample rate and/or duty cycle for the sensor components 7005 based on the state of the end effector determined by the device state monitor 7026. In one aspect, the sample rate monitor 7024 may set the duty cycle to about 10% when the end effector is in State 1, to about 50% when the end effector is in State 2, or about 20% when the end effector is in State 3. In various other embodiments, the duty cycle and/or sample rate set by the sample rate monitor 7024 may take on ranges of values. For example, in another aspect, the sample rate monitor 7024 may set the duty cycle to a value between about 5% to about 15% when the end effector is in State 1, to a value of about 45% to about 55% when the end effector is in State 2, or to a value of about 15% to about 25% when the end effector is in State 3. In various other embodiments, the duty cycle and/or sample rate set by the sample rate monitor 7024 may take on additional ranges of values. For example, in another aspect, the sample rate monitor 7024 may set the duty cycle to a value between about 1% to about 20% when the end effector is in State 1, to a value of about 20% to about 80%

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when the end effector is in State 2, or to a value of about 1% to about 50% when the end effector is in State 3. In various other embodiments, the duty cycle and/or sample rate set by the sample rate monitor 7024 may take on additional ranges of values.

In one aspect, the sample rate monitor 7024 may be implemented by creating a supplementary circuit/software coupled to a main circuit/software. When the supplementary circuit/software determines that the surgical instrument system 7022 is in a non-sensing condition, the sample rate monitor 7024 enters the sensors and/or electronic components 7005 into a reduced sampling or duty cycle mode reducing the power load on the main circuit. The main power supply circuit 7010 will still be active to supply power, so that the protected processor 7008 of the primary circuit can monitor the condition. When the surgical instrument system 7022 enters a condition requiring more rigorous sensing activity the sample rate monitor 7024 increases the supplementary circuit sample rate or duty cycle. The circuit could utilize a mixture of integrated circuits, solid state components, microprocessors, and firmware. The reduced sample rate or duty cycle mode circuit also may be monitored to indicate the condition to the end user of the surgical instrument system 7022. The circuit/software might also be monitored to lockout the firing or function of the device in the event the device is in the power saving mode.

In one embodiment, the sample rate monitor 7024 hardware circuit or software technique reduce the sample rate and/or duty cycle for the sensors and/or electronic components 7005 to reduce power consumption of the surgical instrument. The reduced sample rate and/or duty cycle may be monitored to indicate one or more conditions to the end user of the surgical instrument. In the event of a reduced sample rate and/or duty cycle condition in the surgical instrument the protection circuit/software may be configured to lock-out the surgical instrument from being fired or otherwise operated.

In one embodiment, the present disclosure provides an over current and/or a voltage protection circuit for sensors and/or electronic components of a surgical instrument. FIG. 119 is a block diagram of a surgical instrument electronic subsystem 7028 comprising an over current and/or over voltage protection circuit 7030 for sensors and/or electronic components 7005 of the secondary circuit of a surgical instrument, according to one embodiment. In various embodiments, the surgical instrument electronic subsystem 7028 provides real time feedback about the compressibility and thickness of tissue using the sensors and/or electronic components 7005 of the secondary circuit as previously described herein. The modular architecture of the surgical instrument enables the configuration of custom modular shafts to employ function job specific technologies. To enable the sensors and/or electronic components 7005, additional electronic connection points and components to transfer both power and signal between modular components are added. There is potential for these additional conductors for the sensors and/or electronic components 7005 from the modular pieces to be shorted and or damaged causing large draws of current that could damage fragile processor 7008 circuits or and other electronic components of the primary circuit. In one embodiment, the over current/voltage protection circuit 7030 protects the conductors for the sensors and/or electronic components 7005 on a surgical instrument using a supplementary self-isolating/restoring circuit 7014 coupled to the main power supply circuit 7010. The operation of one embodiment of the supplementary self-isolating/

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restoring circuit 7014 is described in connection with FIG. 117 and will not be repeated here for conciseness and clarity of disclosure.

In one embodiment, to reduce electronic damage during large current draws in a sensing surgical instrument, the electronic subsystem 7028 of the surgical instrument comprises an over current/voltage protection circuit 7030 for the conductors for the sensors and/or electronic components 7005. The over current/voltage protection circuit 7030 may be implemented by creating a supplementary circuit coupled to a main power supply circuit 7010 circuit. In the case that the supplementary circuit electrical conductors 7002, 7004 experience higher levels of current than expected, the over current/voltage protection circuit 7030 isolates the current from the main power supply circuit 7010 circuit to prevent damage. The main power supply circuit 7010 circuit will still be active to supply power, so that the protected main processor 7008 can monitor the condition. When a large current draw in the supplementary power supply circuit 7014 is remedied, the supplementary power supply circuit 7014 rejoins the main power supply circuit 7010 and is available to supply power to the sensors and/or electronic components 7005 (e.g., the secondary circuit). The over current/voltage protection circuit 7030 may utilize a mixture of integrated circuits, solid state components, micro-processors, firmware, circuit breaker, fuses, or PTC (positive temperature coefficient) type technologies.

In various embodiments, the over current/voltage protection circuit 7030 also may be monitored to indicate the over current/voltage condition to the end user of the device. The over current/voltage protection circuit 7030 also may be monitored to lockout the firing of the surgical instrument when the over current/voltage condition event is indicated. The over current/voltage protection circuit 7030 also may be monitored to indicate one or more over current/voltage conditions to the end user of the device. In the event of over current/voltage condition in the device the over current/voltage protection circuit 7030 may lock-out the surgical instrument from being fired or lock-out other operations of the surgical instrument.

FIG. 120 is an over current/voltage protection circuit 7030 for sensors and electronic components 7005 (FIG. 119) of the secondary circuit of a surgical instrument, according to one embodiment. The over current/voltage protection circuit 7030 provides a current path during a hard short circuit (SHORT) at the output of the over current/voltage protection circuit 7030, and also provides a path for follow-through current through a bypass capacitor C_{BYPASS} driven by stray inductance L_{STRAY} .

In one embodiment, the over current/voltage protection circuit 7030 comprises a current limited switch 7032 with autoreset. The current limited switch 7032 comprises a current sense resistor R_{CS} coupled to an amplifier A. When the amplifier A senses a surge current above a predetermined threshold, the amplifier activates a circuit breaker CB to open the current path to interrupt the surge current. In one embodiment, the current limited switch 7032 with autoreset may be implemented with a MAX1558 integrated circuit by Maxim. The current limited switch 7032 with autoreset. Autoreset latches the switch 7032 off if it is shorted for more than 20 ms, saving system power. The shorted output (SHORT) is then tested to determine when the short is removed to automatically restart the channel Low quiescent supply current (45 μ A) and standby current (3 μ A) conserve battery power in the surgical instrument. The current limited switch 7032 with autoreset safety features ensure that the surgical instrument is protected. Built-in thermal-overload

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protection limits power dissipation and junction temperature. Accurate, programmable current-limiting circuits, protects the input supply against both overload and short-circuit conditions. Fault blanking of 20 ms duration enables the circuit to ignore transient faults, such as those caused when hot swapping a capacitive load, preventing false alarms to the host system. In one embodiment, the current limited switch 7032 with autoreset also features a reverse-current protection circuitry to block current flow from the output to the input when the switch 7032 is off.

In one embodiment, the present disclosure provides a reverse polarity protection for sensors and/or electronic components in a surgical instrument. FIG. 121 is a block diagram of a surgical instrument electronic subsystem 7040 with a reverse polarity protection circuit 7042 for sensors and/or electronic components 7005 of the secondary circuit according to one embodiment. Reverse polarity protection is provided for exposed leads (electrical conductors 7002, 7004) of a surgical instrument using a supplementary self-isolating/restoring circuit referred to herein as a supplementary power supply circuit 7014 coupled to the main power supply circuit 7010. The reverse polarity protection circuit 7042 may be monitored to indicate one or more reverse polarity conditions to the end user of the device. In the event of reverse polarity applied to the device the protection circuit 7042 might lock-out the device from being fired or other device critical operations.

In various embodiments, the surgical instruments described herein provide real time feedback about the compressibility and thickness of tissue using sensors and/or electronic components 7005. The modular architecture of the surgical instrument enables the configuration of custom modular shafts to employ job specific technologies. To enable sensors and/or electronic components 7005, both power and data signals are transferred between the modular components. During the assembly of modular components there are typically exposed electrical conductors that when connected are used to transfer power and data signals between the connected components. There is potential for these conductors to become powered with reverse polarity.

Accordingly, in one embodiment, the surgical instrument electronic subsystem 7040 is configured to reduce electronic damage during the application of a reverse polarity connection 7044 in a sensing surgical instrument. The surgical instrument electronic subsystem 7040 employs a polarity protection circuit 7042 inline with the exposed electrical conductors 7002, 7004. In one embodiment, the polarity protection circuit 7042 may be implemented by creating a supplementary power supply circuit 7014 coupled to a main power supply circuit 7010. In the case that the supplementary power supply circuit 7014 electrical conductors 7002, 7004 become powered with reverse polarity it isolates the power from the main power supply circuit 7010 to prevent damage. The main power supply circuit 7010 will still be active to supply power, so that the protected processor 7008 of the main circuit can monitor the condition. When the reverse polarity in the supplementary power supply circuit 7014 is remedied, the supplementary power supply circuit 7014 rejoins the main power supply circuit 7010 and is available to supply power to the secondary circuit. The reverse polarity protection circuit 7042 also may be monitored to indicate that the reverse polarity condition to the end user of the device. The reverse polarity protection circuit 7042 also may be monitored to lockout the firing of the device if a reverse polarity event is indicated.

FIG. 122 is a reverse polarity protection circuit 7042 for sensors and/or electronic components 7005 of the secondary

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circuit of a surgical instrument according to one embodiment. During normal operation, the relay switch S1 comprises output contacts in the normally closed (NC) position and the battery voltage B₁ of the main power supply circuit 7010 (FIG. 121) is applied to V_{OUT} coupled to the secondary circuit. The diode D₁ blocks current from flowing through the coil 7046 (inductor) of the relay switch S₁. When the polarity of the battery B₁ is reversed, diode D₁ conducts and current flows through the coil 7046 of the relay switch S₁ energizing the relay switch S1 to place the output contacts in the normally open (NO) position and thus disconnecting the reverse voltage from V_{OUT} coupled to the secondary circuit. Once the switch S₁ is in the NO position, current from the positive terminal of the battery B₁ flows through LED D₃ and resistor R₁ to prevent the battery B₁ from shorting out. Diode D₂ is a clamping diode to protect from spikes generated by the coil 7046 during switching.

In one embodiment, the surgical instruments described herein provide a power reduction technique utilizing a sleep mode for sensors on a modular device. FIG. 123 is a block diagram of a surgical instrument electronic subsystem 7050 with power reduction utilizing a sleep mode monitor 7052 for sensors and/or electronic components 7005 according to one embodiment. In one embodiment, the sleep mode monitor 7052 for the sensors and/or electronic components 7005 of the secondary circuit may be implemented as a circuit and/or as a software routine to reduce the power consumption of a surgical instrument. The sleep mode monitor 7052 protection circuit may be monitored to indicate one or more sleep mode conditions to the end user of the device. In the event of a sleep mode condition in the device, the sleep mode monitor 7052 protection circuit/software may be configured to lock-out the device from being fired or operated by the user.

In various embodiments, the surgical instruments described herein provide real time feedback about the compressibility and thickness of tissue using electronic sensors 7005. The modular architecture enables the surgical instrument to be configured with custom modular shafts to employ job specific technologies. To enable sensors and/or electronic components 7005, additional electronic connection points and components may be employed to transfer both power and data signal between the modular components. As the number of sensors and/or electronic components 7005 increases, the power consumption of the surgical instrument increases, thus creating a need for techniques to reduce the power consumption of the surgical instrument.

In one embodiment, the electronic subsystem 7050 comprises a sleep mode monitor 7052 circuit and/or software for the sensors 7005 to reduce power consumption of the sensing surgical instrument. The sleep mode monitor 7052 may be implemented by creating a supplementary power supply circuit 7014 coupled to a main power supply circuit 7010. A device state monitor 7054 monitors whether the surgical instrument is in a 1=Unclamped State, 2=Clamping State, or a 3=Clamped State. When the sleep mode monitor 7052 software determines that the surgical instrument is in a non-sensing (1=Unclamped State) condition the sleep mode monitor 7052 enters the sensors and/or electronic components 7005 of the secondary circuit into a sleep mode to reduce the power load on the main power supply circuit 7010. The main power supply circuit 7010 will still be active to supply power, so that the protected processor 7008 of the primary circuit can monitor the condition. When the surgical instrument enters a condition requiring sensor activity the supplementary power supply circuit 7014 is awakened and rejoins the main power supply circuit 7010. The sleep mode

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monitor 7051 circuit can utilize a mixture of integrated circuits, solid state components, micro-processors, and/or firmware. The sleep mode monitor 7051 circuit also may be monitored to indicate the condition to the end user of the device. The sleep mode monitor 7051 circuit may also be monitored to lockout the firing or function of the device in the event the device is in a sleep mode.

In one embodiment the present disclosure provides protection against intermittent power loss for sensors and/or electronic components in modular surgical instruments. FIG. 124 is a block diagram of a surgical instrument electronic subsystem 7060 comprising a temporary power loss circuit 7062 to provide protection against intermittent power loss for sensors and/or electronic components 7005 of the secondary circuit in modular surgical instruments.

In various embodiments, the surgical instruments described herein provide real time feedback about the compressibility and thickness of tissue using sensors and/or electronic components 7005. The modular architecture enables the surgical instrument to be configured with custom modular shafts to employ job specific technologies. To enable sensors and/or electronic components 7005 additional electronic connection points and components may be employed to transfer both power and signal between the modular components. As the number of electrical connection points increase, the potential for sensors and/or electronic components 7005 to experience short term intermittent power loss increases.

In accordance with one embodiment, the temporary power loss circuit 7062 is configured to reduce device operation error from short term intermittent power loss in a sensing surgical instrument. The temporary power loss circuit 7062 has the capacity to deliver continuous power for short periods of time in the event the power from the main power supply circuit 7010 is interrupted. The temporary power loss circuit 7062 may comprises capacitive elements, batteries, and/or other electronic elements capable of leveling, detecting, or storing power.

As shown in FIG. 124, the temporary power loss circuit 7062 may be implemented by creating a supplementary circuit/software coupled to a main circuit/software. In the case that the supplementary circuit/software experiences a sudden power loss from the main power source, the sensors and/or electronic components 7005 powered by the supplementary power supply circuit 7014 would be unaffected for short period times. During the power loss the supplementary power supply circuit 7014 may be powered by capacitive elements, batteries, and/or other electronic elements that are capable of leveling or storing power. The temporary power loss circuit 7062 implemented either in hardware or software also may be monitored to lockout the firing or function of the surgical instrument in the event the device is in the power saving mode. In the event of an intermittent power loss condition in the surgical instrument the temporary power loss circuit 7062 implemented either in hardware or software may lock-out the surgical instrument from being fired or operated.

FIG. 125 illustrates one embodiment of a temporary power loss circuit 7062 implemented as a hardware circuit. The temporary power loss circuit 7062 hardware circuit is configured to reduce surgical instrument operation error from short term intermittent power loss. The temporary power loss circuit 7062 has the capacity to deliver continuous power for short periods of time in the event the power from the main power supply circuit 7010 (FIG. 124) is interrupted. The temporary power loss circuit 7062 employs capacitive elements, batteries, and/or other electronic ele-

ments that are capable of leveling, detecting, or storing power. The temporary power loss circuit **7062** may be monitored to indicate one or more conditions to the end user of the surgical instrument. In the event of an intermittent power loss condition in the surgical instrument, the temporary power loss circuit **7062** protection circuit/software might lock-out the device from being fired or operated.

In the illustrated embodiment, the temporary power loss circuit **7062** comprises an analog switch integrated circuit **U1**. In one embodiment, the analog switch integrated circuit **U1** is a single-pole/single-throw (SPST), low-voltage, single-supply, CMOS analog switch such as the MAX4501 provided by Maxim. In one embodiment, the analog switch integrated circuit **U1** is normally open (NO). In other embodiments, the analog switch integrated circuit **U1** may be normally closed (NC). The input IN activates the NO analog switch **7064** to connect the output of a step-up DC-DC converter **U3** to the input of a linear regulator **U2** via a standby "RESERVE CAPACITOR." The output of the linear regulator **U2** is coupled to the input of the DC-DC converter **U3**. The linear regulator **U2** maximizes battery life by combining ultra-low supply currents and low dropout voltages. In one embodiment, the linear regulator **U2** is a MAX882 integrated circuit provided by Maxim.

The batteries are also coupled to the input of the step-up DC-DC converter **U3**. The step-up DC-DC converter **U3** may be a compact, high-efficiency, step-up DC-DC converter with a built-in synchronous rectifier to improve efficiency and reduce size and cost by eliminating the need for an external Schottky diode. In one embodiment, the step-up DC-DC converter **U3** is a MAX1674 integrated circuit by Maxim.

Smart Cartridge Technology

FIGS. **126A** and **126B** illustrate one embodiment of an end effector **10000** comprising a magnet **10008** and a Hall effect sensor **10010** in communication with a processor **10012**. The end effector **10000** is similar to the end effector **300** described above. The end effector comprises a first jaw member, or anvil **10002**, pivotally coupled to a second jaw member, or elongated channel **10004**. The elongated channel **10004** is configured to operably support a staple cartridge **10006** therein. The staple cartridge **10006** is similar to the staple cartridge **304** described above. The anvil **10008** comprises a magnet **10008**. The staple cartridge comprises a Hall effect sensor **10010** and a processor **10012**. The Hall effect sensor **10010** is operable to communicate with the processor **10012** through a conductive coupling **10014**. The Hall effect sensor **10010** is positioned within the staple cartridge **10006** to operatively couple with the magnet **10008** when the anvil **10002** is in a closed position. The Hall effect sensor **10010** can be configured to detect changes in the magnetic field surrounding the Hall effect sensor **10010** caused by the movement of or location of magnet **10008**.

FIG. **127** illustrates one embodiment of the operable dimensions that relate to the operation of the Hall effect sensor **10010**. A first dimension **10020** is between the bottom of the center of the magnet **10008** and the top of the staple cartridge **10006**. The first dimension **10020** can vary with the size and shape of the staple cartridge **10006**, such as for instance between 0.0466 inches, 0.0325 inches, 0.0154 inches, or 0.0154 inches, or any reasonable value. A second dimension **10022** is between the bottom of the center of the magnet **10008** and the top of the Hall effect sensor **10010**. The second dimension can also vary with the size and shape of the staple cartridge **10006**, such as for instance 0.0666 inches, 0.0525 inches, 0.0354 inches, 0.0347 inches, or any reasonable value. A third dimension **10024** is between the

top of the processor **10012** and the lead-in surface **10028** of the staple cartridge **10006**. The third dimension can also vary with the size and the shape of the staple cartridge, such as for instance 0.0444 inches, 0.0440 inches, 0.0398 inches, 0.0356 inches, or any reasonable value. An angle **10026** is the angle between the anvil **10002** and the top of the staple cartridge **10006**. The angle **10026** also can vary with the size and shape of the staple cartridge **10006**, such as for instance 0.91 degrees, 0.68 degrees, 0.62 degrees, 0.15 degrees, or any reasonable value.

FIGS. **128A** through **128D** further illustrate dimensions that can vary with the size and shape of a staple cartridge **10006** and effect the operation of the Hall effect sensor **10010**. FIG. **128A** illustrates an external side view of an embodiment of a staple cartridge **10006**. The staple cartridge **10006** comprises a push-off lug **10036**. When the staple cartridge **10006** is operatively coupled with the end effector **10000** as illustrated in FIG. **126A**, the push-off lug **10036** rests on the side of the elongated channel **10004**.

FIG. **128B** illustrates various dimensions possible between the lower surface **10038** of the push-off lug **10036** and the top of the Hall effect sensor **10010** (not pictured). A first dimension **10030a** is possible with black, blue, green or gold staple cartridges **10006**, where the color of the body of the staple cartridge **10006** may be used to identify various aspects of the staple cartridge **10006**. The first dimension **10030a** can be, for instance, 0.005 inches below the lower surface **10038** of the push-off lug **10036**. A second dimension **10030b** is possible with gray staple cartridges **10006**, and can be 0.060 inches above the lower surface **10038** of the push-off lug **10036**. A third dimension **10030c** is possible with white staple cartridges **10006**, and can be 0.030 inches above the lower-surface **10038** of the push-off lug **10036**.

FIG. **128C** illustrates an external side view of an embodiment of a staple cartridge **10006**. The staple cartridge **10006** comprises a push-off lug **10036** with a lower surface **10038**. The staple cartridge **10006** further comprises an upper surface **10046** immediately above the Hall effect sensor **10010** (not pictured). FIG. **128D** illustrates various dimensions possible between the lower surface **10038** of the push-off lug **10036** and the upper surface **10046** of the staple cartridge **10006** above the Hall effect sensor **10010**. A first dimension **10040** is possible for black, blue, green or gold staple cartridges **10006**, and can be, for instance, 0.015 inches above the lower surface **10038** of the push-off lug **10036**. A second dimension **10042** is possible for gray staple cartridges **10006**, and can be, for instance, 0.080 inches. A third dimension **10044** is possible for white staple cartridges **10006**, and can be, for instance, 0.050.

It is understood that the references to the color of the body of a staple cartridge **10006** is for convenience and by way of example only. It is understood that other staple cartridge **10006** body colors are possible. It is also understood that the dimensions given for FIGS. **128A** through **128D** are also example and non-limiting.

FIG. **129A** illustrates various embodiments of magnets **10058a-10058d** of various sizes, according to how each magnet **10058a-10058d** may fit in the distal end of an anvil, such as anvil **10002** illustrated in FIGS. **126A-126B**. A magnet **10058a-10058d** can be positioned in the distal tip of the anvil **10002** at a given distance **10050** from the anvil's pin or pivot point **10052**. It is understood that this distance **10050** may vary with the construction of the end effector and staple cartridge and/or the desired position of the magnet. FIG. **129B** further illustrates a front-end cross-sectional view **10054** of the anvil **10002** and the central axis point of the anvil **10002**. FIG. **129A** also illustrates an example

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10056 of how various embodiments of magnets **10058a-10058d** may fit within the same anvil **10002**.

FIGS. **130A-130E** illustrate one embodiment of an end effector **10100** that comprises, by way of example, a magnet **10058a** as illustrated in FIGS. **129A-129B**. FIG. **130A** illustrates a front-end cross-sectional view of the end effector **10100**. The end effector **10100** is similar to the end effector **300** described above. The end effector **10100** comprises a first jaw member or anvil **10102**, a second jaw member or elongated channel **10104**, and a staple cartridge **10106** operatively coupled to the elongated channel **10104**. The anvil **10102** further comprises the magnet **10058a**. The staple cartridge **10106** further comprises a Hall effect sensor **10110**. The anvil **10102** is here illustrated in a closed position. FIG. **130B** illustrates a front-end cutaway view of the anvil **10102** and the magnet **10058a**, in situ. FIG. **130C** illustrates a perspective cutaway view of the anvil **10102** and the magnet **10058a**, in an optional location. FIG. **130D** illustrates a side cutaway view of the anvil **10102** and the magnet **10058a**, in an optional location. FIG. **130E** illustrates a top cutaway view of the anvil **10102** and the magnet **10058a**, in an optional location.

FIGS. **131A-131E** illustrate one embodiment of an end effector **10150** that comprises, by way of example, a magnet **10058d** as illustrated in FIGS. **129A-129B**. FIG. **131A** illustrates a front-end cross-sectional view of the end effector **10150**. The end effector **10150** comprises an anvil **10152**, an elongated channel **10154**, and a staple cartridge **10156**. The anvil **10152** further comprises magnet **10058d**. The staple cartridge **10156** further comprises a Hall effect sensor **10160**. FIG. **131B** illustrates a front-end cutaway view of the anvil **10150** and the magnet **10058d**, in situ. FIG. **131C** illustrates a perspective cutaway view of the anvil **10152** and the magnet **10058d** in an optional location. FIG. **131D** illustrates a side cutaway view of the anvil **10152** and the magnet **10058d** in an optional location. FIG. **131E** illustrates a top cutaway view of the anvil **10152** and magnet **10058d** in an optional location.

FIG. **132** illustrates an end effector **300** as described above, and illustrates contact points between the anvil **306** and either the staple cartridge **304** and/or the elongated channel **302**. Contact points between the anvil **306** and the staple cartridge **304** and/or the elongated channel **302** can be used to determine the position of the anvil **306** and/or provide a point for an electrical contact between the anvil **306** and the staple cartridge **304**, and/or the anvil **306** and the elongated channel **302**. Distal contact point **10170** can provide a contact point between the anvil **306** and the staple cartridge **304**. Proximal contact point **10172** can provide a contact point between the anvil **306** and the elongated channel **302**.

FIGS. **133A** and **133B** illustrate one embodiment of an end effector **10200** that is operable to use conductive surfaces at the distal contact point to create an electrical connection. The end effector **10200** is similar to the end effector **300** described above. The end effector comprises an anvil **10202**, an elongated channel **10204**, and a staple cartridge **10206**. The anvil **10202** further comprises a magnet **10208** and an inside surface **10210**, which further comprises a number of staple-forming indents **10212**. In some embodiments, the inside surface **10210** of the anvil **10202** further comprises a first conductive surface **10214** surrounding the staple-forming indents **10212**. The first conductive surface **10214** can come into contact with second conductive surfaces **10222** on the staple cartridge **10206**, as illustrated in FIG. **107B**. FIG. **107B** illustrates a close-up view of the cartridge body **10216** of the staple cartridge

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10206. The cartridge body **10216** comprises a number of staple cavities **10218** designed to hold staples (not pictured). In some embodiments the staple cavities **10218** further comprise staple cavity extensions **10220** that protrude above the surface of the cartridge body **10216**. The staple cavity extensions **10220** can be coated with the second conductive surfaces **10222**. Because the staple cavity extensions **10222** protrude above the surface of the cartridge body **10216**, the second conductive surfaces **10222** will come into contact with the first conductive surfaces **10214** when the anvil **10202** is in a closed position. In this manner the anvil **10202** can form an electrical contact with the staple cartridge **10206**.

FIGS. **134A-134C** illustrate one embodiment of an end effector **10250** that is operable to use conductive surfaces to form an electrical connection. FIG. **134A** illustrates the end effector **10250** comprises an anvil **10252**, an elongated channel **10254**, and a staple cartridge **10256**. The anvil further comprises a magnet **10258** and an inside surface **10260**, which further comprises staple-forming indents **10262**. In some embodiments the inside surface **10260** of the anvil **10250** can further comprise first conductive surfaces **10264**, located, by way of example, distally from the staple-forming indents **10262**, as illustrated in FIG. **134B**. The first conductive surfaces **10264** are located such that they can come into contact with a second conductive surface **10272** located on the staple cartridge **10256**, as illustrated in FIG. **134C**. FIG. **134C** illustrates the staple cartridge **10256**, which comprises a cartridge body **10266**. The cartridge body **10266** further comprises an upper surface **10270**, which in some embodiments can be coated with the second conductive surface **10272**. The first conductive surfaces **10264** are located on the inside surface **10260** of the anvil **10252** such that they come into contact with the second conductive surface **10272** when the anvil **10252** is in a closed position. In this manner the anvil **10250** can form an electrical contact with the staple cartridge **10256**.

FIGS. **135A** and **135B** illustrate one embodiment of an end effector **10300** that is operable to use conductive surfaces to form an electrical connection. The end effector **10300** comprises an anvil **10302**, an elongated channel **10304**, and a staple cartridge **10306**. The anvil **10302** further comprises a magnet **10308** and an inside surface **10310**, which further comprises a number of staple-forming indents **10312**. In some embodiments the inside surface **10310** further comprises a first conductive surface **10314** surrounding some of the staple-forming indents **10312**. The first conductive surface is located such that it can come into contact with second conductive surfaces **10322** as illustrated in FIG. **109B**. FIG. **109B** illustrates a close-up view of the staple cartridge **10306**. The staple cartridge **10306** comprises a cartridge body **10316** which further comprises an upper surface **10320**. In some embodiments, the leading edge of the upper surface **10320** can be coated with second conductive surfaces **10322**. The first conductive surface **10312** is positioned such that it will come into contact with the second conductive surfaces **10322** when the anvil **10302** is in a closed position. In this manner the anvil **10302** can form an electrical connection with the staple cartridge **10306**.

FIGS. **136A** and **136B** illustrate one embodiment of an end effector **10350** that is operable to use conductive surfaces to form an electrical connection. FIG. **136A** illustrates an end effector **10350** comprising an anvil **10352**, an elongated channel **10354**, and a staple cartridge **10356**. The anvil **10352** further comprises a magnet **10358** and an inside surface **10360**, which further comprises a number of staple-

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forming indents 10362. In some embodiments the inside surface 10360 further comprises a first conductive surface 10364 surrounding some of the staple-forming indents 10362. The first conductive surface is located such that it can come into contact with second conductive surfaces 10372 as illustrated in FIG. 136B. FIG. 136B illustrates a close-up view of the staple cartridge 10356. The staple cartridge 10356 comprises a cartridge body 10366 which further comprises an upper surface 10370. In some embodiments, the leading edge of the upper surface 10327 can be coated with second conductive surfaces 10372. The first conductive surface 10362 is positioned such that it will come into contact with the second conductive surfaces 10372 when the anvil 10352 is in a closed position. In this manner the anvil 10352 can form an electrical connection with the staple cartridge 10356.

FIGS. 137A-137C illustrate one embodiment of an end effector 10400 that is operable to use the proximal contact point 10408 to form an electrical connection. FIG. 137A illustrate the end effector 10400, which comprises an anvil 10402, an elongated channel 10404, and a staple cartridge 10406. The anvil 10402 further comprises pins 10410 that extend from the anvil 10402 and allow the anvil to pivot between an open and a closed position relative to the elongated channel 10404 and the staple cartridge 10406. FIG. 137B is a close-up view of a pin 10410 as it rests within an aperture 10418 defined in the elongated channel 10404 for that purpose. In some embodiments, pin 10410 further comprises a first conductive surface 10412 located on the exterior of the pin 10410. In some embodiments the aperture 10418 further comprises a second conductive surface 10414 on its outside surface. As the anvil 10402 moves between a closed and an open position, the first conductive surface 10412 on the pin 10410 rotates and comes into contact with the second conductive surface 10414 on the surface of the aperture 10418, thus forming an electrical contact. FIG. 137C illustrates an alternate embodiment, with an alternate location for a second conductive surface 10416 on the surface of the aperture 10418.

FIG. 138 illustrates one embodiment of an end effector 10450 with a distal sensor plug 10466. End effector 10450 comprises a first jaw member or anvil 10452, a second jaw member or elongated channel 10454, and a staple cartridge 10466. The staple cartridge 10466 further comprises the distal sensor plug 10466, located at the distal end of the staple cartridge 10466.

FIG. 139A illustrates the end effector 10450 with the anvil 10452 in an open position. FIG. 139B illustrates a cross-sectional view of the end effector 10450 with the anvil 10452 in an open position. As illustrated, the anvil 10452 may further comprise a magnet 10458, and the staple cartridge 10456 may further comprise the distal sensor plug 10466 and a wedge sled, 10468, which is similar to the wedge sled 190 described above. FIG. 139C illustrates the end effector 10450 with the anvil 10452 in a closed position. FIG. 139D illustrates a cross sectional view of the end effector 10450 with the anvil 10452 in a closed position. As illustrated, the anvil 10452 may further comprise a magnet 10458, and the staple cartridge 10456 may further comprise the distal sensor plug 10466 and a wedge sled 10468. As illustrated, when the anvil 10452 is in a closed position relative to the staple cartridge 10456, the magnet 10458 is in proximity to the distal sensor plug 10466.

FIG. 140 provides a close-up view of the cross section of the distal end of the end effector 10450. As illustrated, the distal sensor plug 10466 may further comprise a Hall effect sensor 10460 in communication with a processor 10462. The

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Hall effect sensor 10460 can be operatively connected to a flex board 10464. The processor 10462 can also be operatively connect to the flex board 10464, such that the flex board 10464 provides a communication path between the Hall effect sensor 10460 and the processor 10462. The anvil 10452 is illustrated in a closed position, and as illustrated, when the anvil 10452 is in a closed position the magnet 10458 is in proximity to the Hall effect sensor 10460.

FIG. 141 illustrates a close-up top view of the staple cartridge 10456 that comprises a distal sensor plug 10466. Staple cartridge 10456 further comprises a cartridge body 10470. The cartridge body 10470 further comprises electrical traces 10472. Electrical traces 10472 provide power to the distal sensor plug 10466, and are connected to a power source at the proximal end of the staple cartridge 10456 as described in further detail below. Electrical traces 10472 can be placed in the cartridge body 10470 by various methods, such as for instance laser etching.

FIGS. 142A and 142B illustrate one embodiment of a staple cartridge 10506 with a distal sensor plug 10516. FIG. 142A is a perspective view of the underside of the staple cartridge 10506. The staple cartridge 10506 comprises a cartridge body 10520 and a cartridge tray 10522. The staple cartridge 10506 further comprises a distal sensor cover 10524 that encloses the lower area of the distal end of the staple cartridge 10506. The cartridge tray 10522 further comprises an electrical contact 10526. FIG. 142B illustrates a cross sectional view of the distal end of the staple cartridge 10506. As illustrated, the staple cartridge 10506 can further comprise a distal sensor plug 10516 located within the cartridge body 10520. The distal sensor plug 10516 further comprises a Hall effect sensor 10510 and a processor 10512, both operatively connected to a flex board 10514. The distal sensor plug 10516 can be connected to the electrical contact 10526, and can thus use conductivity in the cartridge tray 10522 as a source of power. FIG. 142B further illustrates the distal sensor cover 10524, which encloses the distal sensor plug 10516 within the cartridge body 10520.

FIGS. 143A-143C illustrate one embodiment of a staple cartridge 10606 that comprises a flex cable 10630 connected to a Hall effect sensor 10610 and processor 10612. The staple cartridge 10606 is similar to the staple cartridge 10606 is similar to the staple cartridge 306 described above. FIG. 143A is an exploded view of the staple cartridge 10606. The staple cartridge comprises 10606 a cartridge body 10620, a wedge sled 10618, a cartridge tray 10622, and a flex cable 10630. The flex cable 10630 further comprises electrical contacts 10632 at the proximal end of the staple cartridge 10606, placed to make an electrical connection when the staple cartridge 10606 is operatively coupled with an end effector, such as end effector 10800 described below. The electrical contacts 10632 are integrated with cable traces 10634, which extend along some of the length of the staple cartridge 10606. The cable traces 10634 connect 10636 near the distal end of the staple cartridge 10606 and this connection 10636 joins with a conductive coupling 10614. A Hall effect sensor 10610 and a processor 10612 are operatively coupled to the conductive coupling 10614 such that the Hall effect sensor 10610 and the processor 10612 are able to communicate.

FIG. 143B illustrates the assembly of the staple cartridge 10606 and the flex cable 10630 in greater detail. As illustrated, the cartridge tray 10622 encloses the underside of the cartridge body 10620, thereby enclosing the wedge sled 10618. The flex cable 10630 can be located on the exterior of the cartridge tray 10622, with the conductive coupling 10614 positioned within the distal end of the cartridge body

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10620 and the electrical contacts 10632 located on the outside near the proximal end. The flex cable 10630 can be placed on the exterior of the cartridge tray 10622 by any appropriate means, such as for instance bonding or laser etching.

FIG. 143C illustrates a cross sectional view of the staple cartridge 10606 to illustrate the placement of the Hall effect sensor 10610, processor 10612, and conductive coupling 10614 within the distal end of the staple cartridge, in accordance with the present embodiment.

FIG. 144A-144F illustrate one embodiment of a staple cartridge 10656 that comprises a flex cable 10680 connected to a Hall effect sensor 10660 and a processor 10662. FIG. 144A is an exploded view of the staple cartridge 10656. The staple cartridge comprises a cartridge body 10670, a wedge sled 10668, a cartridge tray 10672, and a flex cable 10680. The flex cable 10680 further comprises cable traces 10684 that extend along some of the length of the staple cartridge 10656. Each of the cable traces 10684 have an angle 10686 near their distal end, and connect therefrom to a conductive coupling 10664. A Hall effect sensor 10660 and a processor 10662 are operatively coupled to the conductive coupling 10664 such that the Hall effect sensor 10660 and the processor 10662 are able to communicate.

FIG. 144B illustrates the assembly of the staple cartridge 10656. The cartridge tray 10672 encloses the underside of the cartridge body 10670, thereby enclosing the wedge sled 10668. The flex cable 10680 is located between the cartridge body 10670 and the cartridge tray 10672. As such, in the illustration only the angle 10686 and the conductive coupling 10664 are visible.

FIG. 144C illustrates the underside of an assembled staple cartridge 10656, and also illustrates the flex cable 10680 in greater detail. In an assembled staple cartridge 10656, the conductive coupling 10664 is located in the distal end of the staple cartridge 10656. Because the flex cable 10680 can be located between the cartridge body 10670 and the cartridge tray 10672, only the angle 10686 ends of the cable traces 10684 would be visible from the underside of the staple cartridge 10656, as well as the conductive coupling 10664.

FIG. 144D illustrates a cross sectional view of the staple cartridge 10656 to illustrate the placement of the Hall effect sensor 10660, processor 10662, and conductive coupling 10664. Also illustrated is an angle 10686 of a cable trace 10684, to illustrate where the angle 10686 could be placed. The cable traces 10684 are not pictured.

FIG. 144E illustrates the underside of the staple cartridge 10656 without the cartridge tray 10672 and including the wedge sled 10668, in its most distal position. The staple cartridge 10656 is illustrated without the cartridge tray 10672 in order to illustrate a possible placement for the cable traces 10684, which are otherwise obscured by the cartridge tray 10672. As illustrated, the cable traces 10684 can be placed inside the cartridge body 10670. The angle 10686 optionally allows the cable traces 10684 to occupy a narrower space in the distal end of the cartridge body 10670.

FIG. 144F also illustrates the staple cartridge 10656 without the cartridge tray 10672 in order to illustrate a possible placement for the cable traces 10684. As illustrated the cable traces 10684 can be placed along the length of the exterior of cartridge body 10670. Furthermore, the cable traces 10684 can form an angle 10686 to enter the interior of the distal end of the cartridge body 10670.

FIGS. 145A and 145B illustrates one embodiment of a staple cartridge 10706 that comprises a flex cable 10730, a Hall effect sensor 10710, and a processor 10712. FIG. 145A is an exploded view of the staple cartridge 10706. The staple

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cartridge 10706 comprises a cartridge body 10720, a wedge sled 10718, a cartridge tray 10722, and a flex cable 10730. The flex cable 10730 further comprises electrical contacts 10732 placed to make an electrical connection when the staple cartridge 10706 is operatively coupled with an end effector. The electrical contacts 10732 are integrated with cable traces 10734. The cable traces connect 10736 near the distal end of the staple cartridge 10706, and this connection 10736 joins with a conductive coupling 10714. A Hall effect sensor 10710 and a processor 10712 are operatively connected to the conductive coupling 10714 such that the are able to communicate.

FIG. 145B illustrates the assembly of the staple cartridge 10706 and the flex cable 10730 in greater detail. As illustrated, the cartridge tray 10722 encloses the underside of the cartridge body 10720, thereby enclosing the wedge sled 10718. The flex cable 10730 can be located on the exterior of the cartridge tray 10722 with the conductive coupling 10714 positioned within the distal end of the cartridge body 10720. The flex cable 10730 can be placed on the exterior of the cartridge tray 10722 by any appropriate means, such as for instance bonding or laser etching.

FIGS. 146A-146F illustrate one embodiment of an end effector 10800 with a flex cable 10840 operable to provide power to a staple cartridge 10806 that comprises a distal sensor plug 10816. The end effector 10800 is similar to the end effector 300 described above. The end effector 10800 comprises a first jaw member or anvil 10802, a second jaw member or elongated channel 10804, and a staple cartridge 10806 operatively coupled to the elongated channel 10804. The end effector 10800 is operatively coupled to a shaft assembly 10900. The shaft assembly 10900 is similar to shaft assembly 200 described above. The shaft assembly 10900 further comprises a closure tube 10902 that encloses the exterior of the shaft assembly 10900. In some embodiments the shaft assembly 10900 further comprises an articulation joint 10904, which includes a double pivot closure sleeve assembly 10906. The double pivot closure sleeve assembly 10906 includes an end effector closure sleeve assembly 10908 that is operable to couple with the end effector 10800.

FIG. 146A illustrates a perspective view of the end effector 10800 coupled to the shaft assembly 10900. In various embodiments, the shaft assembly 10900 further comprises a flex cable 10830 that is configured to not interfere with the function of the articulation joint 10904, as described in further detail below. FIG. 146B illustrates a perspective view of the underside of the end effector 10800 and shaft assembly 10900. In some embodiments, the closure tube 10902 of the shaft assembly 10900 further comprises a first aperture 10908, through which the flex cable 10908 can extend. The close sleeve assembly 10908 further comprises a second aperture 10910, through which the flex cable 10908 can also pass.

FIG. 146C illustrates the end effector 10800 with the flex cable 10830 and without the shaft assembly 10900. As illustrated, in some embodiments the flex cable 10830 can include a single coil 10832 operable to wrap around the articulation joint 10904, and thereby be operable to flex with the motion of the articulation joint 10904.

FIGS. 146D and 146E illustrate the elongated channel 10804 portion of the end effector 10800 without the anvil 10802 or the staple cartridge 10806, to illustrate how the flex cable 10830 can be seated within the elongated channel 10804. In some embodiments, the elongated channel 10804 further comprises a third aperture 10824 for receiving the flex cable 10830. Within the body of the elongated channel

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10804 the flex cable splits **10834** to form extensions **10836** on either side of the elongated channel **10804**. FIG. **146E** further illustrates that connectors **10838** can be operatively coupled to the flex cable extensions **10836**.

FIG. **146F** illustrates the flex cable **10830** alone. As illustrated, the flex cable **10830** comprises a single coil **10832** operative to wrap around the articulation joint **10904**, and a split **10834** that attaches to extensions **10836**. The extensions can be coupled to connectors **10838** that have on their distal facing surfaces prongs **10840** for coupling to the staple cartridge **10806**, as described below.

FIG. **147** illustrates a close up view of the elongated channel **10804** with a staple cartridge **10806** coupled thereto. The staple cartridge **10804** comprises a cartridge body **10822** and a cartridge tray **10820**. In some embodiments the staple cartridge **10806** further comprises electrical traces **10828** that are coupled to proximal contacts **10856** at the proximal end of the staple cartridge **10806**. The proximal contacts **10856** can be positioned to form a conductive connection with the prongs **10840** of the connectors **10838** that are coupled to the flex cable extensions **10836**. Thus, when the staple cartridge **10806** is operatively coupled with the elongated channel **10804**, the flex cable **10830**, through the connectors **10838** and the connector prongs **10840**, can provide power to the staple cartridge **10806**.

FIGS. **148A-148D** further illustrate one embodiment of a staple cartridge **10806** operative with the present embodiment of an end effector **10800**. FIG. **148A** illustrates a close up view of the proximal end of the staple cartridge **10806**. As discussed above, the staple cartridge **10806** comprises electrical traces **10828** that, at the proximal end of the staple cartridge **10806**, form proximal contacts **10856** that are operable to couple with the flex cable **10830** as described above. FIG. **148B** illustrates a close-up view of the distal end of the staple cartridge **10806**, with a space for a distal sensor plug **10816**, described below. As illustrated, the electrical traces **10828** can extend along the length of the staple cartridge body **10822** and, at the distal end, form distal contacts **10856**. FIG. **148C** further illustrates the distal sensor plug **10816**, which in some embodiments is shaped to be received by the space formed for it in the distal end of the staple cartridge **10806**. FIG. **148D** illustrates the proximal-facing side of the distal sensor plug **10816**. As illustrated, the distal sensor plug **10816** has sensor plug contacts **10854**, positioned to couple with the distal contacts **10858** of the staple cartridge **10806**. Thus, in some embodiments the electrical traces **10828** can be operative to provide power to the distal sensor plug **10816**.

FIGS. **149A** and **149B** illustrate one embodiment of a distal sensor plug **10816**. FIG. **149A** illustrates a cutaway view of the distal sensor plug **10816**. As illustrated, the distal sensor plug **10816** comprises a Hall effect sensor **10810** and a processor **10812**. The distal sensor plug **10816** further comprises a flex board **10814**. As further illustrated in FIG. **149B**, the Hall effect sensor **10810** and the processor **10812** are operatively coupled to the flex board **10814** such that they are capable of communicating.

FIG. **150** illustrates an embodiment of an end effector **10960** with a flex cable **10980** operable to provide power to sensors and electronics **10972** in the distal tip of the anvil **10952** portion. The end effector **10950** comprises a first jaw member or anvil **10962**, a second jaw member or elongated channel **10964**, and a staple cartridge **10956** operatively coupled to the elongated channel **10952**. The end effector **10960** is operatively coupled to a shaft assembly **10960**. The shaft assembly **10960** further comprises a closure tube **10962** that encloses the shaft assembly **10960**. In some

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embodiments the shaft assembly **10960** further comprises an articulation joint **10964**, which includes a double pivot closure sleeve assembly **10966**.

In various embodiments, the end effector **10950** further comprises a flex cable **10980** that is configured to not interfere with the function of the articulation joint **10964**. In some embodiments, the closure tube **10962** comprises a first aperture **10968** through which the flex cable **10980** can extend. In some embodiments, flex cable **10980** further comprises a loop or coil **10982** that wraps around the articulation joint **10964** such that the flex cable **10980** does not interfere with the operation of the articulation joint **10964**, as further described below. In some embodiments, the flex cable **10980** extends along the length of the anvil **10951** to a second aperture **10970** in the distal tip of the anvil **10951**.

FIGS. **151A-151C** illustrate the operation of the articulation joint **10964** and flex cable **10980** of the end effector **10950**. FIG. **151A** illustrates a top view of the end effector **10952** with the end effector **109650** pivoted -45 degrees with respect to the shaft assembly **10960**. As illustrated, the coil **10982** of the flex cable **10980** flexes with the articulation joint **10964** such that the flex cable **10980** does not interfere with the operation of the articulation joint **10964**. FIG. **151B** illustrates a top view of the end effector **10950**. As illustrated, the coil **10982** wraps around the articulation joint **10964** once. FIG. **151C** illustrates a top view of the end effector **10950** with the end effector **10950** pivoted $+45$ degrees with respect to the shaft assembly **10960**. As illustrated, the coil **10982** of the flex cable **10980** flexes with the articulation joint **10964** such that the flex cable **10980** does not interfere with the operation of the articulation joint **10964**.

FIG. **152** illustrates cross-sectional view of the distal tip of an embodiment of an anvil **10952** with sensors and electronics **10972**. The anvil **10952** comprises a flex cable **10980**, as described with respect to FIGS. **150** and **151A-151C**. As illustrated in FIG. **152**, the anvil **10952** further comprises a second aperture **10970** through which the flex cable **10980** can pass such that the flex cable **10980** can enter a housing **10974** in the within the anvil **10952**. Within the housing **10974** the flex cable **10980** can operably couple to sensors and electronics **10972** located within the housing **10974** and thereby provide power to the sensors and electronics **10972**.

FIG. **153** illustrates a cutaway view of the distal tip of the anvil **10952**. FIG. **153** illustrates an embodiment of the housing **10974** that can contain sensors and electronics **10972** as illustrated by FIG. **152**.

In accordance with various embodiments, the surgical instruments described herein may comprise one or more processors (e.g., microprocessor, microcontroller) coupled to various sensors. In addition, to the processor(s), a storage (having operating logic) and communication interface, are coupled to each other.

As described earlier, the sensors may be configured to detect and collect data associated with the surgical device. The processor processes the sensor data received from the sensor(s).

The processor may be configured to execute the operating logic. The processor may be any one of a number of single or multi-core processors known in the art. The storage may comprise volatile and non-volatile storage media configured to store persistent and temporal (working) copy of the operating logic.

In various embodiments, the operating logic may be configured to perform the initial processing, and transmit the

data to the computer hosting the application to determine and generate instructions. For these embodiments, the operating logic may be further configured to receive information from and provide feedback to a hosting computer. In alternate embodiments, the operating logic may be configured to assume a larger role in receiving information and determining the feedback. In either case, whether determined on its own or responsive to instructions from a hosting computer, the operating logic may be further configured to control and provide feedback to the user.

In various embodiments, the operating logic may be implemented in instructions supported by the instruction set architecture (ISA) of the processor, or in higher level languages and compiled into the supported ISA. The operating logic may comprise one or more logic units or modules. The operating logic may be implemented in an object oriented manner. The operating logic may be configured to be executed in a multi-tasking and/or multi-thread manner. In other embodiments, the operating logic may be implemented in hardware such as a gate array.

In various embodiments, the communication interface may be configured to facilitate communication between a peripheral device and the computing system. The communication may include transmission of the collected biometric data associated with position, posture, and/or movement data of the user's body part(s) to a hosting computer, and transmission of data associated with the tactile feedback from the host computer to the peripheral device. In various embodiments, the communication interface may be a wired or a wireless communication interface. An example of a wired communication interface may include, but is not limited to, a Universal Serial Bus (USB) interface. An example of a wireless communication interface may include, but is not limited to, a Bluetooth interface.

For various embodiments, the processor may be packaged together with the operating logic. In various embodiments, the processor may be packaged together with the operating logic to form a SiP. In various embodiments, the processor may be integrated on the same die with the operating logic. In various embodiments, the processor may be packaged together with the operating logic to form a System on Chip (SoC).

Various embodiments may be described herein in the general context of computer executable instructions, such as software, program modules, and/or engines being executed by a processor. Generally, software, program modules, and/or engines include any software element arranged to perform particular operations or implement particular abstract data types. Software, program modules, and/or engines can include routines, programs, objects, components, data structures and the like that perform particular tasks or implement particular abstract data types. An implementation of the software, program modules, and/or engines components and techniques may be stored on and/or transmitted across some form of computer-readable media. In this regard, computer-readable media can be any available medium or media useable to store information and accessible by a computing device. Some embodiments also may be practiced in distributed computing environments where operations are performed by one or more remote processing devices that are linked through a communications network. In a distributed computing environment, software, program modules, and/or engines may be located in both local and remote computer storage media including memory storage devices. A memory such as a random access memory (RAM) or other dynamic storage device may be employed for storing information and instructions to be executed by the processor. The memory

also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by the processor.

Although some embodiments may be illustrated and described as comprising functional components, software, engines, and/or modules performing various operations, it can be appreciated that such components or modules may be implemented by one or more hardware components, software components, and/or combination thereof. The functional components, software, engines, and/or modules may be implemented, for example, by logic (e.g., instructions, data, and/or code) to be executed by a logic device (e.g., processor). Such logic may be stored internally or externally to a logic device on one or more types of computer-readable storage media. In other embodiments, the functional components such as software, engines, and/or modules may be implemented by hardware elements that may include processors, microprocessors, circuits, circuit elements (e.g., transistors, resistors, capacitors, inductors, and so forth), integrated circuits, ASICs, PLDs, DSPs, FPGAs, logic gates, registers, semiconductor device, chips, microchips, chip sets, and so forth.

Examples of software, engines, and/or modules may include software components, programs, applications, computer programs, application programs, system programs, machine programs, operating system software, middleware, firmware, software modules, routines, subroutines, functions, methods, procedures, software interfaces, application program interfaces (API), instruction sets, computing code, computer code, code segments, computer code segments, words, values, symbols, or any combination thereof. Determining whether an embodiment is implemented using hardware elements and/or software elements may vary in accordance with any number of factors, such as desired computational rate, power levels, heat tolerances, processing cycle budget, input data rates, output data rates, memory resources, data bus speeds and other design or performance constraints.

One or more of the modules described herein may comprise one or more embedded applications implemented as firmware, software, hardware, or any combination thereof. One or more of the modules described herein may comprise various executable modules such as software, programs, data, drivers, application APIs, and so forth. The firmware may be stored in a memory of the controller and/or the controller which may comprise a nonvolatile memory (NVM), such as in bit-masked ROM or flash memory. In various implementations, storing the firmware in ROM may preserve flash memory. The NVM may comprise other types of memory including, for example, programmable ROM (PROM), erasable programmable ROM (EPROM), EEPROM, or battery backed RAM such as dynamic RAM (DRAM), Double-Data-Rate DRAM (DDRAM), and/or synchronous DRAM (SDRAM).

In some cases, various embodiments may be implemented as an article of manufacture. The article of manufacture may include a computer readable storage medium arranged to store logic, instructions and/or data for performing various operations of one or more embodiments. In various embodiments, for example, the article of manufacture may comprise a magnetic disk, optical disk, flash memory or firmware containing computer program instructions suitable for execution by a general purpose processor or application specific processor. The embodiments, however, are not limited in this context.

The functions of the various functional elements, logical blocks, modules, and circuits elements described in connec-

tion with the embodiments disclosed herein may be implemented in the general context of computer executable instructions, such as software, control modules, logic, and/or logic modules executed by the processing unit. Generally, software, control modules, logic, and/or logic modules comprise any software element arranged to perform particular operations. Software, control modules, logic, and/or logic modules can comprise routines, programs, objects, components, data structures and the like that perform particular tasks or implement particular abstract data types. An implementation of the software, control modules, logic, and/or logic modules and techniques may be stored on and/or transmitted across some form of computer-readable media. In this regard, computer-readable media can be any available medium or media useable to store information and accessible by a computing device. Some embodiments also may be practiced in distributed computing environments where operations are performed by one or more remote processing devices that are linked through a communications network. In a distributed computing environment, software, control modules, logic, and/or logic modules may be located in both local and remote computer storage media including memory storage devices.

Additionally, it is to be appreciated that the embodiments described herein illustrate example implementations, and that the functional elements, logical blocks, modules, and circuits elements may be implemented in various other ways which are consistent with the described embodiments. Furthermore, the operations performed by such functional elements, logical blocks, modules, and circuits elements may be combined and/or separated for a given implementation and may be performed by a greater number or fewer number of components or modules. As will be apparent to those of skill in the art upon reading the present disclosure, each of the individual embodiments described and illustrated herein has discrete components and features which may be readily separated from or combined with the features of any of the other several aspects without departing from the scope of the present disclosure. Any recited method can be carried out in the order of events recited or in any other order which is logically possible.

It is worthy to note that any reference to "one embodiment" or "an embodiment" means that a particular feature, structure, or characteristic described in connection with the embodiment is comprised in at least one embodiment. The appearances of the phrase "in one embodiment" or "in one aspect" in the specification are not necessarily all referring to the same embodiment.

Unless specifically stated otherwise, it may be appreciated that terms such as "processing," "computing," "calculating," "determining," or the like, refer to the action and/or processes of a computer or computing system, or similar electronic computing device, such as a general purpose processor, a DSP, ASIC, FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein that manipulates and/or transforms data represented as physical quantities (e.g., electronic) within registers and/or memories into other data similarly represented as physical quantities within the memories, registers or other such information storage, transmission or display devices.

It is worthy to note that some embodiments may be described using the expression "coupled" and "connected" along with their derivatives. These terms are not intended as synonyms for each other. For example, some embodiments may be described using the terms "connected" and/or

"coupled" to indicate that two or more elements are in direct physical or electrical contact with each other. The term "coupled," however, also may mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other. With respect to software elements, for example, the term "coupled" may refer to interfaces, message interfaces, API, exchanging messages, and so forth.

It should be appreciated that any patent, publication, or other disclosure material, in whole or in part, that is said to be incorporated by reference herein is incorporated herein only to the extent that the incorporated material does not conflict with existing definitions, statements, or other disclosure material set forth in this disclosure. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

The disclosed embodiments have application in conventional endoscopic and open surgical instrumentation as well as application in robotic-assisted surgery.

Embodiments of the devices disclosed herein can be designed to be disposed of after a single use, or they can be designed to be used multiple times. Embodiments may, in either or both cases, be reconditioned for reuse after at least one use. Reconditioning may include any combination of the steps of disassembly of the device, followed by cleaning or replacement of particular pieces, and subsequent reassembly. In particular, embodiments of the device may be disassembled, and any number of the particular pieces or parts of the device may be selectively replaced or removed in any combination. Upon cleaning and/or replacement of particular parts, embodiments of the device may be reassembled for subsequent use either at a reconditioning facility, or by a surgical team immediately prior to a surgical procedure. Those skilled in the art will appreciate that reconditioning of a device may utilize a variety of techniques for disassembly, cleaning/replacement, and reassembly. Use of such techniques, and the resulting reconditioned device, are all within the scope of the present application.

By way of example only, embodiments described herein may be processed before surgery. First, a new or used instrument may be obtained and when necessary cleaned. The instrument may then be sterilized. In one sterilization technique, the instrument is placed in a closed and sealed container, such as a plastic or TYVEK bag. The container and instrument may then be placed in a field of radiation that can penetrate the container, such as gamma radiation, x-rays, or high-energy electrons. The radiation may kill bacteria on the instrument and in the container. The sterilized instrument may then be stored in the sterile container. The sealed container may keep the instrument sterile until it is opened in a medical facility. A device may also be sterilized using any other technique known in the art, including but not limited to beta or gamma radiation, ethylene oxide, or steam.

One skilled in the art will recognize that the herein described components (e.g., operations), devices, objects, and the discussion accompanying them are used as examples for the sake of conceptual clarity and that various configuration modifications are contemplated. Consequently, as used herein, the specific exemplars set forth and the accompanying discussion are intended to be representative of their more general classes. In general, use of any specific exem-

plar is intended to be representative of its class, and the non-inclusion of specific components (e.g., operations), devices, and objects should not be taken limiting.

With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations are not expressly set forth herein for sake of clarity.

The herein described subject matter sometimes illustrates different components contained within, or connected with, different other components. It is to be understood that such depicted architectures are merely examples and that in fact many other architectures may be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “operably connected,” or “operably coupled,” to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “operably couplable,” to each other to achieve the desired functionality. Specific examples of operably couplable include but are not limited to physically mateable and/or physically interacting components, and/or wirelessly interactable, and/or wirelessly interacting components, and/or logically interacting, and/or logically interactable components.

Some aspects may be described using the expression “coupled” and “connected” along with their derivatives. It should be understood that these terms are not intended as synonyms for each other. For example, some aspects may be described using the term “connected” to indicate that two or more elements are in direct physical or electrical contact with each other. In another example, some aspects may be described using the term “coupled” to indicate that two or more elements are in direct physical or electrical contact. The term “coupled,” however, also may mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

In some instances, one or more components may be referred to herein as “configured to,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that “configured to” can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

While particular aspects of the present subject matter described herein have been shown and described, it will be apparent to those skilled in the art that, based upon the teachings herein, changes and modifications may be made without departing from the subject matter described herein and its broader aspects and, therefore, the appended claims are to encompass within their scope all such changes and modifications as are within the true scope of the subject matter described herein. It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be

interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that when a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

In addition, even when a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flows are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more embodiments has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the

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precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more embodiments were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various embodiments and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

What is claimed is:

1. A surgical stapling assembly, comprising:
a shaft;
an articulation joint;
an electrical connector; and
an end effector attached to a distal end of the shaft, wherein the end effector is articulatable relative to the shaft, and wherein the shaft comprises:

a first jaw;
a second jaw movable relative to the first jaw; and
a replaceable staple cartridge assembly, comprising:

a cartridge body comprising a proximal end;
a plurality of staples; and

electrical contacts positioned at the proximal end of the cartridge body and electrically coupleable to the electrical connector upon installation of the replaceable staple cartridge assembly in the end effector and electrically decouplable from the electrical connector upon removal of the replaceable staple cartridge assembly from the end effector, wherein the electrical contacts are positioned on a proximal face of the cartridge body adjacent a longitudinal slot of the cartridge body, and wherein the electrical connector faces distally toward the proximal face of the cartridge body.

2. The surgical stapling assembly of claim 1, wherein the electrical connector comprises a flex cable attached to a sidewall of the first jaw.

3. The surgical stapling assembly of claim 2, wherein the flex cable comprises a coiled section extending across the articulation joint.

4. The surgical stapling assembly of claim 1, wherein the electrical connector is positioned distal to a pivot about which the second jaw is movable relative to the first jaw.

5. The surgical stapling assembly of claim 1, wherein the replaceable staple cartridge assembly further comprises a cartridge nose and a distal sensor plug positioned within the cartridge nose, wherein the cartridge nose comprises a distal-facing surface comprising distal electrical contacts electrically coupled with the electrical contacts positioned at the proximal end of the cartridge body, and wherein the distal sensor plug is in electrical communication with the distal electrical contacts.

6. The surgical stapling assembly of claim 5, wherein the distal sensor plug comprises a sensor and a processor.

7. A surgical stapling assembly, comprising:
a shaft; and

an end effector attached to the shaft, wherein the end effector comprises:

a first jaw;
a second jaw movable relative to the first jaw;
a channel;

an electrical connector; and

an anvil comprising a proximal end, wherein the proximal end comprises downwardly-extending flanges, wherein each downwardly-extending flange comprises a distal end, and wherein the electrical connector is proximal to the distal ends of the flanges;

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a replaceable staple cartridge assembly configured to be installed into the channel, wherein the replaceable staple cartridge assembly comprises:

a cartridge body, comprising:

a deck surface;

a proximal cartridge body end; and

a plurality of staple cavities defined in the deck surface; and

electrical contacts positioned at the proximal cartridge body end of the cartridge body and electrically coupleable to the electrical connector upon installation of the replaceable staple cartridge assembly into the channel and electrically decouplable from the electrical connector upon removal of the replaceable staple cartridge assembly from the channel, wherein the electrical connector comprises a flex cable attached to a sidewall of the channel, and wherein the flex cable comprises a coiled section.

8. The surgical stapling assembly of claim 7, wherein the electrical contacts are positioned on a proximal face of the cartridge body adjacent a longitudinal slot of the cartridge body, and wherein the electrical connector faces distally toward the proximal face of the cartridge body.

9. The surgical stapling assembly of claim 7, wherein the electrical connector is positioned distal to a pivot about which the second jaw is movable relative to the first jaw.

10. The surgical stapling assembly of claim 7, wherein the replaceable staple cartridge assembly further comprises a cartridge nose and a distal sensor plug positioned within the cartridge nose, wherein the cartridge nose comprises a distal-facing surface comprising distal electrical contacts electrically coupled with the electrical contacts positioned at the proximal cartridge body end of the cartridge body, and wherein the distal sensor plug is in electrical communication with the distal electrical contacts.

11. The surgical stapling assembly of claim 10, wherein the distal sensor plug comprises a sensor and a processor.

12. A replaceable staple cartridge assembly for use with a surgical stapling end effector having a cartridge channel and an electrical connector, wherein the replaceable staple cartridge assembly is configured to be installed into the cartridge channel, the replaceable staple cartridge assembly comprising:

a cartridge body comprising:

a proximal end;

a plurality of staples; and

electrical contacts positioned at the proximal end of the cartridge body and electrically coupleable to the electrical connector of the end effector upon installation of the replaceable staple cartridge assembly into the cartridge channel of the end effector and electrically decouplable from the electrical connector upon removal of the replaceable staple cartridge assembly from the cartridge channel, wherein the replaceable staple cartridge assembly further comprises a cartridge nose and a distal sensor plug positioned within the cartridge nose, wherein the cartridge nose comprises a distal-facing surface comprising distal electrical contacts electrically coupled with the electrical contacts positioned at the proximal end of the cartridge body, and wherein the distal sensor plug is in electrical communication with the distal electrical contacts.

13. The replaceable staple cartridge assembly of claim 12, wherein the electrical contacts are positioned on a proximal face of the cartridge body adjacent a longitudinal slot of the

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cartridge body, and wherein the electrical connector faces distally toward the proximal face of the cartridge body.

14. The replaceable staple cartridge assembly of claim **12**, wherein the distal sensor plug comprises a sensor and a processor.

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